

REPAIR HVAC WIDE BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

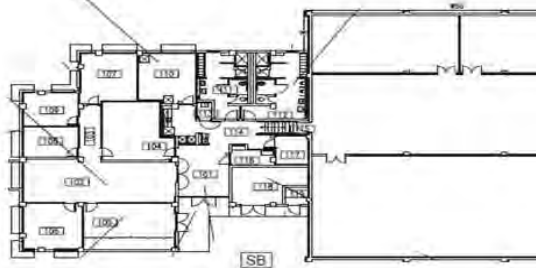
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-81	TYPE OF MATERIAL	:	Vinyl Cove Base w/Mastic
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Ryota Tomoyose
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119



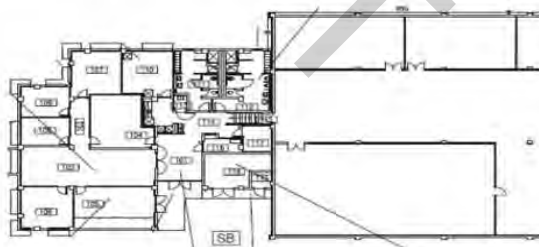
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-82	TYPE OF MATERIAL	:	Vinyl Cove Base w/Mastic
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Ryota Tomoyose
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

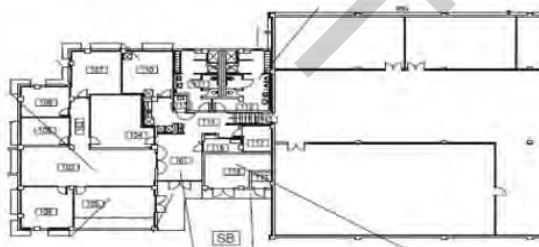
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-83	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



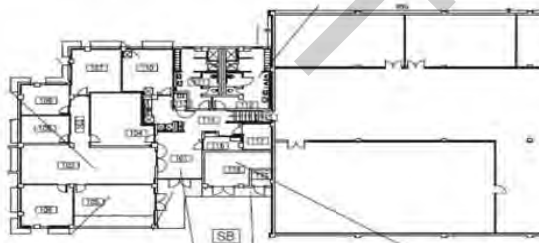
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-84	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

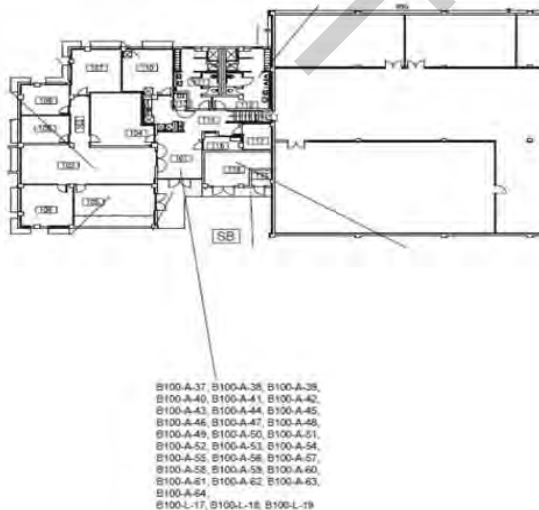
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-85	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



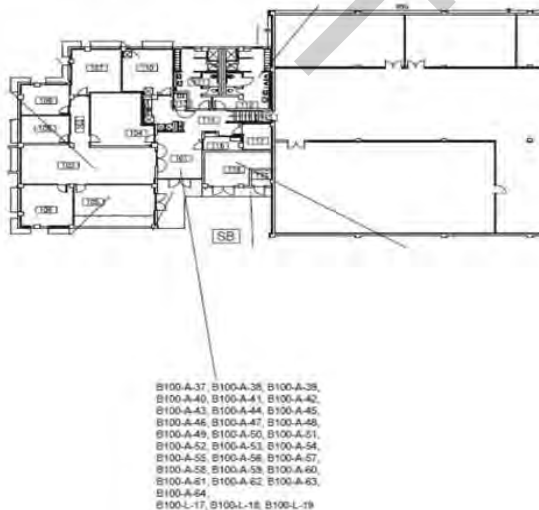
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

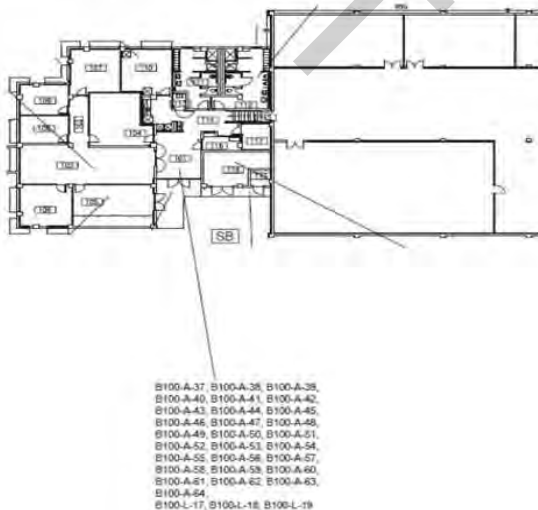
SAMPLE ID NO.	: B100-A-86	TYPE OF MATERIAL	: Wall Insulation
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-87	TYPE OF MATERIAL	:	Wall Insulation
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Ryota Tomoyose
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-88	TYPE OF MATERIAL	: Wall Insulation
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-89	TYPE OF MATERIAL	: Dampproofing Material
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-90	TYPE OF MATERIAL	: Vinyl Floor Tile w/Mastic
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-91	TYPE OF MATERIAL	: Vinyl Floor Tile w/Mastic
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-68, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-92	TYPE OF MATERIAL	:	Vinyl Floor Tile w/Mastic
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-93	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-94	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-95	TYPE OF MATERIAL	:	Concrete Floor Slab
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
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B100-A-95,
B100-L-23, B100-L-24

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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-96	TYPE OF MATERIAL	: Ceramic Tile Tile w/Grout
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

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NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-97	TYPE OF MATERIAL	:	Ceramic Tile Tile w/Grout
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-68, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-98	TYPE OF MATERIAL	: Ceramic Tile Tile w/Grout
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-99	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-100	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-101	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-105	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-106	TYPE OF MATERIAL	:	Concrete Floor Slab
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-107	TYPE OF MATERIAL :	Concrete Floor Slab
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takafumi Noborikawa
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-108	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-109	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-110	TYPE OF MATERIAL :	Ceiling Board
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takafumi Noborikawa
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-111	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-68, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-112	TYPE OF MATERIAL	:	Ceiling Insulation
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-113	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-114	TYPE OF MATERIAL	: Concrete Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-58, B100-A-59, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

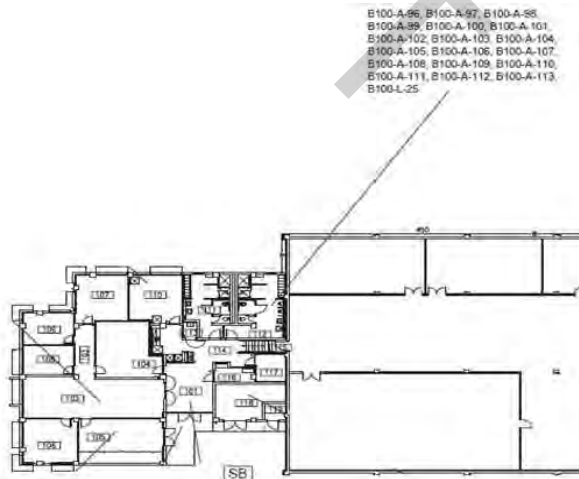
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-115	TYPE OF MATERIAL :	Concrete Slab
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



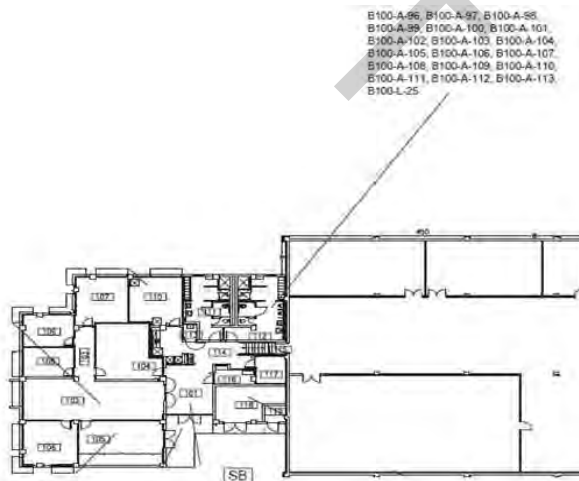
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-116	TYPE OF MATERIAL	: Concrete Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



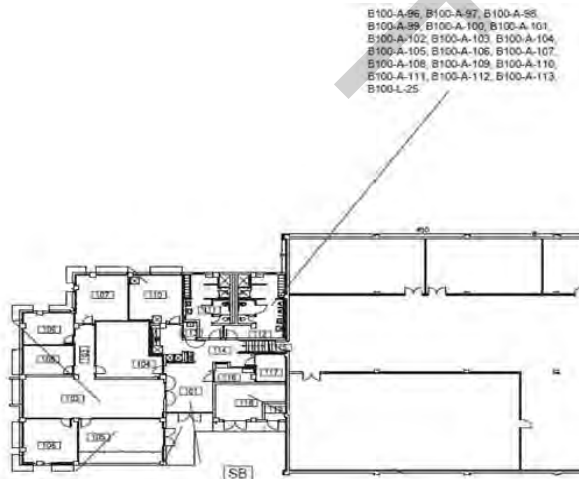
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-117	TYPE OF MATERIAL	: Floor Mortar Repair Seal
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 110	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



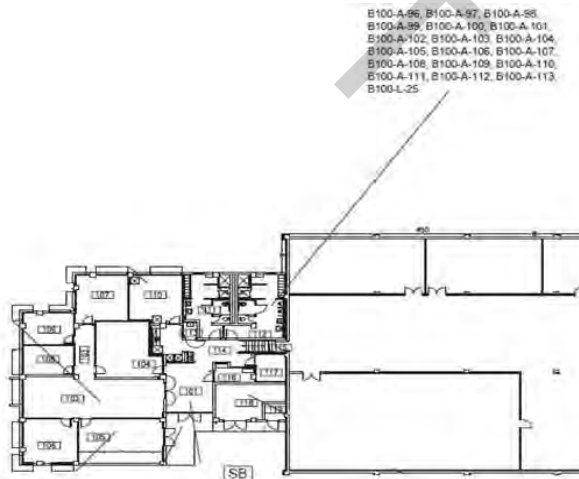
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-118	TYPE OF MATERIAL	: Floor Mortar Repair Seal
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 110	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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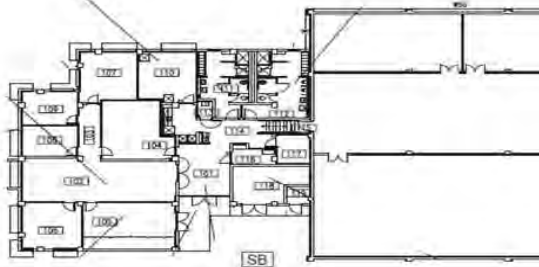
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-119	TYPE OF MATERIAL	: Floor Mortar Repair Seal
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 110	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119



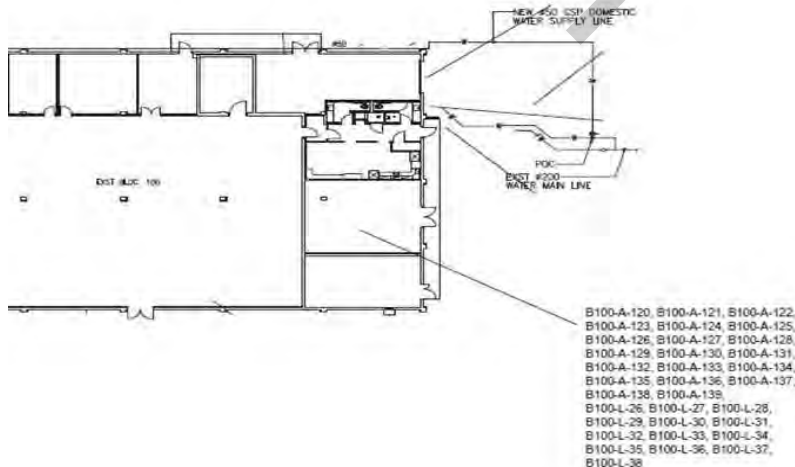
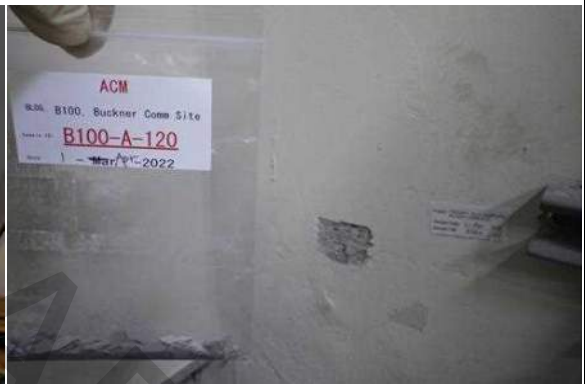
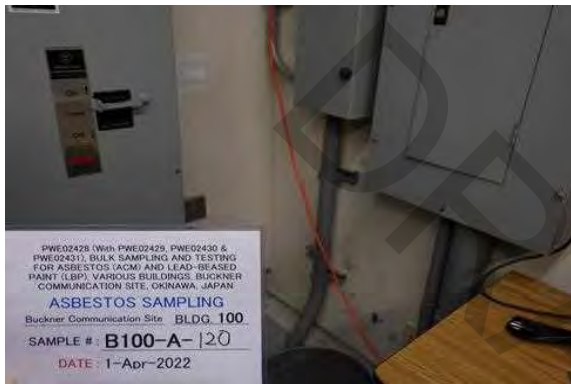
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-120	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: Chrysotile (3%)
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



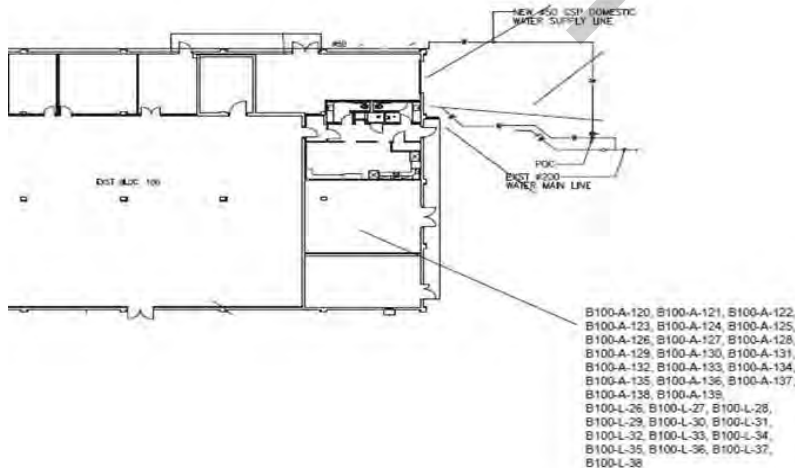
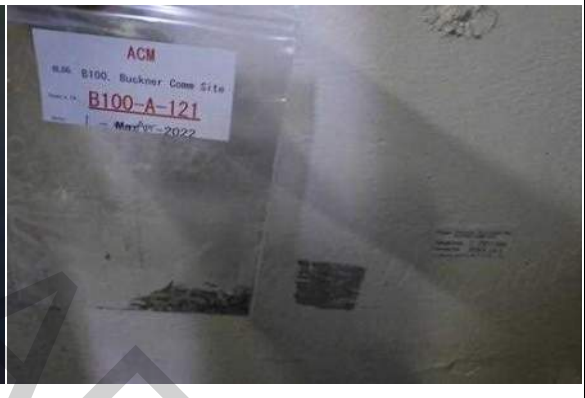
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ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-121	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



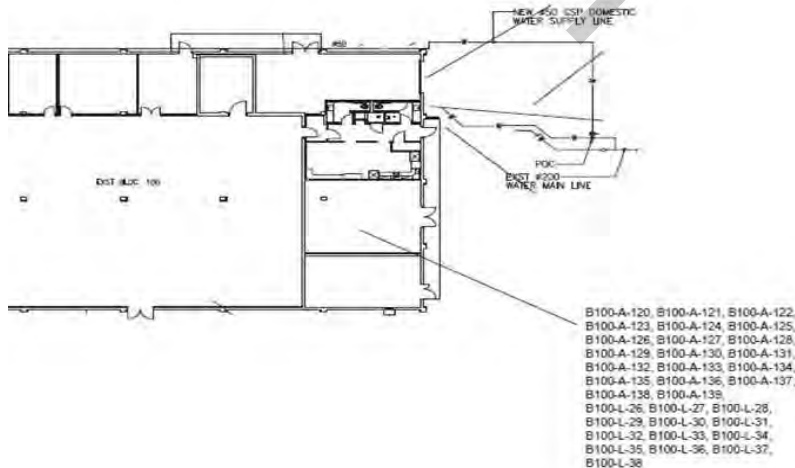
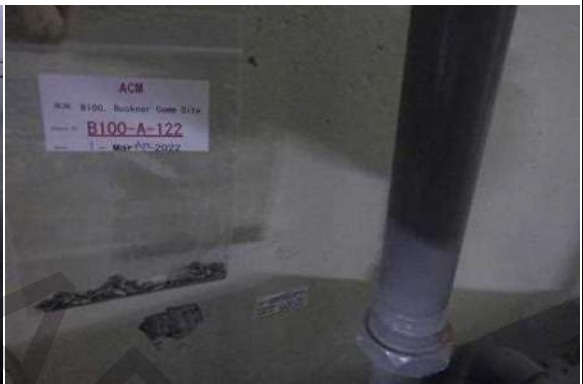
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-122	TYPE OF MATERIAL	:	Interior Concrete Wall
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (1%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takuya Niizato
ROOM NO./NAME	:	Elec Room	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



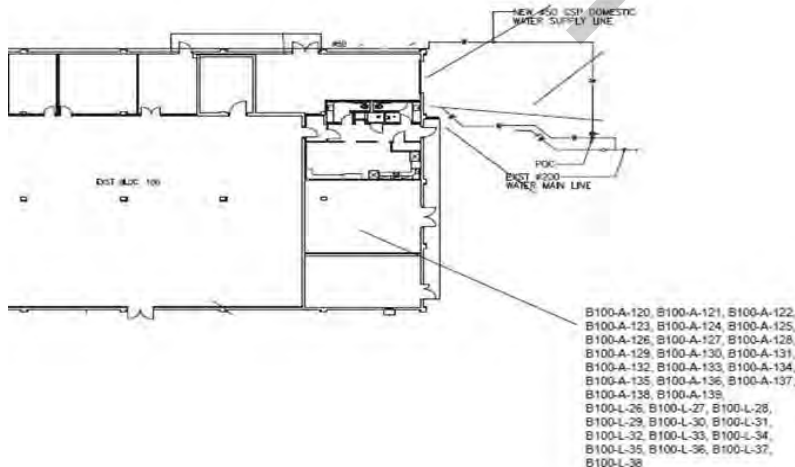
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-123	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



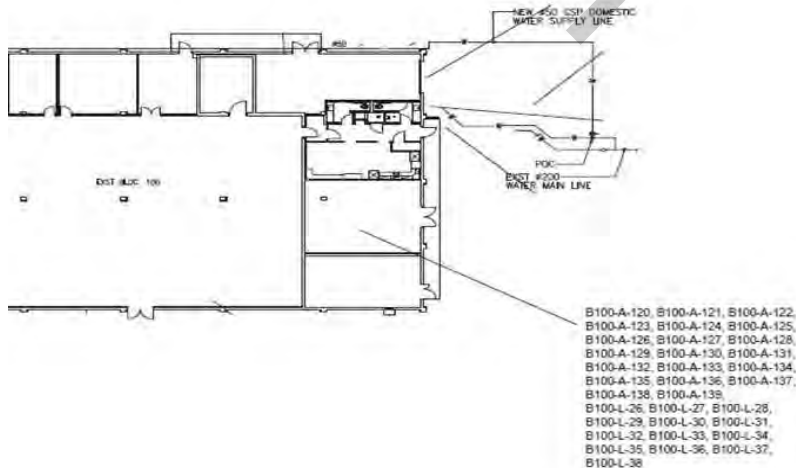
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-124	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



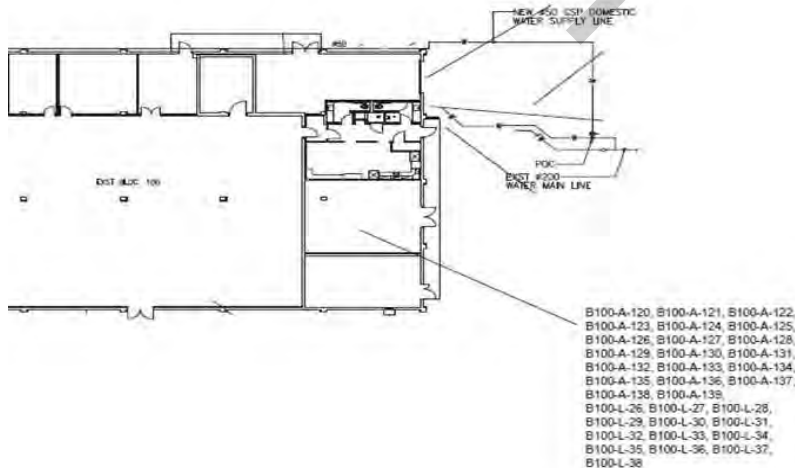
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-126	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



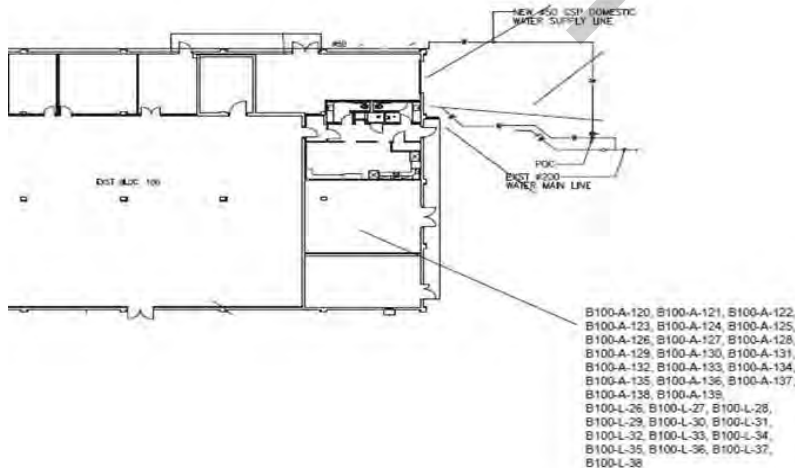
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-127	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



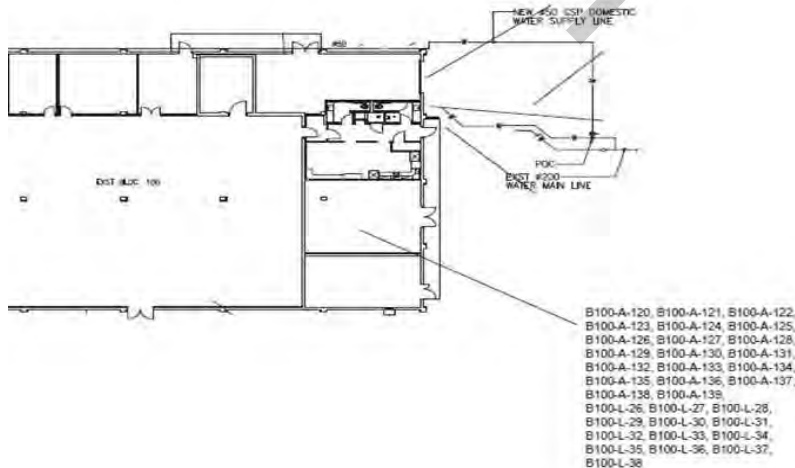
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-128	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



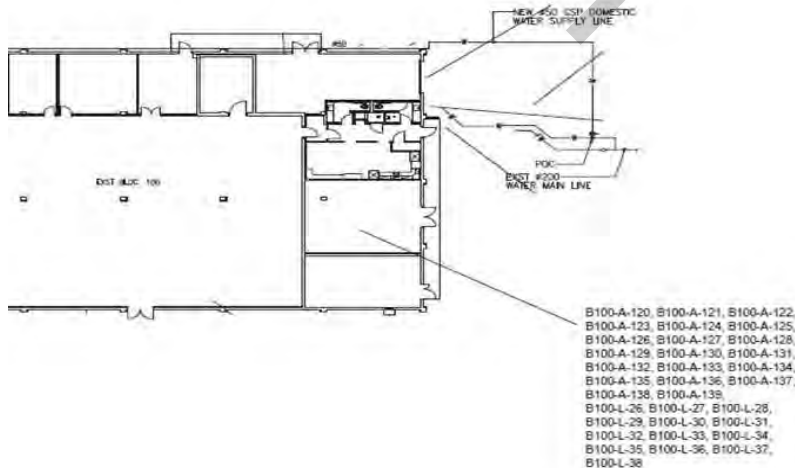
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-129	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takuya Niizato
ROOM NO./NAME :	Elec Room	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



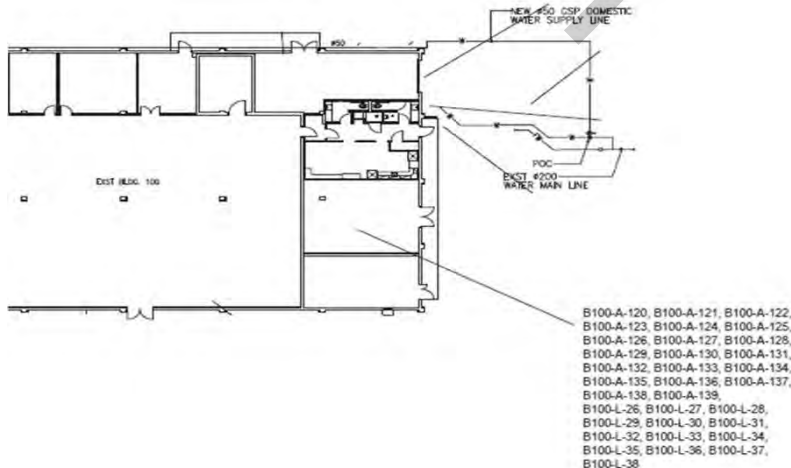
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-130	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



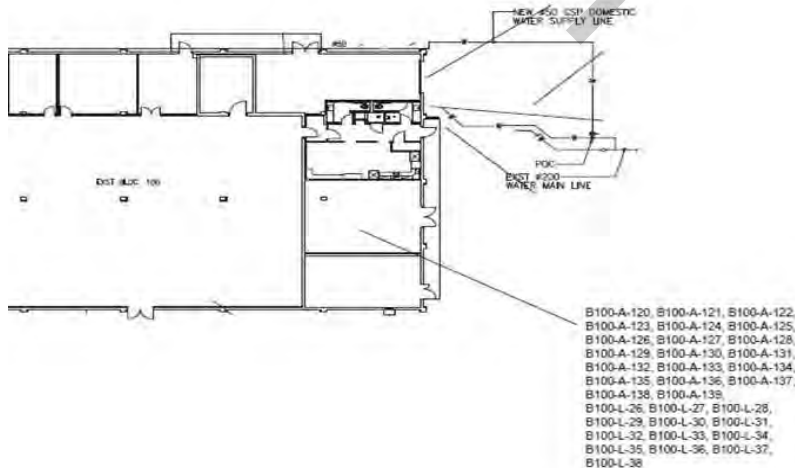
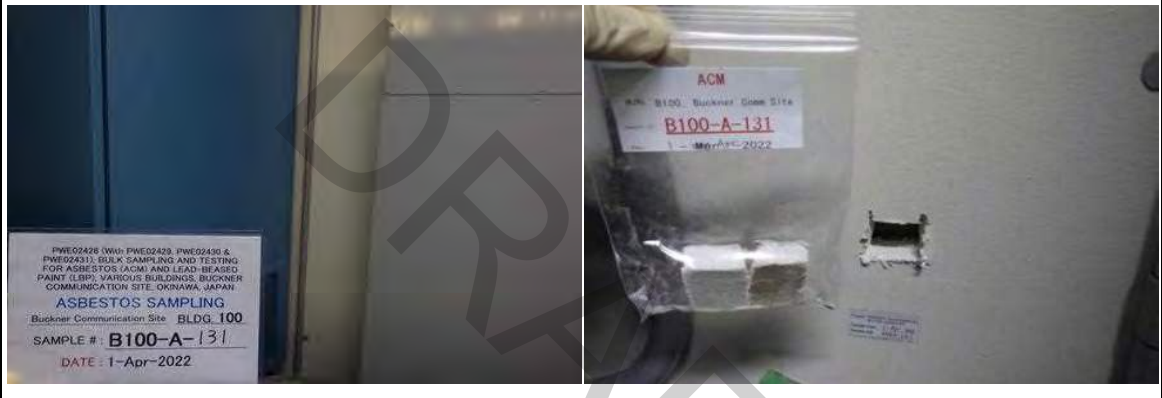
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-131	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



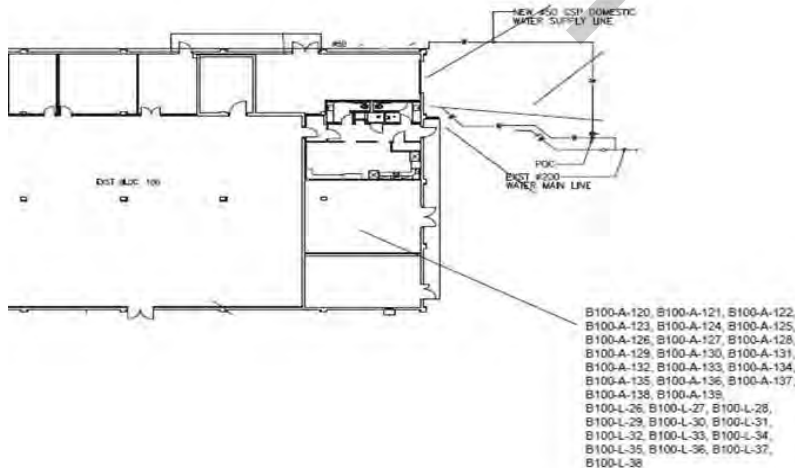
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-132	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



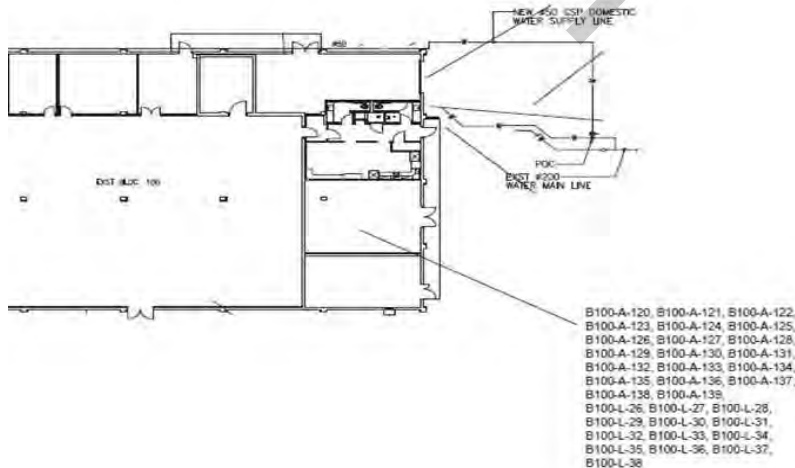
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-133	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



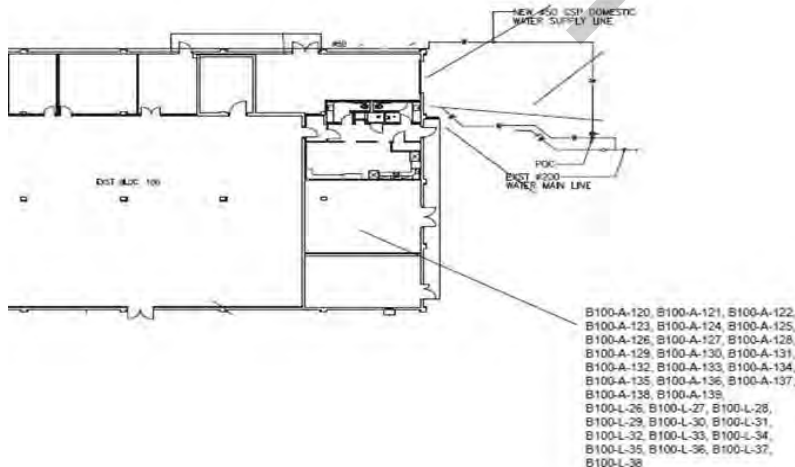
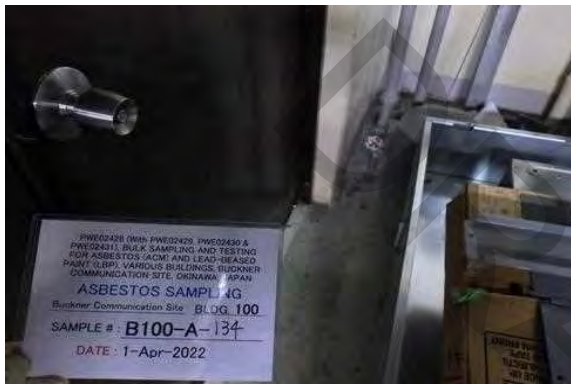
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-134	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



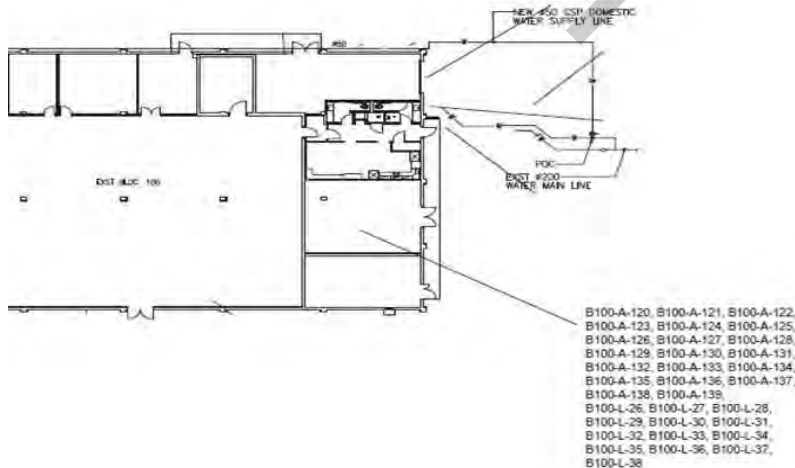
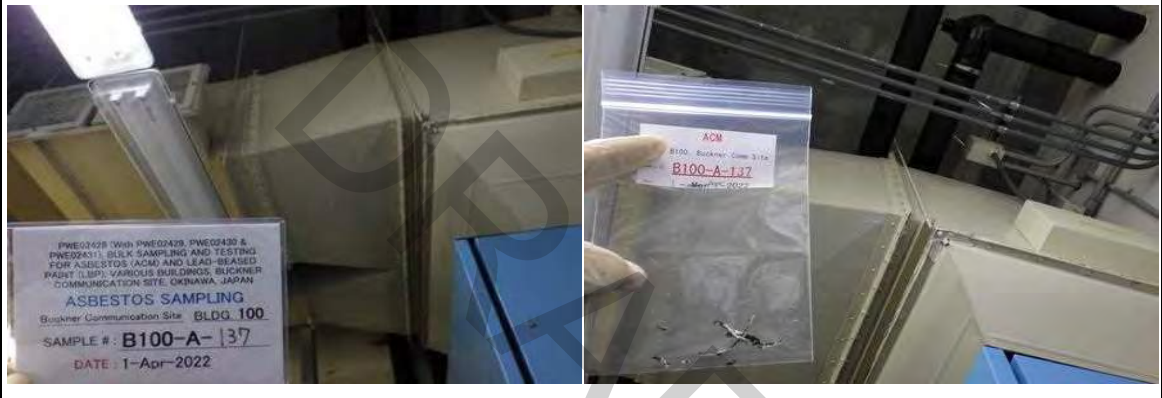
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-137	TYPE OF MATERIAL	: Duct Flange Gasket
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
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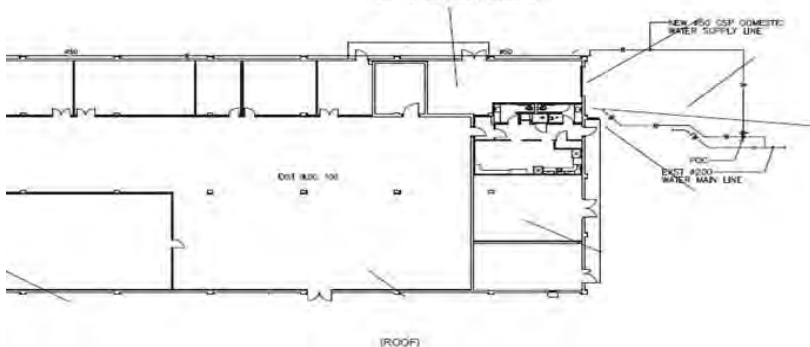
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-140	TYPE OF MATERIAL	:	Interior Concrete Wall
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

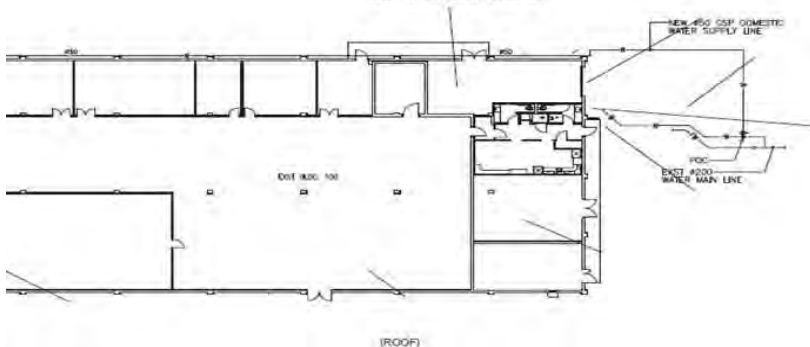
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-141	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

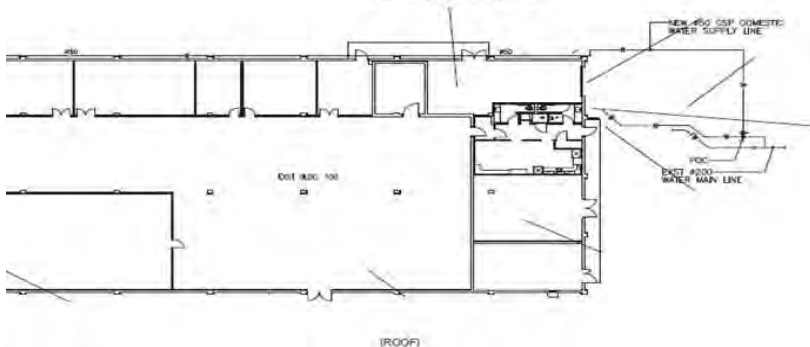
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-142	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: Chrysotile (1%)
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

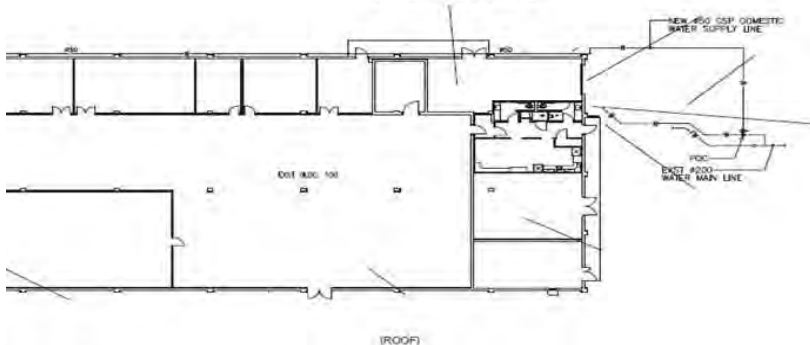
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-143	TYPE OF MATERIAL	:	Ceiling Slab
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



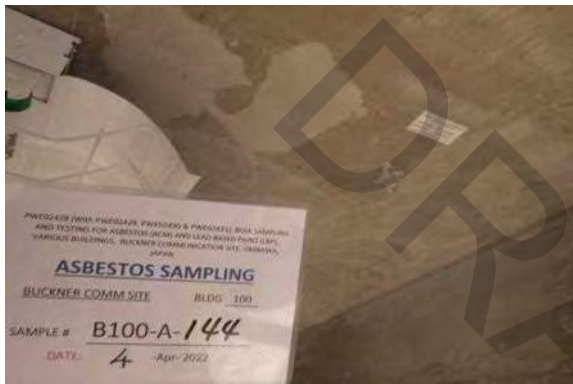
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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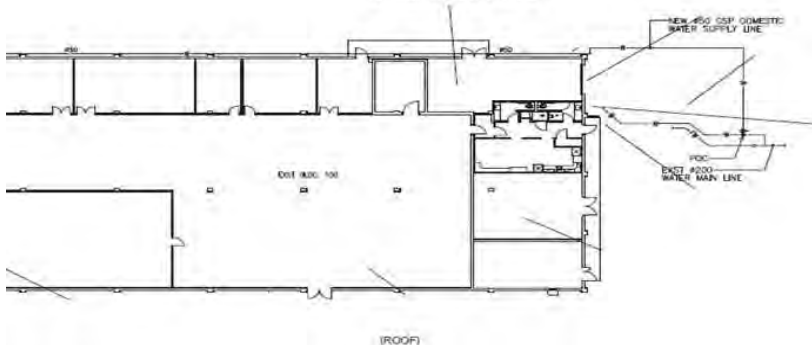
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-144	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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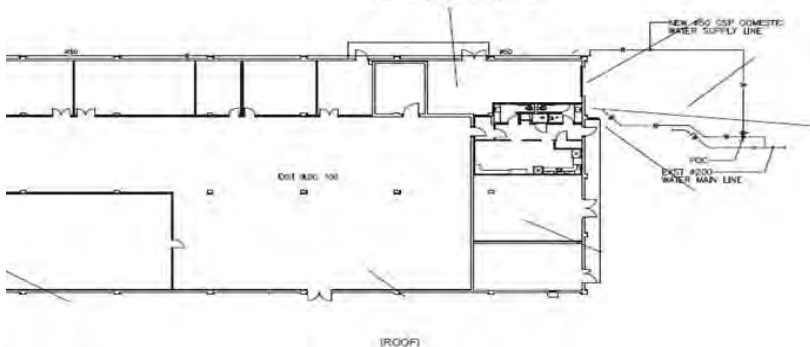
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-145	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
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B100-A-161, B100-A-162, B100-A-163,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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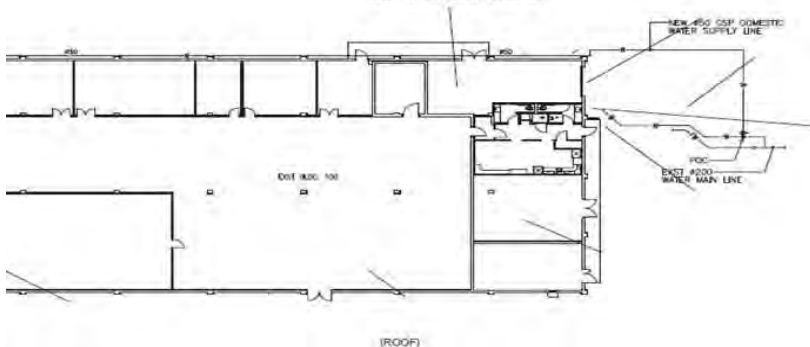
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-149	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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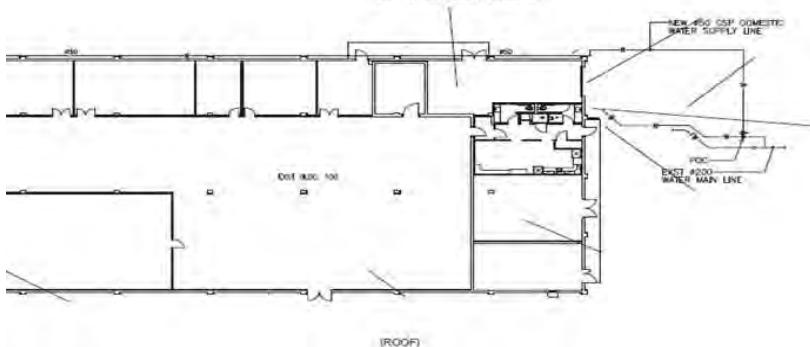
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-150	TYPE OF MATERIAL	:	Wallboard
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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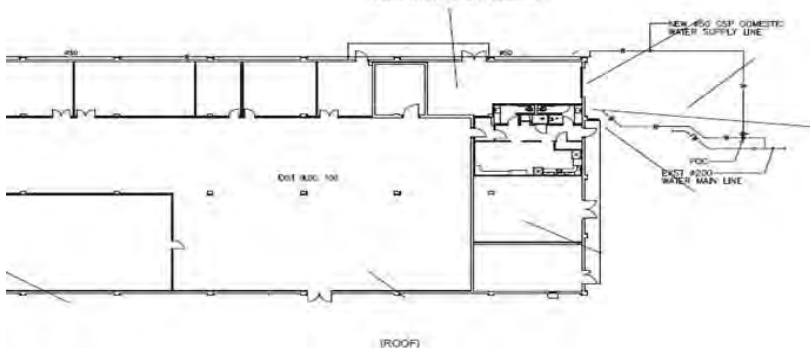
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-151	TYPE OF MATERIAL	:	Wallboard
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-A-167, B100-A-168, B100-A-169,
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B100-L-39, B100-L-40, B100-L-41,
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B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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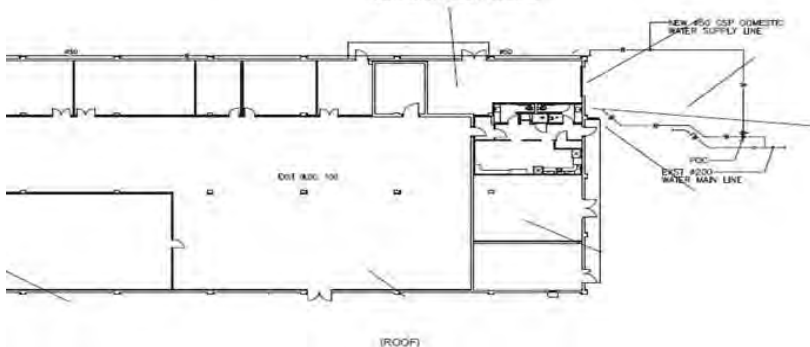
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-152	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (1%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

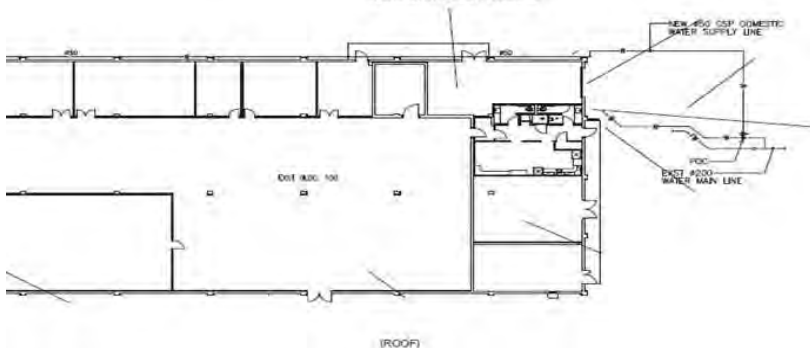
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-153	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (1%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-A-161, B100-A-162, B100-A-163,
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B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
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25SF0004

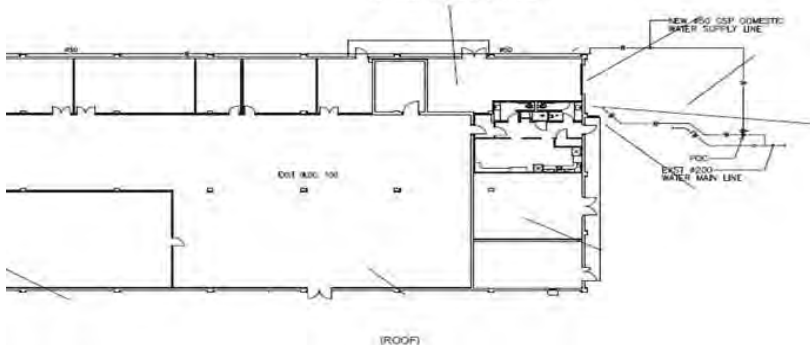
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-154	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
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B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

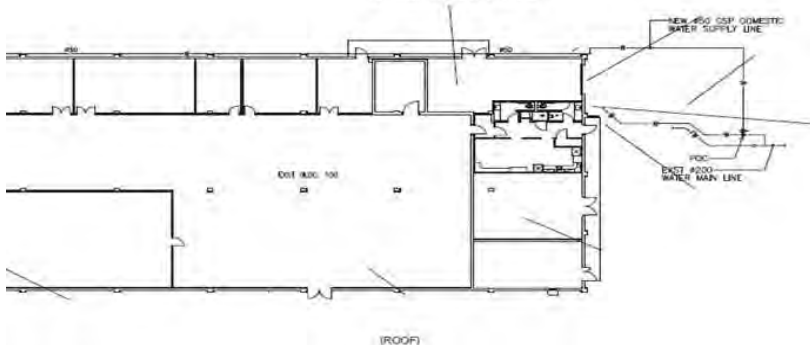
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-155	TYPE OF MATERIAL	: Wall Mortar Repair Seal
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

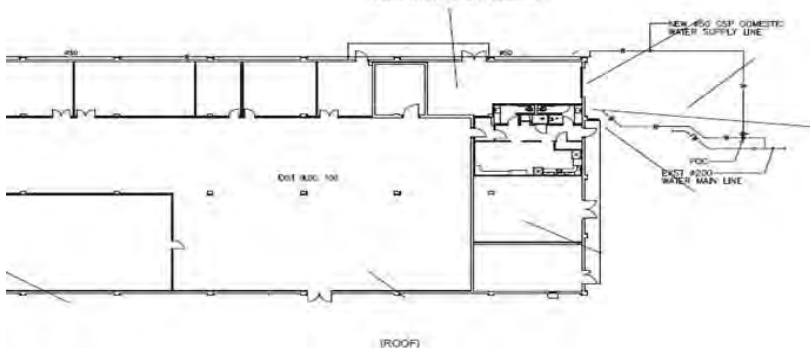
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-156	TYPE OF MATERIAL	:	Floor Mortar Repair Seal
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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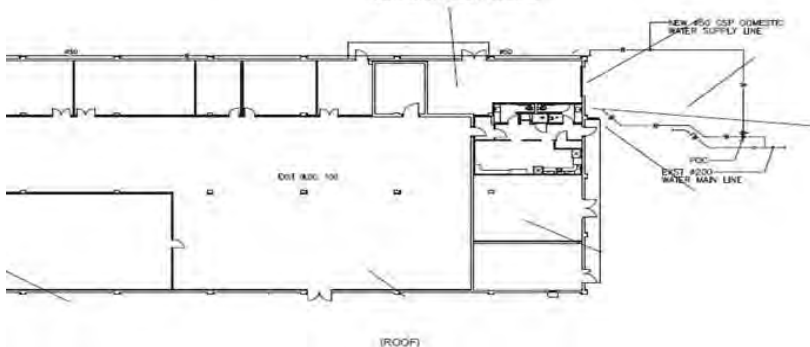
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-157	TYPE OF MATERIAL	: Duct Insulation
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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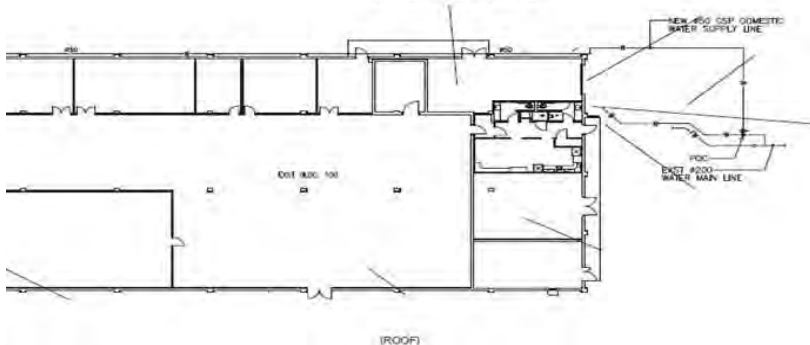
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-158	TYPE OF MATERIAL	: Flexible Duct Joint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Interior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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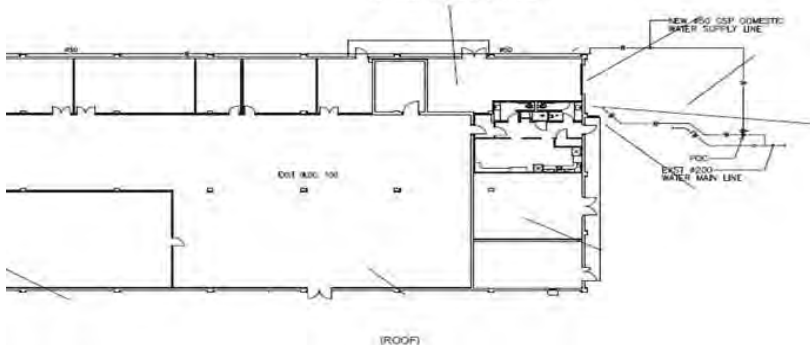
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-159	TYPE OF MATERIAL :	Concrete Pad
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	AHU, Mech	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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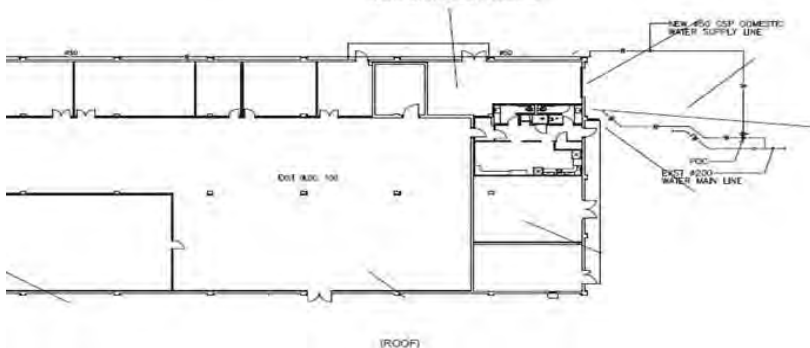
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-160	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: AHU, Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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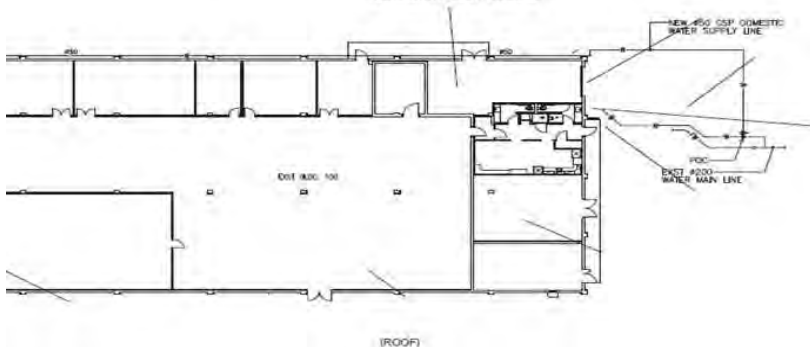
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-161	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	AHU, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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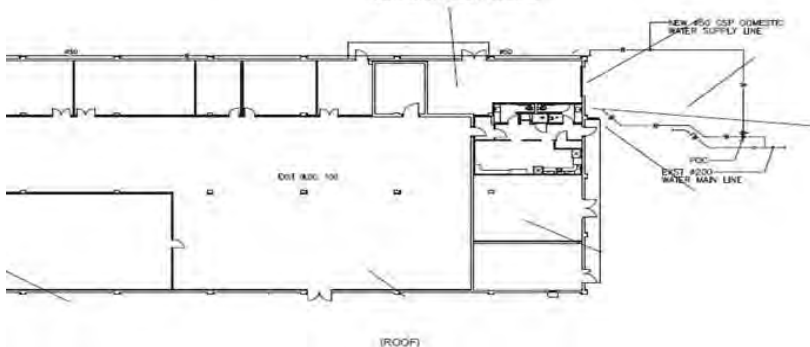
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-162	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Compressor, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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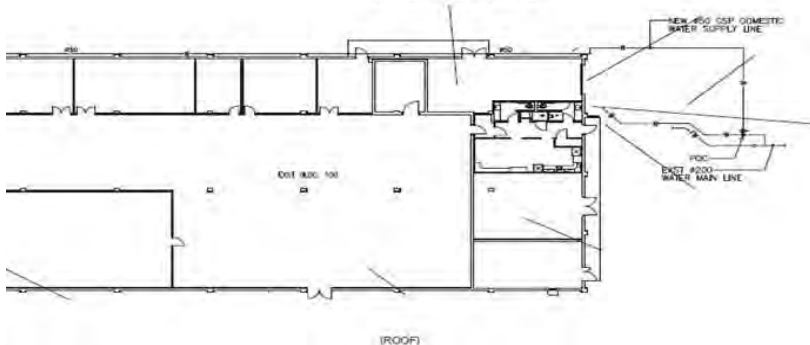
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-163	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Compressor, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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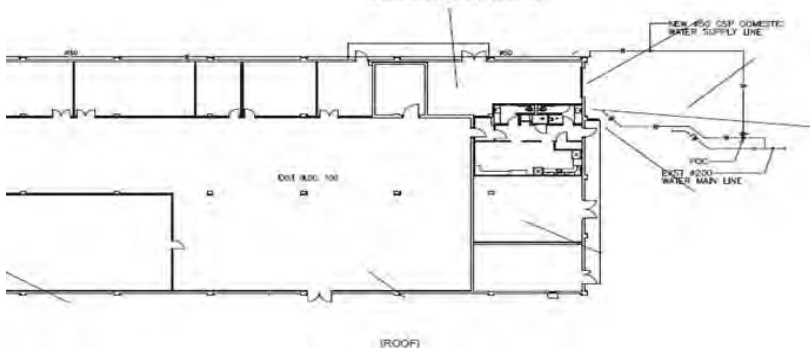
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-164	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Compressor, Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
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ASBESTOS ANALYSIS REPORT

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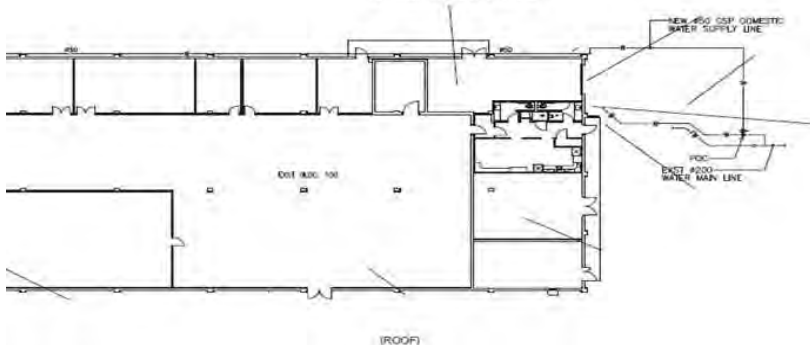
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-165	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Pump, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142, B100-A-143, B100-A-144, B100-A-145, B100-A-146, B100-A-147, B100-A-148, B100-A-149, B100-A-150, B100-A-151, B100-A-152, B100-A-153, B100-A-154, B100-A-155, B100-A-156, B100-A-157, B100-A-158, B100-A-159, B100-A-160, B100-A-161, B100-A-162, B100-A-163, B100-A-164, B100-A-165, B100-A-166, B100-A-167, B100-A-168, B100-A-169, B100-A-170, B100-A-171, B100-L-39, B100-L-40, B100-L-41, B100-L-42, B100-L-43, B100-L-44, B100-L-45, B100-L-46, B100-L-47, B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

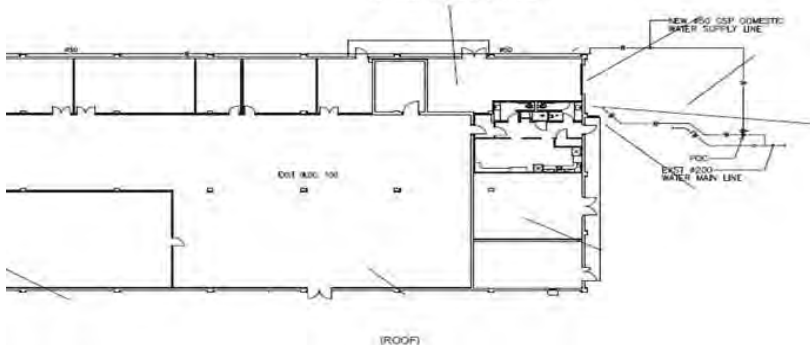
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-166	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Pump, Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

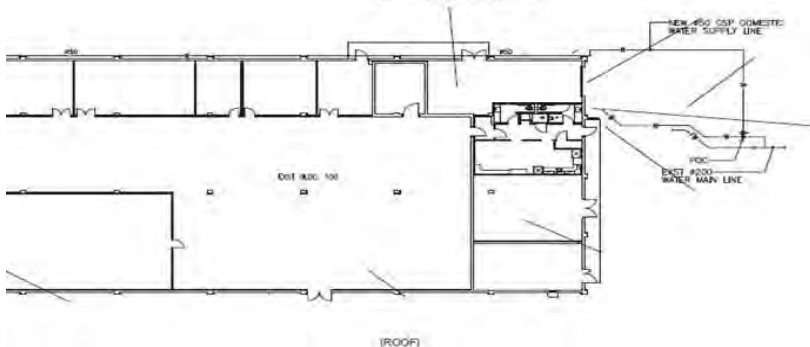
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-167	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Pump, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

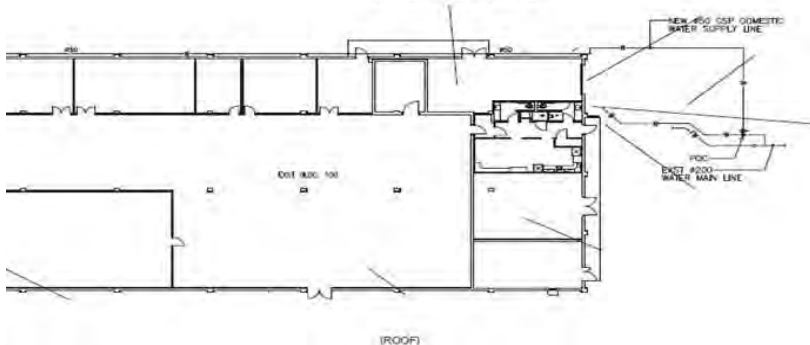
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-168	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Generator, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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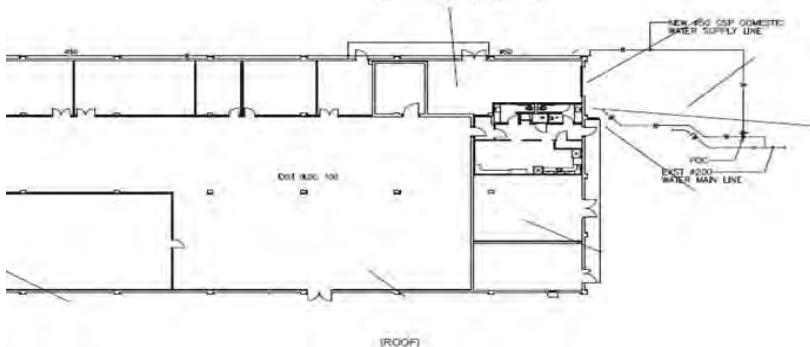
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-169	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Generator, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169;
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

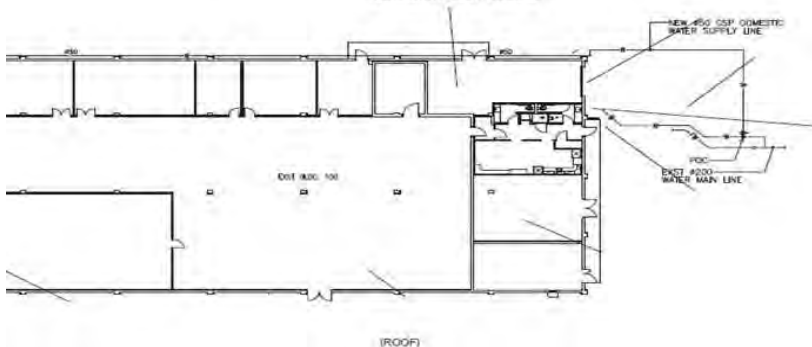
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-170	TYPE OF MATERIAL	:	Concrete Pad
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Generator, Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

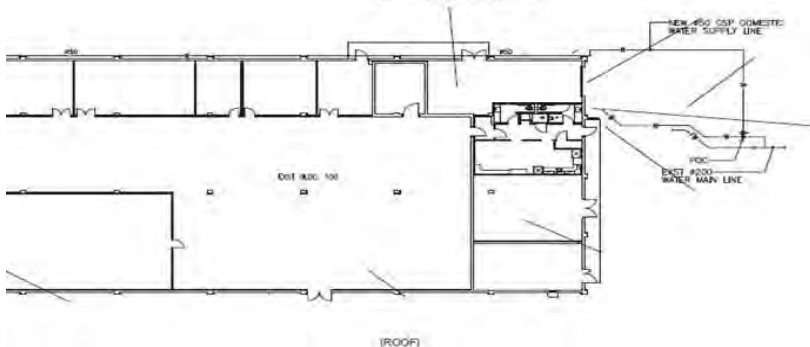
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-171	TYPE OF MATERIAL	: Pipe Flange Gasket
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



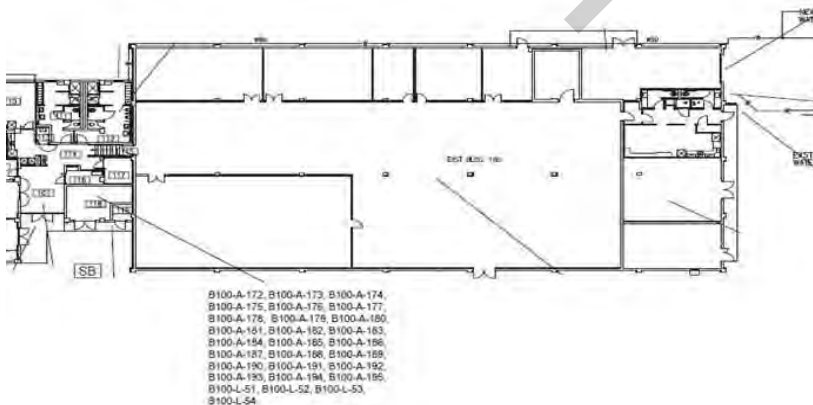
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-172	TYPE OF MATERIAL :	Interior Concrete Wall
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



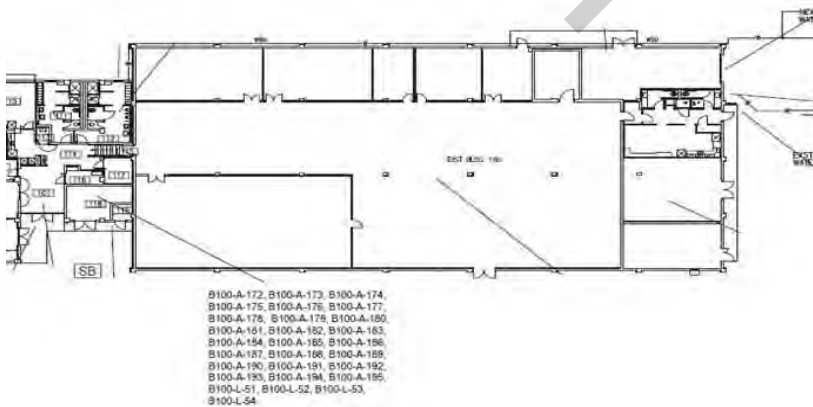
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-173	TYPE OF MATERIAL :	Interior Concrete Wall
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



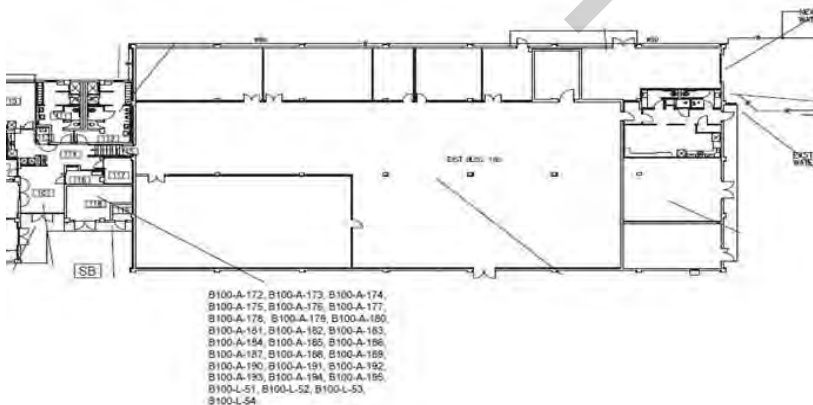
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-174	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



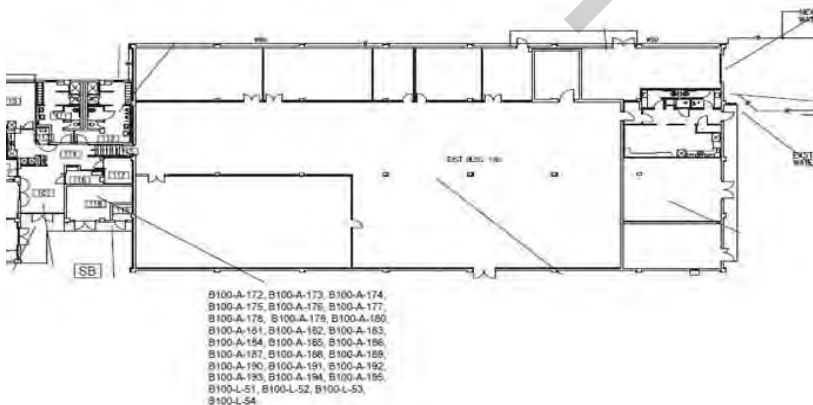
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-175	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



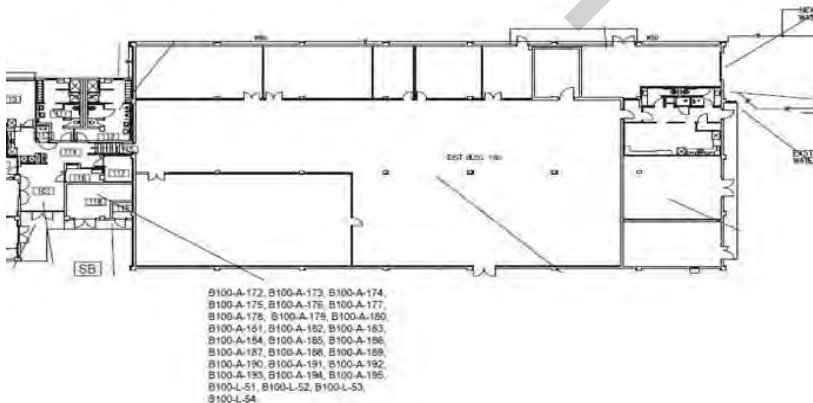
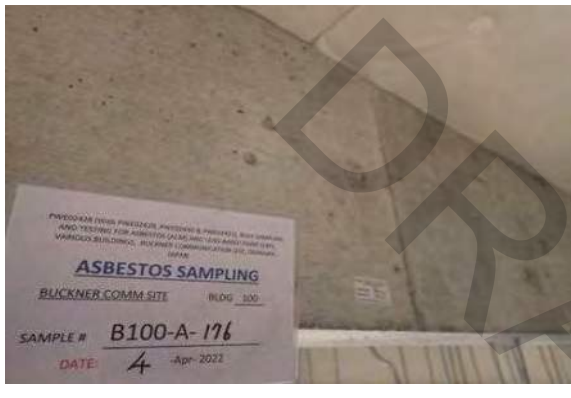
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-176	TYPE OF MATERIAL :	Ceiling Slab
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



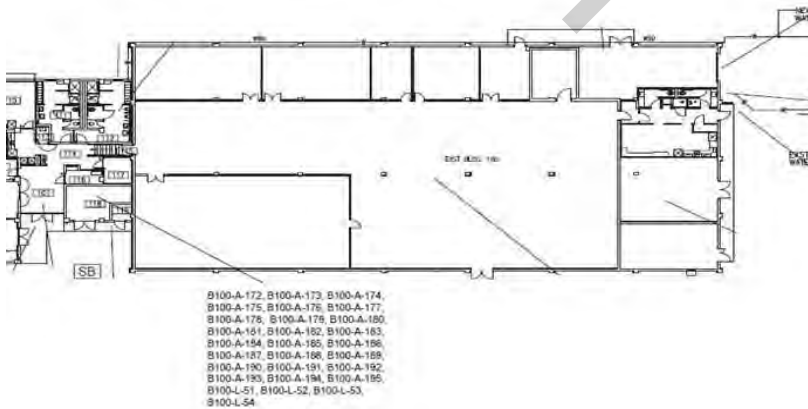
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-177	TYPE OF MATERIAL	:	Ceiling Slab
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tetsuya Yonaha
ROOM NO./NAME	:	Rm 118	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



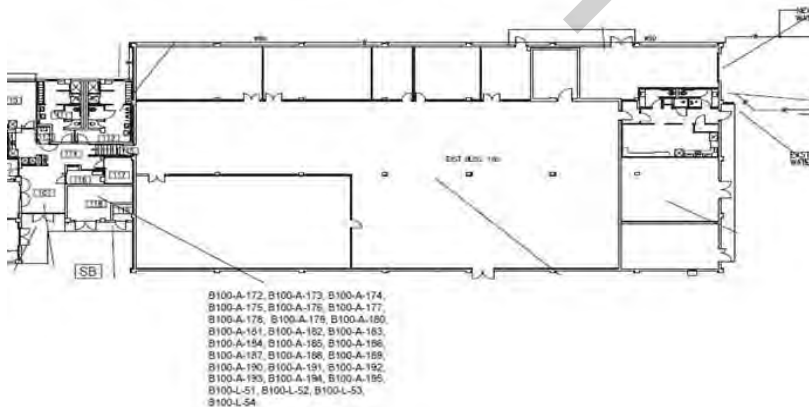
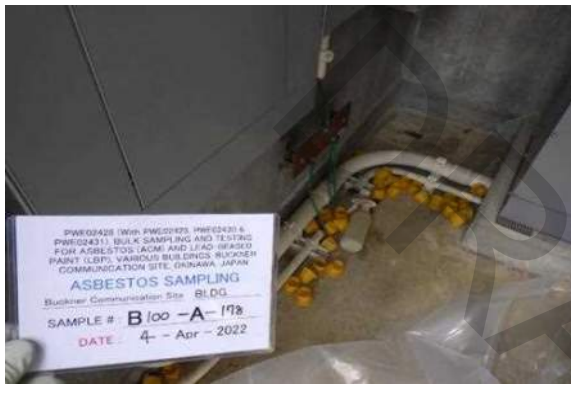
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-178	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



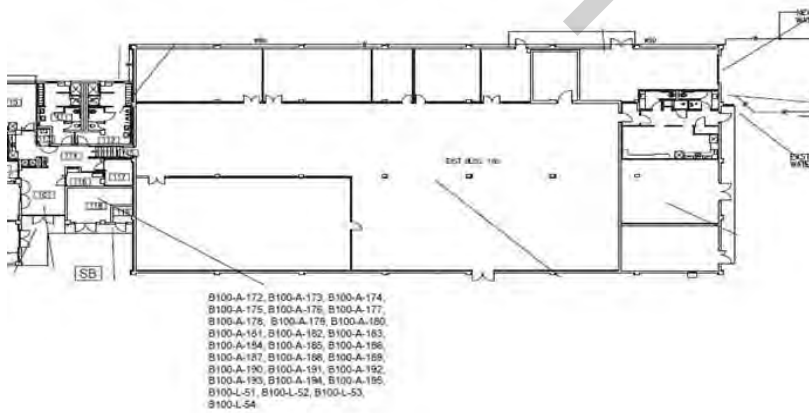
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-179	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



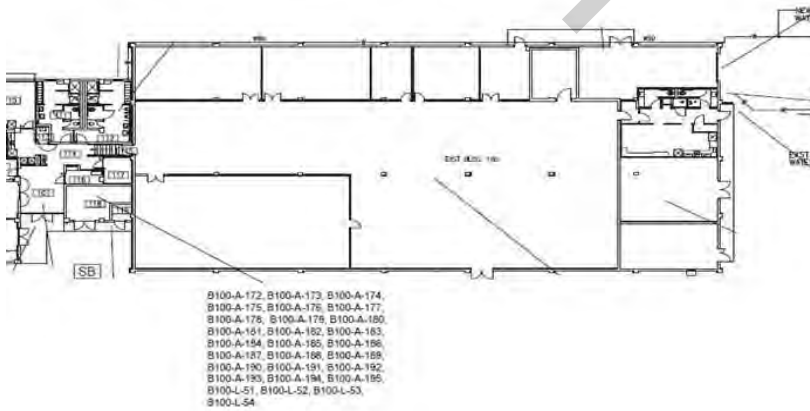
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-180	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Elec Room	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



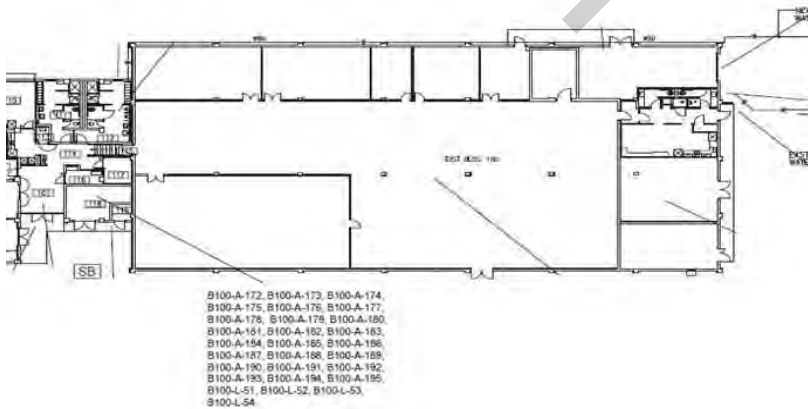
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-181	TYPE OF MATERIAL :	Concrete Floor Slab
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



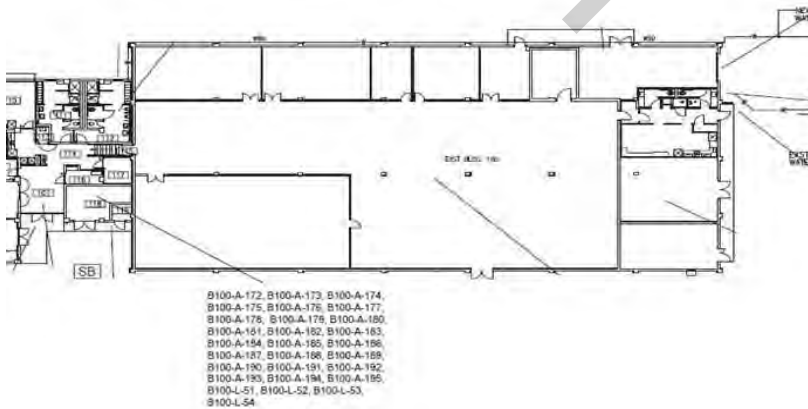
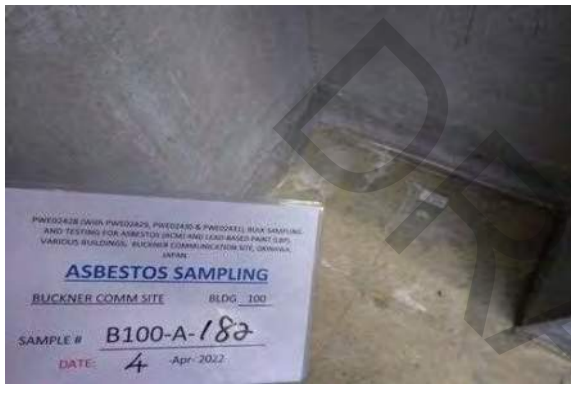
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-182	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



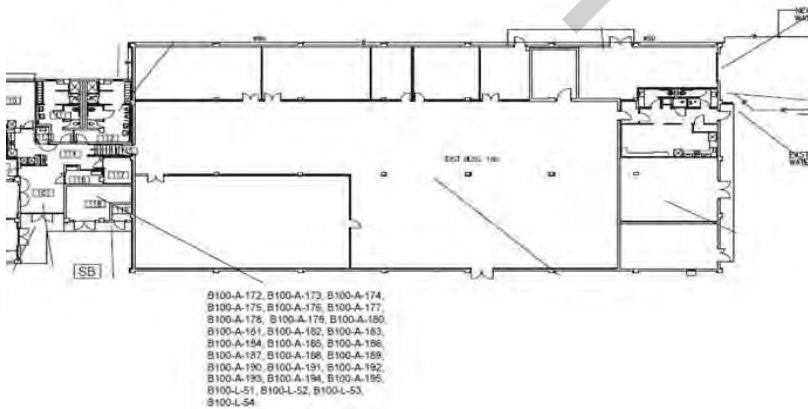
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-183	TYPE OF MATERIAL :	Concrete Floor Slab
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



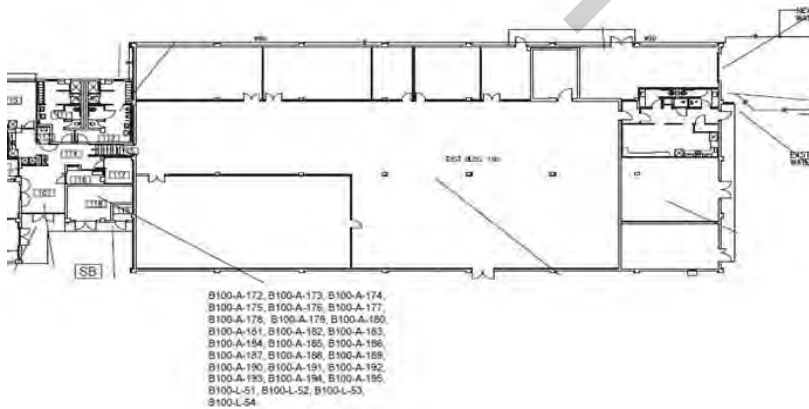
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-184	TYPE OF MATERIAL :	Wall Mortar Repair Seal
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



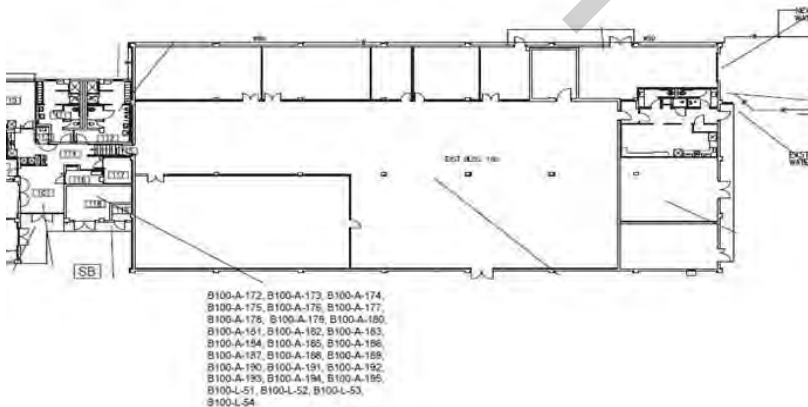
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-185	TYPE OF MATERIAL :	Floor Mortar Repair Seal
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



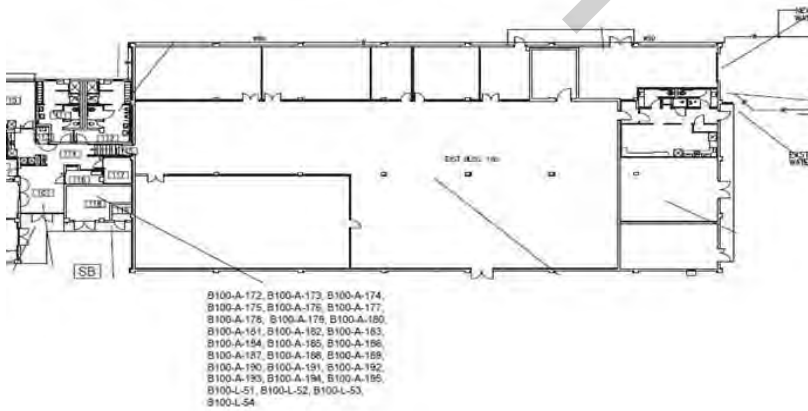
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-186	TYPE OF MATERIAL	: Duct Flange Gasket
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



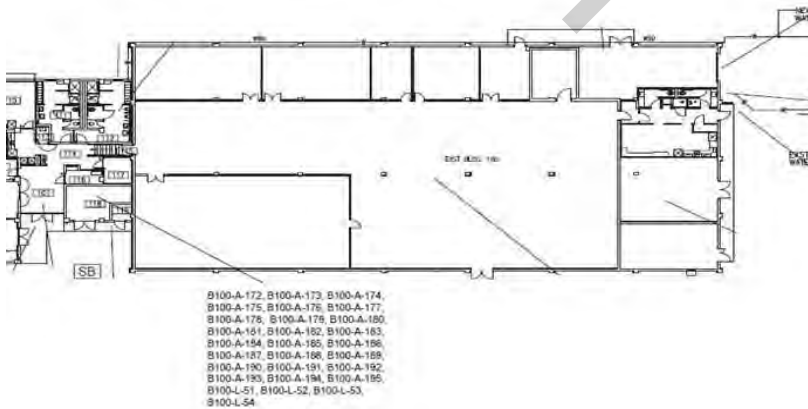
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-188	TYPE OF MATERIAL :	Concrete Pad
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



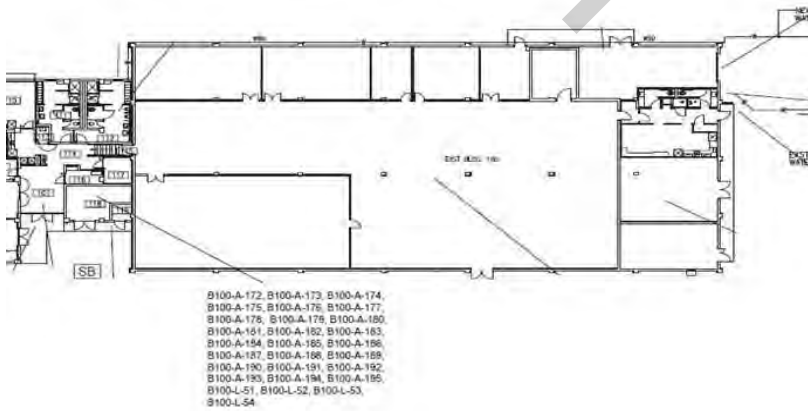
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-189	TYPE OF MATERIAL :	Concrete Pad
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



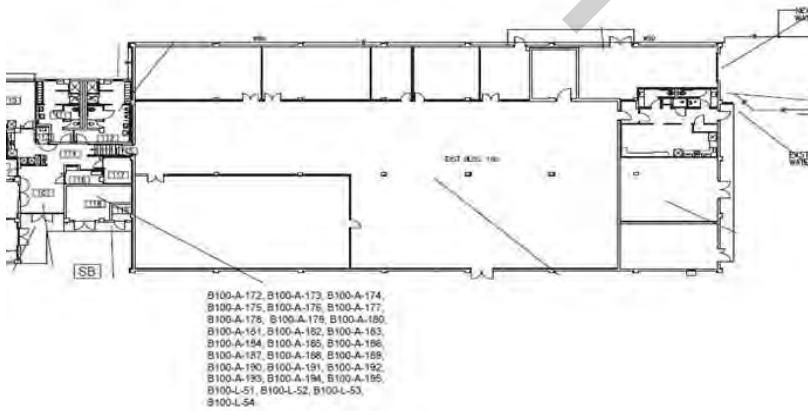
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-190	TYPE OF MATERIAL :	Concrete Pad
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



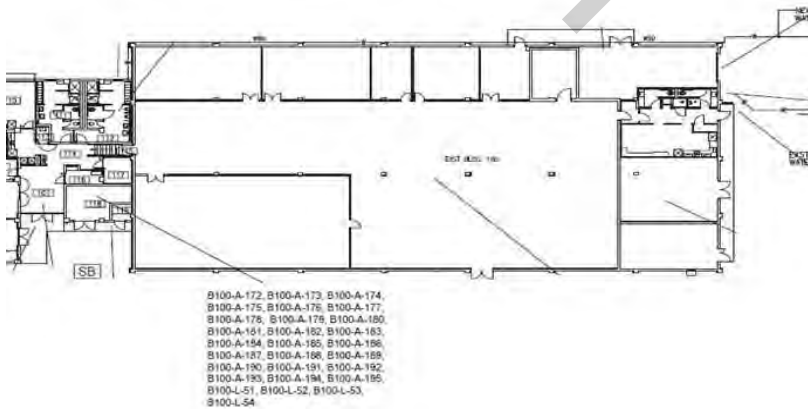
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-191	TYPE OF MATERIAL :	Wall Insulation
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



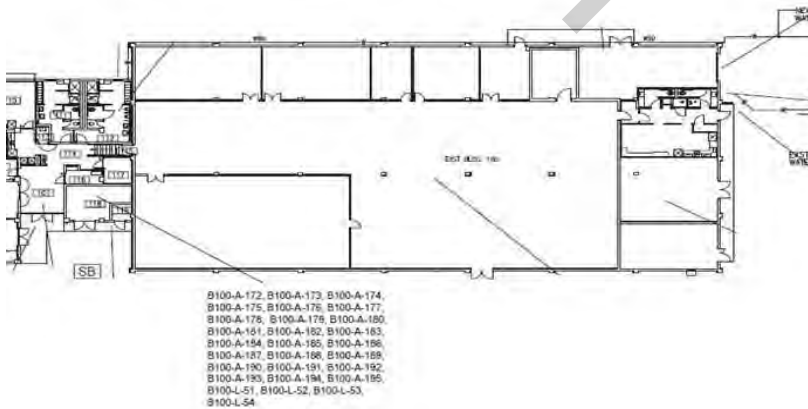
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-192	TYPE OF MATERIAL	: Wall Insulation
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



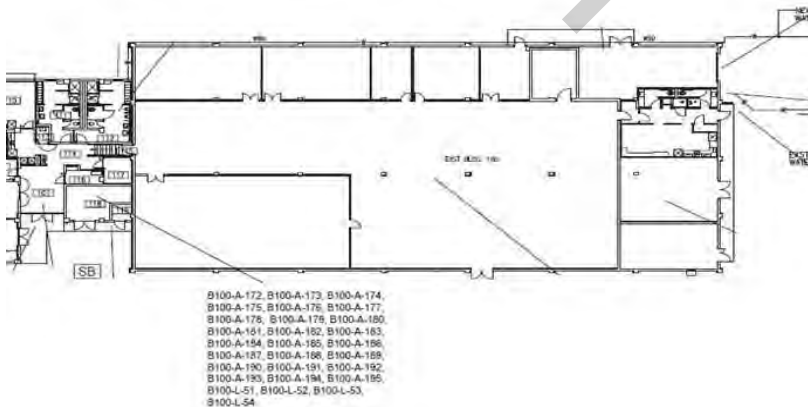
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-193	TYPE OF MATERIAL :	Wall Insulation
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



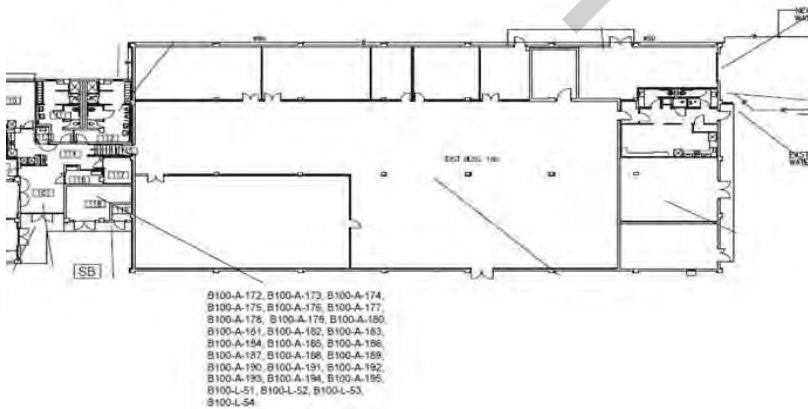
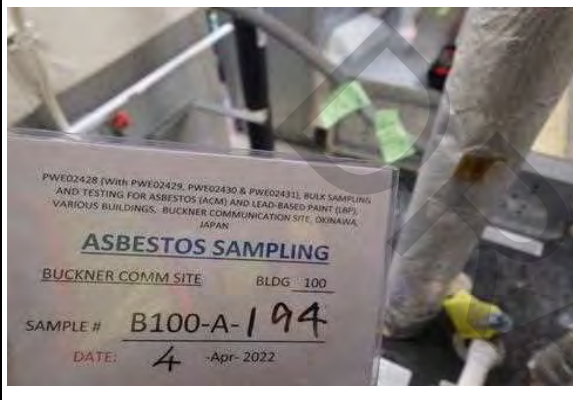
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-194	TYPE OF MATERIAL :	Pipe Insulation
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Straight, Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



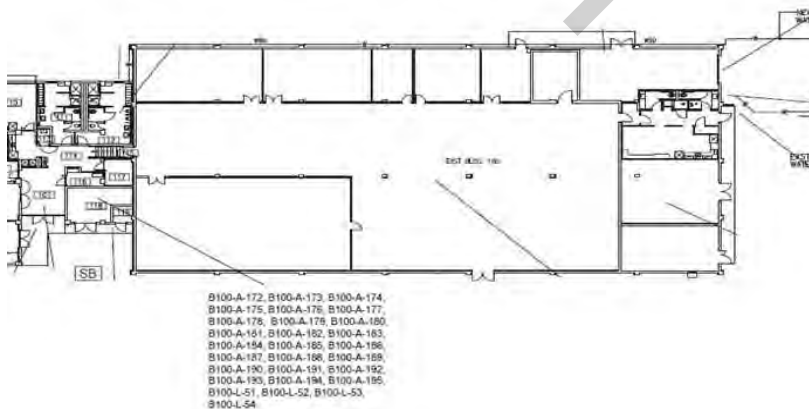
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-195	TYPE OF MATERIAL	: Pipe Insulation
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Elbow, Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



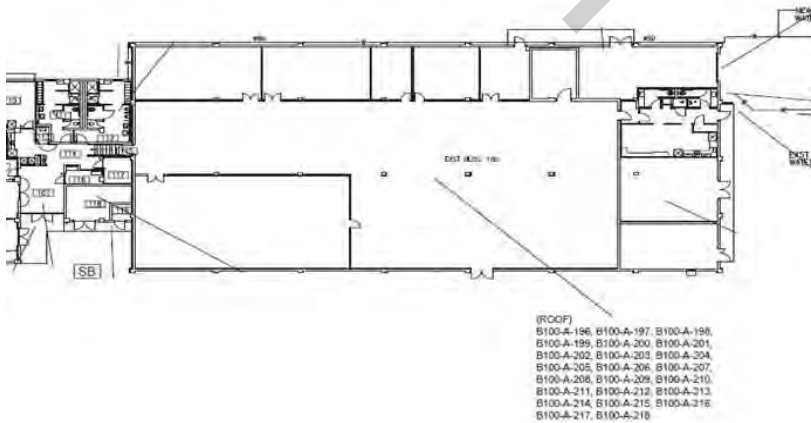
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-196	TYPE OF MATERIAL :	Roofing System
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



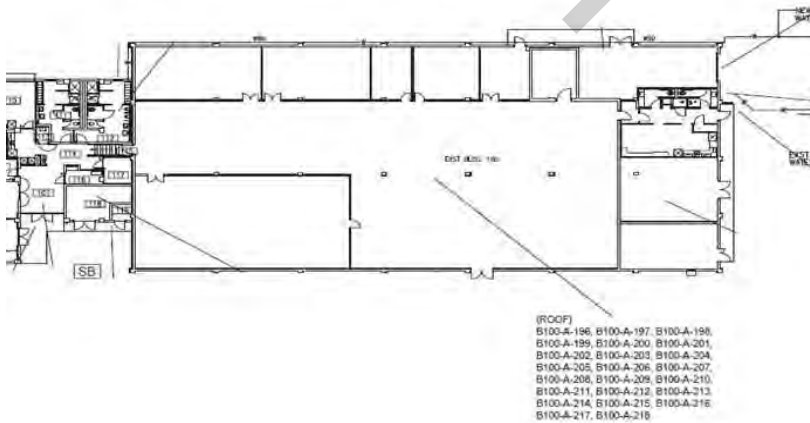
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-197	TYPE OF MATERIAL	: Roofing System
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



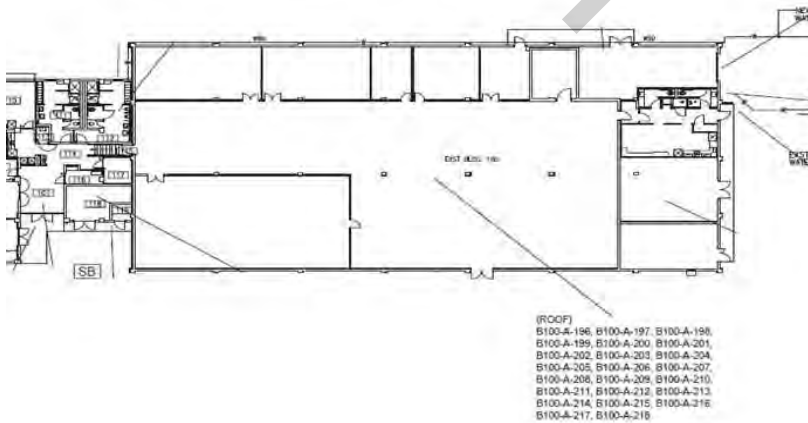
**REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT**

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-198	TYPE OF MATERIAL	: Roofing System
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



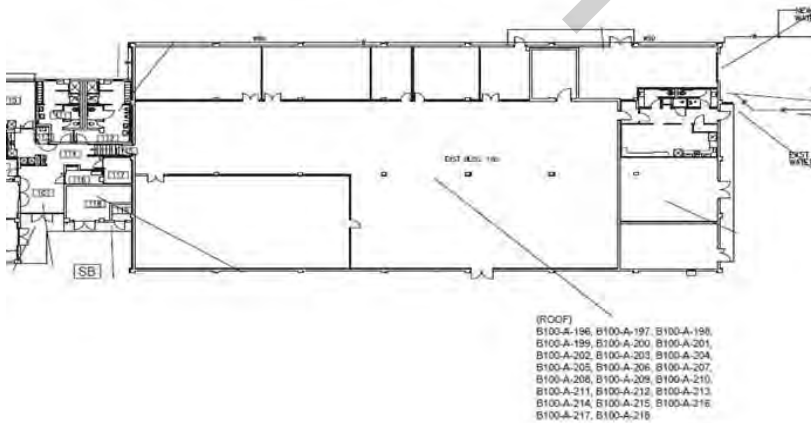
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-199	TYPE OF MATERIAL	: Roofing System
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



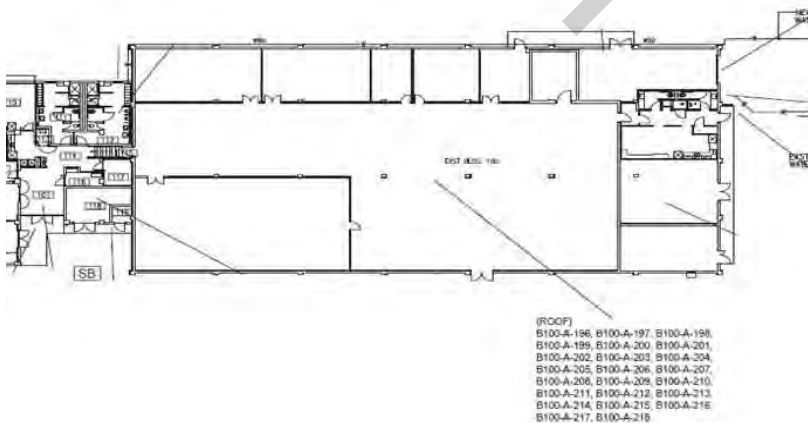
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-200	TYPE OF MATERIAL :	Roofing System
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



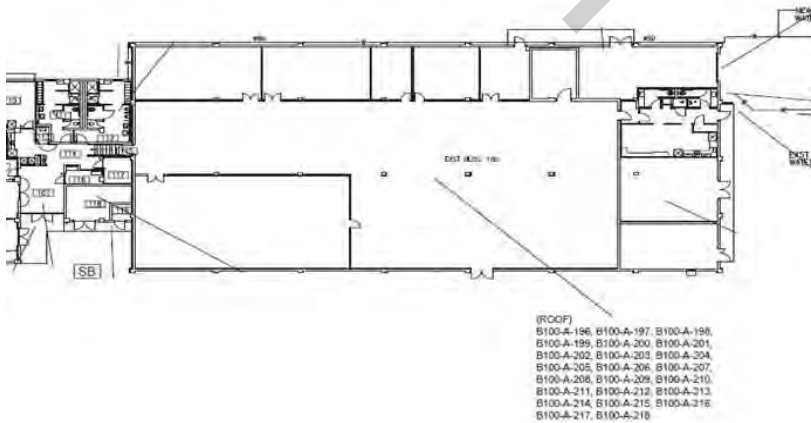
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-201	TYPE OF MATERIAL	: Roofing System
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



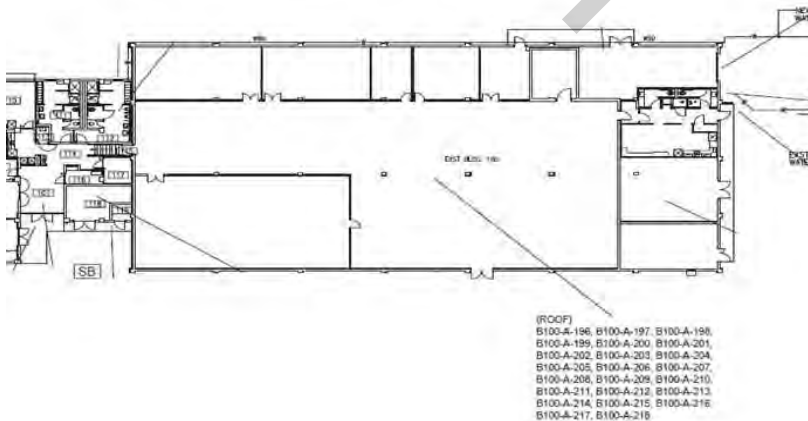
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT**

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-202	TYPE OF MATERIAL :	Roofing System
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



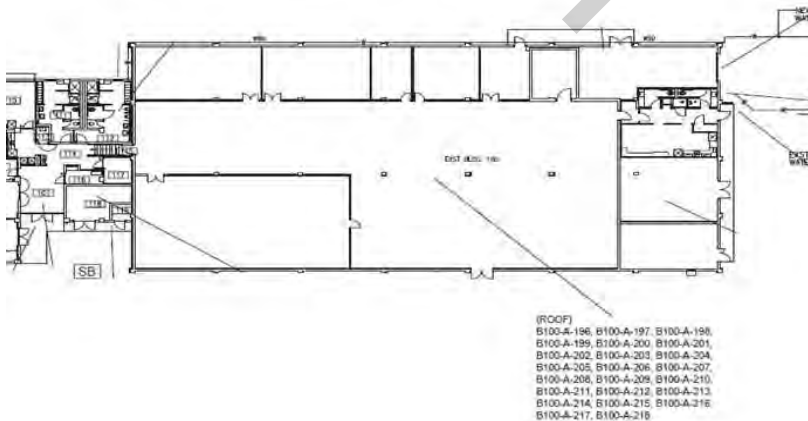
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-203	TYPE OF MATERIAL	: Roof Concrete Surface
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



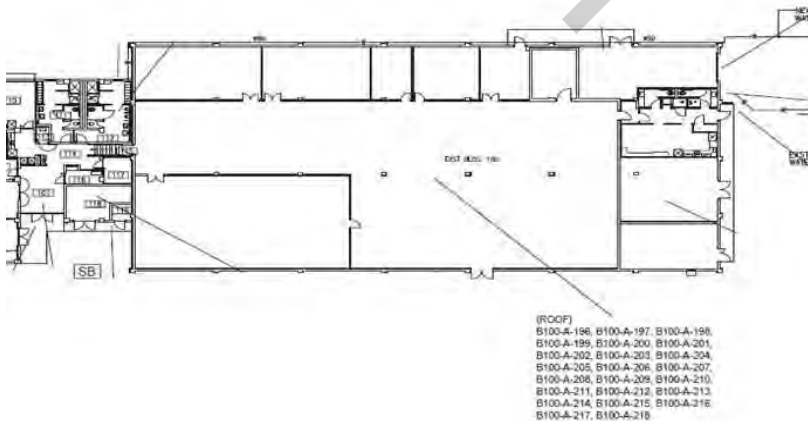
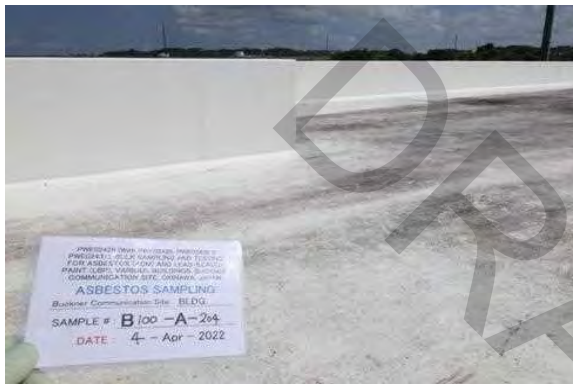
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-204	TYPE OF MATERIAL :	Roof Concrete Surface
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



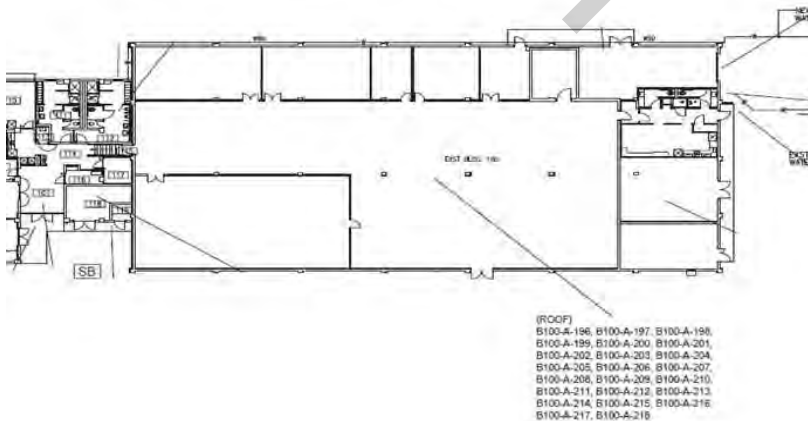
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-205	TYPE OF MATERIAL	: Roof Concrete Surface
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



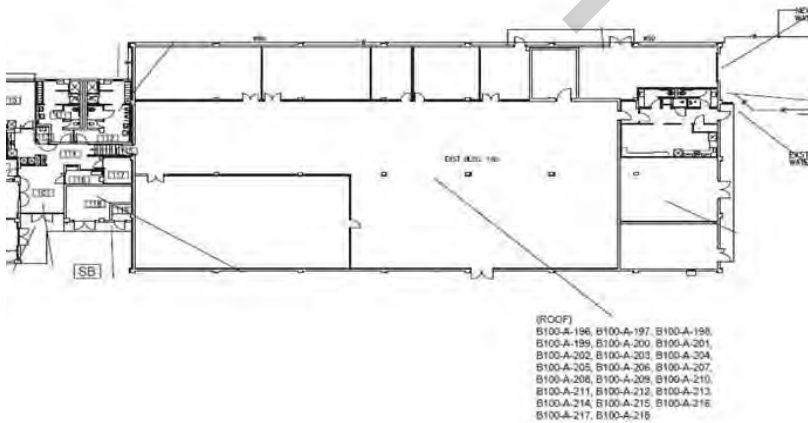
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT**

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-206	TYPE OF MATERIAL	: Roof Concrete Surface
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



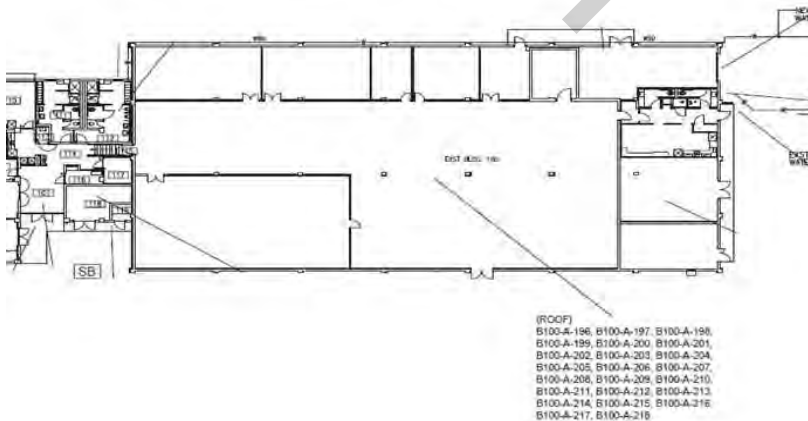
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-207	TYPE OF MATERIAL :	Roof Concrete Surface
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



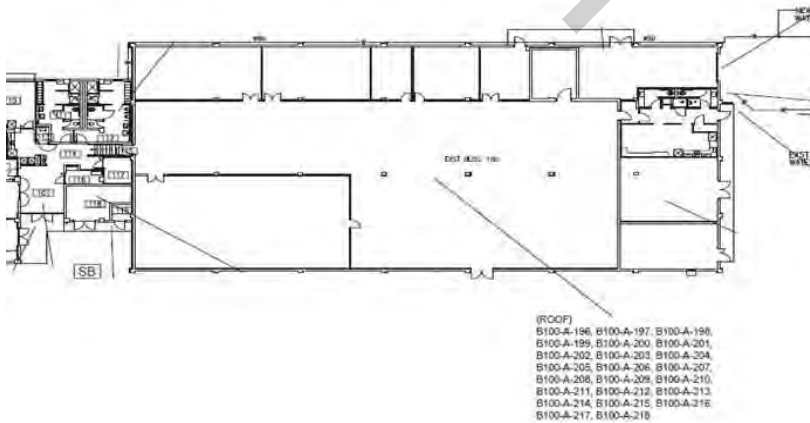
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-208	TYPE OF MATERIAL :	Roof Concrete Surface
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



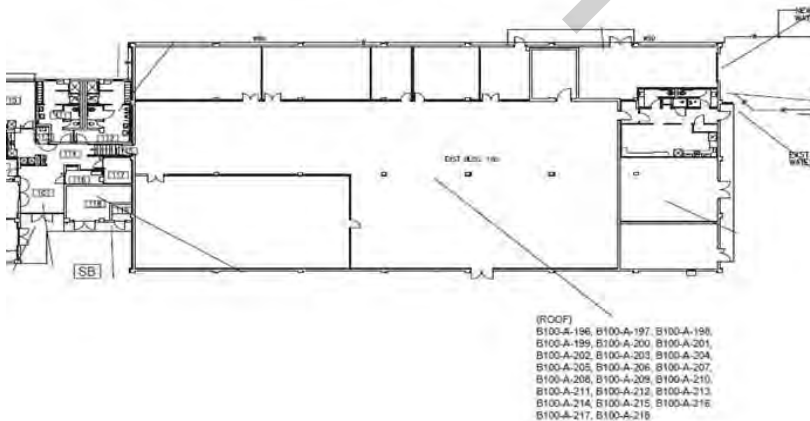
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-209	TYPE OF MATERIAL :	Roof Concrete Surface
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



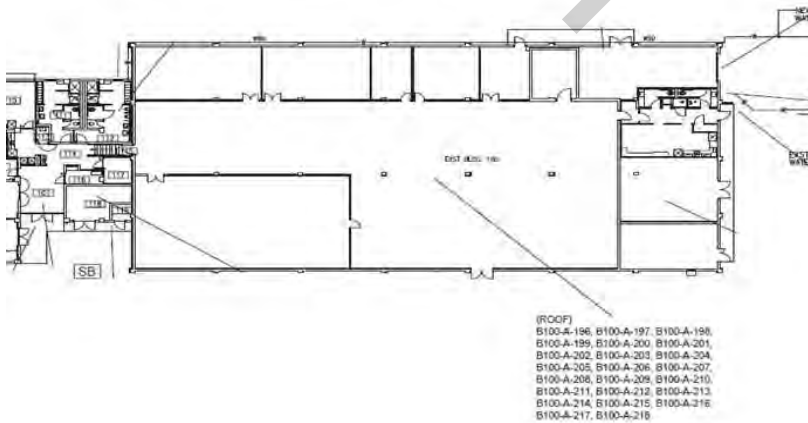
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

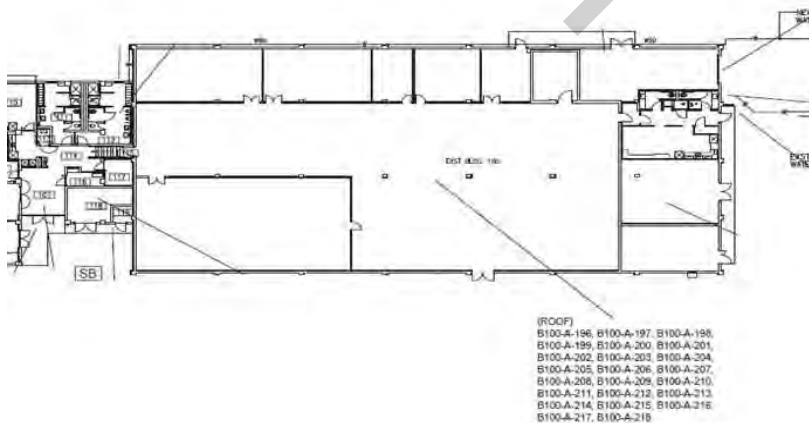
SAMPLE ID NO.	: B100-A-213	TYPE OF MATERIAL	: Coating Material Of Ventilator
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-214	TYPE OF MATERIAL	:	Coating Material Of Ventilator
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tetsuya Yonaha
ROOM NO./NAME	:	Roof	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



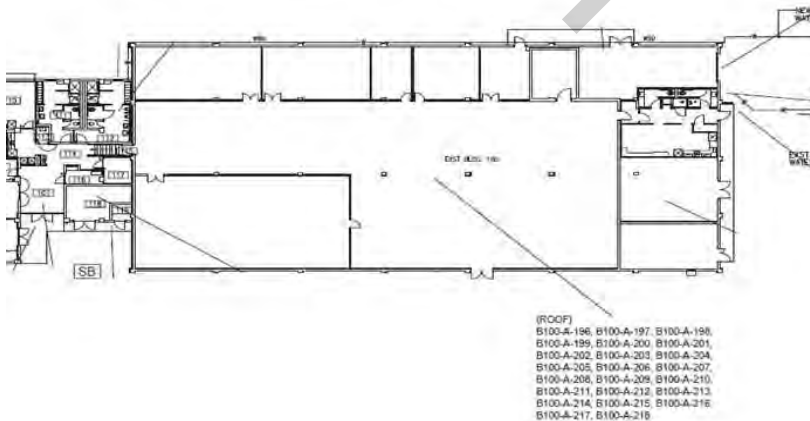
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-215	TYPE OF MATERIAL	: Coating Material Of Ventilator
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



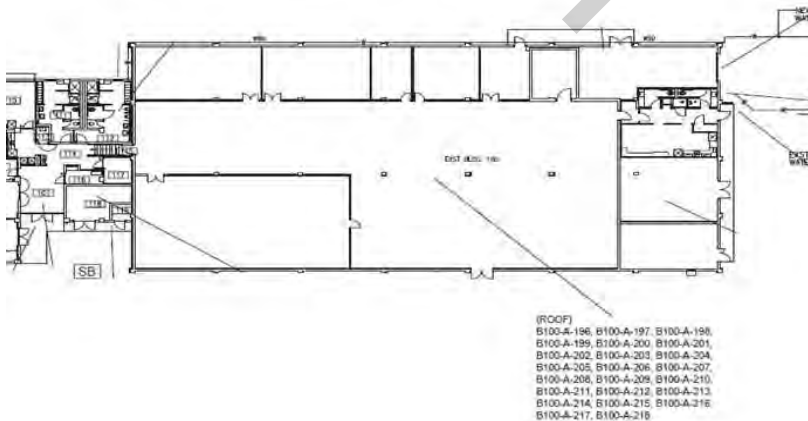
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-216	TYPE OF MATERIAL	: Ventilator Foundation Concrete
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



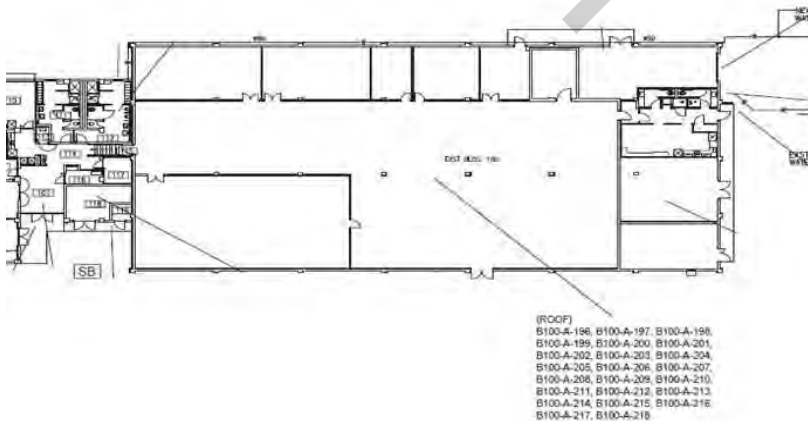
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT**

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-217	TYPE OF MATERIAL :	Ventilator Foundation Concrete
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Roof	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



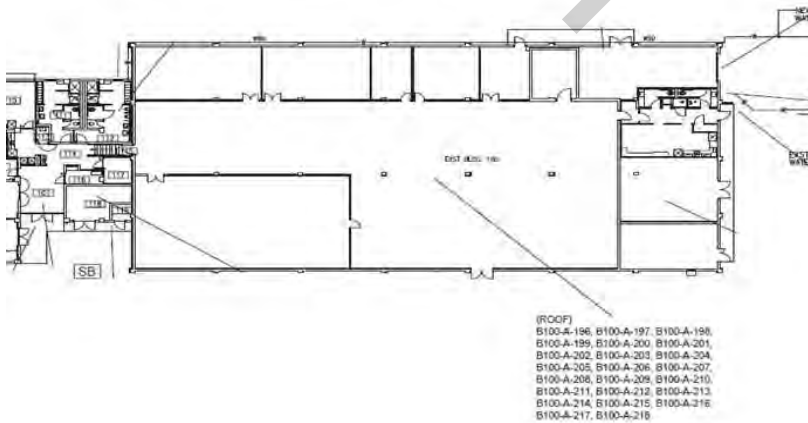
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-218	TYPE OF MATERIAL	: Ventilator Foundation Concrete
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Roof	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



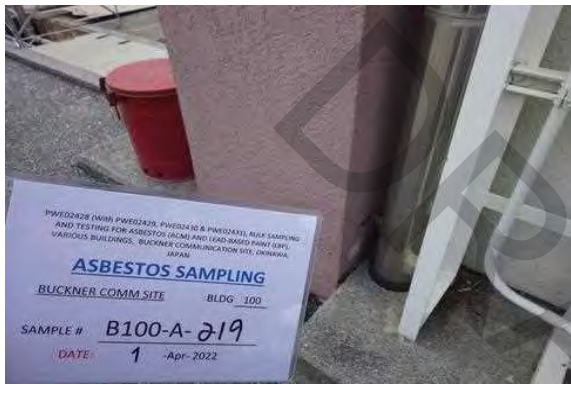
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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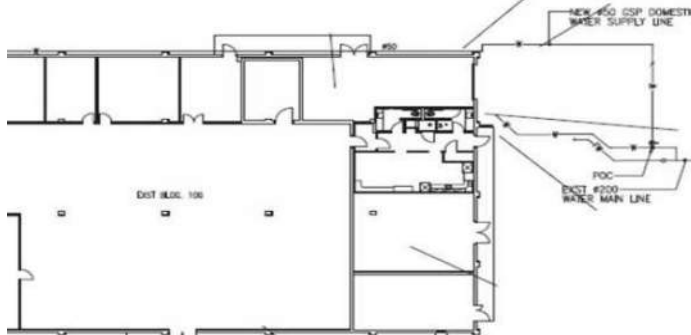
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-219	TYPE OF MATERIAL	: Siding Board (Brown)
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-219



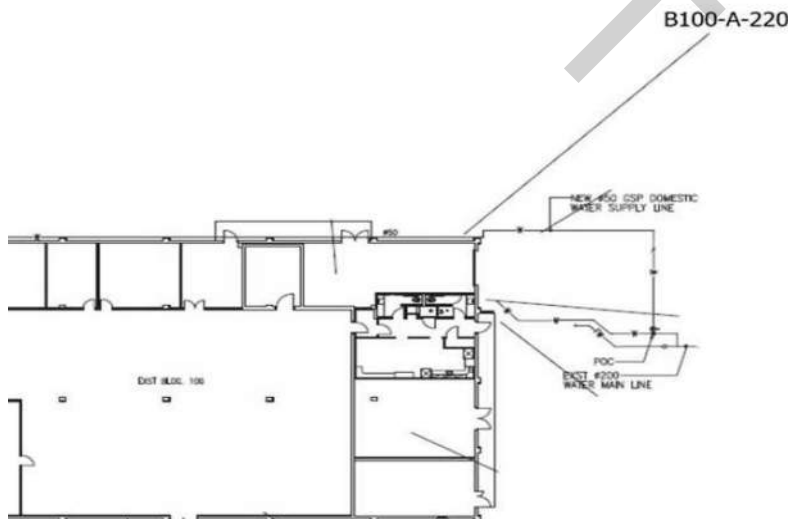
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-220	TYPE OF MATERIAL :	Siding Board (Ivory)
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



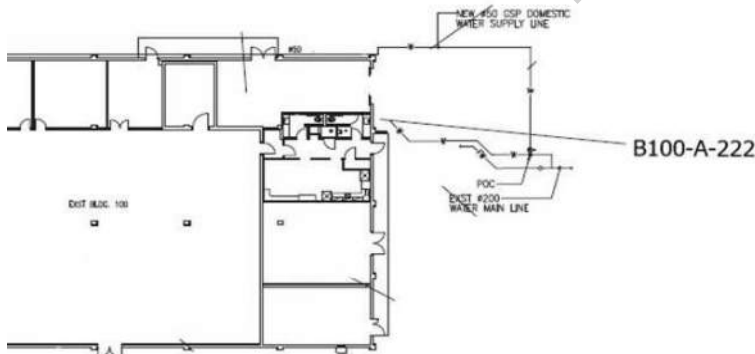
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

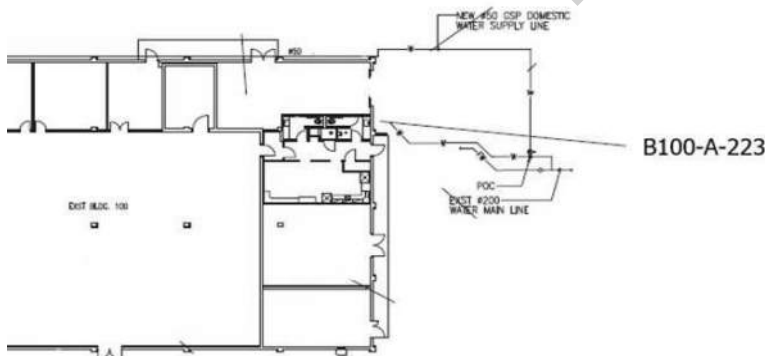
SAMPLE ID NO.	: B100-A-222	TYPE OF MATERIAL	: Flange Gasket
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-223	TYPE OF MATERIAL	:	Flange Gasket
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



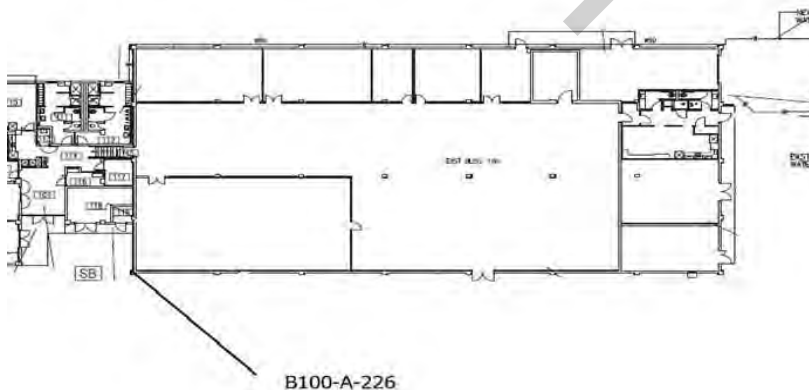
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-226	TYPE OF MATERIAL	: Waterproofing
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



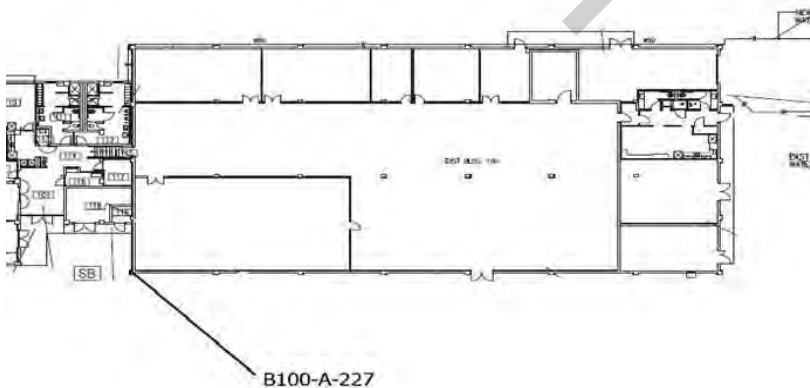
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-227	TYPE OF MATERIAL	:	Concrete Wall
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (5%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



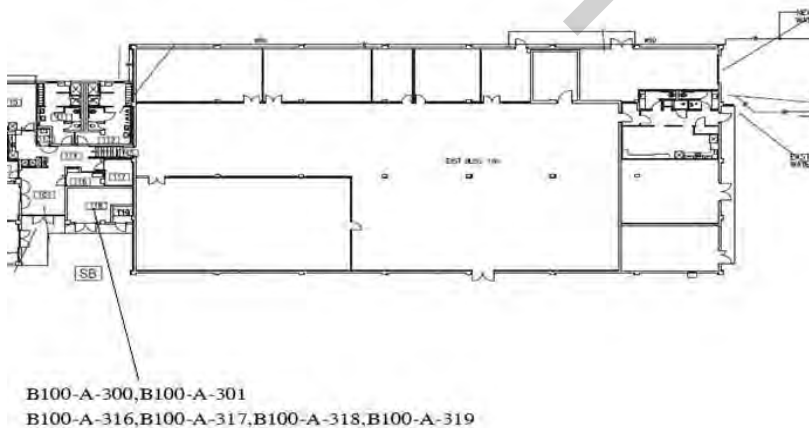
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-300	TYPE OF MATERIAL :	Floor Caulking
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



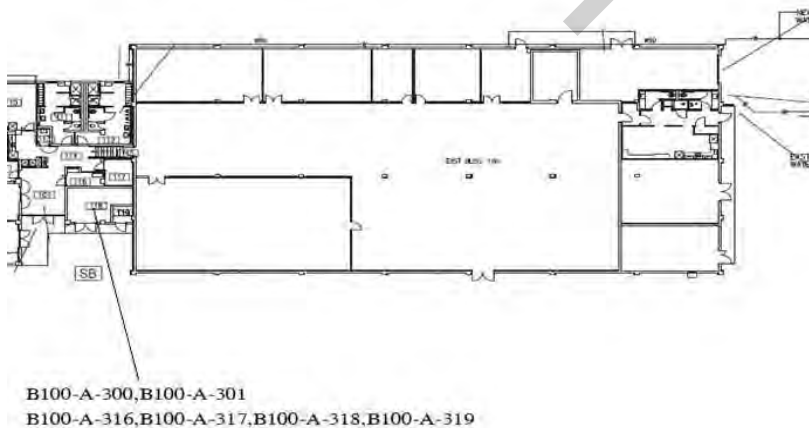
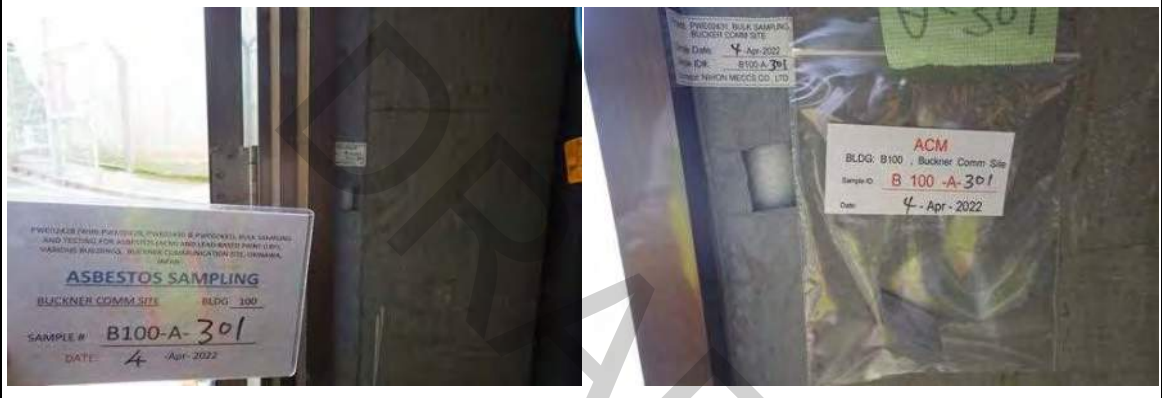
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-301	TYPE OF MATERIAL	: Wall Caulking
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



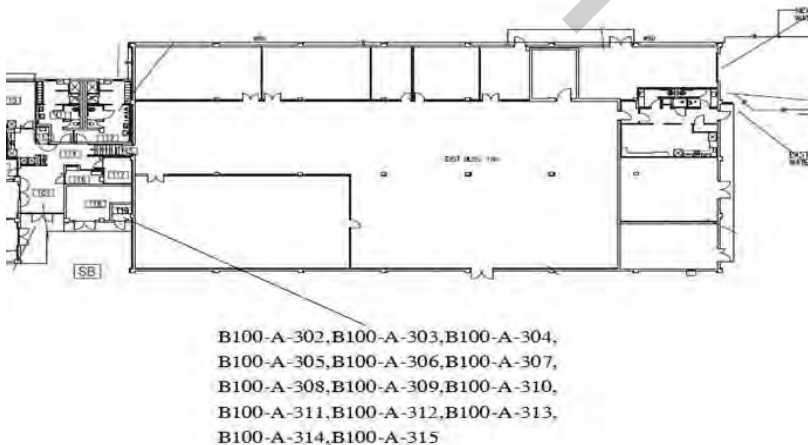
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-302	TYPE OF MATERIAL	: Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



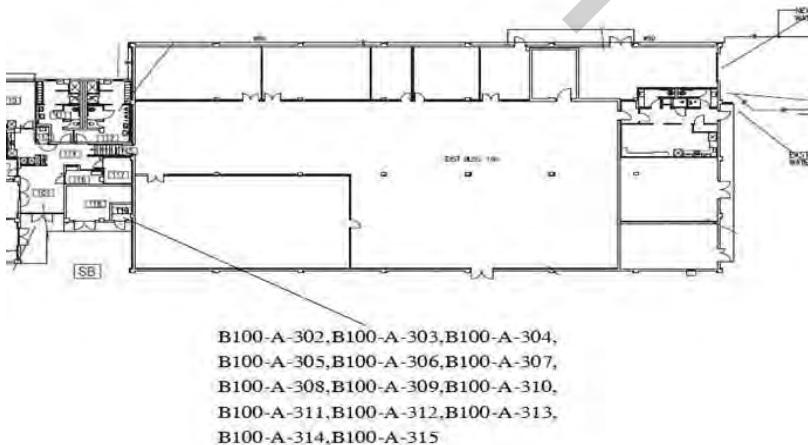
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-303	TYPE OF MATERIAL	: Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



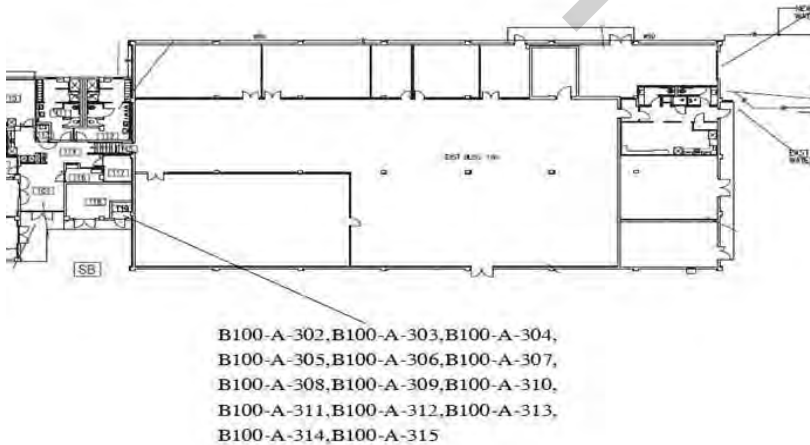
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-304	TYPE OF MATERIAL	: Concrete Wall
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



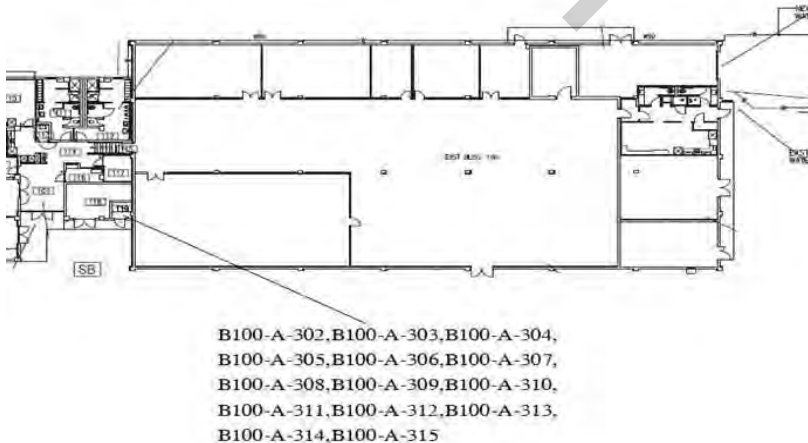
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-305	TYPE OF MATERIAL :	Ceiling Slab
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Rm 119	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



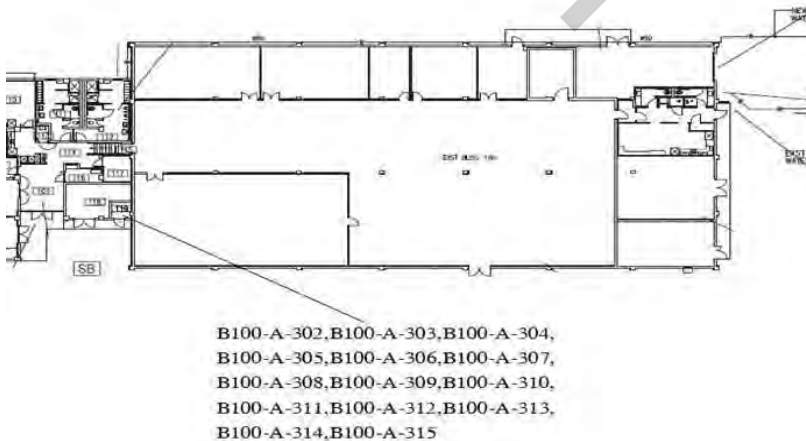
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ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-306	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



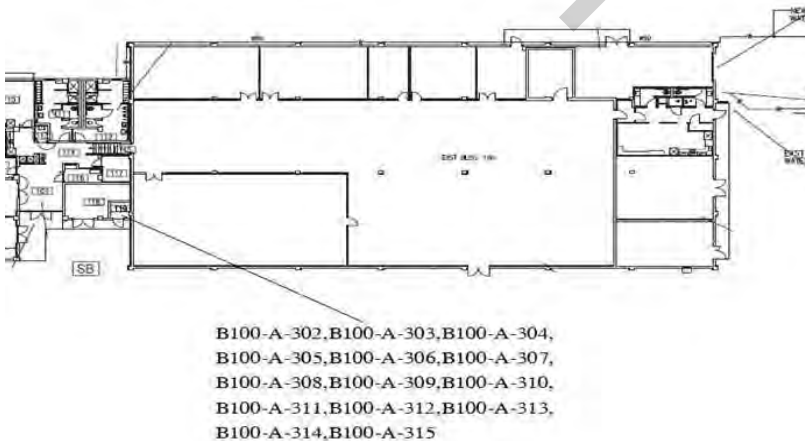
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-307	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



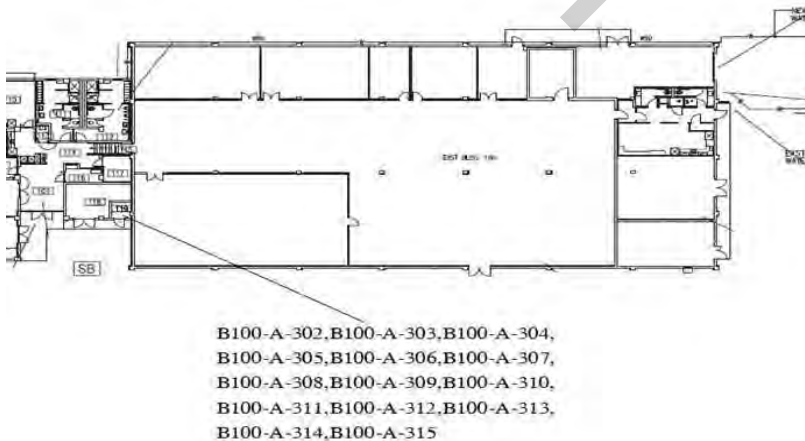
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-308	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



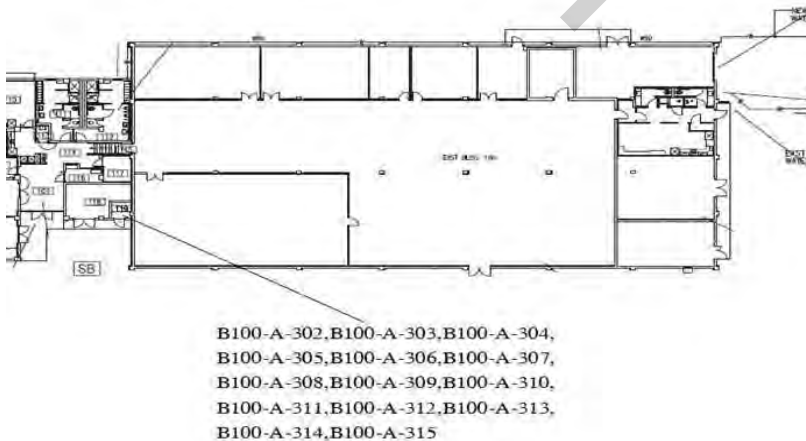
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ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-309	TYPE OF MATERIAL	:	Wallboard
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tetsuya Yonaha
ROOM NO./NAME	:	Rm 119	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



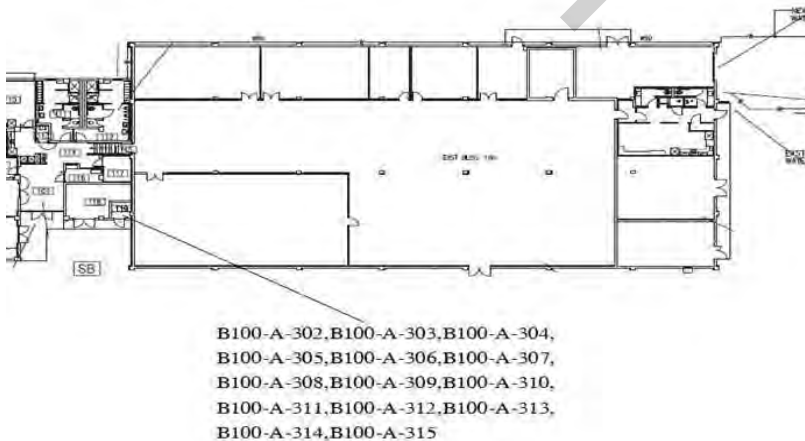
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ASBESTOS ANALYSIS REPORT

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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-310	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Rm 119	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



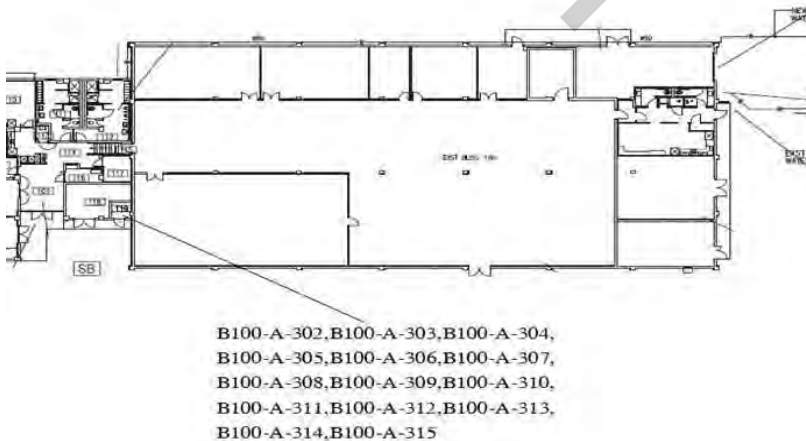
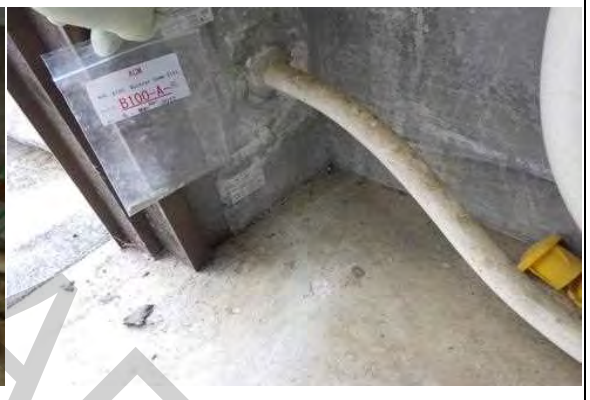
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-311	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



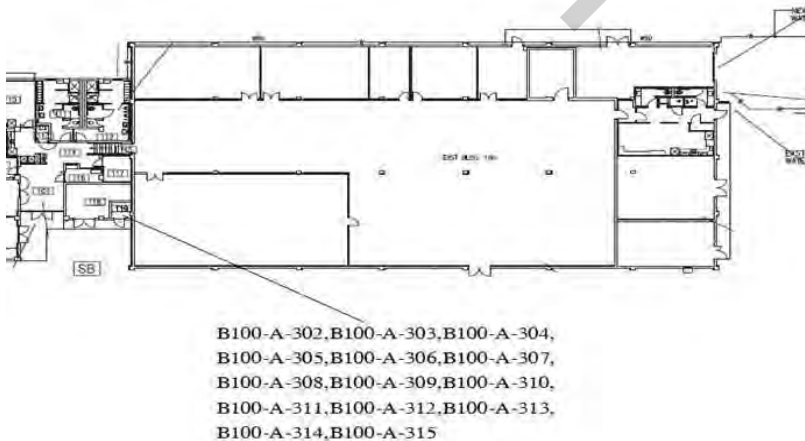
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-312	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



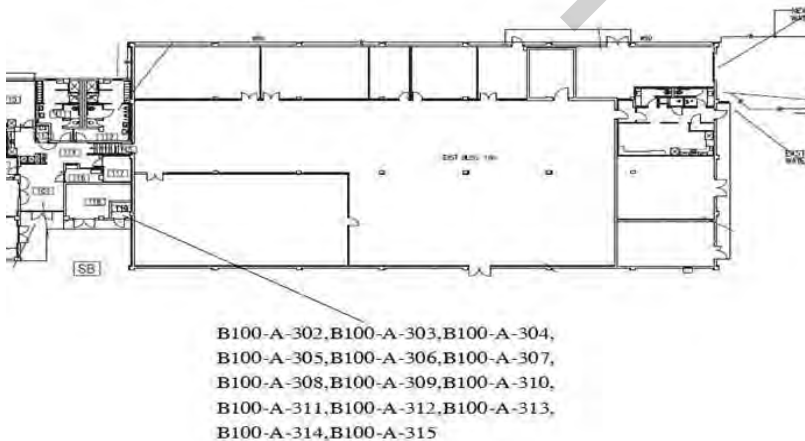
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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-313	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



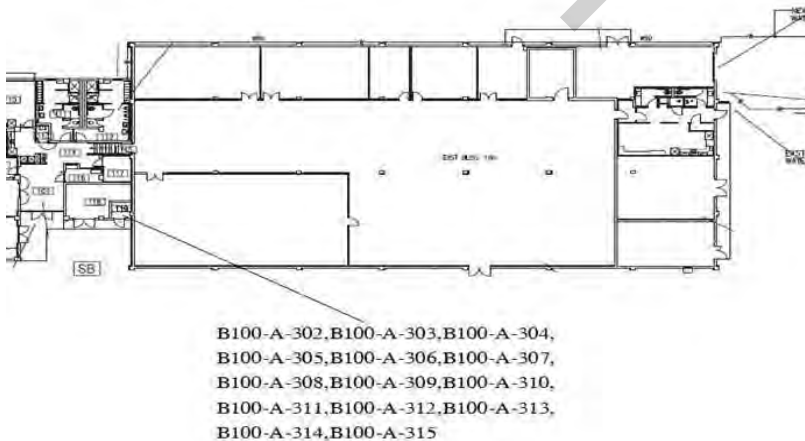
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ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-314	TYPE OF MATERIAL	: Wall Caulking
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



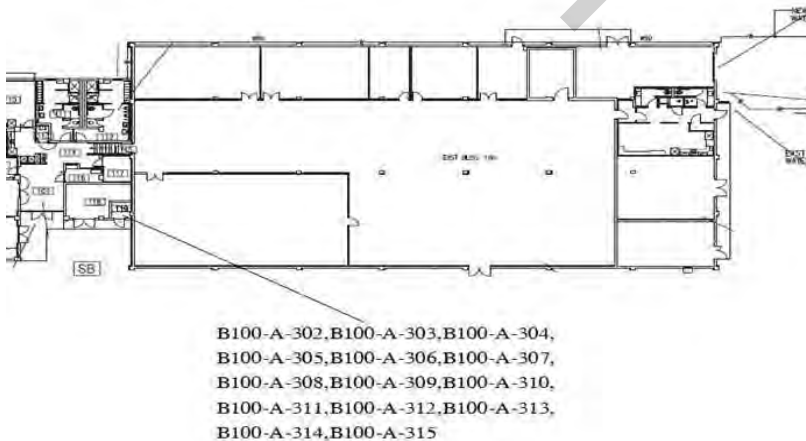
REPAIR HVAC WIDEBAND SATCOM OPS B100
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25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-315	TYPE OF MATERIAL :	Floor Caulking
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Rm 119	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



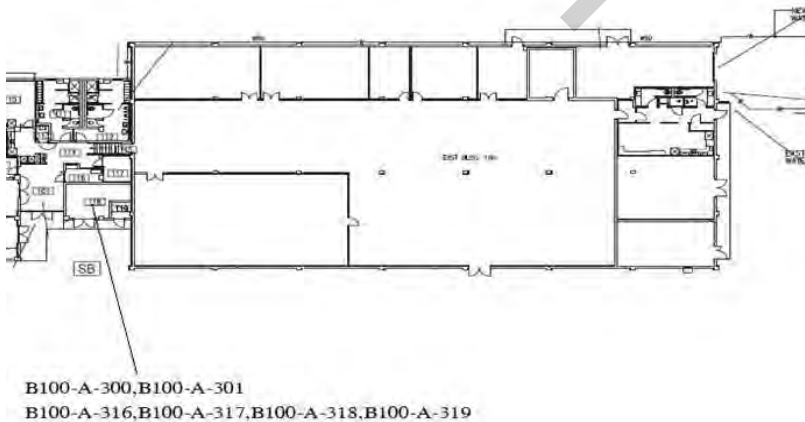
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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-316	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



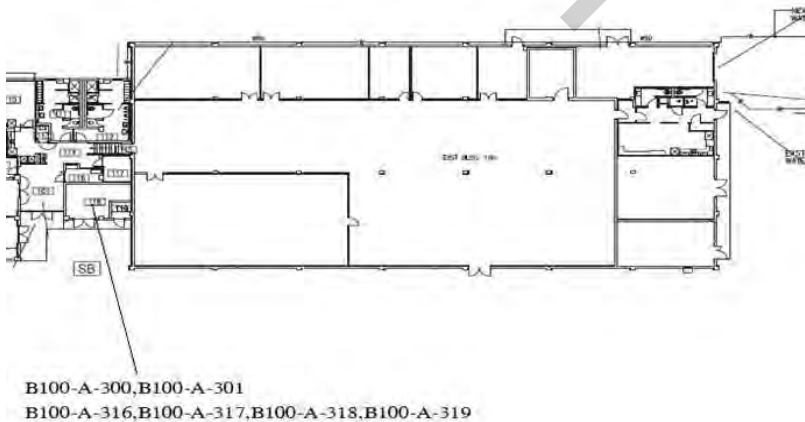
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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-317	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



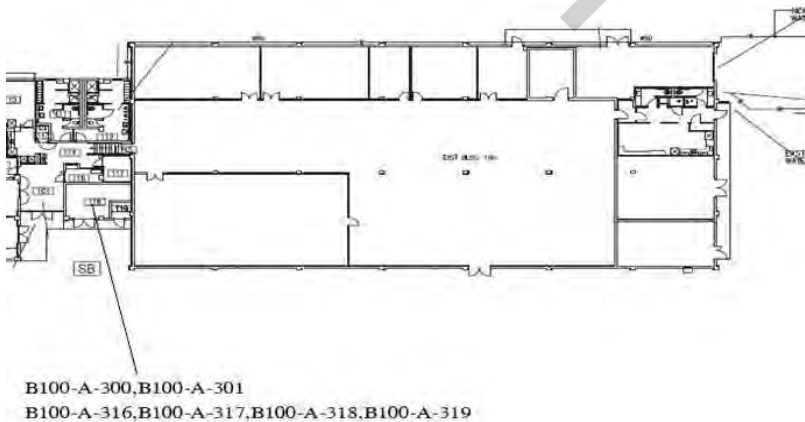
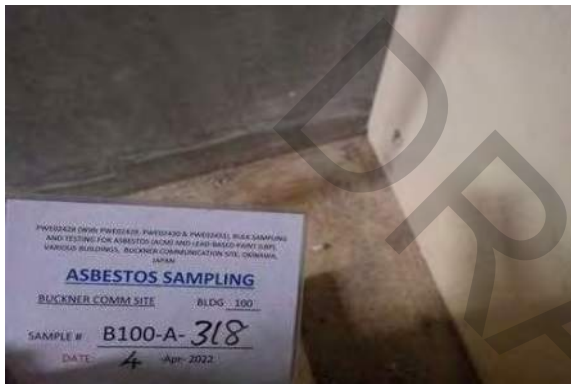
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-318	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



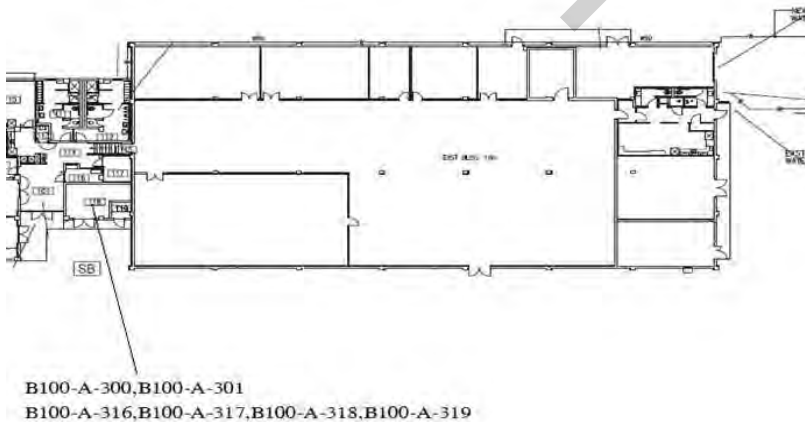
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25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-319	TYPE OF MATERIAL	: Duct Flange Insulation
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

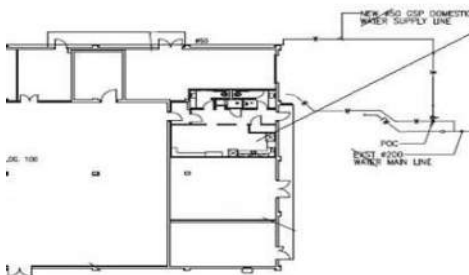
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-400	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

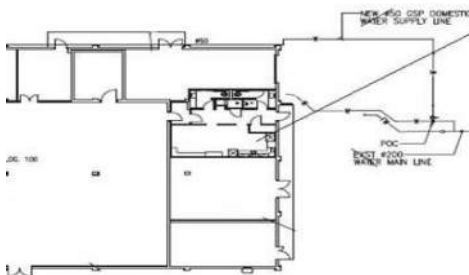
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-401	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN ASBESTOS ANALYSIS REPORT

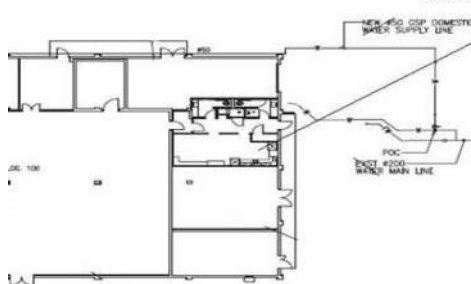
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-402	TYPE OF MATERIAL	:	Ceiling Insulation
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

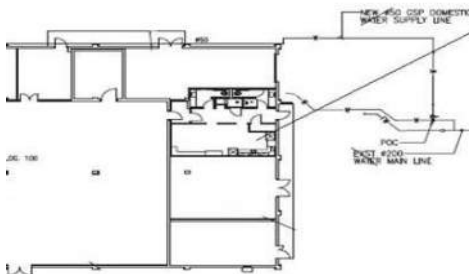
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-403	TYPE OF MATERIAL	:	Ceiling Board
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

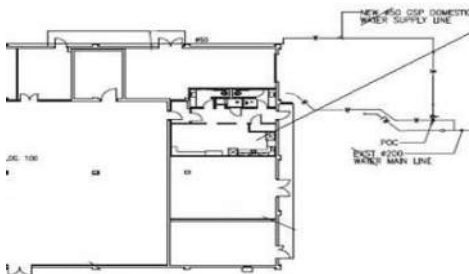
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-404	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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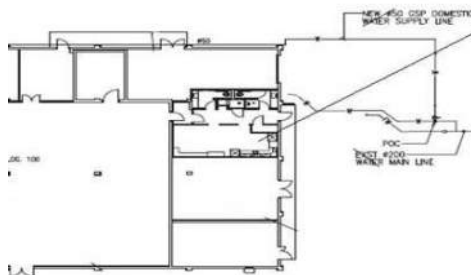
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-405	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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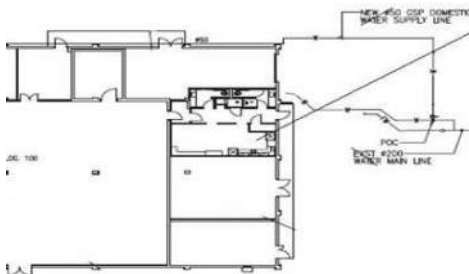
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-406	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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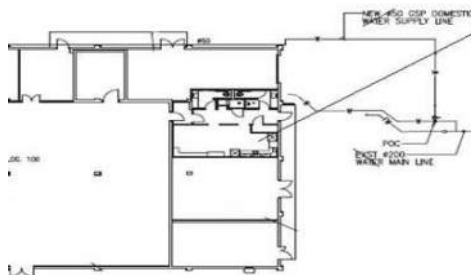
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-407	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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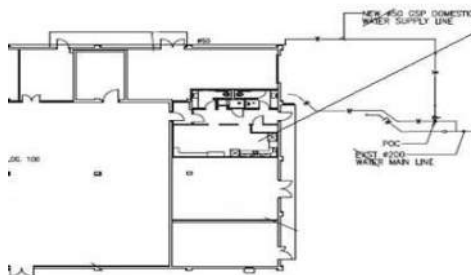
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-408	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

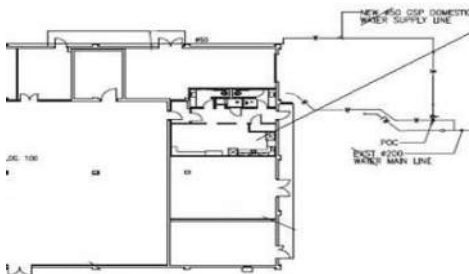
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-409	TYPE OF MATERIAL :	Vinly Cove Base w/Mastic
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Kitchen	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

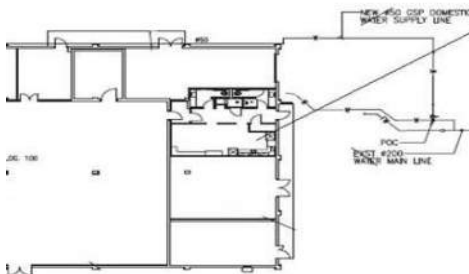
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-410	TYPE OF MATERIAL	: Vinly Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

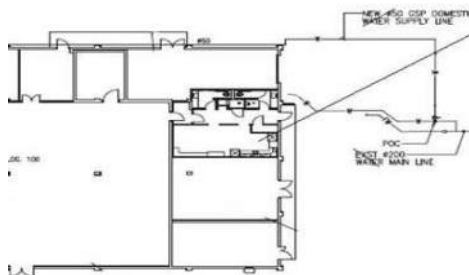
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-411	TYPE OF MATERIAL	: Vinly Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

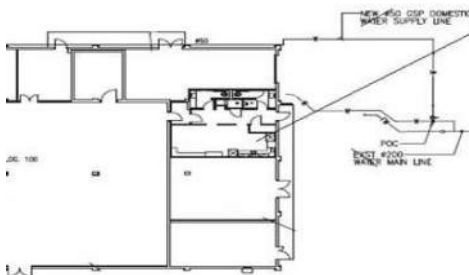
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-412	TYPE OF MATERIAL	:	Wall Tile w/Grout
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

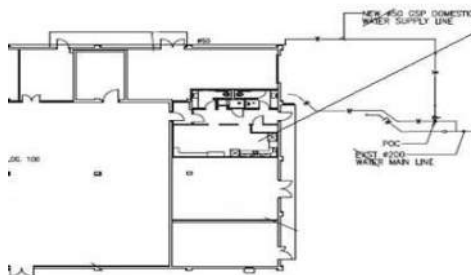
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-415	TYPE OF MATERIAL	: Floor Tile w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

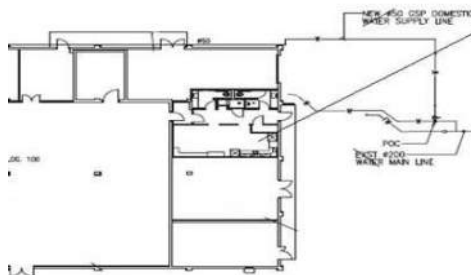
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-416	TYPE OF MATERIAL	: Floor Tile w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



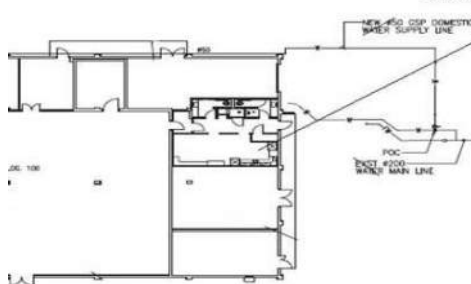
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-417	TYPE OF MATERIAL	:	Floor Tile w/Mastic
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

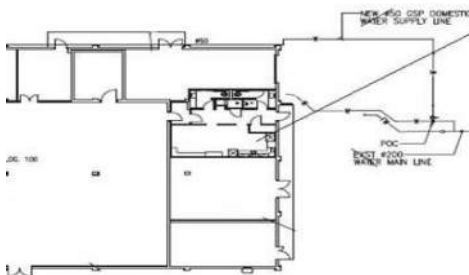
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-418	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

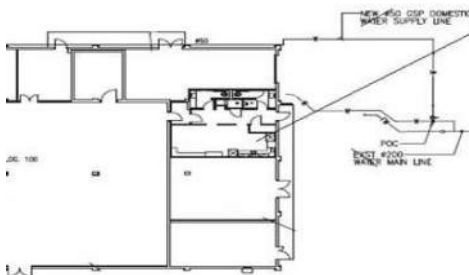
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-419	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

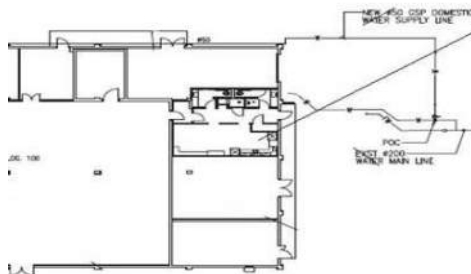
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-420	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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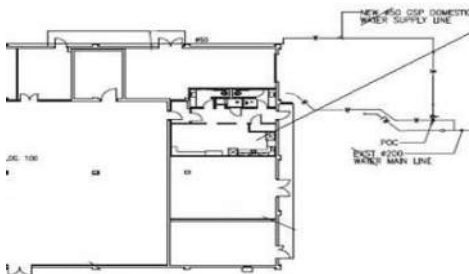
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-421	TYPE OF MATERIAL	: Caulking (Gray)
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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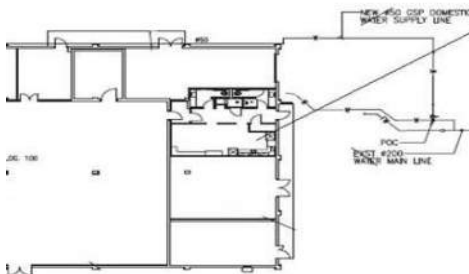
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-422	TYPE OF MATERIAL	: Caulking (White)
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-400,B100-A-401,B100-A-402,
B100-A-403,B100-A-404,B100-A-405,
B100-A-406,B100-A-407,B100-A-408,
B100-A-409,B100-A-410,B100-A-411,
B100-A-412,B100-A-415,B100-A-416,
B100-A-417,B100-A-418,B100-A-419,
B100-A-420,B100-A-421,B100-A-422



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

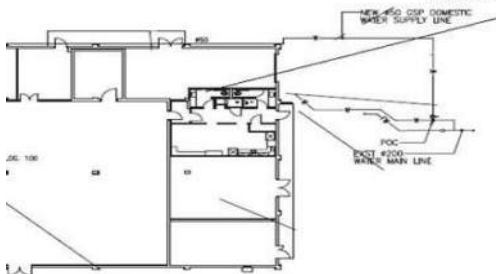
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-425	TYPE OF MATERIAL :	Ceiling Board
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takafumi Noborikawa
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



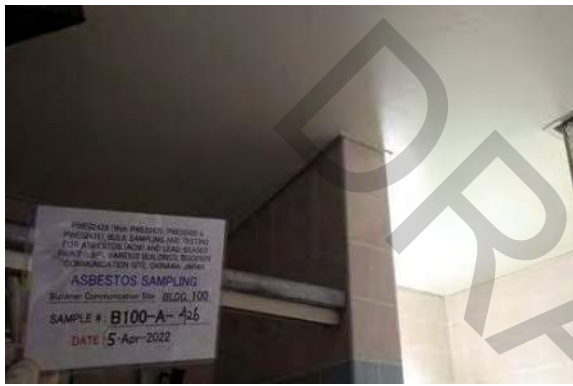
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

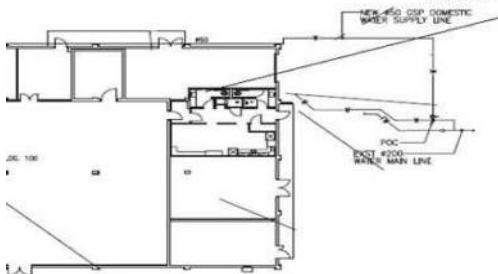
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-426	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

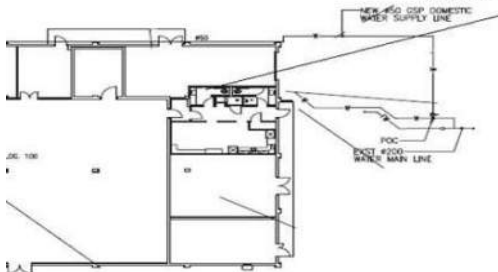
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-427	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

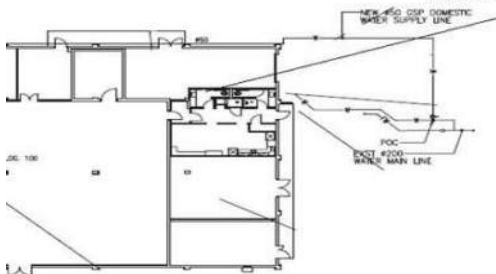
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-428	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

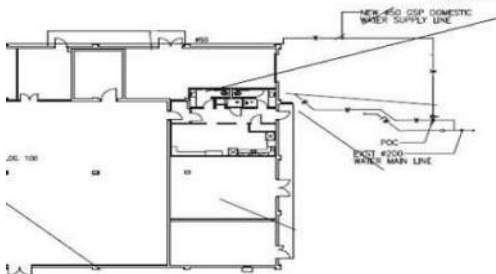
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-429	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takafumi Noborikawa
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

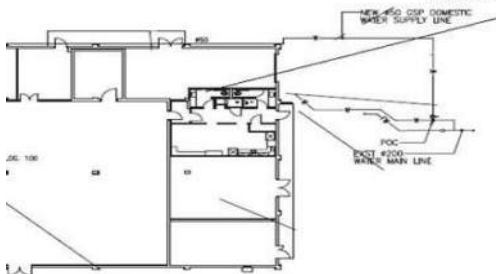
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-430	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



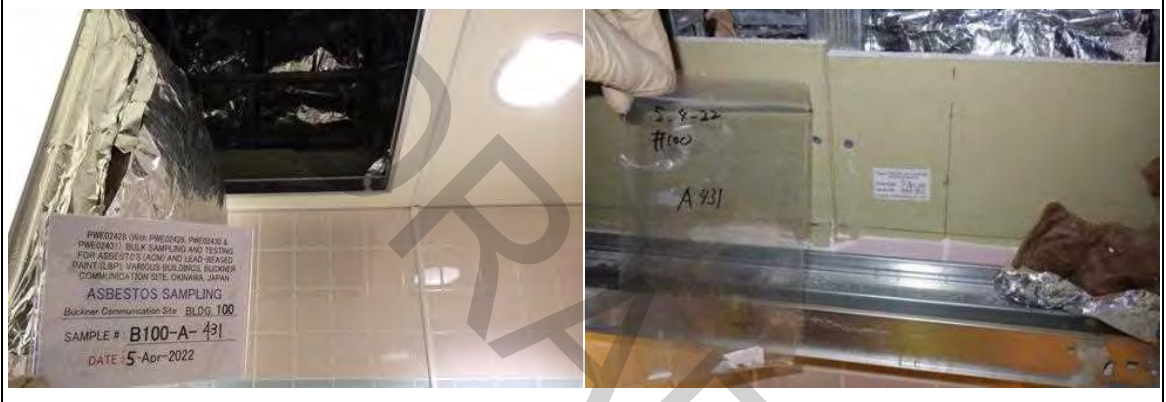
B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



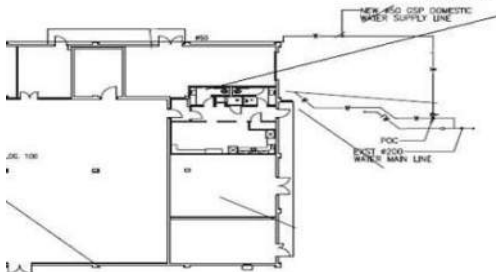
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-431	TYPE OF MATERIAL	:	Wall Ceramic Tile w/Grout
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

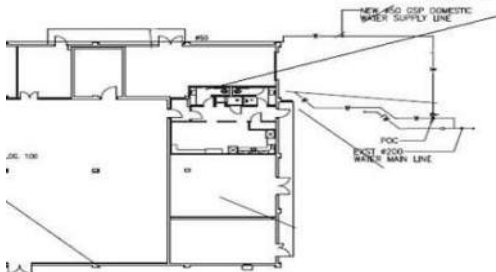
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-432	TYPE OF MATERIAL	: Wall Ceramic Tile w/Grout
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

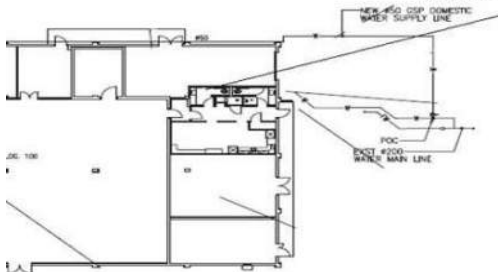
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-433	TYPE OF MATERIAL	:	Wall Ceramic Tile w/Grout
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

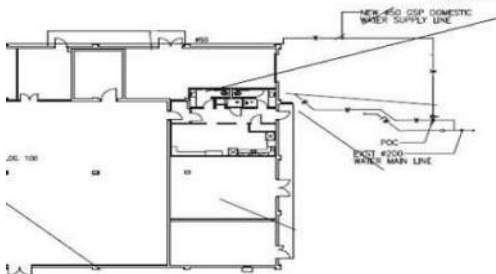
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-434	TYPE OF MATERIAL	: Floor Tile w/Grout
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN ASBESTOS ANALYSIS REPORT

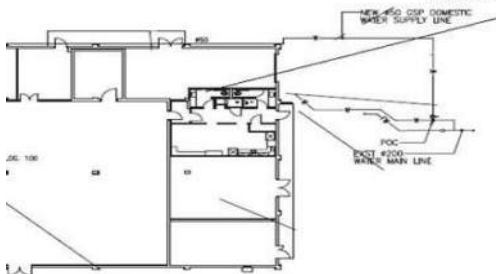
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-435	TYPE OF MATERIAL	:	Floor Tile w/Grout
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

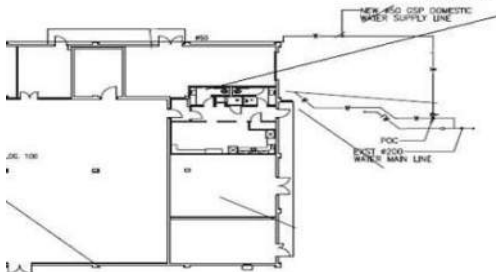
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-436	TYPE OF MATERIAL	:	Floor Tile w/Grout
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

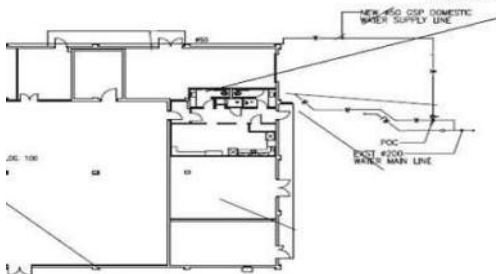
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-437	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439

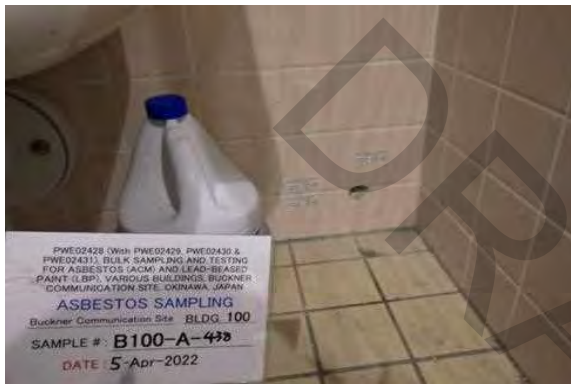


CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN ASBESTOS ANALYSIS REPORT

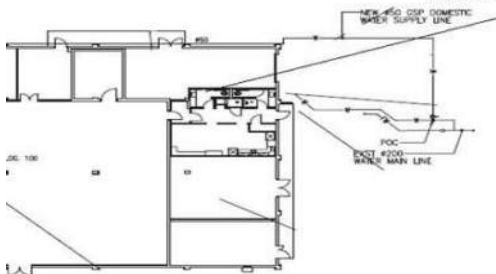
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-438	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Toilet	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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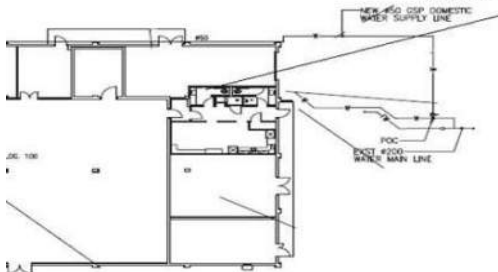
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-439	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-425,B100-A-426,B100-A-427,
B100-A-428,B100-A-429,B100-A-430,
B100-A-431,B100-A-432,B100-A-433,
B100-A-434,B100-A-435,B100-A-436,
B100-A-437,B100-A-438,B100-A-439



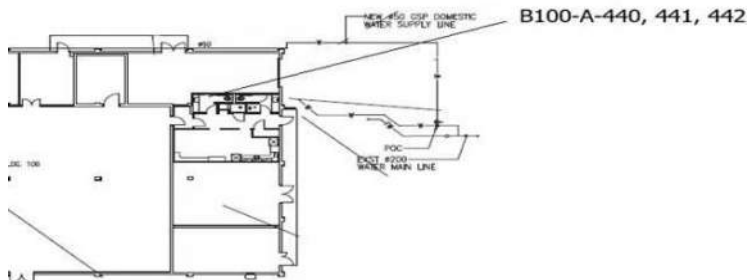
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-440	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



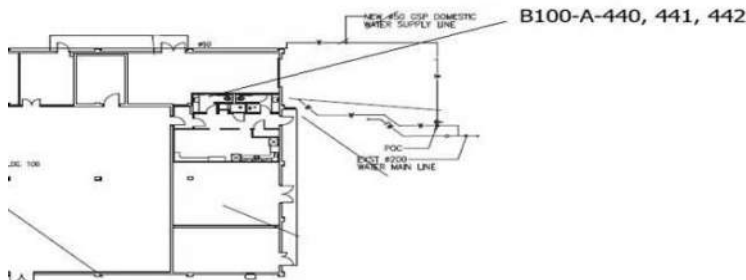
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-441	TYPE OF MATERIAL	:	Pipe Insulation
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Straight, Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



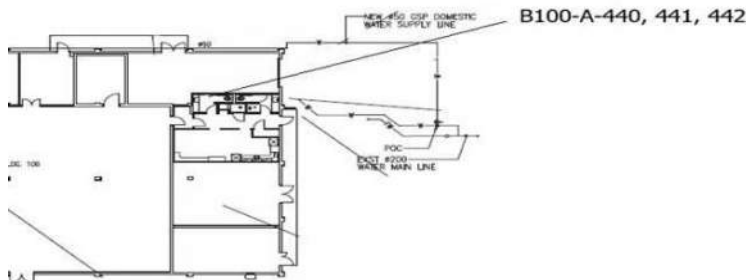
REPAIR HVAC WIDE BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-442	TYPE OF MATERIAL	: Pipe Insulation
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Elbow, Kitchen	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



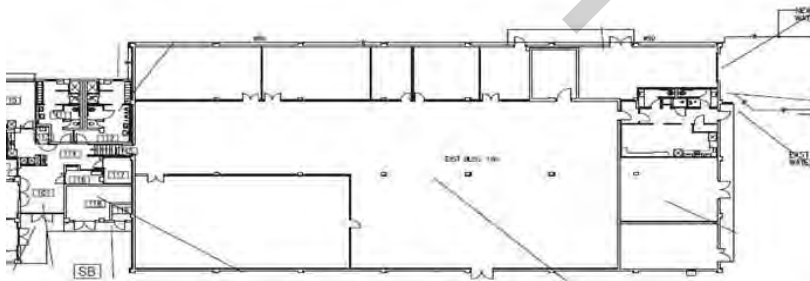
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-500	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

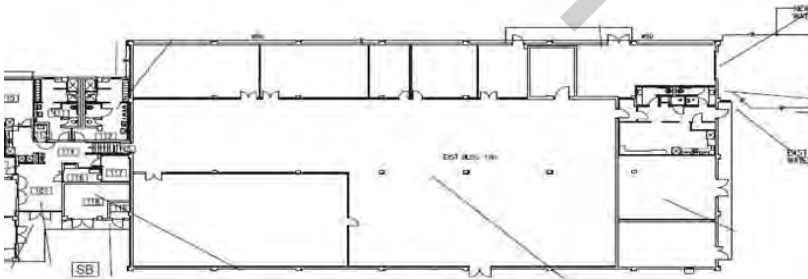
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-501	TYPE OF MATERIAL	:	Ceiling Board
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

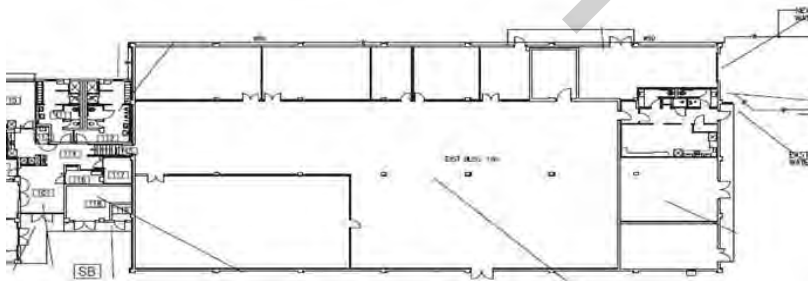
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-502	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

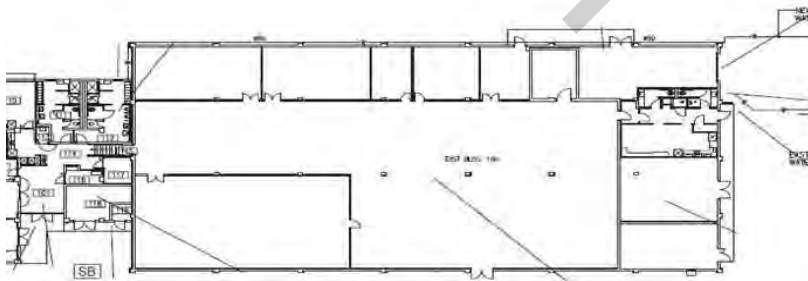
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-503	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
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B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

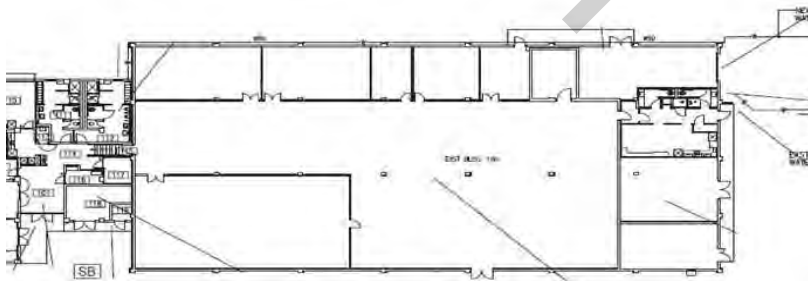
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-504	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

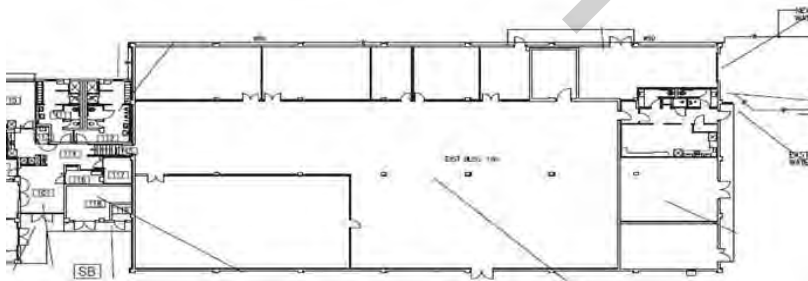
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-505	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

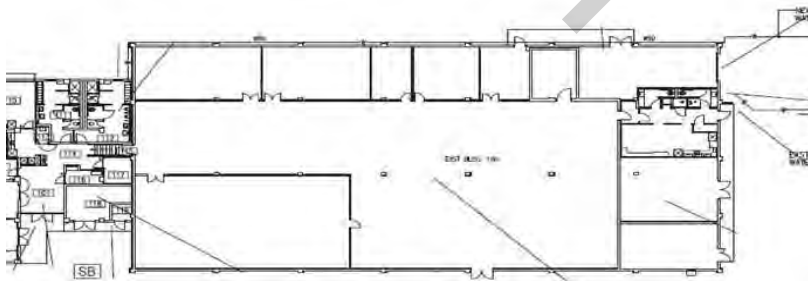
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-506	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

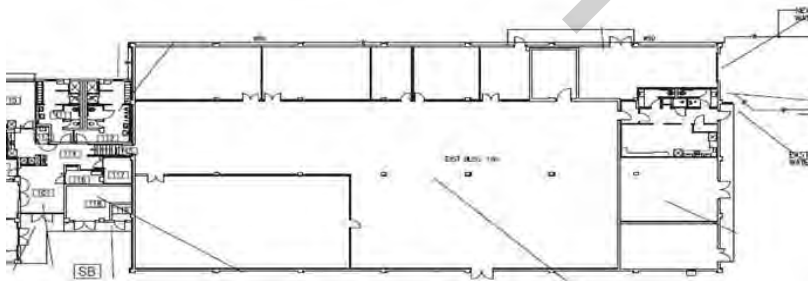
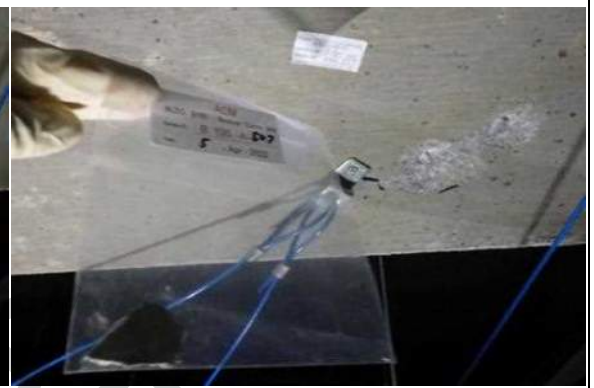
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-507	TYPE OF MATERIAL :	Ceiling Slab
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Takafumi Noborikawa
ROOM NO./NAME :	Server Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

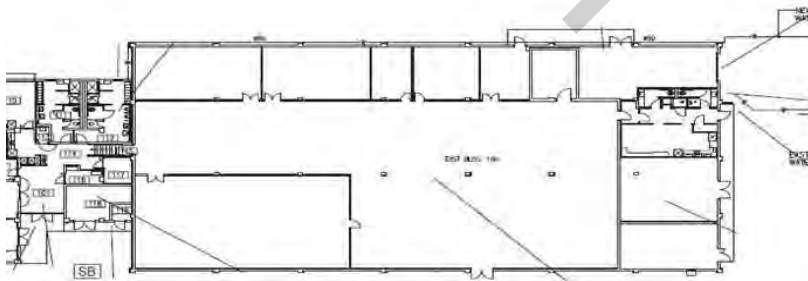
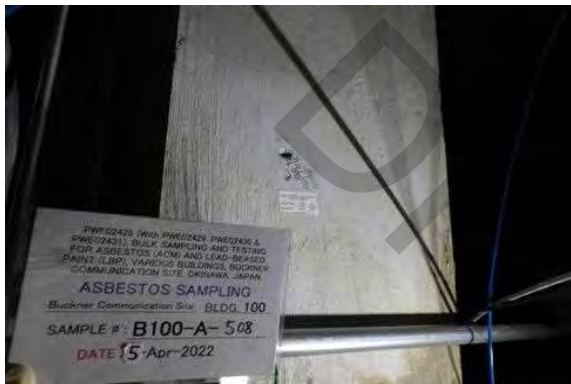
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-508	TYPE OF MATERIAL	:	Ceiling Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Takafumi Noborikawa
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

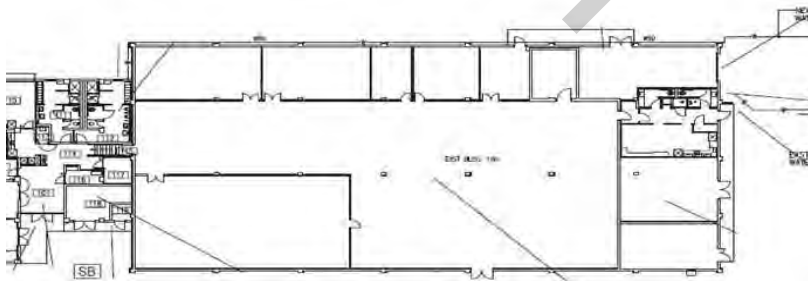
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-510	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

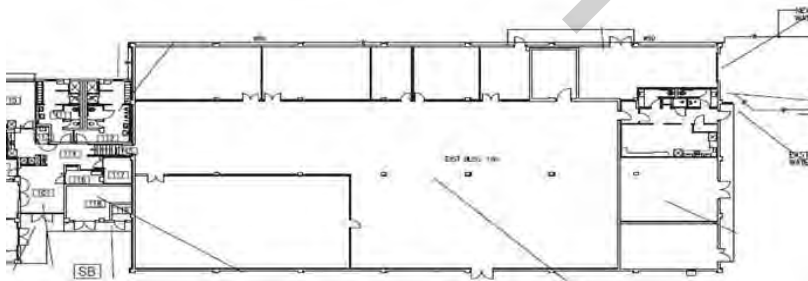
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-511	TYPE OF MATERIAL	:	Wallboard
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (1%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

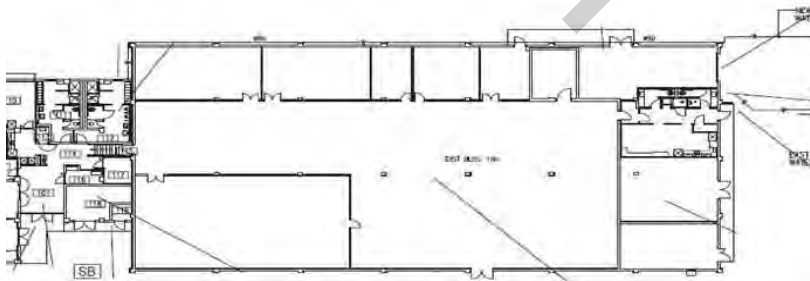
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-512	TYPE OF MATERIAL :	Wallboard
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Server Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

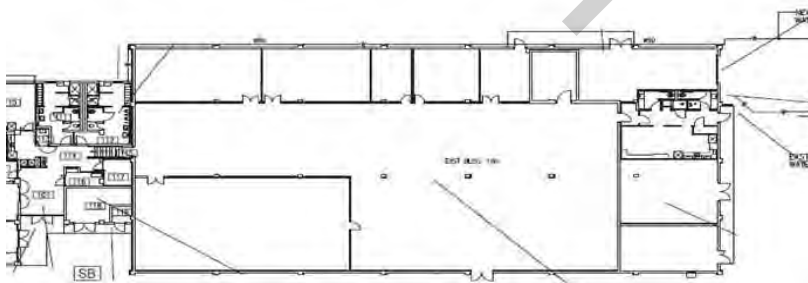
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-513	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
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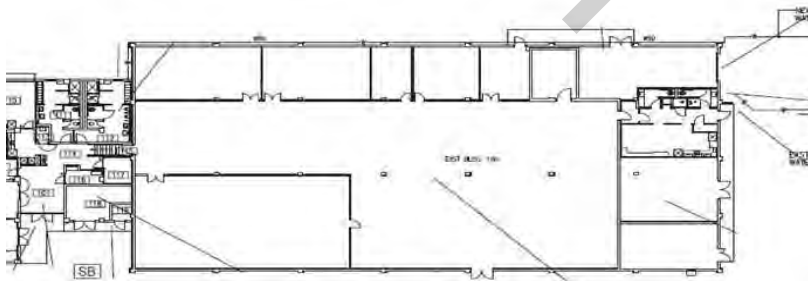
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-514	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: Chrysotile (1%)
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

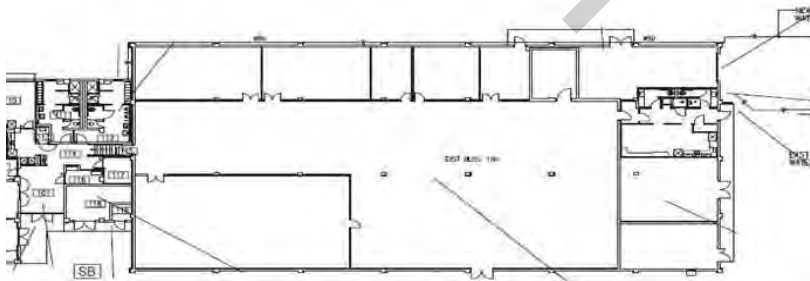
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-515	TYPE OF MATERIAL	: Vinyl Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

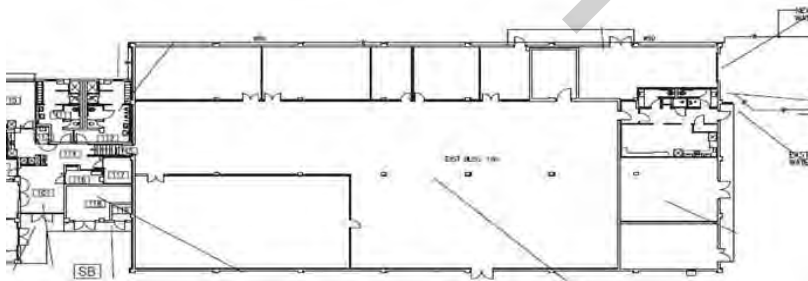
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-516	TYPE OF MATERIAL	: Vinyl Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

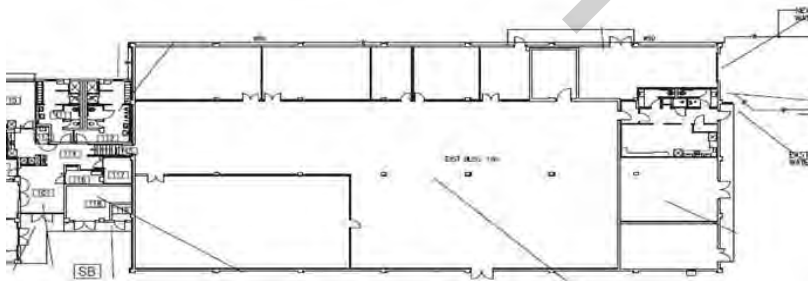
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-517	TYPE OF MATERIAL	: Vinyl Cove Base w/Mastic
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

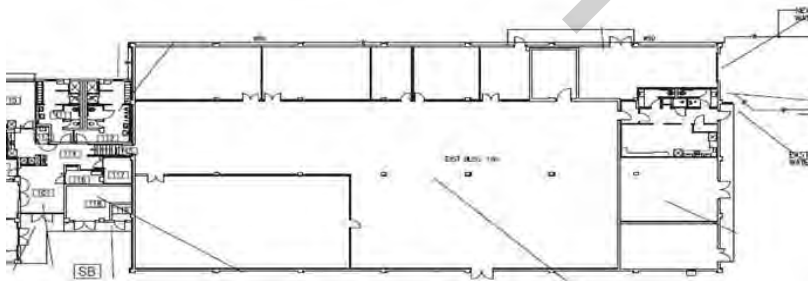
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-518	TYPE OF MATERIAL	: Vinyl Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

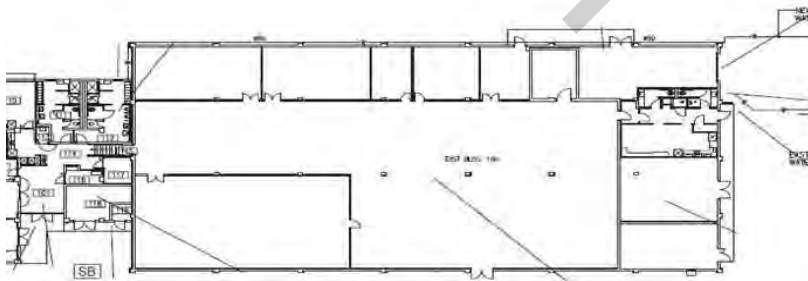
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-519	TYPE OF MATERIAL	: Vinyl Cove Base w/Mastic
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

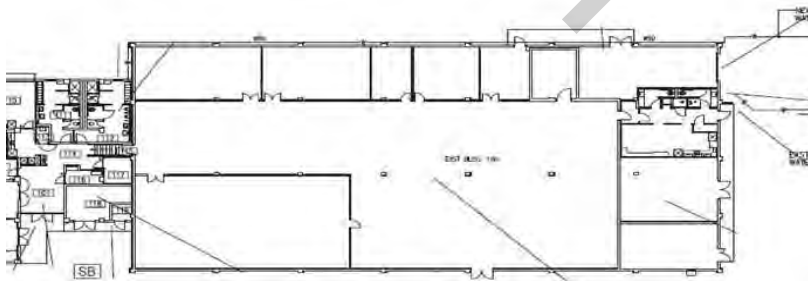
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-520	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

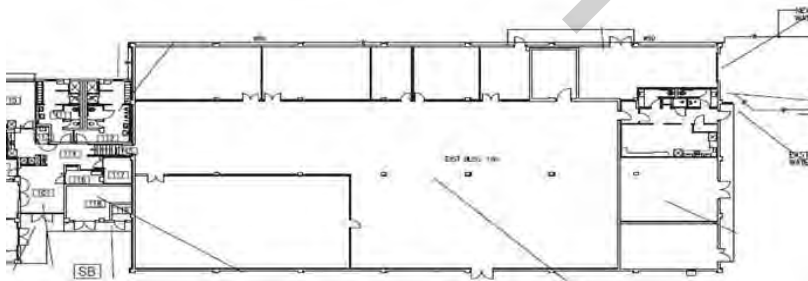
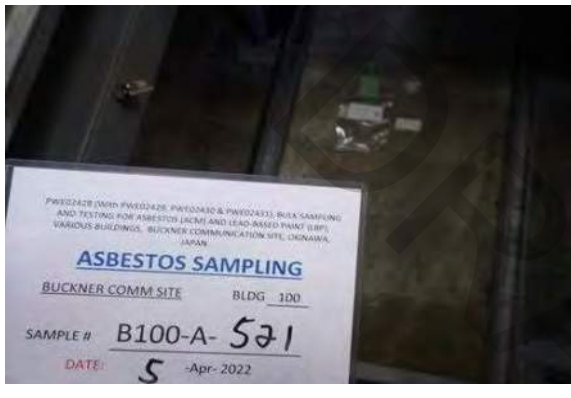
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-521	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
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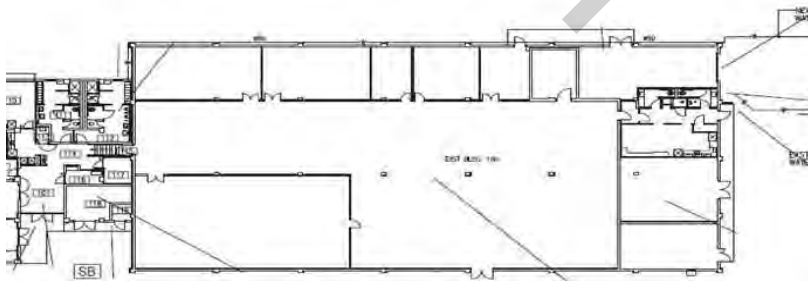
REPAIR HVAC WIDE BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-522	TYPE OF MATERIAL	: Floor Slab
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
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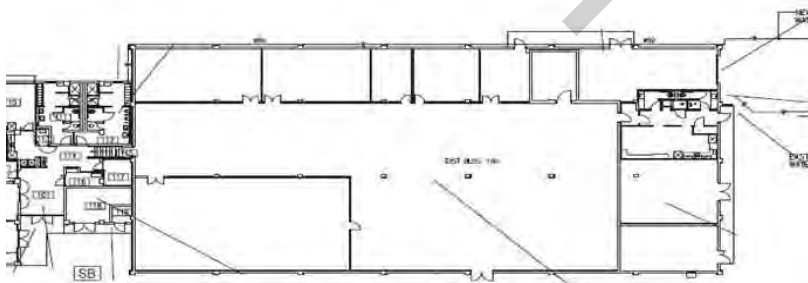
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-523	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

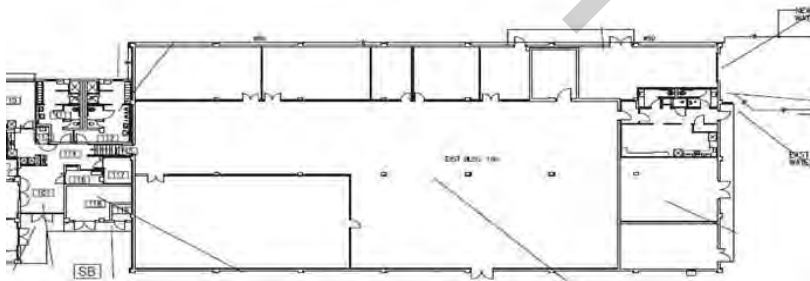
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-524	TYPE OF MATERIAL	:	Floor Slab
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Server Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
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B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

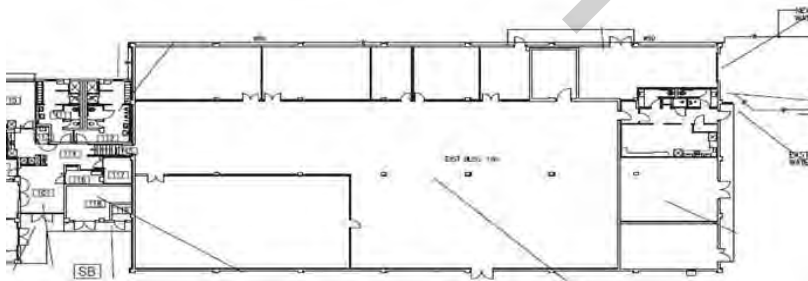
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-530	TYPE OF MATERIAL	: Acoustic Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

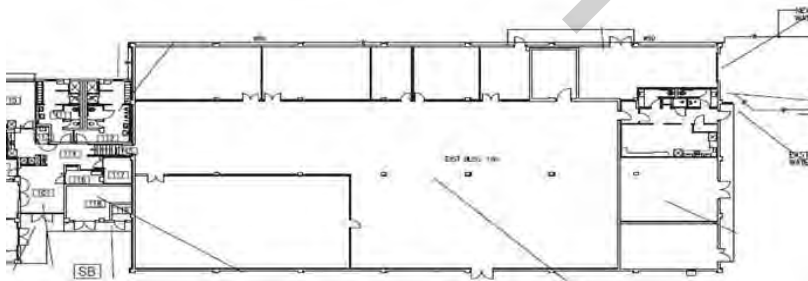
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-531	TYPE OF MATERIAL	: Acoustic Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
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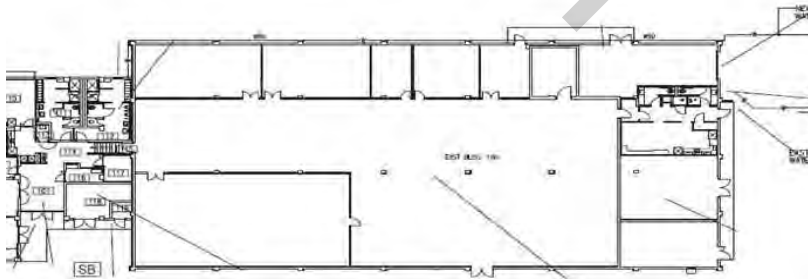
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-532	TYPE OF MATERIAL	: Acoustic Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

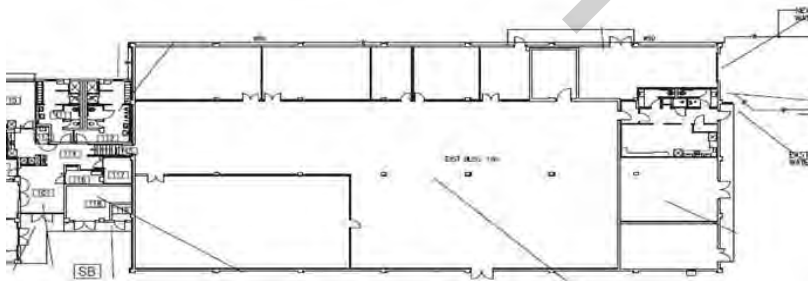
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-533	TYPE OF MATERIAL	: Wainscot Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
B100-A-516,B100-A-517,B100-A-518,B100-A-519,B100-A-520,
B100-A-521,B100-A-522,B100-A-523,B100-A-524,B100-A-530,
B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

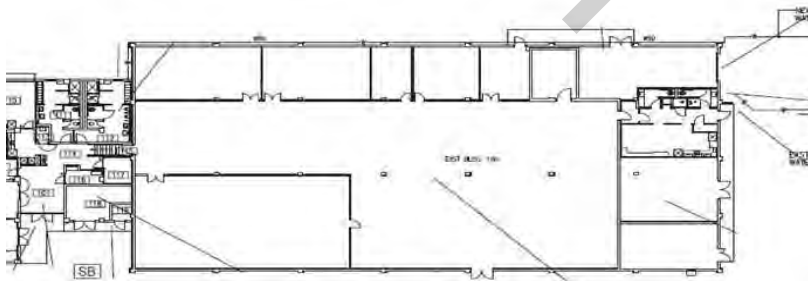
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-534	TYPE OF MATERIAL	: Wainscot Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
B100-A-511,B100-A-512,B100-A-513,B100-A-514,B100-A-515,
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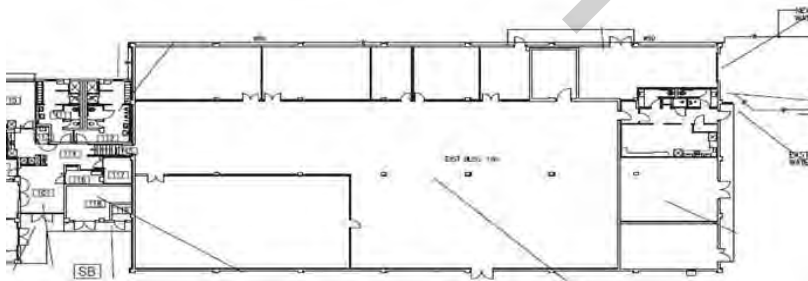
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-535	TYPE OF MATERIAL	: Wainscot Board
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-500,B100-A-501,B100-A-502,B100-A-503,B100-A-504,
B100-A-505,B100-A-506,B100-A-507,B100-A-508,B100-A-510,
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B100-A-531,B100-A-532,B100-A-533,B100-A-534,B100-A-535

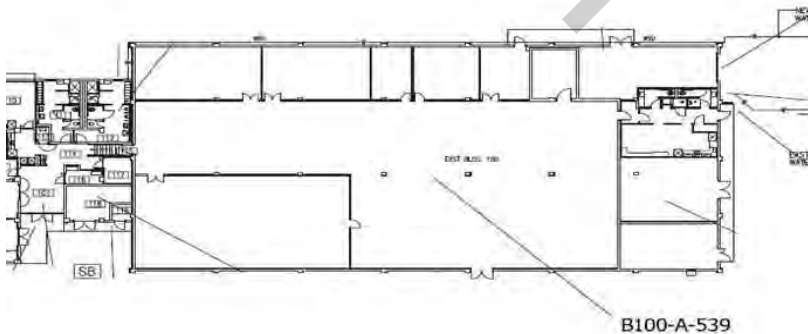
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-539	TYPE OF MATERIAL	: Floor Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



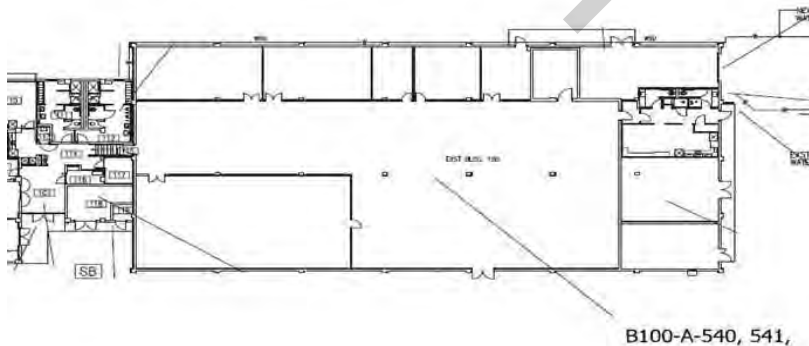
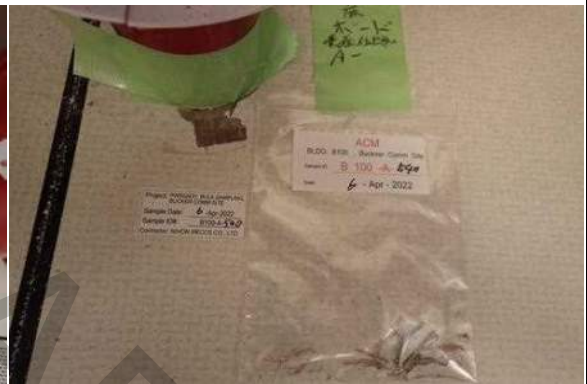
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-540	TYPE OF MATERIAL	: Floor Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



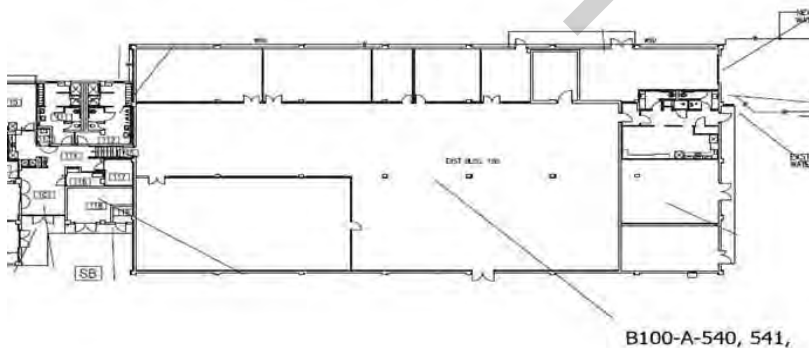
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-541	TYPE OF MATERIAL	: Floor Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Takafumi Noborikawa
ROOM NO./NAME	: Server Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-543	TYPE OF MATERIAL	:	Wall Tile w/Grout
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Kitchen	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



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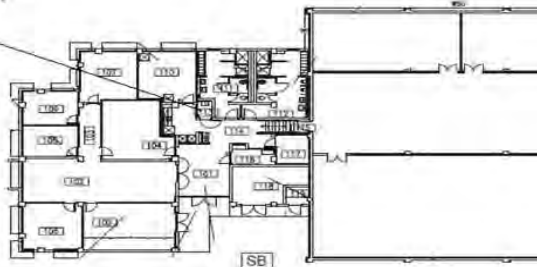
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-544	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Janitor	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-544, B100-A-545,
B100-A-546



REPAIR HVAC WIDEBAND SATCOM OPS B100
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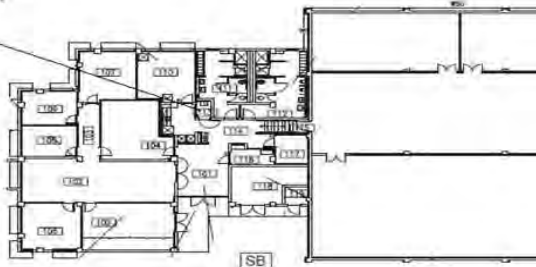
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-545	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Janitor	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-544,B100-A-545,
B100-A-546



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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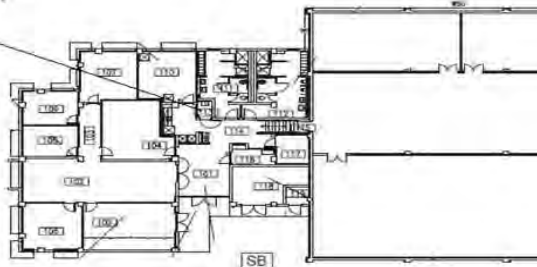
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-546	TYPE OF MATERIAL :	Floor Slab
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Janitor	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-544,B100-A-545,
B100-A-546



REPAIR HVAC WIDE BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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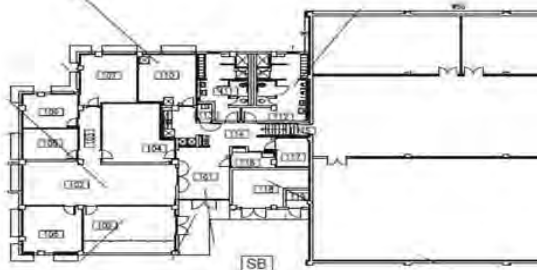
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-547	TYPE OF MATERIAL	: Dampproofing Material
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 110	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-547, B100-A-548



REPAIR HVAC WIDEBAND SATCOM OPS B100
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25SF0004

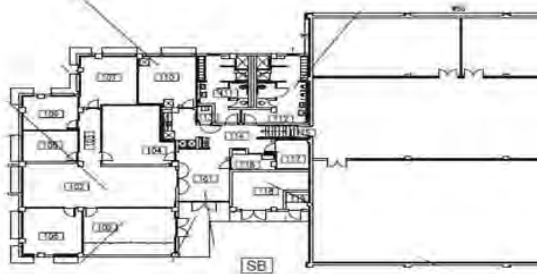
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-548	TYPE OF MATERIAL :	Wall Insulation
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 110	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-547, B100-A-548





Hawaii Analytical Laboratory ANALYTICAL REPORT

Thursday, May 26, 2022

Nanasei Environmental Laboratory, Co., Ltd
4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun
Okinawa Japan .

Phone Number: (098)835-8411

Facsimile:

Email:

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP), Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236023	B100-A-01 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Reddish Brown <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		Fibrous glass (amorphous)	2 Calcite + quartz + foam + paint	5/20/2022
202236024	B100-A-02 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Reddish Brown <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + foam + paint	5/20/2022
202236025	B100-A-03 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/20/2022
202236025	B100-A-03 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Ivory textured paint Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + paint	5/20/2022

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Phone Number: (098)835-8411

Nanasei Environmental Laboratory, Co., Ltd

Facsimile:

4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun

Email:

Okinawa Japan .

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236026	B100-A-04 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/20/2022
202236026	B100-A-04 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Ivory textured paint</u> Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + paint	5/20/2022
202236027	B100-A-05 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/23/2022
202236027	B100-A-05 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Ivory textured paint</u> Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + paint	5/23/2022
202236028	B100-A-06 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
202236029	B100-A-07 Exterior Concrete Wall Surface ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Reddish Brown <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/23/2022

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4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun

Email:

Okinawa Japan .

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236030	B100-A-08 Neo Seal • Elect Box, Exterior, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White neo seal</u> Comments		NONE DETECTED		None detected	Calcite + binder	5/23/2022
202236031	B100-A-10 Caulking Material • Exterior, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White caulk / paint</u> Comments		NONE DETECTED		None detected	Calcite + binder + paint	5/23/2022
202236032	B100-A-11 Concrete Floor Slab • Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
202236033	B100-A-12 Concrete Slab • Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
202236034	B100-A-13 Concrete Slab • Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
202236035	B100-A-14 Floor Mortar Repair Seal • Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Ivory paint / gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz + foam + paint	5/20/2022
202236036	B100-A-15 Expansion Joint • Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Off-white expansion joint / paint</u> Comments		NONE DETECTED		None detected	Binder + other	5/23/2022

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Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236036	B100-A-15 Expansion Joint - Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory Layer <u>Orange adhesive</u>		NONE DETECTED		Synthetic fibers (undulose) + cellulose (undulose)	10 Calcite + binder	5/23/2022
	Comments						
202236037	B100-A-16 Concrete Pad - Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray Layer <u>Gray concrete</u>		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
	Comments						
202236038	B100-A-17 Concrete Pad - Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray Layer <u>Gray concrete</u>		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
	Comments						
202236039	B100-A-18 Concrete Pad - Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray Layer <u>Gray concrete</u>		NONE DETECTED		None detected	Calcite + quartz	5/23/2022
	Comments						
202236040	B100-A-21 Pipe Flange Gasket - Exterior, B100, BUCKNER COMMUNICATION SITE, Blue Layer <u>Blue gasket / red paint</u>		NONE DETECTED		Synthetic fibers (undulose) + fibrous glass (amorphous)	25 Calcite + paint	5/24/2022
	Comments						
202236041	B100-A-22 Concrete Pad - AC, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray Layer <u>Gray concrete</u>		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
	Comments						

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Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236042	B100-A-23 Concrete Pad • AC, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		Wollastonite (+/- optical sign)	2 Calcite + quartz	5/24/2022
202236043	B100-A-24 Concrete Pad • AC, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236044	B100-A-25 Pipe Insulation • Straight, Exterior, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White pipe insulation / paint</u> Comments		NONE DETECTED		None detected	Paint + other	5/24/2022
202236045	B100-A-27 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Brown ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic	5/24/2022
202236045	B100-A-27 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Gray grout</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236045	B100-A-27 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022

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Okinawa Japan .

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236046	B100-A-28 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Brown ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic	5/24/2022
202236046	B100-A-28 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray grout</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236046	B100-A-28 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236047	B100-A-29 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Brown ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic	5/24/2022
202236047	B100-A-29 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Gray grout</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236047	B100-A-29 Ceramic Floor Tile w/Grout • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022

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Email:

Okinawa Japan .

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236048	B100-A-30 Mortar • Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> Brown / gray mortar Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236049	B100-A-31 Mortar • Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Brown / gray mortar Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236050	B100-A-32 Mortar • Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray mortar Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236051	B100-A-33 Concrete Floor Slab • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Brown <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236052	B100-A-34 Concrete Floor Slab • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236053	B100-A-35 Concrete Floor Slab • Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022

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Phone Number: (098)835-8411

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Facsimile:

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Email:

Okinawa Japan .

Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236054	B100-A-36 Expansion Joint • Wall, Exterior, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> White expansion joint / paint Comments		NONE DETECTED		None detected	Calcite + binder + paint	5/24/2022
202236055	B100-A-37 Interior Concrete Wall • Rm 101, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/24/2022
202236056	B100-A-38 Interior Concrete Wall • Rm 101, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/24/2022
202236057	B100-A-39 Interior Concrete Wall • Rm 101, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/24/2022
202236058	B100-A-40 Ceiling Board • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	65 Calcite + perlite + glass beads	5/24/2022
202236059	B100-A-41 Ceiling Board • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	65 Calcite + perlite + glass beads	5/24/2022
202236060	B100-A-42 Ceiling Board • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	65 Calcite + perlite + glass beads	5/24/2022

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Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202236061	B100-A-46 Ceiling Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected		Calcite + quartz + paint	5/24/2022
202236062	B100-A-47 Ceiling Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected		Calcite + quartz + paint	5/24/2022
202236063	B100-A-48 Ceiling Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Ivory paint / gray concrete Comments		NONE DETECTED		None detected		Calcite + quartz + paint	5/24/2022
202236064	B100-A-49 Vinyl Cove Base w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl cove base Comments		NONE DETECTED		None detected		Calcite + vinyl	5/24/2022
202236064	B100-A-49 Vinyl Cove Base w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected		Calcite + quartz + foam	5/24/2022
202236064	B100-A-49 Vinyl Cove Base w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Yellow mastic (limited) Comments		NONE DETECTED		None detected		Calcite + binder	5/24/2022
202236065	B100-A-50 Vinyl Cove Base w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige mastic / white paint Comments	Yes	Chrysotile	2	Cellulose (undulose)	2	Calcite + binder + paint	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236065	B100-A-50 Vinyl Cove Base w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl cove base Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236065	B100-A-50 Vinyl Cove Base w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022
202236066	B100-A-51 Vinyl Cove Base w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige mastic (limited) / white paint Comments	Yes	Chrysotile	2	Cellulose (undulose)	2 Calcite + binder + paint	5/24/2022
202236066	B100-A-51 Vinyl Cove Base w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl cove base Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236066	B100-A-51 Vinyl Cove Base w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022
202236067	B100-A-52 Wallboard - Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White paint / gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236068	B100-A-53 Wallboard • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White paint / gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236069	B100-A-54 Wallboard • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White paint / gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236070	B100-A-55 Vinyl Floor Tile w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige mastic Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + binder	5/24/2022
202236070	B100-A-55 Vinyl Floor Tile w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige vinyl floor tile Comments		NONE DETECTED		None detected	Vinyl	5/24/2022
202236070	B100-A-55 Vinyl Floor Tile w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022
202236071	B100-A-56 Vinyl Floor Tile w/Mastic • Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige mastic Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236071	B100-A-56 Vinyl Floor Tile w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige vinyl floor tile Comments		NONE DETECTED		None detected	Vinyl	5/24/2022
202236072	B100-A-57 Vinyl Floor Tile w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige mastic / gray cementitious material Comments		NONE DETECTED		Cellulose (undulose)	2 Calcite + binder + quartz	5/24/2022
202236072	B100-A-57 Vinyl Floor Tile w/Mastic - Rm 101, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Beige vinyl floor tile Comments		NONE DETECTED		None detected	Vinyl	5/24/2022
202236073	B100-A-58 Concrete Floor Slab - Rm 101, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022
202236074	B100-A-59 Concrete Floor Slab - Rm 101, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022
202236075	B100-A-60 Concrete Floor Slab - Rm 101, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz + foam	5/24/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236076	B100-A-62 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Fibrous glass (amorphous)	60 Foil	5/24/2022
202236076	B100-A-62 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236077	B100-A-63 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Fibrous glass (amorphous)	60 Foil	5/24/2022
202236077	B100-A-63 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236078	B100-A-64 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Fibrous glass (amorphous)	60 Foil	5/24/2022
202236078	B100-A-64 Duct Insulation • Rm 101, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202236079	B100-A-65 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray/black carpet</u> Comments		NONE DETECTED		Synthetic fibers (undulose) + fibrous glass (amorphous)	75	Tar + calcite + other	5/24/2022
202236079	B100-A-65 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	2	Binder + other	5/24/2022
202236079	B100-A-65 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow/gray carpet pad</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	25	Foam	5/24/2022
202236080	B100-A-66 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray/black carpet</u> Comments		NONE DETECTED		Synthetic fibers (undulose) + fibrous glass (amorphous)	75	Tar + calcite + other	5/24/2022
202236080	B100-A-66 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	2	Binder + other	5/24/2022
202236080	B100-A-66 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow/gray carpet pad</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	25	Foam	5/24/2022
202236081	B100-A-67 Floor Carpet w/Mastic ▪ Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray/black carpet</u> Comments		NONE DETECTED		Synthetic fibers (undulose) + fibrous glass (amorphous)	75	Tar + calcite + other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236081	B100-A-67 Floor Carpet w/Mastic • Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	2 Binder + other	5/24/2022
202236081	B100-A-67 Floor Carpet w/Mastic • Rm 102, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow/gray carpet pad</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	25 Foam	5/24/2022
202236082	B100-A-71 Ceiling Board • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236082	B100-A-71 Ceiling Board • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White joint compound / white paint</u> Comments		NONE DETECTED		None detected	Calcite + paint	5/24/2022
202236083	B100-A-72 Ceiling Board • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board / white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236084	B100-A-73 Ceiling Board • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White/gray ceiling tile</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022
202236085	B100-A-75 Ceiling Insulation • Rm 105, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver wrap</u> Comments		NONE DETECTED		Cellulose (undulose)	50 Foil	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236085	B100-A-75 Ceiling Insulation • Rm 105, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236086	B100-A-78 Ceiling Slab • Rm 105, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236087	B100-A-80 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige mastic Comments	Yes	Chrysotile	2	None detected	Calcite + quartz + binder	5/24/2022
202236087	B100-A-80 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236088	B100-A-81 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige mastic Comments	Yes	Chrysotile	2	None detected	Calcite + quartz + binder	5/24/2022
202236088	B100-A-81 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236089	B100-A-82 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige mastic Comments	Yes	Chrysotile	2	None detected	Calcite + quartz + binder	5/24/2022
202236089	B100-A-82 Vinyl Cove Base w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236090	B100-A-83 Wallboard • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236090	B100-A-83 Wallboard • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White joint compound / white paint / mesh Comments		NONE DETECTED		Synthetic fibers (undulose)	10 Gypsum + calcite + paint	5/24/2022
202236091	B100-A-84 Wallboard • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236092	B100-A-85 Wallboard • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board (1) / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236092	B100-A-85 Wallboard • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board (2) Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236093	B100-A-86 Wall Insulation • Rm 105, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236094	B100-A-87 Wall Insulation • Rm 105, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236095	B100-A-88 Wall Insulation • Rm 105, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236096	B100-A-89 Dampproofing Material • Rm 105, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black dampproofing material Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	5 Tar + calcite	5/24/2022
202236097	B100-A-90 Vinyl Floor Tile w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Tan vinyl floor tile Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236097	B100-A-90 Vinyl Floor Tile w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236098	B100-A-91 Vinyl Floor Tile w/Mastic • Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Tan vinyl floor tile Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236098	B100-A-91 Vinyl Floor Tile w/Mastic - Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236099	B100-A-92 Vinyl Floor Tile w/Mastic - Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Tan vinyl floor tile Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236099	B100-A-92 Vinyl Floor Tile w/Mastic - Rm 105, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236100	B100-A-93 Concrete Floor Slab - Rm 105, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236101	B100-A-94 Concrete Floor Slab - Rm 105, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236102	B100-A-95 Concrete Floor Slab - Rm 105, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236103	B100-A-96 Ceramic Tile Tile w/Grout - Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray mortar Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022

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Lab Job No: 202204687

Date Submitted: 5/12/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236103	B100-A-96 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceramic tile Comments		NONE DETECTED		None detected	Ceramic	5/23/2022
202236103	B100-A-96 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite + other	5/23/2022
202236104	B100-A-97 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray mortar Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236104	B100-A-97 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceramic tile Comments		NONE DETECTED		None detected	Ceramic	5/23/2022
202236104	B100-A-97 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite + other	5/23/2022
202236105	B100-A-98 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray mortar Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236105	B100-A-98 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceramic tile Comments		NONE DETECTED		None detected	Ceramic	5/23/2022
202236105	B100-A-98 Ceramic Tile Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite + other	5/23/2022
202236106	B100-A-99 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige ceramic tile Comments		NONE DETECTED		None detected	Ceramic	5/23/2022
202236106	B100-A-99 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Gray grout Comments		NONE DETECTED		None detected	Glass + other	5/23/2022
202236106	B100-A-99 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Gray mortar Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236107	B100-A-100 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige ceramic tile Comments		NONE DETECTED		None detected	Ceramic	5/23/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236107	B100-A-100 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236107	B100-A-100 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray grout</u> Comments		NONE DETECTED		None detected	Glass + other	5/23/2022
202236107	B100-A-100 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236108	B100-A-101 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Beige ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic	5/23/2022
202236108	B100-A-101 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236108	B100-A-101 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray grout</u> Comments		NONE DETECTED		None detected	Glass + other	5/23/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236108	B100-A-101 Ceramic Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236109	B100-A-105 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236109	B100-A-105 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236110	B100-A-106 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236110	B100-A-106 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236111	B100-A-107 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236111	B100-A-107 Concrete Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray mortar</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202236112	B100-A-108 Ceiling Board - Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board / off-white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/23/2022
202236113	B100-A-109 Ceiling Board - Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board / off-white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/23/2022
202236114	B100-A-110 Ceiling Board - Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White ceiling board / off-white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/23/2022
202236115	B100-A-111 Ceiling Insulation - Toilet, B100, BUCKNER COMMUNICATION SITE, Silver - Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose)	40	Aluminum	5/23/2022
202236115	B100-A-111 Ceiling Insulation - Toilet, B100, BUCKNER COMMUNICATION SITE, Silver - Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	95	Other	5/23/2022
202236116	B100-A-112 Ceiling Insulation - Toilet, B100, BUCKNER COMMUNICATION SITE, Silver - Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose)	40	Aluminum	5/23/2022
202236116	B100-A-112 Ceiling Insulation - Toilet, B100, BUCKNER COMMUNICATION SITE, Silver - Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	95	Other	5/23/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236117	B100-A-113 Ceiling Insulation • Toilet, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose)	40 Aluminum	5/23/2022
202236117	B100-A-113 Ceiling Insulation • Toilet, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	95 Other	5/23/2022
202236118	B100-A-114 Concrete Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236119	B100-A-115 Concrete Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236120	B100-A-116 Concrete Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236121	B100-A-117 Floor Mortar Repair Seal • Rm 110, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236122	B100-A-118 Floor Mortar Repair Seal • Rm 110, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236123	B100-A-119 Floor Mortar Repair Seal • Rm 110, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/23/2022
202236124	B100-A-120 Interior Concrete Wall • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236124	B100-A-120 Interior Concrete Wall • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray plaster / white paint</u> Comments	Yes	Chrysotile	3	None detected	Cementitious + paint	5/24/2022
202236125	B100-A-121 Interior Concrete Wall • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray cementitious material / white paint</u> Comments		NONE DETECTED		None detected	Cementitious + paint	5/24/2022
202236126	B100-A-122 Interior Concrete Wall • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray plaster/cementitious material / white paint</u> Comments	Yes	Chrysotile	1	None detected	Cementitious + paint	5/24/2022
202236127	B100-A-123 Ceiling Slab • Elec Room, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236127	B100-A-123 Ceiling Slab • Elec Room, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray plaster material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236128	B100-A-124 Ceiling Slab • Elec Room, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material / white paint</u> Comments		NONE DETECTED		None detected	Cementitious + paint	5/24/2022
202236129	B100-A-126 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Gray neo seal</u> Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236130	B100-A-127 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Gray neo seal</u> Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236131	B100-A-128 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Gray neo seal</u> Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236132	B100-A-129 Wallboard • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White drywall / off-white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236133	B100-A-130 Wallboard • Elec Room, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White drywall / off-white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236134	B100-A-131 Wallboard • Elec Room, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White drywall / off-white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236135	B100-A-132 Floor Slab • Elec Room, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236136	B100-A-133 Floor Slab • Elec Room, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236137	B100-A-134 Floor Slab • Elec Room, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material / paint</u> Comments		NONE DETECTED		None detected	Cementitious + paint	5/24/2022
202236138	B100-A-137 Duct Flange Gasket • Elec Room, B104, BUCKNER COMMUNICATION SITE, White • Black <u>Layer</u> <u>Black gasket / white paint</u> Comments		NONE DETECTED		None detected	Foam + paint	5/24/2022
202236139	B100-A-140 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray cementitious material (1) / off-white paint</u> Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/25/2022
202236139	B100-A-140 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray cementitious material (2)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236140	B100-A-141 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Gray cementitious material / off-white paint</u> Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236141	B100-A-142 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray cementitious material (1) Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236141	B100-A-142 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Gray cementitious material (2) Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236141	B100-A-142 Interior Concrete Wall • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White joint compound (limited) / off-white paint Comments	Yes	Chrysotile	1	None detected	Calcite + paint	5/25/2022
202236142	B100-A-143 Ceiling Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236143	B100-A-144 Ceiling Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material (1) Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236143	B100-A-144 Ceiling Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material (2) Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236144	B100-A-145 Ceiling Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022

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Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236145	B100-A-149 Wallboard • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White drywall / off-white paint Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	15 Gypsum + mica + paint	5/25/2022
202236146	B100-A-150 Wallboard • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white joint compound (limited) / paint Comments		NONE DETECTED		None detected	Calcite + paint	5/25/2022
202236146	B100-A-150 Wallboard • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White drywall Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	15 Gypsum + mica	5/25/2022
202236147	B100-A-151 Wallboard • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white joint compound (limited) / paint Comments		NONE DETECTED		None detected	Calcite + paint	5/25/2022
202236147	B100-A-151 Wallboard • Mech, B104, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White drywall Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	15 Gypsum + mica	5/25/2022
202236148	B100-A-152 Floor Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments	Yes	Chrysotile	1	None detected	Calcite + quartz	5/25/2022
202236149	B100-A-153 Floor Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material (1) / gray paint Comments	Yes	Chrysotile	1	None detected	Calcite + quartz + paint	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236149	B100-A-153 Floor Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (2)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236150	B100-A-154 Floor Slab • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material / gray paint</u> Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/25/2022
202236151	B100-A-155 Wall Mortar Repair Seal • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material / white paint</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	2 Calcite + quartz + paint	5/25/2022
202236152	B100-A-156 Floor Mortar Repair Seal • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (1) / gray paint</u> Comments	Yes	Chrysotile	2	None detected	Calcite + quartz + paint	5/25/2022
202236152	B100-A-156 Floor Mortar Repair Seal • Mech, B104, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (2)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236153	B100-A-157 Duct Insulation • Mech, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver foil wrap</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	85 Other	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236153	B100-A-157 Duct Insulation • Mech, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/25/2022
202236154	B100-A-158 Flexible Duct Joint • Interior, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black flex joint Comments		NONE DETECTED		Synthetic fibers (undulose)	65 Other	5/25/2022
202236155	B100-A-159 Concrete Pad • AHU, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236156	B100-A-160 Concrete Pad • AHU, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236157	B100-A-161 Concrete Pad • AHU, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236157	B100-A-161 Concrete Pad • AHU, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Off-white cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236158	B100-A-162 Concrete Pad • Compressor, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236159	B100-A-163 Concrete Pad • Compressor, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236160	B100-A-164 Concrete Pad • Compressor, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236160	B100-A-164 Concrete Pad • Compressor, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Off-white cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236161	B100-A-165 Concrete Pad • Pump, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236162	B100-A-166 Concrete Pad • Pump, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236163	B100-A-167 Concrete Pad • Pump, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236164	B100-A-168 Concrete Pad • Generator, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236165	B100-A-169 Concrete Pad • Generator, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236166	B100-A-170 Concrete Pad • Generator, Mech, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236167	B100-A-171 Pipe Flange Gasket • Mech, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Black gasket</u> Comments		NONE DETECTED		Synthetic fibers (undulose)	45 Other	5/25/2022
202236168	B100-A-172 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (1)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236168	B100-A-172 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (2)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236168	B100-A-172 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (3)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236169	B100-A-173 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray cementitious material (1)</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236169	B100-A-173 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236169	B100-A-173 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236170	B100-A-174 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236170	B100-A-174 Interior Concrete Wall • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236171	B100-A-175 Ceiling Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236172	B100-A-176 Ceiling Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022
202236173	B100-A-177 Ceiling Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236174	B100-A-178 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Black neo seal</u> Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236175	B100-A-179 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Black neo seal</u> Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236176	B100-A-180 Neo Seal • Elec Room, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White neo seal</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236177	B100-A-181 Concrete Floor Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236178	B100-A-182 Concrete Floor Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236179	B100-A-183 Concrete Floor Slab • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236180	B100-A-184 Wall Mortar Repair Seal • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236181	B100-A-185 Floor Mortar Repair Seal • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236182	B100-A-186 Duct Flange Gasket • Rm 118, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black gasket Comments		NONE DETECTED		None detected	Foam	5/24/2022
202236183	B100-A-188 Concrete Pad • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236184	B100-A-189 Concrete Pad • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236185	B100-A-190 Concrete Pad • Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236186	B100-A-191 Wall Insulation • Rm 118, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236187	B100-A-192 Wall Insulation • Rm 118, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236188	B100-A-193 Wall Insulation • Rm 118, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236189	B100-A-194 Pipe Insulation • Straight, Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver wrap</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	50 Foil	5/24/2022
202236189	B100-A-194 Pipe Insulation • Straight, Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236190	B100-A-195 Pipe Insulation • Elbow, Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver wrap</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	50 Foil	5/24/2022
202236190	B100-A-195 Pipe Insulation • Elbow, Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236191	B100-A-196 Roofing System • Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Off-white patch / white paint</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236191	B100-A-196 Roofing System • Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		None detected	Binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236192	B100-A-197 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white patch / white paint Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236192	B100-A-197 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic / gray cementitious material Comments		NONE DETECTED		None detected	Binder + cementitious	5/24/2022
202236193	B100-A-198 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white patch / white paint Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236193	B100-A-198 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic / gray cementitious material Comments		NONE DETECTED		None detected	Binder + cementitious	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (3) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (4) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (1) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (2) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022

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Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Tan acoustic tile Comments		NONE DETECTED		Cellulose (undulose)	45 Perlite + other	5/24/2022
202236194	B100-A-199 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Yellow foam insulation Comments		NONE DETECTED		None detected	Foam	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (3) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (4) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (1) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (2) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Tan acoustic tile Comments		NONE DETECTED		Cellulose (undulose)	45 Perlite + other	5/24/2022
202236195	B100-A-200 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Yellow foam insulation Comments		NONE DETECTED		None detected	Foam	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (3) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (4) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (1) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (2) Comments		NONE DETECTED		None detected	Tar	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Tan acoustic tile Comments		NONE DETECTED		Cellulose (undulose)	45 Perlite + other	5/24/2022
202236196	B100-A-201 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Yellow foam insulation Comments		NONE DETECTED		None detected	Foam	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (3) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black felt / tar (4) Comments		NONE DETECTED		Fibrous glass (amorphous)	20 Tar	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (1) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black roofing tar (2) Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (1) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Black shingle (2) Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar + aggregate	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Tan acoustic tile Comments		NONE DETECTED		Cellulose (undulose)	45 Perlite + other	5/24/2022
202236197	B100-A-202 Roofing System ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Black ▪ Yellow <u>Layer</u> Yellow foam insulation Comments		NONE DETECTED		None detected	Foam	5/24/2022
202236198	B100-A-203 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236198	B100-A-203 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Binder	5/24/2022
202236199	B100-A-204 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236199	B100-A-204 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Binder	5/24/2022
202236200	B100-A-205 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022

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Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236200	B100-A-205 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Binder	5/24/2022
202236201	B100-A-206 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Black tar Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236201	B100-A-206 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236202	B100-A-207 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Black tar Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236202	B100-A-207 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236203	B100-A-208 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Black tar Comments		NONE DETECTED		None detected	Tar	5/24/2022
202236203	B100-A-208 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236204	B100-A-209 Roof Concrete Surface ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray cementitious material Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236205	B100-A-213 Coating Material Of Ventilator ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white patch / white paint Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236205	B100-A-213 Coating Material Of Ventilator ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic / gray cementitious material Comments		NONE DETECTED		None detected	Binder + cementitious	5/24/2022
202236206	B100-A-214 Coating Material Of Ventilator ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white patch / white paint Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236207	B100-A-215 Coating Material Of Ventilator ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white patch / white paint Comments		NONE DETECTED		Fibrous glass (amorphous)	15 Binder + paint	5/24/2022
202236207	B100-A-215 Coating Material Of Ventilator ▪ Roof, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Yellow mastic / gray cementitious material Comments		NONE DETECTED		None detected	Binder + cementitious	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236208	B100-A-216 Ventilator Foundation Concrete ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236209	B100-A-217 Ventilator Foundation Concrete ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236209	B100-A-217 Ventilator Foundation Concrete ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		None detected	Binder	5/24/2022
202236210	B100-A-218 Ventilator Foundation Concrete ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Cementitious + other	5/24/2022
202236210	B100-A-218 Ventilator Foundation Concrete ▪ Roof, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Yellow mastic</u> Comments		NONE DETECTED		None detected	Binder	5/24/2022
202236211	B100-A-219 Siding Board (Brown) ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Reddish Brown <u>Layer</u> <u>Brown painted siding board</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Glass + paint + other	5/25/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236211	B100-A-219 Siding Board (Brown) ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Reddish Brown <u>Layer</u> <u>White foam insulation</u> Comments		NONE DETECTED		None detected	Foam	5/25/2022
202236212	B100-A-220 Siding Board (Ivory) ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Ivory painted siding board</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Glass + paint + other	5/25/2022
202236212	B100-A-220 Siding Board (Ivory) ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>White foam insulation</u> Comments		NONE DETECTED		None detected	Foam	5/25/2022
202236213	B100-A-222 Flange Gasket ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Beige gasket</u> Comments		NONE DETECTED		Synthetic fiber (undulose) + fibrous glass (amorphous)	65 Binder + other	5/25/2022
202236214	B100-A-223 Flange Gasket ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Red <u>Layer</u> <u>Black gasket / red paint</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Binder + paint	5/25/2022
202236215	B100-A-226 Waterproofing ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Black waterproofing</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	10 Tar	5/25/2022
202236216	B100-A-227 Concrete Wall ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> <u>Gray cementitious material</u> Comments	Yes	Chrysotile	5	None detected	Cementitious + other	5/25/2022

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Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236216	B100-A-227 Concrete Wall ▪ Exterior, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Ivory textured paint Comments		NONE DETECTED		Fibrous glass (amorphous)	2 Glass + paint + other	5/25/2022
202236217	B100-A-300 Floor Caulking ▪ Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray caulk Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236218	B100-A-301 Wall Caulking ▪ Rm 118, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray caulk Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236219	B100-A-302 Concrete Wall ▪ Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236220	B100-A-303 Concrete Wall ▪ Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236221	B100-A-304 Concrete Wall ▪ Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236222	B100-A-305 Ceiling Slab ▪ Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022

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Lab Job No: 202204687

Date Submitted: 5/12/2022

Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236223	B100-A-306 Ceiling Slab • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236224	B100-A-307 Ceiling Slab • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236225	B100-A-308 Wallboard • Rm 119, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board / white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236226	B100-A-309 Wallboard • Rm 119, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board / white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236227	B100-A-310 Wallboard • Rm 119, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board / white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236228	B100-A-311 Floor Slab • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236229	B100-A-312 Floor Slab • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete / yellow adhesive</u> Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236230	B100-A-313 Floor Slab • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236231	B100-A-314 Wall Caulking • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray caulk Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236231	B100-A-314 Wall Caulking • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> White foam Comments		NONE DETECTED		None detected	Foam	5/24/2022
202236232	B100-A-315 Floor Caulking • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray caulk Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236233	B100-A-316 Wallboard • Rm 118, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236234	B100-A-317 Wallboard • Rm 118, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236235	B100-A-318 Wallboard • Rm 118, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236236	B100-A-319 Duct Flange Insulation • Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Fibrous glass (amorphous)	50 Foil	5/24/2022
202236236	B100-A-319 Duct Flange Insulation • Rm 118, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236237	B100-A-400 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Foil	5/24/2022
202236237	B100-A-400 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236238	B100-A-401 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Foil	5/24/2022
202236238	B100-A-401 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236239	B100-A-402 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Silver wrap Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Foil	5/24/2022
202236239	B100-A-402 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Brown <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236240	B100-A-403 Ceiling Board • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling board Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	50 Perlite + calcite + other	5/24/2022
202236241	B100-A-404 Ceiling Board • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling board Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	50 Perlite + calcite + other	5/24/2022
202236242	B100-A-405 Ceiling Board • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling board Comments		NONE DETECTED		Fibrous glass (amorphous) + cellulose (undulose)	50 Perlite + calcite + other	5/24/2022
202236243	B100-A-406 Wallboard • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236244	B100-A-407 Wallboard • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236245	B100-A-408 Wallboard • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236245	B100-A-408 Wallboard • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236246	B100-A-409 Vinly Cove Base w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236246	B100-A-409 Vinly Cove Base w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Off-white mastic Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236247	B100-A-410 Vinly Cove Base w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236247	B100-A-410 Vinly Cove Base w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Off-white mastic Comments		NONE DETECTED		None detected	Binder + other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236248	B100-A-411 Vinly Cove Base w/Mastic ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Beige vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236248	B100-A-411 Vinly Cove Base w/Mastic ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Beige <u>Layer</u> Off-white mastic Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236249	B100-A-412 Wall Tile w/Grout ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Off-white adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236249	B100-A-412 Wall Tile w/Grout ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Pale pink ceramic tile Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022
202236249	B100-A-412 Wall Tile w/Grout ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236250	B100-A-415 Floor Tile w/Mastic ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236250	B100-A-415 Floor Tile w/Mastic ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Pale yellow floor tile Comments		NONE DETECTED		Wollastonite (+/- optical sign)	2 Calcite + vinyl	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236250	B100-A-415 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236251	B100-A-416 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236251	B100-A-416 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Pale yellow floor tile Comments		NONE DETECTED		Wollastonite (+/- optical sign)	2 Calcite + vinyl	5/24/2022
202236251	B100-A-416 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236252	B100-A-417 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236252	B100-A-417 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Pale yellow floor tile Comments		NONE DETECTED		Wollastonite (+/- optical sign)	2 Calcite + vinyl	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236252	B100-A-417 Floor Tile w/Mastic • Kitchen, B100, BUCKNER COMMUNICATION SITE, Pale Yellow <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236253	B100-A-418 Floor Slab • Kitchen, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236254	B100-A-419 Floor Slab • Kitchen, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236255	B100-A-420 Floor Slab • Kitchen, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236256	B100-A-421 Caulking (Gray) • Kitchen, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray caulk Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236257	B100-A-422 Caulking (White) • Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White caulk Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236258	B100-A-425 Ceiling Board • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202236259	B100-A-426 Ceiling Board • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/24/2022
202236260	B100-A-427 Ceiling Board • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/24/2022
202236261	B100-A-428 Wallboard • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	5/24/2022
202236262	B100-A-429 Wallboard • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	5/24/2022
202236263	B100-A-430 Wallboard • Toilet, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum	5/24/2022
202236264	B100-A-431 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Off-white adhesive Comments		NONE DETECTED		None detected		Calcite + quartz + binder	5/24/2022
202236264	B100-A-431 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Pale pink ceramic tile Comments		NONE DETECTED		None detected		Ceramic + quartz	5/24/2022

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Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236264	B100-A-431 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236265	B100-A-432 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236265	B100-A-432 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Pale pink ceramic tile Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022
202236265	B100-A-432 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite	5/24/2022
202236266	B100-A-433 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236266	B100-A-433 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> Pale pink ceramic tile Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022

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Your Project: PWE02430, Bulk Sampling and Testing of Asbestos (ACM) and Lead Based-Paint (LBP),
Various Buildings, Buckner Communication Site, Okinawa, Japan, 4/1/22-4/7/22

Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236266	B100-A-433 Wall Ceramic Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Pale Pink <u>Layer</u> <u>White grout</u> Comments		NONE DETECTED		None detected	Calcite	5/24/2022
202236267	B100-A-434 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Tan ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022
202236268	B100-A-435 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Quartz + calcite	5/24/2022
202236268	B100-A-435 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Tan ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022
202236268	B100-A-435 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>White grout</u> Comments		NONE DETECTED		None detected	Calcite	5/24/2022
202236269	B100-A-436 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Quartz + calcite	5/24/2022
202236269	B100-A-436 Floor Tile w/Grout • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Tan ceramic tile</u> Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236270	B100-A-437 Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236271	B100-A-438 Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236272	B100-A-439 Floor Slab • Toilet, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236273	B100-A-440 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver wrap</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Foil	5/24/2022
202236273	B100-A-440 Ceiling Insulation • Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236274	B100-A-441 Pipe Insulation • Straight, Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Silver wrap</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	50 Foil	5/24/2022
202236274	B100-A-441 Pipe Insulation • Straight, Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236275	B100-A-442 Pipe Insulation • Elbow, Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Silver wrap Comments		NONE DETECTED		Fibrous glass (amorphous)	50 Foil	5/24/2022
202236275	B100-A-442 Pipe Insulation • Elbow, Kitchen, B100, BUCKNER COMMUNICATION SITE, Silver • Yellow <u>Layer</u> Yellow insulation Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022
202236276	B100-A-500 Ceiling Board • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling tile Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022
202236277	B100-A-501 Ceiling Board • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling tile Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022
202236278	B100-A-502 Ceiling Board • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling tile Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022
202236279	B100-A-503 Ceiling Board • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling tile Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022
202236280	B100-A-504 Ceiling Board • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray ceiling tile Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50 Perlite + calcite + other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236281	B100-A-505 Ceiling Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236282	B100-A-506 Ceiling Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236283	B100-A-507 Ceiling Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236284	B100-A-508 Ceiling Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> <u>Gray concrete</u> Comments	Yes	Chrysotile	2	None detected	Calcite + quartz	5/24/2022
202236285	B100-A-510 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board / white paint</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum + paint	5/24/2022
202236286	B100-A-511 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>Blue joint compound</u> Comments	Yes	Chrysotile	1	None detected	Calcite + quartz + other	5/24/2022
202236286	B100-A-511 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> <u>White gypsum board</u> Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236287	B100-A-512 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236288	B100-A-513 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236289	B100-A-514 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236289	B100-A-514 Wallboard • Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White joint compound / white paint Comments	Yes	Chrysotile	1	None detected	Calcite + quartz + paint	5/24/2022
202236290	B100-A-515 Vinyl Cove Base w/Mastic • Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236290	B100-A-515 Vinyl Cove Base w/Mastic • Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236291	B100-A-516 Vinyl Cove Base w/Mastic • Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236291	B100-A-516 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Cream caulk Comments		NONE DETECTED		None detected	Binder + other	5/24/2022
202236291	B100-A-516 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236292	B100-A-517 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236292	B100-A-517 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236293	B100-A-518 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236293	B100-A-518 Vinyl Cove Base w/Mastic ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236294	B100-A-519 Vinyl Cove Base w/Mastic • Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Black vinyl covebase Comments		NONE DETECTED		None detected	Calcite + vinyl	5/24/2022
202236294	B100-A-519 Vinyl Cove Base w/Mastic • Server Rm, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> Yellow mastic Comments		NONE DETECTED		None detected	Calcite + binder	5/24/2022
202236295	B100-A-520 Floor Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236296	B100-A-521 Floor Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236297	B100-A-522 Floor Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022
202236298	B100-A-523 Floor Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments	Yes	Chrysotile	2	None detected	Calcite + quartz	5/24/2022
202236299	B100-A-524 Floor Slab • Server Rm, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete / yellow adhesive Comments		NONE DETECTED		None detected	Calcite + quartz + binder	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v	Matrix	Date Analyzed
202236300	B100-A-530 Acoustic Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray acoustic board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50	Perlite + calcite + other	5/24/2022
202236301	B100-A-531 Acoustic Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray acoustic board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50	Perlite + calcite + other	5/24/2022
202236302	B100-A-532 Acoustic Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White/gray acoustic board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	50	Perlite + calcite + other	5/24/2022
202236303	B100-A-533 Wainscot Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / white paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/24/2022
202236304	B100-A-534 Wainscot Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / silver paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/24/2022
202236305	B100-A-535 Wainscot Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White gypsum board / silver paint Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15	Gypsum + paint	5/24/2022
202236306	B100-A-539 Floor Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white floor board Comments		NONE DETECTED		Cellulose / wood fiber (undulose)	80	Other	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236307	B100-A-540 Floor Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white floor board Comments		NONE DETECTED		Cellulose / wood fiber (undulose)	80 Other	5/24/2022
202236308	B100-A-541 Floor Board ▪ Server Rm, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white floor board Comments		NONE DETECTED		Cellulose / wood fiber (undulose)	80 Other	5/24/2022
202236309	B100-A-543 Wall Tile w/Grout ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> Off-white ceramic tile Comments		NONE DETECTED		None detected	Ceramic + quartz	5/24/2022
202236309	B100-A-543 Wall Tile w/Grout ▪ Kitchen, B100, BUCKNER COMMUNICATION SITE, White <u>Layer</u> White grout Comments		NONE DETECTED		None detected	Calcite	5/24/2022
202236310	B100-A-544 Ceiling Board ▪ Janitor, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> White gypsum board Comments		NONE DETECTED		Cellulose (undulose) + fibrous glass (amorphous)	15 Gypsum	5/24/2022
202236311	B100-A-545 Wallboard ▪ Janitor, B100, BUCKNER COMMUNICATION SITE, Ivory <u>Layer</u> Gray concrete / ivory paint Comments		NONE DETECTED		None detected	Calcite + quartz + paint	5/24/2022
202236312	B100-A-546 Floor Slab ▪ Janitor, B100, BUCKNER COMMUNICATION SITE, Gray <u>Layer</u> Gray concrete Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022

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Bulk Asbestos Determination

Sample No.	Your Sample ID / Description	Asbestos Present?	Type	%v/v	Other Fibrous	%v/v Matrix	Date Analyzed
202236313	B100-A-547 Dampproofing Material • Rm 110, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Black dampproofing material</u> Comments		NONE DETECTED		Cellulose (undulose)	2 Tar + calcite	5/24/2022
202236313	B100-A-547 Dampproofing Material • Rm 110, B100, BUCKNER COMMUNICATION SITE, Black <u>Layer</u> <u>Gray concrete</u> Comments		NONE DETECTED		None detected	Calcite + quartz	5/24/2022
202236314	B100-A-548 Wall Insulation • Rm 110, B100, BUCKNER COMMUNICATION SITE, Yellow <u>Layer</u> <u>Yellow insulation</u> Comments		NONE DETECTED		Fibrous glass (amorphous)	> 99 None detected	5/24/2022

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General Comments

The bulk sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures outlined in the United States Environmental Protection Agency's "Interim Method for the Determination of Asbestos in Bulk Insulation Samples" (EPA-600/M4-82-020, Dec. 1982) and / or "Method for Determination of Asbestos in bulk Building Materials" (EPA-600/R-93-116, July 1993). The analysis of each bulk sample relates only to the material examined, and may or may not represent the overall composition of its original source. Floor tile and other resinously bound materials, when analyzed by the EPA methods referenced above may yield false negative results because of limitations in separating closely bound fibers and in detecting fibers of small length and diameter. Alternative methods of identification, including Transmission Electron Microscopy (TEM) may or may not be applicable. We utilize calibrated visual area estimation on a routine basis and do not conduct point counting unless specifically requested to do so. Estimated error for the visual determinations presented are 75% relative (1 to 2%), 50% relative (3 to 5%); 25% relative (6 to 25%) and 20% (>26% v/v). We will not separate layers which in our opinion are not readily discernable. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report must not be used by the client to claim product certification, approval or endorsement by NVLAP, NIST or any agency of Federal Government. Unless otherwise indicated, the sample condition at the time of receipt was acceptable.

Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

None Detected = asbestos was not observed in the sample. If trace amount of asbestos was detected below our quantifiable limits of 1.0%, <1% (trace) would be indicated and the asbestos type listed. Point counting, where applicable, are recommended to improve accuracy.



Jennifer Hsu Liao
Laboratory Manager

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OFFICIAL LABORATORY CHAIN OF CUSTODY

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN		(1) CUSTOMER INFORMATION お客様情報 (会社名・所在地等): Name お名前: 株式会社南西環境研究所 (Nanasei Environmental Laboratory Co., Ltd) Address 住所: 〒903-0105 沖縄県那中環都西原町字東崎4-4 (4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun, Okinawa.) Tel / Fax: Tel 098-835-8411 / Fax 098-835-8412 E-mail: noborikawa@nansei-kankyo.co.jp Contact Name ご担当者: 登川 貴史	
RESPONSIBLE FOR: (PRINT NAME) サンプル採取責任者: (楷書体) Nanasei Environmental Laboratory Co., Ltd Takafumi Noborikawa	TOTAL SAMPLES 10/292 サンプル総数:		
LANGUAGE 報告言語: ☒ ENG. 英語 ☐ JPN. 日本語 # OF COPIES 報告書部数: 1			
REPORT ATTN TO 報告書宛名: ☐ Same as (1) と同じ ☒ Same as (2) と同じ		(2) Name お名前: NIHON MECES CO., LTD.	
SEND REPORT TO 速報送付先: ☒ Same as (1) と同じ ☐ Same as (2) と同じ		Address 住所: 4-16-7 Minamitobaru, Okinawa City,	
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CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION サンプル名・採取場所	SAMPLE DESCRIPTION サンプル詳細(色・形状・材質等)
B100-A-01	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Reddish Brown
B100-A-02	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Reddish Brown
B100-A-03	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory
B100-A-04	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory
B100-A-05	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory
B100-A-06	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory
B100-A-07	2022/4/1	Exterior Concrete Wall Surface · Exterior, B100, BUCKNER COMMUNICATION SITE	Reddish Brown
B100-A-08	2022/4/1	Neo Seal · Elect Box, Exterior, B100, BUCKNER COMMUNICATION SITE	White
B100-A-10	2022/4/1	Caulking Material · Exterior, B100, BUCKNER COMMUNICATION SITE	White
B100-A-11	2022/4/1	Concrete Floor Slab · Exterior, B100, BUCKNER COMMUNICATION SITE	Gray
Please select REQUIRED ANALYSIS 依頼分析方法: ☒ EPA >1% (600/M4-82-020 PLM) to >0.1% for Trace ☒ EPA >0.1% (600/R-93-116 1,000 Point Count)		Turn-Around-Time 報告希望日: ☒ Standard (5-10 Work Days) 通常分析(5-10営業日) ☐ Priority (1-3 Work Days) 特急分析(1-3営業日)	
RELINQUISHED BY: (SIGNATURE) 依頼者・受渡者(サイン)		DATE / TIME 依頼日時	REMARK 備考:
RECEIVED BY: (SIGNATURE) 受取者(サイン)		DATE / TIME 受取日時	REMARK (備考)
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OFFICIAL LABORATORY CHAIN OF CUSTODY

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 30/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所ジョブ番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細	LAB SAMPLE NUMBER 分析所サンプル番号	
B100-A-12	2022/4/1	Concrete Slab ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236033	
B100-A-13	2022/4/1	Concrete Slab ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236034	
B100-A-14	2022/4/1	Floor Mortar Repair Seal ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236035	
B100-A-15	2022/4/1	Expansion Joint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory	202236036	
B100-A-16	2022/4/1	Concrete Pad ・ Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236037	
B100-A-17	2022/4/1	Concrete Pad ・ Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236038	
B100-A-18	2022/4/1	Concrete Pad ・ Condenser, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236039	
B100-A-21	2022/4/1	Pipe Flange Gasket ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Blue	202236040	
B100-A-22	2022/4/1	Concrete Pad ・ AC, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236041	
B100-A-23	2022/4/1	Concrete Pad ・ AC, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236042	
B100-A-24	2022/4/1	Concrete Pad ・ AC, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236043	
B100-A-25	2022/4/1	Pipe Insulation ・ Straight, Exterior, B100, BUCKNER COMMUNICATION SITE	White	202236044	
B100-A-27	2022/4/1	Ceramic Floor Tile w/Grout ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Brown	202236045	
B100-A-28	2022/4/1	Ceramic Floor Tile w/Grout ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236046	
B100-A-29	2022/4/1	Ceramic Floor Tile w/Grout ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Brown	202236047	
B100-A-30	2022/4/1	Mortar ・ Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Brown	202236048	
B100-A-31	2022/4/1	Mortar ・ Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236049	
B100-A-32	2022/4/1	Mortar ・ Under Tile, Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236050	
B100-A-33	2022/4/1	Concrete Floor Slab ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Brown	202236051	
B100-A-34	2022/4/1	Concrete Floor Slab ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray	202236052	

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN		TOTAL SAMPLES サンプル総数: 50/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%
		LAB JOB NUMBER 分析所ジョブ番号: 202204687	
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細 LAB SAMPLE NUMBER 分析所サンプル#
B100-A-35	2022/4/1	Concrete Floor Slab ・ Front Entrance, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray 202236053
B100-A-36	2022/4/1	Expansion Joint ・ Wall, Exterior, B100, BUCKNER COMMUNICATION SITE	Gray 202236054
B100-A-37	2022/4/6	Interior Concrete Wall ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Ivory 202236055
B100-A-38	2022/4/6	Interior Concrete Wall ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Ivory 202236056
B100-A-39	2022/4/6	Interior Concrete Wall ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Ivory 202236057
B100-A-40	2022/4/6	Ceiling Board ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236058
B100-A-41	2022/4/6	Ceiling Board ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236059
B100-A-42	2022/4/6	Ceiling Board ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236060
B100-A-46	2022/4/6	Ceiling Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray 202236061
B100-A-47	2022/4/6	Ceiling Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray 202236062
B100-A-48	2022/4/6	Ceiling Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray 202236063
B100-A-49	2022/4/6	Vinyl Cove Base w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Beige 202236064
B100-A-50	2022/4/6	Vinyl Cove Base w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Beige 202236065
B100-A-51	2022/4/6	Vinyl Cove Base w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	Beige 202236066
B100-A-52	2022/4/6	Wallboard ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236067
B100-A-53	2022/4/6	Wallboard ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236068
B100-A-54	2022/4/6	Wallboard ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236069
B100-A-55	2022/4/6	Vinyl Floor Tile w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236070
B100-A-56	2022/4/6	Vinyl Floor Tile w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236071
B100-A-57	2022/4/6	Vinyl Floor Tile w/Mastic ・ Rm 101, B100, BUCKNER COMMUNICATION SITE	White 202236072

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN		TOTAL SAMPLES サンプル総数: 70/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所ジョブ番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細	LAB SAMPLE NUMBER 分析所サンプル番号
B100-A-58	2022/4/6	Concrete Floor Slab · Rm 101, B100, BUCKNER COMMUNICATION SITE	Gray	202236073
B100-A-59	2022/4/6	Concrete Floor Slab · Rm 101, B100, BUCKNER COMMUNICATION SITE	Gray	202236074
B100-A-60	2022/4/6	Concrete Floor Slab · Rm 101, B100, BUCKNER COMMUNICATION SITE	Gray	202236075
B100-A-62	2022/4/6	Duct Insulation · Rm 101, B100, BUCKNER COMMUNICATION SITE	Silver · Yellow	202236076
B100-A-63	2022/4/6	Duct Insulation · Rm 101, B100, BUCKNER COMMUNICATION SITE	Silver · Yellow	202236077
B100-A-64	2022/4/6	Duct Insulation · Rm 101, B100, BUCKNER COMMUNICATION SITE	Silver · Yellow	202236078
B100-A-65	2022/4/7	Floor Carpet w/Mastic · Rm 102, B100, BUCKNER COMMUNICATION SITE	Gray	202236079
B100-A-66	2022/4/7	Floor Carpet w/Mastic · Rm 102, B100, BUCKNER COMMUNICATION SITE	Gray	202236080
B100-A-67	2022/4/7	Floor Carpet w/Mastic · Rm 102, B100, BUCKNER COMMUNICATION SITE	Gray	202236081
B100-A-71	2022/4/7	Ceiling Board · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236082
B100-A-72	2022/4/7	Ceiling Board · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236083
B100-A-73	2022/4/7	Ceiling Board · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236084
B100-A-75	2022/4/7	Ceiling Insulation · Rm 105, B100, BUCKNER COMMUNICATION SITE	Silver · Yellow	202236085
B100-A-78	2022/4/7	Ceiling Slab · Rm 105, B100, BUCKNER COMMUNICATION SITE	Gray	202236086
B100-A-80	2022/4/7	Vinyl Cove Base w/Mastic · Rm 105, B100, BUCKNER COMMUNICATION SITE	Beige	202236087
B100-A-81	2022/4/7	Vinyl Cove Base w/Mastic · Rm 105, B100, BUCKNER COMMUNICATION SITE	Beige	202236088
B100-A-82	2022/4/7	Vinyl Cove Base w/Mastic · Rm 105, B100, BUCKNER COMMUNICATION SITE	Beige	202236089
B100-A-83	2022/4/7	Wallboard · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236090
B100-A-84	2022/4/7	Wallboard · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236091
B100-A-85	2022/4/7	Wallboard · Rm 105, B100, BUCKNER COMMUNICATION SITE	White	202236092



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED- PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 90/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所依頼番号 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル#
B100-A-86	2022/4/7	Wall Insulation ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Yellow		202236093
B100-A-87	2022/4/7	Wall Insulation ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Yellow		202236094
B100-A-88	2022/4/7	Wall Insulation ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Yellow		202236095
B100-A-89	2022/4/7	Dampproofing Material ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Black		202236096
B100-A-90	2022/4/7	Vinyl Floor Tile w/Mastic ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	White		202236097
B100-A-91	2022/4/7	Vinyl Floor Tile w/Mastic ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	White		202236098
B100-A-92	2022/4/7	Vinyl Floor Tile w/Mastic ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	White		202236099
B100-A-93	2022/4/7	Concrete Floor Slab ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Gray		202236100
B100-A-94	2022/4/7	Concrete Floor Slab ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Gray		202236101
B100-A-95	2022/4/7	Concrete Floor Slab ・ Rm 105, B100, BUCKNER COMMUNICATION SITE	Gray		202236102
B100-A-96	2022/4/6	Ceramic Tile Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236103
B100-A-97	2022/4/6	Ceramic Tile Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236104
B100-A-98	2022/4/6	Ceramic Tile Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236105
B100-A-99	2022/4/6	Ceramic Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Beige		202236106
B100-A-100	2022/4/6	Ceramic Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Beige		202236107
B100-A-101	2022/4/6	Ceramic Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Beige		202236108
B100-A-105	2022/4/6	Concrete Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236109
B100-A-106	2022/4/6	Concrete Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236110
B100-A-107	2022/4/6	Concrete Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236111
B100-A-108	2022/4/6	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236112



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED- PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 110/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細	LAB SAMPLE NUMBER 分析所サンプル番号	
B100-A-109	2022/4/6	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White	202236113	
B100-A-110	2022/4/6	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White	202236114	
B100-A-111	2022/4/6	Ceiling Insulation ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow	202236115	
B100-A-112	2022/4/6	Ceiling Insulation ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow	202236116	
B100-A-113	2022/4/6	Ceiling Insulation ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow	202236117	
B100-A-114	2022/4/6	Concrete Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray	202236118	
B100-A-115	2022/4/6	Concrete Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray	202236119	
B100-A-116	2022/4/6	Concrete Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray	202236120	
B100-A-117	2022/4/6	Floor Mortar Repair Seal ・ Rm 110, B100, BUCKNER COMMUNICATION SITE	Gray	202236121	
B100-A-118	2022/4/6	Floor Mortar Repair Seal ・ Rm 110, B100, BUCKNER COMMUNICATION SITE	Gray	202236122	
B100-A-119	2022/4/6	Floor Mortar Repair Seal ・ Rm 110, B100, BUCKNER COMMUNICATION SITE	Gray	202236123	
B100-A-120	2022/4/1	Interior Concrete Wall ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	White	202236124	
B100-A-121	2022/4/1	Interior Concrete Wall ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	White	202236125	
B100-A-122	2022/4/1	Interior Concrete Wall ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	White	202236126	
B100-A-123	2022/4/1	Ceiling Slab ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Gray	202236127	
B100-A-124	2022/4/1	Ceiling Slab ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Gray	202236128	
B100-A-126	2022/4/1	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Black	202236129	
B100-A-127	2022/4/1	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Black	202236130	
B100-A-128	2022/4/1	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Black	202236131	
B100-A-129	2022/4/1	Wallboard ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	White	202236132	



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 130/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所依頼番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細	LAB SAMPLE NUMBER 分析所サンプル番号	
B100-A-130	2022/4/1	Wallboard ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	White	202236133	
B100-A-131	2022/4/1	Wallboard ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	White	202236134	
B100-A-132	2022/4/1	Floor Slab ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	Gray	202236135	
B100-A-133	2022/4/1	Floor Slab ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	Gray	202236136	
B100-A-134	2022/4/1	Floor Slab ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	Gray	202236137	
B100-A-137	2022/4/1	Duct Flange Gasket ・ Elec Room, B104, BUCKNER COMMUNICATION SITE	White ・ Black	202236138	
B100-A-140	2022/4/4	Interior Concrete Wall ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236139	
B100-A-141	2022/4/4	Interior Concrete Wall ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236140	
B100-A-142	2022/4/4	Interior Concrete Wall ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236141	
B100-A-143	2022/4/4	Ceiling Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236142	
B100-A-144	2022/4/4	Ceiling Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236143	
B100-A-145	2022/4/4	Ceiling Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236144	
B100-A-149	2022/4/4	Wallboard ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236145	
B100-A-150	2022/4/4	Wallboard ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236146	
B100-A-151	2022/4/4	Wallboard ・ Mech, B104, BUCKNER COMMUNICATION SITE	White	202236147	
B100-A-152	2022/4/4	Floor Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236148	
B100-A-153	2022/4/4	Floor Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236149	
B100-A-154	2022/4/4	Floor Slab ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236150	
B100-A-155	2022/4/4	Wall Mortar Repair Seal ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236151	
B100-A-156	2022/4/4	Floor Mortar Repair Seal ・ Mech, B104, BUCKNER COMMUNICATION SITE	Gray	202236152	



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 150/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所シリアル番号 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細	LAB SAMPLE NUMBER 分析所サンプル番号	
B100-A-157	2022/4/4	Duct Insulation ・ Mech, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow	202236153	
B100-A-158	2022/4/4	Flexible Duct Joint ・ Interior, B100, BUCKNER COMMUNICATION SITE	Silver	202236154	
B100-A-159	2022/4/4	Concrete Pad ・ AHU, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236155	
B100-A-160	2022/4/4	Concrete Pad ・ AHU, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236156	
B100-A-161	2022/4/4	Concrete Pad ・ AHU, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236157	
B100-A-162	2022/4/4	Concrete Pad ・ Compressor, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236158	
B100-A-163	2022/4/4	Concrete Pad ・ Compressor, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236159	
B100-A-164	2022/4/4	Concrete Pad ・ Compressor, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236160	
B100-A-165	2022/4/4	Concrete Pad ・ Pump, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236161	
B100-A-166	2022/4/4	Concrete Pad ・ Pump, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236162	
B100-A-167	2022/4/4	Concrete Pad ・ Pump, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236163	
B100-A-168	2022/4/4	Concrete Pad ・ Generator, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236164	
B100-A-169	2022/4/4	Concrete Pad ・ Generator, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236165	
B100-A-170	2022/4/4	Concrete Pad ・ Generator, Mech, B100, BUCKNER COMMUNICATION SITE	Gray	202236166	
B100-A-171	2022/4/4	Pipe Flange Gasket ・ Mech, B100, BUCKNER COMMUNICATION SITE	Black	202236167	
B100-A-172	2022/4/4	Interior Concrete Wall ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray	202236168	
B100-A-173	2022/4/4	Interior Concrete Wall ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray	202236169	
B100-A-174	2022/4/4	Interior Concrete Wall ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray	202236170	
B100-A-175	2022/4/4	Ceiling Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray	202236171	
B100-A-176	2022/4/4	Ceiling Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray	202236172	



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 170/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所シヨウ番号: 202204687
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル番号
B100-A-177	2022/4/4	Ceiling Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236173
B100-A-178	2022/4/4	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Black		202236174
B100-A-179	2022/4/4	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	Black		202236175
B100-A-180	2022/4/4	Neo Seal ・ Elec Room, B100, BUCKNER COMMUNICATION SITE	White		202236176
B100-A-181	2022/4/4	Concrete Floor Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236177
B100-A-182	2022/4/4	Concrete Floor Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236178
B100-A-183	2022/4/4	Concrete Floor Slab ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236179
B100-A-184	2022/4/4	Wall Mortar Repair Seal ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236180
B100-A-185	2022/4/4	Floor Mortar Repair Seal ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236181
B100-A-186	2022/4/4	Duct Flange Gasket ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Black		202236182
B100-A-188	2022/4/4	Concrete Pad ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236183
B100-A-189	2022/4/4	Concrete Pad ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236184
B100-A-190	2022/4/4	Concrete Pad ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236185
B100-A-191	2022/4/4	Wall Insulation ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Yellow		202236186
B100-A-192	2022/4/4	Wall Insulation ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Yellow		202236187
B100-A-193	2022/4/4	Wall Insulation ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Yellow		202236188
B100-A-194	2022/4/4	Pipe Insulation ・ Straight, Rm 118, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236189
B100-A-195	2022/4/4	Pipe Insulation ・ Elbow, Rm 118, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236190
B100-A-196	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236191
B100-A-197	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236192



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 190/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所シリアル番号: 202204687
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル番号
B100-A-198	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236193
B100-A-199	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	Black ・ Yellow		202236194
B100-A-200	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	Black ・ Yellow		202236195
B100-A-201	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	Black ・ Yellow		202236196
B100-A-202	2022/4/4	Roofing System ・ Roof, B100, BUCKNER COMMUNICATION SITE	Black ・ Yellow		202236197
B100-A-203	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236198
B100-A-204	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236199
B100-A-205	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236200
B100-A-206	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236201
B100-A-207	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236202
B100-A-208	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236203
B100-A-209	2022/4/4	Roof Concrete Surface ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236204
B100-A-213	2022/4/4	Coating Material Of Ventilator ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236205
B100-A-214	2022/4/4	Coating Material Of Ventilator ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236206
B100-A-215	2022/4/4	Coating Material Of Ventilator ・ Roof, B100, BUCKNER COMMUNICATION SITE	White		202236207
B100-A-216	2022/4/4	Ventilator Foundation Concrete ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236208
B100-A-217	2022/4/4	Ventilator Foundation Concrete ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236209
B100-A-218	2022/4/4	Ventilator Foundation Concrete ・ Roof, B100, BUCKNER COMMUNICATION SITE	Gray		202236210
B100-A-219	2022/4/1	Siding Board (Brown) ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Reddish Brown		202236211
B100-A-220	2022/4/1	Siding Board (Ivory) ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236212



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS, NAHA MILITARY PORT, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 210/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所シリアル番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル番号
B100-A-222	2022/4/1	Flange Gasket ・ Exterior, B100, BUCKNER COMMUNICATION SITE	White		202236213
B100-A-223	2022/4/1	Flange Gasket ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Red		202236214
B100-A-226	2022/4/1	Waterproofing ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Black		202236215
B100-A-227	2022/4/1	Concrete Wall ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236216
B100-A-300	2022/4/4	Floor Caulking ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236217
B100-A-301	2022/4/4	Wall Caulking ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Gray		202236218
B100-A-302	2022/4/4	Concrete Wall ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236219
B100-A-303	2022/4/4	Concrete Wall ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236220
B100-A-304	2022/4/4	Concrete Wall ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236221
B100-A-305	2022/4/4	Ceiling Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236222
B100-A-306	2022/4/4	Ceiling Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236223
B100-A-307	2022/4/4	Ceiling Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236224
B100-A-308	2022/4/4	Wallboard ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	White		202236225
B100-A-309	2022/4/4	Wallboard ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	White		202236226
B100-A-310	2022/4/4	Wallboard ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	White		202236227
B100-A-311	2022/4/4	Floor Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236228
B100-A-312	2022/4/4	Floor Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236229
B100-A-313	2022/4/4	Floor Slab ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236230
B100-A-314	2022/4/4	Wall Caulking ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236231
B100-A-315	2022/4/4	Floor Caulking ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236232



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS, NAHA MILITARY PORT, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 230/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所作業番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル#
B100-A-316	2022/4/4	Wallboard ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	White		202236233
B100-A-317	2022/4/4	Wallboard ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	White		202236234
B100-A-318	2022/4/4	Wallboard ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	White		202236235
B100-A-319	2022/4/4	Duct Flange Insulation ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236236
B100-A-400	2022/4/5	Ceiling Insulation ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Brown		202236237
B100-A-401	2022/4/5	Ceiling Insulation ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Brown		202236238
B100-A-402	2022/4/5	Ceiling Insulation ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Brown		202236239
B100-A-403	2022/4/5	Ceiling Board ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236240
B100-A-404	2022/4/5	Ceiling Board ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236241
B100-A-405	2022/4/5	Ceiling Board ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236242
B100-A-406	2022/4/5	Wallboard ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236243
B100-A-407	2022/4/5	Wallboard ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236244
B100-A-408	2022/4/5	Wallboard ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236245
B100-A-409	2022/4/5	Vinyl Cove Base w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Beige		202236246
B100-A-410	2022/4/5	Vinyl Cove Base w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Beige		202236247
B100-A-411	2022/4/5	Vinyl Cove Base w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Beige		202236248
B100-A-412	2022/4/5	Wall Tile w/Grout ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Pale Pink		202236249
B100-A-415	2022/4/5	Floor Tile w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Pale Yellow		202236250
B100-A-416	2022/4/5	Floor Tile w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Pale Yellow		202236251
B100-A-417	2022/4/5	Floor Tile w/Mastic ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Pale Yellow		202236252



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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS, NAHA MILITARY PORT, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 250/292	ANALYSIS REQUIRED 分析方法: ☐ EPA >1% ☒ EPA >0.1%	LAB JOB NUMBER 分析所・シリアル番号: 202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル番号
B100-A-418	2022/4/5	Floor Slab ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Gray		202236253
B100-A-419	2022/4/5	Floor Slab ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Gray		202236254
B100-A-420	2022/4/5	Floor Slab ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Gray		202236255
B100-A-421	2022/4/5	Caulking (Gray) ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Gray		202236256
B100-A-422	2022/4/5	Caulking (White) ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236257
B100-A-425	2022/4/5	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236258
B100-A-426	2022/4/5	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236259
B100-A-427	2022/4/5	Ceiling Board ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236260
B100-A-428	2022/4/5	Wallboard ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236261
B100-A-429	2022/4/5	Wallboard ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236262
B100-A-430	2022/4/5	Wallboard ・ Toilet, B100, BUCKNER COMMUNICATION SITE	White		202236263
B100-A-431	2022/4/5	Wall Ceramic Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Pale Pink		202236264
B100-A-432	2022/4/5	Wall Ceramic Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Pale Pink		202236265
B100-A-433	2022/4/5	Wall Ceramic Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Pale Pink		202236266
B100-A-434	2022/4/5	Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236267
B100-A-435	2022/4/5	Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236268
B100-A-436	2022/4/5	Floor Tile w/Grout ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236269
B100-A-437	2022/4/5	Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236270
B100-A-438	2022/4/5	Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236271
B100-A-439	2022/4/5	Floor Slab ・ Toilet, B100, BUCKNER COMMUNICATION SITE	Gray		202236272



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<アスベスト含有分析依頼書>

OFFICIAL LABORATORY CHAIN OF CUSTODY

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等:			TOTAL SAMPLES サンプル総数:	ANALYSIS REQUIRED 分析方法:	LAB JOB NUMBER 分析依頼番号
PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS, NAHA MILITARY PORT, OKINAWA, JAPAN			270/292	☐ EPA >1% ☒ EPA >0.1%	202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル#
B100-A-440	2022/4/5	Ceiling Insulation ・ Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236273
B100-A-441	2022/4/5	Pipe Insulation ・ Straight, Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236274
B100-A-442	2022/4/5	Pipe Insulation ・ Elbow, Kitchen, B100, BUCKNER COMMUNICATION SITE	Silver ・ Yellow		202236275
B100-A-500	2022/4/5	Ceiling Board ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236276
B100-A-501	2022/4/5	Ceiling Board ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236277
B100-A-502	2022/4/5	Ceiling Board ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236278
B100-A-503	2022/4/5	Ceiling Board ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236279
B100-A-504	2022/4/5	Ceiling Board ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236280
B100-A-505	2022/4/5	Ceiling Slab ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236281
B100-A-506	2022/4/5	Ceiling Slab ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236282
B100-A-507	2022/4/5	Ceiling Slab ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236283
B100-A-508	2022/4/5	Ceiling Slab ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236284
B100-A-510	2022/4/5	Wallboard ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236285
B100-A-511	2022/4/5	Wallboard ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236286
B100-A-512	2022/4/5	Wallboard ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236287
B100-A-513	2022/4/5	Wallboard ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236288
B100-A-514	2022/4/5	Wallboard ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236289
B100-A-515	2022/4/5	Vinyl Cove Base w/Mastic ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Black		202236290
B100-A-516	2022/4/5	Vinyl Cove Base w/Mastic ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Black		202236291
B100-A-517	2022/4/5	Vinyl Cove Base w/Mastic ・ Server Rm, B100, BUCKNER COMMUNICATION SITE	Black		202236292



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OFFICIAL LABORATORY CHAIN OF CUSTODY

PAGE 15 OF 16 -

PROJECT NAME / NUMBER 件名 / プロジェクト番号等:			TOTAL SAMPLES サンプル総数:	ANALYSIS REQUIRED 分析方法:	LAB JOB NUMBER 分析所ジョブ番号
PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS, NAHA MILITARY PORT, OKINAWA, JAPAN			290/292	☐ EPA >1% ☒ EPA >0.1%	202204687
CLIENT SAMPLE # サンプル#	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル#
B100-A-518	2022/4/5	Vinyl Cove Base w/Mastic · Server Rm, B100, BUCKNER COMMUNICATION SITE	Black		202236293
B100-A-519	2022/4/5	Vinyl Cove Base w/Mastic · Server Rm, B100, BUCKNER COMMUNICATION SITE	Black		202236294
B100-A-520	2022/4/5	Floor Slab · Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236295
B100-A-521	2022/4/5	Floor Slab · Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236296
B100-A-522	2022/4/5	Floor Slab · Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236297
B100-A-523	2022/4/5	Floor Slab · Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236298
B100-A-524	2022/4/5	Floor Slab · Server Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236299
B100-A-530	2022/4/5	Acoustic Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236300
B100-A-531	2022/4/5	Acoustic Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236301
B100-A-532	2022/4/5	Acoustic Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236302
B100-A-533	2022/4/5	Wainscot Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236303
B100-A-534	2022/4/5	Wainscot Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236304
B100-A-535	2022/4/5	Wainscot Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236305
B100-A-539	2022/4/6	Floor Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236306
B100-A-540	2022/4/6	Floor Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236307
B100-A-541	2022/4/6	Floor Board · Server Rm, B100, BUCKNER COMMUNICATION SITE	White		202236308
B100-A-543	2022/4/6	Wall Tile w/Grout · Kitchen, B100, BUCKNER COMMUNICATION SITE	White		202236309
B100-A-544	2022/4/6	Ceiling Board · Janitor, B100, BUCKNER COMMUNICATION SITE	Yellow		202236310
B100-A-545	2022/4/6	Wallboard · Janitor, B100, BUCKNER COMMUNICATION SITE	Ivory		202236311
B100-A-546	2022/4/5	Floor Slab · Janitor, B100, BUCKNER COMMUNICATION SITE	Gray		202236312



(ASB ID)

OFFICIAL LABORATORY CHAIN OF CUSTODY

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LEAD-BASED PAINT

Analytical Summary, Photographic Documentation, and
Laboratory Analytical Reports
(Bldg. 100)

DRAFT

Lead-Based Paint Sampling Results Summary - Bldg. 100, Buckner Comm Site

Sample ID No.	LAB ID No.	Location	Material Description	Color	Substrate	Results* (mg/kg)
B100-L-01	202236373	Exterior	Exterior Wall Paint	Ivory	Concrete	< 40 mg/kg
B100-L-02	202236374	Exterior	Exterior Wall Trim Paint	Brown	Concrete	< 40 mg/kg
B100-L-03	202236375	Exterior	Pipe Paint	Ivory	Metal	< 40 mg/kg
B100-L-04	202236376	Exterior	Pipe Paint	Red	Metal	13,000 mg/kg*
B100-L-05	202236377	Exterior	Pipe Paint	Blue	Metal	< 40 mg/kg
B100-L-06	202236378	Exterior	Pipe Paint	Silver	Metal	< 48 mg/kg
B100-L-07	202236379	Exterior	Conduit Paint	Ivory	Metal	230 mg/kg
B100-L-08	202236380	Exterior	Elect Box Paint	Gray	Metal	< 61 mg/kg
B100-L-09	202236381	Exterior	Steel Shutter Door/ Frame Paint	Brown	Metal	1,900 mg/kg
B100-L-10	202236382	Exterior	Support Angle Paint	Ivory	Metal	1,800 mg/kg
B100-L-11	202236383	Exterior	Bollard Paint	Ivory	Metal	150 mg/kg
B100-L-12	202236384	Exterior	Condenser Paint	Gray	Metal	< 40 mg/kg
B100-L-13	202236385	Exterior	Support Angle Paint	Light Gray	Metal	2,000 mg/kg
B100-L-14	202236386	Exterior	Safety Bumper Pole Paint	Yellow	Metal	98 mg/kg
B100-L-15	202236387	Exterior	Chiller Paint	Gray	Metal	< 100 mg/kg
B100-L-16	202236388	Exterior	Conduit Paint	Gray	Metal	5,100 mg/kg*
B100-L-17	202236389	Rm 101	Interior Wall Paint	Ivory	Concrete	< 48 mg/kg
B100-L-18	202236390	Rm 101	Conduit Paint	Ivory	Metal	1,200 mg/kg
B100-L-19	202236391	Rm 101	Elect Box Paint	Red	Metal	< 100 mg/kg
B100-L-20	202236392	Rm 102	Steel Door/Frame Paint	Tan	Steel	15,000 mg/kg*
B100-L-21	202236393	Rm 102	Wood Door/Frame Paint	Tan	Wood	12,000 mg/kg*
B100-L-22	202236394	Rm 105	Interior Wall Paint	Ivory	Board	< 40 mg/kg
B100-L-23	202236395	Rm 105	Wood Door/Frame Paint	Ivory	Wood	22,000 mg/kg*
B100-L-24	202236396	Rm 105	Interior Wall Paint	Ivory	Board	< 49 mg/kg
B100-L-25	202236397	Toilet	Ceiling Paint	Ivory	Board	< 78 mg/kg
B100-L-26	202236398	Elect Rm	Interior Wall Paint	White	Concrete	< 40 mg/kg
B100-L-27	202236399	Elect Rm	Interior Wall Paint	Ivory	Concrete	< 40 mg/kg
B100-L-28	202236400	Elect Rm	Floor Paint	Gray	Concrete	52 mg/kg
B100-L-29	202236401	Elect Rm	Conduit Paint	Gray	Metal	7,800 mg/kg*
B100-L-30	202236402	Elect Rm	Conduit Paint	Dark Gray	Metal	2,400 mg/kg
B100-L-31	202236403	Elect Rm	Pipe Paint	Red	Metal	25,000 mg/kg*

Sample ID No.	LAB ID No.	Location	Material Description	Color	Substrate	Results* (mg/kg)
B100-L-32	202236404	Elect Rm	Pipe Paint	Ivory	Metal	210 mg/kg
B100-L-33	202236405	Elect Rm	Elect Box Paint	Gray	Metal	77 mg/kg
B100-L-34	202236406	Elect Rm	Elect Box Paint	Light Gray	Metal	1,300 mg/kg
B100-L-35	202236407	Elect Rm	Duct Paint	Ivory	Metal	2,200 mg/kg
B100-L-36	202236408	Elect Rm	Duct Paint	Blue	Metal	< 49 mg/kg
B100-L-37	202236409	Elect Rm	Duct Paint	Gray	Metal	11,000 mg/kg*
B100-L-38	202236410	Elect Rm	Pipe Support Angle Paint	Gray	Metal	2,400 mg/kg
B100-L-39	202236411	Mech Rm	Interior Wall Paint	Ivory	Concrete	< 40 mg/kg
B100-L-40	202236412	Mech Rm	Conduit Paint	Gray	Metal	5,700 mg/kg*
B100-L-41	202236413	Mech Rm	Pipe Paint	Red	Metal	2,800 mg/kg
B100-L-42	202236414	Mech Rm	Pipe Paint	Gray	Metal	120 mg/kg
B100-L-43	202236415	Mech Rm	Elect Box Paint	Ivory	Metal	< 64 mg/kg
B100-L-44	202236416	Mech Rm	Elect Box Paint	Gray	Metal	< 53 mg/kg
B100-L-45	202236417	Mech Rm	Floor Paint	Gray	Concrete	280 mg/kg
B100-L-46	202236418	Mech Rm	Pipe Support Angle Paint	Gray	Metal	4,100 mg/kg
B100-L-47	202236419	Mech Rm	Pump Paint	Green	Metal	< 65 mg/kg
B100-L-48	202236420	Mech Rm	Generator Paint	Gray	Metal	< 53 mg/kg
B100-L-49	202236421	Mech Rm	Boiler Paint	Gray	Metal	< 78 mg/kg
B100-L-50	202236422	Mech Rm	Pipe Paint	Silver	Metal	< 160 mg/kg
B100-L-51	202236423	Rm 118	Conduit Paint	Ivory	Metal	3,800 mg/kg
B100-L-52	202236424	Rm 118	Pipe Paint	Red	Metal	< 46 mg/kg
B100-L-53	202236425	Rm 118	Pipe Paint	White	Metal	33,000 mg/kg*
B100-L-54	202236426	Rm 118	Elect Box Paint	Tan	Metal	< 110 mg/kg
B100-L-55	202236427	Rm 118	Support Angle Paint	Ivory	Metal	730 mg/kg
B100-L-100	202236428	Rm 119	Elect Box Paint	Gray	Metal	< 85 mg/kg
B100-L-101	202236429	Rm 119	Wall Paint	Ivory	Board	< 40 mg/kg
B100-L-102	202236430	Rm 119	Elect Box Paint	Ivory	Metal	< 40 mg/kg
B100-L-103	202236431	Rm 118	Wall Paint	Ivory	Board	< 68 mg/kg
B100-L-400	202236432	Interior	Elect Box Paint	Gray	Metal	< 140 mg/kg
B100-L-401	202236433	Interior	Wall Paint	Ivory	Board	< 40 mg/kg
B100-L-402	202236434	Interior	Slide Door Paint	Ivory	Wood	< 40 mg/kg

Sample ID No.	LAB ID No.	Location	Material Description	Color	Substrate	Results* (mg/kg)
B100-L-403	202236435	Interior	Conduit Paint	Ivory	Metal	< 47 mg/kg
B100-L-500	202236436	Rm 108	Elect Box Paint	Gray	Metal	10,000 mg/kg*
B100-L-501	202236437	Rm 108	Elect Box Paint	Tan	Metal	< 64 mg/kg
B100-L-502	202236438	Rm 108	Wall Paint	Ivory	Board	45 mg/kg
B100-L-503	202236439	Rm 108	Frame Paint	Gray	Metal	7,900 mg/kg*
B100-L-504	202236440	Rm 108	Pipe Paint	Red	Metal	610 mg/kg
B100-L-505	202236441	Rm 108	Conduit Paint	Ivory	Metal	3,800 mg/kg
B100-L-506	202236442	Conference Room	Door/ Frame Paint	Orange	Steel	65 mg/kg
B100-L-507	202236443	Conference Room	Wall Paint	Orange	Board	< 40 mg/kg
B100-L-600	202236444	Janitor	Steel Door/Frame Paint	Ivory	Steel	8,000 mg/kg*

* The definition of LBP with EPA is that it is greater than or equal to 0.5% (5,000 mg/kg) by weight.

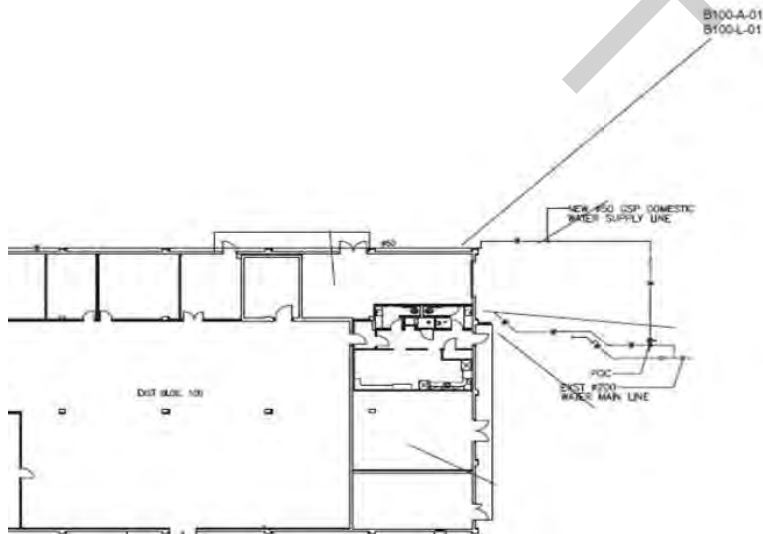
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-01	TYPE OF MATERIAL :	Exterior Wall Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



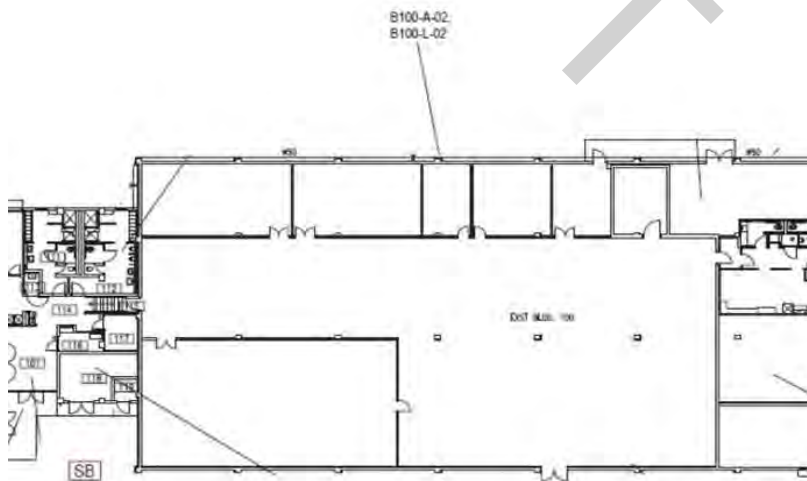
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-02	TYPE OF MATERIAL	: Exterior Wall Trim Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



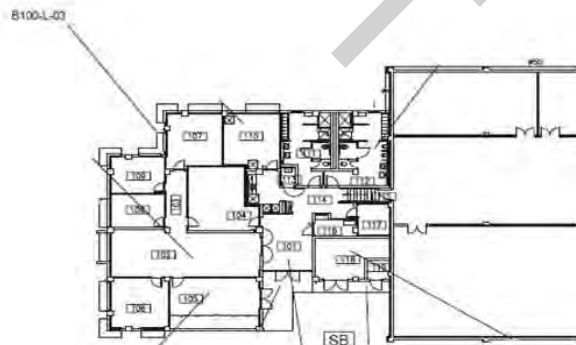
REPAIR HVAC WIDE BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-03	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



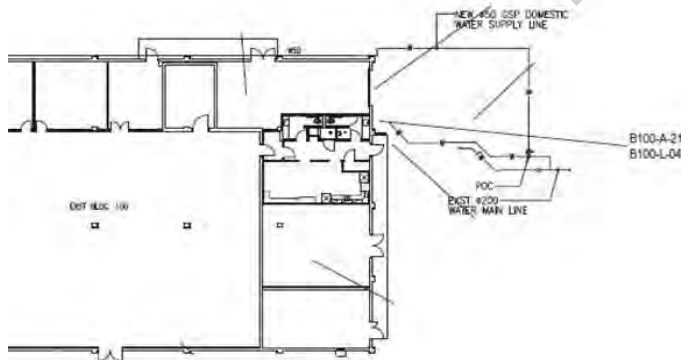
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-04	TYPE OF MATERIAL	:	Pipe Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Damaged
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	13000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



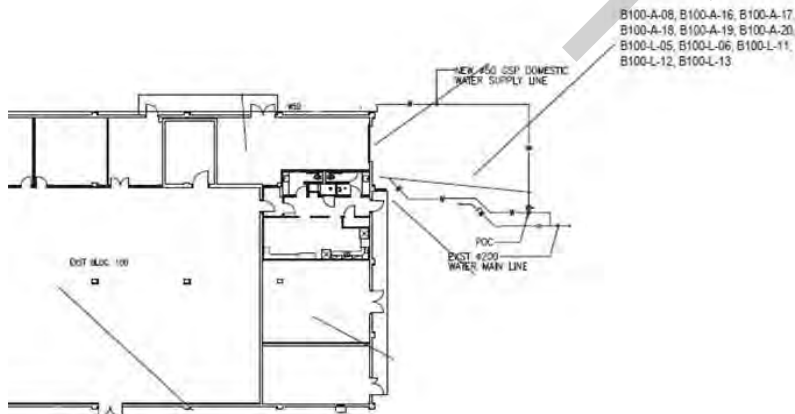
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-05	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN

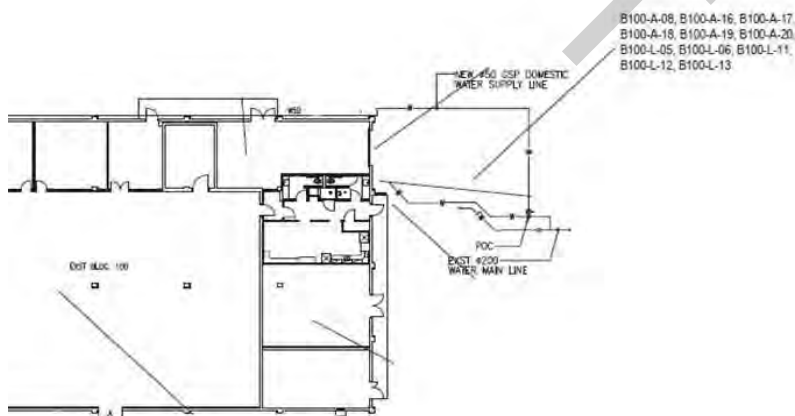
LEAD BASED PAINT ANALYSIS REPORT

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR

PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE. OKINAWA. JAPAN

SAMPLE ID NO.	:	B100-L-06	TYPE OF MATERIAL	:	Pipe Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	<48 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

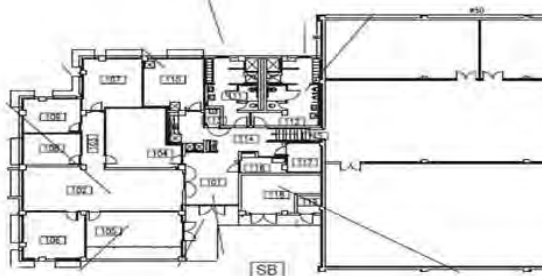
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-07	TYPE OF MATERIAL :	Conduit Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	230 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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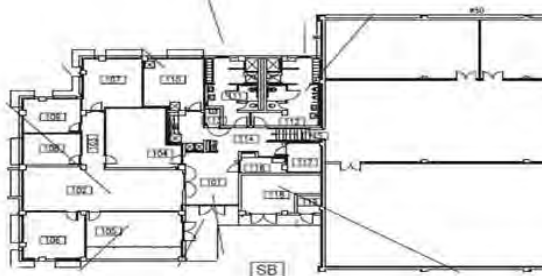
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-08	TYPE OF MATERIAL	: Elect Box Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <61 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



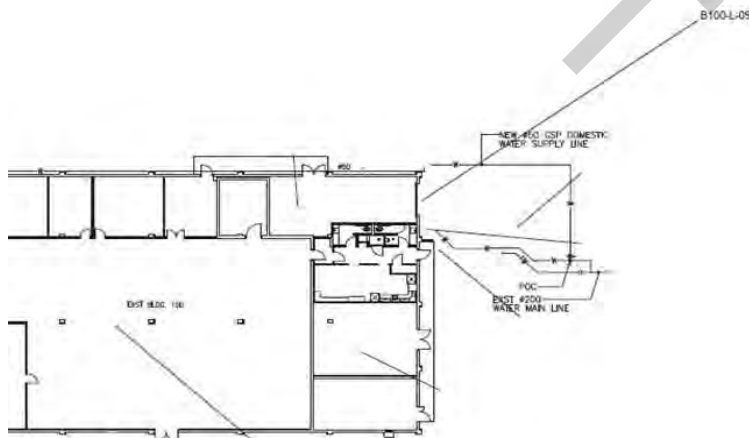
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-09	TYPE OF MATERIAL	: Steel Shutter Door/ Frame Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 1900 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



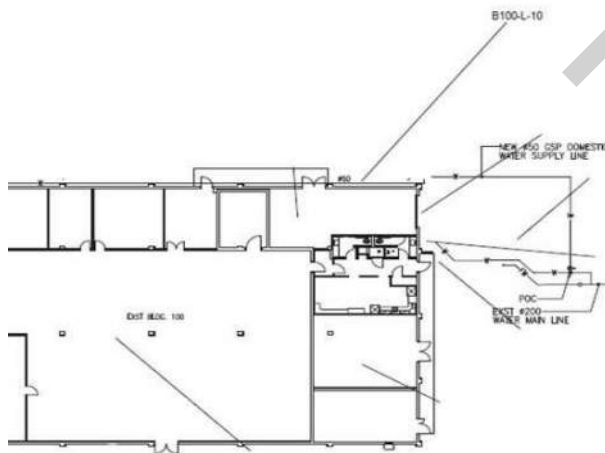
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-10	TYPE OF MATERIAL :	Support Angle Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	1800 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



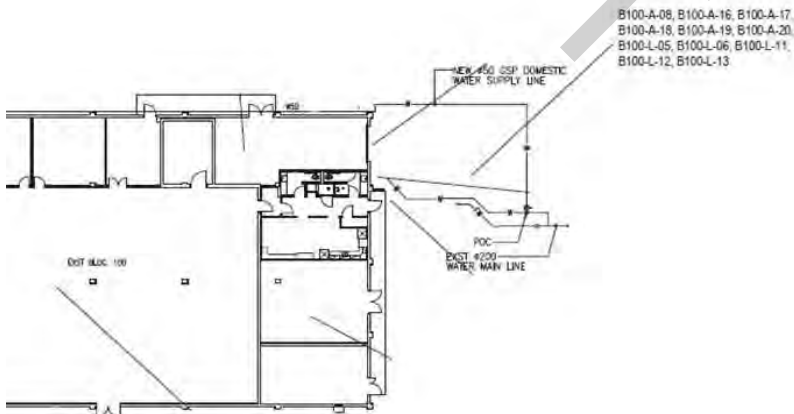
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-11	TYPE OF MATERIAL	:	Bollard Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	150 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



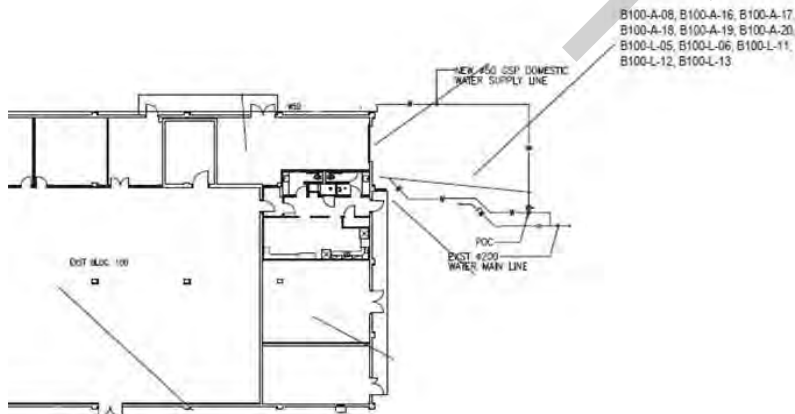
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-12	TYPE OF MATERIAL :	Condenser Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



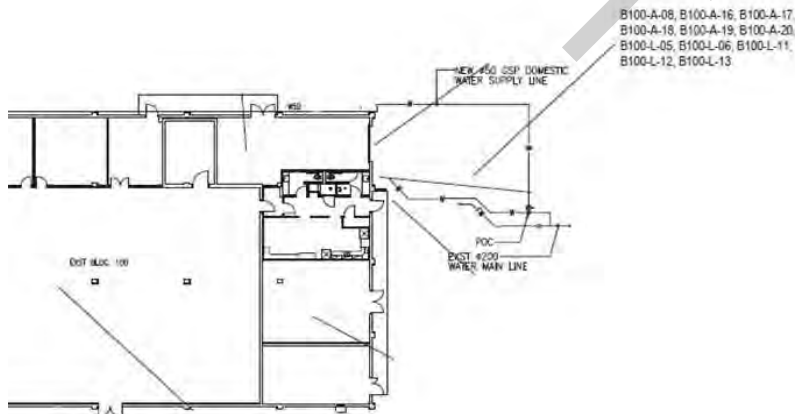
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-13	TYPE OF MATERIAL	: Condenser Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 2000 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



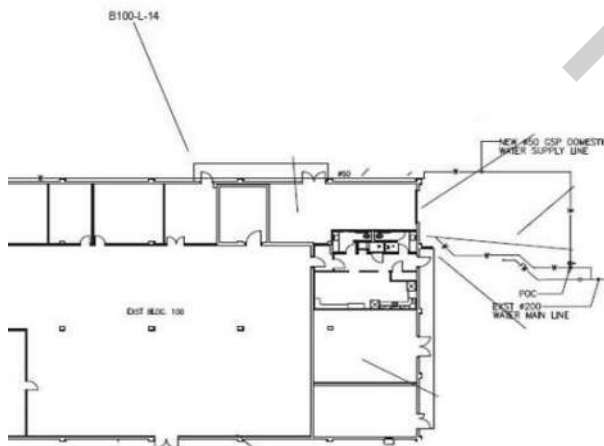
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-14	TYPE OF MATERIAL	: Ssfety Bumper Pole Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 98 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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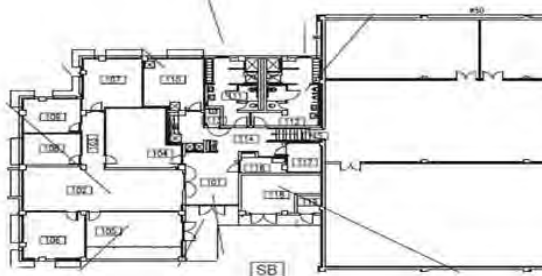
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-15	TYPE OF MATERIAL	: Chiller Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <100 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09 B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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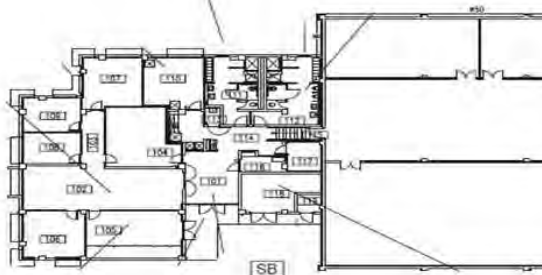
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-16	TYPE OF MATERIAL	:	Conduite Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	5100 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



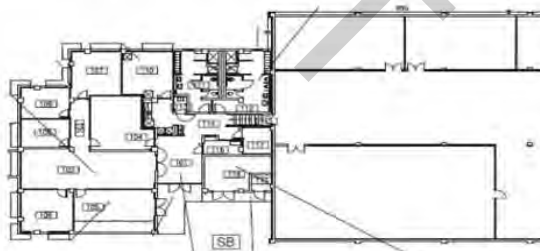
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-17	TYPE OF MATERIAL	: Interior Wall Paint
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 48 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

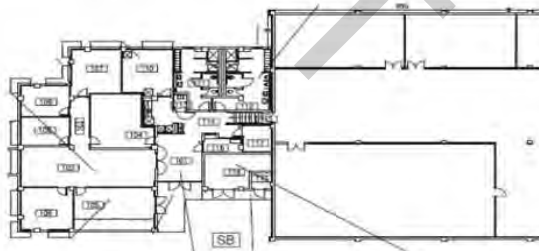
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-18	TYPE OF MATERIAL	: Conduit Paint
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 1200 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

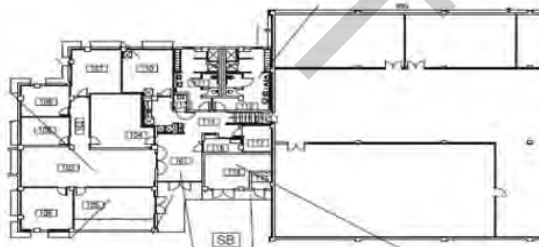
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-19	TYPE OF MATERIAL :	Elec Box Paint
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	100 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 101	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-20	TYPE OF MATERIAL	:	Steel Door/ Frame Paint
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	15000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 102	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



REPAIR HVAC WIDEBAND SATCOM OPS B100
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-21	TYPE OF MATERIAL	:	Wood Door/ Frame Paint
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	12000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 102	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



REPAIR HVAC WIDEBAND SATCOM OPS B100
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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-22	TYPE OF MATERIAL	: Interior Wall Paint
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-65, B100-A-66, B100-A-67,
B100-L-20, B100-L-21, B100-L-22



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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-23	TYPE OF MATERIAL	:	Wood Door/ Frame Paint
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	22000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-68, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73,
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

REPAIR HVAC WIDEBAND SATCOM OPS B100
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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-24	TYPE OF MATERIAL	:	Interior Wall Paint
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	<49 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-66, B100-A-69, B100-A-70,
B100-A-71, B100-A-72, B100-A-73
B100-A-74, B100-A-75, B100-A-76,
B100-A-77, B100-A-78, B100-A-79,
B100-A-80, B100-A-81, B100-A-82,
B100-A-83, B100-A-84, B100-A-85,
B100-A-86, B100-A-87, B100-A-88,
B100-A-89, B100-A-90, B100-A-91,
B100-A-92, B100-A-93, B100-A-94,
B100-A-95,
B100-L-23, B100-L-24

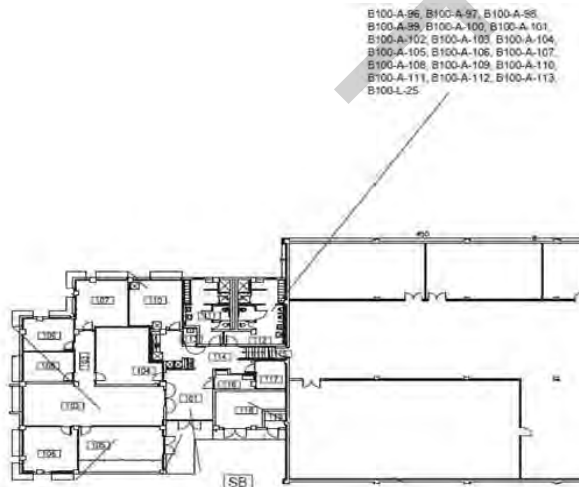
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-25	TYPE OF MATERIAL :	Ceiling Paint
SAMPLE DATE :	7-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<78 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Toilet	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



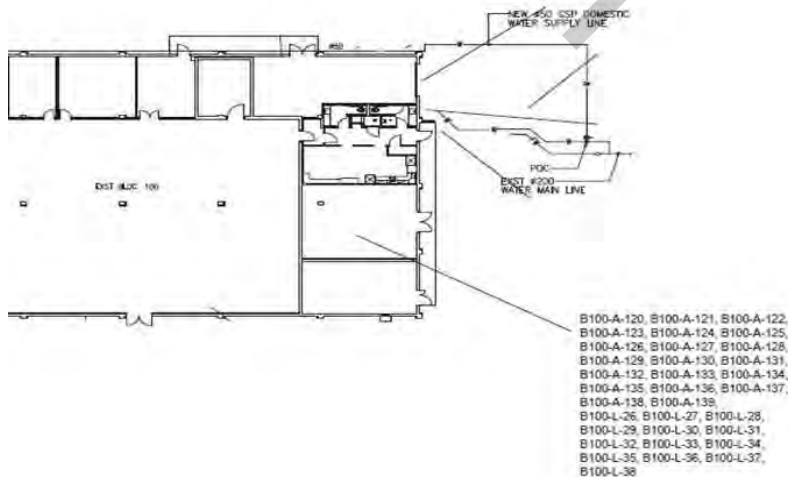
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-26	TYPE OF MATERIAL :	Interior Wall Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Takuya Niizato
ROOM NO./NAME :	Elect Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC

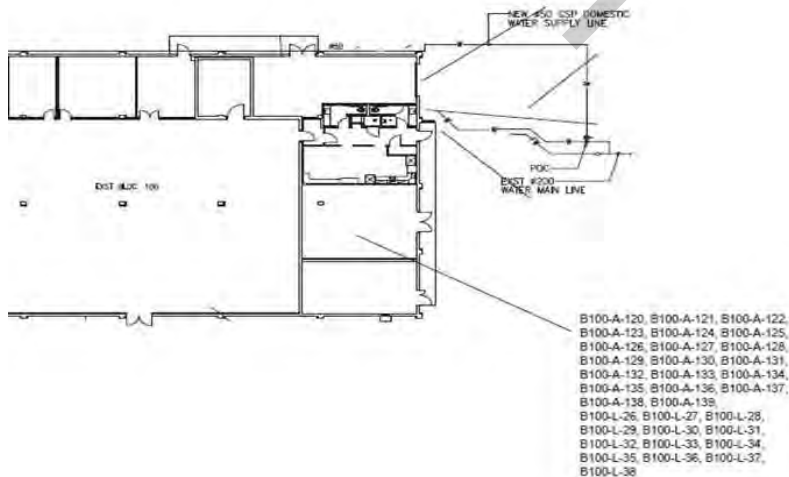


CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-27	TYPE OF MATERIAL	: Interior Wall Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



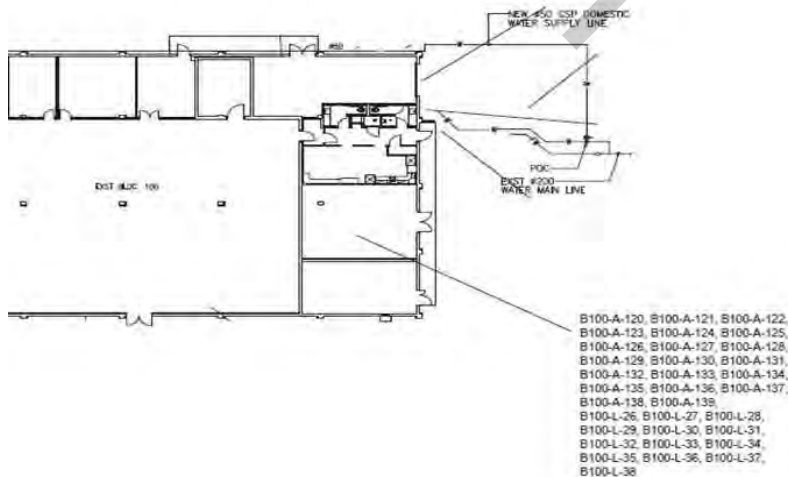
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-28	TYPE OF MATERIAL	: Floor Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 52 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



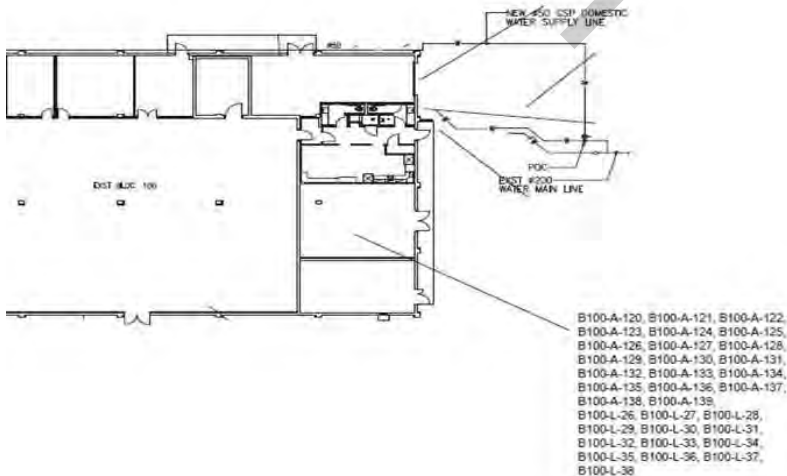
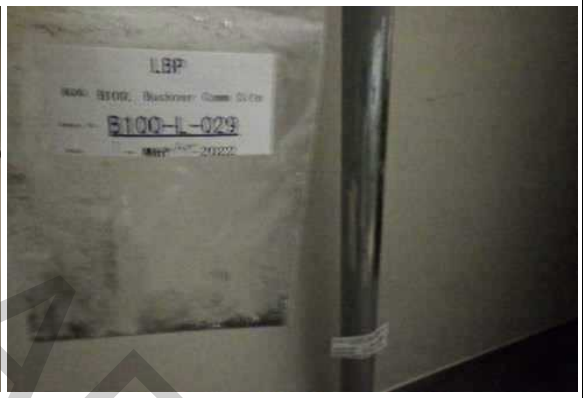
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-29	TYPE OF MATERIAL	:	Conduite Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	7800 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Takuya Niizato
ROOM NO./NAME	:	Elect Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



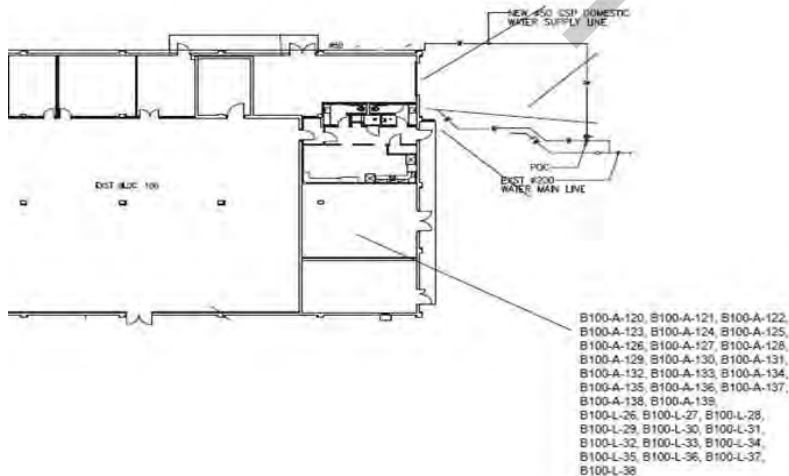
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-30	TYPE OF MATERIAL :	Conduite Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	2400 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Takuya Niizato
ROOM NO./NAME :	Elect Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



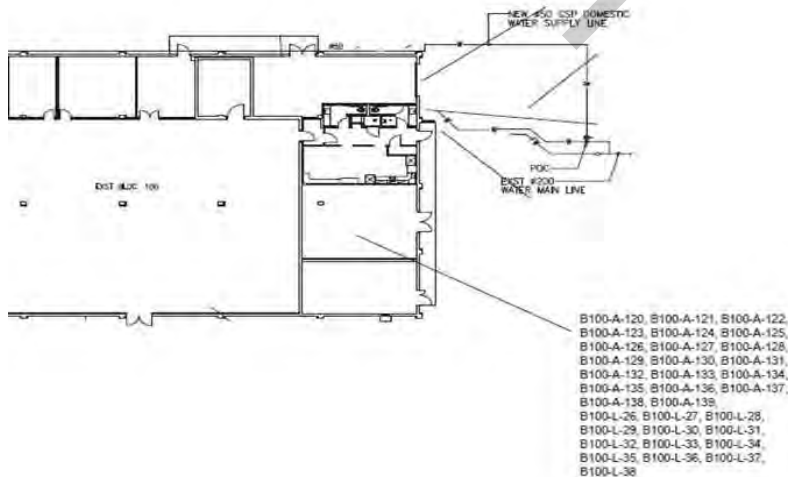
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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-31	TYPE OF MATERIAL	:	Pipe Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	25000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Takuya Niizato
ROOM NO./NAME	:	Elect Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



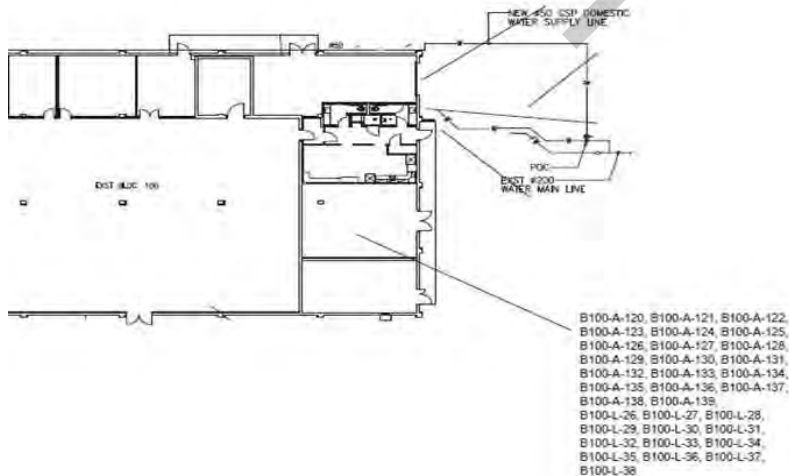
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-32	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 210 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



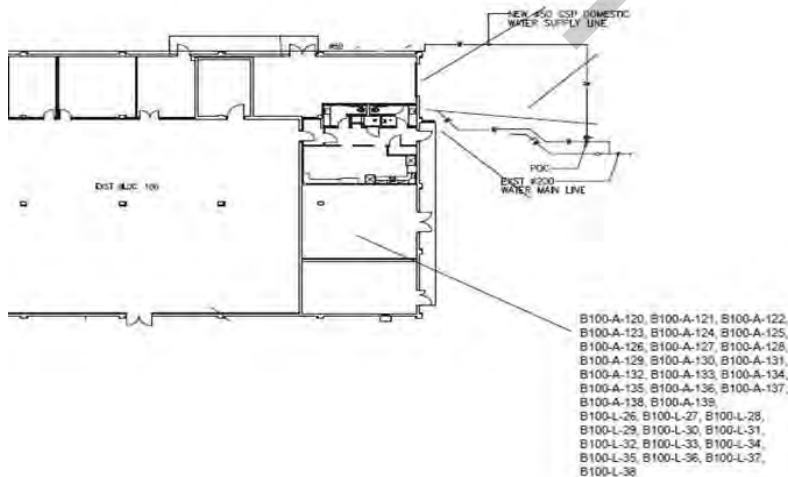
REPAIR HVAC WIDEBAND SATCOM OPS B100
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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-33	TYPE OF MATERIAL :	Elect Box Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	77 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Takuya Niizato
ROOM NO./NAME :	Elect Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



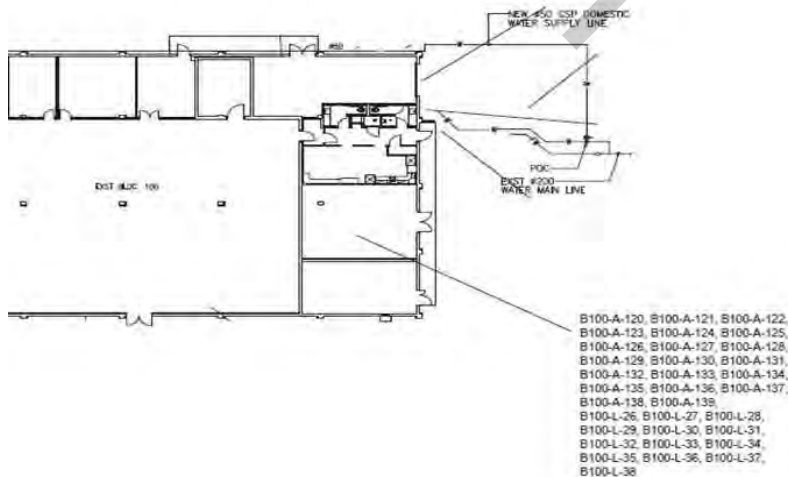
REPAIR HVAC WIDEBAND SATCOM OPS B100
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LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-34	TYPE OF MATERIAL :	Elect Box Paint
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	1300 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Takuya Niizato
ROOM NO./NAME :	Elect Rm	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



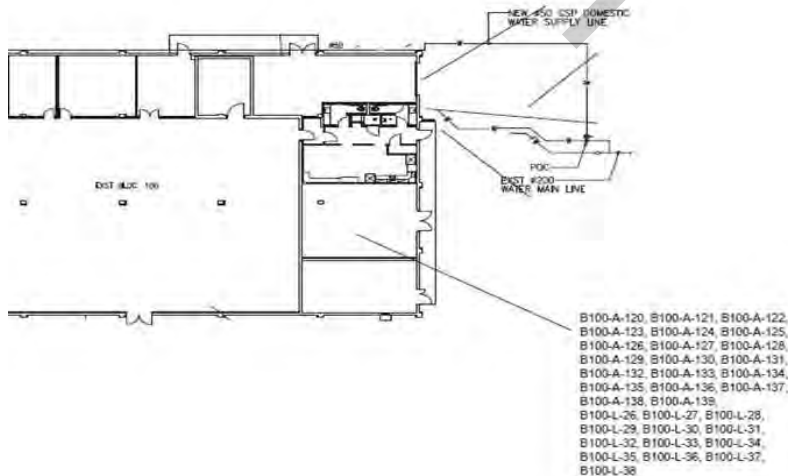
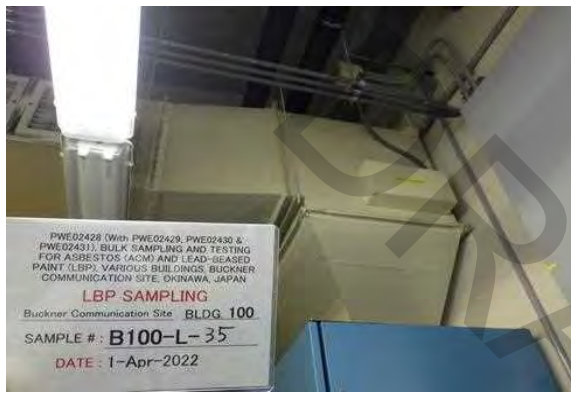
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-35	TYPE OF MATERIAL	: Duct Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 2200 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



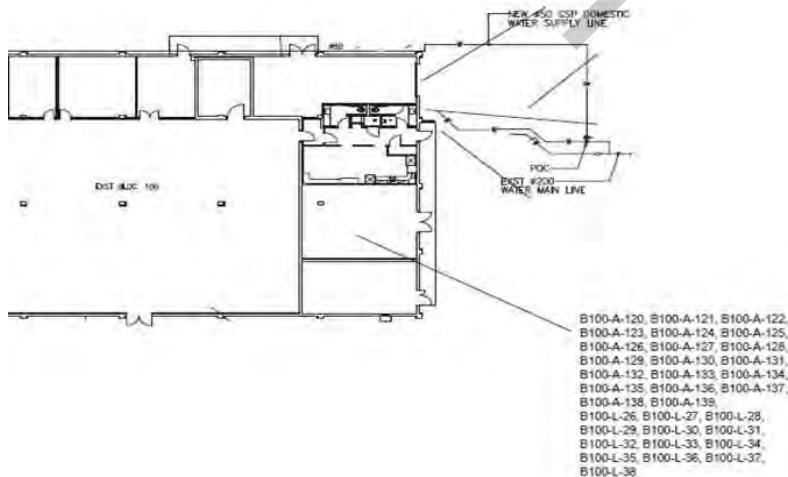
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-36	TYPE OF MATERIAL	: Duct Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <49 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



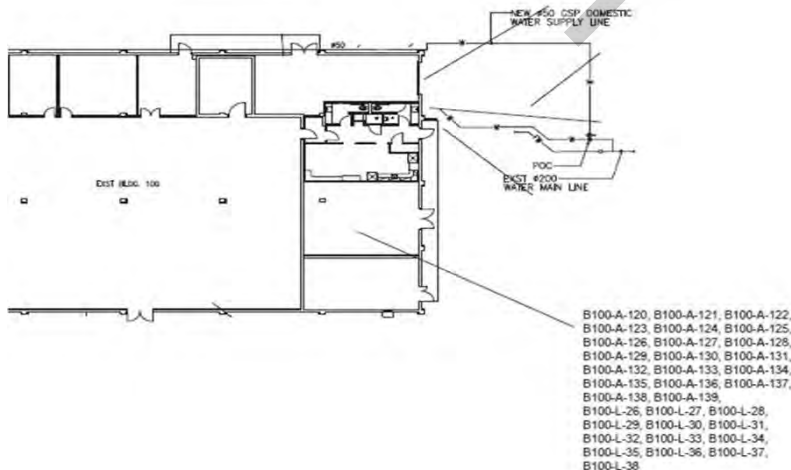
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-37	TYPE OF MATERIAL	:	Pipe Support Angle Paint
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	11000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Takuya Niizato
ROOM NO./NAME	:	Elect Rm	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



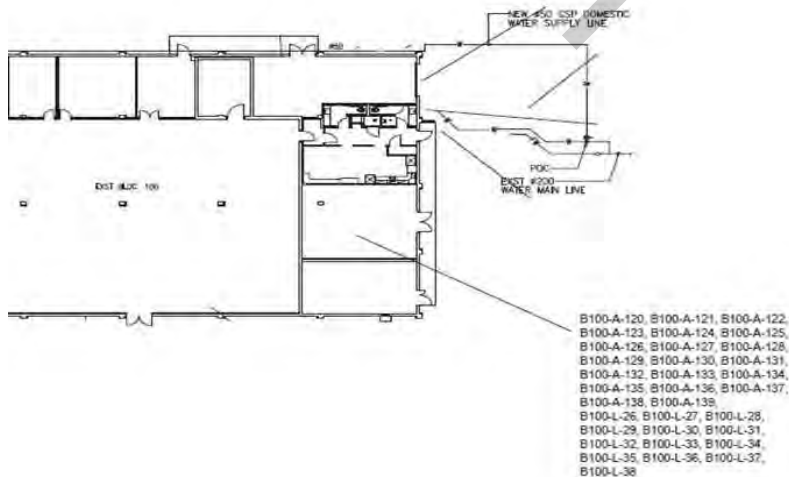
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-38	TYPE OF MATERIAL	: Support Angle Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 2400 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Takuya Niizato
ROOM NO./NAME	: Elect Rm	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



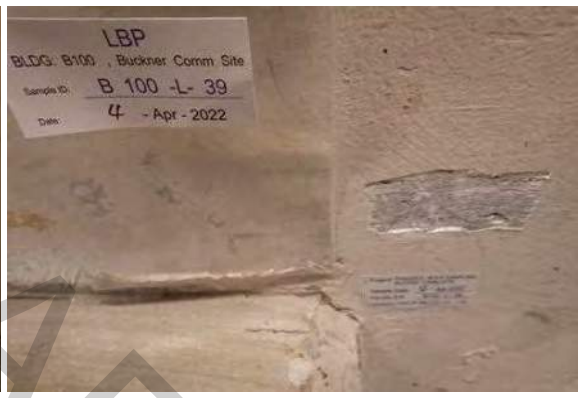
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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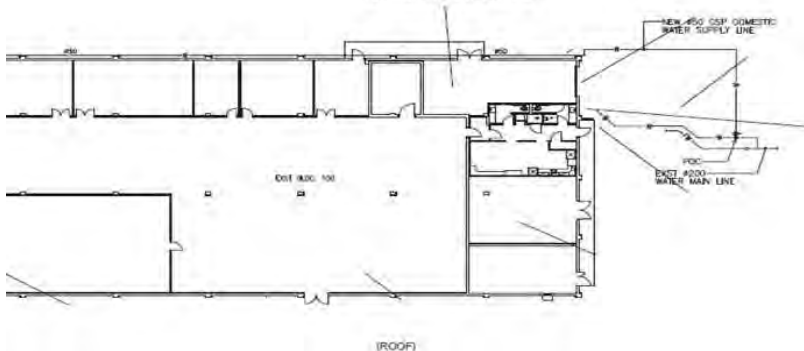
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-39	TYPE OF MATERIAL	:	Interior Wall Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	<40 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
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B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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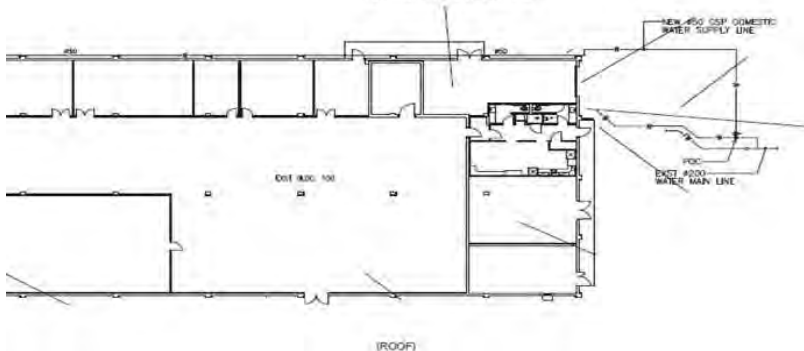
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-40	TYPE OF MATERIAL	:	Conduit Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	5700 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
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B100-L-45, B100-L-46, B100-L-47,
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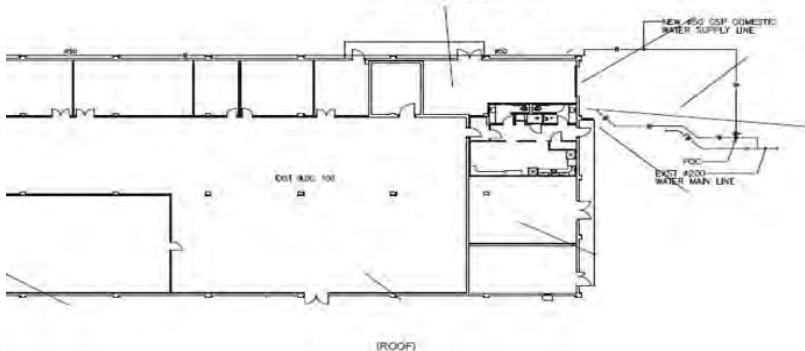
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-41	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 2800 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
B100-A-158, B100-A-159, B100-A-160,
B100-A-161, B100-A-162, B100-A-163,
B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

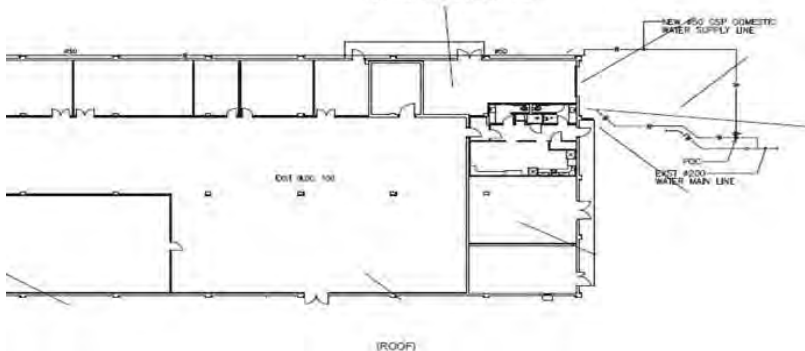
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-42	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 120 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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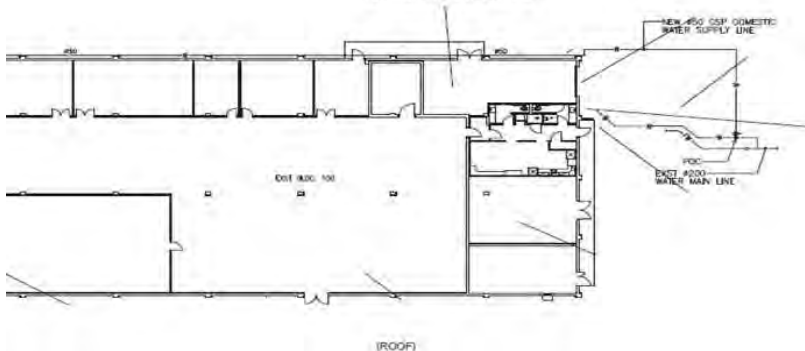
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-43	TYPE OF MATERIAL	: Elect Box Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <64 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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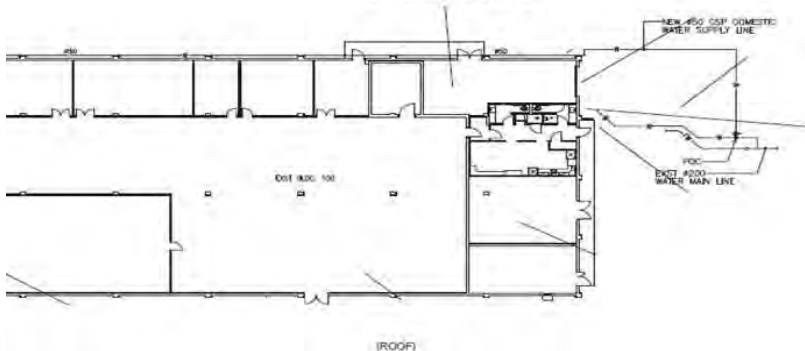
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-44	TYPE OF MATERIAL	:	Elect Box Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	<53 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
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B100-L-48, B100-L-49, B100-L-50



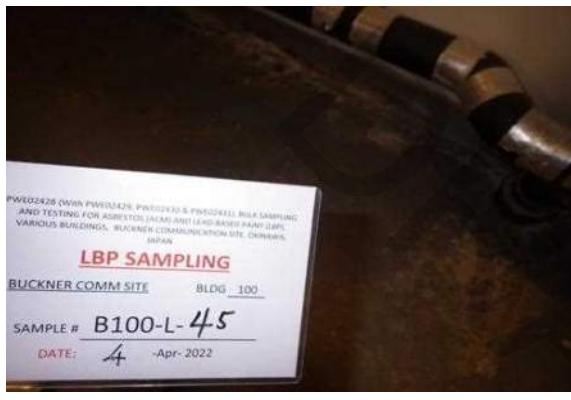
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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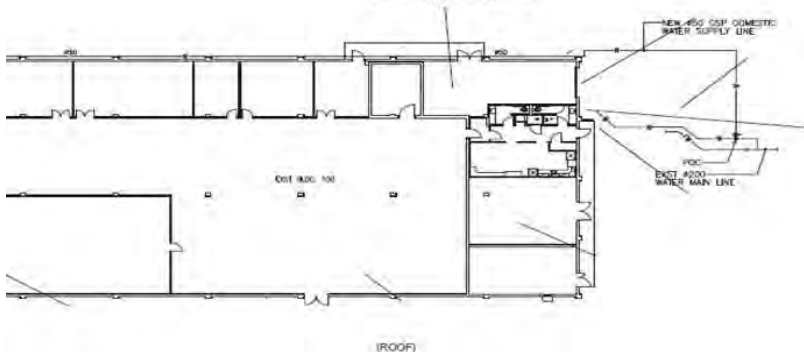
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-45	TYPE OF MATERIAL	:	Floor Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	280 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Mech	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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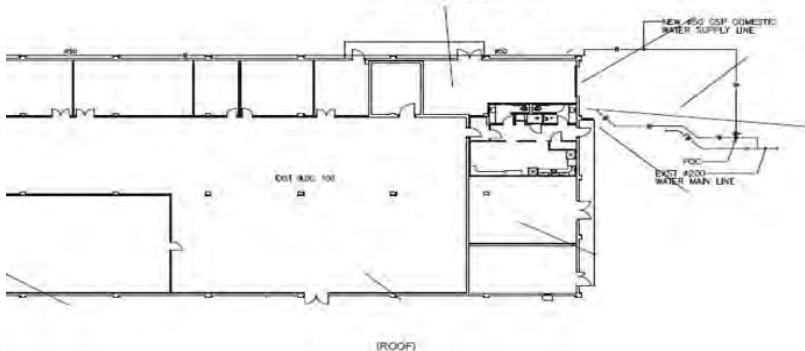
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-46	TYPE OF MATERIAL	: Pipe Support Angle Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 4100 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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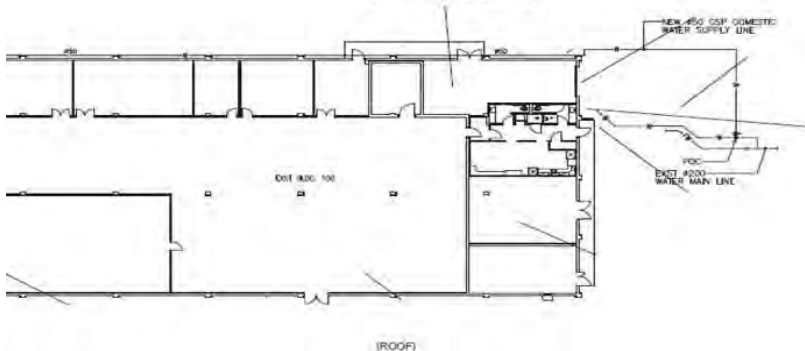
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-47	TYPE OF MATERIAL	: Pump Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <65 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
B100-A-152, B100-A-153, B100-A-154,
B100-A-155, B100-A-156, B100-A-157,
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B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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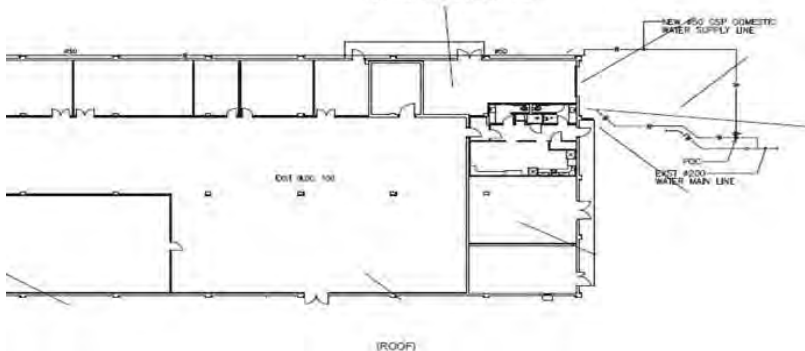
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-48	TYPE OF MATERIAL	: Generator Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <53 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
B100-A-149, B100-A-150, B100-A-151,
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B100-A-164, B100-A-165, B100-A-166,
B100-A-167, B100-A-168, B100-A-169,
B100-A-170, B100-A-171,
B100-L-39, B100-L-40, B100-L-41,
B100-L-42, B100-L-43, B100-L-44,
B100-L-45, B100-L-46, B100-L-47,
B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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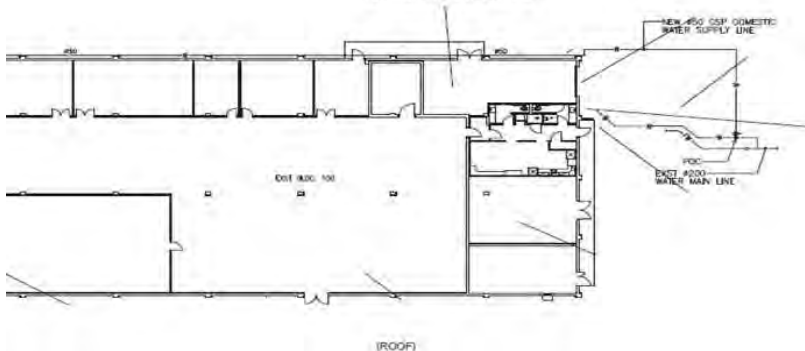
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-49	TYPE OF MATERIAL :	Boiler Paint
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<78 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Mech	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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B100-L-48, B100-L-49, B100-L-50



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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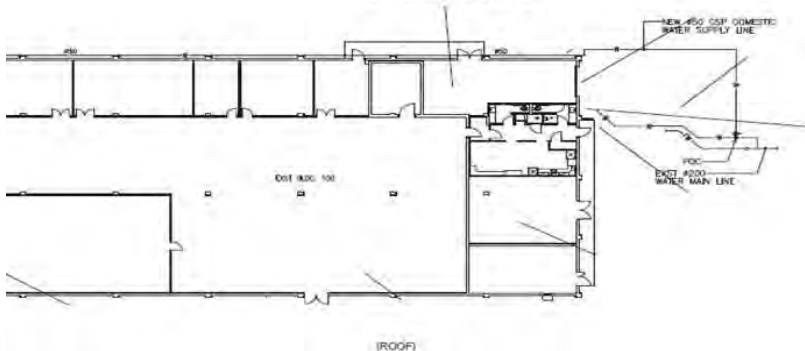
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-50	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <160 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Mech	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-140, B100-A-141, B100-A-142,
B100-A-143, B100-A-144, B100-A-145,
B100-A-146, B100-A-147, B100-A-148,
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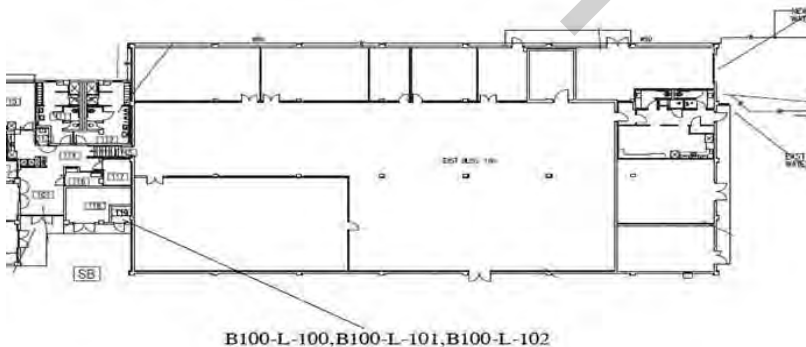
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-51	TYPE OF MATERIAL	: Conduit Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 3800 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



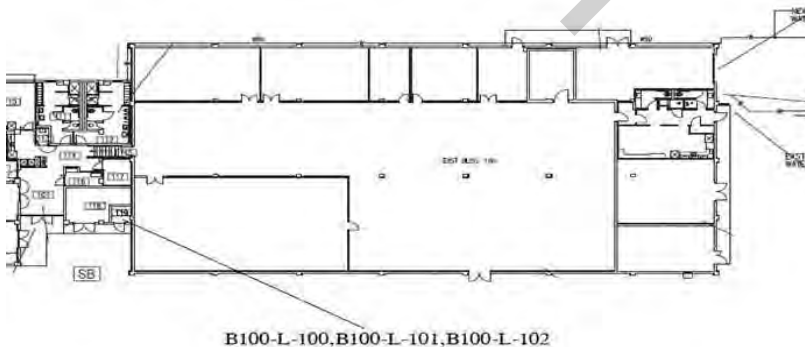
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR

PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-52	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <46 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



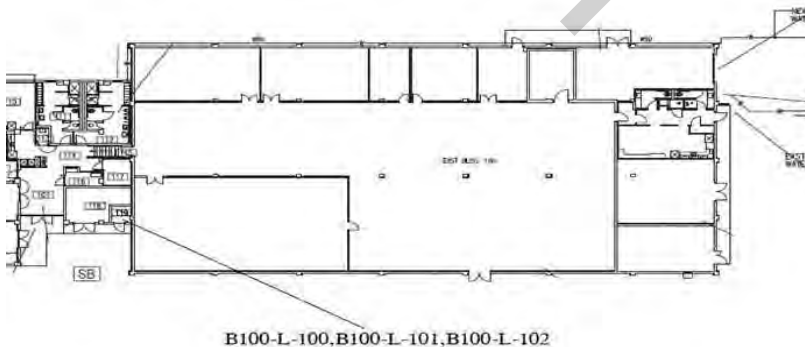
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-53	TYPE OF MATERIAL	:	Pipe Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	33000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 118	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



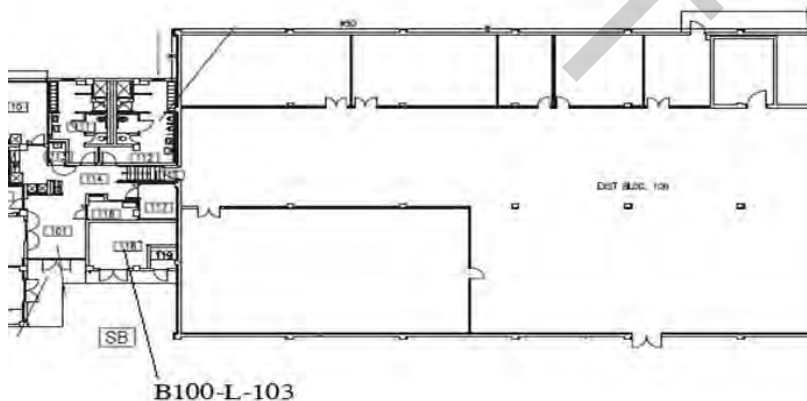
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-54	TYPE OF MATERIAL :	Elect Box Paint
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<110 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



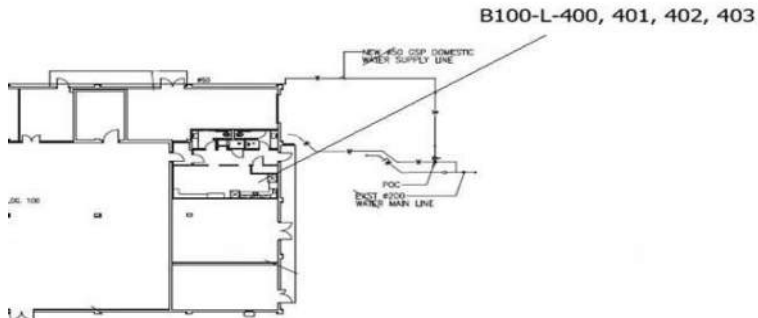
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-55	TYPE OF MATERIAL	: Support Angle Paint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 730 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 118	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



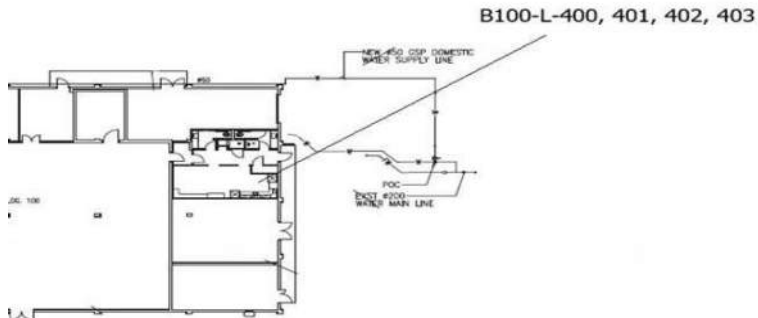
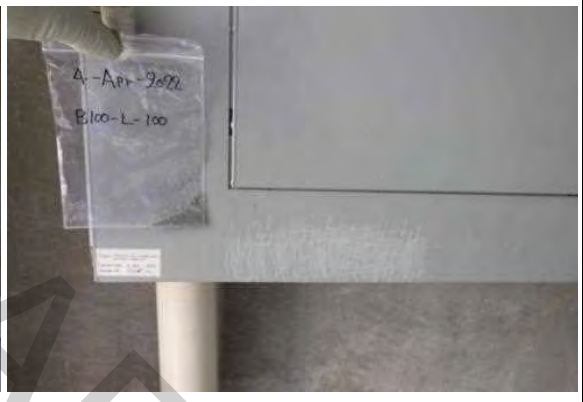
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-100	TYPE OF MATERIAL	: Elect Box Paint
SAMPLE DATE	: 4-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <85 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tetsuya Yonaha
ROOM NO./NAME	: Rm 119	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC

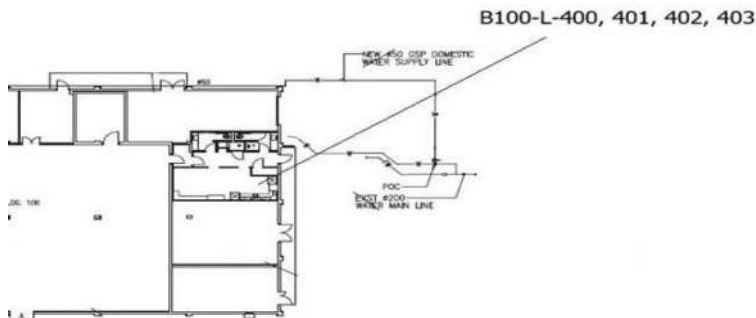


CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN LEAD BASED PAINT ANALYSIS REPORT

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-101	TYPE OF MATERIAL	:	Wall Paint
SAMPLE DATE	:	4-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	<40 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tetsuya Yonaha
ROOM NO./NAME	:	Rm 119	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC

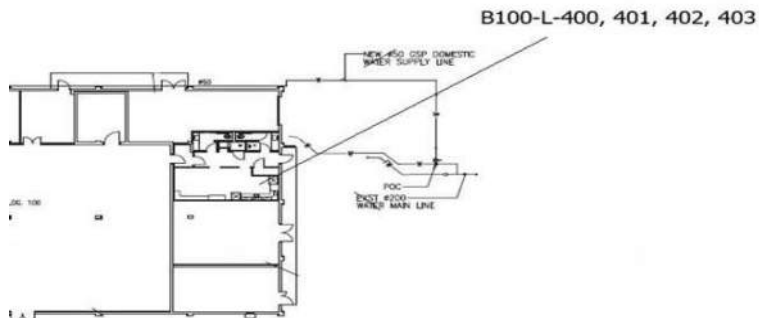


**CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT**

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-102	TYPE OF MATERIAL :	Elect Box Paint
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tetsuya Yonaha
ROOM NO./NAME :	Rm 119	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



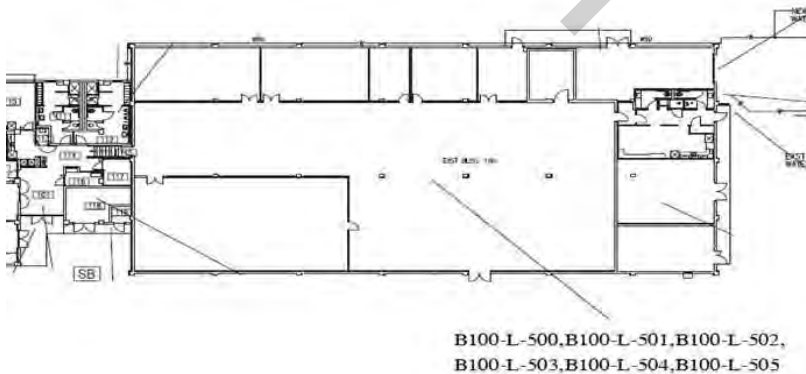
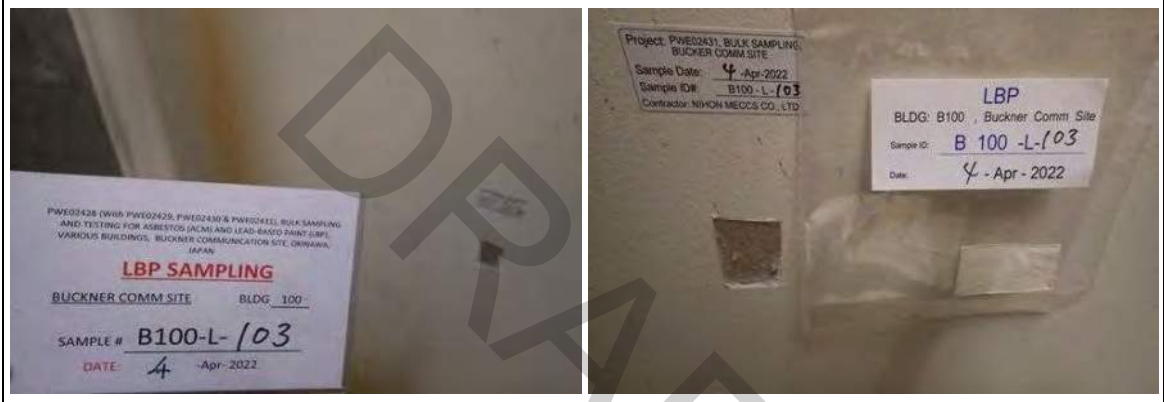
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-103	TYPE OF MATERIAL :	Wall Paint
SAMPLE DATE :	4-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<68 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 118	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



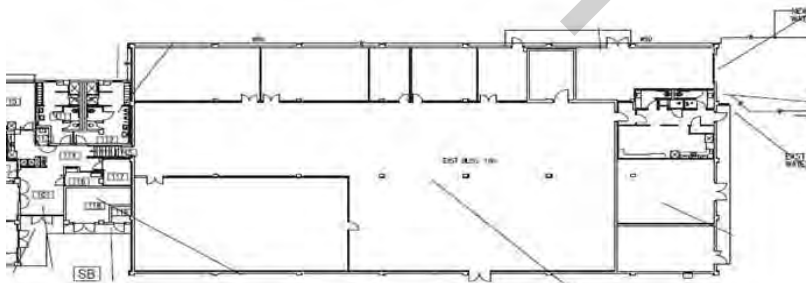
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-400	TYPE OF MATERIAL	: Elect Box Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <140 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Interior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-L-500,B100-L-501,B100-L-502,
B100-L-503,B100-L-504,B100-L-505

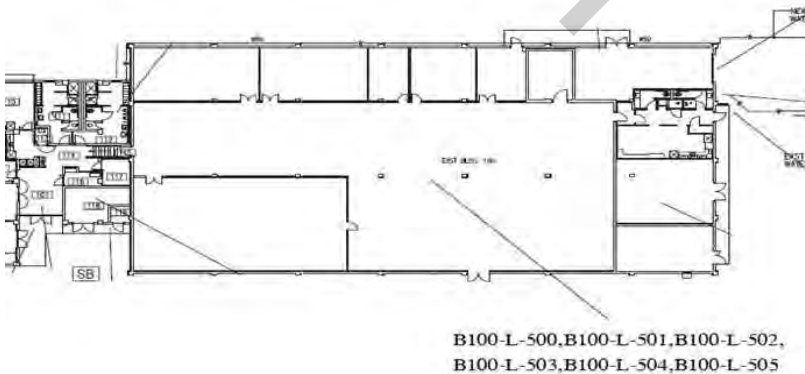
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-401	TYPE OF MATERIAL :	Wall Paint
SAMPLE DATE :	5-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Interior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



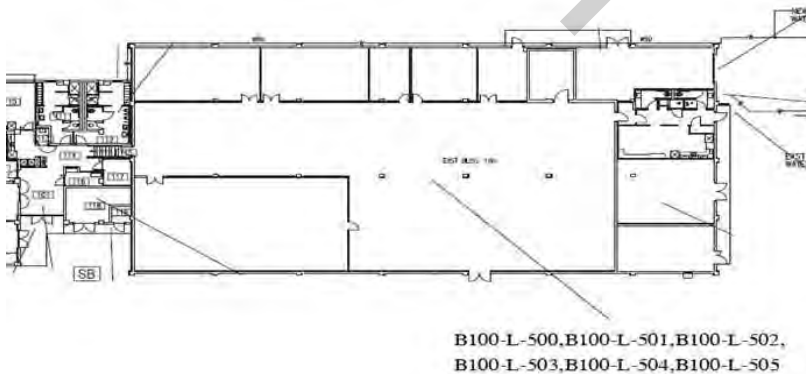
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-402	TYPE OF MATERIAL	: Slide Door Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <40 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Interior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



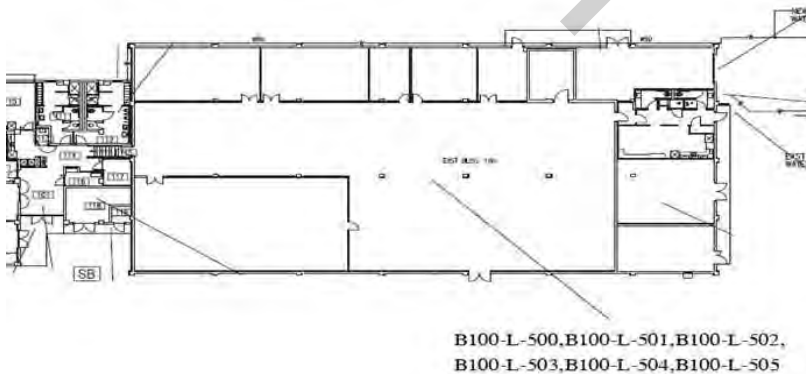
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-403	TYPE OF MATERIAL	: Conduit Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <47 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Interior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



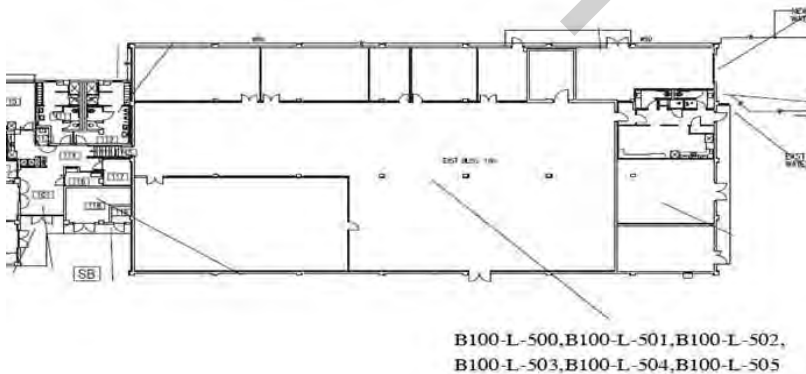
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-500	TYPE OF MATERIAL	:	Elect Box Paint
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	10000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 108	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



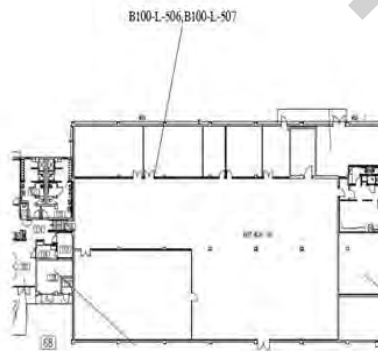
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-501	TYPE OF MATERIAL	: Elect Box Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: <64 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 108	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



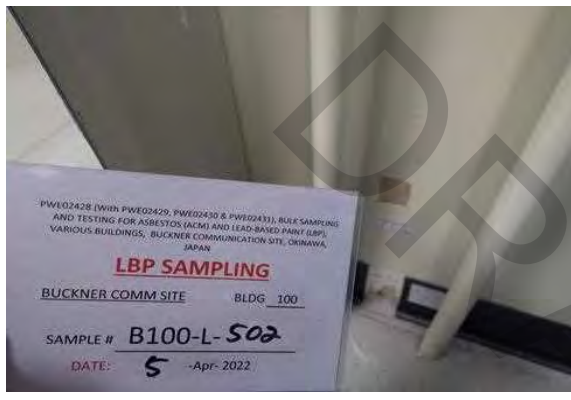
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-502	TYPE OF MATERIAL	: Wall Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 45 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 108	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



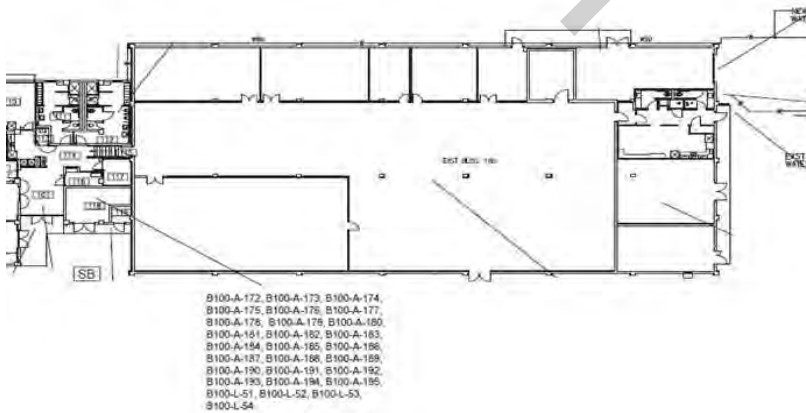
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-503	TYPE OF MATERIAL	:	Frame Paint
SAMPLE DATE	:	5-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	7900 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 108	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



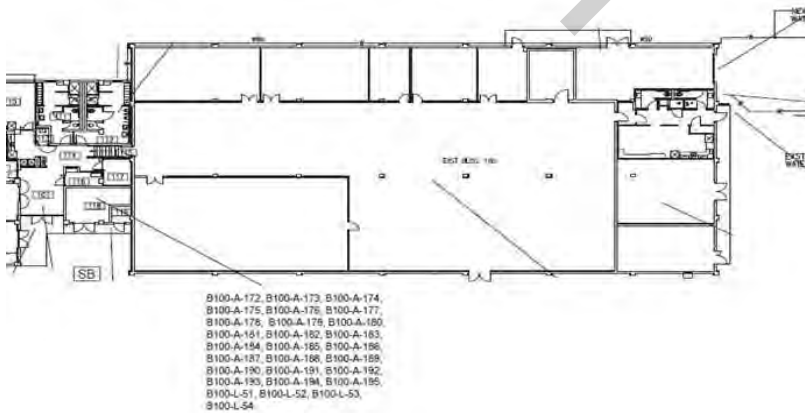
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-504	TYPE OF MATERIAL	: Pipe Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 610 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 108	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



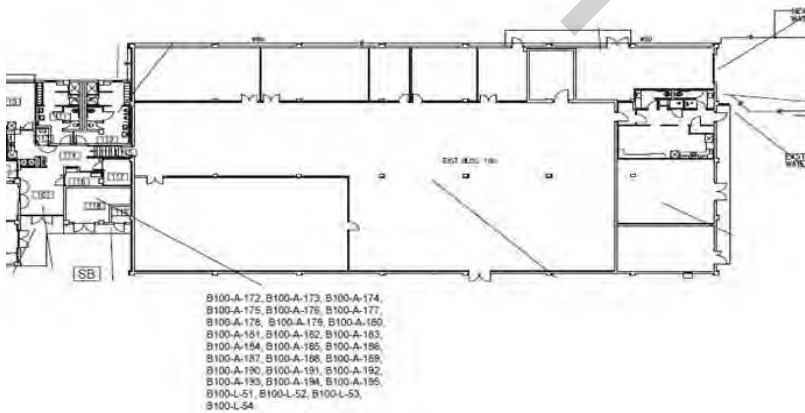
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-L-505	TYPE OF MATERIAL	: Conduit Paint
SAMPLE DATE	: 5-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: 3800 mg/kg
BUILDING NO./LOCATION	: Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 108	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



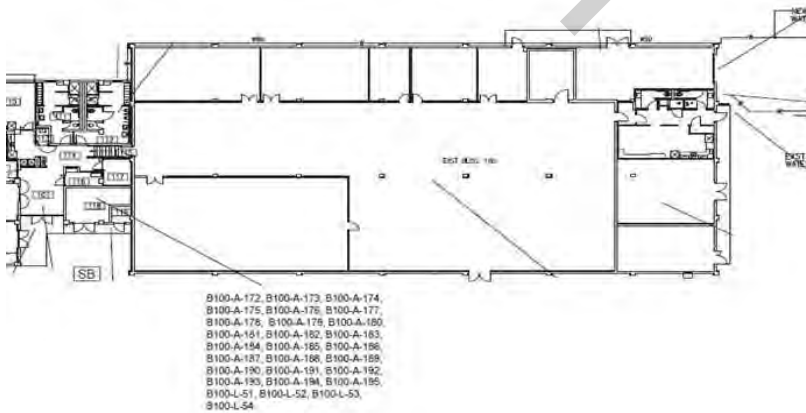
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-506	TYPE OF MATERIAL :	Door/ Frame Paint
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	65 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Conference Room	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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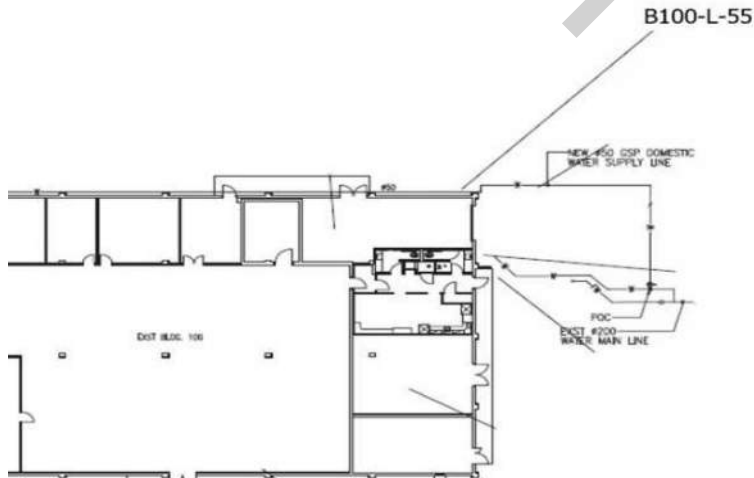
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-L-507	TYPE OF MATERIAL :	Wall Paint
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	<40 mg/kg
BUILDING NO./LOCATION :	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Conference Room	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



※Sample ID is incorreccted. Corect sample ID is "B100-L-507".



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
LEAD BASED PAINT ANALYSIS REPORT

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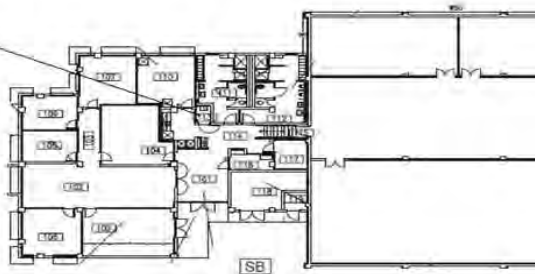
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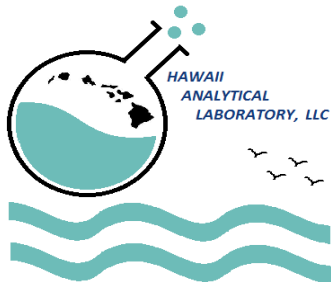
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PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-L-600	TYPE OF MATERIAL	:	Steel Door/ Frame Paint
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	8000 mg/kg
BUILDING NO./LOCATION	:	Bldg. 100	LEAD-BASED PAINT BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Janitor	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-L-600





Hawaii Analytical Laboratory ANALYTICAL REPORT

08 September 2022 issued amended report to replace
original report dated 26 May 2022.

Nanasei Environmental Laboratory, Co., Ltd
4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun
Okinawa Japan .

Phone Number: (098)835-8411

Facsimile:

Email:

Lab Job No: 202204690

Date Submitted: 5/12/2022

Your Project: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-

PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA,
JAPAN, 4/1/22-4/7/22

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236373	B100-L-01 Exterior Wall Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236374	B100-L-02 Exterior Wall Trim Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Brown	< 40	mg/kg	5/25/2022
Comments				
202236375	B100-L-03 Pipe Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236376	B100-L-04 Pipe Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Red	13000	mg/kg	5/25/2022
Comments				
202236377	B100-L-05 Pipe Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Blue	< 40	mg/kg	5/25/2022
Comments				
202236378	B100-L-06 Pipe Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Silver	< 48	mg/kg	5/25/2022
Comments				
202236379	B100-L-07 Conduit Paint • Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory	230	mg/kg	5/25/2022
Comments				

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3615 Harding Avenue, Ste. 308, Honolulu, HI 96816 - Telephone: (808) 735-0422 - Fax: (808) 735-0047
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PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN, 4/1/22-4/7/22

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236380	B100-L-08 Elect Box Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Gray	< 61	mg/kg	5/25/2022
Comments				
202236381	B100-L-09 Steel Shutter Door/ Frame Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Brown	1900	mg/kg	5/25/2022
Comments				
202236382	B100-L-10 Support Angle Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory	1800	mg/kg	5/25/2022
Comments				
202236383	B100-L-11 Bollard Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Ivory	150	mg/kg	5/25/2022
Comments				
202236384	B100-L-12 Condenser Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Gray	< 40	mg/kg	5/25/2022
Comments				
202236385	B100-L-13 Condenser Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Light Gray	2000	mg/kg	5/25/2022
Comments				
202236386	B100-L-14 Safety Bumper Pole Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Yellow	98	mg/kg	5/25/2022
Comments				
202236387	B100-L-15 Chiller Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Gray	< 100	mg/kg	5/25/2022
Comments				
202236388	B100-L-16 Conduite Paint ▪ Exterior , B100, BUCKNER COMMUNICATION SITE, Gray	5100	mg/kg	5/25/2022
Comments				

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PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN, 4/1/22-4/7/22

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236389	B100-L-17 Interior Wall Paint • Rm 101, B100, BUCKNER COMMUNICATION SITE, Ivory	< 48	mg/kg	5/25/2022
Comments				
202236390	B100-L-18 Conduit Paint • Rm 101, B100, BUCKNER COMMUNICATION SITE, Ivory	1200	mg/kg	5/25/2022
Comments				
202236391	B100-L-19 Elec Box Paint • Rm 101, B100, BUCKNER COMMUNICATION SITE, Red	< 100	mg/kg	5/25/2022
Comments				
202236392	B100-L-20 Steel Door/ Frame Paint • Rm 102, B100, BUCKNER COMMUNICATION SITE, Tan	15000	mg/kg	5/25/2022
Comments				
202236393	B100-L-21 Wood Door/ Frame Paint • Rm 102, B100, BUCKNER COMMUNICATION SITE, Tan	12000	mg/kg	5/25/2022
Comments				
202236394	B100-L-22 Interior Wall Paint • Rm 105, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236395	B100-L-23 Wood Door/ Frame Paint • Rm 105, B100, BUCKNER COMMUNICATION SITE, Ivory	22000	mg/kg	5/25/2022
Comments				
202236396	B100-L-24 Interior Wall Paint • Rm 105, B100, BUCKNER COMMUNICATION SITE, Ivory	< 49	mg/kg	5/25/2022
Comments				
202236397	B100-L-25 Ceiling Paint • Toilet, B100, BUCKNER COMMUNICATION SITE, Ivory	< 78	mg/kg	5/25/2022
Comments				

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Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236398	B100-L-26 Interior Wall Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, White	< 40	mg/kg	5/25/2022
Comments				
202236399	B100-L-27 Interior Wall Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236400	B100-L-28 Floor Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Gray	52	mg/kg	5/25/2022
Comments				
202236401	B100-L-29 Conduite Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Gray	7800	mg/kg	5/25/2022
Comments				
202236402	B100-L-30 Conduite Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Dark Gray	2400	mg/kg	5/25/2022
Comments				
202236403	B100-L-31 Pipe Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Red	25000	mg/kg	5/25/2022
Comments				
202236404	B100-L-32 Pipe Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Ivory	210	mg/kg	5/25/2022
Comments				
202236405	B100-L-33 Elect Box Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Gray	77	mg/kg	5/25/2022
Comments				
202236406	B100-L-34 Elect Box Paint ▪ Elect Rm, B100, BUCKNER COMMUNICATION SITE, Light Gray	1300	mg/kg	5/25/2022
Comments				

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Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236407	B100-L-35 Duct Paint • Elect Rm, B100, BUCKNER COMMUNICATION SITE, Ivory	2200	mg/kg	5/25/2022
Comments				
202236408	B100-L-36 Duct Paint • Elect Rm, B100, BUCKNER COMMUNICATION SITE, Blue	< 49	mg/kg	5/25/2022
Comments				
202236409	B100-L-37 Duct Paint • Elect Rm, B100, BUCKNER COMMUNICATION SITE, Gray	11000	mg/kg	5/25/2022
Comments				
202236410	B100-L-38 Pipe Support Angle Paint • Elect Rm, B100, BUCKNER COMMUNICATION SITE, Gray	2400	mg/kg	5/25/2022
Comments				
202236411	B100-L-39 Interior Wall Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236412	B100-L-40 Conduit Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	5700	mg/kg	5/25/2022
Comments				
202236413	B100-L-41 Pipe Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Red	2800	mg/kg	5/25/2022
Comments				
202236414	B100-L-42 Pipe Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	120	mg/kg	5/25/2022
Comments				
202236415	B100-L-43 Elect Box Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Ivory	< 64	mg/kg	5/25/2022
Comments				

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 JAPAN, 4/1/22-4/7/22

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236416	B100-L-44 Elect Box Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	< 53	mg/kg	5/25/2022
Comments				
202236417	B100-L-45 Floor Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	280	mg/kg	5/25/2022
Comments				
202236418	B100-L-46 Pipe Support Angle Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	4100	mg/kg	5/25/2022
Comments				
202236419	B100-L-47 Pump Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Green	< 65	mg/kg	5/25/2022
Comments				
202236420	B100-L-48 Generator Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	< 53	mg/kg	5/25/2022
Comments				
202236421	B100-L-49 Boiler Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Gray	< 78	mg/kg	5/25/2022
Comments				
202236422	B100-L-50 Pipe Paint • Mech, B100, BUCKNER COMMUNICATION SITE, Silver	< 160	mg/kg	5/25/2022
Comments				
202236423	B100-L-51 Conduit Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, Ivory	3800	mg/kg	5/25/2022
Comments				
202236424	B100-L-52 Pipe Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, Red	< 46	mg/kg	5/25/2022
Comments				

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Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236425	B100-L-53 Pipe Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, White	33000	mg/kg	5/25/2022
Comments				
202236426	B100-L-54 Elect Box Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, Tan	< 110	mg/kg	5/25/2022
Comments				
202236427	B100-L-55 Support Angle Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, Ivory	730	mg/kg	5/25/2022
Comments				
202236428	B100-L-100 Elect Box Paint • Rm 119, B100, BUCKNER COMMUNICATION SITE, Gray	< 85	mg/kg	5/25/2022
Comments				
202236429	B100-L-101 Wall Paint • Rm 119, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236430	B100-L-102 Elect Box Paint • Rm 119, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236431	B100-L-103 Wall Paint • Rm 118, B100, BUCKNER COMMUNICATION SITE, Ivory	< 68	mg/kg	5/25/2022
Comments				
202236432	B100-L-400 Elect Box Paint • Interior, B100, BUCKNER COMMUNICATION SITE, Gray	< 140	mg/kg	5/25/2022
Comments				
202236433	B100-L-401 Wall Paint • Interior, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				

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Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236434	B100-L-402 Slide Door Paint ▪ Interior, B100, BUCKNER COMMUNICATION SITE, Ivory	< 40	mg/kg	5/25/2022
Comments				
202236435	B100-L-403 Conduit Paint ▪ Interior, B100, BUCKNER COMMUNICATION SITE, Ivory	< 47	mg/kg	5/25/2022
Comments				
202236436	B100-L-500 Elect Box Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Gray	10000	mg/kg	5/25/2022
Comments				
202236437	B100-L-501 Elect Box Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Tan	< 64	mg/kg	5/25/2022
Comments				
202236438	B100-L-502 Wall Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Ivory	45	mg/kg	5/25/2022
Comments				
202236439	B100-L-503 Frame Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Gray	7900	mg/kg	5/25/2022
Comments				
202236440	B100-L-504 Pipe Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Red	610	mg/kg	5/25/2022
Comments				
202236441	B100-L-505 Conduit Paint ▪ Rm 108, B100, BUCKNER COMMUNICATION SITE, Ivory	3800	mg/kg	5/25/2022
Comments				
202236442	B100-L-506 Door/ Frame Paint ▪ Conference Room, B100, BUCKNER COMMUNICATION SITE, Orange	65	mg/kg	5/25/2022
Comments				

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JAPAN, 4/1/22-4/7/22

Total Lead (paint chips)

NIOSH Method: 7082m LEAD by FAAS

Sample No.	Sample ID / Description	Results	Units	Date Analyzed
202236443	B100-L-507 Wall Paint • Conference Room, B100, BUCKNER COMMUNICATION SITE, Orange	< 40	mg/kg	5/25/2022
Comments				
202236444	B100-L-600 Steel Door/ Frame Paint • Janitor, B100, BUCKNER COMMUNICATION SITE, Ivory	8000	mg/kg	5/25/2022
Comments				

All Quality Control data are acceptable unless otherwise noted.

MRL for lead air is 5ug.

MRL for lead wipe is 10ug.

MRL for lead paint or soil is 40 mg/kg for a 0.25g sample.

General Comments

The sample[s] analysis subject of this analytical report were conducted in general accordance with the procedures associated with the "analytical method" referenced above. Modifications to this methodology may have been made based upon the analyst's professional judgment and / or sample matrix effects encountered. The analysis of sample relates only to the sample analyzed, and may or may not be representative of the original source of the material submitted for our analysis. All analysts participate in interlaboratory quality control testing to continuously document proficiency. This report is not to be duplicated except in full without the expressed written permission of Hawaii Analytical Laboratory. This report should not be construed as an endorsement for a product or a service by the AIHA LAP, LLC or any affiliated organizations. Sample and associated sampling / collection data is reported as provided by client. TWA values have been calculated based on information supplied by the client that the laboratory has not independently verified. Results have not been corrected for blank determinations unless noted in remarks. Unless otherwise indicated the sample condition at the time of receipt was acceptable.

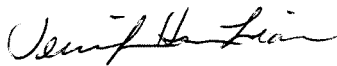
Results and Symbols Definitions

> This testing result is greater than the numerical value listed.

< This testing result is less than the numerical value listed.

= Analytical methods marked with an "#" are not within our AIHA LAP, LLC Scope of Accreditation.

MRL = Method Reporting Limit.



Jennifer Hsu Liao
Laboratory Manager

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(LEAD ID)

<鉛含有分析依頼書>

OFFICIAL LABORATORY CHAIN OF CUSTODY

PAGE 1 OF 5

PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			(1) CUSTOMER INFORMATION お客様情報 (会社名・所在地等): Name お名前: 株式会社南西環境研究所 (Nanasei Environmental Laboratory Co., Ltd) Address 住所: 〒903-0105 沖縄県那中頭郡西原町字東崎4-4 (4-4 Aza Agarizaki, Nishihara-cho, Nakagami-gun, Okinawa,) Tel / Fax : Tel 098-835-8411 / Fax 098-835-8412 E-mail : noborikawa@nansei-kankyo.co.jp Contact Name ご担当者: 登川 貴史		
RESPONSIBLE FOR: (PRINT NAME) サンプル採取責任者: (楷書体) Nanasei Environmental Laboratory Co., Ltd Takafumi Noborikawa		TOTAL SAMPLES 10/72 サンプル総数:			
# OF COPIES 報告書部数: 1					
REPORT ATTN TO 報告書宛名: ☐ Same as (1) と同じ ☒ Same as (2) と同じ			(2) Name お名前: NIHON MECCS CO., LTD.		
SEND REPORT TO 速報送付先: ☒ Same as (1) と同じ ☐ Same as (2) と同じ			Address 住所: 4-16-7 Minamitobaru, Okinawa City,		
SEND BY 速報送付方法: ☒ E-mail ☐ Fax			Tel / Fax:		
MAIL & BILL TO 報告書・ご請求書送付先: ☒ Same as (1) と同じ ☐ Same as (2) と同じ			E-mail:		
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION サンプル名・採取場所	SAMPLE DESCRIPTION サンプル詳細(色・形状・材質等)		LAB JOB NUMBER 分析所ジョブ番号
B100-L-01	2022/4/1	Exterior Wall Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236373
B100-L-02	2022/4/1	Exterior Wall Trim Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Brown		202236374
B100-L-03	2022/4/1	Pipe Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236375
B100-L-04	2022/4/1	Pipe Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Red		202236376
B100-L-05	2022/4/1	Pipe Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Blue		202236377
B100-L-06	2022/4/1	Pipe Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Silver		202236378
B100-L-07	2022/4/1	Conduit Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236379
B100-L-08	2022/4/1	Elect Box Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Gray		202236380
B100-L-09	2022/4/1	Steel Shutter Door/ Frame Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Brown		202236381
B100-L-10	2022/4/1	Support Angle Paint ・ Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236382
ANALYSIS REQUIRED 分析方法: LEAD PAINT CHIPS / AAS		Turn-Around-Time 報告希望日: By JST ☒ UST ☐ 6hrs ☐ 1w.day ☐ 2w.days ☐ 3w.days ☐ 4w.days ☐ 5w.days		Other Info. その他:	
RELINQUISHED BY: (SIGNATURE) 依頼者・受渡者(サイン)	DATE / TIME 依頼日時	REMARK 備考: (#0590)	RECEIVED BY: (SIGNATURE) 受取者(サイン) Anne Antin	DATE / TIME 受取日時 05-12	REMARK 備考: 22A10:55 RCVD 22

SOLICITATION NO.: TBD

ATTACHMENT 01 57-19-A Page 645

BACKCHECK SUBMITTAL

COC-LDID 20140514

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OFFICIAL LABORATORY CHAIN OF CUSTODY

PAGE 2 OF 5

PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 30/72	ANALYSIS REQUIRED 分析方法: LEAD PAINT CHIPS / AAS	LAB JOB NUMBER 分析所番号/番号 202204690
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル #
B100-L-11	2022/4/1	Support Angle Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236383
B100-L-12	2022/4/1	Condenser Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Gray		202236384
B100-L-13	2022/4/1	Condenser Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Light Gray		202236385
B100-L-14	2022/4/1	Safety Bumper Pole Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Yellow		202236386
B100-L-15	2022/4/1	Chiller Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Gray		202236387
B100-L-16	2022/4/1	Conduite Paint * Exterior, B100, BUCKNER COMMUNICATION SITE	Gray		202236388
B100-L-17	2022/4/6	Interior Wall Paint * Rm 101, B100, BUCKNER COMMUNICATION SITE	Ivory		202236389
B100-L-18	2022/4/6	Conduit Paint * Rm 101, B100, BUCKNER COMMUNICATION SITE	Ivory		202236390
B100-L-19	2022/4/6	Elec Box Paint * Rm 101, B100, BUCKNER COMMUNICATION SITE	Red		202236391
B100-L-20	2022/4/7	Steel Door/ Frame Paint * Rm 102, B100, BUCKNER COMMUNICATION SITE	Tan		202236392
B100-L-21	2022/4/7	Wood Door/ Frame Paint * Rm 102, B100, BUCKNER COMMUNICATION SITE	Tan		202236393
B100-L-22	2022/4/7	Interior Wall Paint * Rm 105, B100, BUCKNER COMMUNICATION SITE	Ivory		202236394
B100-L-23	2022/4/7	Wood Door/ Frame Paint * Rm 105, B100, BUCKNER COMMUNICATION SITE	Ivory		202236395
B100-L-24	2022/4/7	Interior Wall Paint * Rm 105, B100, BUCKNER COMMUNICATION SITE	Ivory		202236396
B100-L-25	2022/4/7	Ceiling Paint * Toilet, B100, BUCKNER COMMUNICATION SITE	Ivory		202236397
B100-L-26	2022/4/1	Interior Wall Paint * Elect Rm, B100, BUCKNER COMMUNICATION SITE	White		202236398
B100-L-27	2022/4/1	Interior Wall Paint * Elect Rm, B100, BUCKNER COMMUNICATION SITE	Ivory		202236399
B100-L-28	2022/4/1	Floor Paint * Elect Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236400
B100-L-29	2022/4/1	Conduite Paint * Elect Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236401
B100-L-30	2022/4/1	Conduite Paint * Elect Rm, B100, BUCKNER COMMUNICATION SITE	Dark Gray		202236402



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OFFICIAL LABORATORY CHAIN OF CUSTODY

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PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 50/72	ANALYSIS REQUIRED 分析方法: LEAD PAINT CHIPS / AAS	LAB JOB NUMBER 分析所シフト番号 202204690
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析所サンプル 202236403
B100-L-31	2022/4/1	Pipe Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Red		202236403
B100-L-32	2022/4/1	Pipe Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Ivory		202236404
B100-L-33	2022/4/1	Elect Box Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236405
B100-L-34	2022/4/1	Elect Box Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Light Gray		202236406
B100-L-35	2022/4/1	Duct Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Ivory		202236407
B100-L-36	2022/4/1	Duct Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Blue		202236408
B100-L-37	2022/4/1	Duct Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236409
B100-L-38	2022/4/1	Pipe Support Angle Paint ・ Elect Rm, B100, BUCKNER COMMUNICATION SITE	Gray		202236410
B100-L-39	2022/4/4	Interior Wall Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Ivory		202236411
B100-L-40	2022/4/4	Conduit Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236412
B100-L-41	2022/4/4	Pipe Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Red		202236413
B100-L-42	2022/4/4	Pipe Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236414
B100-L-43	2022/4/4	Elect Box Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Ivory		202236415
B100-L-44	2022/4/4	Elect Box Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236416
B100-L-45	2022/4/4	Floor Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236417
B100-L-46	2022/4/4	Pipe Support Angle Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236418
B100-L-47	2022/4/4	Pump Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Green		202236419
B100-L-48	2022/4/4	Generator Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236420
B100-L-49	2022/4/4	Boiler Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Gray		202236421
B100-L-50	2022/4/4	Pipe Paint ・ Mech, B100, BUCKNER COMMUNICATION SITE	Silver		202236422



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OFFICIAL LABORATORY CHAIN OF CUSTODY

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(LEAD ID)

PROJECT NAME / NUMBER 件名 / プロジェクト番号等: PWE02430, BULK SAMPLING AND TESTING OF ASBESTOS(ACM) & LEAD BASED-PAINT (LBP), VARIOUS BUILDINGS, BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN			TOTAL SAMPLES サンプル総数: 70/72	ANALYSIS REQUIRED 分析方法: LEAD PAINT CHIPS / AAS	LAB JOB NUMBER 分析依頼番号 202204690
CLIENT SAMPLE # サンプル #	SAMPLED DATE 採取日	SAMPLE NAME, LOCATION, ID NUMBER サンプル名・採取場所・識別番号	SAMPLE DESCRIPTION サンプル詳細		LAB SAMPLE NUMBER 分析用サンプル 202236423
B100-L-51	2022/4/4	Conduit Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Ivory		202236423
B100-L-52	2022/4/4	Pipe Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Red		202236424
B100-L-53	2022/4/4	Pipe Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	White		202236425
B100-L-54	2022/4/4	Elect Box Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Tan		202236426
B100-L-55	2022/4/1	Support Angle Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Ivory		202236427
B100-L-100	2022/4/4	Elect Box Paint ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Gray		202236428
B100-L-101	2022/4/4	Wall Paint ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Ivory		202236429
B100-L-102	2022/4/4	Elect Box Paint ・ Rm 119, B100, BUCKNER COMMUNICATION SITE	Ivory		202236430
B100-L-103	2022/4/4	Wall Paint ・ Rm 118, B100, BUCKNER COMMUNICATION SITE	Ivory		202236431
B100-L-400	2022/4/5	Elect Box Paint ・ Interior, B100, BUCKNER COMMUNICATION SITE	Gray		202236432
B100-L-401	2022/4/5	Wall Paint ・ Interior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236433
B100-L-402	2022/4/5	Slide Door Paint ・ Interior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236434
B100-L-403	2022/4/5	Conduit Paint ・ Interior, B100, BUCKNER COMMUNICATION SITE	Ivory		202236435
B100-L-500	2022/4/5	Elect Box Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Gray		202236436
B100-L-501	2022/4/5	Elect Box Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Tan		202236437
B100-L-502	2022/4/5	Wall Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Ivory		202236438
B100-L-503	2022/4/5	Frame Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Gray		202236439
B100-L-504	2022/4/5	Pipe Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Red		202236440
B100-L-505	2022/4/5	Conduit Paint ・ Rm 108, B100, BUCKNER COMMUNICATION SITE	Ivory		202236441
B100-L-506	2022/4/6	Door/ Frame Paint ・ Conference Room, B100, BUCKNER COMMUNICATION SITE	Orange		202236442



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OFFICIAL LABORATORY CHAIN OF CUSTODY

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SECTION 01 74 19

CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS

(Apr 2024) Japan Environmental Governing
Standards

1.2 GOVERNMENT POLICY

Government policy is to apply sound environmental principles in the design, construction and use of facilities. As part of the implementation of that policy, practice efficient waste management when sizing, cutting, and installing products and materials, and use all reasonable means to divert construction and demolition waste from landfills and incinerators and to facilitate their recycling or reuse. Divert a minimum of 60 percent by weight of total project solid waste from the landfill.

1.3 MANAGEMENT

Develop and implement a waste management program. Take a pro-active, responsible role in the management of construction and demolition waste and require all subcontractors, vendors, and suppliers to participate in the effort. The Environmental Manager, as specified in Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS, is responsible for instructing workers and overseeing and documenting results of the Waste Management Plan for the project. Construction and demolition waste includes products of demolition or removal, excess or unusable construction materials, packaging materials for construction products, and other materials generated during the construction process but not incorporated into the work. In the management of waste, consider the availability of viable markets, the condition of the material, the ability to provide the material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates. Implement any special programs involving rebates or similar incentives related to recycling of waste. Revenues or other savings obtained for salvage, or recyclable materials not claimed by the Government will accrue to the Contractor. Appropriately permit firms and facilities used for recycling, reuse, and disposal for the intended use to the extent required by federal, state, and local regulations. Also, provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the project.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit

the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Waste Management Plan; G

SD-11 Closeout Submittals

Records; G

1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Waste Management Plan and to develop a mutual understanding relative to the details of waste management. The requirements for this meeting may be fulfilled during the coordination and mutual understanding meeting outlined in Section 01 45 00.00 10 QUALITY CONTROL. At a minimum, discuss environmental and waste management goals and issues at the following additional meetings:

- a. Pre-bid meeting.
- b. Preconstruction meeting.
- c. Regular QC meetings.
- d. Work safety meetings.

1.6 WASTE MANAGEMENT PLAN

Submit a Waste Management Plan within 15 days after Notice to Proceed. The plan demonstrates how to meet the project waste diversion goal. Also, include the following in the plan:

- a. Name of individuals on the Contractor's staff responsible for waste prevention and management.
- b. Actions that will be taken to reduce solid waste generation, including coordination with Subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of wastes.
- e. Characterization, including estimated types and quantities, of the waste to be generated.
- f. Name of landfill and/or incinerator to be used and the estimated costs for use, assuming that there would be no salvage or recycling on the project.
- g. Identification of local and regional reuse programs, including non-profit organizations such as schools, local housing agencies, and organizations that accept used materials such as materials exchange

networks and Habitat for Humanity. Include the name, location, and phone number for each reuse facility to be used, and provide a copy of the permit or license for each facility.

- h. List of specific waste materials that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, location, and phone number, including a copy of the permit or license for each facility.
- i. Identification of materials that cannot be recycled/reused with an explanation or justification, to be approved by the Contracting Officer.
- j. Description of the means by which any waste materials identified in item (h) above will be protected from contamination.
- k. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- l. Anticipated net cost savings determined by subtracting Contractor program management costs and the cost of disposal from the revenue generated by sale of the materials and the incineration and/or landfill cost avoidance.

Revise and resubmit Plan as required by the Contracting Officer. Approval of Contractor's Plan shall not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Distribute copies of the Waste Management Plan to each Subcontractor, the Quality Control Manager, and the Contracting Officer.

1.7 RECORDS

Maintain records to document the quantity of waste generated; the quantity of waste diverted through sale, reuse, or recycling; and the quantity of waste disposed by landfill or incineration. Quantities may be measured by weight or by volume, but must be consistent throughout. List each type of waste separately noting the disposal or diversion date. Identify the landfill, recycling center, waste processor, or other organization used to process or receive the solid waste. Provide explanations for any waste not recycled or reused. With each application for payment, submit updated documentation for solid waste disposal and diversion, and submit manifests, weight tickets, receipts, and invoices specifically identifying the project and waste material. Make the records available to the Contracting Officer during construction, and deliver to the Contracting Officer upon completion of the construction a copy of the records.

1.8 REPORTS

Provide quarterly reports and a final report to Contracting Officer. Include project name, information for waste generated this quarter, and cumulative totals for the project in quarterly and final reports. Also include in each report, supporting documentation to include manifests, weight tickets, receipts, and invoices specifically identifying the project and waste material. Include timber harvest and demolition information, if any. See Section 01 57 19 TEMPORARY ENVIRONMENTAL

CONTROLS for Nonhazardous Solid Waste Diversion Report requirements.

1.9 COLLECTION

Separate, store, protect, and handle at the site identified recyclable and salvageable waste products in a manner that maximizes recyclability and salvagability of identified materials. Provide the necessary containers, bins and storage areas to facilitate effective waste management and clearly and appropriately identify them. Provide materials for barriers and enclosures around recyclable material storage areas which are nonhazardous and recyclable or reusable. Locate out of the way of construction traffic. Provide adequate space for pick-up and delivery and convenience to Subcontractors. Recycling and waste bin areas are to be kept neat and clean, and handle recyclable materials to prevent contamination of materials from incompatible products and materials. Clean contaminated materials prior to placing in collection containers. Use cleaning materials that are nonhazardous and biodegradable. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods:

1.9.1 Source Separated Method

Separate waste products and materials that are recyclable from trash and sorted as described below into appropriately marked separate containers. Transport materials to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the following category types as appropriate to the project waste and to the available recycling and reuse programs in the project area:

- a. Land clearing debris.
- b. Asphalt.
- c. Concrete and masonry.
- d. Metal (e.g. banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, lead brass, bronze).
 - (1) Ferrous.
 - (2) Non-ferrous.
- e. Wood (nails and staples allowed).
- f. Debris.
- g. Glass (colored glass allowed).
- h. Paper.
 - (1) Bond.
 - (2) Newsprint.

(3) Cardboard and paper packaging materials.

i. Plastic.

(1) Type 1 - Polyethylene Terephthalate (PET, PETE).

(2) Type 2 - High Density Polyethylene (HDPE).

(3) Type 3 - Vinyl (Polyvinyl Chloride or PVC).

(4) Type 4 - Low Density Polyethylene (LDPE).

(5) Type 5 - Polypropylene (PP).

(6) Type 6 - Polystyrene (PS).

(7) Type 7 - Other. Use of this code indicates that the package in question is made with a resin other than the six listed above, or is made of more than one resin listed above, and used in a multi-layer combination.

j. Gypsum.

k. Non-hazardous paint and paint cans.

l. Carpet.

m. Ceiling tiles.

n. Insulation.

o. Beverage containers.

1.9.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to a recycling facility where the recyclable materials are sorted and processed.

1.9.3 Other Methods

Other proposed methods may be used when approved by the Contracting Officer.

1.10 DISPOSAL

Control accumulation of waste materials and trash. Where materials are to be turned over to the Contractor for disposal, every effort shall be made to obtain credit from the disposal to reduce the cost of the Contract. Recycle or dispose of collected materials off of Government property at intervals approved by the Contracting Officer and in compliance with waste management procedures of JEGS and local laws and regulations. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

a. Reuse. Give first consideration to salvage for reuse since little or no re-processing is necessary for this method, and less pollution is created when items are reused in their original form. Consider sale or donation of waste suitable for reuse.

- b. Recycle. Recycle waste materials not suitable for reuse, but having value as being recyclable. Recycle all fluorescent lamps, HID lamps, and mercury-containing thermostats removed from the site. Arrange for timely pickups from the site or deliveries to recycling facilities in order to prevent contamination of recyclable materials.
- c. Compost. Consider composting on-site if a reasonable amount of compostable material will be available. Compostable materials include plant material, sawdust, and certain food scraps.
- d. Waste. Dispose of materials with no practical use or economic benefit to waste-to-energy plants where available. As the last choice, dispose of materials at a landfill or incinerator.
- e. Return. Set aside and protect misdelivered and substandard products and materials and return to supplier for credit.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Stewardship for the
Cleaning of Commercial and Institutional
Buildings

GREEN SEAL (GS)

GS-37 (2017) Cleaning Products for Industrial
and Institutional Use

SPATIAL DATA STANDARDS FOR FACILITIES, INFRASTRUCTURE, AND
ENVIRONMENT (SDSFIE)

SDSFIE 3.1.0.1 (2015) Air Force Data Model

U.S. ARMY CORPS OF ENGINEERS (USACE)

ERDC/ITL TR-12-1 (2015) A/E/C Graphics Standard, Release 2.0

ERDC/ITL TR-12-6 (2015) A/E/C CAD Standard - Release 6.0

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-300-08 (2009, with Change 2, 2011) Criteria for
Transfer and Acceptance of DoD Real
Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.2.3 USACE CAD/BIM Technology Center

The USACE CAD/BIM Technology Center hosts all standard content for USACE. This content can be accessed through the CAD/BIM Technology Center website, <https://cadbimcenter.erdc.dren.mil/>.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse shall be at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan; G

Warranty Tags

Final Cleaning

Spare Parts Data

Spare Parts And Spare Parts Data

SD-08 Manufacturer's Instructions

Posted Instructions

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

As-Built Record of Equipment and Materials; G

Final Approved Shop Drawings; G

Construction Contract Specifications; G

Interim DD FORM 1354; G

Checklist for DD FORM 1354; G

1.5 SPARE PARTS AND SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

- a. Supply the necessary quantity of items of each part for spare parts required by this contract, identified in the technical specification herein. Indicate manufacturer's name, part number, nomenclature, and stock level required for maintenance and repair. List those items that may be standard to the normal maintenance of the system.

1.6 QUALITY CONTROL

Additions and corrections to the contract drawings must be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols must be the same as the original line colors, line weights, lettering, layering conventions, and symbols.

1.7 WARRANTY MANAGEMENT

1.7.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to

which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

- a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.
- b. For each warranty, the name, address, telephone number, and email of each of the guarantor's representatives nearest to the project location.
- c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.
- d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:
 - (1) Name of item.
 - (2) Model and serial numbers.
 - (3) Location where installed.
 - (4) Name and phone numbers of manufacturers or suppliers.
 - (5) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have extended warranties must be indicated with separate warranty expiration dates.
 - (6) Cross-reference to warranty certificates as applicable.
 - (7) Starting point and duration of warranty period.
 - (8) Summary of maintenance procedures required to continue the warranty in force.
 - (9) Cross-reference to specific pertinent Operation and Maintenance manuals.
 - (10) Organization, names and phone numbers of persons to call for warranty service.
 - (11) Typical response time and repair time expected for various

warranted equipment.

- e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.
- f. Procedure and status of tagging of equipment covered by extended warranties.
- g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.7.2 Performance Bond

The Performance Bond must remain effective throughout the construction period.

- a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
- b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.
- c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.7.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.7.4 Contractor's Response to Construction Warranty Service Requirements

Following oral or written notification by the Contracting Officer, respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has

been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.

- a. First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.
- b. Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.
- c. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.
- d. The "Construction Warranty Service Priority List" is as follows:

Code 1-Life Safety Systems

- (1) Fire suppression systems.
- (2) Fire alarm system(s) in place in the building.

Code 1-Air Conditioning Systems

- (1) Recreational support.
- (2) Air conditioning leak in part of building, if causing damage.
- (3) Air conditioning system not cooling properly.

Code 1-Doors

- (1) Overhead doors not operational, causing a security, fire, or safety problem.
- (2) Interior, exterior personnel doors or hardware, not functioning properly, causing a security, fire, or safety problem.

Code 3-Doors

- (1) Overhead doors not operational.
- (2) Interior/exterior personnel doors or hardware not functioning properly.

Code 1-Electrical

- (1) Power failure (entire area or any building operational after 1600 hours).
- (2) Security lights
- (3) Smoke detectors

Code 2-Electrical

- (1) Power failure (no power to a room or part of building).
- (2) Receptacle and lights (in a room or part of building).

Code 3-Electrical

Street lights.

Code 1-Gas

- (1) Leaks and breaks.
- (2) No gas to family housing unit or cantonment area.

Code 1-Heat

- (1) Area power failure affecting heat.
- (2) Heater in unit not working.

Code 2-Kitchen Equipment

- (1) Dishwasher not operating properly.
- (2) All other equipment hampering preparation of a meal.

Code 1-Plumbing

- (1) Hot water heater failure.
- (2) Leaking water supply pipes.

Code 2-Plumbing

- (1) Flush valves not operating properly.
- (2) Fixture drain, supply line to commode, or any water pipe leaking.
- (3) Commode leaking at base.

Code 3 -Plumbing

Leaky faucets.

Code 3-Interior

- (1) Floors damaged.
- (2) Paint chipping or peeling.
- (3) Casework.

Code 1-Roof Leaks

Temporary repairs will be made where major damage to property is occurring.

Code 2-Roof Leaks

Where major damage to property is not occurring, check for location of leak during rain and complete repairs on a Code 2 basis.

Code 2-Water (Exterior)

No water to facility.

Code 2-Water (Hot)

No hot water in portion of building listed.

Code 3-All other work not listed above.

1.7.5 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	

Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time	
priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 RECORD DRAWINGS

Prepare the CAD drawing files in AutoCAD Release 2013 format and Bentley Microstation V8i format compatible with a Windows 7 or higher operating system.

2.1.1 Additional Drawings

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings.

2.1.1.1 Sheet Numbers and File Names

If a sheet needs to be added between two sequential sheets, append a Supplemental Drawing Designator in accordance with ERDC/ITL TR-12-6 Adding a drawing sheet, and ERDC/ITL TR-12-1.

2.2 PDF AS-BUILT FILES

Provide electronic PDF "plots" of all contract drawings sheets associated with the as-built drawing submittal. Compile and organize the PDF set to match the contract drawings.

2.3 REDLINES AND MARKUPS

Provide PDFs of the current working redlines and/or markups complying with the as-builts drawing and markup requirements contained in this specification.

2.4 AS-BUILT RE-SUBMISSION REQUIREMENTS

If elements of an as-built submittal or advanced modeling package are rejected, provide the following for each re-submission, in addition to any information required in Section 01 33 00 SUBMITTAL PROCEDURES:

- a. Re-submit all components required under paragraph As-Builts or Advanced Modeling Package, including a new Advanced Modeling Submittal Checklist and updated content in response to Government comments.

- b. Provide a copy of all Government review comments.
- c. Provide a disposition/response to each Government review comment for a back-check of the re-submission deliverable.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files. Submit As-Built Drawings 30 days prior to Beneficial Occupance Date (BOD).

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:
 - (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
- c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
- d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
- e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
- f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
- g. For deletions, cross out all features, data and captions that relate to that revision.
- h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
- i. Indicate one of the following when attaching a print or sketch to a

markup print:

- 1) Add an entire drawing to contract drawings
- 2) Change the contract drawing to show
- 3) Provided for reference only to further detail the initial design.

j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Revise As-Built Drawings in accordance with ERDC/ITL TR-12-1 and ERDC/ITL TR-12-6. Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the contract drawings which are made during construction or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. Submit the working as-built markup drawings for approval prior to submission of each monthly pay estimate. For failure to maintain the working and final record drawings as specified herein, the Contracting Officer will withhold 10 percent of the monthly progress payment until approval of updated drawings. Show on the as-built drawings, but not limited to, the following information:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.
- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.
- h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.

- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- j. Modifications (include within change order price the cost to change working as-built markup drawings to reflect modifications).
- l. Actual location of anchors, construction and control joints, etc., in concrete.
- m. Unusual or uncharted obstructions that are encountered in the contract work area during construction.
- n. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.1.3 GIS Data

Provide GIS data with as-builts. GIS data shall be in a format accepted by ESRI software. Data shall comply with SDSFIE 3.1.0.1. All GIS data shall be delivered with the following requirements:

- a. Projection: JGD 2000 Zone 15
- b. Datum: WGS 1984
- c. Geoid: EGM 96
- d. Units: Meters

3.2 RECORD DRAWING FILES

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The Contracting Officer will review final PDF record drawings for accuracy and return them to the Contractor for required corrections, changes, additions, and deletions.

3.2.1 Rename the CAD Drawing files

Rename the CAD Drawing files using the as instructed in the Pre-Construction conference. Use only those renamed files for the Marked-up changes. Make all changes on the layer/level as the original item.

- a. For AutoCAD files (DWG), enter all as-built delta changes and notations on the AS-BUILT layer.
- c. When final revisions have been completed, show the wording "RECORD DRAWING AS-BUILTS" followed by the name of the Contractor in letters

at least 5 mm high on the cover sheet drawing. Date RECORD DRAWING AS-BUILTS" drawing revisions in the revision block.

- d. Within 20 days after Government approval of all of the working record drawings for a phase of work, prepare the final CAD record drawings for that phase of work and submit PDF drawing files and two sets of prints for review and approval. The Government will promptly return one set of prints annotated with any necessary corrections. Within 10 days revise the CAD files accordingly at no additional cost and submit one set of final prints for the completed phase of work to the Government. Within 20 days of substantial completion of all phases of work, submit the final record drawing package for the entire project. Submit one set of electronic CAD files, and one set of the approved working record PDF files on an optical disc with two sets of prints. The CAD files must be complete in all details and identical in form and function to the CAD drawing files supplied by the Government. Prepare AutoCAD files for transmittal using e-Transmit. Make any transactions or adjustments necessary to accomplish this. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CAD system. Paper prints, drawing files and storage media submitted will become the property of the Government upon final approval. Failure to submit final record PDF drawing files, CAD files and marked prints as specified will be cause for withholding any payment due under this contract. Approval and acceptance of final record drawings must be accomplished before final payment is made.

3.3 RECORD DRAWINGS

Prepare final record drawings after the completion of each definable feature of work as listed in the Contractor Quality Control Plan (such as Foundations, Utilities, or Structural Steel as appropriate for the project). Transfer the changes from the approved working as-built markup drawings to the original electronic CAD drawing files. Modify the as-built CAD drawing files to correctly show the features of the project as-built by bringing the working CAD drawing set into agreement with approved working as-built markup drawings, and adding such additional drawings as may be necessary. Refer to ERDC/ITL TR-12-1. Jointly review the working as-built markup drawings with printouts from working as-built CAD drawing PDF files for accuracy and completeness. Monthly review of working as-built CAD drawing PDF file printouts must cover all sheets revised since the previous review. These PDF drawing files are part of the permanent records of this project. Any drawings damaged or lost must be satisfactorily replaced at no expense to the Government.

Drawing revisions (include within change order price the cost to change working and final record drawings to reflect revisions) and compliance with the following procedures.

- a. Follow directions in the revision for posting descriptive changes.
- b. The revision delta size must be 8 mm unless the area where the delta is to be placed is crowded. Use a smaller size delta for crowded areas.
- c. Place a revision delta at the location of each deletion.
- d. For new details or sections which are added to a drawing, place a revision delta by the detail or section title.

- e. For minor changes, place a revision delta by the area changed on the drawing (each location).
- f. For major changes to a drawing, place a revision delta by the title of the affected plan, section, or detail at each location.
- g. For changes to schedules or drawings, place a revision delta either by the schedule heading or by the change in the schedule.

3.3.1 Final Record Drawing Package

Submit the final record PDF and CAD drawings package for the entire project within 20 days of substantial completion of all phases of work. Submit one set of A1 size PDF and CAD files on optical disc, read-only memory (ROM), two sets of A1 size prints and one set of the approved working record drawings. The package must be complete in all details and identical in form and function to the contract drawing files supplied by the Government.

3.4 FINAL APPROVED SHOP DRAWINGS

Submit final approved project shop drawings 30 days after transfer of the completed facility.

3.5 CONSTRUCTION CONTRACT SPECIFICATIONS

Submit final PDF file record construction contract specifications, including revisions thereto, 30 days after transfer of the completed facility.

3.6 AS-BUILT RECORD OF EQUIPMENT AND MATERIALS

Furnish one copy of preliminary record of equipment and materials used on the project 15 days prior to final inspection. This preliminary submittal will be reviewed and returned 2 days after final inspection with Government comments. Submit Two sets of final record of equipment and materials 10 days after final inspection. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

3.7 OPERATION AND MAINTENANCE MANUALS

Provide project operation and maintenance manuals as specified in Section 01 78 23 OPERATION AND MAINTENANCE MANUALS DATA. Provide one electronic copy of the Operation and Maintenance Manual files and

three hard copies of the Operation and Maintenance Manuals. Submit to the Contracting Officer for approval within 30 calendar days of the Beneficial Occupancy Date (BOD). Update and resubmit files for final approval at BOD.

3.8 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Clean filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan as required. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Sections 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS, and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

3.9 REAL PROPERTY RECORD

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354. Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Refer to Attachment 01 78 00-A for the Draft DD FORM 1354. For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link: www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

Submit the completed Checklist for DD FORM 1354 of Installed Building Equipment items. Attach this list to the updated DD FORM 1354.

-- End of Section --

BACKCHECK SUBMITTAL

27. CONSTRUCTION DEFICIENCIES <i>(Attach blank sheet for continuations)</i>	28. PROJECT REMARKS <i>(Attach blank sheet for continuations)</i>
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INSTRUCTIONS

GENERAL. This form has been designed and issued for acceptance of real property or transfer between the DoD Components.
For detailed instructions on how to fill out this form, please refer to Unified Facilities Criteria 1-300-08.
Existing instructions issued by the DoD Components relative to the preparation of DD Form 1354 are applicable to this revised form to the extent that the various items and columns on the superseded forms have been retained. The DOD Components may promulgate additional instructions, as appropriate.
For detailed information on data inputs and codes, please refer to the latest version of the Real Property Information Model (RPIM) Report and use the RPIM Pick Lists for Block responses: <https://extranet.acq.osd.mil/ie/>

SPECIFIC DATA ITEMS.

- 1. From.** Name of the DoD Component.
- 2. Date Prepared.** Date of actual preparation. Enter all dates in YYYYMMDD format (Example: March 31, 2010 = 20100331).
- 3. Project/Job Number.** Project number on a DD Form 1391 or Individual Job Order Number.
- 4. Serial Number.** Sequential serial number assigned by the preparing organization (e.g., Item 17. 2010-0001).
- 5. To.** Name and address of the receiving installation, activity, and Service of the Real Property Accountable Officer (RPAO).
- 6. RPSUID/SITENAME/INSTCODE/INSTNAME.** Site Unique Identifier and name or installation code and name where the constructed facility is located.
- 7. Contract Number(s).** Contract number(s) for this project.
- 7a. RPA Placed-In-Service Date.** The calendar date the asset is accepted for use by DoD.
- 8. Transaction Details.**
 - a. Type of Transaction. X only one box.
For Acquisition Type, refer to the Acquisition Method Code pick list in RPIM.
 - b. When/Event. When or event causing preparation of DD Form 1354. X only one box.
 - c. Version. Draft, interim, or final DD Form 1354. X only one box.
- 9. Item Number.** Use a separate item number for each facility, no item number for additional usages.
- 10a. Facility Number.** An installation specific asset identification number.
- 10b. RPUID.** A unique non-intelligent code to permanently identify an asset.
- 11. CATCODE.** The real property Category Code (CATCODE) that classifies the designed use of the asset (original or current) and corresponds to the largest quantity of the asset that is utilized for a single purpose (RPA Predominant Design Use CATCODE).
- 12. CATCODE Description.** The CATCODE name corresponding with Item 11.

- 13. Construction (CONS) Type Code.** The code represents the Construction Type Code and valid values can be found in the RPIM pick list by the same name. Type of construction will be PERM for Permanent or SEMI for Semi-permanent.
- 14. Sustainability (SUST) Code.** : This code is also known as the Facility Sustainability Code and valid values can be found in the RPIM pick list by the same name. It indicates whether a facility meets the Guiding Principles of Federal Leadership in High Performance and Sustainable Buildings for buildings greater than 10,000 sq. ft. The list of valid values is: 1, 2, 3, 4.
- 15. UMA.** Enter the primary unit of measure accountability for each design use category code for each item of real property.
- 16. Quantity.** Enter the total quantity in the unit of measure accountability specified for each item of real property.
- 17. UMO.** Enter the appropriate unit of measure operational for capacity/configuration as applicable for the design use category code.
- 18. Quantity.** Enter the total quantity of capacity/configuration in the unit of measure operational specified for each item of real property.
- 19. Cost.** Enter the total project cost for each item of real property. UFC 1-300-08 contains additional guidance.
- 20. Fund Source.** Identify the fund source in this column for each real property asset or item. UFC 1-300-08 contains additional guidance.
- 21. Funding Organization.** Identify the organization(s) responsible for funding the acquisition of this real property asset.
- 22. RPA Interest Type Code.** This code represents the RPA Interest Type Code and valid values can be found in the RPIM pick list by the same name. The code is used to identify the type of legal interest that DoD holds in a real property asset.
- 23. Item Remarks.** Remarks pertaining only to the item number identified in Item 9; show cost sharing.
- 24. Statement of Completion.** Typed name, signature, title, and date of signature by the responsible transferring individual or agent. In the case of a real property acceptance action this is signed by the DoD Construction Agent. In the case of a real property transfer this is signed by the Installation Management Organization.
- 25. Accepted By.** Typed name, signature, title, and date of signature by the Installation Management Organization RPAO or accepting official.
- 26. Property Voucher Number.** Next sequential number assigned by the RPAO in voucher register.
- 27. Construction Deficiencies.** List construction deficiencies in project during contractor turnover inspection.
- 28. Project Remarks.** Project level remarks and continuation of blocks.

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Stewardship for the Cleaning of Commercial and Institutional Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

O&M Database; G

Training Plan; G

Training Outline; G

Training Content; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

1.3 OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data for the provided equipment, product, or system, defining the importance of system interactions, troubleshooting, and long-term preventive operation and maintenance. Compile, prepare, and aggregate O&M data to include clarifying and updating the original sequences of operation to as-built conditions. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this section and Section 01 33 00 SUBMITTAL PROCEDURES.

1.3.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.3.2 Package Content

Provide data package content in both English and Japanese, and in accordance with paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES. Comply with the data package requirements specified in the individual technical sections, including the content of the packages and addressing each product, component, and system designated for data package submission, except as follows. Use Data Package 5 for commissioned items without a specified data package requirement in the individual technical sections. Provide a Data Package 5 instead of Data Package 1 or 2, as specified in the individual technical section, for items that are commissioned.

1.3.3 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.3.4 Commissioning Authority Review and Approval

Submit the commissioned systems and equipment submittals to the Commissioning Authority (CxA) to review for completeness and applicability. Obtain validation from the CxA that the systems and equipment provided meet the requirements of the Contract documents and design intent, particularly as they relate to functionality, energy performance, water performance, maintainability, sustainability, system cost, indoor environmental quality, and local environmental impacts. The CxA communicates deficiencies to the Contracting Officer. Submit the O&M manuals to the Contracting Officer upon a successful review of the corrections, and with the CxA recommendation for approval and acceptance of these O&M manuals. This work is in addition to the normal review procedures for O&M data.

1.4 O&M DATABASE

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL FILE FORMAT

Assemble data packages into electronic Operation and Maintenance Manuals. Assemble each manual into a composite electronically indexed file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete electronically linked operation and maintenance directory.

1.5.1 Organization

Bookmark Product and Drawing Information documents using the current version of CSI Masterformat numbering system, and arrange submittals using

the specification sections as a structure. Use CSI Masterformat and UFGS numbers along with descriptive bookmarked titles that explain the content of the information that is being bookmarked.

1.5.2 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Building Number.
- b. Project Title.
- c. Activity and Location.
- d. Construction Contract Number.
- e. Prepared For: (Contracting Agency).
- f. Prepared By: (Name, title, phone number and email address).
- g. Include the disk content on the disk label.
- h. Date.
- i. Virus scanning program used.

1.6 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

The following are a detailed description of the data package items listed in paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES.

1.6.1 Operating Instructions

Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.1.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard.

1.6.1.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.1.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.1.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal

Operating Procedures.

1.6.1.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.1.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.1.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.1.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.1.9 Additional Requirements for HVAC Control Systems

Provide Data Package 5 and the following for control systems:

- a. Narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Full as-built sequence of operations.
- c. Copies of checkout tests and calibrations performed by the Contractor (not Cx tests).
- d. Full points list. Provide a listing of rooms with the following information for each room:
 - (1) Floor.
 - (2) Room number.
 - (3) Room name.
 - (4) Air handler unit ID.
 - (5) Reference drawing number.
 - (6) Air terminal unit tag ID.
 - (7) Heating or cooling valve tag ID.

- (8) Minimum cfm.
- (9) Maximum cfm.
- e. Full print out of all schedules and set points after testing and acceptance of the system.
- f. Full as-built print out of software program.
- g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.2 Preventive Maintenance

Provide the following information for preventive and scheduled maintenance to minimize repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.2.1 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.
- c. A Lubrication Schedule showing service interval frequency.

1.6.2.2 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems shall be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each of each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include

electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.2.3 Cleaning Recommendations

Provide environmentally preferable cleaning recommendations in accordance with ASTM E1971.

1.6.3 Repair

Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.3.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.3.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.3.3 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards.

1.6.3.4 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.3.5 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.3.6 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.4 Real Property Equipment

Provide a list of installed equipment furnished under this Contract. Include all information usually listed on manufacturer's name plate. In the "EQUIPMENT-IN-PLACE LIST" include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Submit the final list 30 days after transfer of the completed facility.

Key the designations to the related area depicted on the Contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA

Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used
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1.6.5 Appendices

Provide information required below and information not specified in the preceding paragraphs but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.6.5.1 Product Submittal Data

Provide a copy of SD-03 Product Data submittals documented with the required approval.

1.6.5.2 Manufacturer's Instructions

Provide a copy of SD-08 Manufacturer's Instructions submittals documented with the required approval.

1.6.5.3 O&M Submittal Data

Provide a copy of SD-10 Operation and Maintenance Data submittals documented with the required approval.

1.6.5.4 Parts Identification

Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that shall cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master parts catalog.

1.6.5.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or Contract documents in order to keep warranties in force. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.5.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.5.7 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.5.8 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.5.9 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5.10 Field Test Reports

Provide a copy of Field Test Reports (SD-06) submittals documented with the required approval.

1.6.5.11 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. The information required in each type of data package follows:

1.7.1 Data Package 1

- a. Safety precautions and hazards.
- b. Cleaning recommendations.
- c. Maintenance and repair procedures.
- d. Warranty information.
- e. Extended warranty information.
- f. Contractor information.
- g. Spare parts and supply list.

1.7.2 Data Package 2

- a. Safety precautions and hazards.
- b. Normal operations.
- c. Environmental conditions.
- d. Lubrication data.
- e. Preventive maintenance plan, schedule, and procedures.
- f. Cleaning recommendations.
- g. Maintenance and repair procedures.
- h. Removal and replacement instructions.
- i. Spare parts and supply list.
- j. Parts identification.
- k. Warranty information.
- l. Extended warranty information.
- m. Contractor information.

1.7.3 Data Package 3

- a. Safety precautions and hazards.
- b. Operator prestart.
- c. Startup, shutdown, and post-shutdown procedures.
- d. Normal operations.
- e. Emergency operations.
- f. Environmental conditions.
- g. Operating log.

- h. Lubrication data.
- i. Preventive maintenance plan, schedule, and procedures.
- j. Cleaning recommendations.
- k. Troubleshooting guides and diagnostic techniques.
- l. Wiring diagrams and control diagrams.
- m. Maintenance and repair procedures.
- n. Removal and replacement instructions.
- o. Spare parts and supply list.
- p. Product submittal data.
- q. O&M submittal data.
- r. Parts identification.
- s. Warranty information.
- t. Extended warranty information.
- u. Testing equipment and special tool information.
- v. Testing and performance data.
- w. Contractor information.
- x. Field test reports.

1.7.4 Data Package 4

- a. Safety precautions and hazards.
- b. Operator prestart.
- c. Startup, shutdown, and post-shutdown procedures.
- d. Normal operations.
- e. Emergency operations.
- f. Operator service requirements.
- g. Environmental conditions.
- h. Operating log.
- i. Lubrication data.
- j. Preventive maintenance plan, schedule, and procedures.
- k. Cleaning recommendations.

- l. Troubleshooting guides and diagnostic techniques.
- m. Wiring diagrams and control diagrams.
- n. Repair procedures.
- o. Removal and replacement instructions.
- p. Spare parts and supply list.
- q. Repair work-hours.
- r. Product submittal data.
- s. O&M submittal data.
- t. Parts identification.
- u. Warranty information.
- v. Extended warranty information.
- w. Personnel training requirements.
- x. Testing equipment and special tool information.
- y. Testing and performance data.
- z. Contractor information.
- aa. Field test reports.

1.7.5 Data Package 5

- a. Safety precautions and hazards.
- b. Operator prestart.
- c. Start-up, shutdown, and post-shutdown procedures.
- d. Normal operations.
- e. Environmental conditions.
- f. Preventive maintenance plan, schedule, and procedures.
- g. Troubleshooting guides and diagnostic techniques.
- h. Wiring and control diagrams.
- i. Maintenance and repair procedures.
- j. Removal and replacement instructions.
- k. Spare parts and supply list.
- l. Product submittal data.
- m. Manufacturer's instructions.

- n. O&M submittal data.
- o. Parts identification.
- p. Testing equipment and special tool information.
- q. Warranty information.
- r. Extended warranty information.
- s. Testing and performance data.
- t. Contractor information.
- u. Field test reports.
- v. Additional requirements for HVAC control systems.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training conducted in English for the systems and equipment specified in the technical specifications. The training must be targeted for the Facilities Management Specialist, building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS. Training must include classroom or field lectures based on the system operating requirements. Notify the Contracting Officer in writing seven (7) calendar days prior of the scheduled instructional services. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Commissioning Authority (CxA) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and CxA. Include within the plan the following elements:

- a. Equipment included in training.
- b. Intended audience.
- c. Location of training.
- d. Dates of training.
- e. Objectives.
- f. Outline of the information to be presented and subjects covered

including description.

- g. Start and finish times and duration of training on each subject.
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts).
- i. Instructor names and instructor qualifications for each subject.
- j. List of texts and other materials to be furnished by the Contractor that are required to support training.
- k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The CxA is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.
- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods

with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the CxA in accordance with Section 01 45 00.00 10 QUALITY CONTROL.

-- End of Section --

SECTION 01 91 00.15 10

TOTAL BUILDING COMMISSIONING
05/16

PART 1 GENERAL

1.1 SUMMARY

Commission the building systems listed herein. Employ the services of an independent Commissioning Firm. The Commissioning Firm must be a 1st tier subcontractor of the General or Prime Contractor and must be financially and corporately independent of all other subcontractors. The Commissioning Firm must employ a Lead Commissioning Specialist for the Contractor (CxG) that coordinates all aspects of the commissioning process. Perform Commissioning in accordance with the requirements of the contract specifications and the standard under which the Commissioning Firm's qualifications are approved. In the case of a conflict, the most stringent requirement will prevail.

The Commissioning Specialist for the Government (CxG) is a Government employee. In general, the CxG coordinates the commissioning activities and reports to the Contracting Officer's Representative/CxG, copying the General or Prime Contractor on all results in accordance with Third Party Certification (TPC) requirements. The CxG's responsibilities, along with all other contractors' commissioning responsibilities are detailed in the specifications. All members work together to fulfill their contracted responsibilities and meet the objectives of the Contract Documents.

1.2 SYSTEMS TO BE COMMISSIONED

Commission the following systems:

Heating, Ventilating, Air Conditioning, and Refrigeration Systems (HVAC)
Building Automation System
Mass Notification System
Utility Monitoring and Control System
Integrated Systems Test

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 180 (2012) Standard Practice for Inspection and Maintenance of Commercial Building HVAC Systems

ASHRAE 202 (2013) Commissioning Process for Buildings and Systems

ASSOCIATED AIR BALANCE COUNCIL (AABC)

ACG Commissioning Guideline (2005) Commissioning Guideline

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB Commissioning Standard (2009) Procedural Standards for Whole
Building Systems Commissioning of New
Construction; 3rd Edition

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

SMACNA 1429 (1994) HVAC Systems Commissioning Manual,
1st Edition

1.4 COMMUNICATION WITH THE GOVERNMENT

The Lead Commissioning Specialist (Cx/C) must submit all plans, schedules, reports, and documentation directly to the Contracting Officer's Representative concurrent with submission to the CQC System Manager. The Lead Commissioning Specialist must have direct communication with the Contracting Officer's Representative regarding all elements of the commissioning process; however, the Government has no direct contract authority with the Lead Commissioning Specialist.

1.5 SEQUENCING AND SCHEDULING

1.5.1 Sequencing

Construction will be phased into 3 phases. Phase 1 and 2, will be the east side of building operations, phase 3 will be the west side of building administration. Equipment testing will be performed per the corresponding phase. Integrated systems testing will be performed after Phase 2.

Complete the following prior to starting Functional Performance Tests of mechanical systems:

- a. All equipment and systems have been completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications.
- b. Performance Verification Tests of the controls systems have been completed and the Performance Verification Test Report has been submitted and approved in accordance with Specifications.
- c. Testing, Adjusting, and Balancing has been completed and the Testing, Adjusting, and Balancing Report, has been submitted and approved in accordance with Specification Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.
- d. The building envelope is enclosed according to contract documents with final construction completed.
- e. The Pre-Functional Checklists have been submitted and approved.
- f. The Certificate of Readiness for mechanical systems has been submitted and approved.

1.5.2 Project Schedule

Include the following tasks in the project schedule required by Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES. Ensure sufficient time is scheduled to accommodate the requirements of this specification section. The order of items listed below is not intended to imply a specified sequence:

- a. Submission and approval of the Commissioning Firm and Commissioning Specialist
- b. Submission and approval of the Testing, Adjusting, and Balancing (TAB) Firm and TAB Specialist specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- c. Submission of the Design Review Report specified herein.
- d. Submission of the Design Review Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.
- e. Submission and approval of the Construction Phase Commissioning Plan
- f. Installation of permanent utilities (gas, water, electric)
- g. Drainage and Vent, Building Sewers, Water Supply Systems and Backflow Prevention Assembly Tests specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE
- h. Factory Acceptance Testing for each of the systems to be commissioned as required by technical specifications
- i. Manufacturer's Equipment Start-Up for each of the systems to be commissioned.
- j. Potable Water System Flushing specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE
- k. Operational Tests of the plumbing system specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.
- l. Potable Water System Disinfection specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE
- m. Submission and approval of the TAB Schematic Drawings, Report Forms, and Procedures specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.
- n. Submission and approval of Duct Air Leakage Test Procedures specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- o. Duct Air Leakage Test Execution specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- p. Submission and approval of the Final Duct Air Leakage Test Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- q. Testing, Adjusting, and Balancing (TAB) Field Work required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC

- r. Submission and approval of the TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- s. TAB Field Acceptance Testing required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- t. Submission and approval of the Start-Up Testing Report.
- u. Submission and approval of the Performance Verification Test Procedures.
- v. Performance Verification Tests.
- w. Performance Verification Test Report.
- x. Pre-Functional Checklist Submittal
- y. Functional Performance Testing for each system to be commissioned
- z. Integrated Systems Tests
- aa. Post-Test Deficiency Correction for each system to be commissioned
- bb. Re-Testing
- cc. Endurance Tests
- dd. Training for each of the systems to be commissioned
- ee. Systems Manual/Computerized Maintenance Management System Manual, Maintenance Plan, and Service Life Plan submission and approval
- ff. Submission and approval of the Commissioning Report
- gg. Seasonal Testing

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

- Commissioning Firm; G
- Lead Commissioning Specialist
- Technical Commissioning Specialists
- Commissioning Firm's Contract

SD-06 Test Reports

- Design Review Report; G

Interim Construction Phase Commissioning Plan; G

Final Construction Phase Commissioning Plan; G

Pre-Functional Checklists; G

Issues Log

Commissioning Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate of Readiness; G

SD-10 Operation and Maintenance Data

Training Plan; G

Training Attendance Rosters

Systems Manual; G

Maintenance and Service Life Plans

SD-11 Closeout Submittals

Construction Phase Commissioning Plan; G

Final Commissioning Report; G

1.7 COMMISSIONING FIRM

Provide a Commissioning Firm that is certified in commissioning by one of the following: the AABC Commissioning Group (ACG); the National Environmental Balancing Bureau (NEBB); the International Certification Board/Testing, Adjusting, and Balancing Bureau (ICB/TABB), the Building Commissioning Association (BCA); the Association of Energy Engineers (AEE). The Commissioning Firm must be certified in all systems to be commissioned to the extent such certifications are available from the certifying body. Describe any lapses in certification or disciplinary action taken by the certifying body against the proposed Commissioning Firm or Lead Commissioning Specialist in detail. Any firm or commissioning professional that has been the subject of disciplinary action by the certifying body within the five years preceding contract award is not eligible to perform any duties related to commissioning.

- a. Submit the Commissioning Firm's certification of qualifications including the name of the firm and certifications no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.
- b. The Commissioning Firm's and Commissioning Specialists' certifications must be maintained for the entire duration of the duties specified herein. If, for any reason, the firm or a specialist loses a certification during this period, immediately notify the Contracting Officer's Representative and submit another Commissioning Firm or Commissioning Specialist for approval. All work specified in this

specification section performed by the Commissioning Firm or associated Commissioning Specialists is invalid if the Commissioning Firm or Commissioning Specialist loses its certification prior to contract completion and must be performed by an approved successor.

- c. The Commissioning Firm must oversee and assist the General or Prime Contractor with the work specified herein. Submit the Commissioning Firm's Contract including the Scope of Work associated with the paragraph POST-CONSTRUCTION SUPPORT no later than 30 calendar days after approval of the Commissioning Firm. Submit four hard copies and an electronic copy.

1.7.1 Lead Commissioning Specialist

The Commissioning Firm must provide a Lead Commissioning Specialist (Cx/C) that has a minimum of five years of commissioning experience, including two projects of similar size and complexity, and that is one of the following: a NEBB qualified Systems Commissioning Administrator (SCA); ACG Certified Commissioning Authority (CxA); ICB/TABB Certified Commissioning Supervisor; BCA Certified Commissioning Professional (CCP); AEE Certified Building Commissioning Professional (CBCP); University of Wisconsin-Madison Qualified Commissioning Process Provider (QCxP); ASHRAE Building Commissioning Professional (BCxP).

- a. Submit the Lead Commissioning Specialist's certification of qualifications including the name of the specialist and firm; certifications; years of experience; and a listing of representative projects of similar size and complexity no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.
- b. The Lead Commissioning Specialists certifications must be maintained for the entire duration of the duties specified herein. If, for any reason, the specialist loses a certification during this period, immediately notify the Contracting Officer's Representative and submit another Lead Commissioning Specialist for approval. All work specified in this specification section to be performed by the Lead Commissioning Specialist is invalid if the Lead Commissioning Specialist loses its certification prior to contract completion and must be performed by an approved successor.
- c. The Lead Commissioning Specialist must lead and oversee the commissioning work specified herein and be the primary point of contact for the Government regarding the commissioning work.

1.7.2 Technical Commissioning Specialists

Technical Commissioning Specialists, employed by the Commissioning Firm or financially and corporately independent subcontractor hired by the Commissioning Firm, and that have the following qualifications, must perform the technical work specified herein associated with each system to be commissioned:

- a. The technical work associated with mechanical systems including Heating, Ventilating, Air Conditioning, and Refrigeration Systems; Building Automation System; Utility Monitoring and Control System; Energy and Water Utility Metering Systems must be performed by a Commissioning Specialist certified by NEBB, ACG, ICB/TABB, or BCA in the commissioning of HVAC systems with five years of experience in the

commissioning of HVAC systems.

- b. Submit the Technical Commissioning Specialist's certification of qualifications including the name of the specialist and firm; certifications; years of experience; and a listing of representative projects of similar size and complexity no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.

1.7.3 Commissioning Standard

Comply with the requirements of the commissioning standard under which the Commissioning Firm and Specialists qualifications are approved. When the firm and specialists are certified by BCA, AEE, ASHRAE, or the University of Wisconsin-Madison, comply with the requirements of one of the acceptable standards unless otherwise stated herein. The acceptable standards are ACG Commissioning Guideline, NEBB Commissioning Standard, SMACNA 1429, or ASHRAE 202. Comply with applicable NETA and NICET testing standards for electrical systems.

- a. Implement all recommendations and suggested practices contained in the Commissioning Standard and electrical test standards.
- b. Use the Commissioning Standard for all aspects of Commissioning, including calibration of instruments.
- c. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the Commissioning Standard, adhere to the manufacturer calibration recommendations.
- d. All quality assurance provisions of the Commissioning Standard such as performance guarantees are part of this contract.
- e. The Commissioning Specialists must develop commissioning procedures for any systems or system components not covered in the Commissioning Standard.
- f. Use any new requirements, recommendations, and procedures published or adopted prior to contract solicitation by the body responsible for the Commissioning Standard.

1.8 ISSUES LOG

The Lead Commissioning Specialist must develop and maintain an Issues Log for tracking and resolution of all deficiencies discovered through commissioning review, inspection, and testing. Include the date of final resolution of issues as confirmed by the Commissioning Specialist. Submit the Issues Log on a monthly basis at a minimum. At any point during construction, any commissioning team member finding deficiencies may communicate those deficiencies in writing to the Commissioning Specialist for inclusion into the Issues Log.

Track construction deficiencies identified in the Issues Log using QCS as specified in Specification Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).

1.9 CERTIFICATE OF READINESS

Prior to scheduling Functional Performance Tests for each system, issue a

Certificate of Readiness for the system certifying that the system is ready for Functional Performance Testing. The Certificate of Readiness must include, for each system to be commissioned, all equipment and system start-up reports; Performance Verification Test Reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; HVAC Controls Start-Up Reports; and Diagnostic Test Reports to the extent applicable to the system. The Contractor; the Lead Commissioning Specialist; the Contractor's Quality Control Representative; the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness. Submit the Certificate of Readiness for each system no later than 14 calendar days prior to Functional Performance Tests of that system. Submit four hard copies and an electronic copy. Do not schedule Functional Performance Tests for a system until the Certificate of Readiness for that system receives approval by the Government.

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

3.1 CONSTRUCTION PHASE

Construction will be phased into 3 phases. Phase 1 and 2, will be the east side of building operations, phase 3 will be the west side of building administration. Equipment testing will be performed per the corresponding phase. Integrated systems testing will be performed after Phase 2.

3.1.1 Construction Commissioning Coordination Meeting

The Lead Commissioning Specialist must lead a Construction Commissioning Coordination Meeting no later than 14 days after approval of the Commissioning Firm and Commissioning Specialists to discuss the commissioning process including contract requirements, lines of communication, roles and responsibilities, schedules, documentation requirements, inspection and test procedures, and logistics as specified in this specification section. The Contractor's Superintendent or Project Manager, the Contractor's Quality Control Representative, and the Government must attend this meeting. Invite the User and a Directorate of Public Works Representative, Reserve Support Command Representative, or a Base Civil Engineer Office Representative to attend this meeting.

3.1.2 Construction Phase Commissioning Plan

3.1.2.1 Interim Construction Phase Commissioning Plan

The Lead Commissioning Specialist (Cx) must prepare the Interim Construction Phase Commissioning Plan. Submit the Interim Construction Phase Commissioning Plan no later than 30 calendar days after the Construction Commissioning Coordination Meeting and no later than 14 days prior to the start of construction of the building envelope. Submit four hard copies and an electronic copy.

Identify the commissioning and testing standards and outline the overall commissioning process, the commissioning schedule, the commissioning team members and responsibilities, lines of communication, documentation requirements for the construction phase of the project in the Interim

Construction Phase Commissioning Plan. A Template Construction Phase Commissioning Plan is attached to this specification and is available in word format upon request. Please note that the Template Construction Phase Commissioning Plan is provided as a supplement and contains additional detailed requirements not specifically addressed within this specification section. The CxC is required to develop a commissioning plan that is comprehensive in scope and conforms to the requirements of the contract documents.

3.1.2.1.1 Checklists

Download example Pre-Functional Checklists, Functional Performance Test Checklists, and Integrated Systems Test Checklists for specification section 01 91 00.15 TOTAL BUILDING COMMISSIONING at the following location: <https://www.wbdg.org/dod/ufgs/ufgs-01-91-00-15>. The checklists submitted in the Interim and Final Construction Phase Commissioning Plans must contain the same level of detail shown in the examples. The submitted checklists are not required to match the format of the examples.

3.1.2.1.2 Contents

In addition to the requirements listed above and those included in the Template Commissioning Plan, include the following in the Interim Construction Phase Commissioning Plan:

- a. Plan purpose
- b. Commissioning scope
- c. Systems to be commissioned
- d. Examples and description of development of pre-functional, integrated systems test, and functional performance test checklists
- e. Building information
- f. Contact information for the Commissioning Specialists, the Contracting Officer's Technical Representative, and the Commissioning Team listed in paragraph Commissioning Team
- g. Roles and responsibilities
- h. Management plan
- i. Owner's Project Requirements
- j. Description of design reviews by the Commissioning Specialists
- k. Description and templates for site observation reports and the issues log
- l. Listing and description of required meetings
- m. Identification and sequence of commissioning and acceptance tasks for incorporation into the Project Schedule
- n. Listing of required submittals to Government and Commissioning Specialists

- o. Description of execution of pre-functional checks, integrated systems tests, and functional performance tests
- p. Description of Endurance Tests
- q. Acceptance testing of critical systems as identified in contract documents
- r. Operation and maintenance manual requirements
- s. Description of training requirements
- t. Description of required Systems Manual
- u. Description of the Commissioning Report

3.1.2.2 Final Construction Phase Commissioning Plan

The Lead Commissioning Specialist (CxS) must prepare the Final Construction Phase Commissioning Plan. Submit the Final Construction Phase Commissioning Plan no later than 30 calendar days prior to the start of Pre-Functional Checks. Submit four hard copies and an electronic copy.

Include the information provided in the Interim Construction Phase Commissioning Plan. In addition, the Technical Commissioning Specialist must develop the Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists for each building, for each system required to be commissioned, and for each component for inclusion in the Final Construction Phase Commissioning Plan.

3.1.2.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. See paragraph Pre-Functional Checks for more information.

3.1.2.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. See paragraph Functional Performance and Integrated Systems Tests for more information. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

- a. Notable system features including information about controls to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements. Recommendation must clearly indicate that the system should or should not be accepted by the Government.
- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions
- d. Attendees

- e. Identification of the equipment involved in the test
- f. Control system feature identification
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station
- h. Actuator operation observations demonstrating actuator responses to commands from the control system
- i. As-found condition of the system operation
- j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- k. Space for comments for each test item.

3.1.2.2.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. See paragraph Functional Performance and Integrated Systems Tests for more information. Include the following sections in the Integrated Systems Test Checklists:

- a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements. Recommendation must clearly indicate that the systems should or should not be accepted by the Government
- c. Test conditions including date and beginning and ending time
- d. Attendees
- e. Identification of the equipment and systems involved in the test
- f. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- g. Space for comments for each test item.

3.1.3 Design Review

The Lead Commissioning Specialist and Technical Commissioning Specialists must review the construction contract plans and specifications.

- a. Advise the Contracting Officer's Representative of any deficiencies that would prevent the building systems and features from operating or performing effectively and from being adequately maintainable.
- b. The Commissioning Specialists must provide a Design Review Report individually listing each deficiency and the corresponding proposed

corrective action necessary for proper system operation or performance. Submit four hard copies and an electronic copy of the report to the Contracting Officer's Representative no later than 30 days after approval of the Commissioning Specialists.

- c. The Lead Commissioning Specialist must participate in a meeting to discuss any items contained in the report no later than 14 calendar days after submission of the report.

3.1.4 Construction Submittals

Provide all submittals associated with the systems to be commissioned, including shop drawings; equipment submittals; test plans, procedures, and reports; and resubmittal's to the Commissioning Specialists. The Technical Commissioning Specialist must review the submittals to the extent necessary to verify that the equipment and system installation will comply with the contract requirements.

3.1.5 Inspection and Testing

Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents and the Owner's Project Requirements. Requirements in related specification sections are independent from the requirements of this section and do not satisfy any of the requirements specified in this specification section. Provide all materials, services, and labor required to perform the Pre-Functional Checks, Integrated Systems Tests, and Functional Performance Tests.

3.1.5.1 Commissioning Team

Provide a commissioning representative for each sub-contractor associated with the systems to be commissioned. Each commissioning representative is responsible for coordination of their respective sub-contractor's execution of the commissioning activities and participation in the inspection and testing required by this specification section. The designers listed below are the designers of record for their respective systems. Substitutes must be approved by the Contracting Officer's Representative. The General or Prime Contractor shall provide a detailed schedule of all Functional Performance and Integrated System Testing activities to be accomplished at least 14 calendar days prior to the start date of testing.

3.1.5.1.1 Mechanical System Pre-Functional Checks Team

The following team members must participate in Pre-Functional checks of mechanical systems:

Designation	Function
CxM	Mechanical System Technical Commissioning Specialist
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel

Designation	Function
MC	Contractor's Mechanical Commissioning Representative
EC	Contractor's Electrical Commissioning Representative
CC	Contractor's Controls Commissioning Representative
TABC	Contractor's TAB Commissioning Representative
PC	Contractor's Plumbing Commissioning Representative
IC	Contractor's Irrigation Commissioning Representative

3.1.5.1.2 Mechanical Systems Test Team

The following team members must participate in Functional Performance, Seasonal, and Integrated Systems Testing of mechanical systems:

Designation	Function
CXM	Mechanical System Technical Commissioning Specialist
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel
MC	Contractor's Mechanical Commissioning Representative
EC	Contractor's Electrical Commissioning Representative
CC	Contractor's Controls Commissioning Representative
TABC	Contractor's TAB Commissioning Representative
PC	Contractor's Plumbing Commissioning Representative
IC	Contractor's Irrigation Commissioning Representative

3.1.5.1.3 Other Pre-Functional and Functional Performance Participants

The following may participate as team members during Pre-Functional Checks and Functional Performance Testing:

Designation	Function
{DPW}	{Directorate of Public Works Representative}
{BCE}	{Base Civil Engineer Office Representative}
{RSC}	{Reserve Support Command Representative}
{PWD}	{Public Works Division Representative}

Designation	Function
User	Using Agent's Representative

3.1.5.2 Pre-Functional Checks

Pre-Functional Checklists from the approved Final Construction Phase Commissioning Plan must be completed by the commissioning team. Complete one Pre-Functional Checklist for each individual item of equipment or for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, and controls. Indicate commissioning team member inspection and acceptance of each Pre-Functional Checklist item by initials. Acceptance of each Pre-Functional Checklist item by each team member indicates that item conforms to the construction contract requirements in their area of responsibility. Technical Commissioning Specialist acceptance of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents and the Owner's Project Requirements. CxC will witness 100 percent of the Pre-Functional Checklist for critical equipment and a minimum of a single sample of each type of non-critical equipment. Submit the completed and initialed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklist items for each system. Submit four hard copies and an electronic copy. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists.

3.1.5.3 Testing, Adjusting, and Balancing (TAB) Report and Field Acceptance Testing

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Specification Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. Resolve all deficiencies prior to TAB Field Acceptance Testing.

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. The CxC witness verification of the Testing, Adjusting, and Balancing shall be a 100 percent sample strategy with any failed results requiring retest. Include a certification by the Mechanical Technical Specialist that no outstanding deficiencies exist in the systems relative to Testing, Adjusting, and Balancing with the final TAB Report submittal.

3.1.5.4 HVAC Controls Test Reports

The Mechanical System Technical Commissioning Specialist must review the Performance Verification Testing Plan, Checklists, and Report. Include a certification by the Mechanical System Technical Commissioning Specialist that the submittals contain no deficiencies or that the submittals do not indicate any deficiencies in the HVAC systems or HVAC control systems with each of these submittals.

3.1.5.5 Tests

3.1.5.5.1 Functional Performance and Integrated Systems Tests

Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests. Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

Integrated System Test (IST) must be accomplished without impacting mission equipment power. "Blackout" test shall only impact the power to the HVAC system. Fire system shunt trip may be used for IST if applicable.

- a. Functional Performance Tests and Integrated Systems Tests must be performed with the Contracting Officer's Quality Assurance Representative present.
- b. Abort Functional Performance Tests or Integrated Systems Tests when any system deficiency prevents the successful completion of the test.
- c. Technical Commissioning Specialists must lead and document all Functional Performance Tests and Integrated Systems Tests for the systems to be commissioned with the Contractor and appropriate sub-contractors performing the Functional Performance Tests and Integrated Systems Tests. The representatives listed in the paragraph Commissioning Team must attend the tests. Abort Functional Performance Tests or Integrated Systems Tests when any required commissioning team member is not present for the test.

3.1.5.5.1.1 Checklist

Use the Functional Performance Test and Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan to guide the Functional Performance Tests and Integrated Systems Tests. Functional Performance Tests must be performed for each item of equipment and each system required to be commissioned and verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, lighting levels, and all other performance requirements comply with construction contract regardless of the specific items listed within the Functional Performance Test and Integrated Systems Test Checklists provided. Testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Integrated Systems Tests must be performed for the interactive operation between systems such as HVAC systems, fire protection systems, back-up electrical supply, energy generation systems, and other systems, and verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation, correct operation of uninterruptible power supplies or energy generators and connected systems, or lighting system operation during power outage or emergency system activation. The order of components and systems to be tested must be determined by the Technical Commissioning Specialists.

3.1.5.5.1.2 Acceptance

Indicate acceptance of each item of equipment and systems tested by signature of each commissioning team member for each Functional Performance Test or Integrated Systems Test Checklist. The Contractor's Quality Control Representative and the Technical Commissioning Specialists must indicate acceptance after the equipment and systems are free of deficiencies.

3.1.5.5.2 HVAC Test Methods

Perform Functional Performance Tests in accordance with the following:

3.1.5.5.2.1 Prior to Testing

Prior to testing operating modes, sequences of operation, interlocks, and safeties, complete control point-to-point observations, test sensor calibrations, and test actuator commands.

3.1.5.5.2.2 Simulating Conditions

Over-writing control input values through the controls system is not acceptable, unless approved by the Contracting Officer's Representative . Identify proposed exceptions in a protocol submitted to the Contracting Officer's Representative for approval. Before simulating conditions, overwriting values (if approved), or changing set-points, calibrate all sensors, transducers and devices. Below are several examples of exceptions that would be considered acceptable:

- a. When varying static pressures inside ductwork can not be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).
- b. Dirty filter pressure drops can be simulated using sheets of cardboard at filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. High entering cooling coil temperatures can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer's Representative , as noted above, for special cases.
- g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 13 degrees C, when the outside air temperature is above 13 degrees C, temporarily change the lockout set point to be minus 18 degrees C above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point such that damage to

the components, systems, or the building structure and/or contents will occur.

- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition in ductwork.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer's Representative .

3.1.5.5.2.3 Setup

Perform each test under conditions that simulate actual conditions as close as is practically possible. Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

3.1.5.5.3 Sample Strategy

Perform Functional Performance Tests and Integrated System Tests for all equipment and systems. Prepare and complete a Functional Performance Test checklist for each piece of equipment or system. Prepare and complete an Integrated Systems Test Checklist for all systems and equipment having interactive operation. Test 100 percent of all HVAC Central Plant equipment and primary air handling units. Test 100 percent for all other HVAC equipment and systems.

3.1.5.5.4 Seasonal Tests

3.1.5.5.4.1 Initial Functional Performance Tests

Perform Initial Functional Performance Tests as soon as all contract work is completed, regardless of the season. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads.

3.1.5.5.4.2 Full-Load Conditions

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems under full-load conditions during peak heating and cooling seasons during outdoor air condition design extremes. Test cooling equipment and systems with the building fully occupied when performing the Functional Performance Tests during peak cooling season.

Schedule Seasonal Functional Performance Tests in coordination with the Government.

3.1.5.5.4.3 System Acceptance

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract that can be tested during initial

Functional Performance Tests. All Functional Performance Test procedures must be completed prior to full systems acceptance.

3.1.5.5.5 Aborted Tests and Re-Testing

Abort Functional Performance Tests, Integrated Systems Tests, or Seasonal Tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. reimburse the Government for all costs associated with effort lost due to re-testing due to test failures and aborted tests. These costs must include salary, travel costs, and per diem for Government commissioning team members. Re-test only after all deficiencies identified during the original tests have been corrected.

3.1.5.5.5.1 100 Percent Sample

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.1.5.5.5.2 Less than 100 Percent Sample

No equipment will have a sample rate of less than 100 percent.

3.1.6 Training Plan

Develop a training plan which identifies all training required by specification sections associated with commissioned systems. Include a matrix listing each training requirement, content of the training, the trainer name, trainer contact information, and schedule and location of training. Also include a template attendance roster and attendee training evaluation form. Submit four hard copies and an electronic copy of the Training Plan to the Commissioning Specialists and the Government no later than 30 calendar days prior to the associated training.

Provide two separate training levels, one for User Operators and the other for Operation and Maintenance staff. User Operator training schedule should accommodate attendance of 3 different shifts schedules by providing the same training over the course of 4 days maximum, between the hours of 8am to 5pm. Operation and Maintenance training schedule to provide a single training session per specific equipment and or system that is separate from User Operator training, between the hours of 8am to 5pm.

Document training attendance using training attendance rosters and attendee training evaluation forms, provide completed attendance rosters and attendee completed training evaluation forms to the Commissioning Specialists and the Government no later than 7 calendar days following the completion of training for each system to be commissioned. Submit four hard copies and an electronic copy.

3.1.7 Systems Manual

Prepare and submit a Systems Manual including, for all commissioned systems, the Owner's Project Requirements, system single line diagrams, as-built sequences of operation and controls drawings, as-built control

setpoints (provide a basic version for User Operator and a full version for Operation and Maintenance staff), recommended schedule for sensor and actuator calibration, recommended schedule of maintenance when not in the O&M manuals, recommended re-testing schedule with proposed testing forms, and full equipment warranty information. Update and resubmit the Systems Manual based on any corrective action taken during the warranty period. Include a signed certification or letter from the Lead Commissioning Specialist with the submittal stating that the Systems Manual is complete, clear, and accurate.

Submit Systems Manual no later than 30 calendar days following completion of Functional Performance Tests. Submit four hard copies and an electronic copy.

3.1.8 Maintenance and Service Life Plans

3.1.8.1 Maintenance Plan

Prepare and submit a Maintenance Plan for the project mechanical systems. Prepare the HVAC and refrigeration sections of the Maintenance Plan in accordance with ASHRAE 180. Develop required inspection and maintenance tasks similar to Section 5 of ASHRAE 180 for the other commissioned systems and fire protection systems.

Submit the Maintenance Plan no later than 30 calendar days following the completion of Functional Performance tests. Submit four hard copies and an electronic copy.

3.1.8.2 Service Life Plan

Prepare and submit a Service Life Plan for the building envelope, structural systems, and site hardscape that includes the following for each assembly or component:

- a. A description of each including the materials or products.
- b. The estimated service life, in years.
- c. The estimated maintenance frequency and description of maintenance tasks.
- d. The point of maintenance access for the components with estimated service life less than service life of the building.

Submit the Service Life Plan no later than 30 calendar days following the completion of Functional Performance tests. Submit four hard copies and an electronic copy.

3.2 COMMISSIONING REPORT

Following the completion of Functional Performance Tests and Integrated Systems Tests, with the exception of Seasonal Tests, the Lead Commissioning Specialist must prepare a Commissioning Report.

- a. Include an executive summary describing the overall commissioning process, the results of the commissioning process, any outstanding deficiencies and recommended resolutions, and any seasonal testing that must be scheduled for a later date. Indicate, in the executive summary, whether the systems meet the requirements of the construction

contract and the Owner's Project Requirements.

- b. Detail any deficiencies discovered during the commissioning process and the corrective actions taken in the report. Include the completed Pre-Functional Checklists, Functional Performance Test Checklists, Integrated Systems Test Checklists, the Commissioning Plans, the Issues Log, control sequences, meeting minutes, progress and site visit reports, Performance Verification Test Reports, Training Attendance Rosters, the Design Review Report, the final TAB Report.
- c. Submit the Commissioning Report no later than 14 calendar days following commissioning team acceptance of all Functional Performance Tests and Integrated Systems Tests with the exception of Seasonal Tests. Submit four hard copies and an electronic copy.
- d. Following any Seasonal Tests or Post-Construction Activities, update the Final Commissioning Report to reflect any changes and resubmit.

3.3 POST-CONSTRUCTION SUPPORT

3.3.1 Post-Construction Endurance Test

Perform a one-week Endurance Test once during the peak cooling season during outdoor air condition extremes using the building control system to trend all points necessary to evaluate the installed equipment and system performance or as shown as requiring a trend on the project schedules. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost. Poll all points shown in the project schedules with an alarm condition at 5 minute intervals. Poll all points shown in the project schedules required for trending, overrides, or graphical displays at 15 minute intervals.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract and the Owner's Project Requirements. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit four hard copies and one electronic copy of the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist.

3.3.2 Post-Construction Site Visit

The Commissioning Specialists must visit the building site concurrent with the 9 month warranty inspection to inspect building system equipment and review building operation with the building operating/maintenance staff. The Commissioning Specialists must identify any deficiency of the building systems to operate in accordance with the contract requirements and the Owner's Project Requirements. The Commissioning Specialists must advise the Contracting Officer's Representative of any identified deficiencies and the proposed corrective action. Submit an updated commissioning report and systems manual documenting the results of the post-construction inspection.

3.4 ATTACHMENTS

Appendix A - Owner's Project Requirments
Appendix B - Template Commissioning Plan

-- End of Section --

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Owner's Project Requirements (OPR)

FY25 MILCON LXEZ1076699

**Repair HVAC Wideband SATCOM OPS B100,
Camp Buckner Communication Annex,
Okinawa, Japan**

Prepared By Jacobs

Jacobs

31 July 2025

Document history and status

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Approvals

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1. Project Description

Provide all plant, labor, and materials to repair by replacement the existing HVAC system. This may include new packaged, water chiller unit with concrete block enclosure, Air Handling Units (AHUs), Variable Air Volume (VAV) terminal units, chilled water system, replacement of the building exhaust system, Direct Digital Control (DDC) system and mass notification. Modify the existing fire sprinkler and fire alarm systems affected by mechanical and architectural work if required.

The design must prioritize the seamless continuity of operation within the occupied computer server room throughout the duration of construction activities. The design must encompass work sequencing , ensuring minimal disruption to the functionality of the servers and systems in B100.

Project Name: FY25 Repair HVAC Wideband SATCOM Ops B100

- Location: Camp Foster, Buckner Communication Annex, Japan
- Project PA: \$ [REDACTED]
- Projected Beneficial Occupancy (BoD) Date: 2028
- Operational Schedule: 24x7
- Space Type Usage: Building 100 houses the 5th Detachment of 53SOPS. The equipment floor contains both highly fidelity spectrum monitoring sensors and telemetry monitoring satellite command systems.
- Building 100 is a one-story building for administration, communication, and operation rooms.

CATCODE	Item	U/M	Quantity
Type II-B / Type II (0,0,0), noncombustible / Unrated construction	Buckner Communication Annex B100	SM	19K

2. Quality Requirements of Materials and Construction

2.1. Design Criteria and References

The following criteria apply to the environmental design for the project.

- Department of Defense Japan Environmental Governing Standards (JEGS) (2024)
- Engineer Manual (EM) 385-1-1, U.S. Army Corps of Engineers, Safety and Health Requirements Manual (2024)
- EPA 40 CFR 61, Subpart M, National Emission Standard for Asbestos
- EPA 40 CFR 261 Identification and Listing of Hazardous Waste
- EPA 40 CFR 273 Standards for Universal Waste Management
- Government of Japan, Japan Waste Management and Public Cleansing Law of 1970, updated 2001.

- Government of Japan (GOJ) Regulations Pertaining to Asbestos
 - Ambient Air Pollution Control Law
 - Asbestos Impediment Prevention Regulations
 - Building Standards Law
 - Industrial Safety and Health Law
 - Waste Cleaning and Disposal Law
- OSHA 29 CFR 1910.1000 Air Contaminants
- OSHA 29 CFR 1910.1001 Asbestos
- OSHA 29 CFR 1910.1200 Hazard Communication
- OSHA 29 CFR 1926.1101 Asbestos
- OSHA 29 CFR 1926.62 Lead
- USACE Japan Edited Specifications (JES) 1.6

All phasing for this project needs to be outlined in the construction documents. Consideration of a 24/7/365 use facility with visual construction monitoring shall be taken into consideration. Further development of construction phasing will be provided by the Government.

Construct a concrete enclosure around the chiller yard to include overhead protection from the elements.

Commissioning of the systems is expected to meet requirements of UFC 1- 200-02. The following list includes systems to be commissioned. Additional details will be described in a Commissioning Plan.

Systems to be Commissioned

- **HVAC system**
 - Chilled water system
 - AHUs system
 - VAVs system
 - CRAC units
 - Building Exhaust system
 - Direct Digital Controls System (DDC)
- **Special Systems**
 - Mass Notification
 - BAS System

Commissioning process shall also include:

- Integrated System Test
- HVAC control points trending within 10 months of occupancy
- Review of training requirements, plans, training evaluation forms, etc.

3. Energy Efficiency and Sustainability Goals

Sustainable strategies and features should be integrated into the design to minimize nonrenewable energy consumption; conserve resources; minimize adverse effects to the environment; and improve occupant productivity, health, and comfort. Sustainable design should reduce the total cost of ownership of the project using a whole building, life-cycle approach. The design and construction should incorporate High Performance and Sustainable Building (HPSB) requirements as stated in the UFC 1-200-02 (Dec 2020). The DoD has designated that all new projects shall register with US Green Building Council (USGBC) or Green Building Initiative (GBI) for DoD specific Guiding Principles Compliance (GPC) certification, regardless of funding source. The project is greater than 10,000 square feet and the construction value is greater than \$3 million, but since it is less than 50% of the replacement cost of a new facility of the same category code GPC/GPA certification registration of the project is not required. Depending on the availability of products and processes in Japan, the incorporation of HPSB requirements should be revalidated during the design phase.

4. Systems Integration Requirements

Specifications shall address the technical requirements for design and installation of the following categories. Systems are to comply with location requirements.

- Mechanical Systems
- Metering – Base Wide Energy Management Control System
- Special Systems – Mass Notification System

5. Cybersecurity Requirements

Facility-Related Control Systems (FRCS) shall be designed in accordance with the following policies, standards, and procedures:

- Advanced Cyber Industrial Control System Tactics, Techniques and Procedures (ACI TTP) for Department of Defense (DOD) Industrial Control Systems (ICS) Revision 2 (March 2018)
- Committee on National Security Systems Instructions (CNSSI) 1253, Security Categorization and Control Selection for National Security Systems (July 2022)
- Department of Defense Directive (DoDD) 8140.01 Cyberspace Workforce Management, with Change 1 (31 October 2020)
- Department of Defense Instruction (DoDI) 8500.01, Cybersecurity, with Change 1 (October 2019)
- Department of Defense Instruction (DoDI) 8510.01, Risk Management Framework (RMF) for DoD Systems (July 2022)
- Department of Defense Instruction (DoDI) 8530.01 Cybersecurity Activities Support for DoD Information Network Operations (July 2017)
- Department of Defense (DoD) Handbook for Self-Assessing Security Vulnerabilities & Risks of Industrial Control Systems on DoD Installations (December 2012)
- Federal Information Processing Standard (FIPS) 200 Minimum Security Requirements for Federal Information and Information Systems (March 2006)
- Federal Information Processing Standard (FIPS) 201-3 Personal Identity Verification (PIV) of Federal Employees and Contractors (January 2022)

- National Institute of Standards and Technology (NIST) Special Publication (SP) 800-37, Revision 2, Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy (December 2018)
- National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53 Revision 4, Security and Privacy Controls for Federal Information Systems and Organizations (January 2015)
- National Institute of Standards and Technology (NIST) Special Publication (SP) 800-82 Revision 2, Guide to Industrial Control Systems (ICS) Security (May 2015)
- National Institute of Standards and Technology (NIST) Special Publication (SP) 800-115 Technical Guide to Information Security Testing and Assessment (September 2008)
- UFC 1-200-01 DOD Building Code, with Change 4 (2024)
- UFC 1-200-02, High Performance and Sustainable Building (HPSB) Requirements, with Change 2 (2022)
- UFC 3-410-02, Direct Digital Control for HVAC, and Other Building Control Systems, with Change 2 (2021)
- UFC 3-470-01, Utility Monitoring and Control System (UMCS) Front End and Integration (2018)
- UFC 4-010-06, Cybersecurity of Facility-Related Control Systems (FRCS) (2023)
- Unified Facilities Guide Specifications (UFGS), Division 25, Integrated Automation, Cybersecurity for Facility-Related Control Systems, August 2024
- USACE Japan District Design Guide, V.8 (2024)
- USACE Japan Edited Specifications (JES) 1.6

6. Health, Hygiene, and Indoor Environment Requirements

Creation of good indoor environmental quality requires the coordination of many design parameters and construction activities, including acoustical quality, ventilation rates, materials used to construct the facility, installation sequence, location of makeup air intakes, external and internal pollutant generation, humidity, temperature, and other parameters that may affect occupant comfort. The following are to be specified to comply with project requirements and standards:

- VOC Content in caulks, paints, adhesives, sealants, and cleaning products
- Smoking
- Procedures during construction
- How building materials should be stored in a weather-tight, clean area prior to unpacking for installation. Accumulation of water during construction should be avoided and any porous construction materials such as insulation should be protected from moisture.
- Dust in the construction area
- Finishes that emit volatile organic compounds and other pollutants during curing
- The facility shall be positively pressurized.
- Outside Air Intakes

7. Acoustical Requirements

No external noise sources have been identified. The internal noise sources consisting of normal maintenance and administrative activities are typical throughout the facility. The facility is primarily constructed of cast-in-place concrete providing sufficient acoustical separation between Administration, Operations, and Utility spaces. However, acoustical separation does not exist for individual rooms within the Operations space as partitions/walls are constructed on top of RAF and do not extend to structure above. Secure perimeter wall penetrations will meet minimum adjacent wall STC ratings requirements.

8. Vibration & Seismic Requirements

The existing B-100 Operations Building and the adjoining Company Headquarters Building are both one story concrete buildings. They were designed according to the applicable building code requirements at the time of their construction. A seismic evaluation of both buildings according to ASCE 41-17 were performed in 2021 and found no structural deficiencies with the buildings. Existing CRAC units in the Operations Building are supported on a raised floor system. The new CRAC units will also be supported on the raised floor system. There are no known vibration concerns due to the operation of the CRAC units by the design team.

9. General Building Requirements

SATCOM Ops B100 is an approximate 1,685 square meter (18,132 square feet) one-story cast-in-place concrete communications building originally constructed in 1991 with concrete structure and roof. This project provides facility repairs and equipment upgrades necessary to provide communications to support the Space Force mission at Camp Buckner and will remain fully operational for 24/7/365 shifts.

Materials should be selected for ease and speed of erection and resistance to fire, winds, and tropical conditions. Any field applied paints should be flexible enough to expand and contract with the substrate. The possible need for paint should be avoided by using materials that are in their natural color, are prefinished or are integrally colored.

10. Equipment and System Expectations

10.1. HVAC Requirements

The design objective is to provide a building HVAC system that replaces the existing systems as indicated in the SOW. A building life cycle analysis was performed for the replaced systems.

HVAC systems shall be designed to contribute to overall building energy goals as defined in this document. The HVAC systems will have low life cycle cost and will be capable of providing increased indoor environmental quality to facilitate occupant's productivity while minimizing maintenance requirements.

The HVAC systems should comply with International Mechanical Code, ASHRAE 90.1, UFC 1-200-02, and UFC 3-410-01.

Multiple user requirements are impacting the HVAC system selection, operation, construction phasing and layout for the replaced systems. The mission is required to operate 24/7/365 with any potential of downtime during construction being scheduled and approved in advance. The overall system capacity should remain the same with an anticipated 15% growth over the next few years. This growth is due to a projected increase in servers and additional electrical gains within the spaces. The facility is located within its own secured perimeter fence creating a limited footprint for different equipment. Mission requires 100% redundancy in the main mechanical equipment with an N+2 redundancy in the server rooms. The climate is hot and humid resulting in a highly corrosive environment. HVAC

equipment exposed to the elements must be constructed with coatings and anti-corrosive materials.

10.1.1. Design Conditions (UFC 3-400-02)

The following presents the design conditions for the project per UFC 3-400-02, Design: Engineering Weather Data, with Change 2. Kadena AFB is the closest weather station listed in ASHRAE

- Location: Kadena AFB, Okinawa Weather Station 479310
- Latitude: 26.35 degrees north
- Longitude: 127.77 degrees west
- Elevation: 48 meters
- Heating Degree Days: 216 HDD18.3
- Cooling Degree Days: 4707 CDD10.0
- Climate Zone: 2A
- Design Temperatures:
 - Outside Design (cooling 1.0 percent): 32.2 degrees Celsius (°C) dry bulb (db)/26.7°C mean coincident wet bulb (mcwb)
 - Outside Design (heating 99.0 percent): 10.8°C db/8.3°C
 - Outside Design for OAHU (humidity ratio 0.4 percent): 24.5 g/kg/29.6°C mean coincident dry bulb (mcdb)
 - Administration: 25.5°C db/12.8°C dewpoint
 - Communications Room: 22.2°C db/50 percent relative humidity (RH)
 - Operations Rooms: 22.2°C db/50 percent RH
 - Battery/UPS Storage: 20°C db/50 percent RH

Mechanical systems will be provided in accordance with the Supplemental Statement of Work dated 09 July 2024 to repair by replacement. As this contract approaches fruition- approximately 5 years, the existing equipment will reach the end of their published life span. One of the two primary chillers that provides redundancy to the building has already failed. There is no current redundancy in the main equipment providing chilled water to the building.

The systems being replaced include the chillers, condensing units, chilled water pumps, ductless split systems, computer room air conditioners, air handling units, split systems, heat recovery unit, ceiling cassette split system, and the dedicated outside air unit. All refrigerant lines associated with the equipment will be replaced except for the lines routed over the hard ceiling in the restroom of the Administration portion of Building 100. Those will be capped and abandoned in place over the inaccessible ceiling. All other locations will be removed in its entirety.

Two dedicated outside air units (DOAS) will be provide in accordance with UFC 3-410-01, Section 3-2 to supply treated outside air. Typically, the design would utilize one unit. This facility requires separation between the two functions of the building for security purposes. As such, one will be provided for the Operations portion of the building and one for the Administration portion.

Outside air will be provided and calculated per UFC 3-410-01, UFC 4-010-01 and in accordance with ASHRAE 62.1. Existing outside air intakes that are below 3 meters will not be relocated. The project does not meet the upgrade requirements when the intakes are not being revised. This is in accordance with UFC 4-010-01, 1-6.2.1.

Emergency Air Distribution Shutoff: The facility already has emergency shutoff switches that deactivates ventilation equipment and closes low leakage motorized dampers to eliminate the flow of airborne contaminants within the building and between the building and outdoors per UFC 4-010-01.

Commissioning per UFGS 01 91 00.15 Building Commissioning will be provided. Seismic and AT bracing for all mechanical equipment and piping will be provided as required.

Corrosion protection is a considerable portion of this project. All HVAC components serving unconditioned air or located in unconditioned and/or exterior locations will be fabricated from stainless steel. All exhaust components will be aluminum or stainless steel. Components located within conditioned spaces and handling conditioned air will be galvanized steel. Internal duct liner is not allowed in accordance with UFC 3-410-01. Air cooled condensing units 17.6 Kw and below will be provided copper tube with copper fin condenser coils tested with a minimum 10,000-hour salt spray test as recommended by the Marianas Navy and Marine Corps Design and Construction Standards or better for the condenser coil, cabinet casings, chassis, and internal components. Spray-on application for corrosion protection coating is acceptable for this size equipment.

Air-cooled condensers dedicated outdoor air systems (DOAS), and air-cooled chillers above 17.6 kW will be provided copper tube with copper fin condenser coils tested with a minimum 10,000-hour salt spray test as recommended by the Marianas Navy and Marine Corps Design and Construction Standards. Cabinet casings, chassis, and internal components will be provided with a 10,000-hour ASTM B117 standard spray-on application by a certified applicator.

Building and Related Mechanical System Commissioning, Testing, Adjusting, and Balancing of Mechanical Systems.

The building is required to be commissioned. The mechanical systems commissioning procedures are provided in UFC 1-200-02, IgCC, and JES 01 91 00.15 10.

10.1.2. Operation Portion of Building 100

The two existing 138-ton Carrier, screw liquid chillers with associated condensers currently provide chilled water to the building. They are configured for a 2N redundancy where one matched set provides 100% of the cooling to the building. One of the systems is not operational and therefore, no redundancy is currently provided. Each matched set will be replaced whereas the non-operational pair will be first. A new typhoon rated enclosure will be constructed around the condensers in accordance with the SSOW. Each condenser will meet the corrosion resistant recommendations provided in the Marianas Navy and Marine Corps Design and Construction Standards.

The building contains two base mounted chilled water pumps, of which one is significantly corroded, and the other was repainted whereas the condition is unknown. Both pumps will be replaced along with the entire chilled water piping. A new chilled water distribution loop will be provided in the new design to provide additional chilled water redundancy for the CRACs.

Phasing will be required to maintain the 24/7 operation of the spaces. The intent is to remove the chiller and condenser that is not operational and replace it with a similar system type. That includes a chilled water pump, chilled water piping, chemical treatment, and controls. Once installed, CRAC units can be installed which tie into the new system. The remaining system can be replaced once the system is fully operational.

The Vault will be constructed around the security penetration requirements of ICD/ICS 705. All ductwork penetrations will contain non-conductive breaks, security bars and wave guides where required.

The Marianas Navy and Marine Corps Design and Construction Standards recommendations include corrosion resistant materials and/or a 10,000-hour salt spray rated coating on all materials exposed to the ambient conditions. The most corrosion-resistant metal materials for building components and assemblies shall be used and shall include but not be limited to (in descending order of preference) anodized aluminum, stainless steel, and hot-dipped

galvanized steel. Stainless steel shall be used where aluminum does not provide the required performance characteristics. These include but are not limited to fasteners, anchors, miscellaneous metal fabrications, and other exposed building components and assemblies. Specify 18-8, austenitic stainless-steel type 316. If special order cannot be obtained for type 316 stainless steel, the next best corrosive-resistant stainless steel (e.g. 18-8 austenitic stainless-steel type 304) shall be specified. If the stainless-steel components will be welded, specify the low-carbon content type, usually designated with the letter "L" following the number, e.g. type 316L. Unless otherwise noted in this document, hot-dipped galvanized shall only be used at interior conditioned spaces where aluminum or steel coated per UFGS 09 90 00 Paints and Coatings do not provide the required strength and corrosion performance characteristics. Hot dipped galvanized steel is preferred over electroplating zinc metal spraying or other methods of providing zinc coating to ferrous metals. Exterior chain link fencing shall be hot-dipped galvanized steel. UV-resistant vinyl coating of fence shall be considered as an added measure.

Material selection is key to success in hot humid design. Exterior material coatings and surfaces must resist mold/moisture penetration and be self-cleaning through rainfall. Potential exterior and interior corrosion need to be addressed. Reinforced concrete is the structural material to be used unless otherwise indicated. Exterior metal framing and connections shall use selected stainless steel (or equal) corrosion protection. Exterior fasteners shall be stainless steel. Within interior conditioned spaces exposed steel elements at structural connections must be hot-dipped galvanized steel. Other interior metal surfaces also must address corrosion.

The CRAC units will be replaced per the heat loads provided by the Government. If none are provided, the unit capacity will meet the existing capacities of 18 tons each.

All mechanical equipment that is exposed to the exterior shall be ground mounted on reinforced concrete pads, surrounded by solid 6 inch (153 millimeter) reinforced concrete walls, or 8 inch (204 millimeter) reinforced fully grouted CMU walls with gate(s) that protects mechanical equipment from typhoon flying debris.

The chiller manufacturer will be determined from the Installation requirements. The determination has not been provided at the time of this report.

Chilled water will be supplied at 5.5°C and returned at 12.2°C. Chillers will be each sized for 100 percent of the building load to provide a 2N redundancy.

Heating will be provided by electricity in accordance with Executive Order 14057, Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability.

The office areas, shop spaces, utility rooms, restrooms, shall be provided with outside air as required by UFC 3-410-01, Section 3-2 and ASHRAE 62.1. The outside air will be provided by a single DOAS that will meet the flow rate calculated during the design process. Existing ductwork shall be reused as much as possible and shall be connected to new air devices sized for the calculated flowrates.

The Battery/UPS room shall be analyzed for battery storage and the required ventilation requirements. Outside air intakes shall remain unless the system changes.

A new DDC EMCS shall be provided with a workstation. The location will be determined during the design process. The building EMCS will not connect to anything outside of the building due to the security requirements of the user. All controls will adhere to UFC 4-101-06.

10.1.3. Administration Portion of Building 100

The building contains the following equipment to be replaced per the 2013 As-built Drawings:

- Split system heat pump- 12.5 tons
- Packaged AHU with remote condenser- 10 tons
- Multi-split heat pump- 14 KW cooling/ 16 KW heating
- Ceiling mounted cassette ductless- 1 ton
- Heat recovery make-up air unit- 16 MBH cooling/19 MBH heating
- Ductless split- 10 MBH cooling

This equipment will be replaced in accordance with the SSOW. The new equipment and layout will meet the requirements of UFC 3-410-01, UFC 4-010-01 and in accordance with ASHRAE 62.1. Each condenser will meet the corrosion resistant recommendations provided in the Marianas Navy and Marine Corps Design and Construction Standards.

The Conference Room ceiling will be removed in its entirety to replace the ceiling cassette and remove the heat recovery unit. A DOAS unit will replace the heat recovery unit and will distribute the outside air to the air handling unit per UFC 3-410-01, 3-2.

Heating will be provided by electricity in accordance with Executive Order 14057, Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability.

A new DDC EMCS will be provided with a workstation. The location will be determined during the design process. The building EMCS will not connect to anything outside of the building due to the security requirements of the user. All controls will adhere to UFC 4-101-06.

10.1.4. Mechanical Ventilation and Special Exhaust

Ventilation requirements for the project were obtained from ASHRAE 62.1. The building will be maintained at a slightly positive pressure (0.05-inch w.g.) to the exterior. Restrooms and janitor's closet are required to be exhausted to maintain negative pressurization regarding the surrounding areas. Filtration will be provided at a minimum MERV 8 prefilter and MERV 13 final filter for all occupied areas to improve indoor air quality.

All restrooms and janitorial closets will be mechanically exhausted by a central, continuously operating system interlocked with the associated air handling unit. Janitor's closet will be exhausted at 1 cfm/sf. These areas will be maintained at a negative pressure relative the adjacent spaces.

10.2. Plumbing Requirements

The available site utilities are domestic water and electricity. All metering is existing and will not be modified in this project. All plumbing drain lines will be reconfigured to meet the design and layout of the new systems during the design process in accordance with the above criteria. There is no anticipation to replace any other plumbing piping or fixtures.

10.3. Energy Management and Control System Requirements

A building energy management and control system (EMCS) should be provided for the facility. The EMCS should utilize BACNet protocol.

10.4. Life Safety System Requirements

This project should be designed and constructed in conformance with UFC 3-600-01 and UFC 1-200-01. UFC 3-600-01 and UFC 1-200-01 reference International Building Code for type of construction, allowable height and area limitations, and building separation requirements. UFC 3-600-01 references NFPA 101, for building construction related to egress and safety to life. In conflicts between the IBC and NFPA 101 related to fire-resistance ratings, UFC 3-600-01 requires conformance to NFPA 101, and applicable criteria contained in the UFC.

10.5. Electrical System Requirements

The service for B100 is rated at 480Y/277 volts, 3-phase, 4-wire, 60Hz with 208Y/120 volts, 3-phase, 4-wire, 60Hz also available for lighting and house loads.

The loads for the renovation of this facility consist of HVAC equipment, maintenance receptacles, and an overhead door motor. All HVAC equipment replacements are to be like-for-like, and no significant increase in loading of the existing HVAC panels, Motor Control Center (MCC), or upstream transformer is anticipated.

The MCC located in Mechanical Room 120 and the panels the MCC serves will supply the mechanical loads for the original portion of the building. Specifically, the MCC fed AC panel in WSOC Equipment Room 108 will provide power to all CRAC units serving the WSOC Operations and Equipment areas, while any additional large HVAC loads, such as DOAs, AHUs, and chillers, will be powered directly from the MCC. Smaller HVAC loads, the potential overhead door motor, and power to exterior maintenance receptacles for the original portion of the building will be provided by a 208Y/120-volt panel integral to the MCC.

10.5.1. Exterior Systems

The electrical system design for these facilities should be in accordance with host nation codes and standards, including the Japanese Industrial Standards (JIS), the Japan Electrical Manufacturers' Association (JEMA), the Japan Electrical Construction Association (JECA), and the Energy Conservation Law of Japan (ECL). The design should also be in accordance with UFC 3-501-01, UFC 3-520-01, UFC 3-530-01, UFC 3-550-01 and NFPA as applicable. The hangar design should meet the requirements of UFC 4-211-01 and UFC 4-211-02. Additionally, this design should meet the minimum requirements of UFC 1-202-01.

B100 is fed a secondary voltage of 480Y/277 volts, 60Hz from Switchgear A and B located in B104. The incoming feeder is not expected to be affected by renovation of the HVAC system.

10.5.2. Interior Systems

Panel P4 and P2-1 located in Electrical Room 131 will supply mechanical loads for the administrative portion of the building. Panel D located in Electrical Room 131 will supply power to exterior maintenance receptacles for the administrative portion of the facility.

The UPS panel and associated cable ladders located on the western wall of Electrical Room 122 will be removed, rehosted, and rewired off the wall to create working space for the introduction of a Fiber-glass Reinforced Plastic (FRP) supplement to the deficit wall. Out of use RF filters located on the western wall will must be removed and

retained for future use by the building owner.

VFDs will must have harmonic mitigation where required for conformance to IEEE 519 Tables 10.1 and 10.3. If VFDs are networked together, cybersecurity measures will be implemented per paragraph section CYBERSECURITY.

The design shall be based on Japanese EM or CET insulation wiring and types E, G, and VP conduit. All conductors must be copper. Flexible metal conduit will must be allowed for lighting whips and motor connections with a maximum length of 6 feet.

Maintenance receptacles must be provided as required by UFC 3-520-01 to service existing and new HVAC equipment. Electrical equipment, devices and wiring methods must be compatible with host nation codes and standards.

All components of the interior and exterior distribution systems must be bonded per host nation codes and the National Electrical Code (NEC).

Where mechanical work disturbs ceilings in the administrative portion of B100, remove, reinstall, and rewire existing fluorescent fixtures per JDDG section 14.5.2.

11. Building Occupant and O&M Personnel Requirements

Training for the O&M staff and Occupant personnel shall provide two separate training levels, one for Occupant User Operators and the other for Operation and Maintenance staff.

11.1. O&M staff training and orientation for building systems

Level of training and orientation is based on equipment type and the complexity of the system, as specified in the contract specifications. At a minimum, training for the O&M staff and personnel shall be provided with a minimum of two (2) total days of combined instruction and field visits to familiarize themselves with the systems and its operation. The Operation and Maintenance training schedule shall provide a single training session per specific equipment and or system that is sperate from User Operator training, between the hours of 8am to 5pm.

11.2. Occupant training and orientation for building systems

Level of training and orientation is based on equipment type and the complexity of the system, as specified in the contract specifications. At a minimum, training for the building occupants shall be provided with a minimum of two (2) total days of combined instruction and field visits to familiarize themselves with the systems and its operation. The Occupant User Operator training schedule should accommodate attendance of 3 different shifts schedules by providing the same training over the course of 4 days maximum, between the hours of 8am to 5pm.

11.3. Commissioning Procedures after Warranty Period

The commissioning process activities continue through the end of the contractual warranty period. During the occupancy phase, the commissioning process is completed and documented, equipment trending data review and a 10-month warranty review is conducted.

12. Applicable References, Codes and Standards

Identify all references, codes that were used to develop the OPR.

- Department of Defense Japan Environmental Governing Standards (JEGS) (2024)
- Engineer Manual (EM) 385-1-1, U.S. Army Corps of Engineers, Safety and Health Requirements Manual (2024)
- EPA 40 CFR 61, Subpart M, National Emission Standard for Asbestos
- EPA 40 CFR 261 Identification and Listing of Hazardous Waste
- EPA 40 CFR 273 Standards for Universal Waste Management
- Government of Japan, Japan Waste Management and Public Cleansing Law of 1970, updated 2001.
- Government of Japan (GOJ) Regulations Pertaining to Asbestos
- Ambient Air Pollution Control Law
- Asbestos Impediment Prevention Regulations
- Building Standards Law
- Industrial Safety and Health Law
 - Waste Cleaning and Disposal Law
- OSHA 29 CFR 1910.1000 Air Contaminants
- OSHA 29 CFR 1910.1001 Asbestos
- OSHA 29 CFR 1910.1200 Hazard Communication
- OSHA 29 CFR 1926.1101 Asbestos
- OSHA 29 CFR 1926.62 Lead
- USACE Japan Edited Specifications (JES) 1.6

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REPAIR HVAC WIDEBAND SATCOM OPS B100 Camp Foster, Okinawa, Japan

TEMPLATE - Construction Phase Commissioning Plan

Prepared by: JACOBS

March 2025

TEMPLATE - Construction Phase Commissioning Plan

Summary

The Construction Phase Commissioning Plan is developed in draft form for the specific project during the design phase. During the design phase, the plan provides direction for the development of the site-specific commissioning specifications by the design team. During the construction phase, the plan provides direction for the commissioning tasks during construction. The plan focuses on providing support for the specifications and provides forms for the application of the commissioning process.

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Commissioning Plan- Construction Phase

Project: REPAIR HVAC WIDEBAND SATCOM OPS
B100

Location: Camp Foster, Okinawa, Japan

Date: 3/2025

Cx Plan Version

X Design Phase / Template
Construction Phase

___ Final Construction Phase Plan

Revision History

The commissioning plan is considered a “living” document that will be updated throughout the project duration. When updates are completed, copies of the revised plan will be submitted to the commissioning team members.

Summarized here are the Plan updates along with revision dates:

Rev	By	Date	Description
0	Jacobs	03/2025	DRAFT – Design Phase Cx
0	Jacobs	06/2025	DRAFT – Design Phase Cx

1. Overview

1.1 Abbreviations and Definitions

The following are common abbreviations used in this document.

A/E	Architect and design engineers	FPT	Function Performance Test
BOD	Basis of Design	HVAC	Heating, Ventilating and Air Conditioning
CxG	Commissioning Specialist for the Government	GC	General contractor
CC	Controls contractor	MC	Mechanical contractor
COR	Contracting Officer's Representative	PC	Prefunctional Checklist
Cx	Commissioning	PE	Project Engineer – Gov't
Cx Plan	Commissioning Plan document	PM	Project Manager – Gov't
CxC	Commissioning Specialist for the Contractor	Subs	Subcontractors to General
DA	Design Agent	TAB	Test and Balance Contractor
DOR	Designer of Record	USACE	US Army Corps of Engineers
EC	Electrical contractor	User	Facility User/Occupant – Gov't
FPC	Fire Protection Contractor		

1.2 Purpose of the Commissioning Plan

The purpose of the construction phase commissioning plan is to document a plan for the commissioning process during construction.

This plan does not provide a detailed explanation of required testing procedures. In general, this plan identifies roles and responsibilities, systems to be commissioned, commissioning standards and procedures, and sample forms to indicate the level of detail desired. Additionally, this plan does not provide extensive narrative on all commissioning concepts, as this information can be found in commissioning guide references.

Construction will be phased into 3 phases. Phase 1 and 2, will be the east side of building operations, phase 3 will be the west side of building administration. Equipment testing will be performed per the corresponding phase. Integrated systems testing will be performed after phase 2.

1.3 Commissioning Scope

Commissioning is a systematic process of ensuring that all building systems perform interactively according to the Owner's Project Requirements (OPR) and contract documents. This is achieved by, beginning in the design phase, documenting the design intent and continuing through construction, acceptance and the warranty period with actual verification of performance.

Commissioning during the construction and acceptance phase of this project is intended to achieve the following specific objectives:

- Ensure that applicable equipment and systems are installed properly and receive adequate operational checkout by installing contractors.
- Verify and document proper performance of equipment and systems.

- Ensure that O&M documentation is complete and is in custody of the owner.
- Ensure that the Owner's operating personnel are adequately trained.

1.4 1.4 Commissioned System

The following systems will be commissioned in this project (shall include, but not limited to):

HVAC System (and all integral equipment controls)

- ☒ Computer Room Air Conditioning Units*
- ☒ Chilled Water System*
- ☒ Pumps*
- ☐ Variable speed drives
- ☐ Piping, cleaning and flushing
- ☒ Ductwork
- ☒ Air handling units (including energy recovery modules, MUAs, coils, etc)*
- ☒ Fan Coil Units
- ☒ Condensing Units
- ☒ Testing, Adjusting and Balancing work
- ☒ Exhaust Fans
- ☒ Building Automation System (controlled devices, control loops and system integration)* -
There are two systems Admin & Operations system

Electrical Systems

None in CxC Scope

Plumbing

None in CxC Scope

Other

- ☒ Utility Monitoring and Control System
- ☒ Mass Notification System
- ☒ Integrated Systems Test

* Denotes critical equipment

2. General Building Information

Owner: **FEDERAL GOVERNMENT**

Project: **REPAIR HVAC WIDEBAND SATCOM OPS B100**

Location: **CAMP FOSTER, OKINAWA, JAPAN**

Building Type (office, court, etc.): Type II-B / Type II (0,0,0), noncombustible / Unrated construction

Square Footage: 18,500 GSF Number of stories: 1

3. Cx Team Data (primary parties)

Team Member	Contact Name	Email Address	Phone
Project Manager-Gov	Greaves, Shane	Shane.M.Greaves@usace.army.mil	+1 315-645-9353
Design Team Leader	Guerra, Ana	ana.guerra@jacobs.com	+1 214-708-2435
Project Engineer			
General Contractor			
Commissioning Specialist Government			
Commissioning Specialist Contractor			
Mechanical Designer/Eng.	Allerheiligen, Brian	Brian.Allerheiligen@jacobs.com	+1-970-343-0657
Electrical Designer/Eng.	Zabransky, Chris	Chris.Zabransky@jacobs.com	+1 817 347 5379
Tenant Representative	Yacono, Vincent	vincent.yacono.1@spaceforce.mil	+1 315-652-2602
Mechanical Contractor			
Electrical Contractor			
TAB Contractor			
Controls Contractor			
Mechanical Independent Technical Review			
Architect	Metzler, Jake	jake.metzler@jacobs.com	+1 940-902-0599

4. Roles and Responsibilities

4.1 Team Members

The members of the commissioning team consist of the CxC, CxG, PE/COR, GC, A/E (particularly the mechanical engineer), the mechanical contractor, electrical contractor, TAB representative, controls contractor, any other installing subcontractors or suppliers of equipment and those performing technical design review of the construction documents. If known, the Owner's building or plant operator/ engineer is also a member of the commissioning team. Additional information on roles and responsibilities of the CxC and CxG is provided in APPENDIX D.

4.2 General Management Plan

The CxC will be hired by the General Contractor. The CxG is a Government employee. In general, the CxC coordinates the commissioning activities and reports to the CxG and Contracting Officer's Representative copying the GC on all results. The CxC's responsibilities, along with all other contractors' commissioning responsibilities are outlined herein or detailed in the specifications. The Specifications

will take precedence over this Cx Plan. All members work together to fulfill their contracted responsibilities and meet the objectives of the Contract Documents. Refer to the management protocols section below.

4.3 Descriptions of Roles and Responsibilities

General descriptions of the commissioning roles are as follows:

Commissioning Specialist for the Contractor (CxC):

- a. Directs/Coordinates the Cx process and activities.
- b. Coordinates all commissioning schedule and activities with the CxG.
- c. Obtain copies of all shop drawings, manufacturer's literature, maintenance information or other information as may be needed for systems to be commissioned.
- d. Collect the information needed for development of a complete Commissioning Plan and functional performance tests.
- e. Obtain all proposed start-up and Prefunctional Checklists documentation.
- f. Obtain updates to all project documentation to reflect all supplemental instructions, addenda or other revisions to the project construction documents.
- g. Obtain submittals for all systems to be commissioned including controls system and wiring diagrams and narrative sequences of operation, in time for use in preparing the Functional Test Procedures.
- h. Obtain preliminary TAB report, indicating all actual field values recorded, prior to initiation of functional testing.
- i. Obtain complete operation and maintenance information and as-built drawings for verification, organization and distribution.
- j. Update and finalize the Draft Construction Phase Commissioning Plan.
- k. As part of Final Commissioning Plan, develop Prefunctional Checklists and Functional Test Procedures from Contract Documents and final equipment submittals including narrative sequences of operation, control diagrams and software code for execution with the assistance of Contractor staff as required. Sample PC and FPT documents located under Appendix C are examples representing the scope and rigor of the commissioning procedures required, and shall be used as the basis for developing the detailed checklists and functional performance test procedures for all equipment requiring commissioning.
- l. Perform site observations to follow installation progress and to verify system installation and readiness for testing.
- m. Reviews and implements the contractor Start-up Procedures and Forms
- n. Review submittal of all required prefunctional and start-up documentation provided by Contractor for completeness and reasonableness. This includes installation documentation, start-up documentation, point-to-point checklists and preliminary TAB report, prior to initiation of functional testing.

- o. Schedule, direct and witness complete functional testing as defined in the Commissioning Plan and Functional Test Procedures. All testing shall be performed by the Contractors and subcontractors, and documented by the Commissioning Specialist.
- p. Conduct commissioning meetings.
- q. Provide site observation, functional tests or other project reports in a timely manner.
- r. Identify and document inconsistencies or deficiencies in system operations and system compliance. System deficiencies shall be forwarded to the Contractor and Contracting Officer and documented in a Communication Log and the CQC system.
- s. Provides limited problem resolution assistance.
- t. Coordinate the participation of Government's personnel with equipment, component and systems performance verification and participation in required training.
- u. When commissioning has been successfully completed, recommend acceptance to the Government.
- v. Once all functional tests have been successfully completed and all outstanding issues resolved, the Commissioning Specialist will provide the Contracting Officer with a Final Commissioning Report of all commissioning activities and test results that occurred during the project.
- w. Review training documents and document training of government personnel on commissioning systems and equipment.
- x. Develop Systems Manual and obtain all necessary information/documentation needed for inclusion.
- y. Perform Post Occupancy Activities.

Owner/Government Representative (CxG, COR, PE, or PM):

- a. Facilitates and supports the Cx process assuring that all commissioning activities are performed in accordance with commissioning guideline requirements and contract documents.
- b. Approves the Final Commissioning Plan including PCs and FPTs.
- c. Approves the Commissioning Report.
- d. Performs construction observation.
- e. Has final authority in decision making processes

General Contractor:

- a. Facilitate the Cx process.
- b. Ensure CxC receives copies of all shop drawings, manufacturer's literature, maintenance information or other information as may be needed for systems to be commissioned.
- c. Ensure CxC is provided necessary information for updating the Commissioning Plan and development of checklists and functional tests. The Contractor shall review these documents and confirm in writing to the CxC, PE/COR and CxG any known areas of conflict or areas requiring clarifications.
- d. Assists in development of Prefunctional Checklists (PC).
- e. Ensure all proposed start-up and Prefunctional Checklists documentation is completed by qualified personnel and provided to the CxC.
- f. Assists in the development of the Functional Performance Testing Procedures and Forms (FPT).
- g. Plan for and incorporate all commissioning activities into the construction schedule.
- h. Provide a fully operational system per Specifications, started, verified, debugged, calibrated, balanced, tested and under automatic control.
- i. Provide qualified personnel to participate in the commissioning tests, including seasonal testing.
- j. Provide updates to all project documentation to reflect all supplemental instructions, addenda or other revisions to the project construction documents. Updates and supplemental instructions must be posted to the master set of documentation for review and reference by all Contractors and for the Commissioning Specialist's use.
- k. Provide adequate time and resources to support CxC with functional testing of systems to be commissioned.
- l. Coordinate participation of the mechanical, electrical, controls and TAB subcontractors, and all Contractor Quality Control personnel in the commissioning process. Ensure that Subs perform their responsibilities.
- m. Ensure CxC and CxG receive submittals for all systems to be commissioned including controls system and wiring diagrams and narrative sequences of operation, in time for use in preparing the Functional Test Procedures.
- n. Participate in any efforts to finalize sequences of operations with Government and CxC.
- o. Verify that coordination, installation, quality control and final testing have been completed such that installed systems and equipment comply with construction documents.
- p. Review the Commissioning Plan, interim reports such as issues log and progress reports, and Commissioning Report to include test results and submit comments to the CxC.
- q. In a timely manner, address issues identified during construction that may affect the commissioning process or final system performance.
- r. Perform start-up and testing of mechanical and electrical equipment and systems and document as required with start-up reports and completion of Prefunctional Checklists. These checklists include installation documentation, start-up documentation, controls point-to-point

documentation and calibration documentation, verification that controls sequence of operations meets design intent and TAB final documentation. Reports will be stored in the Contractor's field trailer. Contractor will coordinate efforts to complete the prefunctional documentation.

- s. Ensure preliminary TAB report, indicating all actual field values recorded is provided to the CxC and CxG, prior to initiation of functional testing. These reports shall be incorporated in the commissioning field notebook. The final TAB report is distinguished from the preliminary TAB report by the fact that all submittals and corrections shall be approved by the issuance of the final TAB report. All balancing issues and corrections shall have been resolved to the satisfaction of all parties by the final TAB report.
- t. Issue a written Certificate of Readiness for each system to CxC, CxG and PE/COR upon completion of all systems work, start-up and Prefunctional Checklists requirements by trade contractors.
- u. Responsible for and demonstrate proper system installation and performance (typically through the use of installers, subs and/or manufacturers). Operate equipment and systems as required for functional performance testing. This includes, but is not limited to; manipulating the appropriate controls systems to execute the Functional Test Procedures.
- v. Participate in the fine-tuning or troubleshooting of system performance, if either of these measures becomes necessary.
- w. Ensure complete operation and maintenance information and as-built drawings is provided to the CxC and CxG for verification, organization and distribution.
- x. Provide documentation of training for the systems specified.
- y. Provide test equipment required to test all the systems and equipment in this project.
- z. Review operating and maintenance data for verification, organization, distribution and conformance to requirement of the Contract Documents.
- aa. Provide necessary information/documentation to CxC for inclusion in the Systems Manual.

Subcontractors:

- Demonstrate proper system installation and performance.
- Assists in development and execution of Prefunctional Checklists (PC).
- Assists in the development and execution of the Functional Performance Testing Procedures and Forms (FPT).

Architect/Engineer (A/E or DOR):

- Develops and updates Basis of Design.
- Incorporates Commissioning requirements in construction documents.
- Performs construction observation.
- Assists in resolving problems.

Project Manager:

- Facilitates and supports the Cx process.

5. Construction Phase Commissioning Activities

This section sequentially details the commissioning process by commissioning task or activity.

5.1 Construction Commissioning Coordination Meeting

A commissioning scoping meeting is planned and conducted by the CxC early during the construction phase. Cx Team members such as the GC, CxG, PE/COR, A/E, Technical Commissioning Specialists, and the mechanical, electrical, controls, and TAB subs shall attend. The agenda should include but is not limited to:

- Cx party introductions
- Cx process overview
- Management and lines of reporting and communication are determined
- Draft Cx Plan is reviewed
- Roles and Responsibilities
- Proposed Cx Schedule
- Documentation flow and submittal data the CxG will receive
- Review of building systems to be commissioned including intended operation
- Process questions are addressed

The outcome of the meeting is increased understanding by all parties of the commissioning process and their respective responsibilities. The meeting provides the CxC additional information needed to update the Cx Plan, including the commissioning schedule.

Prior to this meeting the CxC is given, by the GC, all drawings and specifications and the construction schedule by trade. The CxC keeps notes from the meeting and distributes them, along with the updated Cx Plan, to each team member.

5.2 Final Commissioning Plan and Schedule - Construction Phase

The CxC develops a final Construction Phase Cx Plan using the information gathered from the scoping meeting. The initial commissioning schedule is also developed and integrated into the construction schedule. The Cx schedule will be adjusted as construction progresses. Reference Project Schedule in Total Building Commissioning Specification.

5.3 Site Observation

The CxC will make periodic site visits to observe equipment and system installations. The CxG, selected maintenance personnel and the PE/COR should be notified and may accompany the CxC during these visits. Site visits shall begin regularly by the time 40% of the commissioned systems are installed. The frequency of these visits shall be monthly at a minimum. As construction progresses and mechanical

system installation activity increases, CxC representatives will need to increase site observation frequency. The CxC shall provide a site observation report for each visit. Additional CxC visits should include, but not limited to, selective witnessing of DALT, hydrostatic pressure testing, startup, controls startup/point verification, TAB, and spot-checking contractor performed Prefunctional Checkout for completeness and accuracy. CxC will witness 100 percent of the Prefunctional Checklist for critical equipment and a minimum of a single sample of each type of noncritical equipment. CxC witness verification of the Testing, Adjusting, and Balancing shall be a 100 percent sample strategy with any failed results requiring retest. Observed deficiencies shall be documented and tracked in a construction Cx issues log. The GC will maintain the master issue log.

5.4 Periodic Meetings

The CxC attends selected planning and job-site meetings while on site in order to remain informed on construction progress and to update parties involved in commissioning. The PE/COR and GC provide the CxC and CxG with information regarding substitutions, change orders and any supplemental instructions that may affect commissioning equipment, systems or the commissioning schedule. The CxC and CxG may review construction meeting minutes, change orders or requests for information for the same purpose. Later during construction, necessary meetings between various commissioning team parties will be scheduled by the CxC, through the GC, as required to meet the project's needs. These periodic meetings shall initially be held monthly (and may coincide with the CxC site visits) and then more frequently as project nears completion or as requested by the COR or CxG. The COR and CxG should be notified of all meetings. The CxC will provide minutes for these meetings.

5.5 Miscellaneous Management Protocols

The following protocols will be used on this project.

<u>Issue</u>	<u>Protocol</u>
Requests for Information (RFI) or formal documentation requests:	The CxC goes first through the GC and CxG.
Minor or verbal information and clarifications:	The CxC communicates directly with subcontractors while keeping the CxG, PE/COR and GC informed.
Deficiencies notification:	The CxC documents deficiencies through the CxG and PE/COR, but may discuss deficiency issues with GC prior to notifying the CxG and PE/COR. Copies of deficiency reports shall be provided to CxG and PE/COR.
Functional Test Scheduling or Training:	The CxC may provide input for (and do some coordination of) testing and training, but does not schedule testing or training.
Scheduling Commissioning meetings:	The CxC coordinates date and schedules through GC, CxG, and PE/COR. The GC notifies attendees directly.
Significant Change requests:	The CxC has no authority to issue change orders.
For making small changes in specified sequences of operations:	The CxC may suggest (to the DOR, for approval by the PE/COR) changes in sequences of operations to improve efficiency or to control or to correct deficiencies.

Disagreements between contractors and CxC's interpretations:

Try and resolve with the CxC first. Then work through CxG and GC who will work with CxC directly.

5.6 Progress Reporting and Logs

At the beginning of construction, the CxC provides the PE/COR with monthly Commissioning Progress Reports. Thirty (30) days prior to the startup of the first piece of major equipment, the frequency of progress reports is increased to twice per month, until startup is completed. Thirty (30) days before functional testing of equipment begins, weekly progress reports are required until functional testing and all non-conformance issues are resolved. The CxC may increase the reporting frequency as needed. The progress reports shall include, but are not limited to, an update of the schedule with list of requested schedule changes and new items added to the schedule, a list of new and outstanding deficiencies, and a description of commissioning progress corresponding to the plan. The CxC keeps a log of all commissioning-related issues that require current or future attention using a Commissioning Issues Log. Samples of the Commissioning Progress Report and Commissioning Issues Log are included in Appendix A.

The CxC regularly communicates with all members of the commissioning team, keeping them apprised of outstanding issues, commissioning progress, and scheduling through memos, progress reports, etc. The CxC maintains all commissioning related materials in an organized notebook to become part of the final Cx report.

5.7 Cx Related Submittals and Documentation

5.7.1 Standard Submittals

The CxC provides all Subs responsible for commissioned equipment with commissioning documentation requirements for their respective equipment and systems through the GC. Subcontractor documentation submissions typically coincide with the normal submittal process. At a minimum, this documentation includes manufacturer's installation instructions, start-up and check-out procedures, O&M data, performance data and control drawings. The CxC reviews and approves submissions relative to commissioning issues expressed in the contract documents. CxC recommendations are provided to the DOR, CxG, owner and/or PE/COR as directed.

5.7.2 Special Submittals, Notifications and Clarifications

The Subs, GC or DOR shall notify the CxC within one (1) week of approval of any design intent or operating parameter changes, added control strategies and sequences of operation, or other change orders that may affect commissioned systems. Thirty (30) days prior to performing any applicable tests, the Subs must provide the CxC full procedure details. As TAB phases are completed, a draft TAB report is provided to the CxC with full explanations of approach, methods, results, data table legends, etc. The final TAB report is provided to the CxC and CxG upon completion.

These submittals to the CxC do not constitute O&M manual submittal compliance. Documentation requirements for the O&M manuals are discussed in Section 5.11, herein.

The CxC may request additional design narrative from the DOR and/or the contractor depending on completeness of original bid documents. The CxC may submit written RFIs to contractors through the standard process.

5.8 Prefunctional Checklists, Tests and Startup

Prefunctional checklists (PC) are important to ensure that the equipment and systems are installed and operational and that functional performance testing (FPT) may proceed without unnecessary delays. The CxC will assist/develop with commissioning team members detailed prefunctional checklists and start up plans. The CxC shall provide prefunctional checklists and start up plans to CxG for review. Each piece of equipment receives full prefunctional checkout by the GC/Subs. No sampling strategies are used. The prefunctional checkout for a given system must be successfully completed prior to formal functional performance testing of equipment or subsystems of the given system.

Prefunctional checklists are primarily static inspections and procedures to prepare the equipment or system for initial operation (oil levels OK, fan belt tension, labels affixed, gages in place, sensor calibration, etc.). Some prefunctional checklist items entail simple testing of the function of a component, a piece of equipment or system (such as measuring the voltage imbalance on a three phase pump motor of a chiller system). Prefunctional checklists augment and are combined with the manufacturer's start-up checklist.

Contractors typically already perform some, if not all, of the prefunctional checklist items the commissioning authority will recommend. Few contractors actually document in writing the execution of these checklist items. This project requires that the procedures be documented in writing by the installing technician. To document the prefunctional testing and startup process, site technicians performing line item tasks, initial and date each checklist item. Only individuals having direct knowledge of a line item being completed shall check and initial forms. The CxC may not witness all of the prefunctional checks, but will perform spot checks and will witness prefunctional testing of larger or more critical pieces of equipment. Critical equipment requiring witnessing by the CxC is identified in paragraph 1.4 Commissioned Systems.

Subcontractors will execute and sign the prefunctional checks and start up procedures. The GC ensures that each PC is properly completed and provides them to the CxC.

5.8.1 Start-up Plan

The CxC assists the commissioning team members responsible for startup in developing detailed start-up plans for all equipment. The parties responsible for each part of startup and initial checkout are identified on the prefunctional checklists by the GC.

The following procedures will be used for this project: (the Subs are responsible for the plan development)

1. The CxC adapts and enhances, if necessary, the representative prefunctional checklists (PC) and procedures and develops original lists, as necessary.
2. The CxC transmits them to the GC who designates which trade or contractor is responsible to fill out each line item on the Prefunctional Checklist from the CxC. The GC then transmits the checklist to the responsible Subs.
3. The Sub designated to develop the Start-up Plan obtains manufacturer installation, start-up and checkout data, including actual field checkout sheets used by the field technicians.
4. The Sub copies all pages with important instructional data and procedures (not covered in manufacturer field checkout sheets) from the start-up and checkout manuals and adds a signature line in the column by each procedure.

5. The copied pages, along with the prefunctional checklist provided by the GC (originally from the CxC) and the manufacturer field checkout sheets become the “Start-up and Checkout Plan.” For systems that may not have adequate manufacturer start-up and checkout procedures, particularly for components being integrated with other equipment, the Sub should provide the added necessary detail and documenting format to the CxC for approval, prior to execution.
6. The GC/Subs transmit the full Start-up Plan to the CxC and CxG for review and approval.
7. The CxC and CxG review and approve the procedures and the format for documenting them, using a standard form, noting any procedures that need to be added, and conveys to the GC. The GC then transmits the full start-up plan to the Subs for their review and use. (This usually means that the Prefunctional Checklist, alone, will go to more than one Sub, while the full plan will go to the primary installing contractor.)

5.8.2 Execution of Checklists and Startup

Four weeks prior to startup, the GC and Subs shall incorporate the startup and initial checkout activities into the overall construction schedule and notify all applicable parties. The startup and initial checkout plan shall be executed. The CxC, and CxG if available, will observe, at minimum, the procedures for each piece of primary equipment. For components of equipment, (e.g., VAV boxes), the CxC observes a sampling of the prefunctional and start-up procedures.

To document the process of startup and checkout, the site technician performing the line item task initials and dates each paragraph of procedures in the “Startup Plan” and checks off items on the prefunctional and manufacturer field checkout sheets, as they are completed. Only individuals having direct knowledge of a line item being completed shall check or initial the forms.

The Subs and vendors execute the checklists and tests and submit a signed copy of the completed start-up and prefunctional tests and checklists to the CxC. The CxC and CxG may review prefunctional checklists in progress, as necessary.

5.8.3 Sampling Strategy for CxC Observation of Prefunctional Checkout and Startup

<u>Equipment or System</u>	<u>Fraction To Be Observed by CxC</u>
Chillers	100%
Air Handling Units	100%
Pumps, VFD's	100%
Pipe flushing	At beginning and end
Computer Room AC Units	100%
Condensing Units	25%
DX FCU	One of type
Exhaust Fans	One of type
Building automation system	100% review of PVT Report
TAB work	100% witness and review of TAB Report
Other misc. equipment/systems	25%

5.8.4 Deficiencies and Non-Conformance

The Subs clearly list any outstanding items of the initial start-up and prefunctional procedures that were not completed successfully at the bottom of the procedures form or on an attached sheet. The procedures form and deficiencies are provided to the CxC within 48 hours of test completion. The CxC works with the Subs and vendors to correct and retest deficiencies or uncompleted items, involving the CxG, PE/COR and others as necessary. The GC, through installing Subs or vendors, corrects all areas that are deficient or incomplete according to the checklists and tests in a timely manner, and shall notify the CxC as soon as outstanding items have been corrected and resubmit an updated report. Upon satisfactory completion, the CxC recommends approval of the execution of PCs and startup and initial checkout of each system to the CxG and PE/COR using a standard form.

5.8.5 TAB

The TAB contractor submits the outline of the TAB plan and approach to the CxC and the controls contractor eight weeks prior to starting the TAB. Included in the approach, is an explanation of the intended use of the building control system. The CxC reviews the plan and approach for understanding and coordination issues and may comment, but does not "approve." The controls contractor reviews the feasibility of using the building control system for assistance in the TAB work. The TAB submits weekly written reports of discrepancies, contract interpretation requests and lists of completed tests to the GC, CxC, and CxG. This facilitates quicker resolution of problems and will result in a more complete TAB

before functional testing begins. TAB work will not begin until the control system has been prefunctionally tested and approved by the CxC.

5.8.6 Controls Checkout Plan

The controls contractor develops and submits a written step-by-step plan to the CxC, CxG, and PE/COR through the GC which describes the process they intend to follow in checking out the control system for proper operation. This Controls Checkout Plan shall include forms for the documentation of individual component verification which is considered the Performance Verifications Test (PVT) or Point-to-Point test. Components that are tested and verified on an individual basis are part of the Prefunctional Checks and are documented in conjunction with these forms.

The GC shall verify that the controls contractor reviews the TAB plan with the TAB contractor prior to starting TAB work in order to determine the control system capabilities to aid in executing TAB work. The controls contractor will provide the TAB with any necessary unique instruments for setting terminal unit boxes and instruct TAB in their use (handheld control system interface for use around the building during TAB, etc.). The controls contractor shall also provide a technician qualified to operate the controls to assist the TAB contractor in performing TAB and the CxC in performing all PCC checkouts.

All CxC required controls prefunctional checklists, calibrations, start-up and selected functional tests of the system shall be completed and approved by the CxC prior to TAB. The GC through the controls contractor shall execute tests and trend logs and remain on site for assistance for mechanical system functional tests.

5.8.7 Certificate of Readiness

The GC shall issue a Certificate of Readiness certifying that all equipment, systems, and controls are complete and ready for Functional Performance Testing. The Certificate of Readiness shall include all equipment and system start-up reports, Performance Verification Test Reports, Prefunctional Checklists, TAB Report, and the Building Air Tightness Report. The GC and the Mechanical, Electrical, Controls, and Testing, Adjusting, and Balancing Representatives shall sign and date the Certificate of Readiness.

5.9 Development of Functional Performance Test and Verification Procedures

5.9.1 Overview

Functional performance testing (or functional testing) is the dynamic testing of systems (rather than just components) under full automatic operation (e.g., the chiller pump is tested interactively with the chiller functions to see if the pump ramps up and down to maintain the differential pressure setpoint, etc). Systems are tested under various modes, such as during low cooling or heating loads, high loads, component failures, unoccupied, varying outside air temperatures, fire alarm, power failure, etc. The systems are run through all of the control system's sequences of operation and components are verified to be responding as the sequences state. The CxC develops the functional performance test procedures in a sequential written format and coordinates, oversees and documents the actual testing, which is performed by the installing contractor or vendor.

5.9.2 Development Process

Before test procedures are written, the CxC obtains all requested documentation and a current list of change orders affecting equipment or systems, including an updated points list, control sequences and setpoints. The CxC develops specific test procedures to verify proper operation of each piece of equipment and system listed in the Cx Plan and Specifications. The CxC obtains clarification, as needed, from contractors and the DOR regarding sequences and operation to develop these tests. Prior to

execution, the CxC provides a copy of the equipment and systems functional testing to the installing Sub (through the GC) who reviews the tests for feasibility, safety, warranty and equipment protection. Blank copies of the procedures are input into the O&M manuals for later use by operations staff. The test procedure forms shall include the following, at a minimum:

- a. System and equipment or component name(s) and configuration(s).
- b. Equipment location and ID number.
- c. Unique test ID number, and reference to unique PC and startup documentation ID numbers.
- d. Date.
- e. Project name.
- f. Participating parties.
- g. A copy of the specific sequence of operations or other specified parameters being verified.
- h. Formulas used in any calculations.
- i. Required pre-test field measurements.
- j. Instructions for setting up the test, including special cautions, alarm limits, or other equipment-specific information.
- k. Specific step-by-step procedures to execute the test in a clear, sequential, and repeatable format.
- l. Acceptance criteria of proper performance with a Yes / No check box to allow for clear marking of whether or not proper performance of each part of the test was achieved.
- m. A section for comments.
- n. Signature and date blocks for the CxC, CxG, Contractor, PE and DOR.

Functional testing and verification shall be achieved by manual testing (persons manipulate the equipment and observe performance) and by monitoring the performance and analyzing the results using the control system's trend log capabilities.

5.9.3 Functional Testing Plan Overview

The CxC develops a testing plan overview to provide the Cx Team with a better idea of where functional testing lies in the schedule, what issues prevent the start of testing, which contractors are needed for each test and how much time might be expected from them. This is developed after most equipment has been started up and when functional testing dates are not too far off. The testing plan overview is provided to the contractors to assist in moving more efficiently to functional testing.

5.10 Execution of Functional Testing Procedures

5.10.1 Overview and Process

Functional Performance Testing shall be performed after PCs, startup, calibration, control checkout and TAB are complete and Certificate of Readiness has been submitted for a given system. The CxC

schedules functional tests through the GC, CxG, and PE/COR. The CxC directs, witnesses, and documents the functional testing of all equipment and systems according to the Specifications and the Cx Plan. The CxG will oversee/witness the functional testing. The Subs execute the tests, unless otherwise specified. The control system shall be successfully tested before it can be used to verify dependent system and/or component performance. Air and water system balancing shall be complete and reconciled prior to functional testing of related equipment and /or systems. Testing proceeds from components to subsystems to systems and finally to interlocks and connections between systems. The CxC documents the results of the test.

Functional Performance Tests shall be demonstrated to the satisfaction of the CxC. FPTs shall be witnessed and endorsed by the CxC upon satisfactory completion. The CxG recommends acceptance of each test to the PE/COR. The PE/COR gives final approval on each test. The actual testing program shall be conducted in accordance with the approved FPT procedures and shall be documented as required. The GC shall notify the CxG and PE/COR at least 14 calendar days prior to date of each functional performance tests.

Integrated System Test must be accomplished without impacting mission equipment power. "Blackout" test shall only impact the power to the HVAC system. Fire system shunt trip may be used for IST if applicable.

5.10.2 Deficiencies and Retesting

Every effort shall be made to expedite the testing process and minimize unnecessary delays, while not compromising the integrity of the procedures. Nonconformance and deficiencies observed in materials, installation, or operation shall be addressed immediately, in terms of notification to responsible parties, and providing recommended actions to correct deficiencies. The GC shall have responsibility for resolving construction deficiencies. Corrections of minor deficiencies identified are made during the tests at the discretion of the CxC and CxG. The CxC shall maintain a master deficiency and resolution log, and shall provide the PE and CxG with written progress reports and test results with recommended actions. The CxC shall also document the deficiency and resolution on the test procedure form. Subs correct deficiencies and notify the CxC of corrective action. The CxC schedules retesting through the GC, CxG and PE/COR. Decisions regarding deficiencies and corrections are made at as low a level as possible, preferably between CxC, CxG or PE/COR and the Sub. For areas in dispute, final authority resides with the PM.

5.10.3 Facility Staff Participation

The Owner's facilities operating staff are encouraged to attend and participate in the testing process. The CxC will notify the PE/COR, who will then notify the facility staff when the commissioning events will occur.

5.10.4 Sampling

All equipment will be functionally tested using a 100% sampling strategy. Critical Equipment, requiring 100% of systems/components to be functionally tested, is identified in 1.4 Commissioned Systems. The following table provides additional information on the sampling strategy to be used on this project:

<u>Equipment or System</u>	<u>Fraction To Be Observed by CxC</u>
Chillers	100%
Air Handling Units	100%
Pumps, VFD's	100%
Computer Room AC Units	100%
Condensing Units	100%
DX FCU	100%
Exhaust Fans	100%
Building Automation Systems	100%
Utility Monitoring and Control Systems	100%
Mass Notification System	100%
Integrated Systems Test	100%
Other misc. equipment/systems	100%

5.11 Deferred Testing

5.11.1 Unforeseen Deferred Tests

If any check or test cannot be completed due to the building structure, required occupancy condition, or other deficiency, a request for delay execution of checklists and FPT may be delayed contingent on approval of the PM. These tests shall be conducted as soon as possible in the same manner as seasonal testing.

5.11.2 Seasonal Testing

The CxC shall schedule, coordinate, and observe additional testing for seasonal variation in operations and control strategies during the opposite season to verify performance of the HVAC system and controls. The GC shall execute and document tests and correct deficiencies with facilities staff and the CxC, CxG, and/or PE witnessing. Testing shall be completed during the warranty period to fully test all sequences of operation. The GC shall make necessary revisions to O&M manuals and records due to the testing.

- a. Seasonal commissioning pertains to testing under full-load conditions during peak cooling seasons, as well as part-load conditions in the spring and fall. Initial commissioning will be done as soon as contract work is completed, regardless of season. For the major heating and cooling generation and distribution systems, means of artificial loading shall be developed by the CxC as a means of demonstration, to a reasonable level of confidence, the ability to handle larger peak seasonal loads. Subsequent commissioning shall be undertaken at the appropriate time thereafter to ascertain adequate performance during the different seasons.
- b. All equipment and systems shall be tested and commissioned in a peak season to observe full-load performance. Heating equipment shall be tested during winter design extremes. Cooling equipment shall be tested during summer design extremes, with a fully occupied building. Each Contractor and supplier will be responsible to participate in the initial and the alternate peak season test of the systems required to demonstrate performance.
- c. All equipment and systems affected by occupancy variations will be tested and commissioned at the minimum and peak loads to observe system performance. Each Contractor and supplier will be responsible to participate in the occupancy sensitive testing of systems to provide verification of adequate performance.
- d. Based on the scheduling of seasonal testing, the Contractor and COR shall discuss/coordinate Beneficial Occupancy and start of warranty period for affected systems.
- e. Perform a one-week Endurance Test once during the peak cooling season during outdoor air condition extremes using the building control system to trend all points necessary to evaluate the installed equipment and system performance or as shown as requiring a trend on the project schedules.

5.11.3 Short-Term Diagnostic Testing

After initial occupancy, the GC shall perform short-term diagnostic testing, using data acquisition equipment or the building automation system to record system operation over a two- to three-week period. The dynamic interactions between components in the building system shall be investigated. The scheduling, interaction between heating and cooling, and effectiveness of the HVAC system in meeting the comfort requirements and design conditions shall be evaluated. The GC shall document tests and findings, and correct deficiencies according to the original testing requirements.

6. Occupancy and Operation Phase Commissioning Activities

6.1 O&M Manuals and Warranties

6.1.1 Standard O&M Manuals

The CxC, CxG, and DOR review the O&M manuals, documentation and redline as-builts for systems that were commissioned to verify compliance with the Specifications. They recommend approval and acceptance of these sections of the O&M manuals to the PE/COR. The PE/COR gives final approval on these documents. The CxC, CxG, DOR and PE/COR review each equipment warranty and verify that all requirements to keep the warranty valid are clearly stated.

6.1.2 Systems Manual

The CxC will compile, organize and index the following commissioning data by equipment into labeled, indexed and tabbed, three-ring binders and deliver it to the GC, to be included with the O&M manuals.

The Systems Manual shall include the following (ASHRAE Guideline 0, Annex O shall be used for format and content):

- a. Index of Systems Manual with notation as to content storage location if not in actual manual.
- b. Executive Summary.
- c. Owner's Project Requirements.
- d. Construction Record Documents, specifications, and approved submittals.
- e. A list of recommended operational record-keeping procedures, including sample forms, logs, or other means, and a rationale for each.
- f. Ongoing optimization guidance.
- g. . Operations and maintenance manuals (includes operating procedures for all normal, abnormal, and emergency modes of operation; as-built control setpoints (provide a basic version for User Operator and a full version for Operation and Maintenance staff); maintenance procedures; parts and recommended spare parts list; troubleshooting guide; and systems schematics (one-line diagrams).
- h. Training materials.
- i. Commissioning Report.

6.2 Training and Orientation of Owner Personnel

The GC shall provide Owner/Maintenance personnel training and orientation for all commissioned equipment and systems as required by the contract documents. Training will be coordinated by the GC.

6.3 Warranty Period

During the warranty period, seasonal testing and other deferred testing required is completed according to the Specifications. The CxC coordinates this activity. Tests are executed and deficiencies corrected by the appropriate Subs, witnessed by CxC, facilities staff, the PE/COR and the CxG. Any final adjustments to the O&M manuals and as-builts due to the testing are made. In addition, the CxC will return to the project approximately 9 months into the 12 month warranty period. During this visit(s) the CxC will review with facility staff the current building operation and the condition of outstanding issues related to the original and seasonal commissioning and review and analyze trend data for compliance to the construction documents. The CxC will also interview facility staff and identify problems or concerns they have operating the building as originally intended. The CxC will make suggestions for improvements and for recording these changes in the O&M manuals. The CxC will identify areas that may come under warranty or under the original construction contract. The CxC will also assist facility staff in developing reports and documents and requests for services to remedy outstanding problems. Results of this inspection shall be documented in the final summary Commissioning Report.

6.4 System Performance Verification

The CxC shall review the operation of the building with the operation and maintenance staff and occupants within nine months of facility acceptance by the Government. The CxC shall resolve all outstanding commissioning related issues during the warranty period.

The CxC shall implement a thermal comfort survey of building occupants within a period of six to ten months of facility acceptance by the Government. This survey shall collect anonymous responses about

thermal comfort in the buildings, including an assessment of overall satisfaction with thermal performance and identification of thermal comfort-related problems. The CxC shall develop a plan for corrective action if the survey results indicate that more than 20% of occupants are dissatisfied with thermal comfort in the building. This plan shall include measurement of relevant environmental variables in problem areas in accordance with ASHRAE Standard 55.

The main parameter to be measured in the thermal comfort survey shall be satisfaction with thermal environment. The answer shall be posed in a seven-point scale format running from very satisfied (+3) to very dissatisfied (-3) with the center (0) signifying the neutral point. The percent dissatisfied shall be the percentage of respondents who answer "dissatisfied" (any of the lower three points of the seven point scale). The survey shall identify each thermal zones by room number(s) and ask the respondent to identify his/her thermal zone. Survey shall include follow-up questions that are asked if the respondent indicates dissatisfaction to identify the nature and cause of the problem. The survey shall be administered in person, over the phone, over networked computers, or on paper. The CxC shall be responsible for collecting each completed survey. The survey shall be consistently applied and available for participation by all regular occupants.

The CxC shall develop a corrective action plan. This plan shall identify each question and the number of responses for each answer of each question. The corrective action plan shall identify the percent dissatisfied for each question. The corrective action plan shall identify the nature and location of any thermal environmental problems. The plan shall suggest directions for corrective actions based on the follow-up questions that identify the nature and cause of the problem.

7. Written Work Products

The written work products are described in Appendix B, Formal Written Work Products. The table describes each product, responsible party, due date, recipients, and approving authority.

7.1 Final Commissioning Report

A final summary report by the CxG will be provided to the PE/COR within thirty (30) days of substantial completion. The report shall include an executive summary, list of participants and roles, brief building description, overview of commissioning and testing scope and a general description of testing and verification methods. For all commissioned equipment, the report should contain the disposition of the commissioning specialist regarding the adequacy of the equipment, documentation and training meeting the contract documents in the following areas: 1) Equipment meeting the equipment specifications, 2) Equipment installation, 3) Functional performance and efficiency, 4) Equipment documentation and design intent, and 5) Operator training. All outstanding non-compliance items shall be specifically listed. Recommendations for improvement to equipment or operations, future actions, commissioning process changes, etc. shall also be listed. Each non-compliance issue shall be referenced to the specific functional test, inspection, trend log, etc. where the deficiency is documented. The functional performance and efficiency section for each piece of equipment shall include a brief description of the verification method used (manual testing, BAS trend logs, data loggers, etc.) and include observations and conclusions from the testing.

Appendices shall contain control sequences, meeting minutes, progress reports, issues logs, site visit reports, findings, unresolved issues, communications, etc. The commissioning plan, prefunctional checklists and functional tests (along with blanks for the operators) and monitoring data and analysis will be provided.

8. Commissioning Schedule

8.1 General Issues

The following sequential priorities are followed:

1. Equipment shall not be used for heating/cooling during construction until pre-start checklist items and all manufacturers' pre-start procedures have been completed. Moisture, dust and other environmental and building integrity issues must also have been addressed.
2. Equipment/system functional testing shall begin only after prefunctional, startup, and TAB work has been completed.
3. Functional testing of controls system and its controlled equipment shall begin only after all points have been calibrated and prefunctional testing has been completed.
4. TAB shall begin only after the controls system has been approved by the CxC for use for TAB work.
5. TAB work shall begin only after envelope work and ceiling installation have been completed.

Attached Appendixes

- **Appendix A - Example Construction Phase Forms**
- **Appendix B - Commissioning-Related Formal Written Work Products and Submittals**
- **Appendix C - Example Prefunctional Checklists, Functional Performance Tests, Integrated System Tests, and Building Envelope Inspection Checklists**
- **Appendix D - CxC and CxG Roles and Responsibilities**

APPENDIX A

Example Construction Phase Forms

COMMISSIONING ISSUES LOG

Project: _____

Location: _____

Prepared by: _____

Page _____ of _____

Attach additional pages as necessary for issues requiring more explanation and tracking.

#	Issue	Date Found	Effects	Possible Cause	Recommendations	Actions Taken	O&M Doc. Issue?	Open/ Closed (Date)

COMMISSIONING PROGRESS REPORT

Project: _____ Date: _____ Prepared by: _____

Reporting Period: _____ Report #: _____

Commissioning tasks worked on since last report and general progress:

Areas where schedule is not being met: _____

Recommended actions: _____

Requested schedule adjustments: _____

Next steps: _____

Other comments (include general comments and field notes): _____

Non-Compliance Report attached. (Y/N) _____

Issues Log Attached. (Y/N) _____

Misc. comment sheets attached. (Y/N) _____

Commissioning Agent

APPENDIX B**Commissioning-Related Formal Written
Work Products and Submittals**

Product	Created By	Product Description and Form	Approved By	Product Assigned In
Final Cx Plan	CxC	Final Cx plan for const. phase	CxG recommends approval to PE/COR	Cx Plan 5.2
Cx Schedule	GC/CxC/Subs	Schedule of Cx Activities	PE/COR	Cx Plan 7.2
Cx related submittals	GC/Subs	Detailed data on all commissioned equipment	CxC/CxG recommends approval to PE/COR	Cx Plan 5.7
Prefunctional Tests and Checklists	CxC, GC and Subs	List by equipment of prefunctional checklists and prefunctional tests	CxG recommends approval to PE/COR	Cx Plan 5.8
Start-up and initial checkout plans	GC and Subs	Specific listing of procedures for combining CxG prefunctional checklists with Sub's startup and checkout.	CxC/CxG	Cx Plan 5.8.1
Start-up and initial checkout reports	GC and Subs	Filled out prefunctional checklists, tests, startup and initial checkout	CxC/CxG recommends approval to PE/COR	Cx Plan 5.8.2
Controls Contractor Initial Check-out Plan	Controls Contr.	Complete step-by-step plan on checkout and calibration procedures, including forms for documentation	CxC/CxG recommends approval to PE/COR	Cx Plan 5.8.6
TAB plan and approach	TAB	Outline of TAB plan, approach and schedule	CxC; PE/COR	Cx Plan 5.8.5, Specifications
TAB progress reports	TAB	List of discrepancies, interpretations needed, tests completed	none	Cx Plan 5.8.5
Draft TAB report	TAB	Draft TAB report with method and results	CxC/CxG (Required for Cx)	Cx Plan 5.7.2
Final TAB report	TAB	Final TAB report with method and results	CxC/CxG (Required for Cx)	Cx Plan 5.7.2
Change orders	GC; PE/COR	Change orders that affect Commissioned equipment	N/A	Cx Plan 5.7.2
Issues Log	CxC	Record / track of all issues and deficiencies	N/A	Cx Plan 5.6
Systems Manual	CxC	Record / track of all submittals, checklists, tests, etc.	N/A	Cx Plan 6.1.2

Template Construction Phase Commissioning Plan

Cx progress reports	CxC	Gives scheduling needs and update, deficiency report, Cx progress	N/A	Cx Plan 5.6
Functional Performance Test forms	CxC	Full description of test procedures in “form” format	CxG recommends approval to PE/COR	Cx Plan 5.9
Completed Functional Performance Test forms	CxC	Recorded documentation of the test on the form	CxG recommends approval to PE/COR	Cx Plan 5.9
Final Cx report	CxC	Summary report with status of each system and important findings, etc. Compilation of all important Cx data (checks, tests, issues, reviews, etc.)	CxG recommends approval to PC/COR	Cx Plan 7.1
O&M manuals	GC and Subs	Documentation of design, equipment, operations and maintenance, as-builts, etc.	CxC/CxG recommends approval to PE/COR (required for Cx)	Cx Plan 6.1 and Specifications
Thermal Comfort Survey	CxC	Survey used to collect data from building occupants regarding thermal comfort and satisfaction	N/A	Cx Plan 6.3
Corrective Action Plan	CxC	Plan recommending correction action for issues identified during occupancy phase commissioning and thermal comfort survey	N/A	Cx Plan 6.3

APPENDIX C

Example Prefunctional Checklists, Functional Performance Tests, Integrated System Tests, and Building Envelope Inspection Checklists

Download example Building Envelope Inspection Checklists, Pre-Functional Checklists, Functional Performance Test Checklists, and Integrated Systems Test Checklists for specification section 01 91 00.15

BUILDING COMMISSIONING and section 01 91 19 BUILDING ENCLOSURE COMMISSIONING at the following location:

<https://www.wbdg.org/dod/ufgs/ufgs-01-91-00-15> and **<https://www.wbdg.org/dod/ufgs/ufgs-01-91-19>**. The checklists submitted in the Interim and Final Construction Phase Commissioning Plans must contain the same level of detail shown in the examples. The submitted checklists are not required to match the format of the examples.

APPENDIX D

CxC and CxG Roles and Responsibilities

DESIGN PHASE	CxG	CxC
Ensure that design team has OPR	X	
Develop Cx plan (DESIGN PHASE include draft CONSTRUCTION PHASE)	X	
Review concept level design submission against OPR	X	
Start developing Cx specification	X	
Ensure that design concept is in compliance with OPR	X	
Develop format for the systems manual	X	
Review midpoint design submission against OPR	X	
Update and incorporate comments into Cx specification	X	
Review completed design documents prior to release for construction against OPR	X	
Finalize Cx specification	X	
Ensure the completed design documents are in compliance with the OPR prior to release for construction	X	
Review Basis of Design (BOD) and concur that it is compliance with the requirements in the OPR	X	
Develops format for Construction Phase Cx issues log	X	
Develop draft Sample Prefunctional Checklists (PC)	X	
Develop draft Sample Functional Performance Tests (FPT)	X	
Compile Design Phase Cx report	X	
CONSTRUCTION PHASE	CxG	CxC
Review construction submittals and shop drawings against contract documents and maintain Cx Submittal Log		X
Assure construction submittals and shop drawings have been reviewed by CxC in accordance with Cx Spec, conducting random audits of Cx Submittal log and construction submittals	X	
Reviews and recommend approval of Cx schedule	X	X
Finalize commissioning plan in accordance with Cx spec		X
Review and recommend approval of Final Commissioning Plan	X	
Perform routine inspections	X	X
Chairs Cx construction progress meetings		X
Maintains Cx issue log		X
Review Cx issue log	X	
Finalize PFC for installed equipment using draft PFC as template		X

Template Construction Phase Commissioning Plan

Review and recommend approval of finalized PFC	X	
Executes PFC as required (CxG must witness in accordance with Cx Plan)		X
Review and recommend approval of completed PFC	X	
Develops FPT for installed equipment using draft FPT as starting point		X
Review and recommend approval of finalized FPT	X	
Executes FPT as required by Cx Plan		X
Witness representative sample of FPT during execution	X	
Review and recommend approval of completed FPT	X	
Compile Construction Phase Cx report		X
Review and approve Construction Phase Cx Report	X	
Compile Systems Manual		X
Review and approve Systems Manual	X	
Ensure turnover of Systems O&M Manual (by Contractor) to Facility Staff	X	X
Submits all Cx documentation required per LEED	X	X
Ensure proper Systems O&M training has been provided (by Contractor) to Facility Staff	X	X
OCCUPANCY AND OPERATIONS PHASE	CxG	CxC
Verify any seasonal testing of the facility.	X	X
Review the facility and its performance within 10 months of substantial completion.	X	X
Within 10 months of substantial completion, review building operation with operations staff and occupants in order to identify any problems in operating the building as originally intended.	X	X
Verify training effectiveness of O&M staff within 10 months after substantial completion	X	X
Document results of post occupancy review		X
Review post occupancy review documentation	X	

ROLES AND RESPONSIBILITIES MATRIX

(excerpt from ER 1110-345-723, Total Building Commissioning Procedures)

DRAFT

Pre-Design Phase(D-B-B)/RFP Development(D-B)		CxG = Gov't Comm Specialist COR = Contracting Officer Rep DOR= Designer of Record Cx D = Design Comm Specialist Cx C = Construction Comm Specialist O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional N/A = Not Applicable	
Commissioning Roles & Responsibilities								
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Meetings	Pre-Design Kick-Off Meeting	P	P	P	L	N/A	P	
	Owner's Project Requirements Meeting	P	P	P	L	N/A	P	
	Commissioning Planning Meetings	P	P	L	P	N/A	O	
Coordination	Coordinate with [COR, DOR, etc.] to ensure that Cx is incorporated into project planning and documents.	L	P	P	P	N/A	O	
Cx Plan & Spec	Draft Commissioning Plan	R	A	L	R	N/A	R	
Schedules	Preliminary Commissioning Schedule	R	A	L	R	N/A	O	
OPR	Prepare Owner's Project Requirements	R	A	P	L	N/A	P	
Commissioning Documents	Identify systems to be commissioned	P	P	L	P	N/A	R	
	Preliminary acceptance criteria	R	P	L	P	N/A	R	
	Commissioning Documents preliminary templates	R	R	L	R	N/A	R	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

Design Phase (D-B-B)		CxG = Gov't Comm Specialist COR = Contracting Officer Rep DOR = Designer of Record CxD = Design Comm Specialist CxC = Construction Comm Specialist O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional N/A = Not Applicable	
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Coordination Cx Plan & Spec Schedules	Coordinate with [COR, AHJ, Vendors, etc.] to ensure that Cx interacts properly with other systems as needed to support OPR and BoD	P	P	P	L	N/A	P	
	Preliminary Commissioning Plan	R	A	L	R	N/A	R	
	Preliminary Cx Specifications	R	A	L	R	N/A	R	
	Design Phase Commissioning Schedule	R	A	P	L	N/A	O	
OPR and BoD	Maintain OPR on behalf of Owner	P	P	P	L	N/A	P	
	Review Basis of Design Document vs. OPR	R	P	L	P	N/A	P	
	Maintain BoD on behalf of Owner	P	P	P	L	N/A	P	
Reviews	Focused Concept Design Review	R	L	P	P	N/A	P	
	Focused Design Development (35-50%) Review	R	L	P	P	N/A	P	
	Focused Construction Document Review	R	L	P	P	N/A	P	
	Focused Pre-Final Construction Document	R	L	P	P	N/A	P	
	Focused Final Construction Document	R	L	P	P	N/A	P	
	Final Construction Document Comment Backcheck	R	L	P	P	N/A	P	
Functional	Draft Pre-Functional Checklists (PFC)	R	A	L	R	N/A	R	
Test Protocols	Draft System Functional Performance Tests (FPT)	R	A	L	R	N/A	R	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

Design Phase (D-B)		CxG = Gov't Comm Specialist COR = Contracting Officer Rep DOR = Designer of Record CxD = Design Comm Specialist CxC = Construction Comm Specialist O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional N/A = Not Applicable	
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Coordination Cx Plan & Spec Schedules	Coordinate with [COR, AHJ, Vendors, etc.] to ensure that Cx interacts properly with other systems as needed to support OPR and BoD	P	P	N/A	L	P	P	
	Preliminary Commissioning Plan	R	A	N/A	R	L	R	
	Preliminary Cx Specifications	R	A	N/A	R	L	R	
	Design Phase Commissioning Schedule	R	A	N/A	L	P	O	
OPR and BoD	Maintain OPR on behalf of Owner	P	P	N/A	L	P	P	
	Review Basis of Design Document vs. OPR	R	P	N/A	P	L	P	
	Maintain BoD on behalf of Owner	P	P	N/A	L	P	P	
Reviews	Focused Concept Design Review	R	L	N/A	P	R	P	
	Focused Design Development (35-50%) Review	R	L	N/A	P	R	P	
	Focused Construction Document Review	R	L	N/A	P	R	P	
	Focused Pre-Final Construction Document	R	L	N/A	P	R	P	
	Focused Final Construction Document	R	L	N/A	P	R	P	
	Final Construction Document Comment Backcheck	R	L	N/A	P	R	P	
Functional	Draft Pre-Functional Checklists (PFC)	R	A	N/A	R	L	R	
Test Protocols	Draft System Functional Performance Tests (FPT)	R	A	N/A	R	L	R	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

Construction Phase (D-B-B & D-B)		CxG= Gov't Comm Specialist COR = Contracting Officer Rep CxD = Design Comm Specialist DOR = Designer of Record CxC=Construction Comm Specialist O&M =Gov't Facility O&M					L=Lead P = Participate A = Approve R=Review O=Optional N/A=Not Applicable	
Commissioning Roles & Responsibilities								
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Meetings*	Construction Commissioning Kick Off meeting	P	P	N/A	O	L	P	
	Commissioning/Issues Resolution Meetings	P	P	N/A	O	L	O	
	Project Commissioning Progress Meetings	P	P	N/A	O	L	O	
	Controls Meeting	P	P	N/A	P	L	P	
Coordination	Coordinate with [COR, AHJ, Vendors, etc.] to ensure that Cx interacts properly with other systems as needed to support OPR and BoD	P	P	N/A	P	L	P	
Cx Plan	Final Commissioning Plan w/ final PFCs and FPTs	R	A	N/A	R	L	R	
Schedules*	Duration Schedule for Commissioning Activities	R	A	N/A	O	L	R	
Doc Reviews*	TAB Plan Review	P	A	N/A	R	P	O	
	Submittal and Shop Drawing Review	P	A	N/A	R	P	O	
	Review Contractor Equipment Startup Checklists	P	A	N/A	O	P	O	
	Review Change Orders, ASI, and RFI	P	A	N/A	R	P	O	
Site Observation*	Witness Factory Testing	P	P	N/A	O	P	O	
	Construction Observation Site Visits	P	P	N/A	O	P	O	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

**For D-B-B acquisition, DOR participation or review for tasks indicated as optional for the DOR should be determined on a case-by-case basis with consideration of scope and complexity of the project/system, availability of resources, and potential benefit of DOR participation/review; these additional services require funding consideration.*

Acceptance Phase (D-B-B & D-B)		CxA = Commissioning Authority COR = Contracting Officer Rep CxG= Gov't Comm Consultant A/E = Designer of Record CxC = Construction Comm Specialist O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional N/A = Not Applicable	
Commissioning Roles & Responsibilities								
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Meetings*	Pre-Test Coordination Meeting	P	P	N/A	O	L	P	
	Issues Resolution and Commissioning Report Review Meetings	P	P	N/A	O	L	P	
Document Reviews*	Review Completed Pre-Functional Checklists	R	A	N/A	R	L	P	
	Pre-Functional Checklist Verification	R	A	N/A	O	L	P	
	Review Operations & Maintenance Manuals	R	A	N/A	O	L	P	
	Training Plan Review	R	A	N/A	O	L	P	
	Warranty Review	R	P	N/A	O	L	P	
	Review TAB Report	R	A	N/A	R	P	P	
	Systems Manual	R	A	N/A	O	L	R	
Site Observations*	Construction Observation Site Visits	P	P	N/A	O	L	P	
	Witness Selected Equipment Startup	P	P	N/A	O	P	P	
Functional Test Protocols*	TAB Verification	P	A	N/A	O	P	P	
	Systems Functional Performance Testing	P	A	N/A	O	L	P	
Reports & Logs*	Final Commissioning Report	R	A	N/A	R	L	R	
	Submit Final Systems Manuals	R	A	N/A	O	L	R	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

**For D-B-B acquisition, DOR participation or review for tasks indicated as optional for the DOR should be determined on a case-by-case basis with consideration of scope and complexity of the project/system, availability of resources, and potential benefit of DOR participation/review; these additional services require funding consideration.*

Post Occupancy Phase (D-B-B & D-B)		CxG= Gov't Comm Specialist COR = Contracting Officer Rep CxD = Design Comm Specialist DOR = Designer of Record CxC = Construction Comm Specialist O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional N/A = Not Applicable	
Commissioning Roles & Responsibilities								
Category	Task Description	CxG	COR	CxD	DOR	CxC	O&M	Notes
Document Reviews*	Systems Monitoring Trended Data	R	R	N/A	R	L	P	
	Systems Manuals Update	R	A	N/A	O	L	R	
Site Observations*	Periodic Warranty (4 & 9 Month) Site Visits	P	P	N/A	O	P	P	
Functional Test Protocols*	Deferred and/or seasonal Testing	P	A	N/A	O	L	P	
Technical Activities*	Commissioning/Issues Resolution/Lessons Learned Meetings	P	P	N/A	O	L	P	
	Post-Occupancy Warranty Checkup and review of Significant Outstanding Issues	P	P	N/A	O	L	P	
Reports and Logs*	Final Commissioning Report Amendment	R	A	N/A	R	L	R	
	Issues Logs Closure Report	R	R	N/A	O	L	R	

Note: (Appendix C) is just a sample to depict how such a matrix would be developed. There are many more commissioning tasks to be identified beyond what the sample identifies. The type of involvement shown is for instruction purposes only and would need to be developed on a project by project basis for the scope, size, and complexity of the project, the degree of rigor required, and with consideration of available District resources and expertise.

**For D-B-B acquisition, DOR participation or review for tasks indicated as optional for the DOR should be determined on a case-by-case basis with consideration of scope and complexity of the project/system, availability of resources, and potential benefit of DOR participation/review; these additional services require funding consideration.*

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SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K (2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

CARPET AND RUG INSTITUTE (CRI)

CRI 104 (2015) Carpet Installation Standard for Commercial Carpet

CRI 105 (2015) Carpet Installation Standard for Residential Carpet

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing Standards

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25 (Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders;
<https://www.dla.mil/Portals/104/Documents/Disposition/dds/ddsdocs/cylinderjointpub.pdf>

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M (2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures

MIL-STD-129 (2014; Rev R; Change 1 2018; Change 2

2019; Change 3 2023) Military Marking for
Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous
Air Pollutants

49 CFR 173.301 Shipment of Compressed Gases in Cylinders
and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a Demolition Plan and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. Remove rubbish and debris from the project site; do not allow accumulations inside or outside the building. The work includes demolition, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 17, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Provide new supports and reinforcement for existing construction weakened by demolition or removal work. Repairs,

reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

1.3.3 Trees

Protect trees within the project site which might be damaged during demolition and which are indicated to be left in place, by a 1.8 m high fence. Erect and secure fence a minimum of 1.5 m from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.4 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations.

1.3.5 Facilities

Protect electrical and mechanical services and utilities.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

SD-11 Closeout Submittals

Receipts

1.7 QUALITY ASSURANCE

Submit timely notification of demolition and renovation projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the State's environmental protection agency local air pollution control district/agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris to occupied portions of the building and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily.

1.8 PROTECTION

1.8.1 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.10 EXISTING CONDITIONS

Before beginning any demolition work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer or the Contracting Officer's Representative showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304

pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Utilities and Related Equipment

3.1.1.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2 Carpet

Remove existing carpet for reclamation in accordance with manufacturer recommendations and as follows. Remove used carpet in large pieces, roll tightly, and pack neatly in a container. Remove adhesive according to recommendations of the Carpet and Rug Institute (CRI). Provide adhesive removal solvents in compliance with CRI 104/CRI 105. Recycle removed carpet cushion.

3.1.3 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, give priority to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

3.1.4 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a

manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

- a. Concrete and Masonry: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals in existing masonry walls to remain, with an approved masonry patching material, applied in accordance with the manufacturer's printed instructions.
- b. Where existing partitions have been removed leaving damaged or missing resilient tile flooring, patch to match the existing floor tile.
- c. Patch acoustic lay-in ceiling where partitions have been removed. Make the transition between the different ceiling heights by continuing the higher ceiling level over to the first runner on the lower ceiling and closing the vertical opening with a painted sheet metal strip.

3.1.5 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990. Recover all refrigerants prior to removing air conditioning, refrigeration, and other equipment containing refrigerants and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)." Turn in salvaged Class I ODS refrigerants as specified in paragraph, "Salvaged Materials and Equipment."

3.1.6 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.7 Locksets on Swinging Doors

Remove all locksets from all swinging doors indicated to be removed and disposed of. Deliver the locksets and related items to a designated location for receipt by the Contracting Officer after removal.

3.1.8 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Salvage each item of equipment and fixtures as a whole unit; listed, indexed, tagged, and stored. Salvage each unit with its normal operating auxiliary equipment. Transport salvaged equipment and fixtures, including motors and machines, to a designated on station storage area as directed by the Contracting Officer. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse; provide to recycling service for disassembly and recycling of parts.

3.1.8.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; drain tanks, piping and fixtures; if previously used to store flammable, explosive, or

other dangerous liquids, steam clean interiors. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.8.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.8.3 Ducts

Classify removed duct work as scrap metal.

3.1.8.4 Fixtures, Motors and Machines

Remove and salvage fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Salvage, box and store auxiliary units and accessories with the main motor and machines. Tag salvaged items for identification, storage, and protection from damage. Classify non-porcelain broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor. Salvage and crush porcelain plumbing fixtures unsuitable for reuse.

3.1.9 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.9.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.9.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.9.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.9.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.2 DISPOSITION OF MATERIAL

3.2.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.2.2 Reuse of Materials and Equipment

Remove and store materials and equipment listed in the Demolition Plan indicated to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.2.3 Salvaged Materials and Equipment

Remove materials and equipment that are listed in the Demolition Plan indicated and indicated to be removed and/or salvaged in the Drawings.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers. Deliver the following items reserved as property of the Government to the areas

designated.

- d. Remove and capture all Class I ODS refrigerants in accordance with the Clean Air Act Amendment of 1990, and turn in to the Navy as directed by the Commanding Officer. (1)

3.2.4 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Recovered ODS must be removed from Government property and disposed of in accordance with JEGS and applicable installation requirements. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with JEGS and applicable installation requirements. Submit Receipts or bills of lading, as specified. Confirm with Contracting Officer if ozone depleting substance (ODS) is to be shipped to the Defense Depot, Richmond, Virginia.

3.2.4.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

- a. Activity name and unit identification code
- b. Activity point of contact and phone number
- c. Type of ODS and pounds of ODS contained
- d. Date of shipment
- e. National stock number (for information, call (804) 279-4525).

3.2.4.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.2.5 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.3 CLEANUP

Remove debris and rubbish from project site. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.4 DISPOSAL OF REMOVED MATERIALS

3.4.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan. Storage of removed materials on the project site is prohibited.

3.4.2 Burning on Government Property

Burning of materials removed from demolished structures will not be permitted on Government property.

3.4.3 Removal to Spoil Areas on Government Property

Transport noncombustible materials removed from demolition structures to designated spoil areas on Government property.

3.4.4 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil as directed .

3.5 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION 02 82 00
ASBESTOS REMEDIATION
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSE/SAFE Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM C732 (2017; R 2022 Standard Test Method for Aging Effects of Artificial Weathering on Latex Sealants

ASTM D2794 (1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)

ASTM D4397 (2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications

ASTM D522/D522M (2017) Mandrel Bend Test of Attached Organic Coatings

ASTM E119 (2024) Standard Test Methods for Fire Tests of Building Construction and Materials

ASTM E1368 (2014) Visual Inspection of Asbestos Abatement Projects

ASTM E736/E736M (2019) Standard Test Method for Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

ASTM E96/E96M (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7 (2014) Compressed Air for Human
Respiration; 6th Edition

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1 (2020) Occupational and Educational
Personal Eye and Face Protection Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2023; ERTA 1 2023) Standard Methods of
Fire Tests for Flame Propagation of
Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM (2016; 5th Ed) NIOSH Manual of Analytical
Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing
Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147 The Control of Hazardous Energy (Lock
Out/Tag Out)

29 CFR 1926.103 Respiratory Protection

29 CFR 1926.1101 Asbestos

29 CFR 1926.200 Accident Prevention Signs and Tags

29 CFR 1926.51 Sanitation

29 CFR 1926.59 Hazard Communication

40 CFR 61-SUBPART A General Provisions

40 CFR 61-SUBPART M National Emission Standard for Asbestos

40 CFR 763 Asbestos

42 CFR 84 Approval of Respiratory Protective Devices

49 CFR 107 Hazardous Materials Program Procedures

49 CFR 171 General Information, Regulations, and
Definitions

49 CFR 172 Hazardous Materials Table, Special

Provisions, Hazardous Materials
Communications, Emergency Response
Information, and Training Requirements

49 CFR 173

Shippers - General Requirements for
Shipments and Packagings

UL SOLUTIONS (UL)

UL 586

(2009; Reprint Sep 2022) UL Standard for
Safety High-Efficiency Particulate, Air
Filter Units

1.2 DEFINITIONS

1.2.1 Amended Water

Water containing a wetting agent or surfactant with a maximum surface tension of 2.9 Pa.

1.2.2 Area Sampling

Sampling of asbestos fiber concentrations which approximates the concentrations of asbestos in the theoretical breathing zone but is not actually collected in the breathing zone of an employee.

1.2.3 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, and actinolite asbestos and any of these minerals that has been chemically treated or altered.

1.2.3.1 Asbestos-Containing Material (ACM)

Any materials containing more than one tenth of one percent (0.1%) asbestos.

1.2.4 Asbestos Control Area

That area where asbestos removal operations are performed which is isolated by physical boundaries which assist in the prevention of the uncontrolled release of asbestos dust, fibers, or debris.

1.2.5 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5 micrometers as determined by National Institute for Occupational Safety and Health (NIOSH) Method 7400.

1.2.6 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average measured in the breathing zone as defined by 29 CFR 1926.1101 or other Federal legislation having legal jurisdiction for the protection of workers health.

1.2.7 Authorized Person

Any person authorized by the Contractor and required by work duties to be

present in the regulated areas.

1.2.8 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

1.2.9 Building Inspector

Individual who inspects buildings for asbestos and has EPA Model Accreditation Plan (MAP) "Building Inspector" training; accreditation required by 40 CFR 763, Subpart E, Appendix C, has EPA/State certification/license as a "Building Inspector".

1.2.10 Competent Person (CP)

A person meeting the requirements for competent person as specified in 29 CFR 1926.1101 including a person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for contractor/supervisor, and is AHERA certified as a contractor/supervisor.

1.2.11 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work.

1.2.12 Class I Asbestos Work

Activities defined by OSHA involving the removal of thermal system insulation (TSI) and surfacing ACM.

1.2.13 Class II Asbestos Work

Activities defined by OSHA involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos - containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastic. Certain "incidental" roofing materials such as mastic, flashing and cements when they are still intact are excluded from Class II asbestos work. Removal of small amounts of these materials which would fit into a glovebag may be classified as a Class III job.

1.2.14 Class III Asbestos Work

Activities defined by OSHA that involve repair and maintenance operations, where ACM, including TSI and surfacing ACM, is likely to be disturbed. Operations may include drilling, abrading, cutting a hole, cable pulling, crawling through tunnels or attics and spaces above the ceiling, where asbestos is actively disturbed or asbestos-containing debris is actively disturbed.

1.2.15 Class IV Asbestos Work

Maintenance and custodial construction activities during which employees contact but do not disturb ACM and activities to clean-up dust, waste and debris resulting from Class I, II, and III activities. This may include dusting surfaces where ACM waste and debris and accompanying dust exists and cleaning up loose ACM debris from TSI or surfacing ACM following construction

1.2.16 Contractor/Supervisor

Individual who supervises asbestos abatement work and has EPA Model Accreditation Plan "Contractor/Supervisor" training; and is AHERA certified as a "Contractor/Supervisor".

1.2.17 Disposal Bag

A 0.15 mm thick, leak-tight plastic bag, pre-labeled in accordance with 29 CFR 1926.1101, used for transporting asbestos waste from containment to disposal site.

1.2.18 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 1.5 m in length and width in order to access a building component.

1.2.19 Encapsulation

The abatement of an asbestos hazard through the appropriate use of chemical encapsulants.

1.2.20 Encapsulants

Specific materials in various forms used to chemically or physically entrap asbestos fibers in various configurations to prevent these fibers from becoming airborne. There are four types of encapsulants as follows which must comply with performance requirements as specified herein.

- a. Removal Encapsulant (can be used as a wetting agent)
- b. Bridging Encapsulant (used to provide a tough, durable surface coating to asbestos containing material)
- c. Penetrating Encapsulant (used to penetrate the asbestos containing material encapsulating all asbestos fibers and preventing fiber release due to routine mechanical damage)
- d. Lock-Down Encapsulant (used to seal off or "lock-down" minute asbestos fibers left on surfaces from which asbestos containing material has been removed).

1.2.21 Friable Asbestos Material

A term defined in the JECS meaning any material which contains more than 0.1 percent asbestos, as determined using the method specified in the JECS, that when dry, can be crumbled, pulverized, or reduced to powder by hand

pressure.

1.2.22 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

1.2.23 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.24 Industrial Hygienist (IH)/Private Qualified Person (PQP)

A registered Architect, Professional Engineer (licensed), or U.S. Certified Industrial Hygienist who has successfully completed training in, and is accredited under a legitimate Model Accreditation Plan, as described in 40 CFR 763, as a certified Building Inspector, Contractor/Supervisor, and Project Designer. The IH/PQP shall have working knowledge of applicable U.S. Federal and Japanese occupational safety and health regulations for asbestos in construction.

The IH/PQP shall be considered a competent person as defined in 29 CFR 1926.1101. The IH/PQP shall be a third party Contractor and have no employee/employer relationship or financial relationship with the abatement Contractor, which could constitute a conflict of interest.

1.2.25 Industrial Hygiene Technician

The Industrial Hygiene Technician (IHT) shall be an employee of the IH/PQP or the testing laboratory, have a minimum of 2 years experience in the industrial hygiene field working under the direction of the IH/PQP, and have completed the following courses: "Contractor Supervisor Abatement Worker" and "Asbestos Project Designer".

1.2.26 Negative Initial Exposure Assessment

A demonstration by the Contractor to show that employee exposure during an operation is expected to be consistently below the OSHA Permissible Exposure Limits (PELs).

1.2.27 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.28 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.29 Nonfriable Asbestos Material

A term defined in the JEGS meaning a material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating,

binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.30 Permissible Exposure Limits (PELs)

1.2.30.1 PEL-Time Weighted Average(TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.30.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.31 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.32 Private Qualified Person (PQP)

A registered Architect, Professional Engineer (licensed), or U.S. Certified Industrial Hygienist who has successfully completed training in, and is accredited under a legitimate Model Accreditation Plan, as described in 40 CFR 763, as a certified Building Inspector, Contractor/Supervisor, and Project Designer. The IH/PQP shall have working knowledge of applicable U.S. Federal and Japanese occupational safety and health regulations for asbestos in construction.

The IH/PQP shall be considered a competent person as defined in 29 CFR 1926.1101. The IH/PQP shall be a third party Contractor and have no employee/employer relationship or financial relationship with the abatement Contractor, which could constitute a conflict of interest.

1.2.33 Regulated Asbestos Containing Material

Friable asbestos material; Category I nonfriable ACM that has become friable; Category I nonfriable ACM that will be or has been subjected to sanding grinding, cutting, or abrading; or Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations.

1.2.34 TEM

Refers to Transmission Electron Microscopy.

1.2.35 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

1.2.36 Transite

A generic name for asbestos cement wallboard and pipe.

1.2.37 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most 2.9 Pa.

1.2.38 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed.

1.3 REQUIREMENTS

1.3.1 Description of Work

The work covered by this section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The asbestos work includes the demolition and removal of mastic on covebase material, gray plaster/cementitious material/joint compound on concrete walls, joint compound on wallboards, and cementitious material on ceiling and floor slab which is governed by 40 CFR 763. Under normal conditions non-friable or chemically bound materials containing asbestos would not be considered hazardous; however, this material may release airborne asbestos fibers during demolition and removal and therefore must be handled in accordance with the removal and disposal procedures as specified herein and/or developed by the IH/PQP and pre-approved by the Government. Provide techniques as outlined in this specification. The building will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

1.3.1.1 Wallboard/Joint Compound

Discrete samples of the components (wallboard and joint compound) have been tested and results are shown in the attached Hazardous Material Survey.

1.3.2 Unexpected Discovery of Asbestos

The Contractor shall be responsible to review the available asbestos survey reports for this project, including, but not limited to: Hazmat Materials Inspection Report - FY25 Repair HVAC Wideband OPS B100, by ESC, dated January 2025.

The Contractor's IH/PQP shall review the existing asbestos survey report(s) to develop the appropriate work response to ACM which may be disturbed as part of this project. Should the contract require the Contractor to perform any hazardous material survey, the Contractor shall provide a copy of all surveys/sampling activities and analytical results

regarding asbestos to the Contracting Officer. The Asbestos Abatement Contractor shall also submit an Asbestos Hazard Abatement Plan, reviewed and certified by the IH/PQP, to the Contracting Officer for review before asbestos abatement work begins.

The Contractor shall review the existing Hazardous Material Reports and familiarize themselves on the specific homogeneous materials that were tested and all materials that were determined to be ACM. The Contractor shall be responsible for visually confirming the presence of these materials. The Contractor shall remove ACM which would otherwise be disturbed during normal work practices to execute the work required by this project:

The Contractor shall notify the Contracting Officer if any previously untested building components suspected to contain asbestos are discovered and will be impacted by the work. The material(s) in question shall not be disturbed until a response is received by the Contracting Officer. For any previously untested building components suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer (CO) who will order up to 20 bulk samples to be obtained and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary. This cost shall be included in the Contractor's bid and will be performed at no additional cost to the Government. The laboratory shall have a working definition of "Trace" amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1% as a "Trace" concentration. If PLM does not detect the presence of asbestos (e.g. "non-detect"), the material shall be considered <0.1% asbestos. If PLM analysis detects asbestos in any discernible amount (to include "trace" or "less than 1%"), the material shall be considered >0.1% asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1%, such as Transmission Electron Microscopy (TEM) or 1000 point counting. The CO will order up to 20 samples to be analyzed further by TEM or 1000 point counting counting at no additional cost to the Government. Any additional components identified as ACM (not identified as ACM at the time of bidding) that will be disturbed will be paid for by an equitable adjustment to the contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is EPA/State certified/licensed as a "Building Inspector".

1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101.

1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or other pertinent local, prefectural or Government of Japan (GOJ) directives. This requirement must have been satisfied within the 12 months prior to the start of work on this contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving

asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of as required by Government of Japan law after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Assistant Secretary of Labor for Occupational Safety and Health (OSHA), or authorized representatives of them, and an employee's physician upon the request of the employee or former employee.

1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification. Train personnel involved in the asbestos control work in accordance with United States Environmental Protection Agency (USEPA) Asbestos Hazard Emergency Response Act (AHERA) training criteria. Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures.

1.3.5 Permits and Notifications

Prior to the start of work, obtain necessary permits in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by local, prefectural and GOG regulations. Notify the Contracting Officer in writing 20 working days prior to commencement of work in accordance with 40 CFR 61-SUBPART M and applicable laws, ordinances, criteria, rules, and regulations. In accordance with the JEGS, prior to demolition or renovation of a facility that involves removing or disturbing friable ACM, a written assessment of the action will be prepared and submitted to the installation commander via the Contracting Officer.

The Asbestos assessment shall include the following information:

1. Type of Operation: demolition or renovation.
2. Description of the facility or affected part of the facility, including the age, size, and present and past use of the facility.
3. Procedure including analytical methods, use to detect the presence of regulated ACM and category I and category II nonfriable ACM.
4. Estimated amount of a)regulated ACM to be removed from the facility

- and b) Category I and II nonfriable ACM in the affected part of the facility that will not be removed before demolition.
5. Location, street address, and building number of the facility being demolished or renovated.
 6. Scheduled starting and completion dates of asbestos removal work, or any other activity such as site preparation, that would break up, dislodge or similarly disturb asbestos material.
 7. Scheduled starting and completion dates of demolition or renovation.
 8. Description of planned demolition or renovation work and method(s) to be used, including a description of affected facility components.
 9. Description of work practices and engineering controls, including asbestos removal and waste-handling emission control procedures.
 10. Name and location of the waste disposal site where the asbestos-containing waste material will be deposited.
 11. Description of procedures to follow if unexpected regulated ACM is found or Category II nonfriable ACM becomes crumbled, pulverized, or reduced to powder.

Submit copies of all Notifications to the Contracting Officer who will need to provide to the Installation Asbestos Manager for their records in accordance with the JEGS.

1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of local prefectural and GOJ regulations regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, and 40 CFR 763.

Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply. The following laws, ordinances, criteria, rules and regulations regarding removal, handling, storing, transporting and disposing of asbestos materials apply:

- a. Japan Environmental Governing Standards (JEGS), most recent version.

1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be

performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health NIOSH, under the provisions of 42 CFR 84, for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained using the model Contractor accreditation plan as indicated in the Federal statutes for all portions of the herein listed work.

1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the removal and demolition of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection, the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, removal method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in writing prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor, CP and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the

Contracting Officer prior to starting work.

1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the analysis, and reporting of airborne concentrations of asbestos fibers along with evidence that each laboratory selected holds the appropriate State license and permits and certification that each laboratory is American Industrial Hygiene Association (AIHA) accredited and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR) and successful participation of the laboratory in the Proficiency Analytical Testing (PAT) Program. Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited by the National Institute of Science and Technology (NIST) under National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.3.12 Landfill Approval

Submit written evidence that the landfill is approved for asbestos disposal by local prefectural and GOJ regulatory agencies. Within three working days after delivery, submit detailed delivery tickets, prepared, signed, and dated by an agent of the landfill, certifying the amount of asbestos materials delivered to the landfill. Submit a copy of the waste shipment records within one day of the shipment leaving the project site.

1.3.13 Transporter Certification

Submit written evidence that the transporter is approved to transport asbestos waste in accordance with the DOT requirements of 49 CFR 171, 49 CFR 172 and 49 CFR 173 as well as registration requirements of 49 CFR 107 and all other GOJ and local prefectural regulatory agency requirements.

1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in 29 CFR 1926.1101 and 29 CFR 1926.103 as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

1.4 SUBMITTALS

The Contractor's PQP shall, at a minimum, review and certify all submittals required under this section prior to submission to the Government. Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Asbestos Hazard Abatement Plan; G

SD-03 Product Data

Amended Water

Safety Data Sheets (SDS) for All Materials

Encapsulants

Respirators

Local Exhaust Equipment

Pressure Differential Automatic Recording Instrument

Vacuums

SD-06 Test Reports

Air Sampling Results; G

Pressure Differential Recordings for Local Exhaust System

Clearance Sampling; G

Asbestos Disposal Quantity Report

SD-07 Certificates

Employee Training; G

Notifications

Respiratory Protection Program

Testing Laboratory; G

Landfill Approval; G

Delivery Tickets

Waste Shipment Records

Transporter Certification; G

Medical Certification

Private Qualified Person Documentation; G

Competent Person; G

Worker's License; G

Contractor's License; G

Encapsulants

Equipment Used to Contain Airborne Asbestos Fibers

Water Filtration Equipment

Vacuums

Ventilation Systems

SD-11 Closeout Submittals

Permits

Notifications

Respirator Program Records

Rental Equipment

1.5 QUALITY ASSURANCE

1.5.1 Additional Requirements

In addition to detailed requirements of this specification, work performed under this contract shall comply with the JEGS, EM 385-1-1, applicable federal, GOJ, and local laws, ordinances, criteria, rules and regulations regarding handling, storing, transporting, and disposing of asbestos waste materials. Matters of interpretation of standards shall be submitted to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements shall apply. The following GOJ and local laws, rules and regulations regarding demolition, removal, encapsulation, construction alteration, repair, maintenance, renovation, spill/emergency cleanup, housekeeping, handling, storing, transporting and disposing of asbestos material apply:

1. Japan Environmental Governing Standards (JEGS), most recent version.

1.5.2 Private Qualified Person Documentation

Submit the name, address, telephone number, and resume of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, direct air monitoring and assist the Contractor's Competent Person in implementing and ensuring that safety and health requirements are complied with during the performance of all required work, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector and Contractor/Supervisor Abatement Worker. As described by 40 CFR 763. Submit certification to show that the PQP is a Registered Architect, Professional Engineer (licensed) or US Certified Industrial Hygienist determined and documented by the American Board of Industrial Hygiene (ABIH). A copy of the PQP's current valid ABIH certification and/or documentation that the PQP is a registered Architect or Professional Engineer shall be provided. The PQP and the asbestos contractor must not have an employee/employer relationship or financial relationship which could constitute a conflict of interest. The PQP must be a first tier subcontractor. The Designated IH/PQP shall be onsite at all times for the duration of asbestos activities and shall be available

for emergencies. In addition, submit resumes of additional IH/PQP's and industrial hygiene technicians (IHT) who will be assisting the Designated IH/PQP in performing onsite tasks. IH/PQPs and IHTs supporting the Designated IH/PQP shall have a minimum of 2 years of practical onsite asbestos abatement experience. Indicate the formal reporting relationship between the Designated IH/PQP and the support IH/PQPs and IHTs, the Designated Competent Person, and the Contractor.

1.5.3 Competent Person Documentation

The Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, notification of other employees onsite. The Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current EPA MAP "Contractor/Supervisor" training certificates.. Submit evidence that the Competent Person has a minimum of 2 years of comprehensive experience in planning and overseeing asbestos abatement activities and is experienced in the administration and supervision of asbestos abatement projects relevant to OSHA competent person requirements, including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, notification of other employees onsite, etc. The Designated Competent Person shall be responsible for compliance with the JEGS, applicable U.S. federal, GOJ, and local requirements, the Contractor's Accident Prevention Plan (APP) and Asbestos Hazard Abatement Plan (AHAP).

1.5.4 Worker's License

Submit documentation that workers meet the requirements of 29 CFR 1926.1101, 40 CFR 61-SUBPART M and other applicable federal, GOJ, and local requirements. Worker training documentation shall be provided as required on the "Certificate of Workers Acknowledgment". Training documentation is required for each employee who will perform OSHA Class I, Class II, Class III, or Class IV asbestos abatement operations. Such documentation shall be submitted on a Contractor generated form titled "Certificate of Workers Acknowledgment", to be completed for each employee in the same format. Training course completion certificates (initial and most recent update refresher) required by the information checked on the form shall be attached.

1.5.5 Contractor's License

Submit a copy of the asbestos contractor's AHERA Asbestos Worker(s) certification. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART M, EM 385-1-1, and the Federal, JEGS, GOJ and local prefecture requirement for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the

Company's President or Chief Executive.

1.5.6 Air Sampling Results

Complete fiber counting and provide results to the PQP for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor and the Contracting Officer immediately of any results in excess of 0.01 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.1 fibers per cubic centimeter.

1.5.7 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.51 mm of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the asbestos control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer within 24-hours from the end of each work day. Notify the Contractor and the Contracting Officer immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.01 fibers per cubic centimeter or background whichever is higher.

1.5.8 G0J, Prefectural or Local Citations on Previous Projects

Keep a statement, signed by an officer of the company, containing a record of any citations issued by G0J, Prefectural or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations) in the Environmental Records Binder on the project site. If there are none, a negative declaration signed by an officer of the company must be placed in the binder.

1.5.9 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan reviewed and certified by the Contractor's PQP, Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. The safety preconstruction conference must include the Contractor and their Designated Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 SECURITY

All work areas shall be secured to keep unauthorized personnel out of the area. This may be by locking of doors, installation of barricades, or other means. However, the contractor shall ensure that the security

measures are adequate to prevent access 24 hours a day.

1.7 EQUIPMENT

1.7.1 Rental Equipment

Provide a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

PART 2 PRODUCTS

2.1 ENCAPSULANTS

Encapsulants must conform to current USEPA requirements, contain no toxic or hazardous substances as defined in 29 CFR 1926.59, and conform to the following performance requirements. Submit certificates stating that encapsulants meet the applicable specified performance requirements.

2.1.1 Removal Encapsulants

<u>Requirement</u>	<u>Test Standard</u>
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3 hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.2 Bridging Encapsulant

<u>Requirement</u>	<u>Test Standard</u>
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M

<u>Requirement</u>	<u>Test Standard</u>
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.3 Penetrating Encapsulant

<u>Requirement</u>	<u>Test Standard</u>
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Cohesion/Adhesion Test - 50 pounds of force/foot	ASTM E119
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 43 in/lb	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.4 Lock-down Encapsulant

<u>Requirement</u>	<u>Test Standard</u>
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test

Requirement	Test Standard
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Tested with fireproofing over encapsulant applied directly to steel member)	ASTM E119
Bond Strength: 100 pounds of force/foot	ASTM E736/E736M
(Tests compatibility with cementitious and fibrous fireproofing)	

2.2 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

2.3 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

- a. Disposal Bags
- b. Fiberboard Drums
- c. Cardboard Boxes

2.4 SHEET PLASTIC

Sheet plastic must be polyethylene of 0.15 mm minimum thickness and must be provided in the largest sheet size necessary to minimize seams. Film must be clearfrosted or black and conform to ASTM D4397, except as specified below.

2.4.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided. Film must be frosted or black and must conform to the requirements of NFPA 701.

2.4.2 Reinforced

Reinforced sheets must be provided where high skin strength is required, such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film. Film must meet flame resistant standards of NFPA 701.

2.5 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 60 degrees C.

2.6 LEAK-TIGHT WRAPPING

Two layers of 0.15 mm minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

2.7 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 3 mm thick, acrylic sheet, 450 by 610 mm, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

2.8 WETTING AGENTS

Removal encapsulant (a penetrating encapsulant) must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

PART 3 EXECUTION

3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least two complete sets of personal protective equipment as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all equipment used to contain airborne asbestos fibers.

3.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

- a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.
- b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5 liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter

surface.

- c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.
- d. Single use standard 25 mm diameter cassette, open face, 0.45 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive cowl, and shrink bands when conducting environmental area sampling using NIOSH NMAM Methods 7400 and 7402, (and the transmission electric microscopy method specified at 40 CFR 763 if required).
- e. A flow calibrator capable of calibration to within plus or minus 2 percent of reading over a temperature range of minus 20 to plus 60 degrees C and traceable to a NIST primary standard.

3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services.

3.1.2.1 Respirators for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, removal and or demolition of asbestos materials with respiratory protection as indicated in 29 CFR 1926.1101 and 29 CFR 1926.103. Breathing air must comply with CGA G-7.

3.1.3 Exterior Whole Body Protection

3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable," whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.

3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as recommended by the PQP after each use.

3.1.3.3 Personal Decontamination Unit

Provide a temporary, negative pressure unit with a separate decontamination locker room and clean locker room with a shower that complies with 29 CFR 1926.51(f)(4)(ii) through (V) in between for personnel required to wear whole body protective clothing. Provide two separate lockers for each asbestos worker, one in each locker room. Keep street clothing and street shoes in the clean locker. HEPA vacuum and remove asbestos contaminated disposable protective clothing while still wearing respirators at the boundary of the asbestos work area and seal in

impermeable bags or containers for disposal. Do not wear work clothing between home and work. Locate showers between the decontamination locker room and the clean locker room and require that all employees shower before changing into street clothes. Collect used shower water and filter with approved water filtration equipment to remove asbestos contamination. Wastewater filters must be installed in series with the first stage pore size 20 microns and the second stage pore size of 5 microns. Dispose of filters and residue as asbestos waste. Discharge clean water to the sanitary system. Dispose of asbestos contaminated work clothing as asbestos contaminated waste. Keep the floor of the decontamination unit's clean room dry and clean at all times. Proper housekeeping and hygiene requirements must be maintained. Provide soap and towels for showering, washing and drying. Cloth towels provided must be disposed of as ACM waste or must be laundered in accordance with 29 CFR 1926.1101. Physically attach the decontamination units to the asbestos control area. Construct both a personnel decontamination unit and an equipment decontamination unit onto and integral with each asbestos control area.

3.1.3.4 Eye Protection

Provide eye protection that complies with ANSI/ISEA Z87.1 when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area and access tunnel. Attach the load-out unit in a leak-tight manner to each regulated area.

3.1.6 Warning Signs and Labels

Provide bilingual warning signs printed in English and Japanese at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable

3.1.6.1 Warning Sign

Provide vertical format conforming to 29 CFR 1926.200, and 29 CFR 1926.1101 minimum 500 by 355 mm displaying the following legend in the lower panel:

Legend	Notation
DANGER	one inch Sans Serif Gothic or Block
ASBESTOS	one inch Sans Serif Gothic or Block
MAY CAUSE CANCER	one inch Sans Serif Gothic or Block
CAUSES DAMAGE TO LUNGS	1/4 inch Sans Serif Gothic or Block
AUTHORIZED PERSONNEL ONLY	1/4 inch Sans Serif Gothic or Block
WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA	1/4 inch Sans Serif Gothic or Block

Spacing between lines must be at least equal to the height of the upper of any two lines.

3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST AVOID CREATING DUST

3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSE/SAFE Z9.2 and 29 CFR 1926.1101 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the asbestos control area. Filters on exhaust equipment must conform to ASSE/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSE/SAFE Z9.2 and UL 586. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

3.1.10 Single Stage Decontamination Area

A decontamination area (equipment room/area) must be provided for Class I work involving less than 7.5 m or 0.9 square meters of TSI or surfacing ACM, and for Class II and Class III asbestos work operations where exposures exceed the PELs or where there is no negative exposure assessment. The equipment room or area must be adjacent to the regulated area for the decontamination of employees, material, and their equipment which could be contaminated with asbestos. The area must be covered by an impermeable drop cloth on the floor or horizontal working surface. The area must be of sufficient size to accommodate cleaning of equipment and removing personal protective equipment without spreading contamination beyond the area.

3.1.11 Decontamination Area Exit Procedures

Ensure that the following procedures are followed:

- a. Before leaving the regulated area, remove all gross contamination and debris from work clothing using a HEPA vacuum.
- b. Employees must remove their protective clothing in the equipment room and deposit the clothing in labeled impermeable bags or containers for disposal or laundering.
- c. Employees must not remove their respirators until showering.
- d. Employees must shower prior to entering the clean room. If a shower has not been located between the equipment room and the clean room or the work is performed outdoors, ensure that employees engaged in Class I asbestos jobs: a) Remove asbestos contamination from their work suits in the equipment room or decontamination area using a HEPA vacuum before proceeding to a shower that is not adjacent to the work area; or b) Remove their contaminated work suits in the equipment room, without cleaning worksuits, and proceed to a shower that is not adjacent to the work area.

3.2 WORK PROCEDURE

Perform asbestos related work in accordance with the JEGS, 29 CFR 1926.1101, 40 CFR 61-SUBPART M and as specified herein. Use wet removal procedures and negative pressure enclosure techniques. Wear and utilize protective

clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the removal and demolition of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel. Seal all roof top penetrations, except plumbing vents, prior to asbestos roofing work. Shut down the building heating, ventilating, and air conditioning system, cap the openings to the system, prior to the commencement of asbestos work. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147. Disconnect electrical service when wet removal is performed and provide temporary electrical service with verifiable ground fault circuit interrupter (GFCI) protection prior to the use of any water.

All electrical work must be performed by a licensed electrician. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.01 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the PQP, including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer once the PQP has certified that the condition has been corrected. Corrective actions must be documented. If an asbestos fiber release or spill occurs outside of the asbestos control area, stop work immediately, correct the condition to the satisfaction of the PQP. Submit the documents certified by the PQP to the Contracting Officer including clearance sampling, and obtain the Contracting Officer's approval prior to resumption of work.

3.2.1 Building Ventilation System and Critical Barriers

Building ventilation system supply and return air ducts in a regulated area must be isolated by airtight seals to prevent the spread of contamination throughout the system. The airtight seals must consist of 2 layers of polyethylene. Edges to wall, ceiling and floor surfaces must be sealed with industrial grade duct tape.

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.2 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection and air sampling results are obtained from the PQP work may proceed at the discretion of the Contracting Officer.

3.2.3 Furnishings

Furniture and equipment will be removed from the area of work by the Government before asbestos work begins.

3.2.4 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

3.2.5 Asbestos Control Area Requirements

3.2.5.1 Negative Pressure Enclosure

Removal of mastic on covebase material, gray plaster/cementitious material/joint compound on concrete walls, joint compound on wallboards, and cementitious material on ceiling and floor slab require the use of a negative pressure enclosure. Block and seal openings in areas where the release of airborne asbestos fibers can be expected. Establish an asbestos negative pressure enclosure with the use of curtains, portable partitions, or other enclosures in order to prevent the escape of asbestos fibers from the contaminated asbestos work area. Negative pressure enclosure development must include protective covering of uncontaminated walls, and ceilings with a continuous membrane of two layers of minimum 0.15 mm plastic sheet sealed with tape to prevent water or other damage. Provide two layers of 0.15 mm plastic sheet over floors and extend a minimum of 300 mm up walls. Seal all joints with tape. Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

3.2.5.2 Regulated Area for Class II Removal

Removal of covebase/mastic, sealants, are Class II removal activities. Establish designated limits for the asbestos regulated work area with the use of red barrier tape; install critical barriers, splash guards and signs, and maintain all other requirements for asbestos control area except local exhaust. Place impermeable dropcloths on surfaces beneath removal activity extending out 3 feet in all directions. A detached decontamination system may be used. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If workers the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.6 Removal Procedures

Wet asbestos material with a fine spray of amended water during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 0.15 mm plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 0.15 mm plastic bags, submit an alternate proposal for containment of asbestos fibers to

the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in 40 CFR 61-SUBPART M.

3.2.6.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos lock-down encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants must comply with the performance requirements specified herein.

3.2.6.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 6 mm thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 100 mm. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

3.2.7 Methods of Compliance

3.2.7.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

- a. Vacuum cleaners equipped with HEPA filters.
- b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.
- c. Prompt clean-up and disposal.
- d. Inspection and repair of polyethylene.
- e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

3.2.7.2 Control Methods

Use the following control methods:

- a. Local exhaust ventilation equipped with HEPA filter;
- b. Enclosure or isolation of processes producing asbestos dust;
- c. Where the feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PELs, use them to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

3.2.7.3 Unacceptable Practices

The following work practices must not be used:

- a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.
- b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.
- c. Dry sweeping, shoveling, or other dry clean up.
- d. Employee rotation as a means of reducing employee exposure to asbestos.

3.2.8 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the installation and operation of the control methods.
- b. For jobs involving the removal of more than 7.5 m or 0.9 square m of TSI or surfacing material, place critical barriers over all openings to the regulated area.
- c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.
- d. Impermeable dropcloths (0.15 mm or greater thickness) must be placed on surfaces beneath all removal activity.
- e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

3.2.9 Specific Control Methods for Class I Work

Use Class I work procedures, control methods and removal methods for the following ACM:

- a. Spray Applied Fireproofing
- b. Gypsum Wallboard and Joint Compound
- c. Thermal System Insulation and Mudded Pipe Fittings

d. Plaster and Textured Ceilings and Walls

e. Vermiculite

3.2.9.1 Negative Pressure Enclosure (NPE) System

The system must provide at least four air changes per hour inside the containment. The local exhaust unit equipment must be operated 24-hours per day until the containment is removed. The NPE must be smoke tested for leaks at the beginning of each shift and be sufficient to maintain a minimum pressure differential of minus 0.5 mm of water column relative to adjacent, unsealed areas. Pressure differential must be monitored continuously, 24-hours per day, with an automatic manometric recording instrument and Records must be provided daily on the same day collected to the Contracting Officer. The Contracting Officer must be notified immediately if the pressure differential falls below the prescribed minimum. The building ventilation system must not be used as the local exhaust system for the regulated area. The NPE must terminate outdoors unless an alternate arrangement is allowed by the Contracting Officer. All filters used must be new at the beginning of the project and must be periodically changed as necessary and disposed of as ACM waste.

3.2.9.2 Mini-Enclosure

Single bulkhead containment Double bulkhead containment or Mini-containment (small walk-in enclosure) to accommodate no more than two persons, may be used if the disturbance or removal can be completely contained by the enclosure. The mini-enclosure must be inspected for leaks and smoke tested before each use. Air movement must be directed away from the employee's breathing zone within the mini-enclosure.

3.2.9.3 Class I Removal Method

Class I ACM must be removed using a control method described above. Prepare work area as previously specified. Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area. Spread one layer of 0.15 mm seamless plastic sheeting on the floor below the work area. Cut manageable sections of gypsum wallboard and joint compound and immediately place into a 0.15 mm minimum thickness disposal bag or other approved container. Make every effort to keep the material from falling to the floor of the work area. Use a wire brush and wet clean to remove residual material from studs. Continue wet cleaning until the surface is clean of visible material and encapsulate stud walls. Remove ACM plaster ceilings or walls using mechanical means and adequately wet methods and immediately place into 0.15 mm thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris. Remove ACM textured ceiling finish using a scraper and wet methods and immediately place into 0.15 mm thickness disposal bag. Floors are considered contaminated from fallen textured ceiling finish. Clean up debris on floor and dispose of as asbestos contaminated material. After removal of the material use a wire brush to clean the exposed concrete ceiling to remove residual material. Continue wet cleaning until surfaces are free of visible debris. Bag all asbestos debris which has fallen to the floor as asbestos-containing debris. Place all debris in plastic disposal bags of 0.15 mm minimum thickness. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged

asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers or the designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work, and immediately correct the situation.

3.2.10 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.11 Specific Control Methods for Class II Work

3.2.11.1 Vinyl and Asphaltic Flooring Materials Covebase and Mastic

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. When removing covebase and mastic which contains ACM, use the following practices. Remove covebase and mastic using adequately wet methods. Remove covebase and mastic intact (if possible). Do not sand flooring or its backing. Scrape residual adhesive and backing using wet methods. Mechanical chipping is prohibited unless performed in a negative pressure enclosure. Dry sweeping is prohibited. Use vacuums equipped with HEPA filter, disposable dust bag, and metal floor tool (no brush) to clean floors. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If workers the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.2 Sealants and Mastic

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers and signs, and maintain all other requirements for asbestos control area except local exhaust. Spread 0.15 mm plastic sheeting on the ground around the perimeter of the work area extending out in all directions. Using adequately wet methods, carefully remove the ACM sealants and mastics using a scraper of knife blade. As it is removed place the material into a disposal bag. Make every effort to keep the asbestos material from falling to the ground or work area floor below. Dry sweeping is prohibited. Use vacuums equipped with HEPA filter and disposable dust bag. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or at designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.3 Suspect Fire Doors

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. Spread 0.15 mm plastic sheeting on the ground beneath the work area and around the perimeter of the work area extending out in all directions. Remove door intact from hinges and wrap with 6-mil plastic sheeting. Inspect the interior areas of the door to determine if ACM is present. If ACM is not present the door may be disposed of as general construction debris. If ACM is present place whole door in enclosed container for disposal. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.4 Cementitious Siding and Shingles or Transite Panels

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing cementitious asbestos-containing siding, shingles or Transite panels use the following work practices. Intentionally cutting, abrading or breaking is prohibited. Each panel or shingle must be sprayed with amended water prior to removal. Nails must be cut with flat, sharp instruments. Unwrapped or unbagged panels or shingles must be immediately lowered to the ground via covered dust-tight chute, crane or hoist, or placed in an impervious waste bag or wrapped in plastic sheeting and lowered to the ground no later than the end of the work shift. Place

debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.5 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken. . Any scraping to remove residue must be performed wet. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.12 Abatement of Asbestos Contaminated Soil

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Asbestos contaminated soil must be removed from areas to a minimum depth of 50 mm. Soil must be thoroughly dampened with amended water and then removed by manual shoveling into labeled containers. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.13 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101 must be performed by the PQP. Unless otherwise specified, use NIOSH Method 7400 for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and made available to the Contracting Officer. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results.

3.2.13.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each demolition removal site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

3.2.13.2 Sampling During Asbestos Work

The PQP must provide personal sampling as indicated in 29 CFR 1926.1101. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and 72 hours (environmental/clearance monitoring) after completion of a sampling period. At the same time the GC will provide area sampling close to the work inside the enclosure, outside the clean room entrance to the enclosure, and at the exhaust opening of the local exhaust system. In addition, provided the same type of work is being performed, the GC will provide area sampling once every work shift close to the work inside the enclosure, outside the clean room entrance to the enclosure, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.01 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer immediately. Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels. The written results must be signed by testing laboratory analyst, testing laboratory principal and the Contractor's PQP. The air sampling results must be documented on a Contractor's daily air monitoring log.

3.2.13.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using NIOSH NMAM Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.01 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.01 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using NIOSH NMAM Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.01 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the

Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.13.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in 40 CFR 763, abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.13.5 Sampling After Final Clean-Up (Clearance Sampling)

The Contractor must perform any Clearance Sampling utilizing a third-party independent person who has no ties to the Contractor (except as being a subcontractor), the Asbestos Abatement Subcontractor, and the IH/PQP for the project. Provide area sampling of asbestos fibers and establish an airborne asbestos concentration of less than 0.01 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos work control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the PQP and GC must perform a visual inspection in accordance with ASTM E1368 to ensure that the asbestos control and work area is free of any accumulations of dirt, dust, or debris. Prepare a written report signed and dated by the PQP documenting that the asbestos control area is free of dust, dirt, and debris and all waste has been removed. The asbestos fiber counts from these samples must be less than 0.01 fibers per cubic centimeter or be not greater than the background, whichever is greater. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling and analysis at the Contractor's expense.

3.2.13.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

3.2.14 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the PQP must conduct a visual inspection of all areas affected by the removal in accordance with ASTM E1368. Inspect for any visible fibers, and to ensure that encapsulants were applied evenly and appropriately. Spray apply a post removal (lock-down) encapsulant to ceiling, walls, floors and other areas exposed in the removal area. The exposed area includes but is not limited to plastic barriers, furnishings and articles to be discarded as well as dirty change room, air locks for bag removal and decontamination chambers.

3.2.15 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3 CLEAN-UP AND DISPOSAL

3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The Contracting Officer will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in accordance with the JEGS, applicable U.S. Federal, GJO, and local regulations.

3.3.3 Disposal of Asbestos

3.3.3.1 Procedure for Disposal

Coordinate all waste disposal manifests with the Contracting Officer and NAVFAC EV. Collect asbestos waste, contaminated waste water filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 0.15 mm thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with 40 CFR 61-SUBPART M. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 0.15 mm thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of

each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete. Dispose of waste asbestos material at an approved asbestos landfill off Government property. For temporary storage, store sealed impermeable bags in asbestos waste drums or skids. An area for interim storage of asbestos waste-containing drums or skids will be assigned by the Contracting Officer or his authorized representative. Comply with 40 CFR 61-SUBPART M, G0J, regional, and local standards for hauling and disposal. Sealed plastic bags may be dumped from drums into the burial site unless the bags have been broken or damaged. Damaged bags must remain in the drum and the entire contaminated drum must be buried. Uncontaminated drums may be recycled. Workers unloading the sealed drums must wear appropriate respirators and personal protective equipment when handling asbestos materials at the disposal site.

3.3.3.2 Asbestos Disposal Quantity Report

Allow the GC to inspect, record and report the amount of asbestos containing material removed and released for disposal on a daily basis.

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SECTION 02 83 00

LEAD REMEDIATION
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSE/SAFE Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103 Respiratory Protection

29 CFR 1926.1126 Chromium

29 CFR 1926.1127 Cadmium

29 CFR 1926.21 Safety Training and Education

29 CFR 1926.33 Access to Employee Exposure and Medical Records

29 CFR 1926.55 Gases, Vapors, Fumes, Dusts, and Mists

29 CFR 1926.59 Hazard Communication

29 CFR 1926.62 Lead

UL SOLUTIONS (UL)

UL 586 (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

1.2 DEFINITIONS

1.2.1 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.2 Area Sampling

Sampling of lead, cadmium, chromium concentrations within the lead, cadmium, chromium control area and inside the physical boundaries which is representative of the airborne lead, cadmium, chromium concentrations but is not collected in the breathing zone of personnel (approximately 1.5 to 1.8 meters above the floor).

1.2.3 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.4 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium hazards in accordance with current U.S. federal, GOJ national, prefectural, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer.

1.2.5 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.6 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.7 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium or to plan such activities in commercial buildings, bridges or other structures.

1.2.8 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, chromium to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.2.9 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, chromium contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.10 Industrial Hygienist/Private Qualified Person (IH/PQP)

The IH/PQP shall have demonstratable experience in lead air monitoring techniques and in the establishment of a respiratory protection program for employees. The IH/PQP shall have working knowledge of applicable Federal and Japanese regulations occupational safety and health regulations for lead in construction. The IH/PQP shall have attended and completed Lead Abatement Supervision and Monitoring training and shall be considered a competent person as defined in 29 CFR 1926.62. The IH/PQP shall be a registered Architect, Professional Engineer, or Certified Industrial Hygienist, who has successfully completed training and is therefore accredited under a legitimate Model Accreditation Plan. The IH/PQP must be qualified to perform visual inspections.

1.2.11 Industrial Hygiene Technician (IHT)

The Industrial Hygiene Technician (IHT) shall be an employee of the IH/PQP or the testing laboratory and shall have a minimum of 2 years experience in the industrial hygiene field working under the direction of the IH/PQP and has completed the following course: Lead Abatement Supervision and Monitoring - Covering practices and procedures in lead abatement, lead air sampling and abatement monitoring.

1.2.12 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium. For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium.

1.2.13 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.14 Lead, Cadmium, Chromium Control Area

A system of control methods to prevent the spread of lead, cadmium, chromium dust, paint chips or debris to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.15 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

1.2.16 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this section does not apply if no detectable levels of lead are found using a quantitative method for analyzing paint or MCL using laboratory instruments with specified limits of detection (usually 0.01 percent). An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.17 Personal Sampling

Sampling of airborne lead, cadmium, chromium concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 150 to 225 mm and centered at the nose or mouth of an employee.

1.2.18 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, chromium control area to limit unauthorized entry of personnel.

1.3 DESCRIPTION

Construction activities impacting PWL or material containing lead, cadmium, chromium which are covered by this specification include the demolition or removal of material containing lead, cadmium, chromium intact condition. The work covered by this section includes work tasks and the precautions specified in this section for the protection of building occupants and the environment during and after the performance of the hazard abatement activities.

The Contractor shall assume that all coatings will contain some level of lead, cadmium, and Chromium. The Contractor shall review existing Lead Paint survey data.

1.3.1 Protection of Existing Areas To Remain

Project work including, but not limited to, lead, cadmium, chromium hazard abatement work, storage, transportation, and disposal must be performed without damaging or contaminating adjacent work and areas. Where such work or areas are damaged or contaminated, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, Chromium Compliance Plan and describe how the Contractor will prevent

lead, cadmium and chromium exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium activities.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Occupational and Environmental Assessment Data Report

Medical Examinations

Licenses, Permits and Notifications; G

Written Evidence of TSD Approval; G

Lead, Cadmium, Chromium Compliance Plan; G

SD-03 Product Data

Respirators

Vacuum Filters

Negative Air Pressure System

Materials and Equipment

Expendable Supplies

Local Exhaust Equipment

Pressure Differential Automatic Recording Instrument

Pressure Differential Log

SD-06 Test Reports

Occupational and Environmental Assessment Data Report; G

Sampling Results; G

SD-11 Closeout Submittals

Hazardous Waste Manifest

Turn-In Documents or Weight Tickets

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address,

and telephone number of the CP selected to perform responsibilities specified in paragraph COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, chromium; experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 3 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and certified in accordance with U.S. federal, GOJ national, prefectural and local laws. The competent person must be a trained and certified lead-based paint abatement Supervisor/Project Designer.

1.5.1.2 Training Certification

Include in the Lead, Cadmium, Chromium Compliance plan, a certificate for each worker and supervisor, signed and dated by the training provider, stating that the employee has received the required lead, cadmium and chromium training specified in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.5.1.3 Testing Laboratory

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address, and telephone number of the testing laboratory selected to perform the air analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium. Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved.

1.5.1.4 Third Party Consultant Qualifications

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address and telephone number of the third party consultant selected to perform the air sampling for determining concentrations of lead, cadmium and chromium in dust. Submit proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the USEPA authorized certification and accreditation program.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all U.S. federal, GOJ national, prefectural, and local requirements.
- b. Review and approve Lead, Cadmium, Chromium Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect work area for conformance with the approved plan.
- d. Perform (or oversee performance of) air sampling. Recommend upgrades

or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.

- e. Ensure work is performed in strict accordance with specifications at all times.
- f. Control work to prevent hazardous exposure to human beings and to the environment at all times.

1.5.2.2 Lead, Cadmium, Chromium Compliance Plan (LCCCP)

The purpose of the plan for this project is to verify that the Contractor understands that they will encounter paint with lead (PWL), cadmium and Chromium. The Contractor shall assume that all coatings will contain some level of lead, cadmium and chromium. The Contractor's LCCCP shall identify the Contractor's process to comply with all applicable regulations depending on the method of work the Contractor choose when encountering paints and coatings.

The Lead, Cadmium, Chromium Compliance Plan shall be reviewed and approved (with signature, date of approval, and certification number) by the IH/PQP.

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium, LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, chromium control areas, physical boundaries, and mechanical ventilation systems (when used). Include a description of equipment and materials, work practices, controls and job responsibilities for each activity from which lead, cadmium, chromium is emitted. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, chromium related work (if applicable), air sampling, respirators, personal protective equipment, and a detailed description of the method used to ensure that lead, cadmium, chromium containing dust is not released outside of the lead, cadmium, chromium control area. Include site preparation, cleanup procedures. Include occupational sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the air sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksites to inform affected employees and to clarify responsibilities to control exposures.

The plan must be developed and signed by the IH/PQP. The plan must include the name, certificate, and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to notify and protect the building occupants.

1.5.2.3 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational sampling results to the Contracting Officer within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 the Contractor must provide

documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, Chromium Compliance Plan.

- a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, chromium for stated work.
- b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 with a complete process description. This includes manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium containing coatings are present.
- c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, chromium compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.5.2.4 Medical Examinations

Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, chromium at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, chromium as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.5 Training

Train each employee performing work that disturbs lead, cadmium, chromium, who performs LBP/MCL/PWL disposal, and air sampling operations prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and GJO and local regulations where appropriate.

1.5.2.6 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.55.

1.5.2.7 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.8 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of U.S. federal, GOJ national, prefectural, and local authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and EM 385-1-1. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply. The following local laws, ordinances, criteria, rules and regulations regarding removing, handling, storing, transporting, and disposing of lead, cadmium and chromium-contaminated materials apply:

- a. Waste Disposal and Public Cleaning Law - licensing and certification in accordance with Japanese law

1.5.3 Pressure Differential Recordings for Local Exhaust System

When significant levels of dust are expected, or as required by the IH/PQP, provide a local exhaust system that creates a negative pressure of at least 0.51 mm of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, chromium control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer within 24-hours from the end of each work day.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the contract price.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and any other GOJ and local prefectural requirements on respirators.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, chromium contaminated dust must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining

approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during PWL or MCL handling and disposal, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

UL 586 labeled HEPA filters.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of the lead, cadmium and chromium removal work within the lead, cadmium and chromium controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machine for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ASSE/SAFE Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of minus 0.005 kPa relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.
- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 0.625 kPa static pressure differential on a magnehelic gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.005 kPa and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the

readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.

- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.
- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.
- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 590 degrees C. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 0.15 mm thick (0.1 mm thick if double bags are used) or any other thick plastic

material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, chromium contaminated debris must be polyethylene plastic that is a minimum of 0.15 mm thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 0.15 mm, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out of the ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 50 mm wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT, GOJ, local prefecture, and OSHA standards.

2.1.1.6 Chemical Paint Strippers

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium control work using cleaning techniques.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Lead, Cadmium, Chromium Control Area

- a. Physical Boundary: Provide physical boundaries around the lead, cadmium, chromium control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium will not escape outside of the lead, cadmium and chromium control area. Prohibit the general public from accessing the lead, cadmium, chromium control areas.
- b. Warning Signs: Provide bilingual warning signs in English and Japanese at approaches to lead, cadmium, chromium control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.2 Furnishings

The Government will remove furniture and equipment from the building before lead, cadmium and chromium work begins.

3.1.1.3 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, chromium control areas. Seal intake and exhaust vents in the lead, cadmium, chromium control area with 0.15 mm plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, chromium control area. Provide temporary HVAC system for areas in which HVAC has been shut down outside the lead, cadmium, chromium control area.

3.1.1.4 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, chromium control area in accordance with ASSE/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the lead, cadmium, chromium control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, chromium control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, chromium control area. Filters on exhaust equipment must conform to ASSE/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.5 Negative Air Pressure System Containment

- a. When used, operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the

local exhaust unit equipment continuously until the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium contaminated waste.

- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during abatement activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Dismantling of the negative air pressure systems must conform to written decontamination procedures be as presented in the Lead, Cadmium, Chromium Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination.

3.1.1.6 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.7 Mechanical Ventilation System

- a. Use adequate ventilation to control personnel exposure to lead, cadmium and chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. To the extent feasible, use local exhaust ventilation or other collection systems, approved by the CP. Evaluate and maintain local exhaust ventilation systems in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.
- b. Vent local exhaust outside the building and away from building ventilation intakes or ensure system is connected to HEPA filters.
- c. Use locally exhausted, power actuated tools or manual hand tools.

3.1.1.8 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, chromium control area. No one will be permitted in the lead, cadmium, chromium control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, Chromium Control Area Requirements

Establish a lead, cadmium, chromium control area by completely establishing barriers and physical boundaries around the area or structure

where PWL or MCL removal operations will be performed.

3.3 APPLICATION

3.3.1 Lead, Cadmium, Chromium Work

Perform lead, cadmium, chromium work in accordance with approved Lead, Cadmium, Chromium Compliance Plan. Use procedures and equipment required to limit occupational exposure and environmental contamination with lead, cadmium, chromium when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with the JEGS. U.S. federal, GOJ national, prefectural laws and regulations, and local requirements.

3.3.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium Removal

Manual or power sanding or grinding of lead, cadmium, chromium surfaces or materials is not permitted unless tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead, cadmium, chromium is prohibited. Provide methodology for removing lead, cadmium, chromium in the Lead, Cadmium, Chromium Compliance Plan. Select lead, cadmium, chromium removal processes to minimize contamination of work areas outside the control area with lead, cadmium, chromium contaminated dust or other lead, cadmium, chromium contaminated debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, chromium. Describe this removal process in the Lead, Cadmium, Chromium Compliance Plan.

Provide methodology for lead, cadmium and chromium, LBP/PWL control and processes to minimize contamination of work areas outside the control area with lead, cadmium, chromium contaminated dust or other lead, cadmium, chromium contaminated debris/waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, chromium. Describe this lead, cadmium and chromium, LBP/PWL removal/control process in the Lead, Cadmium, Chromium Compliance Plan.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, chromium controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing.
- b. Remove protective clothing in the specified change room, and place them in an approved impermeable disposal bag.
- c. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air Sampling

Conduct sampling for lead, cadmium, chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and as specified

herein. Air sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing the air sampling and inspecting the PWL or MCL removal work to ensure that the requirements of the contract have been satisfied during the entire PWL or MCL operation.
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.
- c. Submit results of air samples, signed by the CP, within 72-hours after the air samples are taken.
- d. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air is reached or exceeded, stop work, correct the condition(s) causing the increased levels. Notify the Contracting Officer immediately. Determine if condition(s) require any further change in work methods. Resume removal work only after the CP and the Contracting Officer give approval.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, chromium control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead, cadmium, chromium operation has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area as indicated by the Lead, Cadmium, Chromium Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become contaminated at any time during the work, clean, visually inspect, and then wipe sample all contaminated areas. The CP must then certify in writing that the area has been cleaned of lead, cadmium and chromium contamination before clearance testing.

3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with all laws and provisions and all JEGS, U.S. federal, GOJ national, prefectural laws and regulations or local regulations. Ensure all waste is properly characterized. The result of each waste characterization (TCLP for RCRA materials) will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, chromium contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, chromium contaminated clothing that may produce airborne concentrations of lead, cadmium, chromium particles. Label the containers in accordance with the JEGS, 29 CFR 1926.62,

29 CFR 1926.1126, 29 CFR 1926.1127.

- c. Dispose of lead, cadmium, chromium contaminated material classified as hazardous waste at a locally approved hazardous waste treatment, storage, or disposal facility off Government property.
- d. Accumulate waste materials in drums or appropriately sized containers. Properly label each drum to identify the type of hazardous material. For hazardous waste, the collection container requires marking/labeling in accordance with the JEGS and corresponding GOJ national and local prefectural regulations during the accumulation/collection timeframe. The Contracting Officer or an authorized representative will assign an area for accumulation of waste containers. Coordinate authorized accumulation volumes and time limits with the host installation environmental function.
- e. All lead, cadmium, and chromium waste generation, management, and disposal will be coordinated with the host installation environmental function.

3.5.2.1 Disposal Documentation

Coordinate all disposal or off-site shipments of lead, cadmium, and chromium waste with the host installation environmental function. Submit written evidence of TSD approval to demonstrate the hazardous waste treatment, storage, or disposal facility (TSD) is approved for lead, cadmium, chromium disposal by the EPA, State or local regulatory agencies. Submit one copy of the completed hazardous waste manifest, signed and dated by the initial transporter in accordance JEGS, GOJ National, prefectural laws and local regulations. Provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Hazardous Waste

Payment for disposal of hazardous and non-hazardous waste will not be made until a signed copy of the manifest from the treatment or disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of lead, cadmium, chromium containing materials or non-hazardous waste delivered to the treatment or disposal facility.

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SECTION 02 84 16

HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

MINISTRY OF THE ENVIRONMENT GOVERNMENT OF JAPAN (MOE)

Act No. 137 (1970, Amended 1991 and 2006) Waste
Management and Public Cleansing Law

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing
Standards

1.2 REQUIREMENTS

Removal and disposal of PCB containing lighting ballasts and associated mercury-containing lamps. Contractor may encounter leaking PCB ballasts. PCB ballasts and/or mercury containing lighting fixtures or other equipment may be encountered in additional locations not listed in the drawings or the project Hazardous Materials Inspection Report(s). Inspect, confirm, and identify all PCB-containing ballasts and any other PCB-containing, mercury-containing, or otherwise hazardous material as part of project work.

1.3 DEFINITIONS

1.3.1 IH/PQP

An industrial hygienist/private qualified person (IH/PQP) hired by the Contractor shall be a registered Architect, Professional Engineer (licensed), or US Certified Industrial Hygienist and having demonstrable experience in hazardous materials management (i.e. lighting ballasts and lamps containing PCBs or mercury), who is trained in the recognition and control of hazardous chemical related hazards, and has the authority to take prompt corrective actions to control the hazard.

1.3.2 Leak

Leak or leaking means any instance in which a PCB article, PCB container, or PCB equipment has any PCBs on any portion of its external surface.

1.3.3 Lamps

Lamp is defined as the bulb or tube portion of an electric lighting device. A lamp is specifically designed to produce radiant energy, most often in the ultraviolet, visible, and infra-red regions of the electromagnetic spectrum. Examples of common universal waste electric lamps include,

but are not limited to, fluorescent, high intensity discharge, neon, mercury vapor, high pressure sodium, and metal halide lamps.

1.3.4 Polychlorinated Biphenyls (PCBs)

PCBs as used in this specification shall mean the same as PCBs, PCB containing lighting ballast, and PCB container, as defined in JEGS, Chapter 7, Polychlorinated Biphenyls.

1.3.5 Spill

Spill means both intentional and unintentional spills, leaks, and other uncontrolled discharges when the release results in any quantity of PCBs running off or about to run off the external surface of the equipment or other PCB source, as well as the contamination resulting from those releases.

1.4 QUALITY ASSURANCE

1.4.1 Regulatory Requirements

Perform PCB related work in accordance with JEGS, Act No. 137, and other applicable US Federal, GOJ, and local laws and regulations. Perform mercury-containing lamps storage and transport in accordance with JEGS, Act No. 137, and other applicable US Federal, GOJ, and local laws and regulations.

1.4.1.1 PCB and Lamp Management and Disposal Plan

Prior to handling any PCB items, the Contractor shall submit a PCB and Lamp Management and Disposal Plan. The submitted plan shall include the PCB and Lamp Removal Work Plan, PCB and Lamp Disposal Plan, Qualifications of the IH/PQP, and Japan Industrial Waste Collection and Transport Permit.

1.4.2 Training

Industrial Hygienist/Private Qualified Person (IH/PQP) shall instruct and certify the training of all persons involved in the removal of PCB containing lighting ballasts and mercury-containing lamps. The instruction shall include: The dangers of PCB and mercury exposure, decontamination, safe work practices, and applicable OSHA and EPA, JEGS, GOJ, and prefectural regulations.

The IH/PQP shall review and approve the PCB and Mercury-Containing Lamp Removal Work Plans and temporary on-site Storage Plans.

1.4.3 Regulation Documents

Maintain at all times one copy each at the office and one copy each in view at the job site of pertinent JEGS Chapter 16, Hazardous Waste and Chapter 7, Polychlorinated Biphenyls, JEGS Appendix 1, and a copy of the Contractor PCB and Lamp Removal and Disposal Work Plan.

1.4.4 Qualifications of IH/PQP

An industrial hygienist/private qualified person (IH/PQP) hired by the Contractor shall be a registered Architect, Professional Engineer (licensed), or US Certified Industrial Hygienist and having demonstrable experience in hazardous materials management (i.e. lighting ballasts and

lamps containing PCBs or mercury), who is trained in the recognition and control of hazardous chemical related hazards, and has the authority to take prompt corrective actions to control the hazard. An IH/PQP must have working knowledge of applicable GOJ, Federal, local prefecture, and JEGS regulations as well as occupational safety and health regulations and shall be capable of recognizing chemical hazards associated with lighting ballasts, lamps, and other similar building equipment or materials.

The IH/PQP shall have at least 2 years of on the job experience in hazardous materials and hazardous waste management and have working knowledge of the JEGS, GOJ, as well as US and Japanese occupational safety and health regulations. The IH/PQP shall have knowledge in detecting, identifying, and evaluating existing or potential hazardous conditions related to PCB-containing light ballasts and mercury-containing lamps and other similar building equipment or materials. The IH/PQP may be considered qualified if he/she is either a certified Industrial Hygienist or is a Private Qualified Person with adequate prerequisite experience and knowledge as listed in this section.

1.4.5 PCB and Lamp Removal Work Plan

Submit a job-specific plan to be incorporated within the PCB and Lamp Management and Disposal Plan within 20 calendar days after prior to removal work explaining the work procedures to be used in the removal, packaging, and storage of PCB-containing lighting ballasts and associated mercury-containing lamps.

Include in the plan: Requirements for Personal Protective Equipment (PPE), spill cleanup procedures and equipment, eating, smoking and restroom procedures.

The Plan shall comply with applicable requirements of Federal, JEGS, GOJ and prefectural PCB and mercury-related regulations. The plan shall be approved and signed by the IH/PQP. Obtain approval of the plan by the Contracting Officer prior to the start of PCB and/or lamp removal work.

1.4.6 PCB and Lamp Disposal Plan

Submit a PCB and lamp Disposal Plan to be incorporated within the PCB and Lamp Management and Disposal Plan, within 45 calendar days after prior to removal work. The PCB and Lamp Disposal Plan shall comply with applicable requirements of the JEGS and applicable U.S. federal, GOJ, and local prefecture and JEGS PCB regulations and address:

- a. Estimated quantities of wastes to be generated, disposed of, and recycled.
- b. Names and qualifications of each Contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location. Furnish two copies of PCB and mercury-containing lamp waste permit applications and pertinent identification numbers, as required.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with PCB and mercury-containing lamp wastes.
- d. Spill prevention, containment, and cleanup contingency measures to be implemented.
- e. Work plan and schedule for PCB and mercury-containing lamp waste

removal, containment, storage, transportation, disposal and or recycling. Wastes shall be cleaned up and containerized daily.

1.4.7 Notification

Notify the Contracting Officer 20 days prior to the start of PCB removal work.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Notification

PCB and Lamp Management and Disposal Plan; G

SD-11 Closeout Submittals

Disposal Request Form

Transporter Certification; G

Certification Of Decontamination

1.6 ENVIRONMENTAL REQUIREMENTS

Use special clothing:

- a. Disposable gloves (polyethylene)
- b. Eye protection
- c. PPE as required by IH/PQP

1.7 SCHEDULING

Notify the Contracting Officer 20 days prior to the start of PCB and mercury-containing lamp removal work.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 WORK PROCEDURE

Furnish labor, materials, services, and equipment necessary for the removal of PCB containing lighting ballasts, associated mercury-containing fluorescent lamps, and high intensity discharge (HID) lamps in accordance with the JEGS, applicable GOJ and local, prefectural regulations. Do not expose PCBs to open flames or other high temperature sources since toxic decomposition by-products may be produced. Do not break mercury

containing fluorescent lamps or high intensity discharge lamps.

3.1.1 Work Operations

Ensure that work operations or processes involving PCB or PCB-contaminated materials are conducted in accordance with the JEGS, applicable U.S. Federal, GOJ, and local laws and regulations, and the applicable requirements of this section, including but not limited to:

- a. Obtaining suitable PCB and mercury-containing lamp storage sites.
- b. Notifying Contracting Officer prior to commencing the operation.
- c. Reporting leaks and spills to the Contracting Officer.
- d. Cleaning up spills.
- e. Inspecting PCB and PCB-contaminated items and waste containers for leaks and forwarding copies of inspection reports to the Contracting Officer.
- f. Maintaining inspection, inventory and spill records.

3.2 PCB SPILL CLEANUP REQUIREMENTS

3.2.1 PCB Spills

The Contractor shall respond to spills of PCB-containing materials by following Attachment 02 84 16-A EMP 07.3 PCB Item Management For Contractors SOP. Immediately report to the Contracting Officer any PCB spills. The Contractor is responsible to keep emergency spill kits kept within walking distance from the job site in accordance to requirements in Attachment 02 84 16-A.

3.2.2 PCB Spill Control Area

Rope off an area around the edges of a PCB leak or spill and post a bilingual "PCB Spill Authorized Personnel Only" caution sign in English and Japanese. Immediately transfer leaking items to a drip pan or other container.

3.2.3 PCB Spill Cleanup

Initiate cleanup of spills as soon as possible, but no later than 24 hours of its discovery. Mop up the liquid with rags or other conventional absorbent. The spent absorbent shall be properly contained and turned over to the MCB Butler Environmental Support Team (EST) as solid PCB waste. Follow all requirements in Attachment 02 84 16-A.

3.2.4 Records and Certification

Document the cleanup with records of decontamination in accordance with the JEGS Chapter 7, Polychlorinated Biphenyls and applicable US Federal, GOJ, and prefectural requirements for PCB Spill Cleanup. Provide test results of cleanup and certification of decontamination.

3.3 REMOVAL

3.3.1 Ballasts

Light ballasts may still contain PCBs even where a "No PCB" label exists on the ballast due to regulatory differences between U.S. and G0J definitions for "PCB Free". Regardless of PCB concentration, manufacturer, country of origin, all light ballasts shall be segregated by country of manufacture/origin, removed, containerized, and turned into the Installation Environmental Division, in accordance with Attachment 02 84 16-A. PCB abatement or handling shall be in accordance with JEGS Chapter 7 and Attachment 02 84 16-A at the end of this section.

The Contractor shall identify all project ballasts that contain and do not contain PCBs or are suspect to contain PCBs. The Contractor shall submit the PCB ballast list (to include suspect PCB ballasts) to the Contracting Officer Representative (COR). PCB containing ballasts and suspect PCB containing ballasts shall be segregated by country of manufacture.

The following procedures for US manufactured and Japan manufactured ballasts shall be used in the identification of PCB ballasts and described in this section.

3.3.1.1 For US Manufactured Ballasts

US manufactured ballasts shall not be mixed with Japan manufactured ballasts. The Contractor shall segregate and palletize US manufactured ballasts separately from Japan manufactured ballasts at the point of generation for subsequent turn in to the MCB Butler EST (HW/PCB Program Manager, 098-970-3139 or 098-970-5790). All ballasts, including "No PCB" labeled ballasts and non-PCB containing ballasts, shall be containerized and turned into the MCB Butler EST.

3.3.1.2 For Japan Manufactured Ballasts

The Contractor shall carefully examine the ballast to identify the manufacturer name and year made. Consult with the manufacturer to identify any presence of PCBs. If absence of PCB cannot be confirmed, assume and treat the ballast as a PCB item.

Japan manufactured ballasts shall not be mixed with US manufactured ballasts. The Contractor shall segregate and palletize Japan manufactured ballasts separately from US manufactured ballasts at the point of generation for subsequent turn in to the MCB Butler EST (HW/PCB Program Manager, 098-970-3139 or 098-970-5790).

3.3.1.3 For Unmarked/Unlabeled Ballasts

The Contractor shall treat all unmarked/unlabeled ballasts as suspect PCB-containing materials. Segregate unmarked/unlabeled ballasts from other US or Japanese manufactured items. The Contractor shall contact the MCB Butler EST (HW/PCB Program Manager, 098-970-3139 or 098-970-5790) to establish proper procedures for the management of unmarked/unlabeled ballasts.

3.3.2 Fluorescent Light Tubes (Bulbs) - Fluorescent Lamps

The Contractor shall carefully remove and store fluorescent light bulbs in a manner such that they remain intact. Fluorescent light tubes suspect

for containing mercury shall be handled and disposed of by the Contractor in accordance to all GOJ, Federal, local prefectural laws and regulations and the JEGS. In the event of a lighting tube/lamp breaking, sweep and place waste in double plastic taped bags and dispose of as appropriate to GOJ, Federal, local prefecture laws and regulations and the JEGS.

3.4 STORAGE FOR DISPOSAL

3.4.1 Storage Containers for PCBs

Store ballasts in accordance with JEGS and this section. Ballasts must be in secondary containment that is not exposed to rainfall.

3.4.2 Storage Containers for Lamps

Store mercury containing lamps in appropriate containers. The boxes shall be stored and labeled for transport in accordance with JEGS, Act No. 137, the MOE Waste Management, and Public Cleansing Law, and applicable US Federal, GOJ, and local requirements.

3.4.3 Labeling of Waste Containers

In accordance with JEGS, Chapter 7, prominently label in both English and Japanese with the following:

- a. Date the item was placed in storage and the name of the cognizant activity/building.
- b. Identification of items, warning against improper disposal and handling, and phone number in case of spills, conforming to JEGS. Affix labels to PCB waste containers.
- c. Label mercury-containing lamp waste in accordance with JEGS and Act No. 137, the MOE Waste Management and Public Cleansing Law. Affix labels to all lighting waste containers.

3.5 DISPOSAL / TURNOVER

3.5.1 Japan Specially Controlled Hazardous Waste Disposal Permit

In the submitted PCB and Lamp Management and Disposal plan, include a copy of the Japan Specially Controlled Industrial Waste (SCIW) Permit ("Tokubetsu Kanri Haikibutsu Shobungyou Kyoka") for the disposal or recycling facility where the mercury-containing lamps will be disposed. The permit must specifically say that the firm is authorized to accept mercury.

3.5.2 Japan Industrial Waste Collection and Transport Permit

In the submitted PCB and Lamp Management and Disposal plan, include a copy of the Japan Industrial Waste Collection and Transport Permit ("Sangyou Haikibutsu Shuushuu Unpangyou Kyoka") for the transportation of mercury-contaminated wastes from the installation.

Turn in , ballasts, and other PCB or mercury-containing materials to the Installation Environmental Division. Contact Installation Environmental Division at least 5 working days in advance to make arrangements for delivery of these materials to the storage site.

Do not dispose of PCB related materials at an off Government facility. Do not consider US manufactured light ballasts with "No PCBs" labeling to contain PCBs less than Japan regulatory limits and always acquire PCB content related information from the manufacturer. All ballasts, shall be turned into the Installation Environmental Division per the turnover procedures stated in paragraph TURNOVER PROCEDURES.

3.5.3 Turnover Procedures

- a. Segregate ballasts by manufacturer.
- b. Visit the websites of the manufacturers to verify if the ballasts contain PCB or not by product model number, product color, etc. Certificates shall be obtained stating that the ballast does not contain PCB. If the ballast does not contain PCB in accordance with the certificate, then the ballast should not be turned over to the installation.
- c. Sort the ballast in cardboard cases by PCB containing or unknown for each manufacturer and country of origin.
- d. Bring ballasts to the Installation Environmental Division with certificates and supporting documents.

3.5.4 Transporter Certification

Comply with disposal and transportation requirements outlined in the JEGS. Before transporting the PCB waste, prepare a Disposal Request Form for acknowledgement of acceptance of the PCB waste from the Government. Return a duly signed copy to the Government before leaving the Installation Environmental Division. Ensure that the Disposal Request Form accompanies the PCB or mercury-containing waste at all times.

3.5.5 Disposal by the Government

Comply with disposal and transportation requirements outlined in the JEGS Chapter 16 and Chapter 7. Load and haul PCBs to the Installation Environmental Division.

3.5.5.1 Delivery

Contact Installation Environmental Division at least 5 working days in advance to make arrangements for delivery of PCB or mercury-containing items to the storage site.

3.5.5.2 Disposal Request Form

Prepare Disposal Request Form, which will accompany the PCB to the storage site. Contact Installation Environmental Division for a copy of the latest Disposal Request Form version. Ensure that a responsible person from the activity that owns the PCB signs the Disposal Request Form.

-- End of Section --

We are committed to promoting responsible stewardship while continually improving environmental performance through an effective and efficient Environmental Management System

1. Purpose. The purpose of this Standard Operating Procedure (SOP) is to establish procedures and guidelines for the removal from service and turn-in for disposal of Polychlorinated Biphenyls (PCBs) and items that could potentially contain PCBs.
2. Application. This guidance applies to all contractors and/or subcontractors providing services aboard Marine Corps Installations on Okinawa under US funded projects. This SOP does not cover projects under the jurisdiction of the Government of Japan or Government of Japan-funded projects.
3. References.
 - (a) Japan Environmental Governing Standards (JEGS)
 - (b) 40 CFR 761
 - (c) EMSP 06.8 Contractor-Vendor Training Packet
 - (d) EMP 07.1.1 PCB Spill Response and Notification Procedures
 - (e) Emergency Response Guide Ver 2020 guide number 171
 - (f) United States Environmental Protection Agency Method 8082A
 - (g) EMP 16.1.1 Camp to Camp Manifest
4. Overview. Care shall be taken when personnel handle, remove, and turn-in items that may contain PCBs. Contractors may be required to remove, sample, handle, label, manifest/transport, and turn-in items including, but not limited to - Transformers, Switches, Medium/High Voltage Cables, Paper Insulate Lead Cable (PILC), Potheads, Capacitors, Ballasts, Lamp Ignitors, and Sealed Transformers. These items have the potential to contain PCBs. This SOP outlines procedures for contractor handling of each type of PCB containing article.
5. Procedure.
 - 5.1. Operational Control.
 - a. Item Handling and Storage.
 - (1) Tarps, containments, absorbent berms, and spill kits shall always be used to provide protection for personnel and the environment during operations.
 - (2) Contracting officers or representatives are required to provide supervision during removal operations; and G-F Environmental Affairs Branch personnel may be consulted to provide guidance or clarification when needed.
 - (3) Contractor is responsible to fund and conduct sampling and analysis on any potential PCB-containing items or equipment being removed or demolished. A list of laboratories capable of analyzing PCBs may be provided upon request. Laboratories conducting analysis shall be trained and qualified in Japanese and US EPA methodologies.
 - (a) Sampling shall be completed using a combined laboratory method report with the statement the analysis was conducted by "Simplified Manual for Determination of Trace Amount of PCBs in Insulation Oil Manual", Third Edition and the remark of "The above analytical method is equivalent to US EPA test method 600/4-81-045".
 - (b) All samples will be noted in a log (digital or hard copy); this will be maintained by the contractor and available for review by the MCB Butler PCB Program Manager.

Approved By: ROEN.DAVID.T.
II.1160921486

Digitally signed by
ROEN.DAVID.T.II.1160921486
Date: 2023.10.12 16:50:19
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(4) The Contracting Officer Representative (COR) will forward all sample results to the MCB Butler Hazardous Waste / PCB Program Manager and Environmental Support Team (EST), including sample results Non-Detect (ND) for the presence of PCBs. MCB Butler EAB will maintain sampling records and provide concurrence for removal and disposal methodology at no additional cost to the contractor.

(5) All potential PCB-containing equipment will be segregated by type and further segregated into make, model, and manufacturer for items such as ballasts. MCB Butler HW/PCB PM may be contacted for a list of ballasts already sampled.

(6) Negligent spills or illegal disposal methods as determined by the Marine Corps or COR will be the responsibility of the contractor for clean-up and management.

(7) Labeling and marking of drums will follow examples provided on Pages 6 & 7 – Proper UN POP Container Labeling and Marking for Turn-in.

b. PCB Containing Materials.

(1) Transformers/Switches (free flowing liquid types). Most transformers contain free flowing dielectric oil. Some, classified as hermetically sealed, may contain dry dielectric potting material or semi-solid tar (see Ballast/Sealed Transformer on Section 5.1.b (4).

(a) Oil-filled transformers will be placed on metal drip pans with absorbent sheets placed between the transformer base and drip pan when not in service.

(b) Drip pans will be placed to ensure all four sides of the transformer, including all parts (e.g., insulators) are contained in the event of leaks.

(c) During storage, the transformer and drip pan will be strapped or banded together and placed indoors for protection from weather and the elements.

(d) During transport, the transformer and drip pan will be strapped or banded together, placed on a truck rated to support the load, and secured with straps to prevent movement.

(2) PILC Cable/Splices. PILC does not have free flowing liquid; however, the paper insulation will leak at open ends, even at ambient temperature.

(a) Blue polyethylene tarps that are in good condition with a minimum of 10 mm thickness shall be placed around manholes and under spools of PILC during temporary storage and removal.

(b) Cables shall be placed on a plastic spool during removal and handled to minimize cuts. MCB Butler EAB Environmental Support Team (EST) personnel office will provide plastic spools for contracting crews prior to removal activities commence.

(c) Cable ends, or any area where sheathing is compromised, are to be wrapped and sealed with electrical tape to prevent leaks.

(d) Spools shall be marked to identify which manhole the PILC originated from. Place spools on a polyethylene tarp and secure them inside an enclosed structure for protection from the elements.

(e) Arrange turn in schedule of items with the EST office prior to removal to minimize temporary storage in the construction areas.

(f) Splices will be placed in United Nations Performance-Oriented Packaging (UN POP) certified open-top drums; and drums shall be marked to identify the manhole location where the splices originated. Drums may not be filled to more than 700 lbs.

(g) All debris items, including but not limited to tarps, rags, gloves, and absorbents, that come in contact with the cable during cable removal operations will be placed in UN POP certified open-top drums and marked with the contract number. Debris will be containerized daily to prevent cross contamination, as in the case of high winds.

(h) Tools must be stored in UN POP certified containers daily and stored in secure location out of inclement weather. Upon completion of project work all tools known to have come into contact with PCBs will be decontaminated by the contractor and reanalyzed for PCB surface contamination prior to return to use.

(3) Potheads (Bell Assemblies). Potheads contain a semi-solid tar that will leak when exposed to excessive ambient temperatures. These items will be placed in UN POP certified open-top drums and labeled as "POTHEAD". Drums will be stored in a secured location out of the weather. These items will be segregated by manufacturer per drum, i.e., Japanese and American.

(4) Ballasts and Sealed Transformers. These items contain semi-solid tars hermetically sealed within the casing.

(a) Ballasts will be segregated by model and manufacturer, i.e., Japanese, and American. Personnel should use caution when removing and handling these items so as not to damage the manufacturer's label.

(b) Electrical wiring will be cut off at the casing and the ballasts then placed in a single layer on wooden pallets, segregated by country of origin.

(c) Ballasts in temporary storage will be placed in a secure location and protected against weather.

(d) Sealed transformers will be segregated by country of manufacture and placed in appropriate UN POP certified open-top drums and marked as "SEALED TRANSFORMER" with the corresponding contract number. Temporary storage will be in a secure location and protected against the weather. Drums may not be filled to more than 700 lbs.

c. Procurement.

(1) Contractors that require purchasing of items such as transformers, switches, circuit breakers, capacitors, and light ballasts to complete a renovation or establish a new facility infrastructure are required to obtain a manufacturer's certification stating, in English & Japanese, the item is "PCB Free" as required by JEGS.

(a) Ensure all procurement of transformers or any other equipment containing dielectric or hydraulic fluid includes a manufacturer's certification that the equipment contains no detectable PCBs (less than 0.5 ppm).

(b) Manufacturer statements that show "PCB FREE" or data plate statements with "less than 1 ppm at the time of manufacture" are NOT acceptable.

(c) Copies of the certificate, item serial number, and installed location will be forwarded to the Hazardous Waste/PCB Program Manager and to the Facilities Maintenance Officer. Records will be uploaded into USMCMMax asset management for the duration of the item's service life.

d. Waste turn-in/Transportation.

(1) Contractors will call to pre-arrange a turn-in date and time for accumulated PCB-containing waste. In Okinawa, contractors will contact Environmental Support Team to arrange for turn-in at Camp Kinser, Building #407.

(2) All PCB-related items turned-in will be documented with a waste disposal manifest (EMP 16.1.1 Camp-to-Camp Manifest). The form will accompany the waste from the time the waste leaves the point of origin and will remain with the vehicle driver until reaching the final destination.

(3) Contractor vehicles transporting waste are required to have an emergency spill kit (see Section 5.3). Loads need to be secured to minimize movement in all four directions with appropriate weight-acceptable straps or chains.

(4) Vehicle drivers are required to have a means to communicate in case of an emergency, i.e., cell phone or radio transmitter.

(5) Contractors must utilize EMP 16.1.1 Camp to Camp Manifesting for all transportation.

5.2. Inspection: Spill kits should be inspected on a weekly basis to verify that the kit is complete. An inventory checklist must be posted within the container. Adequate and appropriate response equipment will be replenished immediately by the contractor.

5.3. Emergency Preparedness and Response Procedures.

a. On-site.

(1) Emergency spill kits are required when construction projects begin operations on potential PCB-containing items. The quantity of spill supplies within the spill kits must be adequate to contain a release depending on the size of the equipment being moved.

(2) Spill kits must be within walking distance from the job site and placed in an open-top drum or roll out box for easy access. Spill kit items shall include, but are not limited to, the following:

- Dry sweep
- Oil only absorbent pads or sheets
- Oil only absorbent socks
- All-purpose absorbent or sheets (for use with oil/water mix)
- All-purpose absorbent socks (for use with oil/water mix)
- Nitrile inner gloves
- Butyl outer gloves
- Tyvek inner suits
- Saranex splash guard suits
- Face shields

b. During Transport

- (1) Spill kits are mandatory in vehicles transporting waste on-base and on local roads.
- (2) In the event of a release, contractors are only authorized to stop, dike, and dam the spill. EST must be contacted immediately and will perform clean-up. Caution must be used when applying absorbents to minimize oil contact with the environment, equipment, and personnel.
- (3) Report the spill by calling 9-1-1 from on-base, and 098-911-1911 from off-base telephone or cell phone. The EST Supervisor can be contacted by cell at 090-6861-5818.
- (4) Contractors must note all items that come into contact with PCB related oils/tar and provide this and the data plate information to the on-scene emergency response personnel at arrival.

c. In case of a fire, the contractor will evacuate all personnel to a minimum of 800 meters (1/2 mile) in all directions.

5.4. Training. It will be the responsibility of the contractor to provide PCB awareness training before operations begin. Training on this SOP is mandatory. MCB Butler Environmental can provide limited guidance as requested.

5.5. Documentation and Record Keeping. Contractors must keep a copy of the manifests and training records. Hazardous Waste/PCB Program Manager and the Facilities Maintenance Officer must keep a copy of the certificate, item serial number, and installed location of PCB-related items provided by contractors.

6. If you have any questions or need additional information, please call your respective Camp/Station Environmental office.

EST Team	645-7189/3139/2083
HW/PCB Program Manager	645-1060
Camp Foster/Lester	645-5970/0789
MCAS Futenma	636-2066/2067
Camp Kinser	637-4405/1885
Camp Hansen/Courtney/CTA /Ie-Shima	623-4495/7223
Camp Schwab/JWTC	625-2683/7194

Proper UN POP Container Labeling and Marking for Turn-in

CAUTION CONTAINS PCBs (POLYCHLORINATED BIPHENYLS) SOVEREIGN LAW PROHIBITS IMPROPER DISPOSAL A toxic environmental contaminant requiring special handling and disposal. For disposal information or in case of accident or spill, contact below or emergency contact : 注 意 PCB (ポリ塩化ビフェニル) 含有物 不適切処分禁止 有毒環境汚染物質につき特別な取り扱い及び処分を要す 処分方法、事故又は流出時の連絡先は下記又は 非常時の連絡先: Point of Contact (担当者): <u>MCBB PCB Program Manager</u> Phone (電話番号): <u>DSN 315-645-1060</u> USFJ LABEL No. 4 APR 2022
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The diagram shows a cylindrical container with three main sections. The top section is a black oval labeled "Contract #". The middle section is a white cylinder with a black outline, labeled "WT _____" and "Nomenclature of Item (e.g., Ballasts JN)". The bottom section is a black oval. A yellow warning label is affixed to the middle section. The label contains the following text:

CAUTION
CONTAINS
PCBs
(POLYCHLORINATED BIPHENYLS)
A toxic environmental contaminant, requiring special handling and disposal.
For disposal information or in case of accident or spill, contact local or emergency contact.
注意
PCB (ポリ塩化ビフェニル) 含有物
本製品は有害物質に分類され、取り扱いに注意が必要です。
処分方法、事故又は漏洩時の連絡先は下記又は
作業員の連絡先
Point of Contact (担当者) _____
Phone (電話番号) _____
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SECTION 03 01 00

REHABILITATION OF CONCRETE
02/18

PART 1 GENERAL

1.1 SCOPE

This specification governs the rehabilitation of structural concrete.

1.2 DEFINITIONS

1.2.1 Bracing

Temporary supplemental members used to avoid local or global instability during construction, evaluation, or repair that are intended to be removed after completion of construction.

1.2.2 Delamination

A planar separation in a material that is roughly parallel to the surface of the material.

1.2.3 Rehabilitation

Repairing or modifying an existing structure to a desired useful condition.

1.2.4 Repair

The reconstruction or renewal of concrete parts of an existing structure for its maintenance or to correct deterioration, damage, or faulty construction of members or systems of a structure.

1.2.5 Shoring

Props or posts of timber or other material in compression used for the temporary support of excavations, formwork, or unsafe structures; the process of erecting shores.

1.2.6 Termination Joint

The interface where a placement of repair material meets existing concrete, the edge of an expansion joint, or other existing surfaces.

1.2.7 Unsound Concrete

Concrete that is fractured, delaminated, spalled, deteriorated, defective, contaminated or otherwise damaged.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; R 2015) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 503.2-503.4	(2010, R 2003) Three Epoxy Specifications
ACI 503.7	(2007) Specification for Crack Repair by Epoxy Injection
ACI 548.4	(2011) Standard Specification for Latex-Modified Concrete (LMC) Overlays
ACI 548.10	(2010) Specification for Type MMS (Methyl Methacrylate Slurry) Polymer Overlays for Bridge and Parking Garage Decks

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE/SEI 37	(2015) Design Loads on Structures During Construction
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ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C42/C42M	(2020) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C387/C387M	(2023) Standard Specification for Packaged, Dry, Combined Materials for Concrete and High Strength Mortar
ASTM C496/C496M	(2017) Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens
ASTM C881/C881M	(2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C882/C882M	(2024) Standard Test Method for Bond Strength of Bonding Systems Used With Concrete by Slant Shear
ASTM C928/C928M	(2020a) Standard Specification for Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs
ASTM C1059/C1059M	(2024) Standard Specification for Latex Agents for Bonding Fresh to Hardened Concrete
ASTM C1077	(2024) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for

Testing Agency Evaluation

ASTM C1438	(2013; R 2017) Standard Specification for Latex and Powder Polymer Modifiers for use in Hydraulic Cement Concrete and Mortar
ASTM C1583/C1583M	(2013) Standard Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method)
ASTM C1600/C1600M	(2017) Standard Specification for Rapid Hardening Hydraulic Cement
ASTM C1602/C1602M	(2022) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D93	(2019) Standard Test Methods for Flash-Point by Pensky-Martens Closed Cup Tester
ASTM D323	(2015a) Vapor Pressure of Petroleum Products (Reid Method)
ASTM D542	(2014) Index of Refraction of Transparent Organic Plastics
ASTM D1078	(2011) Standard Test Method for Distillation Range of Volatile Organic Liquids
ASTM D3418	(2015) Transition Temperatures of Polymers by Differential Scanning Calorimetry
ASTM D4016	(2014) Viscosity of Chemical Grouts by Brook field Viscometer (Laboratory Method)
ASTM D4580/D4580M	(2012) Standard Practice for Measuring Delaminations in Concrete Bridge Decks by Sounding
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
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1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL

PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

Work Plan; G

Quality Control Plan; G

SD-03 Product Data

Conventional Concrete

Polymers

Miscellaneous Materials And Equipment

SD-04 Samples

Reinforcement And Reinforcement Supports

Polymers

Miscellaneous Materials And Equipment

SD-05 Design Data

Formwork And Shoring; G

Repair Procedures; G

Mixture Proportioning; G

SD-06 Test Reports

Mixture Proportioning

Quality Control

Tolerance Report

Reinforcement And Reinforcement Supports

Conventional Concrete

Polymers

Miscellaneous Materials And Equipment

SD-07 Certificates

Qualifications

Reinforcement And Reinforcement Supports

Conventional Concrete

Polymers

SD-08 Manufacturer's Instructions

Equipment For Concrete Preparation

Conventional Concrete

Polymers

Miscellaneous Materials And Equipment

1.5 QUALITY ASSURANCE

1.5.1 General Requirements

- a. Follow the requirements of Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE for Work involving portland cement concrete.
- b. To protect personnel from overexposure to toxic materials, conform to the applicable manufacturer's Safety data sheets or local regulations. Submit manufacturer's Safety Data Sheets for all polymers as well as other potentially hazardous materials.
- c. Submit the repair procedures for executing the work as well as the test data and documentation on materials used for repair. Submittal must include component materials, mixture proportions, and supplier's quality control program.
- d. Inspection and testing of surface preparation as well as placement of reinforcing steel must be in accordance with provisions included herein and the Contract Document.
- e. Sampling and testing of materials, as well as inspection and testing of work, must be in accordance with established procedures, manufacturer's instructions, specific instructions from the Contracting Officer if given, or recommended practices as referenced herein and the Contract Documents.
- f. Trial batches and testing requirements for various repair materials specified are the responsibility of the Contractor.
- g. The testing agency must inspect, sample, and test repair materials and concrete production as required. When it appears that material furnished or work performed by Contractor fails to conform to Contract Documents the testing agency will immediately report such deficiency.

1.5.2 Quality Control Plan

Submit a quality control plan as specified in Sections 01 45 00.00 10 QUALITY CONTROL and 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.

1.5.3 Qualifications

The submittals must where applicable, identify agencies and individuals who will be working on this contract and their relevant experience. Do not make changes in approved agencies or personnel without prior approval of the Contracting Officer.

1.5.3.1 Testing Agencies

In addition to the requirements of Section 01 45 00.00 10 QUALITY CONTROL, agencies that test concrete materials must meet the requirements of ASTM C1077. Testing agencies that test or inspect placement of reinforcing steel must meet the requirement of ASTM E329. Submit data on qualifications of Contractor's proposed testing agency for acceptance.

1.5.3.2 Quality Control Personnel

Field tests of repair materials required must be made by an ICRI Concrete Surface Repair Technician Tier 2. Submit resumes, pertinent information, past experience, training and education of all operators of specialized demolition equipment if needed for this and the three paragraphs above.

1.5.3.3 Contractor Qualifications

The contractor performing the repair work must have been involved in a minimum of three concrete repair projects similar in size and scope to this project for at least five years. Submit information, including name, dollar value, date, and point-of-contact for similar projects which demonstrates the required experience and/or training.

1.5.3.4 Worker Qualifications

- a. Each worker engaged in the use of specialized removal or application equipment, including saw operators, milling machine operators, hydromilling equipment operators, and epoxy injection, must have satisfactorily completed an instruction program and three years of experience in the operation of the equipment. The worker must have active experience with the equipment within five years of the project.
- b. Workers installing adhesive anchors must be ACI Adhesive Anchor Installer certified or equivalent.

1.5.3.5 Regulatory Requirements

Perform all work in accordance with applicable Federal, State, and local safety, health, and environmental requirements. The Contractor is responsible for obtaining all permits required by Federal, State, and local agencies for the performance of the work.

1.5.4 Pre-Construction Conference

Conduct a pre-construction conference to discuss repair materials performance requirements, control provisions, and roles and responsibilities for the work to ensure that the Contractor's personnel understand all aspects of the repair material, its properties and application procedures. The conference must include the Contracting Officer or authorized representative, the Contractor's field superintendent and foreman, and a competent Technical Representative of the material manufacturer, and other involved trades or supplier representatives. The Technical Representative must be fully qualified to perform the work.

1.5.5 Work Plan

Prepare a work plan describing the methods of concrete removal and repair, including methods, equipment and materials to be used for each feature.

Submit the work plan for approval at least 30 days prior to the start of the work. The plan must include, but not be limited to, repair materials to be used with specific information on products and/or constituents, and requirements for handling, storage, etc., equipment to be used, surface preparation, and requirements for placement, finishing, curing and protection specific to the materials used. Include a description of field demonstrations in the work plan. Do not commence work until the work plan and field demonstration representative of the type of work are approved.

1.6 ACCEPTANCE OF REHABILITATION WORK

1.6.1 General Requirements

- a. Completed concrete rehabilitation work must conform to applicable requirements of Contract Document and this specification. The Contractor is responsible to bring Work into compliance with requirements of Contract Documents if the Concrete repair work fails to meet one or more requirements of Contract Documents.
- b. Correct rejected repair work by removing and replacing or by strengthening with additional construction acceptable to the Contracting Officer. Use repair methods that meet applicable requirements for function, durability, dimensional tolerances, and appearance.
- c. Submit proposed work plan, repair methods, materials, and modifications to the Work needed to correct rejected repair work to meet the requirements of Contract Documents.

1.6.2 Tolerances

- a. Construction tolerances for repairs must conform to ACI 117. Where existing conditions do not allow tolerances to conform to ACI 117, use the details and materials for such conditions as indicated in the Contract Documents. For conditions not shown or that are different than indicated in the Contract Documents, notify the Contracting Officer before proceeding with the work at those locations. Provide a tolerance report as required by Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- b. Inaccurately formed concrete surfaces resulting in concrete members with dimensions that exceed ACI 117 tolerances are subject to rejection.

1.6.3 Appearance

Concrete surfaces not meeting the requirements of the Contract Documents must be brought into compliance.

1.7 PROTECTION OF COMPLETED REHABILITATION WORK

- a. Do not allow construction loads to exceed the loads that a structural member or structure is safely capable of supporting without damage. Provide supplemental support if construction loads are expected to exceed safe load capacity.
- b. Protect repaired and adjacent areas from damage by construction traffic, equipment, and materials. During the curing period, protect repair materials from damage by mechanical disturbances, including

load-induced stresses, shock, and vibration.

- c. Protect repair materials from environmental damage by weather events during the length of the curing period.

PART 2 PRODUCTS

Products or materials used must conform to the requirements included herein as well as the Contract Documents. The usage of other products or materials not covered by this requirement or specified in the Contract Documents are permitted upon approval by the Contracting Officer. Additional information and submittals for products and materials not included in this document including product data, samples, design data, test reports, certificates, manufacturer's instructions, and field reports must be submitted as requested by the Contracting Officer.

2.1 MATERIALS FOR SHORING AND BRACING

2.1.1 Shoring and Bracing Systems

Use commercially manufactured and engineered shoring and bracing systems and components, except where custom built assemblies of lumber or other suitable materials are permitted by the Contracting Officer.

2.1.2 Design Requirements

The design of the bracing and shoring must be based on ASCE/SEI 37.

- a. Non-manufactured shoring and bracing systems must have calculations signed and sealed by a Licensed Design Professional.
- b. Members of non-manufactured shoring systems, must be designed in accordance with the provisions of the governing building code for the specific material of the member.
- c. Members of manufactured shoring systems, consisting of pre-engineered components designed and produced specifically for structural shoring, must be used in accordance with the manufacturer's recommendations.

2.2 EQUIPMENT FOR CONCRETE PREPARATION

Means and methods used for concrete removal and surface preparation must be selected and used such as to minimize damage to the structure and to the concrete substrate that remains.

2.2.1 Equipment for Concrete Removal

Removal equipment and techniques must be suitable to produce concrete surface profiles and level of cleanliness in designated areas as required by this specification and the contract Documents.

2.2.1.1 Cutting Equipment

- a. The following cutting equipment are permitted: High-pressure water jet without abrasives, saw cutting and stitch drilling.
- b. Cutting, lifting, and transporting equipment must be adequate to cut, support, and transport concrete sections without incurring any damage to the existing structure.

2.2.1.2 Concrete Breakers

- a. Provide sharp tips on breaker equipment to minimize microcracking damage in partial depth removal.
- b. The use of the following impact equipment and methods is permitted: Hand-held breakers, scabblers, needle scalers, scarifiers and milling methods.

2.2.1.3 Hydromilling Equipment

- a. Hydromilling equipment must include a trailer-mounted water tank, pumps, high-pressure hose, wand with safety release cutoff control, nozzle, and auxiliary water re-supply equipment. The water tank and auxiliary re-supply equipment must be of sufficient capacity to permit continuous operations.
- b. Hydrodemolition for concrete removal is permitted in the following locations: Deep spalls.
- c. Use protective covers and barriers to protect adjacent surfaces not intended to be repaired from water blasting and over-spray.
- d. Use equipment capable of delivering pressures of 35 MPa to 275 MPa at 7.5 liters/min to 190 liters/min for concrete removal and surface preparation.

2.2.2 Surface preparation and cleaning equipment

2.2.2.1 Abrasive Blasting

- a. Use dry or wet oil-free abrasive blasting capable of removing loose micro-fractured (bruised) or otherwise damaged or pulverized concrete surfaces, and rust from exposed steel reinforcement, and providing a surface profile in compliance with the Contract Documents.
- b. Use the following abrasive blasting methods: Sandblasting or s hotblasting

2.2.2.2 Low Pressure Water Cleaning

Use equipment capable of delivering 7 MPa to 35 MPa to at 7.5 liters/min to 38 liters/min for cleaning loose material from repair areas.

2.2.2.3 Other Cleaning Equipment

Use equipment that delivers oil free air capable of cleaning loose material and debris from repair areas. If necessary to dry the concrete surface, clean, dry, compressed air may be used. Also, use vacuums capable of removing loose material and debris.

2.3 MATERIALS FOR FORMWORK AND EMBEDDED ITEMS

- a. Formwork and embedded items must meet the requirements specified in Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.

- b. Install and remove formwork without damaging or staining the existing structure or repair material.
- c. Forms used for polymer concrete/mortars must be tight enough to hold the material that is used without leaking. All surfaces where bond is not desired, but which are exposed to the monomer or resin, must be treated with a form release agent.

2.4 REINFORCEMENT AND REINFORCEMENT SUPPORTS

2.4.1 Steel Bars, Wires, and Fiber-reinforced Concrete

- a. Reinforcement and reinforcement support must meet the requirements specified in Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- d. Submit mill certificates and shop drawings as requirement by Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.

2.5 CONVENTIONAL CONCRETE

- a. Portland cement concrete materials must meet the requirements specified in Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- b. Use cementitious materials indicated in the Contract Documents. Use cementitious materials of the same brand and type from the same manufacturing plant as the cementitious materials used in the concrete represented by the submitted field test records or used in trial mixtures.
- c. Refer to Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE for details on submittals involving conventional concrete.

2.6 POLYMERS

- a. The requirements for the properties of polymers and aggregates used in polymers must meet the requirements specified in this paragraph as well as the properties specified in the referenced specifications and the Contract Documents.
- b. Polymers used must be compatible with other polymers and materials used on the project. Unless repair materials are specified in the contract documents, the Contractor is responsible for verifying material compatibilities.
- c. Submit product data, manufacturer's Safety Data Sheets, samples, design data, test reports, certificates, manufacturer's instructions, and field reports for materials as required by this document as well as the referenced specifications and the Contract Documents.

2.6.1 Epoxies

- a. Epoxy mortars and epoxy compounds must conform to ASTM C881/C881M, grade 2 or 3 and class B or C, as required by repair type and conditions.
- b. Epoxy mortars used for repairing defects in hardened portland cement concrete must meet the requirements of ACI 503.2-503.4.
- c. Epoxy used for crack repair must meet the requirements of ACI 503.7.

2.6.2 Latexes

- a. Latex used in polymer modified portland cement concrete/mortar must meet the requirements of ASTM C1438.
- b. Latex used in polymer modified portland cement concrete overlays must meet the requirements of ACI 548.4.
- c. Latex used for bonding freshly mixed concrete and hardened concrete must meet the requirements of ASTM C1059/C1059M, Type II.

2.6.3 Methacrylates

- a. Methyl methacrylate slurry (MMS) used for overlays must meet the requirements of ACI 548.10.
- b. High molecular weight methacrylate (HMWM) must be a 2-component, rapid curing, and solvent-free system.
- c. HMWM monomers must be a high molecular weight or substituted methacrylate that conforms the following properties:
- d.

Physical Properties of HMWM Monomer		
Property	Test Method	Criteria
Vapor Pressure Flash Point Density	ASTM D323 ASTM D93	Less than 0.02 psi at 77 degrees F Greater than 200 degrees F Greater than 8.4 lbs. per gal. at 77 degrees F
Viscosity Index of Refraction Boiling point @ 0.02 psi Shrinkage on cure	ASTM D4016 ASTM D542 ASTM D1078	12 + 4 cps at 73 degrees F 1.470 + 0.002 158 degrees F Less than 11 percent
Glass Transition Temperature (DSC)	ASTM D3418	158 degrees F
Curing Time (100 g mass)	ASTM D3418	Greater than 40 minutes at 73 degrees F, with 4 percent cuemene hydroperoxide

Physical Properties of HMWM Monomer		
Bond Strength	ASTM C882/C882M	Greater than 1,500 psi

- e. The initiator/promoter system for HMWM must be capable of providing a surface cure time of not less than 40 minutes nor more than 3 hours at the surface temperature of the concrete during application. The initiator/promoter system must be such that the gel time may be adjusted to compensate for changes in temperature that may occur throughout the treatment application.
- f. The initiator/promoter system for HMWM must meet the following criteria:
(1)

Initiator Cumene Hydroperoxide	78 percent
Promoter Cobalt Napthenate	6 percent

2.6.4 Aggregate

- a. Unless otherwise specified or recommended by the polymer material manufacturer, aggregate used with polymers must meet ASTM C33/C33M or JIS A 5005 requirements.
- b. Aggregate properties and proportions used with polymers must meet the requirements of the polymer material manufacturer, the requirements of the referenced polymer standard, and the Contract Documents.
- c. Aggregate used with polymers must be dry and free of dirt, asphalt, and other organic materials. Aggregate moisture content must be less than 1 percent by weight.
- d. For patch repairs, the maximum-sized aggregate must not be greater than one third the depth of the patch area.

2.7 MISCELLANEOUS MATERIALS AND EQUIPMENT

2.7.1 Packaged and proprietary materials

The required properties for the materials listed in this paragraph must meet the properties specified in the Contract Documents. Submit Product data, design data, test reports, certificates, manufacturer's instructions, and field reports as required by the Contracting Officer and the Contract Documents.

- a. Packaged, rapid hardening concrete repair materials must conform to ASTM C928/C928M.
- b. Packaged, mortar and concrete must conform ASTM C387/C387M.
- c. Rapid hardening cement must conform to ASTM C1600/C1600M.

Water used with packaged and proprietary materials must meet ASTM C1602/C1602M requirements. Aggregates must meet the repair material manufacturer's requirements if available and ASTM C33/C33M or JIS A 5005

if such requirements are not specified.

2.7.2 Concrete Accessories

All concrete accessories not included in this document must meet the requirements specified in 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.

2.7.3 Miscellaneous Equipment

- a. Equipment designed specifically for the application of repair materials must be used as required by the repair material manufacturer and the referenced specification.
- b. Equipment not listed in this specification but referenced or used for repairs must be clean and in good operating condition.
- c. All supplies and equipment must be available in sufficient quantities to allow continuity in the installation project and quality assurance.

2.8 MIXTURE PROPORTIONING

- a. Portland cement-based concrete mixtures must be in accordance with the requirements of Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- b. Polymer concrete/mortar/resin/monomer proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements of the referenced material specifications and the repair material manufacturer's directions.
- c. Polymer-modified portland cement concrete proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements provided by the repair material manufacturer as well as ACI 548.4 when such materials are used for overlays.
- d. Proportioning and mixing materials not specified above must follow the requirements provided by the repair material manufacturer.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Examination

Locate area of unsound concrete or delamination using hammer sounding or chain drag sound methods in accordance to ASTM D4580/D4580M. Denote and mark perimeter boundaries and notify the Contracting Officer to approve the unsound concrete layout boundaries.

3.1.2 Protection

Protect pedestrians, motorized traffic, mechanical, electrical, and plumbing equipment, surrounding construction, project site, landscaping, and surrounding buildings from damage or injury resulting from concrete rehabilitation work.

- a. Construct dust and debris barriers surrounding repair work perimeter to control dust and to protect and control construction traffic.

- b. Dispose of runoff from wet demolition or surface preparation operations in accordance with all local ordinances. Disposal methods must avoid soil erosion, avoid undermining pavements and foundations, damage to landscaping and vegetation, and minimize water penetration through other parts of buildings.
- c. Collect and neutralize alkaline wastes and acid wastes and dispose in accordance with local, state, and federal regulations.
- d. Comply with local noise ordinances during demolition operations.
- e. Perform demolition work and surface preparation work in a manner that minimizes disturbances of operations. Coordinate work with the Contracting Officer.
- f. Submit a proposed protection plan for approval by owner representative and Licensed Design Professional.

3.1.3 Formwork and Shoring

Execution of formwork and shoring must meet the requirements specified in Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.

3.1.3.1 Formwork

- a. Construct forms to sizes, shapes, lines, and dimensions to match existing adjacent surfaces and textures. Provide forms that match openings, offsets, chamfers, anchorages, inserts and other features as described on Contract Documents. Construct forms to accommodate installation of products by other trades. Provide forms for easy removal to minimize damage to concrete surfaces and adjacent surfaces. Apply form release coating over formwork surfaces prior to each concrete placement. Form release agents must not be applied to or come in contact with the repair area concrete substrate or reinforcement.
- b. Do not damage repair material during removal of formwork for columns, walls, sides of beams, and other parts not supporting weight of concrete or repair material. Perform needed repair and treatment required on vertical surfaces at once and follow immediately with specified curing. Remove all formwork anchors embedded in existing concrete. Fill anchor holes and repair all damage to existing concrete at anchor holes.

3.1.3.2 Shoring

- a. Provide shoring in accordance with the shoring drawings prior to performing work to brace the substrate structure temporarily while repair work is proceeding. Shoring must be designed, documented, and stamped by a Licensed Design Professional. Shoring designs must be submitted to and approved by the Contracting Officer prior to work commencing.
- b. Leave formwork and shoring in place to support existing loads, construction loads and weight of repair material in beams, slabs, and other structural members until in-place strength of repair material determined in accordance with the Contract Documents. For post-tensioned construction, leave formwork and shoring in place until stressing is complete. When shores and other supports are arranged to allow removal of form-facing material without allowing structural slab

or member to deflect, form-facing material and its horizontal supporting members may be removed at an earlier age.

3.1.4 Concrete preparation

- a. Remove concrete as needed per the removal requirements of this section. Limits on removal equipment are specified in the paragraph titled EQUIPMENT FOR CONCRETE PREPARATION.
- b. Remove foreign material, such as dirt, oil, grease, or other chemicals, from the cracks before injection using compressed air, low-pressure water, or vacuuming. Allow wet surfaces to dry at least 24 hours.
- c. Immediately before placing the repair material or installing formwork, make the repair area available for inspection by the Contracting Officer. Obtain acceptance by the Contracting Officer of surface preparation before proceeding with Work. If the Work is rejected, perform additional operations to the satisfaction of Contracting Officer.
- d. Perform tensile pull-off tests in accordance with ASTM C1583/C1583M. Pull-off strength must meet or exceed 1.7 MPa. Test a minimum of 3 specimens at locations no greater than 420 square meters of prepared surface.

3.1.5 Quality Control

3.1.5.1 Quality control of surface preparation

Evaluation of prepared substrate must be continuously monitored to assure that the prepared substrate surface meets project requirements.

3.1.5.2 Quality control of repair overlays

All components of overlay PPCC materials must be certified by the material manufacturer or aggregate supplier to meet all project testing requirements. During the PPCC overlay, take mixed samples and check that the materials are mixed properly. Confirm that the right PC overlay thickness was applied by recording the volume of PC overlay materials and the substrate surface area covered by the overlay.

3.1.6 Curing

- a. For portland cement concrete Work, follow the requirements indicated in 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- b. For polymer concrete/mortar Work, follow manufacturer's requirements for curing.
- c. For polymer modified portland cement concrete Work follow manufacturer's requirements for curing.

3.1.7 Clean up

- a. Clean and remove all spills and leaks of injection adhesive and stains caused by the injection adhesives.
- b. Dispose wastewater used for cutting and cleaning without staining or damaging the existing surfaces of the structure or the environment of

the project area. The method of disposal must meet all the requirements of Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

3.1.8 Safety

- a. Provide Material Safety Data Sheets (MSDS) for products on site reviewing them before work begins.
- b. Provide safety guards, maintenance, and warnings for all machinery and equipment.
- c. Have personal protection equipment practice in place - eye protection and face guards.
- d. Have all workers in contact with wet cementitious material wear protective gloves and clothing.
- e. Provide eyewash facilities on-site with location signage.
- f. Provide dust masks for workers operating mixers.
- g. Have confined space procedures in place including adequate ventilation in closed spaces before operating equipment or using products that emit potentially dangerous or toxic exhaust, fumes, or dust.
- h. Provide secured storage available for all hazardous or flammable materials.
- i. Conduct safety meetings prior to beginning repair operations.

3.2 CRACK REPAIR

3.2.1 Preparation

3.2.1.1 General Requirements

- a. Clean all cracks in accordance with the paragraph titled Concrete Preparation.
- b. Do not repair cracks when the temperature of the concrete is below freezing and moisture conditions indicate the possibility of ice on the internal surfaces of the crack.
- c. Do not apply adhesive if the temperature of the concrete is not within the range of application temperatures recommended by the manufacturer of the adhesive.

3.2.1.2 Crack routing

Inspect surfaces adjacent to crack to receive repair material. If deteriorated, route a V-groove section at the crack face until sound concrete is reached.

3.2.1.3 Sealing

- a. For epoxy injection, apply a surface seal over all exterior faces of the crack that can be reached to contain the injection adhesive in the crack.

- b. For gravity fill repairs, apply a surface seal along the bottom surface of the element that can be reached to contain the repair material in the crack.

3.2.2 Application

3.2.2.1 Epoxy Injection

- a. Install the injection entry and venting ports using flush mounted or drilled fittings per proprietary manufacturer's instructions.
- b. Space the ports at a distance equal to the thickness of the member 200 mm maximum.
- c. Inject the epoxy using material manufacturer's recommended equipment.
- d. Apply recommended manufacturer's injection pressure.
- e. For vertical or inclined cracks, apply injection by pumping epoxy into entry ports at the lowest elevation, cap, and move upward.
- f. For horizontal cracks, apply injection by proceeding from one end of the crack to the other until the crack is fully sealed.
- g. After 10 min., repeat injection procedure until all ports refuse injection.
- h. Remove ports and remove the surface seal by chipping, or grinding or other acceptable means after the injected epoxy has cured.

3.2.2.2 Gravity fill

- a. Mix resin or monomer per material manufacturer's instructions.
- b. Pre-fill cracks at least 3 mm wide with aggregate.
- c. Pour resin or monomer onto the surface, over the cracks and spread with brooms, rollers, or squeegees.
- d. Work material back and forth over the cracks to maximize fill in crack.
- e. Allow at least 20 minutes for material to penetrate cracks.
- f. Remove excess material once cracks have been filled to refusal.
- g. Allow material to cure per material manufacturer's recommendations.
- h. Remove sealant and grind smooth.

3.2.3 Quality Control

- a. Conduct quality and control tests for metering accuracy and mixing effectiveness of the continuous mixing pump in accordance with ACI 503.7.
- b. Qualify the test injection procedures in accordance with ACI 503.7.

3.2.4 Acceptance Criteria

3.2.4.1 Core Sampling

- a. Obtain core samples in accordance with ASTM C42/C42M.
- b. Allow 24 hours after injection before coring.
- c. Obtain cores in a manner that includes as much of the bond line of the repaired concrete as possible. Replace cores that do not intersect the crack for at least 75 percent of the length of the core.
- d. Obtain three diameter core from first 30 m and one core for each 30 m thereafter.
- e. If cores would sever reinforcing steel or other embedded items, do not core, and notify the Contracting Officer so that an alternative location can be chosen.
- f. Obtain cores at least 50 mm in diameter for visual inspections and at least 100 mm in diameter for the splitting tensile test. Perform a splitting tensile test on one core from the first 30 m and one core for each 75 m thereafter.
- g. Fill core holes with non-shrink grout.

3.2.4.2 Core Testing

- a. Test a portion of the core samples for the splitting tensile strength in accordance with ASTM C496/C496M.
- b. Allow 72 hours after injection before beginning splitting tensile tests
- c. Prepare core sample per ASTM C42/C42M.
- d. Align the core so that the crack is in a plane as close to vertical as possible.

3.2.4.3 Acceptance

Work is acceptable if at least 90 percent of the depth of the crack in each core is filled with adhesive and a or b is met.

- a. The splitting tensile strength of the core is at least 90 percent of the splitting tensile strength of a core taken from an uncracked area within 300 mm of the repaired crack.
- b. A splitting tensile test of the core indicates that no more than 10 percent of the bonded area of the crack in each core exhibits combined areas of separation of the adhesive from the concrete or cohesive failure within the adhesive.

3.3 CORROSION AND SURFACE REPAIR

3.3.1 Preparation

3.3.1.1 Identification of Extent of Concrete Removal

- a. Configure geometry of removal area to maximize the use of right-angle

geometry, avoiding reentrant corners, and to obtain uniformity of depth. Determine the depth, location, and size of reinforcing bars prior to removal of concrete.

- b. Perform visual inspection and hammer tapping, chain drag sounding, or other methods acceptable by the Contracting Officer to identify cracked, delaminated, spalled, disintegrated, and otherwise unsound concrete for removal. Mark boundaries of repair area before concrete removal.
- c. Inspect the marked boundaries with the Contracting Officer prior to commencing with the concrete removal. Revise the repair area boundaries as instructed by the Contracting Officer.

3.3.1.2 Shoring and Formwork

- a. Provide shoring and formwork per the paragraph titled Formwork and Shoring.
- b. For post-tensioned concrete, detension strands and wires as required by Contract Documents prior to repair.

3.3.1.3 Concrete Removal

- a. Remove concrete from repair areas to indicated depth and profile. Notify Contracting Officer if additional delaminated, fractured, or unsound concrete is present.
- b. Do not damage embedded reinforcing and adjacent concrete. The removal methods must produce minimal microcracking (bruising) of the prepared substrate surfaces. Avoid directly striking reinforcing steel with impact tools used for concrete removal.
- c. Provide perpendicular edges at perimeter of repair area. The perimeter of the repair areas must be saw cut to a depth of 15 to 20 mm. For vertical or overhead surfaces, provide 45-degree slope at repair boundaries to facilitate air and rebound escape. Do not cut or damage embedded reinforcement or other embedded items. If embedded reinforcing steel or other embedded items are too close to the surface to provide the perpendicular edge cut, notify the Contracting Officer for direction before proceeding.
- d. Extend concrete removal along the corroded reinforcing steel to a point where there is no further delamination, concrete cracking, or reinforcing steel corrosion, and where the reinforcement is bonded to the surrounding concrete.
- e. Remove concrete around the exposed layer of reinforcement to a uniform depth beyond within the repair areas to provide a minimum clearance between exposed reinforcing steel and surrounding concrete of 20 mm, or at least 5 mm larger than the maximum nominal size of the coarse aggregate in the repair material.
- f. Do not remove concrete behind vertical reinforcing bars in columns.

3.3.1.4 Preparation of Concrete Substrate Surface

- a. Confirm perpendicular edges at repair area perimeter, and reinstate if damaged by concrete removal process. Remove loosely bonded concrete,

bruised or fractured concrete, and bond-inhibiting materials such as dirt, concrete slurry, or any other detrimental materials from the concrete substrate using approved methods. Where concrete has been removed by impact methods, abrasive blasting must be used to prepare the surface and remove bruised concrete.

- b. Provide substrate surface profiles as specified in the Contract Documents.
- c. Visually inspect and sound substrate surface to confirm that no further delaminations or otherwise unsound concrete remains. If encountered, notify the Contracting Officer.
- d. Clean the substrate per the paragraph titled Concrete preparation.

3.3.2 Application

3.3.2.1 Existing Reinforcement Preparation

- a. Clean existing reinforcement that will remain. Remove corrosion and/or other laitance and notify the Contracting Officer if section loss is greater than 20 percent.
- b. Permit evaluation of existing reinforcement and placement of new reinforcement by the Contracting Officer.

3.3.2.2 Placement of New Reinforcement

Placement of new reinforcement

- a. Placement of new reinforcement to replace or strengthen existing reinforcement is like new construction. Placement, splicing, and handling of new reinforcement must meet the requirements specified in Section 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE.
- b. Reinforcement must be free of materials deleterious to bond. New reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

3.3.2.3 Placement of Concrete

- a. If portland cement concrete is used as the repair material, follow the requirements indicated in 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE as well the Contract Document for proportioning, mixing, and placing concrete. For all other materials, follow material manufacturer's recommendations.
- b. A bonding agent must be used.
- c. Finish the surface to match surface finish and texture requirements indicated in the Contract Document.

3.3.2.4 Placement of Other Repair Materials

- a. Equilibrate repair material(s) and substrate to the temperature, cleanliness of substrate and reinforcement, and moisture requirements

of the repair material manufacturer's requirements.

- b. Comply with the repair material manufacturer's requirements for batching, mixing, placing and curing repair materials.
- c. Review consistency of the mixed repair material(s) relative to the parameters documented in the repair material manufacturer product data sheet. If non-conforming, adjust consistency in compliance with the repair material manufacturer's requirements.
- d. Apply or install repair material(s) within the application time frame (pot life) requirements of the repair material manufacturer's requirements, and place and consolidate to provide well-compacted repair.
- e. Finish and tool repair materials, finished in accordance with the repair material manufacturer's written instructions and as indicated in Contract Documents.
- f. Protect installed repair material(s) from damage, exposure to environmental conditions that are detrimental to the uncured or cured properties of the material. Cure in accordance with the requirements of the repair material manufacturer's requirements.

3.3.3 Quality Control

- a. Protect concrete surfaces, beyond limits of surfaces receiving bonding agent adhesive, against spillage. Immediately remove any bonding agent adhesive that has spilled beyond desired area. Perform cleanup with material designated by bonding agent adhesive manufacturer. Avoid contamination of work area.
- b. The bond strength between the existing concrete and the repair material must be a minimum of 1.7 MPa per ASTM C1583/C1583M. Test a minimum of 3 specimens at locations no greater than 420 square meters of prepared surface.

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SECTION 03 30 53

MISCELLANEOUS CAST-IN-PLACE CONCRETE
05/14

PART 1 GENERAL

1.1 SUMMARY

Perform all work in accordance with ACI 318M.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; R 2015) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 301	(2020) Specifications for Structural Concrete
ACI 301M	(2020) Metric Specifications for Structural Concrete
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 318	(2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)
ACI 318M	(2019; Errata 2022) Building Code Requirements for Structural Concrete & Commentary
ACI SP-66	(2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C31/C31M	(2025a) Standard Practice for Making and Curing Concrete Test Specimens in the Field

ASTM C39/C39M	(2024) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C94/C94M	(2024d) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C231/C231M	(2024) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2024) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C494/C494M	(2024) Standard Specification for Chemical Admixtures for Concrete
ASTM C618	(2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C685/C685M	(2024) Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing
ASTM C989/C989M	(2024) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1064/C1064M	(2023) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM C1602/C1602M	(2022) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM E1155	(2020) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers
ASTM E1155M	(2014) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers (Metric)
ARCHITECTURAL INSTITUTE OF JAPAN (AIJ)	
JASS 5	(2015) Reinforced Concrete Work

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1101	(2014) Method of Test for Slump of Concrete
JIS A 1108	(2018) Method of Test for Compressive Strength of Concrete
JIS A 1115	(2014) Method of Sampling Fresh Concrete
JIS A 1118	(2017) Method of Test for Air Content of Fresh Concrete by Volumetric Method
JIS A 1128	(2014) Method of Test for Air Content of Fresh Concrete by Pressure Method
JIS A 1132	(2014) Method of Making and Curing Concrete Specimens
JIS A 1804	(2009) Methods of Test for Production Control of Concrete - Method of Rapid Test for Identification of Alkali-Silica Reactivity of Aggregate
JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
JIS A 5011	(2013) Slag Aggregate for Concrete
JIS A 5308	(2014) Ready-Mixed Concrete
JIS A 6201	(2015) Fly Ash for Use in Concrete
JIS A 6204	(2011) Chemical Admixtures for Concrete
JIS G 3112	(2010) Steel Bars for Concrete Reinforcement
JIS G 3551	(2005) Welded Wire Mesh and Rebar Grid
JIS R 5210	(2009) Portland Cement
JIS R 5211	(2019) Portland Blast-Furnace Slag Cement
JIS R 5213	(2019) Portland Fly Ash Cement

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-SS Chapter 6	MLIT Architectural Standard Specifications, Chapter 6 Concrete
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1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation Drawings; G

SD-03 Product Data

Air-Entraining Admixture

Water-Reducing or Retarding Admixture

Curing Materials

Batching and Mixing Equipment

Conveying and Placing Concrete

Formwork

Ready-Mix Concrete

Curing Compound

Mechanical Reinforcing Bar Connectors

SD-06 Test Reports

Aggregates

Concrete Mixture Proportions; G

Compressive Strength Testing; G

Slump; G

Air Content

Water

SD-07 Certificates

Cementitious Materials

Aggregates

Delivery Tickets

SD-08 Manufacturer's Instructions

Curing Compound

1.4 QUALITY ASSURANCE

Indicate specific locations of Concrete Placement, Forms, Steel Reinforcement, Accessories, Expansion Joints, Construction Joints, Contraction Joints, Control Joints on installation drawings and include, but not be limited to, square meters of concrete placements, thicknesses and widths, plan dimensions, and arrangement of cast-in-place concrete section.

1.4.1 Flatness and Levelness of Floor Slabs

Conduct floor flatness and levelness test, (FF and FL respectively), on floor slabs in accordance with the provisions set forth in ASTM E1155M or ASTM E1155. Make floor tolerance measurements by the approved laboratory and inspection service within 24 hours after completion of final troweling operation and before forms and shores have been removed. Provide results of floor tolerance tests, including formal notice of acceptance or rejection of the work, to the Contracting Officer within 24 hours after data collection.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

The Government retains the option to sample and test aggregates and concrete to determine compliance with the specifications. Provide facilities and labor as may be necessary to assist the Government in procurement of representative test samples. Obtain samples of aggregates at the point of batching in accordance with ASTM D75/D75M. Sample concrete in accordance with ASTM C172/C172M or JIS A 1115. Determine slump and air content in accordance with ASTM C143/C143M or JIS A 1101 and ASTM C231/C231M or JIS A 1128, respectively, when cylinders are molded. Prepare, cure, and transport compression test specimens in accordance with ASTM C31/C31M or JIS A 1132. Test compression test specimens in accordance with ASTM C39/C39M or JIS A 1108. Take samples for strength tests not less than once each shift in which concrete is produced from each strength of concrete required. Provide a minimum of four 150 x 300 mm or six 100 x 200 mm specimens from each sample; two 150 x 300 mm or three 100 x 200 mm to be tested at 28 days (90 days if pozzolan or slag cement is used) for acceptance. Two 150 x 300mm or three will be tested at 7 days for information.

2.1.1 Strength

Acceptance test results are the average strengths of two 150 x 300 mm or three 100 x 200 mm specimens tested at 28 days (90 days if pozzolan or slag cement is used). The strength of the concrete is considered satisfactory so long as the average of all three consecutive acceptance test results equal or exceed the specified compressive strength, f'_c , and no individual acceptance test result falls below f'_c by more than 3.4 MPa.

2.1.2 Construction Tolerances

Apply a Class "C" finish to all surfaces except those specified to receive a Class "D" finish. Apply a Class "D" finish to all post-construction surfaces which will be permanently concealed. Surface requirements for the classes of finish required are as specified in ACI 117 or JASS 5 and MLIT-SS Chapter 6.

2.1.3 Concrete Mixture Proportions

Concrete mixture proportions are the responsibility of the Contractor. Mixture proportions must include the dry weights of cementitious material(s); the nominal maximum size of the coarse aggregate; the specific gravities, absorptions, and saturated surface-dry weights of fine and coarse aggregates; the quantities, types, and names of admixtures; and quantity of water per cubic meter of concrete. Provide materials included in the mixture proportions of the same type and from the same source as

will be used on the project. The specified compressive strength f'_c is 20.7 MPa at 28 days (90 days if pozzolan is used). The maximum nominal size coarse aggregate is 25 mm, in accordance with ACI 304R. The air content must be between 4.5 and 7.5 percent with a slump between 50 and 125 mm. The maximum water-cementitious material ratio is 0.50. Submit the applicable test reports and mixture proportions that will produce concrete of the quality required, ten days prior to placement of concrete.

2.1.4 Slag Cement

Slag Cement must conform to ASTM C989/C989M or JIS A 5011.

2.2 MATERIALS

Submit manufacturer's literature from suppliers which demonstrates compliance with applicable specifications for the specified materials.

2.2.1 Cementitious Materials

Submit Manufacturer's certificates of compliance, accompanied by mill test reports, attesting that the concrete materials meet the requirements of the specifications in accordance with the Special Clause "CERTIFICATES OF COMPLIANCE". Provide cementitious materials that conform to the appropriate specifications listed:

2.2.1.1 Portland Cement

Unless otherwise specified, provide cement that conforms to JIS R 5210, Ordinary Portland Cement, and meets low alkali content requirements.

2.2.1.2 Blended Hydraulic Cement

Blended cements must conform to JIS R 5211, Type A or B, or JIS R 5213, Type A, B, or C.

2.2.1.3 Pozzolan

Provide pozzolan that conforms to ASTM C618, Class F, including requirements of Tables 1A and 2A or JIS A 6201.

2.2.2 Aggregates

- a. Aggregates must conform to JIS A 5005 unless otherwise specified in the Contract Documents or approved by the Contracting Officer.
- b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.
- c. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.
- d. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with JIS A 5005 and JIS A 1804 expansion data not more than 18 months old.

2.2.3 Admixtures

Provide admixtures, when required or approved, in compliance with the appropriate specification listed. Retest chemical admixtures that have been in storage at the project site, for longer than 6 months or that have been subjected to freezing, at the expense of the Contractor at the request of the Contracting Officer and will be rejected if test results are not satisfactory.

2.2.3.1 Air-Entraining Admixture

Provide air-entraining admixture that meets the requirements of ASTM C260/C260M or JIS A 6204.

2.2.3.2 Water-Reducing or Retarding Admixture

Provide water-reducing or retarding admixture meeting the requirements of ASTM C494/C494M, Type A, B, or D or JIS A 6204. High-range water reducing admixture Type F may be used only when approved, approval being contingent upon particular placement requirements as described in the Contractor's Quality Control Plan.

2.2.4 Water

Mixing and curing water in compliance with the requirements of ASTM C1602/C1602M or JIS A 5308, Annex C; potable, and free of injurious amounts of oil, acid, salt, or alkali. Submit test report showing water complies with ASTM C1602/C1602M or JIS A 5308, Annex C.

2.2.5 Reinforcing Steel

Provide reinforcing bars conforming to the requirements of ASTM A615/A615M, Grade 60, deformed or JIS G 3112, Grade SD390 ($F_y = 390$ MPa) for D29 and larger bars, Grade SD345 ($F_y = 345$ MPa) for D19 to D25 bars, and Grade SD295 ($F_y = 295$ MPa) for D6 to D16 bars. Provide welded steel wire reinforcement conforming to the requirements of ASTM A1064/A1064M or JIS G 3551. Detail reinforcement not indicated in accordance with ACI 301M and ACI SP-66. Provide mechanical reinforcing bar connectors in accordance with ACI 301M and provide 125 percent minimum yield strength of the reinforcement bar.

2.2.6 Formwork

Design formwork, shores, reshores, and backshores to support all vertical and lateral loads transmitted to them and to comply with applicable building code requirements. Fabricate of wood, steel, or other approved material. Submit formwork design prior to the first concrete placement.

2.2.7 Form Coatings

Provide form coating in accordance with ACI 301M.

2.2.8 Curing Materials

Provide curing materials in accordance with JASS 5 and MLIT-SS Chapter 6, Section 5.

2.3 READY-MIX CONCRETE

Provide concrete that meets the requirements of JIS A 5308.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by JIS A 5308:

- a. Type and brand cement
- b. Cement content in 43 kilogram bags per cubic meter of concrete
- c. Maximum size of aggregate
- d. Amount and brand name of admixture
- e. Total water content expressed by water cementitious material ratio

2.4 ACCESSORIES

2.4.1 Curing Compound

Provide concrete curing material in accordance with JASS 5 and MLIT-SS Chapter 6. Submit manufacturers instructions for placing curing compound.

PART 3 EXECUTION

3.1 PREPARATION

Prepare construction joints to expose coarse aggregate. The surface must be clean, damp, and free of laitance. Construct ramps and walkways, as necessary, to allow safe and expeditious access for concrete and workmen. Remove snow, ice, standing or flowing water, loose particles, debris, and foreign matter. Satisfactorily compact earth foundations. Make spare vibrators available. Placement cannot begin until the entire preparation has been accepted by the Government.

3.1.1 Embedded Items

Secure reinforcement in place after joints, anchors, and other embedded items have been positioned. Arrange internal ties so that when the forms are removed the metal part of the tie is not less than 50 mm from concrete surfaces permanently exposed to view or exposed to water on the finished structures. Prepare embedded items so they are free of oil and other foreign matters such as loose coatings or rust, paint, and scale. The embedding of wood in concrete is permitted only when specifically authorized or directed. Provide all equipment needed to place, consolidate, protect, and cure the concrete at the placement site and in good operating condition.

3.1.2 Formwork Installation

Forms must be properly aligned, adequately supported, and mortar-tight. Provide smooth form surfaces, free from irregularities, dents, sags, or holes when used for permanently exposed faces. Chamfer all exposed joints and edges, unless otherwise indicated.

3.1.3 Production of Concrete

3.1.3.1 Ready-Mixed Concrete

Provide ready-mixed concrete conforming to ASTM C94/C94M or JIS A 5308 except as otherwise specified.

3.1.3.2 Concrete Made by Volumetric Batching and Continuous Mixing

Conform to ASTM C685/C685M.

3.1.3.3 Batching and Mixing Equipment

The option of using an on-site batching and mixing facility is available. The facility must provide sufficient batching and mixing equipment capacity to prevent cold joints. Submit the method of measuring materials, batching operation, and mixer for review, and manufacturer's data for batching and mixing equipment demonstrating compliance with the applicable specifications.

3.2 CONVEYING AND PLACING CONCRETE

Convey and place concrete in accordance with ACI 301M, Section 5.

3.2.1 Hot-Weather Requirements

Place concrete in hot weather in accordance with ACI 305R

3.3 FINISHING

3.3.1 Temperature Requirement

Do not finish or repair concrete when either the concrete or the ambient temperature is below 10 degrees C.

3.3.2 Finishing Formed Surfaces

Remove all fins and loose materials, and surface defects including filling of tie holes. Repair all honeycomb areas and other defects. Remove all unsound concrete from areas to be repaired. Ream or chip surface defects greater than 13 mm in diameter and holes left by removal of tie rods in all surfaces not to receive additional concrete and fill with dry-pack mortar. Brush-coat the prepared area with an approved epoxy resin or latex bonding compound or with a neat cement grout after dampening and filling with mortar or concrete. Use a blend of portland cement and white cement in mortar or concrete for repairs to all surfaces permanently exposed to view so that the final color when cured is the same as adjacent concrete.

3.3.3 Finishing Unformed Surfaces

Finish unformed surfaces in accordance with ACI 301M, Section 5.

3.4 CURING AND PROTECTION

Cure and protect in accordance with ACI 301M, Section 5 or JASS 5 and MLIT-SS Chapter 6.

3.5 FORM WORK

Provide form work in accordance with ACI 301M, Section 2 and Section 5.

3.5.1 Removal of Forms

Remove forms in accordance with ACI 301M, Section 2.

3.6 STEEL REINFORCING

Reinforcement must be free from loose, flaky rust and scale, and free from oil, grease, or other coating which might destroy or reduce the reinforcement's bond with the concrete.

3.6.1 Fabrication

Shop fabricate steel reinforcement in accordance with ACI 318 and ACI SP-66 or JASS 5 and MLIT-SS Chapter 6. Provide shop details and bending in accordance with ACI 318 and ACI SP-66 or JASS 5 and MLIT-SS Chapter 6.

3.6.2 Splicing

Lap splice lengths and locations as indicated in the Contract Documents. For splice locations not indicated follow JASS 5 and MLIT-SS Chapter 6, at no additional cost to the Government and subject to approval. Splicing must be by lapping, by gas pressure welding, or by mechanical or welded connection, except that lap splices must not be used for bars larger than D35. Do not splice at points of maximum stress and stagger splices a minimum of 1200 mm or as otherwise indicated so no more than half of the bars are spliced at any one section. Overlap welded wire reinforcement the spacing of the cross wires, plus 50 mm. Approve welded splices prior to use.

3.6.3 Supports

Secure reinforcement in place by the use of metal or concrete supports, spacers, or ties.

3.7 EMBEDDED ITEMS

Before placing concrete, take care to determine that all embedded items are firmly and securely fastened in place. Provide embedded items free of oil and other foreign matter, such as loose coatings of rust, paint and scale. Embedding of wood in concrete is permitted only when specifically authorized or directed.

3.8 TESTING AND INSPECTING

Report the results of all tests and inspections conducted at the project site informally at the end of each shift. Submit written reports weekly. Deliver within three days after the end of each weekly reporting period. See Section 01 45 00.00 10 QUALITY CONTROL.

3.8.1 Field Testing Technicians

The individuals who sample and test concrete must have demonstrated a knowledge and ability to perform the necessary test procedures equivalent to the ACI minimum guidelines for certification of Concrete Field Testing Technicians, Grade I.

3.8.2 Preparations for Placing

Inspect foundation or construction joints, forms, and embedded items in sufficient time prior to each concrete placement to certify that it is ready to receive concrete.

3.8.3 Sampling and Testing

- a. Obtain samples and test concrete for quality control during placement. Sample fresh concrete for testing in accordance with ASTM C172/C172M or JIS A 1115. Make six test cylinders.
- b. Test concrete for compressive strength at 7 and 28 days for each design mix and for every 77 cubic meters of concrete. Test two cylinders at 7 days; two cylinders at 28 days; and hold two cylinders in reserve. Conform test specimens to ASTM C31/C31M or JIS A 1132. Perform compressive strength testing conforming to ASTM C39/C39M or JIS A 1108.
- c. Test slump at the site of discharge for each design mix in accordance with ASTM C143/C143M or JIS A 1101. Perform tests at commencement of concrete placement, when test cylinders are made, and for each batch (minimum) or every 16 cubic meters (maximum) of concrete.
- d. Test air content for air-entrained concrete in accordance with ASTM C231/C231M or JIS A 1118 or JIS A 1128. Test air-entrained concrete for air content at the same frequency as specified for slump tests.
- e. Determine temperature of concrete at time of placement in accordance with ASTM C1064/C1064M. Check concrete temperature at least once during each shift that concrete is placed for each strength of concrete required.

3.8.4 Action Required

3.8.4.1 Placing

Do not begin placement until the availability of an adequate number of acceptable vibrators, which are in working order and have competent operators, has been verified. Discontinue placing if any lift is inadequately consolidated.

3.8.4.2 Air Content

Whenever an air content test result is outside the specification limits, adjust the dosage of the air-entrainment admixture prior to delivery of concrete to forms.

3.8.4.3 Slump

Whenever a slump test result is outside the specification limits, adjust the batch weights of water and fine aggregate prior to delivery of concrete to the forms. Make the adjustments so that the water-cementitious material ratio does not exceed that specified in the submitted concrete mixture proportion and the required concrete strength is still met.

-- End of Section --

DRAFT

SECTION 06 10 00

ROUGH CARPENTRY
08/16, CHG 2: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)

ALSC PS 20 (2015) American Softwood Lumber Standard

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)

ASME B18.2.2 (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B18.5.2.1M (2006; R 2011) Metric Round Head Short Square Neck Bolts

ASME B18.5.2.2M (1982; R 2010) Metric Round Head Square Neck Bolts

AMERICAN WOOD COUNCIL (AWC)

AWC WFCM (2012) Wood Frame Construction Manual for One- and Two-Family Dwellings

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA M6 (2013) Brands Used on Preservative Treated Materials

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA F405 (19) Product Guide: Performance Rated Panels

APA L870 (2010) Voluntary Product Standard, PS 1-09, Structural Plywood

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM D2898 (2010; R 2017) Standard Practice for Accelerated Weathering of

Fire-Retardant-Treated Wood for Fire
Testing

ASTM D5516 (2018) Standard Test Method for Evaluating
the Flexural Properties of Fire-Retardant
Treated Softwood Plywood Exposed to
Elevated Temperatures

ASTM E84 (2023) Standard Test Method for Surface
Burning Characteristics of Building
Materials

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS K 1570 (2018) Wood Preservatives

MINISTRY OF AGRICULTURE, FORESTRY AND FISHERIES (MAFF)

JAS no. 1920 Japanese Agricultural Standard for Lumber
Notification

JAS no. 1083 Japanese Agricultural Standard for Sawn
Lumber

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification.
Submittals not having a "G" classification are for Contractor Quality
Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-03 Product Data

Fire-retardant Treatment

1.3 DELIVERY AND STORAGE

Deliver materials to the site in an undamaged condition. Store, protect
and handle, material in accordance with manufacturer's instructions and as
specified. Store materials off the ground to provide proper ventilation,
with drainage to avoid standing water, and protection against ground
moisture and dampness. Store materials with a moisture barrier at both
the ground level and as a cover forming a well ventilated enclosure.
Adhere to requirements for stacking, lifting, bracing, cutting, notching,
and special fastening requirements. Do not use materials that have visible
moisture or biological growth. Remove defective and damaged materials and
provide new materials. Store separated reusable wood waste convenient to
cutting station and area of work.

1.4 GRADING AND MARKING

1.4.1 Lumber

Mark each piece of framing and board lumber or each bundle of small pieces
of lumber with the grade mark of a recognized association or independent
inspection agency in accordance with JAS no. 1083. Such association or
agency must be certified by the Board of Review, American Lumber Standards
Committee or Japan Plywood Inspection Corporation (JPIC), to grade the
species used. Surfaces that are to be exposed to view must not bear

grademarks, stamps, or any type of identifying mark.

1.4.2 Plywood

Mark each sheet with the mark of a recognized association or independent inspection agency that maintains continuing control over the quality of the plywood. The mark must identify the plywood by species group or span rating, exposure durability classification, grade, and compliance with APA L870. Surfaces that are to be exposed to view must not bear grademarks or other types of identifying marks.

1.4.3 Fire-Retardant Treated Lumber

Mark each piece in accordance with AWP A M6 or JIS K 1570 , except pieces that are to be natural or transparent finished. In addition, exterior fire-retardant lumber must be distinguished by a permanent penetrating blue stain. Labels of a nationally recognized independent testing agency will be accepted as evidence of conformance to the fire-retardant requirements of AWP A M6 or JIS K 1570.

1.5 SIZES AND SURFACING

ALSC PS 20 or JAS no. 1920 for dressed sizes of yard lumber. Lumber must be surfaced four sides. Size references, unless otherwise specified, are nominal sizes, and actual sizes must be within manufacturing tolerances allowed by the standard under which the product is produced. Other measurements are IP or SI standard.

1.6 MOISTURE CONTENT

Air-dry or kiln-dry lumber. Kiln-dry treated lumber after treatment. Maximum moisture content of wood products must be as follows at the time of delivery to the job site:

- a. Framing lumber and board, 19 percent maximum
- b. Fire-retardant-treated plywood, 15 percent maximum
- c. Materials other than lumber; moisture content must be in accordance with standard under which the product is produced

1.7 FIRE-RETARDANT TREATMENT

Fire-retardant treated wood must be pressure treated Treatment and performance inspection must be by an independent and qualified testing agency that establishes performance ratings. Each piece or bundle of treated material must bear identification of the testing agency to indicate performance in accordance with such rating. Treated materials to be exposed to rain wetting must be subjected to an accelerated weathering technique in accordance with ASTM D2898 prior to being tested. Such items which will not be inside a building, and such items which will be exposed to heat or high humidity, must receive exterior fire-retardant treatment. Fire-retardant-treated wood products must be free of halogens, sulfates, ammonium phosphate, and formaldehyde.

PART 2 PRODUCTS

2.1 MATERIALS

2.2 PLYWOOD PANELS

APA L870 and APA F405 respectively.

2.2.1 Subflooring

2.2.1.1 Plywood

C-D Grade, Exposure 1 durability classification, Span rating of 24/16 or greater Fire-retardant-treated (FRTW) Class A, Class 1 in accordance with ASTM D5516 and ASTM E84.

2.2.2 Other Uses

2.2.2.1 Plywood

Plywood for electrical back boards. Fire retardant-treated (FRTW) C-D Grade, Exposure 1.

2.3 OTHER MATERIALS

2.3.1 Miscellaneous Wood Members

2.3.1.1 Blocking

Blocking must be standard or number 2 grade, plywood, metal flat strapping or backing bards, same gauge as metal partition or as specified in Section 09 22 00 SUPPORTS FOR GYPSUM BOARD.

2.4 ROUGH HARDWARE

Unless otherwise indicated or specified, rough hardware must be of the type and size necessary for the project requirements. Sizes, types, and spacing of fastenings of manufactured building materials must be as recommended by the product manufacturer unless otherwise indicated or specified. Rough hardware exposed to the weather or embedded in or in contact with concrete walls or slabs must be hot-dip zinc-coated in accordance with ASTM A153/A153M. Fastenings for fire-retardant treated lumber and woodwork exposed to the weather must be copper alloy or hot-dipped galvanized fasteners as recommended by the treated wood manufacturer.

2.4.1 Bolts, Nuts, Studs, and Rivets

ASME B18.2.1, ASME B18.5.2.1M, ASME B18.5.2.2M and ASME B18.2.2.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Conform to AWC WFCM unless otherwise indicated or specified. Select lumber sizes to minimize waste. Fit framing lumber and other rough

carpentry, set accurately to the required lines and levels, and secure in place in a rigid manner.

3.1.1 Subflooring

3.1.1.1 Plywood

Install fire-retardant-treated plywood used in structural applications in accordance with the conditions and limitations listed in ICC ESR-4244 and in compliance with the requirements of the applicable building codes and product recommendations.

3.2 MISCELLANEOUS

3.2.1 Rough Wood Bucks

50 mm nominal thickness. Set wood bucks true and plumb. Anchor bucks to concrete or masonry with steel straps extending into the wall 200 mm minimum. Place anchors near the top and bottom of the buck and space uniformly at 600 mm maximum intervals.

3.2.2 Blocking

Provide proper sizes and shapes at proper locations for the installation and attachment of finish materials, fixtures, equipment, and items indicated or specified.

-- End of Section --

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SECTION 06 41 16.00 10

PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS
08/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5905	(2014) Fiberboards
JIS B 1099	(2019) Construction Standard Specification
JIS K 6903	(2008) Laminated Thermosetting High-Pressure Decorative Sheets

1.2 SYSTEM DESCRIPTION

Work in this section includes laminate clad custom casework vanities as shown on the drawings and as described in this specification. This Section includes high-pressure laminate surfacing and cabinet hardware. All exposed and semi-exposed surfaces, whose finish is not otherwise noted on the drawings or finish schedule, shall be sanded smooth and shall receive a clear finish of polyurethane. Wood finish may be shop finished or field applied in accordance with Section 09 90 00 PAINTS AND COATINGS.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings

Installation

SD-04 Samples

Plastic Laminates

1.4 QUALITY ASSURANCE

1.4.1 General Requirements

Unless otherwise noted on the drawings, all materials, construction methods, and fabrication shall conform to and comply with the premium grade quality standards as outlined in this specification. Contractors and their personnel engaged in the work shall be able to demonstrate successful experience with work of comparable extent, complexity and

quality to that shown and specified. Submit a quality control statement which illustrates compliance with and understanding of project requirements. The quality control statement shall also certify a minimum of ten years Contractor's experience in laminate clad casework fabrication and construction. The quality control statement shall provide a list of a minimum of five successfully completed projects of a similar scope, size, and complexity.

1.5 DELIVERY, STORAGE, AND HANDLING

Casework may be delivered knockdown or fully assembled. Deliver all units to the site in undamaged condition, stored off the ground in fully enclosed areas, and protected from damage. The storage area shall be well ventilated and not subject to extreme changes in temperature or humidity.

1.6 SEQUENCING AND SCHEDULING

Coordinate work with other trades. Units shall not be installed in any room or space until painting, and ceiling installation are complete within the room where the units are located. Floor cabinets shall be installed before finished flooring materials are installed.

PART 2 PRODUCTS

2.1 WOOD MATERIALS

2.1.1 Lumber

All framing lumber shall be kiln-dried with moisture content 6 to 8 percent, to dimensions as shown on the drawings. Frame front, where indicated on the drawings, shall be nominal 15 mm to 19 mm hardwood.

2.1.2 Panel Products

2.1.2.1 Medium Density Fiberboard

Medium density fiberboard (MDF) shall be an acceptable panel substrate where noted on the drawings. Medium density fiberboard shall meet the minimum standards listed in JIS A 5905.

2.2 HIGH PRESSURE DECORATIVE LAMINATE (HPDL)

All plastic laminates shall meet the requirements of JIS K 6903 for high-pressure decorative laminates. Design, colors, surface finish and texture, and locations shall be as indicated on the drawings. Submit two samples of each plastic laminate pattern and color. Samples shall be a minimum of 120 by 170 mm in size. Plastic laminate types and nominal minimum thicknesses for casework components shall be as indicated in the following paragraphs.

2.2.1 Vertical General Purpose Standard (VGS) Grade

Vertical general purpose standard grade plastic laminate shall be 0.55 mm to 0.7 mm in thickness. This laminate grade is intended for exposed exterior vertical surfaces of casework components where postforming is not required.

2.3 EDGE BANDING

Edge banding for casework doors and drawer fronts shall be PVC vinyl and shall be 0.5 mm thick. Material width shall be as indicated on the drawings. Color and pattern shall match exposed door and drawer front laminate pattern and color.

2.4 FASTENERS

Nails, screws, and other suitable fasteners shall be the size and type best suited for the purpose and shall conform to JIS B 1099 where applicable.

2.5 ADHESIVES, CAULKS, AND SEALANTS

2.5.1 Adhesives

Adhesives shall be of a formula and type recommended by manufacturer. Adhesives shall be selected for their ability to provide a durable, permanent bond and shall take into consideration such factors as materials to be bonded, expansion and contraction, bond strength, fire rating, and moisture resistance.

2.5.1.1 Laminate Adhesive

Adhesive used to join high-pressure decorative laminate to wood shall be a water-based contact adhesive consistent with AWI and laminate manufacturer's recommendations. PVC edgebanding shall be adhered using a polymer-based hot melt glue.

2.5.2 Caulk

Caulk used to fill voids and joints between laminated components and between laminated components and adjacent surfaces shall be clear, 100 percent silicone.

2.5.3 Sealant

Sealant shall be of a type and composition recommended by the substrate manufacturer to provide a moisture barrier at sink cutouts and all other locations where unfinished substrate edges may be subjected to moisture.

2.6 FABRICATION

Verify field measurements as indicated in the shop drawings before fabrication. Fabrication and assembly of components shall be accomplished at the shop site to the maximum extent possible. Construction and fabrication of cabinets and their components shall meet or exceed the requirements for AWI premium grade unless otherwise indicated in this specification.

2.6.1 Base and Wall Cabinet Case Body

2.6.2 Vanity Apron Base

Base assembly components shall be treated lumber. Finished height for each apron base shall be as indicated on the drawings.

2.6.3 Vanity Apron

Components shall be constructed from the following materials and thicknesses:

2.6.3.1 Body Members (Bottoms and Front)

15 mm to 20 mm medium density fiberboard (MDF) panel product

2.6.4 Laminate Application

Laminate application to substrates shall follow the recommended procedures and instructions of the laminate manufacturer and JIS K 6903, using tools and devices specifically designed for laminate fabrication and application. Provide a balanced backer sheet (Grade BK) wherever only one surface of the component substrate requires a plastic laminate finish. Apply required grade of laminate in full uninterrupted sheets consistent with manufactured sizes using one piece for full length only, using adhesives specified herein or as recommended by the manufacturer. Fit corners and joints hairline. All laminate edges shall be machined flush, filed, sanded, or buffed to remove machine marks and eased (sharp corners removed). Clean up at easing shall be such that no overlap of the member eased is visible. Fabrication shall conform to laminate types and grades for component surfaces and shall be as follows unless otherwise indicated on the drawings:

2.6.4.1 Vanity Body

Exterior (exposed) surfaces to include exposed and semi-exposed face frame surfaces: HPDL Grade VGS.

2.6.4.2 Tolerances

Flushness, flatness, and joint tolerances of laminated surfaces shall meet the premium grade requirements.

PART 3 EXECUTION

3.1 INSTALLATION

Installation shall comply with applicable requirements for premium quality standards. Countertops and fabricated assemblies shall be installed level, plumb, and true to line, in locations shown on the drawings. Cabinets and other laminate clad casework assemblies shall be attached and anchored securely to the walls with mechanical fasteners that are appropriate for the wall construction.

3.1.1 Anchoring Systems

3.1.1.1 Wall

Vanity apron to be wall mounted shall utilize minimum 13 mm thick lumber or panel product hanging strips, minimum 60 mm width; securely attached to the wall side of the cabinet back, both top and bottom.

-- End of Section --

SECTION 06 61 16

SOLID SURFACING FABRICATIONS

08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D570	(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D696	(2016) Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30 degrees C and 30 degrees C With a Vitreous Silica Dilatometer
ASTM D790	(2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D2583	(2013a) Indentation Hardness of Rigid Plastics by Means of a Barcol Impressor
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM G21	(2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

CSA GROUP (CSA)

CSA B45.5-17/IAPMO Z124	(2017; Errata 2017; Errata 2018) Plastic Plumbing Fixtures
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INTERNATIONAL CAST POLYMER ASSOCIATION (ICPA)

ICPA SS-1	(2001) Performance Standard for Solid Surface Materials
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JAPANESE STANDARDS ASSOCIATION (JSA)

JIS K 6902	(2022) Testing Method for Laminated Thermosetting High-Pressure Decorative Sheets
JIS K 7060	(1995) Testing Method for Barcol Hardness of Glass Fiber Reinforced Plastics
JIS K 7161-1	(2014) Plastics - Determination of Tensile Properties - Part 1: General Principles
JIS K 7161-2	(2014) Plastics - Determination of Tensile Properties - Part 2: Test Conditions for Moulding and Extrusion Plastics
JIS K 7209	(2000) Plastics - Determination of Water Absorption
JIS Z 2801	(2012) Antibacterial Processed Products - Antibacterial Test Method / Antibacterial Effect

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3	(2005) Standard for High-Pressure Decorative Laminates
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NSF INTERNATIONAL (NSF)

NSF/ANSI 51	(2023) Food Equipment Materials
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1.2 SYSTEM DESCRIPTION

- a. Work under this section includes and other items utilizing solid surfacing material fabrications as indicated on the drawings and as described in this specification. Do not change source of supply for materials after work has started, if the appearance of finished work would be affected.
- b. In most instances, installation of solid surfacing material fabricated components and assemblies requires strong correctly located structural support provided by other trades. To provide a stable, sound, secure installation, close coordination is required between the solid surfacing material fabricator/installer and other trades to ensure that necessary structural wall support, cabinet counter top structural support, proper clearances, and other supporting components are provided for the installation of wall panels, counter tops, shelving, and all other solid surfacing material fabrications to the degree and extent recommended by the solid surfacing material manufacturer.
- c. Provide appropriate staging areas for solid surfacing material fabrications. Allow variation in component size and location of openings of plus or minus 3 mm.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality

Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation; G

SD-03 Product Data

Solid Polymer; G

SD-04 Samples

Material; G

Counter Tops; G

SD-06 Test Reports

Test Report Results

SD-07 Certificates

Qualifications

SD-10 Operation and Maintenance Data

Solid Polymer, Data Package 1; G

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

To ensure warranty coverage, provide manufacturer certified solid surfacing fabricators to fabricate the solid surfacing material being utilized. Mark all fabrications with the fabricator's certification label affixed in an inconspicuous location. Minimum of 5 years of experience working with solid surfacing materials is required of fabricators. Submit solid surfacing material manufacturer's certification attesting to fabricator qualification approval.

1.5 DELIVERY, STORAGE, AND HANDLING

Do not deliver materials to project site until areas are ready for installation. Deliver components and materials to the site undamaged, in containers clearly marked and labeled with manufacturer's name. Store materials indoors and take adequate precautions to prevent damage to finished surfaces. Provide protective coverings to prevent physical damage or staining following installation, for duration of project.

1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials, excluding damages caused by physical or chemical abuse or excessive heat, and workmanship for a period of 10 years from date of final acceptance of the work.

PART 2 PRODUCTS

2.1 MATERIAL

Submit detail fabrication drawings and installation drawings of each solid surfacing fabrication indicated. Include elevations, dimensions, clearances, details of construction and anchorage, and details of joints and connections.

Submit manufacturers' descriptive product data for solid polymer fabrication indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for solid polymer fabrication in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

2.1.1 Solid Surfacing Material

Provide solid polymer that is a homogeneous filled solid polymer; not coated, laminated or of a composite construction, complying with ICPA SS-1. Provide material that meets or exceeds the minimum physical and performance properties specified. Superficial damage to a depth of 0.25 mm must be repairable by sanding or polishing. Material thickness is as indicated below; required minimum thickness is 6 mm. Submit a minimum 102 by 102 mm sample of each color and pattern for approval; include full range of color and pattern variation. Retain approved samples as a standard for this work. Submit test report results from an independent testing laboratory attesting that the submitted solid surfacing materials meet or exceed each of the specified performance requirements.

a. Horizontal Surfaces: 19 mm thick material

2.1.2 Cast, 100 Percent Acrylic Polymer Solid Surfacing Material

Cast, 100 percent acrylic solid polymer material composed of acrylic polymer, mineral fillers, and pigments. Provide acrylic polymer that meets or exceeds the following minimum performance requirements:

PROPERTY	REQUIREMENT (min. or max.)	TEST PROCEDURE
Tensile Strength	4000 psi (max.)	ASTM D638 or JIS K 7161-1 /JIS K 7161-2
Hardness	55-Barcol Impressor (min.)	ASTM D2583 or JIS K 7060
Thermal Expansion	.000023 in/in/F (max.)	ASTM D696
Boiling Water Surface Resistance	No Change	ANSI/NEMA LD 3-3.05 or JIS K 6902
High Temperature Resistance	No Change	ANSI/NEMA LD 3-3.06 or JIS K 6902

PROPERTY	REQUIREMENT (min. or max.)	TEST PROCEDURE
Impact Resistance (Ball drop)		ANSI/NEMA LD 3-303 <u>or</u> <u>JIS K 6902</u>
1/4 inch sheet	36-inches, 1/2 lb ball, no failure	
1/2 inch sheet	140-inches, 1/2 lb ball, no failure	
3/4 inch sheet	200-inches, 1/2 lb ball, no failure	
Mold & Mildew Growth	No growth	ASTM G21
Bacteria Growth	No growth	ASTM G21
Liquid Absorption (Weight in 24 hrs.)	0.1 percent max.	ASTM D570 <u>or</u> JIS K 7209
Flammability		ASTM E84
Flame Spread	25 max.	
Smoke Developed	30 max.	
Sanitation	"Food Contact" approval	NSF/ANSI 51 <u>or</u> JIS Z 2801
Flexural Strength	6,800 [10,400] psi (min.)	ASTM D790

2.1.3 Material Patterns and Colors

Provide pattern and color for all solid surfacing material components and fabrications as indicated on the Drawings. Provide products with consistent patterned color throughout thickness of the product.

2.1.4 Surface Finish

Provide a uniform appearance on exposed finished surfaces and edges. Exposed surface finish is matte; gloss rating of 5-20.

2.2 ACCESSORY PRODUCTS

Provide accessory products, as specified below, as manufactured by the solid surfacing material manufacturer or as approved by the solid surfacing material manufacturer for use with the solid surfacing materials being specified.

2.2.1 Adhesives

Provide a two-part seam adhesive kit to create permanent, inconspicuous, non-porous, hard seams and joints by chemical bond between solid surfacing materials and components to create a monolithic appearance of the fabrication. Provide adhesive approved by the solid surfacing material manufacturer. Color-match adhesive to the surfaces being bonded where

solid-colored, solid surfacing materials are being bonded together. Provide clear or color matched seam adhesive where particulate patterned, solid surfacing materials are being bonded together.

2.2.2 Silicone Sealant

Provide silicone sealant, mildew-resistant, single-component, nonsag, plus 25 percent and minus 25 percent movement capability, acid-curing; ASTM C920, Type S, Grade NS, Class 25, Use NT; clear formulation; approved for use by the solid surfacing material manufacturer.

2.2.3 Mounting Hardware

Provide mounting hardware, including sink/bowl clips, inserts and fasteners for attachment of undermount sinks and lavatories.

2.3 FABRICATIONS

Provide factory or shop fabricate components to sizes and shapes indicated, to the greatest extent practical, in accordance with approved Shop Drawings and manufacturer's requirements. Provide factory cutouts for sinks, lavatories, and plumbing fixtures where indicated on the drawings. Contours and radii must be routed to template, with edges smooth. Defective and inaccurate work will be rejected. Submit product data indicating product description, fabrication information, and compliance with specified performance requirements for solid surfacing material, joint adhesive, and sealants.

2.3.1 Joints and Seams

Form joints and seams between solid surfacing material components using manufacturer's approved seam adhesive. Provide inconspicuous joints in appearance without voids to create a monolithic appearance.

2.3.2 Edge Finishing

Rout and finish component edges to a smooth, uniform appearance and finish. Provide edge shapes and treatments, including any inserts, as detailed on the drawings. Rout all cutouts, then sand all edges smooth. Repair or reject defective or inaccurate work.

2.3.3 Counter Top Splashes

Fabricate backsplashes and end splashes from 13 mm thick solid surfacing material to be 102 mm high. Provide backsplashes and end splashes for all counter tops. Shop fabricate backsplashes and provide loose, to be field attached.

2.3.3.1 End Splashes

Provide end splashes loose for installation at the jobsite after horizontal surfaces to which they are to be attached have been installed.

2.3.4 Counter Tops

Fabricate all solid surfacing material, counter top components from 19 mm thick material. Indicate details, dimensions, locations, and quantities on the drawings. Provide counter tops with 102 mm high loose at all locations. Attach 51 mm wide reinforcing strip of solid surfacing

material under each horizontal counter top seam. Submit a minimum 305 mm wide by 152 mm deep, full size sample for each type of counter top shown on the project drawings; include the edge profile and backsplash as detailed on the drawings and at least one seam. Retain approved sample as standard for this work. Provide bullnose edge profile.

2.3.4.1 Counter Tops with Bowls

- a. Provide manufacturer's standard solid polymer bowls, pre-molded product specifically designed for attachment to solid surfacing material counter tops. See paragraph SOLID POLYMER BOWLS for additional requirements

2.3.4.2 Cafeteria Counter Tops

Include cutouts for cold or hot appliances to templates furnished by the equipment manufacturers. Reinforce joints and cutouts as recommended by the solid surfacing material manufacturer. Provide insulation between the solid surface material and all appliances, hot or cold. Thermally isolate hot applications from cold applications in accordance with the solid surfacing material manufacturer's recommendations. Provide expansion joints as necessary to accommodate hot appliances. Provide adequate ventilation for cabinets beneath counter tops to prevent heat build-up.

2.3.5 Solid Polymer Bowls

Provide solid polymer bowls that are a standard product of the solid polymer manufacturer, in compliance with CSA B45.5-17/IAPMO Z124 requirements, designed specifically to be installed in solid surfacing material counter tops. Provide bowls of the same polymer composition as the adjoining counter top. Bowl design must support a seam adhesive undermount installation method. Bowl dimensions must be manufacturer's standard, in configuration and locations as indicated on the Drawings.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Components

Install all components and fabricated units plumb, level, and rigid. Make field joints between solid surfacing material components using solid surfacing material manufacturer's approved seam adhesives, to provide a monolithic appearance with joints inconspicuous in the finished work. Attach metal or vitreous china sinks and lavatory bowls to counter tops using solid surfacing material manufacturer's recommended clear silicone sealant and mounting hardware. Install solid polymer sinks and bowls using a color-matched seam adhesive.

3.1.1.1 Loose Counter Top Splashes

Mount loose splashes in the locations noted on the drawings. Adhere loose splashes to the counter top with a color matched silicone sealant when the solid surfacing material components are solid colors. Use a clear silicone sealant to provide adhesion of particulate patterned solid surfacing material splashes to counter tops.

3.1.2 Silicone Sealant

Use specified silicone sealant to seal all expansion joints between solid surfacing material components and all joints between solid surfacing material components and other adjacent surfaces such as walls, floors, ceiling, and plumbing fixtures. Provide sealant bead smooth and uniform in appearance and minimum size necessary to bridge any gaps between the solid surfacing material and the adjacent surface. Provide continuous bead and run the entire length of the joint being sealed.

3.1.3 Plumbing

Make plumbing connections to sinks and lavatories in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.2 CLEAN-UP

Components must be cleaned after installation and covered to protect against damage during completion of the remaining project items. Damaged components must be repaired or replaced at the Contractor's sole expense.

-- End of Section --

SECTION 07 21 13

BOARD INSULATION
08/23

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Standard Details; G

SD-03 Product Data

Board Insulation; G

Protection Board or Coating; G

Accessories including sealants; G

SD-08 Manufacturer's Instructions

Insulation Installation and Handling

Protection Board or Coating Installation

Adhesive

SD-11 Closeout Submittals

Draft Warranty; G

Final Warranty; G

1.2 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

1.3 PRODUCT DATA

Include descriptions, recommendations for product shelf life, requirements for protection board or coatings, and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery

Deliver materials to the site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.4.2 Storage

Inspect materials delivered to the site for damage and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Comply with manufacturer's recommendations for handling, storage, and protection of materials before and during installation.

1.5 SPECIAL WARRANTIES

1.5.1 Warranty

Provide manufacturer's material warranty for all system components for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft warranty and final warranty in accordance with Sections 01 78 00 CLOSEOUT SUBMITTALS and 01 78 23 OPERATION AND MAINTENANCE DATA.

PART 2 PRODUCTS

2.1 BOARD INSULATION

Provide thermal insulating materials as recommended by manufacturer for each type of application indicated. Provide insulation with the following physical properties:

2.1.1 Phenolic-Foam Insulation

- a. Pure Substand or Mixture: Mixture.
- b. Chemical Name:
 - (1) Base Material: Phenolic resin foam.
 - (2) Face Material: Mineral filled, glass fiber mixed paper.
- c. Hazard Identification
 - (1) Hazard Classification: Not applicable to classification criteria.
 - (2) Hazardous Properties: Not recognized.
 - (3) Toxicity: May cause irritation when skin or eye comes into contact.

(4) Environmental Impact: None known.

d. Physical and Chemical Properties

(1) Appearance: Brownish-red foam.

(2) Volatility: None.

(3) Melting Point: No Data.

(4) Bulk Specific Gravity: 0.04-0.08.

(5) Solubility: Not soluble in water and organic solvents.

e. Hazard Information

(1) Flash Point: Equal to or greater than 220 degrees C.

(2) Auto-ignition Temperature: Equal to or greater than 500 degrees C.

(3) Flammability: Oxygen index equal to or greater than 30.

(4) Oxidation Properties: None.

(5) Flammability (Pyrophoricity, Water Reactivity): None.

(6) Corrosive Properties: May corrode metal in contact when exposed to high humidity for a long period of time.

(7) Self-reactive Properties, Explosive Properties: None.

(8) Stability and Reactivity: Chemical stable.

(9) Dust Explosion Properties: Resin foam dust may form explosive dust-air mixtures.

2.1.2 Prohibited Materials

Do not provide materials containing asbestos.

2.2 PROTECTION BOARD OR COATING

As recommended by insulation manufacturer.

2.3 ACCESSORIES

2.3.1 Adhesive

As recommended by insulation manufacturer.

2.3.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Prior to installation, ensure all areas that will be in contact with the insulation are dry and free of projections that could cause voids or compressed insulation.

3.2 INSTALLATION

3.2.1 Insulation Installation and Handling

Provide insulation in accordance with the manufacturer's printed installation instructions. Keep material dry and free of extraneous materials.

3.2.2 Electrical Wiring

Do not install insulation in a manner that would enclose electrical conductors, that are not installed in conduit, between two layers of insulation.

3.2.3 Continuity of Insulation

Butt tightly against adjoining boards, studs, headers, and obstructions.

3.2.4 Coordination

Verify final installed insulation thicknesses comply with thicknesses to match existing.

3.3 INSTALLATION ON WALLS

3.3.1 Installation using Furring Strips

Install insulation members as recommended by insulation manufacturer based on existing project conditions.

3.3.2 Protection Board or Coating Installation

Install protection board or coating in accordance with manufacturer's printed instructions. Install protection over all exterior exposed insulation and to 300 mm below grade.

-- End of Section --

SECTION 07 21 16

MINERAL FIBER BLANKET INSULATION
11/11, CHG 4: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C553	(2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
ASTM C665	(2017) Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing
ASTM C930	(2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134	Respiratory Protection
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JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 9504	(2021) Man Made Mineral Fibre Thermal Insulation Materials
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Blanket Insulation

SD-08 Manufacturer's Instructions

Insulation

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery

Deliver materials to site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.3.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling.

1.4 SAFETY PRECAUTIONS

1.4.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) in accordance with 29 CFR 1910.134.

1.4.2 Other Safety Concerns

Consider other safety concerns and measures as outlined in ASTM C930.

PART 2 PRODUCTS

2.1 BLANKET INSULATION

ASTM C553 or JIS A 9504, Type VII, Non-combustible, lightweight, semi-rigid mineral wool batt insulation blankets without membrane coverings; Class A, with a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84.

2.1.1 Thermal Conductivity Value (W/m2)

The W/m2 must be as indicated on drawings.

2.1.2 Prohibited Materials

Do not provide asbestos-containing materials.

2.2 BLOCKING

Wood, metal, unfaced mineral fiber blankets in accordance with ASTM C665 or JIS A 9504, Type VII, or other approved materials.

2.3 ACCESSORIES

2.3.1 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation

manufacturer.

2.3.2 Wire Mesh

Corrosion resistant and as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Before installing insulation, ensure that areas that will be in contact with the insulation are dry and free of projections which could cause voids, compressed insulation, or punctured vapor retarders. If moisture or other conditions are found that do not allow the workmanlike installation of the insulation, do not proceed but notify Contracting Officer of such conditions.

3.2 INSTALLATION

3.2.1 Insulation

Install and handle insulation in accordance with manufacturer's instructions. Keep material dry and free of extraneous materials. Any materials that show visual evidence of biological growth due to presence of moisture must not be installed on the building project. Ensure personal protective clothing and respiratory equipment is used as required. Observe safe work practices.

3.2.1.1 Electrical wiring

Do not install insulation in a manner that would sandwich electrical wiring between two layers of insulation.

3.2.1.2 Continuity of Insulation

Install blanket insulation to butt tightly against adjoining blankets and to studs, headers and any obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joints, roof, and floor. Avoid creating thermal bridges.

3.2.1.3 Sizing of Blankets

Provide only full width blankets when insulating between studs. Size width of blankets for a snug fit where studs are irregularly spaced.

-- End of Section --

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SECTION 07 24 00

EXTERIOR INSULATION AND FINISH SYSTEMS
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM C67/C67M	(2024) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C578	(2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM C847	(2018; R 2024) Standard Specification for Metal Lath
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D968	(2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive
ASTM D2247	(2015; R 2020) Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity
ASTM D3273	(2021) Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E330/E330M	(2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E331	(2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows,

Skylights, Doors, and Curtain Walls by
Uniform Static Air Pressure Difference

ASTM E2098/E2098M	(2025) Standard Test Method for Determining Tensile Breaking Strength of Glass Fiber Reinforcing Mesh for Use in Class PB Exterior Insulation and Finish Systems (EIFS), after Exposure to a Sodium Hydroxide Solution
ASTM E2486/E2486M	(2022) Standard Test Method for Impact Resistance of Class PB and PI Exterior Insulation and Finish Systems (EIFS)
ASTM E2570/E2570M	(2007; R 2019) Standard Test Methods for Evaluating Water-Resistive Barrier (WRB) Coatings Used under Exterior Insulation and Finish Systems (EIFS) or EIFS with Drainage
ASTM G153	(2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
INTERNATIONAL CODE COUNCIL (ICC)	
ICC IBC	(2024) International Building Code
NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)	
NFPA 268	(2022) Standard Test Method for Determining Ignitability of Exterior Wall Assemblies Using a Radiant Heat Energy Source
NFPA 285	(2023) Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Wall Assemblies Containing Combustible Components

1.2 SYSTEM DESCRIPTION AND REQUIREMENTS

The exterior insulation and finish system (EIFS) must be a job-fabricated, drainable, exterior wall covering consisting of sheathing, air and moisture barrier, insulation board, reinforcing fabric, base coat, finish coat, adhesive and mechanical fasteners as applicable. The system components must be compatible with each other and with the substrate as recommended or approved by, and the products of, a single manufacturer regularly engaged in furnishing Exterior Insulation and Finish Systems. All materials must be installed by an applicator approved by the system manufacturer. EIFS must be Class PB in color and texture to match existing.

1.2.1 System Requirements and Tests

The system must meet the performance requirements as verified by the tests listed below. Where a wall system of similar type, size, and design as specified for this project has been previously tested under the condition

specified herein, the resulting test reports may be submitted in lieu of job specific tests.

1.2.1.1 Water Penetration

Test the system for water penetration by uniform static air pressure in accordance with ASTM E331. There must be no penetration of water beyond the plane of the base coat/EPS board interface after 15 minutes at 300 Pa, or 20 percent of positive design wind pressure, whichever is greater.

1.2.1.2 Wind Load Test

Test the system for wind load by uniform static air pressure in accordance with ASTM E330/E330M (procedure A) to the design pressure as indicated on the drawings. There must be no permanent deformation, delamination, or other deterioration.

1.2.1.3 Full Scale or Intermediate Scale Fire Test

Conduct wall fire test using apparatus, specimen, performance criteria, and procedure in accordance with NFPA 285 when required by ICC IBC 2603.5.5. The following requirements must be met:

- a. No vertical spread of flame within core of panel from one story to the next.
- b. No flame spread over the exterior surface.
- c. No vertical flame spread over the interior surface from one story to the next.
- d. No significant lateral spread of flame from compartment of fire origin to adjacent spaces.

1.2.2 Component Requirements and Tests

The components of the system must meet the performance requirements as verified by the tests listed below.

1.2.2.1 Surface Burning Characteristics

Conduct ASTM E84 test on samples consisting of base coat, reinforcing fabric, and finish coat. Cure for 28 days. The flame spread index must be 25 or less and the smoke developed index must be 450 or less.

1.2.2.2 Radiant Heat

The system must be tested in accordance with NFPA 268 on both the minimum and maximum thickness of insulation intended for use with no ignition during the 20-minute period.

1.2.2.3 Impact Resistance

- a. Class PB Systems: Hemispherical Head Test; 28 day cured specimen of PB EIFS in accordance with ASTM E2486/E2486M. The test specimen must exhibit no broken reinforcing fabric in accordance with ASTM E2486/E2486M at an impact of 17.1 J and above.

1.2.3 Sub-Component Requirements and Tests

Unless otherwise stated, the test specimen must consist of reinforcing mesh, base coat, and finish coat applied in accordance with manufacturer's printed recommendations to the insulation board to be used on the building. For mildew resistance, only the finish coat is applied onto glass slides for testing. The specimen must be suitably sized for the apparatus used and be allowed to cure for a minimum of 28 days prior to testing.

1.2.3.1 Abrasion Resistance

Test in accordance with ASTM D968, Method A. Test a minimum of two specimens. After testing, the specimens must show only very slight smoothing, with no loss of film integrity after 500 liters of sand.

1.2.3.2 Accelerated Weathering

Test in accordance with ASTM G153, Cycle 1. After 2000 hours specimens must exhibit no visible cracking, flaking, peeling, blistering, yellowing, fading, or other such deterioration.

1.2.3.3 Mildew Resistance

Test in accordance with ASTM D3273. The specimen must consist of the finish coat material, applied to clean 75 mm by 100 mm glass slides and must be allowed to cure for 28 days. After 28 days of exposure, the specimen must not show any growth.

1.2.3.4 Salt Spray Resistance

Test in accordance with ASTM B117. The specimen must be a minimum of 100 mm by 150 mm and must be tested for a minimum of 300 hours. After exposure, the specimen must exhibit no observable deterioration, such as chalking, fading, or rust staining.

1.2.3.5 Water Resistance

Test in accordance with ASTM D2247. The specimen must be a minimum of 100 mm by 150 mm. After 14 days, the specimen must exhibit no cracking, checking, crazing, erosion, blistering, peeling, or delamination.

1.2.3.6 Absorption-Freeze-Thaw

Class PB systems must be tested in accordance with ASTM E2570/E2570M for 60 cycles of freezing and thawing. After testing, the specimen must exhibit no cracking, checking, or splitting, and negligible weight gain. Class PM systems must be tested in accordance with ASTM C67/C67M for 50 cycles of freezing and thawing. After testing, the specimens must exhibit no cracking or checking and have negligible weight gain.

1.2.3.7 Sample Boards

Unless otherwise stated, provide sample EIFS Component 300 by 600 mm, on sheathing board, including finish color and texture, typical joints and sealant. If more than one color, finish, or pattern is used, provide one sample for each. The test specimen must consist of reinforcing mesh, base coat, and finish coat applied in accordance with manufacturer's printed recommendations to the insulation board to be used on the building.

1.2.4 Moisture Analysis

Perform a job specific vapor transmission analysis based on project specific climate and specified wall components and materials. Indicate the temperatures and relative humidities for the inside and outside of the building; a complete listing of the building components, their thickness, thermal resistance and permeance, as well as building location and use. If a mathematical model was used for the analysis, include the name of the model and the supplier/developer.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Thermal Insulation

Adhesive

Mechanical Fasteners

Accessories

Base Coat

Portland Cement

Reinforcing Fabric

Finish Coat

Joint Sealant

Sealant Primer

Bond Breaker

Backer Rod

Insulation Board

Warranty

SD-04 Samples

Sample Boards; G

Color and Texture

SD-05 Design Data

Moisture Analysis Calculations

SD-06 Test Reports

Abrasion Resistance

Accelerated Weathering

Impact Resistance

Mildew Resistance

Salt Spray Resistance

Water Vapor Transmission

Absorption-Freeze-Thaw

Wall Fire Test

Water Penetration

Water Resistance

Full Scale or Intermediate Scale Fire Test

Surface Burning Characteristics

Radiant Heat

Substrate

Wind Load Test

SD-07 Certificates

Qualifications of EIFS Manufacturer

Qualification of EIFS Installer

Qualification of Sealant Applicator

Qualifications of Third-Party Inspector

Inspection Checklist; G

SD-08 Manufacturer's Instructions

Manufacturer's Instructions

1.4 QUALITY CONTROL

1.4.1 Qualifications of EIFS Manufacturer

The EIFS must be the product of a manufacturer who has been in the practice of manufacturing and designing EIFS for a period of not less than

3 years, and has been involved in at least five projects similar to this project in size, scope, and complexity, in the same or a similar climate as this project.

1.4.2 Qualification of EIFS Installer

The EIFS Installer must be trained by the EIFS manufacturer to perform the installation of the System and must have successfully installed at least five projects at or near the size and complexity of this project. The Contractor must employ qualified workers trained and experienced in installing the manufacturer's EIFS.

1.4.3 Qualification of Sealant Applicator

The sealant applicator must be experienced and competent in the installation of high performance industrial and commercial sealants and must have successfully installed at least five projects at or near the size and complexity of this project. Also provide certification that sealant applicator is approved by the EIFS Manufacturer.

1.4.4 Qualifications of Third-Party Inspector

Submit evidence that third-party inspector has current certification from the Exterior Design Institute or equal inspector certification as inspector for the installation of EIFS.

1.4.5 Insulation Board

Insulation Board must be approved and labeled under third party quality program as required by applicable building code.

1.4.6 Shop Drawings

Submit shop drawings for EIFS. Show wall layout, construction and expansion joints, thermal insulation board, and reinforcing mesh and strip reinforcing fabric; joint and flashing details; details at wall penetrations; types and location of fasteners; details at wall louvers and repair/patch at doors, windows, base, corners and any other effected areas as determined from site inspection.

1.4.7 Pre-Installation Conference

After approval of submittals and before commencing any work on the EIFS, including installation of any insulation, and associated work, the Contracting Officer will hold a pre-installation conference to review:

- a. Drawings, specifications, and samples;
- b. Procedure for on-site inspection and acceptance of EIFS substrate and pertinent details (for example, mock-up installation);
- c. Contractor's plan for coordination of work of the various trades involved in providing EIF system and other components;
- d. Inspection procedures; and
- e. Safety requirements.

Pre-installation conference must be attended by the Contractor and all

personnel directly responsible for installation of the EIF system, including sealant applicator, and personnel responsible for related work, such as flashing and sheet metal, windows and doors, and a representative of the EIFS manufacturer. Before beginning EIFS work, the Contractor must confirm in writing the resolution of conflicts among those attending the pre-installation conference.

1.5 DELIVERY AND STORAGE

Deliver materials to job site in original unopened packages, marked with manufacturer's name, brand name, and description of contents. Store materials off the ground and in accordance with the manufacturer's recommendations in a clean, dry, well-ventilated area. Protect stored materials from rain, sunlight, and excessive heat. Keep coating materials which would be damaged by freezing at a temperature not less than 4 degrees C. Do not expose insulation board to flame or other ignition sources.

1.6 ENVIRONMENTAL CONDITIONS

- a. Do not prepare materials or apply EIFS during inclement weather unless appropriate protection is provided. Protect installed materials from inclement weather until they are dry.
- b. Apply sealants and wet materials only at ambient temperatures of 4 degrees C or above, unless supplemental heat is provided. The system must be protected from inclement weather and maintain this temperature for a minimum of 24 hours after installation.
- c. Do not leave insulation board exposed to sunlight after installation.

1.7 WARRANTY

Furnish manufacturer's standard warranty for the EIFS. Warranty must run directly to Government and cover a period of not less than 5 years from date Government accepted the work.

Warranty submittal must include joint and other details, such as end conditions, wall louvers and repair/patch at doors, windows, base, corners and any other effected areas as determined from site inspection. Include shelf life and recommended cleaning solvents in data for sealants. Include Safety Data Sheets (SDS) for all components of the EIFS. The SDS must be available at the job site.

PART 2 PRODUCTS

2.1 COMPATIBILITY

Provide all materials compatible with each other and with the substrate, and as recommended by EIFS manufacturer.

2.2 ADHESIVE

Manufacturer's standard product, including primer as required, must be compatible with substrate and insulation board to which the system is applied.

2.3 LATHING AND FURRING

Conform to ASTM C847, 1.4 kg/sqm, self-furring, galvanized.

2.4 MECHANICAL FASTENERS

Corrosion resistant and as approved by EIFS manufacturer. Select fastener type and pattern based on applicable wind loads and substrate into which fastener will be attached, to provide the necessary pull-out, tensile, and shear strengths.

2.5 THERMAL INSULATION

2.5.1 Manufacturer's Recommendations

Provide only thermal insulation recommended by the EIFS manufacturer for the type of application intended.

2.5.2 Insulation Board

Insulation board must be standard product of manufacturer and must be compatible with other system components. Boards must be factory marked individually with the manufacturer's name or trademark, the material specification number, the R-value at 24 degree C, and thickness. No layer of insulation can be less than 20 mm thick. The maximum thickness of all layers must not exceed 100 mm. Insulation Board must be certified as aged, in block form, prior to cutting and shipping, a minimum of 6 weeks by air drying, or equivalent.

- a. Thermal resistance: As indicated on drawings.
- b. Insulating material: ASTM C578 Type I as recommended by the EIFS manufacturer and treated to be compatible with other EIFS components. Age insulation by air drying a minimum of 6 weeks prior to cutting and shipping.
- c. Provide drainage system in accordance with manufacturer's written recommendations and approved shop drawings.

2.6 BASE COAT

Manufacturer's standard product and compatible with other system components.

2.7 PORTLAND CEMENT

Conform to ASTM C150/C150M, Type I as required, fresh and free of lumps, and approved by the systems manufacturer.

2.8 REINFORCING FABRIC

Ultra-high impact reinforcing fabric mesh must be alkali-resistant, balanced, open weave, glass fiber fabric made from twisted multi-end strands specifically treated for compatibility with the other system materials, and comply with ASTM E2098/E2098M and as recommended by EIFS manufacturer.

2.9 FINISH COAT

Manufacturer's standard product conforming to the requirements in the paragraph on Sub-Component Requirements and Tests. For color consistency, use materials from the same batch or lot number.

2.10 SEALANT PRIMER

Non-staining, quick-drying type recommended by sealant manufacturer and EIFS manufacturer.

2.11 ACCESSORIES

Conform to recommendations of EIFS manufacturer, including trim, edging, anchors, and expansion joints. All metal items and fasteners to be corrosion resistant.

2.12 JOINT SEALANT

Non-staining, quick-drying type meeting ASTM C920, as Type S or M, minimum Grade NS, minimum Class 25 and compatible with the finish system type and grade, and recommended by both the sealant manufacturer and EIFS manufacturer.

2.13 BOND BREAKER

As required by EIFS manufacturer and recommended by sealant manufacturer and EIFS manufacturer.

2.14 BACKER ROD

Closed cell polyethylene free from oil or other staining elements and as recommended by sealant manufacturer and EIFS manufacturer. Do not use absorptive materials as backer rod. Size the backer rod 25 percent larger than the width of the joint.

PART 3 EXECUTION

3.1 EXAMINATION

Examine substrate and existing conditions to determine that the EIFS can be installed as required by the EIFS manufacturer and that all work related to the EIFS is properly coordinated. Surface must be sound and free of oil, loose materials or protrusions which will interfere with the system installation. If deficiencies are found, notify the Contracting Officer and do not proceed with installation until the deficiencies are corrected. The substrate must be plane, with no deviation greater than 6 mm when tested with a 3 m straightedge. Determine flatness, plumbness, and any other conditions for conformance to manufacturer's instructions.

3.2 SURFACE PREPARATION

Prepare existing surfaces for application of the EIFS to meet flatness tolerances and surface preparation according to manufacturer's installation instructions. Provide clean surfaces free of oil and loose material without protrusions adversely affecting the installation of the insulation board. For adhesively attached EIFS, existing deteriorated paint must be removed. Due to substrate conditions or as recommended by the system manufacturer, a primer may be required. Apply the primer to

existing surfaces as recommended by the manufacturer. Use masking tape to protect areas adjacent to the EIFS to prevent base or finish coat to be applied to areas not intended to be covered with the EIFS. The Contractor must not proceed with the installation until all noted deficiencies of the substrate are corrected.

3.3 INSTALLATION

Install EIFS as indicated, comply with manufacturer's instructions except as otherwise specified, and in accordance with the shop drawings. Submit manufacturer's standard printed instructions for the installation of the EIFS; include requirements for condition and preparation of substrate, installation of EIFS, and requirements for sealants and sealing. Also include detailed finish repair procedures and information regarding compatibility of sealants with base and finish coatings.

EIFS must be installed only by an applicator trained by the EIFS manufacturer. Specifically, include all manufacturer recommended provisions regarding flashing and treatment of wall penetrations. Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

3.3.1 Insulation Board

Unless otherwise specified by the system manufacturer, place the long edge horizontally from level base line. Stagger vertical joints and interlock at corners. Butt joints tightly. Provide flush surfaces at joints. Offset insulation board joints from joints in sheathing by at least 200 mm. Align drainage channels of integral drainage system or provide polypropylene drainage lath space to provide a path for any water weeped from behind the insulation to escape wall construction. Use L-shaped insulation board pieces at corners of openings. Joints of insulation must be butted tightly. Surfaces of adjacent insulation boards must be flush at joints. Gaps greater than 1.6 mm between the insulation boards must be filled with slivers of insulation. Uneven board surfaces with irregularities projecting more than 1.6 mm must be rasped in accordance with the manufacturer's instructions to produce an even surface. Attach insulation board as recommended by manufacturer. The adhered insulation board must be allowed to remain undisturbed for 24 hours prior to proceeding with the installation of the base coat/reinforcing mesh, or longer if necessary for the adhesive to dry. However, do not leave insulation board exposed longer than recommended by insulation manufacturer.

3.3.2 Base Coat and Reinforcing Fabric Mesh

3.3.2.1 Class PB Systems

Allow the adhered insulation board to dry for 24 hours, or longer if necessary, prior to proceeding with the installation of the base coat/reinforcing fabric mesh. Install reinforcing fabric in accordance with manufacturer's instructions. Mix base coat in accordance with the manufacturer's instructions and apply to insulated wall surfaces to the thickness specified by the system manufacturer and provide any other reinforcement recommended by EIFS manufacturer. Trowel the reinforcing fabric mesh into the wet base coat material. Fully embed the mesh in the base coat. When properly worked-in, the pattern of the reinforcing fabric mesh must not be visible. Provide diagonal reinforcement at opening corners. Back-wrap or edge wrap all terminations of the EIFS. Overlap the

reinforcing fabric mesh a minimum of 60 mm on previously installed mesh, or butted, in accordance with the manufacturer's instructions.

3.3.3 Finish Coat

The base coat/reinforcing mesh must be allowed to dry a minimum of 24 hours prior to application of the finish coat. Surface irregularities in the base coat, such as trowel marks, board lines, and reinforcing mesh laps, must be corrected prior to the application of the finish coat. Apply and level finish coat in one operation. Obtain final texture by trowels, floats, or by spray application as necessary to achieve the required finish matching approved sample. Apply the finish coat to the dry base coat maintaining a wet edge at all times to obtain a uniform appearance. The thickness of the finish coat must be in accordance with the system manufacturer's current published instructions. Apply finish coat so that it does not cover surfaces to which joint sealants are to be applied.

3.4 JOINT SEALING

Seal EIFS at openings as recommended by the system manufacturer. Apply sealant only to the base coat with EIFS Manufacturer's color coating. Do not apply sealant to the finish coat.

3.4.1 Surface Preparation, Backer Rod, and Primer

Immediately prior to application, remove loose matter from joint. Ensure that joint is dry and free of finish coat, or other foreign matter. Install backer rod. Apply primer as required by sealant and EIFS manufacturer. Check that joint width is as shown on drawings but in no case can it be less than 13 mm for perimeter seals and 20 mm for expansion joints. The width must not be less than 4 times the anticipated movement. Check sealant manufacturer's recommendations regarding proper width to depth ratio.

3.4.2 Sealant

Do not apply sealant until all EIFS coatings are fully dry. Apply sealant in accordance with sealant manufacturer's instructions with gun having nozzle that fits joint width. Do not use sealant that has exceeded shelf life or cannot be discharged in a continuous flow. Completely fill the joint solidly with sealant without air pockets so that full contact is made with both sides of the joint. Tool sealant with a round instrument that provides a concave profile and a uniformly smooth and wrinkle free sealant surface. Do not wet tool the joint with soap, water, or any other liquid tooling aid. During inclement weather, protect the joints until sealant application. Use particular caution in sealing joints between window and door frames and the EIFS wall and at all other wall penetrations. Clean all surfaces to remove excess sealant.

3.5 FIELD QUALITY CONTROL

Throughout the installation, the Contractor must establish and maintain an inspection procedure to assure compliance of the installed EIFS with Contract requirements. Work not in compliance must be removed and replaced or corrected in an approved manner. The inspection procedures, from acceptance of deliveries through installation of sealants and final acceptance must be performed by a qualified inspector trained by the manufacturer. No work on the EIFS is allowed unless the inspector is

present at the job site.

3.5.1 Third-Party Inspection

Provide full-time third-party inspection during the entire process of installing the EIFS, from examination through cleanup. The third-party inspector must be certified by the Exterior Design Institute (EDI), AWC, or by an equivalent independent party and must be trained in the proper installation of EIFS.

3.5.2 Inspection Checklist

During the installation and at the completion of installation, perform inspections covering at the minimum all applicable items enumerated on the attached checklist. The inspector must initial and date all applicable items, sign the checklist, and submit to the Contracting Officer daily.

CHECKLIST

<u>Item</u>	<u>Description</u>	<u>Appr'd/Date</u>
a.	Materials are handled and stored correctly.	_____
b.	Environmental conditions are within specified limits, including temperature not below 4 degrees C (40 degrees F), and the work is protected from the elements as required.	_____
c.	Preparation and installation is performed by qualified personnel using the correct tools.	_____
d.	Adjacent areas to which EIFS is not to be applied (such as on window and door frames) are protected with masking tape, plastic films, drop cloths, or other covering to prevent accidental application of EIFS materials.	_____
e.	Control, expansion and aesthetic joints are installed as indicated or recommended. Accessories are properly installed.	_____
f.	Substrate is in-plane, properly attached, clean, dry, and free of contaminants. Concrete substrate is free of efflorescence.	_____
g.	Materials are mixed thoroughly and in proper proportions.	_____
h.	Adhesive is applied in sufficient quantity with proper-size notched trowel.	_____
i.	Mechanical attachments have proper spacing, layout and fastener depth.	_____
j.	Insulation boards are tightly abutted, in running bond pattern, with joints staggered with the sheathing, board corners interlocked, L-shaped boards around openings, edges free of adhesive, and provision for joints. Gaps are filled and surfaces rasped.	_____
k.	Insulation adhesive must be allowed to dry (a minimum of	_____

CHECKLIST

<u>Item</u>	<u>Description</u>	<u>Appr'd/Date</u>
	24-hours) prior to the application of the base coat.	
l.	Reinforcing fabric mesh is properly back-wrapped at terminations.	=====
m.	Reinforcing fabric mesh is fully embedded and properly placed. Corners are reinforced. Openings are diagonally reinforced. Mesh overlaps minimum 65 mm (2-1/2 inches).	=====
n.	Base coat thickness is within specified limits.	=====
o.	The base coat/reinforcing fabric mesh must be allowed to dry (a minimum of 24-hours) prior to the application of the finish coat.	=====
p.	Finish coat is applied with sufficient number of personnel and stopped at suitable points. Floats and methods of texturing are uniform.	=====
q.	All flashings are properly installed.	=====
r.	All joints are properly sealed in their entire length at time and under environmental conditions as specified by the manufacturer.	=====
s.	All scaffolding, equipment, materials, debris and temporary protection are removed from site upon completion.	=====
Name of Inspector:_____ Signed:_____		
Date:_____		

3.6 CLEANUP

Upon completion, remove all scaffolding, equipment, materials and debris from site. Remove all temporary protection installed to facilitate installation of EIFS.

-- End of Section --

SECTION 07 57 12.42

LIQUID APPLIED ELASTOMERIC SILICONE WALL COATING
04/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D 1644	(2012) Standard Test Methods for Nonvolatile Content of Varnishes
ASTM D 2240	(2005; R 2010) Standard Test Method for Rubber Property - Durometer Hardness
ASTM D 2697	(2003; R 2008) Volume Nonvolatile Matter in Clear or Pigmented Coatings
ASTM D 412	(2006ae2) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D 56	(2005; R 2010) Flash Point by Tag Closed Cup Tester
ASTM D 822	(2001; R 2006) Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings
ASTM D 93	(2010) Flash-Point by Pensky-Martens Closed Cup Tester
ASTM E 96/E 96M	(2010) Standard Test Methods for Water Vapor Transmission of Materials

1.2 DESCRIPTION OF WALL SYSTEM

The wall system shall consist of a sprayed or rolled waterproof elastomeric silicone protective coating and surfaced with ceramic granules in areas indicated.

1.2.1 Performance Requirements

The installed wall system shall be watertight; free of defects including coating delamination, blistering, or voids; suitable for the climatic and service conditions of the installation.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a code following the "G" classification identifies the office that

will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Protective Water-based Elastomeric Silicone Coating

Primer

Sealants

SD-04 Samples

Elastomeric Silicone Wall System Mock-up

Sample Warranty

SD-07 Certificates

Qualification of Manufacturer

Qualification of Applicator

SD-08 Manufacturer's Instructions

Protective Coating

Manufacturer's Application Instructions

SD-11 Closeout Submittals

Warranty

1.4 QUALITY ASSURANCE

1.4.1 Qualification of Manufacturer

Sprayed elastomeric coating products manufacturer shall have a minimum of 10 years experience in the manufacture of elastomeric coating products.

1.4.1.1 Manufacturer's Technical Representative

Manufacturer's technical representative shall have a minimum of 10 years experience with elastomeric silicone wall system products and installations and be thoroughly familiar with the products to be installed, installation requirements and practices, quality control of the installation, and with any special considerations in the geographical area and climate where construction will take place. The representative shall be available to perform field inspections and attend meetings as specified.

1.4.2 Qualification of Applicator

The wall system applicator shall have prior manufacturer training in the application of elastomeric silicone coating materials. Applicator shall have a minimum of 5 years experience in application of the specified materials. Mechanics applying the coating materials shall have minimum 3 years prior experience in handling and spraying the type of materials specified and spray equipment must be operated by or under the direct full-time supervision of manufacturer-trained personnel. The applicator

shall supply the names, locations and client contact information of 5 projects of similar size and scope that the applicator has constructed using the manufacturer's wall products submitted for this project within the previous three years.

1.4.3 Pre-construction Conference

After approval of submittals and before performing wall system installation work, hold a pre-construction conference to review the following:

- a. Drawings and specifications and submittals related to the wall work;
- b. Wall system components installation;
- c. Procedure for the wall manufacturer's technical representative's onsite inspection and acceptance of the wall substrate, the name of the manufacturer's technical representatives, the frequency of the onsite visits, distribution of copies of the inspection reports from the manufacturer's technical representative;
- d. Contractor's plan for coordination of the work of the various trades involved in providing the wall system and other components impacting the walls;
- e. Quality control plan for the wall system installation;
- f. Property protection measures.
- g. Safety requirements.

Pre-construction conference scheduling shall be coordinated with the Contracting Officer. The conference shall be attended by the Contractor, the Contracting Officer's designated personnel, personnel directly responsible for the installation of wall system, related sheet metal work, mechanical and electrical work, other trades interfacing with the wall work, and representative of the roofing materials manufacturer. Before beginning all work, provide a copy of meeting notes and action items to all attending parties. Note action items requiring resolution prior to start of work.

1.4.4 Elastomeric Silicone Wall System Mock-Up

Apply the elastomeric silicone roofing system, in a designated test area of not less than 4 square meters. Notify the Contracting Officer a minimum of 48 hours prior to the test application. The applied test system shall conform with applicable wall details. Equipment used in the construction shall be used in the application of the test wall system.

1.4.5 Sample Warranty

Submit sample warranty for the complete wall system meeting the specified warranty requirements. Sample warranty shall be submitted and approved by the Contracting Officer prior to commencement of work.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

Deliver and store materials in sufficient quantity to allow for

uninterrupted flow of work. Materials shall be delivered to the jobsite in their original unopened packages, clearly marked with the manufacturer's name, brand name, description of contents, and shelf life of containerized materials.

1.5.2 Storage

Materials shall be stored in clean, dry areas, away from excessive heat, sparks, and open flame. Storage area shall be ventilated to prevent build-up of flammable gases. Maintain temperatures in the storage area below the materials' flash point and within limits recommended by the manufacturer's printed instructions.

1.5.3 Handling

Handle materials and containers during application work safely and in accordance with manufacturer recommendations. Store liquids in airtight containers and keep containers closed except when removing materials. Do not use equipment or containers containing remains of dissimilar materials. Mark and remove from job site materials which have been exposed to moisture or that exceed shelf life limits. Not more than half the shelf life shall have expired when materials are applied.

1.6 ENVIRONMENTAL CONDITIONS

Do not apply system materials during inclement weather, frost, surface moisture, or visible dampness is present on the surface to be covered, or when precipitation is imminent. Use moisture-measuring methods and equipment as required to verify that the moisture conditions of substrate surfaces are in accordance with system materials manufacturer requirements prior to application of coating materials. Substrate temperatures shall be within limits recommended by the manufacturer's printed instructions, unless specified otherwise. Use wind screen protection for all spray applications when wind speeds exceed 10 miles per hour.

1.6.1 Elastomeric Coating

Apply wall coating between the temperature ranges of 10 and 43 degrees C, ambient.

1.7 COORDINATION

Wall operations shall be coordinated with work of other trades to ensure that components are installed as required to permit continuous self-flashing of the elastomeric silicone coating system. The installed roofing system shall be protected from damage. Damaged areas shall be repaired.

1.8 SPECIAL SAFETY PROVISIONS

During application, comply with the manufacturer's recommendations for the safe application of the elastomeric silicone product.

1.9 WARRANTY

Provide system material and workmanship warranties meeting specified requirements. Revision or amendment to standard manufacturer warranty shall be provided as required to comply with the specified requirements.

1.9.1 Wall System Manufacturer Warranty

Furnish a single-source wall system manufacturer's 10-year no dollar limit materials and installation workmanship warranty, for a watertight wall system construction. The warranty shall run directly to the Government and commence at time of Government's acceptance of the wall work. The warranty shall state that:

- a. If within the warranty period the wall system, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes non-watertight, shows evidence of moisture intrusion into the wall system, blisters, cracks, ruptures, splits, delaminates, disbonds, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged coating materials of the wall system and correction of defective workmanship shall be the responsibility of the coating manufacturer. All costs associated with the repair or replacement work shall be the responsibility of the manufacturer.
- b. When the manufacturer or his approved applicator fail to perform the repairs within 72 hours of notification, emergency temporary repairs performed by others shall not void the warranty.

1.9.2 Wall System Installer Warranty

The wall system installer shall warrant for a period of not less than two years that the wall system, as installed, is free from defects in installation workmanship, to include the coating applications, accessories, and attachments. The warranty shall run directly to the Government. Correction of defective workmanship and replacement of damaged or affected materials shall be the responsibility of the wall system installer. All costs associated with the repair or replacement work shall be the responsibility of the installer.

1.9.3 Continuance of Warranty

Repair or replacement work that becomes necessary within the warranty period shall be approved, as required, and accomplished in a manner so as to restore the integrity of the wall system assembly and validity of the manufacturer warranty for the remainder of the manufacturer warranty period.

1.10 CONFORMANCE AND COMPATIBILITY

The entire wall system shall be in accordance with specified and indicated requirements, including fire resistance requirements. Work not specifically addressed and any deviation from specified requirements shall be in general accordance with applicable recommendations of the Waterproofing Manual, reference standards, membrane manufacturer published recommendations and details, and compatible with surrounding components and construction. Any deviation from specified or indicated requirements shall be submitted to the Contracting Officer for approval prior to installation.

PART 2 PRODUCTS

2.1 PROTECTIVE COATING

SELECTION GUIDANCE TABLE

COATING DESCRIPTION	PERFORMANCE FEATURES	RECOMMENDED APPLICATIONS
Type A (silicone)	One component system. High permeability (breathable). Good weatherability. Withstands temperature extremes well.	Exterior above grade.

Protective water-based elastomeric silicone coating shall consist of separately applied base, and finish coat of one-component silicone (Type A). Submit literature including material description, physical properties, recommended storage conditions, Material Safety Data Sheets, and shelf life expiration date. Coating shall have the following minimum properties:

2.1.1 Type A Coating

Silicone rubber, single component. Uncured silicone rubber shall have the following values when tested for specified properties in accordance with specified test methods:

<u>Properties in Metric Units</u>			
<u>Properties</u>	<u>ASTM Test</u>	<u>Value</u>	<u>Units</u>
Solids Content	ASTM D 2697	50, minimum	percent by volume
Solids Content	ASTM D 1644	58, minimum	percent by weight
Flash Point	ASTM D 56 or ASTM D 93	115, minimum	degrees C

Cured silicone rubber shall have the following values when tested for specified properties in accordance with specified test methods:

<u>Properties in Metric Units</u>			
<u>Properties</u>	<u>ASTM Test</u>	<u>Value</u>	<u>Units</u>
Elongation	ASTM D 412, 24 degrees C, 50 percent relative humidity	190, minimum	percentage
Tensile Strength	ASTM D 412, die C, 24 degrees C, 50 percent relative humidity	2900, minimum	kPa

UV Exposure	ASTM D 822, 6000 hours in atlas carbon arc, Type E weatherometer	No cracking checking, or significant discoloration	--
Hardness, Shore A	ASTM D 2240	38, minimum	points
Permeability	ASTM E 96/E 96M Procedure E 0.50 mm thickness	10 maximum. Class III Vapor Retarder	perms

Add mildewcide to silicone coating as per manufacturer's recommendations.

2.1.2 Coating Color

Coating layers shall be of distinguishably different color such that dry film thickness of each layer of the completed application can be determined. Finish coat color shall be as indicated on drawing A-200.

2.1.3 Manufacturer's Application Instructions

Submit manufacturer's complete application instructions and details, and to include storage, handling, and warnings or precautions on flammability and toxicity. Include manufacturer's written recommendations for primers and for surface preparation of concrete and wall materials.

2.2 PRIMER

Primers used shall be as required and recommended by the coating manufacturer for the substrate to be covered. Submit literature including material description, physical properties, recommended storage conditions, Material Safety Data Sheets, and shelf life expiration date.

2.3 CERAMIC GRANULES

Texture to match existing building.

2.4 SEALANTS

Sealants shall be as recommended by the coating manufacturer. Submit literature including material description, physical properties, recommended storage conditions, Material Safety Data Sheets, and shelf life expiration date.

2.5 INSPECTION TOOLS

Maintain the following inspection tools on site for use in evaluating conditions and quality:

- Moisture meter to measure degree of moisture within or on the substrate surface.
- Sling Psychrometer and psychrometric chart, or electronic psychrometer or hygrometer to measure ambient temperature, humidity and dew point.
- Surface thermometer to read temperature of a surface.
- Optical comparator to read dry film thickness.

- e. Magnifying glass, minimum 75 mm to inspect surface conditions.
- f. Wet film thickness gauge to read wet film thickness.

PART 3 EXECUTION

3.1 PROTECTION OF PROPERTY

Protect the building structure, equipment, and other surfaces adjacent to the work from overspray from coating materials. Protective coverings shall be secured and vented to prevent collection of moisture on covered surfaces. Use protective shields or barriers when spraying walls to prevent uncontrolled overspray. Any surfaces damaged by wall system products shall be restored or replaced to the satisfaction of the Government at no additional expense to the Government.

3.1.1 Masking

Provide masking protection to protect surfaces immediately adjacent to coating terminations at time of application. Adjust or provide new masking protection at roof perimeter to protect surfaces immediately adjacent to coating terminations and to provide for clean smooth coating termination lines.

3.1.2 Warning Signs

Post warning signs at ground level in the adjacent to the work area and a minimum of 45.72 meters from the application area stating the area is off limits to unauthorized persons and warning of potential overspray hazard. Place clearly visible and legible warning sign at entrance to primary road leading to the project facility warning of presence of flammable materials, irritating fumes, and potential of overspray damage.

3.2 SPECIAL PRECAUTIONS AND INSTRUCTIONS

3.2.1 Primers

Do not dilute primers or other materials unless required and recommended by the manufacturer. Do not use cleaning solvents for thinning primers or other materials.

3.2.2 Material Handling

Handle materials and containers during application work safely and in accordance with recommendations of the manufacturer. Store liquids in airtight containers and keep containers closed except when removing materials. Do not use equipment or containers containing remains of dissimilar materials or products.

3.2.3 Fire and Explosion Hazards

Prohibit open flames, sparks, welding, and smoking in the application area. Provide and maintain a fire extinguisher of appropriate type and size in the application area.

3.3 WALL AREA PREPARATION

3.3.1 Preapplication Inspection

Ensure the perimeter walls are in place prior to the application of the elastomeric silicone coating.

3.3.1.1 Surface Examination

Examine surfaces and correct defects that may adversely affect the wall system application or performance.

3.3.2 Close Intake Vents

Seal off and shutdown air intake vents during coating application. Coordinate with the Contracting Officer. Provide portable ventilation and air conditioning during shutdown as required by the Contracting Officer

3.4 SURFACE PREPARATION FOR ELASTOMERIC SILICONE COATING APPLICATION

Apply wall coating on surfaces free of water, grease, oils, dirt, debris, and other foreign materials, and cured completely.

3.4.1 Concrete Surfaces

Wire brush concrete surfaces which are to receive coatings, and remove resulting dust.

3.4.2 Primer

Apply as recommended by the coating manufacturer.

3.4.3 Sealant

Apply as recommended by the coating manufacturer to include application in V-groove at eave and rake terminations.

3.5 COATING APPLICATION

Check coating wet film thickness during application to ensure that the wet mil thickness required for each coat is provided. For each coat, provide a dry film thickness not less than specified in paragraph entitled "Minimum Dry Film Thickness (DFT) in Mils." Provide smooth coatings free of runs, dry spray, or overspray, and provide a uniform film. Coating applications shall consider possible coverage loss due to overspray, waste, texture, wind, and other factors that may affect coverage rate. Adjust methods and quantities to provide the minimum DFT for each coating layer free of blowholes, pinholes, voids, blisters, and other conditions detrimental to coating performance.

3.5.1 Base Coat Application

Apply coating as the base coat. Apply in accordance with manufacturer's printed installation instructions in a manner and rate to provide full coverage at the specified DFT. Apply coating uniformly and in one direction. Inspect the completed surface for blowholes, pinholes, cracks, breaks, voids, blisters, and other defects and repair with base coat material. Allow base coat to cure as recommended by the coating manufacturer prior to application of intermediate coat, except that the

base coat shall cure a minimum of 2 hours. Check cured base coat to determine dry film thickness before applying the next coat. Clean areas of base coat which become dirty with a clean wet cloth prior to application of intermediate coat.

3.5.2 Intermediate Coat Application

Apply coating in same manner as base coat except apply in a direction perpendicular to the direction in which base coat was applied. Apply intermediate coat within 48 hours of base coat application. Apply in accordance with manufacturer's printed installation instructions in a manner and rate to provide full coverage at the specified DFT. Allow intermediate coat to cure as recommended by the coating manufacturer prior to application of finish coat, except that the base coat shall cure a minimum of 2 hours. Check cured intermediate coat to determine dry film thickness and inspect surface for blowholes, pinholes, cracks, breaks, voids, blisters, and other defects. Repair with intermediate coat material before applying finish coat. If base or intermediate coat thickness is deficient, apply additional intermediate coat prior to applying finish coat to bring total dry film thickness up to that required.

3.5.3 Finish Coat Application

Apply coating as finish coat in accordance with manufacturer's printed installation instructions in a manner and rate to provide full coverage at the specified DFT, exclusive of granules. Apply in a direction perpendicular to the direction in which intermediate coat was applied. Provide finish coat free of blowholes, pinholes, cracks, breaks, voids, blisters, and other defects that may affect performance, durability, and longevity of the coating.

3.5.4 Minimum Dry Film Thickness (DFT)

Type	Base Coat	Intermediate Coat	Finish Coat	Total System
A-Silicone	0.138 mm	3 coats - , ,	2 coats - ,	1.6 mm

3.5.5 Granule Surfacing

Spray-apply mineral granules to finish coat, coating within 2 to 5 minutes after application of finish coating, depending on temperature and humidity, as required by roof coating manufacturer's printed installation instructions. Spray applicator equipment shall be a sand blaster with output pressure reduced to 69 to 138 kPa. Apply granules to indicated areas evenly at a rate of 23 kg per 9.3 square meters.

3.5.6 Penetrations

Apply an additional 0.25 mm finish coat for one meter all around roof access locations and 0.61 meters around all penetrations.

3.5.7 Coating Application Clean-Up

Clean surfaces that receive coating materials which are not designated to receive such materials. Remove overspray masking materials and coverings

upon completion of the coating work.

3.6 FIELD QUALITY CONTROL

3.6.1 Construction Monitoring

During progress of the work, Contractor shall make visual inspections as necessary to insure compliance with specified parameters. Additionally, verify the following:

- a. Protection measures are in place.
- b. Equipment is in working order. Metering devices are accurate.
- c. Materials are not installed in adverse weather.
- d. Surfaces are cleaned and primed and substrates are in acceptable condition prior to application of materials.
- e. Materials comply with specified requirements.
- f. All materials are properly stored, handled and protected from moisture or other damages.
- g. Coating is applied in a minimum of three coats as specified. Each coat is applied perpendicular in direction to the preceeding coat.
- h. Wind screens are utilized as necessary.

Granules, where utilized, are embedded as specified.

3.6.2 Manufacturer's Field Inspection

Manufacturer's technical representative shall visit the site as necessary during the installation process to ensure wall system materials are being applied in a satisfactory manner. As a minimum, manufacturer's technical representative shall be present at work start-up and perform field inspection of the first day's completed application and at substantial completion, prior to demobilization of contractor. Inspections shall be conducted in the presence of Government representatives. After each inspection, an inspection report signed by the manufacturer's technical representative, shall be submitted to the Contracting Officer within 2 working days. The inspection report shall note overall quality of work, deficiencies and any other concerns, and recommended corrective actions in detail. Notify the Contracting Officer a minimum of 2 working days prior to site visit by manufacturer's technical representative.

3.6.3 Visual Inspection and Moisture Scanning

The Government may conduct a detailed visual inspection and nondestructive moisture scan of the completed installation prior to expiration of the Contractor warranty. Any wall system deficiencies or moisture found in the wall system installation shall be repaired by the Contractor.

3.7 CORRECTION OF DEFICIENCIES

Correction of deficiencies shall be as directed by the Contracting Officer at no additional cost to the Government.

3.8 CLEAN-UP AND DISPOSAL

All waste material, material containers, and debris shall be cleaned up daily and placed in appropriate trash containers. At completion of the work all waste material, debris, and containers shall be removed from the job site and disposed of as required by local regulations.

-- End of Section --

DRAFT

SECTION 08 11 13

STEEL DOORS AND FRAMES
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/SDI A250.8 (2017) Specifications for Standard Steel
Doors and Frames

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 4702 (2021) Doorsets

JIS Z 3420 (2003) Specification and Approval of
Welding Procedures for Metallic Materials
- General Rules

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT SS Chapter 16 (2019) Building Construction Standard
Specifications - Chapter 16 Opening
Construction

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Doors; G

Frames; G

Show elevations, construction details, metal gages, hardware provisions, method of glazing, and installation details.

Schedule of Doors; G

Schedule of Frames; G

Submit door and frame locations.

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver doors, frames, and accessories undamaged and with protective wrappings or packaging. Provide temporary steel spreaders securely

fastened to the bottom of each welded frame. Store doors and frames on platforms under cover in clean, dry, ventilated, and accessible locations, with 6 mm airspace between doors. Remove damp or wet packaging immediately and wipe affected surfaces dry. Replace damaged materials with new.

PART 2 PRODUCTS

2.1 STANDARD STEEL DOORS

JIS A 4702, except as specified otherwise. Prepare doors to receive door hardware as specified in Section 08 71 00 DOOR HARDWARE. Undercut where indicated. Provide exterior doors with top edge closed flush and sealed to prevent water intrusion. Provide doors at 44.5 mm thick, unless otherwise indicated. Provide door material that uses a minimum of 25 percent recycled content.

2.1.1 Classification - Level, Performance, Model

2.1.1.1 Maximum Duty Doors (Level 4)

45 mm door thickness, 1.3 mm door faces and frames steel thickness, full flush with core construction as required by the manufacturer for interior doors, of size(s) and design(s) indicated. Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation. Provide door classification label per JIS A 4702.

2.2 INSULATED STEEL DOOR SYSTEMS

Provide insulated steel doors with a core of polyurethane foam; face sheets, edges, and frames of galvanized steel not lighter than 0.7 mm thick, 1.5 mm thick, and 1.5 mm respectively; magnetic weatherstripping; nonremovable-pin hinges; thermal-break aluminum threshold; and vinyl door bottom. Provide to doors and frames a phosphate treatment and rust-inhibitive primer. Door shall meet JIS A 4702 and tested for 500,000 cycles. Prepare doors to receive specified hardware. Provide doors 44.5 mm thick.

2.3 SOUND RATED STEEL DOORS

Provide sound rated doors with a Reverberation Time Rating (Rt) as indicated on the drawings.

2.4 INSULATION CORES

Provide insulating cores of the type specified, and provide an apparent per insulation requirements in accordance with JIS A4702.

2.5 STANDARD STEEL FRAMES

Provide hardware reinforcing thickness per ANSI/SDI A250.8 requirement. Form frames to sizes and shapes indicated, with welded corners.

2.5.1 Welded Frames

Continuously weld frame faces at corner joints. Mechanically interlock or continuously weld stops and rabbets. Grind welds smooth.

Weld frames in accordance with the recommended practice of Welding Code

per JIS Z 3420 and in accordance with the practice specified by the producer of the metal being welded.

2.5.2 Anchors

Provide anchors to secure the frame to adjoining construction. Provide steel anchors, zinc-coated not lighter than 1.2 mm thick.

2.5.2.1 Wall Anchors

Provide at least three anchors for each jamb. For frames which are more than 2285 mm in height, provide one additional anchor for each jamb for each additional 760 mm or fraction thereof.

- a. Stud partitions: Weld or otherwise securely fasten anchors to backs of frames. Design anchors to be fastened to closed steel studs with sheet metal screws, and to open steel studs by wiring or welding.

2.6 EXTERIOR FRAMES

Provide thermal insulation in all exterior frames. Provide frames of a minimum Level 4, with frames of a minimum thickness of 1.7 mm, 14 gage.

2.7 WEATHERSTRIPPING

As specified in Section 08 71 00 DOOR HARDWARE.

2.8 HARDWARE PREPARATION

Provide minimum hardware reinforcing gages as specified in JIS A 4702. Drill and tap doors and frames to receive finish hardware. Prepare doors and frames for hardware in accordance with the applicable requirements of JIS A 4702. Drill and tap for surface-applied hardware at the project site. Build additional reinforcing for surface-applied hardware into the door at the factory. Locate hardware in accordance with the requirements of JIS A 4702, as applicable. Punch door frames, with the exception of frames that will have weatherstripping or soundproof gasketing, to receive a minimum of two rubber or vinyl door silencers on lock side of single doors. Set lock strikes out to provide clearance for silencers.

2.9 FINISHES

2.9.1 Factory-Primed Finish

Thoroughly clean all surfaces of doors and frames then chemically treat and factory prime with a rust inhibiting coating as specified in JIS A 4702 or paintable A25 galvanized steel without primer. Where coating is removed by welding, apply touchup of factory primer.

2.10 FABRICATION AND WORKMANSHIP

Provide finished doors and frames that are strong and rigid, neat in appearance, and free from defects, waves, scratches, cuts, dents, ridges, holes, warp, and buckle. Provide molded members that are clean cut, straight, and true, with joints coped or mitered, well formed, and in true alignment. Dress exposed welded and soldered joints smooth. Design door frame sections for use with the wall construction indicated. Corner joints must be well formed and in true alignment. Conceal fastenings where practicable.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Frames

Set frames in accordance with JIS A 4702 and MLIT SS Chapter 16. Plumb, align, and brace securely until permanent anchors are set. Anchor bottoms of frames with expansion bolts or powder-actuated fasteners. Build in or secure wall anchors to adjoining construction.

3.1.2 Doors

Hang doors in accordance with clearances specified in JIS A 4702 and MLIT SS Chapter 16. After erection, clean and adjust hardware.

3.2 PROTECTION

Protect doors and frames from damage. Repair damaged doors and frames prior to completion and acceptance of the project or replace with new, as directed. Wire brush rusted frames until rust is removed. Clean thoroughly. Apply an all-over coat of rust-inhibitive paint of the same type used for shop coat.

3.3 CLEANING

Upon completion, clean exposed surfaces of doors and frames thoroughly. Remove mastic smears and other unsightly marks.

-- End of Section --

SECTION 08 33 23

OVERHEAD COILING DOORS
08/20, CHG 1: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE FUN SI (2021) Fundamentals Handbook, SI Edition

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings

ASTM A48/A48M (2022) Standard Specification for Gray Iron Castings

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A666 (2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar

ASTM A780/A780M (2020) Standard Practice for Repair of

Damaged and Uncoated Areas of Hot-Dip
Galvanized Coatings

ASTM A924/A924M (2022a) Standard Specification for General
Requirements for Steel Sheet,
Metallic-Coated by the Hot-Dip Process

ASTM B221M (2021) Standard Specification for Aluminum
and Aluminum-Alloy Extruded Bars, Rods,
Wire, Profiles, and Tubes (Metric)

ASTM E84 (2023) Standard Test Method for Surface
Burning Characteristics of Building
Materials

ASTM E330/E330M (2014; R 2021) Standard Test Method for
Structural Performance of Exterior
Windows, Doors, Skylights and Curtain
Walls by Uniform Static Air Pressure
Difference

ASTM F568M (2007) Standard Specification for Carbon
and Alloy Steel Externally Threaded Metric
Fasteners

DOOR AND ACCESS SYSTEM MANUFACTURERS ASSOCIATION (DASMA)

ANSI/DASMA 108 (2017) Standard Method for Testing
Sectional Garage Doors, Rolling Doors and
Flexible Doors: Determination of
Structural Performance Under Uniform
Static Air Pressure Difference

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Overhead Coiling Doors; G

Counterbalancing Mechanism; G

Manual Door Operators; G

Bottom Bars; G

Guides; G

Mounting Brackets; G

Hood; G

Installation Drawings; G

SD-03 Product Data

Overhead Coiling Doors; G

Hardware; G

Counterbalancing Mechanism; G

Manual Door Operators; G

SD-05 Design Data

Overhead Coiling Doors; G

Hardware; G

Counterbalancing Mechanism; G

Manual Door Operators; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals, Data Package 2; G

SD-11 Closeout Submittals

Warranty; G

1.3 QUALITY CONTROL

1.3.1 Warranty

Furnish a written guarantee that the helical spring and counterbalance mechanism are free from defects in material and workmanship for not less than two years after completion and acceptance of the project.

Warrant that upon notification by the Government, any defects in material, workmanship, and door operation are immediately correct within the same time period covered by the guarantee, at no cost to the Government.

1.3.2 Operation And Maintenance Submittals

Submit electronic copy of the operation and maintenance manuals 30 calendar days prior to testing the Overhead Coiling Door Assemblies. Update and resubmit data for final approval no later than 30 calendar days prior to cContract completion.

Submit Operation and Maintenance Manuals for Overhead Coiling Door Assemblies, including the following items:

Manual Door Operators

Hood

Counterbalancing Mechanism

Painting

Provide operation and maintenance manuals which are consistent with

manufacturer's standard brochures, schematics, printed instructions, operating procedures, and safety precautions.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver doors to the jobsite wrapped in a protective covering with the brands and names clearly marked thereon. Store doors in an adequately ventilated dry location that is free from dirt and dust, water, or other contaminants. Store in a manner that permits easy access for inspection and handling. Handle doors carefully to prevent damage. Remove damaged items that cannot be restored to like-new condition and provide new items.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide overhead coiling doors with interlocking slats, complete with anchoring and door hardware, guides, hood, and operating mechanisms, and designed for use on openings as indicated. Doors must be spring counterbalanced, rolling type, and designed for use on exterior openings, as indicated. Doors must be operated by hand chain with gear or sprocket reduction by hand crank with gear or sprocket reduction. Doors to be surface-mounted type with guides at jambs set back a sufficient distance to provide a clear opening when door is in open position.

2.1.1 Design Requirements

2.1.1.1 Door Detail Shop Drawings

Provide installation drawings for door assemblies which show: elevations of each door type, shape and thickness of materials, finishes, details of joints and connections, details of guides and fittings, rough opening dimensions, location and description of hardware, anchorage locations, and counterbalancing mechanism and door operator details. Include a schedule showing the location of each door with the drawings.

2.1.2 Performance Requirements

2.1.2.1 Wind Loading

Design and fabricate door assembly to withstand the wind loading pressure of at least as indicated on the Drawings and in accordance with ANSI/DASMA 108. Provide test data showing compliance with ASTM E330/E330M. Sound engineering principles may be used to interpolate or extrapolate test results to door sizes not specifically tested. Ensure that the complete assembly meets or exceeds the requirements of ASCE 7-16.

2.1.2.2 Operational Cycle Life

Design all portions of the door, hardware and operating mechanism that are subject to movement, wear, or stress fatigue to operate through a minimum number of 10 cycles per day. One complete cycle of door operation is defined as when the door is in the closed position, moves to the fully open position, and returns to the closed position.

2.2 COMPONENTS

2.2.1 Overhead Coiling Doors

2.2.1.1 Curtain Materials and Construction

Provide curtain slats fabricated from Type 304 stainless steel sheets conforming to ASTM A666; sheet thickness as required by the size of the door to meet the required windload.

Fabricate doors from interlocking cold-rolled slats, with section profiles as specified, designed to withstand the specified wind loading. Ensure the provided slats are continuous without splices for the width of the door.

Provide slats filled with foamed-in-place polyurethane insulation, complying with the maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, according to ASTM E84. Enclose the insulation completely within the slat faces on the interior surface of the slats.

2.2.1.2 Insulated Curtains

Form curtains from manufacturer's standard shapes of interlocking slats of a cross section not less than 75 mm wide by 25 mm deep. Supply a slat system with a minimum R-value of 4 when calculated in accordance with ASHRAE FUN SI. Slats to consist of a polyurethane core not less than 17 mm thick, completely enclosed within metal facings. Slat steel thickness as required by the size of the door to meet specified performance requirements. The insulated slat assembly requires a flame spread rating of not more than 25 and a smoke development factor of not more than 50 when tested in accordance with ASTM E84.

2.2.1.3 Curtain Bottom Bar

Install curtain bottom bars as pairs of angles or using extrusions from the manufacturer's standard steel not less than 50 by 50 millimeter by 4.8 millimeter. Stainless steel extrusions conforming to ASTM A666, Type 304. Aluminum extrusions conforming to ASTM B221M. Galvanize angles and fasteners in accordance with ASTM A653/A653M and ASTM A924/A924M. Coat welds and abrasions with paint conforming to ASTM A780/A780M.

Provide two minimum 50 mm by 50 mm by 3.2 mm structural steel angles.

2.2.1.4 Endlocks (and Windlocks)

Provide endlocks of Grade B cast steel conforming to ASTM A47/A47M, galvanized in accordance with ASTM A153/A153M. Secure locks at every other curtain slat. In addition to endlocks, exterior doors which are more than 4877 mm wide or which have a design wind load of more than 0.96 kilopascal, must have windlocks of manufacturer's standard design. Windlocks must prevent curtain from leaving guide because of deflection from wind pressure or other forces.

2.2.1.5 Weather Stripping

Provide a hood baffle inside the hood that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber or equivalent. Provide guide weather stripping that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber, or equivalent.

Provide bottom bar weather-stripping that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber, or equivalent.

2.2.1.6 Locking Devices

Ensure that the slide bolt engages through slots in tracks for locking by padlock, located on both left and right jamb sides, operable from coil side.

Provide a chain lock keeper suitable for a standard padlock.

2.2.2 Hardware

Ensure that all hardware conforms to ASTM A153/A153M, ASTM A307, and ASTM F568M.

2.2.2.1 Guides

Fabricate curtain jamb guides from the manufacturer's standard angles or channels of same material and finish as curtain slats unless otherwise indicated. Provide guides with sufficient depth or incorporate a steel locking bar to retain the curtain in place under the wind pressure specified. Ensure curtain operates smoothly. Slot bolt holes for track adjustment. Securely attach guides to adjoining construction with not less than 10 mm diameter bolts, spaced near each end and not over 762 mm apart.

Ensure guides are roll-formed steel channel bolted to angle or structural grade, three angle assembly of stainless steel to form a slot of sufficient depth to retain curtains in guides to achieve 13.8 kilopascal windload standard. Guides may be provided with integral windlock bars and removable bottom bar stops.

Fabricate with stainless steel angles. Provide windlock bars of same material when windlocks are required to meet specified wind load. Flare the top of inner and outer guide angles outwards to form bellmouth for smooth entry of curtain into guides. Provide removable guide stoppers to prevent over travel of curtain and bottom bar.

2.2.2.2 Hood

Provide a hood with a minimum stainless steel sheet metal, flanged at top for attachment to header and flanged at bottom to provide longitudinal stiffness. The hood encloses the curtain coil and counterbalance mechanism.

Provide a weather baffle at the lintel or inside the hood of each exterior door.

2.2.3 Counterbalancing Mechanism

Counterbalance doors by means of manufacturer's standard mechanism with an adjustable-tension, steel helical torsion spring mounted, around a steel shaft and contained in a spring barrel connected to top of curtain with barrel rings. Use grease-sealed or self-lubricating bearings for rotating members.

2.2.3.1 Brackets

Provide the manufacturer's standard mounting brackets with one located at each end of the counterbalance barrel conforming to ASTM A36/A36M. Provide brackets of hot-rolled steel.

Brackets will be of 5 mm thick steel plates, with permanently sealed ball bearings. Designed to enclose ends of coil and provide support of counterbalance pipe at each end.

2.2.3.2 Counterbalance Barrels

Curtain must roll up on a barrel supported at head of opening on brackets and be balanced by a torsion spring system in the barrel. Fabricate spring barrel of manufacturer's standard hot-formed, structural-quality, welded or seamless carbon-steel pipe, conforming to ASTM A53/A53M or equivalent. Ensure the barrel is of sufficient diameter and wall thickness to support rolled-up curtain without distortion of slats. Limit barrel deflection to not more than 2.5 mm per meter of span under full load.

a. Barrel

Provide steel pipe capable of supporting curtain load with maximum deflection of 2.5 mm per meter of width.

b. Spring Balance

Provide an oil-tempered, heat-treated steel helical torsion spring assembly designed for proper balance of door. Ensure that effort to operate manually operated units does not exceed 110 N. At least 80 percent of the door weight must be counterbalanced at any position. Provide wheel for applying and adjusting spring torque.

2.2.3.3 Spring Balance

Install one or more oil-tempered, heat-treated steel helical torsion springs within the barrel, capable of producing sufficient torque to assure easy operation of the door curtain. Provide and size springs to counterbalance weight of curtain, with uniform adjustment accessible from outside barrel. Secure ends of springs to barrel and shaft with cast-steel barrel plugs.

2.2.3.4 Torsion Rod for Counter Balance

Fabricate rod from the manufacturer's standard cold-rolled steel, sized to hold fixed spring ends and carry torsional load.

2.2.4 Manual Door Operators

2.2.4.1 Manual Chain-Hoist Door Operators

Provide door operators which consist of an endless steel hand chain, chain-pocket wheel, guard, and a geared reduction unit of at least a 3 to 1 ratio. Required pull for operation cannot exceed 16 kilogram. Chain must extend to within 914 mm of floor.

Provide chain hoists with a mechanism allowing the curtain to be stopped at any point in its upward or downward travel and to remain in that

position until moved to the fully open or closed position. Provide hand chains of galvanized steel. Ensure that the yield point of the chain is at least three times the required hand-chain pull.

Provide chain sprocket wheels of cast iron conforming to ASTM A48/A48M.

2.2.5 Surface Finishing

Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes. Noticeable variations in the same metal component are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved samples and are assembled or installed to minimize contrast.

PART 3 EXECUTION

3.1 INSTALLATION

Install overhead coiling door assembly, anchors and inserts for guides, brackets, hardware, and other accessories in accordance with approved detail drawings and manufacturer's written instructions. Upon completion of installation, ensure doors are free from all distortion.

Install overhead coiling doors and hoods at the mounting locations as indicated for each door in the Contract Documents and as required by the manufacturer.

3.2 ADJUSTING AND CLEANING

3.2.1 Acceptance Provisions

After installation, adjust the hardware and moving parts. Lubricate bearings and sliding parts as recommended by manufacturer to provide smooth operating functions for ease movement, free of warping, twisting, or distortion of the door assembly.

Adjust seals to provide a weather-tight fit around entire perimeter.

3.2.1.1 Maintenance and Adjustment

Not more than 90 calendar days after completion and acceptance of the project, examine, lubricate, test, and re-adjust doors as required for proper operation.

3.2.1.2 Cleaning

Clean doors in accordance with manufacturer's approved instructions.

-- End of Section --

SECTION 08 34 73

SOUND CONTROL DOOR ASSEMBLIES

11/19, CHG 1: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A108	(2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A568/A568M	(2019a) Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for
ASTM A1008/A1008M	(2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM A1011/A1011M	(2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C476	(2023) Standard Specification for Grout for Masonry
ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM D6386	(2022) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting
ASTM E90	(2023) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
ASTM E336	(2024) Standard Test Method for

Measurement of Airborne Sound Attenuation
between Rooms in Buildings

ASTM E413 (2022) Classification for Rating Sound
Insulation

ASTM E1289 (2008; R 2022) Standard Specification for
Reference Specimen for Sound Transmission
Loss

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101 (2024) Life Safety Code

NFPA 252 (2022) Standard Methods of Fire Tests of
Door Assemblies

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA)
Accessibility Guidelines for Buildings and
Facilities; Architectural Barriers Act
(ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings

SD-03 Product Data

Hollow Metal Sound Retardant Doors; G

Door Frames; G

Door Hardware; G

Door Frame Sound Infill; G

Gasketing; G

Thresholds; G

SD-06 Test Reports

Acoustical Tests; G

Positive Pressure Tests; G

SD-07 Certificates

Hollow Metal Sound Retardant Doors; G

Door Frames; G

Door Hardware; G

Gasketing and Door Bottoms; G

Thresholds; G

Assembly Test Reports

1.3 QUALITY CONTROL

Ensure work within this section is designed and furnished by one manufacturer, who has been engaged in the manufacture of Sound Retardant Hollow Metal Door and Frame systems for at least five years prior to the start of this work.

Provide acoustic assemblies manufactured by a single source specializing in the production of this type work for a minimum of five years.

1.3.1 Compliance and Labeling

1.3.1.1 Compliance with Accessibility Requirements

Americans with Disabilities Act/Architectural Barriers Act (ADA/ABA)
36 CFR 1191

Accessibility Guidelines for Buildings and Facilities (ADAAG) 36 CFR 1191

1.3.1.2 Category B Positive Pressure Fire Door Construction

Conform all door openings to the applicable portions of NFPA 101 and NFPA 252. Incorporate field applied intumescent materials, applied by a licensed installer according to the manufacturers' instructions. Keep instructions on file. Additional gasketing may be required to meet the 'S' smoke rating. Submit Certificate for gasketing and door bottoms.

1.3.1.3 Labeling

Ensure all positive pressure door assemblies carry the fire label for the complete opening, clearly identifying the:

- a. Manufacturer
- b. Third party testing and certification agency
- c. Fire door rating
- d. Installation limitations
- e. Compatible frame, hardware component ratings
- f. Compatible lite or vision panel component ratings
- g. Required building code information, including temperature and smoke rating
- h. Rt rating as indicated on the Drawings.

Indicate fire-ratings of applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.4 DELIVERY, STORAGE, AND HANDLING

Ship all doors in the manufacturer's undamaged individual cartons, securely bundled and wrapped with moisture-resistant covers and stored in accordance with the manufacturer's printed instructions in a dry, clean, and ventilated area.

Store all materials on planks in a dry location. Store doors and frames vertically with minimum 25 mm airspace between. Store doors on the edge to eliminate any potential damage to the door bottom seal. Cover all material to protect from damage but in a manner to allow proper circulation.

Maintain relative humidity in the building between 30 and 65 percent.

Perform final adjustment of seals when temperatures and humidity conditions replicate the interior conditions that will exist when the building is occupied.

1.5 WARRANTY

Manufacturer's warranty for 5 years from date of supply, covering material and workmanship. Failures include, but are not limited to, the following:

- a. Failure to meet sound rating requirements
- b. Faulty operation of sound seals
- c. Deterioration of metals, metal finishes, and other materials beyond normal use or weathering.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide sound retardant door assemblies of the thickness, width, and height indicated, complete with perimeter seals, seal housings, gasketing, automatic door bottoms, thresholds, and door frames as required to conform to the specified STC per ASTM E90 and ASTM E1289.

Submit fabrication drawings for Hollow Metal Sound Retardant Doors, Door Frames and Door Frame Sound Infill.

Submit certificates showing conformance with the referenced standards in this section, and manufacturer's catalog data including STC ratings and UL fire rating, where applicable, for the following items: Hollow metal sound retardant doors; door frames; door hardware; gasketing; thresholds.

Provide assemblies that are complete with metal frame, metal door(s), sealing system, and Cam-lift hinges (when required).

2.1.1 Design Requirements

2.1.1.1 Door Design

Provide sound Retardant Metal Swinging Doors that are a 4.445 centimeter thickness construction with sizes as indicated on drawings. No visible seams are permitted on door faces. Provide face gauges, internal sound retardant core and perimeter door edge construction per manufacturer's standard for the specified STC rating. No lead or asbestos is permitted in door construction to achieve STC performance.

2.1.1.2 Frame Design

Provide sound Retardant Metal Frames conforming to ASTM A1008/A1008M, not less than 1.90 millimeter thick, and free from pitting, scale, stretcher strains, fluting, and surface defects with integral trim and shipped with temporary spreader. Knockdown frames are not acceptable.

Provide frames with 50 millimeter faces, profiles and dimensions as indicated, with mitered reinforced corners, welded the full depth of frame and trim, with exposed surfaces ground smooth and flush. Close contact edges to hairline joints.

2.1.2 Performance Requirements

2.1.2.1 Reverberation Time (Rt) Rating

Provide doors with an Rt per the door schedule.

2.2 FABRICATION

Provide doors that are minimum 16 gauge, 4.445 centimeter thick with welded, seamless construction. No visible joints are permitted on the exposed faces or edges. Join door skins at vertical edges by continuous welds, ground and dressed smooth to provide a flush finish. Reinforce top and bottom with 1.52 millimeter continuous inverted steel channels spot welded to both faces. Finish both top and bottom to provide a smooth flush condition. Bevel both vertical edges .3175 centimeter in 5.08 centimeter.

2.2.1 Hollow Metal Sound Retardant Doors

2.2.1.1 Construction

Conform to ASTM A1008/A1008M for door construction utilizing steel facing sheets. Conform stretcher level flatness to ASTM A568/A568M; not less than 1.52 millimeter thick; free from pitting, scale, and surface defects; separated by a core construction designed to meet the required STC; and tested and rated in accordance with ASTM E90.

Provide doors that have flush seamless face sheets and vertical edges, with continuous welded and smooth joints. Provide edges that are flush or rabbeted as required for perimeter seals.

Provide door surfaces that are visually flat and free from warp, waviness, and other surface irregularities and defects. Maximum allowable warp or twist-can not exceed 3 millimeter when measured with a 2100 millimeter straightedge along the diagonal and not exceed 1.5 millimeter when measured with a 2100 millimeter straightedge in the width or in any

position along the length of the door.

Provide hardware reinforcement that is steel drilled, tapped to template requirements and welded in place. Provide minimum thicknesses as follows:

- a. Butts, 4.7 millimeter
- b. Lock strike, 3.04 millimeter
- c. Surface applied hardware 1.90 millimeter

2.2.1.2 Coating

Thoroughly clean all mill scale, rust, oil, grease, dirt, and other foreign materials from surfaces before the application of the shop coat of paint.

After cleaning, provide galvanized surfaces free of paint in accordance with ASTM D6386, Method A, B, C, or D.

Apply to clean prepared dry surfaces one shop coat of rust inhibitive metallic oxide or synthetic resin primer by brush, dipping, or other approved method to provide a continuous minimum dry film thickness (dft) of 0.023 millimeter.

2.3 COMPONENTS

2.3.1 Frames

Construct frames for Sound Retardant Wood Swinging Doors from formed sheet steel or structural shapes and bars. Provide sheet steel that is commercial quality, level, cold rolled steel conforming to ASTM A1008/A1008M or hot rolled, pickled and oiled steel conforming to ASTM A1011/A1011M. Comply steel shapes with ASTM A36/A36M and steel bars with ASTM A108, Grade 1018.

2.3.2 Door Frame Sound Infill

Grout: Comply with ASTM C476, with a slump of not more than 102 mm as measured according to ASTM C143/C143M.

Corrosion-Resistant Coating: Cold-applied asphalt mastic, compounded for 0.381 millimeter dry film thickness per coat. Provide inert-type noncorrosive compound free of asbestos fibers, sulfur components, and other deleterious impurities.

2.3.3 Hardware Reinforcements

Factory mortise, reinforce, drill and tap frames for all mortise hardware as required by hardware manufacturer's template. Provide necessary reinforcement plates as required for surface mounted hardware; installer to perform all field drilling and tapping. Provide dust cover boxes on all frame mortises. Provide minimum thicknesses as follows:

- a. Butts, 4.7 millimeter
- b. Lock strike, 3.04 millimeter
- c. Surface applied hardware 1.90 millimeter

2.3.4 Jamb Anchors

Provide number and spacing of anchors as follows:

2.3.4.1 Stud-Wall Type

Locate anchors not more than 457 mm from top and bottom of frame. Space anchors not more than 813 mm o.c. and as follows:

- a. Three anchors per jamb up to 1524 mm in height.
- b. Four anchors per jamb from 1524 to 2286 mm in height.
- c. Five anchors per jamb from 2286 to 2438 mm in height.
- d. Two anchors per head for frames more than 1066 mm wide and mounted in metal-stud partitions.

2.3.5 Door Hardware

Provide the following STC related hardware with the door; cam-lift hinges, perimeter seals, door bottoms, thresholds, hardware standoff brackets and other hardware as required to meet the STC requirements.

Include on Fabrication drawings a finish hardware schedule for each door and a hollow metal door frame schedule for each door indicating profile, dimensions, hardware reinforcement, and frame anchorage. Also indicate perimeter seals, door-bottom devices and other hardware items that are assembled in the shop.

Refer to Section 08 71 00 DOOR HARDWARE for remaining hardware requirements.

2.3.6 Head and Jamb Seals

Provide a closed-cell, expanded cellular rubber Seal material conforming to ASTM D1056, Type S, Grade SBE-42 or SCE-42 for heads, jambs, and door bottoms.

Install seals in formed steel or extruded aluminum shapes designed to receive and hold seals and to provide concealed adjustable attachment to door frames. Provide concealed adjustment screws that are not more than 300 millimeter on center and provide at least 10 millimeter adjustment.

2.3.7 Door Bottoms

Neoprene or silicone gasket held in place by metal housing; mortised into bottom edge of door.

2.3.7.1 Automatic Door Bottoms

Neoprene or silicone gasket, held in place by metal housing, that automatically drops to form seal when door is closed; mounted to bottom edge of door with screws.

Mounting: Mortised or semimortised into bottom of door or surface mounted on face of door as required by testing to achieve STC rating indicated.

2.3.8 Thresholds

Provide metal thresholds where indicated. Provide thresholds that are extruded aluminum, 6063-T5 alloy, mill finish, not less than 3 millimeter thick, with integral seal grooves formed to the indicated section.

Provide flat, smooth, unfluted thresholds as recommended by manufacturer; fabricated from aluminum.

- a. Finish: Color anodic finish.
- b. Color: As selected by Architect from full range of industry colors and color densities.

2.4 TESTS, INSPECTIONS, AND VERIFICATIONS

2.4.1 Reverberation Time (Rt) Rating

Provide test reports prepared by a nationally recognized, independent laboratory for Acoustical Tests, Air Infiltration Tests, Wind Loading Tests, and Water Leakage Tests indicating that the reverberation time (Rt) rating of the proposed door, based on tests at 16 third-octave band frequencies from 125 to 4,000 hertz, is no less than the specified STC when tested in accordance with ASTM E90, and that the door tested is hung in substantially the type of wall and frame as indicated and is fully operable with hardware and perimeter seals installed.

2.4.2 Positive Pressure

Provide test reports, prepared by a nationally recognized, independent laboratory for Positive Pressure Tests, for all fire rated door assemblies, including Gasketing and Door Bottoms.

2.4.3 Cam Lift Hinges

When required to achieve STC, manufacturer to furnish laboratory test data certifying hinges have been cycled a minimum of 1,000,000 while supporting a minimum door weight of 159 kg.

Full-mortise template type that raises the door 13 mm when door is fully open; with hardened pin; fabricated from stainless steel.

2.4.4 Guarantee

Provide written guarantee that each door delivered to the project is equal in construction, sound transmission classification (STC), and positive pressure test rating where applicable, with appropriate labeling and markings, to that of the sample door tested. Clearly state in written guarantee that each door assembly, when installed in accordance with the manufacturer's printed instructions, has an in-place STC within 3 decibels of the specimen tested. Submit the following test data and Certificates with the written Guarantee:

- a. Acoustical Tests
- b. Positive Pressure Tests

PART 3 EXECUTION

3.1 PREPARATION

Upon receipt of material, thoroughly inspect all frames, doors and accessories. Verify quantities and tag numbers according to the packing list provided. Report all discrepancies, deficiencies and/or damages immediately to Contracting Officer.

Prior to installation check all doors and frames for correct size and swing. Verify that frames are plumb, square and aligned without twist in accordance with tolerances published by NAAMM/HMMA and SDI.

3.1.1 Frame Painting and Cleaning

Clean thoroughly all surfaces of all mill scale, rust, oil, grease, dirt, and other foreign materials before the application of the shop coat of paint.

Apply one shop coat of rust inhibitive metallic oxide or synthetic resin primer applied to clean, dry, and prepared surfaces by brush, dipping, or other approved method to provide a continuous minimum dry film thickness of 0.023 millimeter.

3.2 INSTALLATION

3.2.1 Frame

Install frames plumb and true with not more than 0.8 millimeter deviation in vertical alignment in 2440 millimeter. Anchor to the wall in accordance with the manufacturer's instructions. Spot grout frames in dry wall partitions with mortar at the jamb anchor clips; fill the space between metal frame and stud partition solidly with fiberglass or mineral wool insulation.

3.2.2 Door

Install and adjust all doors, hardware, and seals in accordance with the approved drawings, hardware schedules, and the printed instructions of the door manufacturer.

Install and adjust perimeter seals and automatic door bottom seals to provide positive compression contact with the entire sealing surface with no gaps, openings, or breaks. Hinges or hardware which distort or pinch the perimeter seal during operation of the door will be rejected.

Field apply perimeter seal housings with mitered corners and with flush, aligned hairline joints.

Install components to manufacturer's written instructions. Coordinate with gypsum board wall construction for anchor placement. Set frames plumb, square, level and at correct elevation. Adjust operable parts for correct clearances and function. Install and adjust perimeter and bottom acoustic seals.

3.3 FIELD QUALITY CONTROL

Provide third party testing in accordance with ASTM E336. Verify in writing that installed product performs no less than five (5) ASTC or NIC

rating points below the specified laboratory STC rating. Examine, adjust, and retest any installation not meeting that criteria until compliance is obtained.

3.3.1 Testing and Performance

Provide assemblies that are identical to those tested at an independent acoustical laboratory qualified under the National Voluntary Laboratory Accreditation Program (NVLAP) by the National Institute for Science and Technology (NIST) in accordance with ASTM E90 and ASTM E413. For the assembly test reports include the laboratory name, test report number and date of test.

-- End of Section --

DRAFT

SECTION 08 71 00

DOOR HARDWARE
02/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1510-2	(2019) Test Method for Door Fittings - Part 2: Fittings for Doors
JIS A 1510-3	(2001) Test Methods for Door Fittings - Part 3: Floor Concealed Door Closers, Door Closers and Hinge Closers
JIS A 1516	(1998) Windows and Doorsets - Air Permeability Test
JIS A 1541-1	(2016) Building Hardware - Locks and Latches - Part 1: Test Methods for Locks and Latches
JIS A 1541-2	(2016) Building Hardware - Locks and Latches - Part 2: Methods of the Presentation and Grade of Criteria for Practical Performance Item
JIS A 4702	(2021) Doorsets
JIS A 5756	(2013) Preformed Gaskets Used in Buildings - Classification, Specifications and Test Methods

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101	(2024) Life Safety Code
NFPA 252	(2022) Standard Methods of Fire Tests of Door Assemblies
NFPA 72	(2022; ERTA 22-1) National Fire Alarm and Signaling Code
NFPA 80	(2022) Standard for Fire Doors and Other Opening Protectives

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191	Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act
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(ABA) Accessibility Guidelines

UL SOLUTIONS (UL)

UL Bld Mat Dir (updated continuously online) Building
Materials Directory

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Detail Drawings; G

Verification of Existing Conditions; G

Hardware Schedule; G

Keying System; G

SD-08 Manufacturer's Instructions

Installation

SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G

SD-11 Closeout Submittals

Key Bitting

1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction. Base shop drawings on verified field measurements and include verification of existing conditions.

1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.5 HARDWARE SCHEDULE

Prepare and submit hardware schedule in the following form:

Hardware Item	Quantity	Size	Reference Publication Type No.	Finish	Mfr Name and Catalog No.	Key Control Symbols	UL Mark (If fire-rated and listed)	BHMA Finish Designation

In addition, submit hardware schedule data package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6 KEY BITTING CHART REQUIREMENTS

1.6.1 Requirements

Submit key bitting charts to the Contracting Officer prior to completion of the work. Include:

- Complete listing of all keys.
- Complete listing of all key cuts.
- Tabulation showing which key fits which door.
- Copy of floor plan showing doors and door numbers.
- Listing of 20 percent more key cuts than are presently required in each master system.

1.7 QUALITY ASSURANCE

1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges, and closers of one lock, hinge, or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule. Deliver permanent keys and removable cores to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.

PART 2 PRODUCTS

2.1 TEMPLATE HARDWARE

Hardware applied to metal doors must be manufactured using a template.

Provide templates to door and frame manufacturers in accordance with JIS A 1510-2 for template hinges. Coordinate hardware items to prevent interference with other hardware.

2.2 HARDWARE FOR FIRE DOORS AND EXIT DOORS

Provide all hardware necessary to meet the requirements of NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, NFPA 252 for fire tests of door assemblies, ABA/ADA accessibility requirements, and all other requirements indicated, even if such hardware is not specifically mentioned in paragraph HARDWARE SCHEDULE. Provide Underwriters Laboratories, Inc. labels for such hardware in accordance with UL Bld Mat Dir or equivalent labels in accordance with another testing laboratory approved in writing by the Contracting Officer.

2.3 HARDWARE ITEMS

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover. Coordinate electrified door hardware components with corresponding components specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

2.3.1 Hinges

Provide in accordance with JIS A 1510-3. Provide hinges that are 114 by 114 mm unless otherwise indicated. Construct loose pin hinges for interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

2.3.1.1 Protection Devices

Provide full height hand and finger protection device at the hinge-side area opening of doors and gates. Provide hinge-side protection devices on both sides of doors and gates, covering hinges and space between door and frame when doors are in the open position. The installed device must push hand and fingers out of the opening and away from a crushing hazard.

2.3.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with JIS A 1510-2.

2.3.3 Locks and Latches

2.3.3.1 Bored Locks and Latches

Provide in accordance with JIS A 4702 and to match facility standard "Best Access Systems".

2.3.4 Exit Devices

Provide in accordance with ANSI/BHMA A156.3, Grade 1. Provide stainless steel "T" style touchpad with ends and endcaps ramped in direction of egress to prevent hooking of carts, ruck sacks and similar. Exposed components which are painted, powder coated, plastic or aluminum are not acceptable. Lever shall match lockset lever design. All non-fire rated

electrified and mechanical devices shall have masterkeyed coremax cylinder dogging. Provide strikes as required by conditions.

2.3.5 Cylinders and Cores

Provide small format interchangeable cormax patented cores with seven pin tumblers and sidebar milling to prevent key duplication. Provide all cylinders and cores from one manufacturer. Stamp and serialize each interchangeable core with a key control symbol in a concealed place on the core and coordinated with the Owners existing Best Coremax statewide system. Key blanks shall be brass/bronze material. Cores shall not project beyond the face of the cylinder. Cylinders shall not project beyond the face of the exit device. Cylinders shall be installed in lockset whereas only the core is visible.

2.3.6 Keying System

Provide an extension of the existing keying system. Existing locks were manufactured by "Best Access Systems".

2.3.7 Lock Trim

Provide cast, forged, or heavy wrought construction and commercial plain design for lock trim.

2.3.7.1 Lever Handles

Provide lever handles. Provide lever handle locks with a breakaway feature (such as a weakened spindle or a shear key) to prevent irreparable damage to the lock when force in excess of that specified in JIS A 1541-1 is applied to the lever handle. Provide lever handles return to within 13 mm of the door face.

2.3.7.2 Texture

Provide knurled or abrasive coated knobs or lever handles for doors which are accessible to blind persons and which lead to dangerous areas.

2.3.8 Keys

Provide one file key, one duplicate key, and one working key for each key change and for each master keying system. Provide a quantity of key blanks equal to 20 percent of the total number of file keys. Stamp each key with appropriate key control symbol and "U.S. property - do not duplicate." Do not place room number on keys.

2.3.9 Closers

Provide in accordance with JIS A 1510-3 Series C72000, cast iron, Grade 1, with PT 4C and 1 1/2 inch minimum bore. Provide with brackets, arms, mounting devices, fasteners, full size covers and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

2.3.10 Door Protection Plates

Provide in accordance with JIS A 4702.

2.3.10.1 Sizes of Kick Plates

50 mm less than door width for single doors; 25 mm less than door width for pairs of doors. Provide 200 mm kick plates for flush doors.

2.3.11 Door Stops and Silencers

Provide in accordance with JIS A 4702. Provide three silencers for each single door, two for each pair.

2.3.12 Padlocks

Provide padlock of stainless steel. Shackle to be cut-resistant. Units shall be masterkeyed into the building and shall have a brass/bronze chain permanently attached to the padlock body. Provide key retention function so key can not be removed without shackle being secured to prevent loss or theft.

2.3.13 Thresholds

Use vinyl or silicone rubber insert in face of stop, for exterior doors opening out, unless specified otherwise.

2.3.14 Weatherstripping Gasketing

Provide in accordance with JIS A 5756. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals, sweep strips, and, for pairs of doors, astragals. Air leakage of weatherstripped doors not to exceed 2.19 by 10-5 cms per minute of air per square meter of door area when tested in accordance with JIS A 1516. Provide weatherstripping with one of the following:

2.3.14.1 Extruded Aluminum Retainers

Extruded aluminum retainers not less than 1.25 mm wall thickness with vinyl, neoprene, silicone rubber, or polyurethane inserts. Provide clear (natural) anodized aluminum.

2.3.15 Rain Drips

Provide extruded aluminum rain drips, not less than 2.03 mm thick, clear anodized finish. Set drips in sealant and fasten with stainless steel screws.

2.3.15.1 Door Rain Drips

Approximately 38 mm high by 16 mm projection. Align bottom with bottom edge of door.

2.3.16 Auxiliary Hardware (Other than locks)

Provide in accordance with JIS A 4702.

2.3.17 Special Tools

Provide special tools, such as spanner and socket wrenches and dogging keys, as required to service and adjust hardware items.

2.4 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

2.5 FINISHES

Provide in accordance with JIS A 1541-2. Provide hardware in satin bronze, unless specified otherwise. Provide steel hinges in satin bronze plated. Provide exposed parts of concealed closers finish to match lock and door trim. Match hardware finish for aluminum doors to match the doors.

PART 3 EXECUTION

3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

3.1.1.1 Stop Applied Weatherstripping

Fasten in place with color matched sheet metal screws not more than 225 mm on center after doors and frames have been finish painted.

3.1.2 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws in expansion sleeves. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies.

3.3 HARDWARE LOCATIONS

Provide as indicated or specified otherwise.

- a. Kick and Armor Plates: Push side of single-acting doors. Both sides

of double-acting doors.

3.4 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining work.

3.5 HARDWARE SETS

Provide hardware for aluminum doors under this section. Deliver Hardware templates and hardware, except field applied hardware, to the aluminum door and frame manufacturer for use in fabricating doors and frames.

HW1			
Door: 108A			
Each Opening To Receive:			
2 ea.	Continuous Hinges	661HD	628
1 ea.	Removable Mullion	KR822	600
2 ea.	GSA Exit Device	FF-L-2890C TYPE X	
2 ea.	Closer	C72021 PT4C/D/G/H X 3.81cm Bore	689
2 ea.	Kick Plate	J102 20.32cm	630
1 ea.	Threshold	896 S	628
1 set	Gasketing	700 s	628
1 ea.	Rain Drip	16A	628
2 ea.	Sweep	C627A	628
1 set	Compensating Astragal	110 S X 110 S	628
HW2			
Door: 120A			
Each Opening To Receive:			

1 ea.	Padlock	41B-772T M1	606
HW3			
Door: 106A			
Each Opening To Receive:			
3 ea.	Hinges	A5112 4.5 x 4.5	630
1 ea.	Lockset	F82 x LEVER X CORMAX	626
1 ea.	Closer	C72041 PT4C/D X 3.81cm BORE	689
1 ea.	Kick Plate	J102 20.32cm	639
1 set	Gasketing	2525 C	
HW4			
Door: 107A			
Each Opening To Receive:			
3 ea.	Hinges	A5112 4.5 x 4.5	630
1 ea.	Lockset	F82 x LEVER X CORMAX	626
1 ea.	Closer	C72021 PT4C/D/H X 3.81cm BORE	689
1 ea.	Kick Plate	J102 20.32cm	639
1 set	Gasketing	2525 C	
1 ea.	Auto Door Bottom	320 S	628

-- End of Section --

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SECTION 08 91 00

METAL WALL AND DOOR LOUVERS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-D	(2018) Laboratory Methods of Testing Dampers for Rating
AMCA 500-L	(2015) Laboratory Methods of Testing Louvers for Rating
AMCA 511	(2010; R 2016) Certified Ratings Program for Air Control Devices
AMCA 540	(2023) Test Method for Louvers Impacted by Wind Borne Debris
AMCA 550	(2022) Listed Wind-Driven-Rain-Resistant Louvers

ALUMINUM ASSOCIATION (AA)

AA DAF45	(2003; Reaffirmed 2009) Designation System for Aluminum Finishes
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AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 611	(2014) Voluntary Specification for Anodized Architectural Aluminum
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ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A1008/A1008M	(2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength

Low-Alloy with Improved Formability,
Solution Hardened, and Bake Hardenable

ASTM B209M (2014) Standard Specification for Aluminum
and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221M (2021) Standard Specification for Aluminum
and Aluminum-Alloy Extruded Bars, Rods,
Wire, Profiles, and Tubes (Metric)

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS H 4000 (2017) Aluminium and Aluminium Alloy
Sheets, Strips and Plates (Amendment 1)

JIS H 4100 (2015) Aluminum and Aluminum Alloy
Extruded Profiles

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Wall Louvers

SD-03 Product Data

Metal Wall Louvers

SD-04 Samples

Wall Louver Samples; G

1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

1.5 COLOR SAMPLES

Colors of finishes for wall louver samples must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Galvanized Steel Sheet

ASTM A653/A653M, coating designation Z275.

2.1.2 Aluminum Sheet

ASTM B209M or JIS H 4000, alloy 3003 or 5005 with temper as required for forming.

2.1.3 Extruded Aluminum

ASTM B221M or JIS H 4100, alloy 6063-T5 or -T52.

2.1.4 Cold Rolled Steel Sheet

ASTM A1008/A1008M, Class 1, with matte finish. Use for interior louvers only.

2.2 METAL WALL LOUVERS

2.2.1 Exterior Louver

Fixed aluminum typhoon proof type, with sight-proof, double drainable, horizontally mounted blades at 2 inch o.c. with bird screens and forced entry components; made to withstand a wind load of not less than 30 pounds per square foot and a rainfall rate pf 8.0 inches per hours with a wind directed at the face of the louver at a velocity of 50 miles per hour. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L, AMCA 511, AMCA 540 and AMCA 550. The rating must show a water penetration of 0.20 or less ounce per square foot of free area at a free velocity of 800 feet per minute. Provide size as indicated on the Drawings. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.06 kilograms or less per square meter of free area at a free velocity of 244 meters per minute.

2.2.1.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 2 mm.

2.2.2 Interior Louver

Fixed vertical mounted galvanized steel barometric relief, eccentrically pivoted back-draft damper for low velocity systems and designed to allow horizontal airflow and prevent reverse airflow. On-blade counterweights are provided to fine tune start-to-open and full open blade operation with ball bearings to minimize friction. Wall louvers must bear the AMCA 500-D certified ratings program seal for air performance. Provide size as indicated on the Drawings. Performance ratings as follows:

- a. Back pressure to be a maximum rating of 0.5 kPa.
- b. Start-to-open pressure to be 0.1 kPa.

c. Velocity to be a maximum 10.2 m/s.

d. Temperature to be 82 degrees C.

2.2.2.1 Construction

2.2.2.1.1 Frame

16 ga. Galvanized steel damper frame.

2.2.2.1.2 Blades

Damper blades to be a minimum 0.063 inch thick aluminum.

2.2.2.1.3 Blade Stops

Each blade stop (at top and bottom of damper frame) to occupy no more than 89 millimeters of the damper opening area to allow for maximum free area and to minimize pressure loss across the damper.

2.2.2.1.4 Seals

a. Blade edge seals to be thermoplastic elastomer rubber-like plastic component to fit over the edge of damper blades to create a tight, air blocking barrier.

b. Jamb to be manufacturer's standard.

2.2.2.1.5 Linkage

External, steel tie bars.

2.2.2.1.6 Axles

Manufacturers standard plated steel stub axles.

2.2.2.1.7 Bearings

Galvanized steel press fit.

2.2.2.1.8 Counterbalance

Blade mounted with adjustable weights.

2.2.3 Formed Metal Louvers

Formed of steel sheet not thinner than 16 U.S. gage, or aluminum sheet not less than 2 mm thick.

2.2.4 Screens and Frames

For aluminum louvers, provide 12.5 mm square mesh, 1.8 or 1.5 mm aluminum or 6 mm square mesh, 1.5 mm aluminum bird screening. For steel louvers, provide 12.5 mm square mesh, 2.5 or 1.5 mm zinc-coated steel; or 6 mm square mesh, 1.5 mm thick zinc-coated steel. Mount screens in removable, rewirable frames of same material and finish as the louvers.

2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

2.4 FINISHES

2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an anodic coating. Color must be as indicated on the drawings. Louvers must have the same finish.

2.4.1.1 Anodic Coating

Clean exposed aluminum surfaces and provide an anodized finish conforming to AA DAF45 and AAMA 611. Finish must be:

- a. Architectural Class I (0.0175 mm or thicker), designation AA-M10-C22-A42, integral color anodized on exterior louvers.

2.4.2 Steel

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating for louvers located in conditioned spaces on interior of the building may be Z180 (G60). The weight of zinc coatings must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied before forming, in lieu of prime coat specified.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

3.2.1 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

-- End of Section --

DRAFT

SECTION 09 22 00

SUPPORTS FOR PLASTER AND GYPSUM BOARD
02/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 6517	(2020) Steel Furrings for Wall and Ceiling in Buildings
JIS G 3302	(2022) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3314	(2010) Hot-Dip Aluminum-Coated Steel Sheet and Strip
JIS G 3321	(2012) Molten 55 Percent Aluminum Zinc Alloy Plated Steel Sheet and Strip

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Metal Support Systems; G

Submit for the erection of metal framing. Indicate materials, sizes, thicknesses, and fastenings.

SD-03 Product Data

Metal Support Systems

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the job site and store in ventilated dry locations permitting easy access for inspection and handling. If materials are stored outdoors, stack materials off the ground, supported on a level platform, and fully protected from the weather. Handle materials carefully to prevent damage. Remove damaged items and provide new items.

PART 2 PRODUCTS

2.1 MATERIALS

Provide steel materials for metal support systems with galvanized coating

JIS G 3302, Z18; aluminum coating JIS G 3314, with a plating adhesion amount of 80, or JIS G 3321. Provide support systems, bracing and attachments per Building Standard Law of Japan.

2.1.1 Materials for Attachment of Gypsum Wallboard

2.1.1.1 Furred Ceiling Systems

JIS A 6517.

2.1.1.2 Nonload-Bearing Wall Framing

JIS A 6517, but not thinner than 0.45 mm thickness, with 0.85 mm minimum thickness supporting wall hung items such as cabinetwork, equipment and fixtures.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Systems for Attachment of Gypsum Board

Follow manufacturer's instructions.

3.2 ERECTION TOLERANCES

Provide framing members which will be covered by finish materials such as wallboard, plaster, or ceramic tile set in a mortar setting bed, within the following limits:

- a. Layout of walls and partitions: 6 mm from intended position;
- b. Plates and runners: 5 mm in 1.9 meters from a straight line;
- c. Studs: 5 mm in 1.9 meters out of plumb, not cumulative; and
- d. Face of framing members: 5 mm in 1.9 meters from a true plane.

-- End of Section --

SECTION 09 29 00

GYPSUM BOARD
08/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------|---|
| ASTM C919 | (2024) Standard Practice for Use of Sealants in Acoustical Applications |
| ASTM E84 | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials |

GYPSUM BOARD ASSOCIATION OF JAPAN

- | | |
|------|---------------------------------|
| GRAM | Gypsum Board Application Manual |
|------|---------------------------------|

JAPANESE STANDARDS ASSOCIATION (JSA)

- | | |
|------------|---|
| JIS A 1901 | (2015) Determination of the Emission of Volatile Organic Compounds and Aldehydes by Building Products - Small Chamber Test Method |
| JIS A 5430 | (2018) Fiber Reinforced Cement Boards |
| JIS A 5758 | (2022) Sealants for Sealing and Glazing in Buildings |
| JIS A 6901 | (2019) Gypsum Boards |
| JIS A 6914 | (2017) Jointing Materials for Gypsum Plasterboards |
| JIS A 9504 | (2021) Man Made Mineral Fibre Thermal Insulation Materials |
| JIS B 1125 | (2020) Self Drilling Tapping Screws |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-08 Manufacturer's Instructions

Safety Data Sheets

SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

1.3.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range. Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants. Do not store panels near materials that may off gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

1.3.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

1.4 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

1.5 SCHEDULING

The gypsum wallboard must be taped, finished and primed before the installation of the highly emitting materials.

The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 10 degrees C and not more than 27 degrees C for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist

air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 75 to 100 mm. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

2.1.1 Gypsum Board

JIS A 6901.

2.1.1.1 Regular

900 mm wide, 12.5 or 15 mm thick, tapered and featured edges.

2.1.2 Regular Water-Resistant Gypsum Backing Board

2.1.2.1 Regular

900 mm wide, 15 mm thick, tapered or beveled edges.

2.1.3 Cementitious Backer Units

In accordance with JIS A 5430.

2.1.4 Joint Treatment Materials

JIS A 6914. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound. Use all-purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.

2.1.4.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

2.1.4.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

2.1.4.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

2.1.4.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

2.1.4.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

2.1.5 Fasteners

2.1.5.1 Screws

JIS B 1125 steel drill screws for fastening gypsum board steel framing members less than 0.84 mm thick. JIS B 1125 steel drill screws for fastening gypsum board to steel framing members 0.84 to 2.84 mm thick. Provide cementitious backer unit screws with a polymer coating.

2.1.6 Accessories

Fabricate from corrosion protected steel or plastic designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

2.1.7 Sound Attenuation Batt Insulation

Provide 40.05 kg/m³, nominal density sound attenuation batt insulation with a RSI- value of 0.65 per 25 mm thickness conforming to JIS A 9504.

2.1.8 Sealants

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied, and comply with MLIT SS Chapter 9.

2.1.9 Interior Sealants

Provide sealant products per JIS A 5758, used on the interior of the building (defined as inside of the weatherproofing system) meeting VOC content requirements to attain F4-Stars rating or VOC content requirements per JIS A 1901, and meet Japan Sealants Industry Association requirement. Color to be selected from manufacturer's full range of color options.

2.1.10 Floor Joint Sealants

Provide sealant products per JIS A 5758 used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements to attain F4-stars rating or VOC content requirements per JIS A 1901, and meet Japan Sealant Industry Association requirement. Color to be selected from manufacturer's full range of color options

2.1.11 Acoustical Sealants

Rubber or polymer based acoustical sealant in accordance to ASTM C919, to have a flame spread of 25 or less and a smoke developed rating of 50 or

less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) to have low or no pollutant emissions.

2.1.12 Water

Provide clean, fresh, and potable water.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Framing and Furring

Verify that framing is securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing is acceptable for application of gypsum board and cementitious backer units.

3.1.2 Building Construction Materials

Do not install building construction materials that show visual evidence of biological growth.

3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing members in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturers instructions and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may not be bonded together with an adhesive, except where prohibited by fire rating(s). Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. Minimize framing by floating corners with single studs and drywall clips. Install gypsum over framing at 610 mm on center. Provide type of gypsum board for use in each system specified herein as indicated.

3.2.1 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.

3.2.2 Gypsum Board for Wall Tile or Tile Base Applied with Adhesive

In dry areas (areas other than tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply glass mat water-resistant gypsum tile backing board in accordance with Gypsum Board Application Manual (GRAM),

Gypsum Board Association of Japan or Manufacturer's Instructions.

3.2.3 Floating Interior Angles

Minimize framing by floating corners with single studs and drywall clips. Locate the attachment fasteners adjacent to ceiling and wall intersections in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions, for single-ply and two-ply applications of gypsum board to metal framing.

3.2.4 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions. Fill control joints between studs in fire-rated construction with firesafing insulation to match the fire-rating of construction.

3.3 APPLICATION OF CEMENTITIOUS BACKER UNITS

3.3.1 Application

In wet areas (tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply cementitious backer units. Place a 7.6 kg asphalt impregnated, continuous felt paper membrane behind cementitious backer units, between backer units and studs or base layer of gypsum board. Place membrane with a minimum 150 mm overlap of sheets laid shingle style.

3.3.2 Joint Treatment

Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.

3.4 FINISHING OF GYPSUM BOARD

Finish plenum areas above ceilings to Level 1. Finish water resistant gypsum backing board to receive ceramic tile to Level 2. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4. Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

3.4.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to Level 5. Apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

3.4.2 Gypsum Board Finish Levels

3.4.2.1 Level 1

All joints and interior angles shall have tape set in joint compound. Surface shall be free of excess joint compound. Tool marks and ridges are acceptable.

3.4.2.2 Level 3

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. One additional coat of joint compound shall be applied over all joints and interior angles. Fastener heads and accessories shall be covered with two separate coats of joint compound. All joint compound shall be smooth and free of tool marks and ridges. Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of final finishes.

3.4.2.3 Level 4

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. Two separate coats of joint compound shall be applied overall flat joints and one separate coat of joint compound shall be applied over interior angles. Fastener heads and accessories shall be covered with three separate coats of joint compound. All joint compound shall be smooth and free of tool marks and ridges. Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of final finishes.

3.4.2.4 Level 5

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. Two separate coats of joint compound shall be applied over all flat joints and one separate coat of joint compound shall be applied over interior angles. Fastener heads and accessories shall be covered with three separate coats of joint compound. A thin skim coat of joint compound trowel applied, or a material manufactured especially for this purpose and applied in accordance with manufacturer's recommendations, applied to the entire surface. The surface shall be smooth and free of tool marks and ridges.

Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of finish paint.

3.5 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified herein. Apply material with exposed surface flush with gypsum board or cementitious backer units.

3.6 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

-- End of Section --

DRAFT

SECTION 09 30 10

CERAMIC, QUARRY, AND GLASS TILING
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920 (2018) Standard Specification for
Elastomeric Joint Sealants

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1509-3 (2014) Test Method for Ceramic Tiles -
Part 3: Determination of Water Absorption,
Apparent Porosity and Bulk Density

JIS A 1509-4 (2008) Test Methods for Ceramic Tiles -
Part 4: Determination of Modulus of
Rupture and Breaking strength

JIS A 1509-8 (2014) Test Methods for Ceramic Tiles -
Part 8: Determination for crazing
resistance for glazed tiles

JIS A 1509-9 (2014) Test Methods for Ceramic Tiles -
Part 9: Determination for frost resistance

JIS A 5209 (2010) Ceramic Tiles

JIS A 6022 (2010) Stretchy Asphalt Roofing Felts
(Synthetic Fiber Base)

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-SS Chapter 11 (2019) Public Building Construction
Standard Specifications - Ch.11 Tile Work
and Wood Work

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G

SD-10 Operation and Maintenance Data

Installation; G

1.3 QUALITY ASSURANCE

Provide installers having a minimum of two years experience with a company specializing in performing the type of work described. Each type and color of tile to be provided from a single source. Each type and color of mortar, adhesive, and grout to be provided from the same source.

1.4 DELIVERY, STORAGE, AND HANDLING

Ship tiles in sealed packages and clearly marked with the grade, type of tile, producer identification, and country of origin. Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and hallmarks intact. Protect materials from weather, and store them under cover in accordance with manufacturer's printed instructions.

1.5 ENVIRONMENTAL REQUIREMENTS

Do not perform ceramic tile work unless the substrate and ambient temperature is at least 10 degrees C and rising. Maintain temperature above 10 degrees C while the work is being performed and for at least 7 days after completion of the work. When temporary heaters are used, ventilate the area to the outside to avoid carbon dioxide damage to new tilework.

1.6 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a 1-year period.

1.7 EXTRA MATERIALS

Supply an extra 2 percent of each type tile used in clean and marked cartons.

PART 2 PRODUCTS

2.1 TILE

Provide tiles that comply with JIS A 5209 and are standard grade tiles. Provide a minimum breaking strength for floor tile in accordance with JIS A 1509-4. Provide exterior building tile for cold climate projects that is approved by the manufacturer for exterior use in accordance with JIS A 1509-9 and JIS A 5209. Provide floor tiles with a wet dynamic coefficient of friction value per JIS A 5209 requirements. Provide glazed floor tile with commercial rating by the manufacturer per JIS A 1509-8, for visible abrasion resistance as related to foot traffic. For materials like tile, accessories, and transition strips submit samples of sufficient size to show color range, pattern, type and joints. Submit manufacturer's catalog data. Tiles and tile works shall comply with MLIT-SS Chapter 11.

2.1.1 Quarry Tile

Furnish an unglazed quarry tile, cove base and trim pieces. Provide tile with smooth surface. Provide nominal tile size(s) of 150 by 150 mm and 13 mm thick. Provide maximum water absorption in accordance with JIS A 5209

or JIS A 1509-3.

2.1.2 Glazed Wall Tile

Furnish glazed wall tile that has cushioned edges and trim with lead-free bright finish. Provide nominal tile size(s) of 106 by 106 mm, match existing size and finish.

2.2 WATER

Provide potable water.

2.3 MORTAR, GROUT, AND ADHESIVE

2.3.1 Dry-Set Portland Cement Mortar

MLIT-SS Chapter 11 Tile Work.

2.3.2 Latex-Portland Cement Mortar

MLIT-SS Chapter 11 Tile Work.

2.3.3 Ceramic Tile Grout

MLIT-SS Chapter 11 Tile Work; petroleum-free and plastic-free sand portland cement grout.

2.3.4 Sealants

2.3.4.1 Interior Sealants

Provide ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Color(s) of sealant to be selected from manufacturer's full range of color options.

2.4 SUBSTRATES

2.4.1 Cementitious Backer Board

Refer to Section 09 29 00 GYPSUM BOARD for tile substrate requirements.

2.5 TRANSITION STRIPS

Provide clear anodized aluminum transitions between tile and carpet or resilient flooring. Provide types as recommended by flooring manufacturer for both edges and transitions of flooring materials appropriate for new and existing conditions.

2.6 MEMBRANE MATERIALS

Conform to JIS A 6022, Type 1 for 33 kg waterproofing membrane, asphalt-saturated building felt.

2.7 COLOR, TEXTURE, AND PATTERN

Provide color, pattern and texture in accordance with as indicated on the Drawings or to match existing.

PART 3 EXECUTION

3.1 PREPARATORY WORK AND WORKMANSHIP

Inspect surface to receive tile in conformance to the requirements of MLIT-SS Chapter 11 Tile Work for surface conditions for the type setting bed specified and for workmanship. Provide variations of tiled surfaces that fall within maximum values shown below:

TYPE	WALLS	FLOORS
Dry-Set Mortar	1/8 inch in 8 ft.	1/8 inch in 10 ft.

3.2 GENERAL INSTALLATION REQUIREMENTS

Do not start tile work until roughing in for mechanical and electrical work has been completed and tested, and built-in items requiring membrane waterproofing have been installed and tested. Close space, in which tile is being set, to traffic and other work. Keep closed until tile is firmly set. Do not start floor tile installation in spaces requiring wall tile until after wall tile has been installed. Apply tile in colors and patterns indicated in the area shown on the drawings. Install tile with the respective surfaces in true even planes to the elevations and grades shown. Provide special shapes as required for sills, jambs, recesses, offsets, external corners, and other conditions to provide a complete and neatly finished installation. Solidly back tile bases and coves with mortar. Do not walk or work on newly tiled floors without using kneeling boards or equivalent protection of the tiled surface. Keep traffic off horizontal portland cement mortar installations for at least 72 hours. Keep all traffic off epoxy installed floors for at least 40 hours after grouting, and heavy traffic off for at least 7 days, unless otherwise specifically authorized by manufacturer. Dimension and draw detail drawings at a minimum scale of 6 mm = 300 mm. Include drawings of pattern at inside corners, outside corners, termination points and location of all equipment items such as thermostats, switch plates, mirrors and toilet accessories mounted on surface. Submit drawings showing ceramic tile pattern elevations and floor plans. Submit manufacturer's preprinted installation instructions.

Do not install building construction materials that show visual evidence of biological growth.

3.3 INSTALLATION OF WALL TILE

Install wall tile in accordance with the MLIT-SS Chapter 11 Tile Work, method and with grout joints as recommended by the manufacturer for the type of tile. Install thinner wall tile flush with thicker wall tile applied on same wall and provide installation materials as recommended by the tile and setting materials manufacturer's to achieve flush installation.

3.3.1 Dry-Set Mortar and Latex-Portland Cement Mortar

Use Latex-Portland Cement to install tile in accordance with MLIT-SS Chapter 11 Tile Work. Use Latex Portland Cement when installing porcelain ceramic tile.

3.3.2 Ceramic Tile Grout

Prepare and install ceramic tile grout in accordance with MLIT-SS Chapter 11 Tile Work. Provide and apply manufacturer's standard product for sealing grout joints in accordance with manufacturer's recommendations.

3.4 INSTALLATION OF FLOOR TILE

Install floor tile in accordance with MLIT-SS Chapter 11 Tile Work method and with grout joints as recommended by the manufacturer for the type of tile.

3.4.1 Dry-Set and Latex-Portland Cement

Use dry-set mortar to install tile directly over properly cured, plane, clean concrete slabs in accordance with Install in accordance with MLIT-SS Chapter 11 Tile Work.

3.4.2 Ceramic Tile Grout

Prepare and install ceramic tile grout in accordance with MLIT-SS Chapter 11 Tile Work. Provide and apply manufacturer's standard product for sealing grout joints in accordance with manufacturer's recommendations.

3.5 INSTALLATION OF TRANSITION STRIPS

Install transition strips where indicated, in a manner similar to that of the ceramic tile floor and as recommended by the manufacturer. Provide thresholds full width of the opening. Install head joints at ends not exceeding 6 mm in width and grouted full.

3.6 EXPANSION JOINTS

Verify existing wall/floor expansion joints and replace and/or repair in accordance with MLIT-SS Chapter 11 Tile Work.

3.7 CLEANING AND PROTECTING

Upon completion, thoroughly clean tile surfaces in accordance with manufacturer's approved cleaning instructions. Do not use acid for cleaning glazed tile. Clean floor tile with resinous grout or with factory mixed grout in accordance with printed instructions of the grout manufacturer. After the grout has set, provide a protective coat of a noncorrosive soap or other approved method of protection for tile wall surfaces. Cover tiled floor areas with building paper before foot traffic is permitted over the finished tile floors. Provide board walkways on tiled floors that are to be continuously used as passageways by workmen. Replace damaged or defective tiles. Submit copy of manufacturer's printed maintenance instructions.

-- End of Section --

DRAFT

SECTION 09 51 00

ACOUSTICAL CEILINGS

08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E580/E580M (2020) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5758 (2022) Sealants for Sealing and Glazing in Buildings

JIS A 1409 (1998) Method for Measurement of Sound Absorption Coefficients in a Reverberation Room

JIS A 1445 (2007) Testing Methods of Materials for Ceiling Suspension System

JIS A 6301 (2007) Sound Absorbing Materials

JIS B 1168 (1994) Eyebolts

JIS G 3537 (2011) Zinc-coated steel wire strands

JIS G 4309 (2013) Stainless Steel Wires

JIS G 5111 (1991) High Tensile Strength Carbon Steel castings and low alloy steel castings for structural purposes

JIS H 8610 (1999) Electroplated-Coatings of Zinc on Iron or Steel

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT Notification No. 771 (2013) Establishment of specified ceilings and a construction method that is effective for structural resistance of specified ceilings

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-310-04 (2013; with Change 1, 2016) Seismic Design of Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Approved Detail Drawings; G

1.3 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

1.4 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 16 degrees C nor more than 29 degrees C and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

1.5 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

1.6 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a one year period. Include an agreement to repair or replace acoustical panels that fail within the warranty period in the standard performance guarantee or warranty. Failures include, but are not limited to, sagging and warping of panels; rusting and manufacturers defects of grid system.

1.7 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 5 tiles for each 1000 tiles installed.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. The unit size, texture, finish, and color must be as specified. Coordinate the whole ceiling system with other details, like the location of access panels and ceiling penetrations, etc., shown on the drawings. The Contractor is responsible for all associated labor and materials and for the final assembly and

performance of the specified work and products are used. The location and extent of acoustical treatment must be as shown on the approved detail drawings. Submit drawings showing suspension system, method of anchoring and fastening, details, and reflected ceiling plan. Coordinate with paragraph RECLAMATION PROCEDURES for reclamation of mineral fiber acoustical ceiling panels to be removed from the job site.

2.1.1 Ceiling Attenuation Class and Test

Provide a ceiling system with an attenuation class (CAC) as specified herein. Provide fixture attenuators over light fixtures and other ceiling penetrations, and provide acoustical blanket insulation adjacent to partitions, as required to achieve the specified CAC. Provide test ceiling continuous at the partition and assembled in the suspension system in the same manner that the ceiling will be installed on the project.

2.1.2 Ceiling Sound Absorption

Determine the sound reduction rate in accordance with JIS A 1409.

2.1.3 Light Reflectance

Light reflectance to be 0.75 or above.

2.2 ACOUSTICAL UNITS

Submit two samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to JIS A 6301 and the following requirements:

2.2.1 Units for Exposed-Grid System

2.2.1.1 Type

III (non-asbestos mineral fiber with painted finish).

2.2.1.2 Flame Spread

Class A, 25 or less

2.2.1.3 Pattern

E

2.2.1.4 Minimum NRC

0.55 in open office areas and other rooms as designated; 0.50 in all other rooms and areas.

2.2.1.5 Minimum Light Reflectance Coefficient

LR-1, 0.75 or greater

2.2.1.6 Nominal Size

600 by 600 mm

2.2.1.7 Edge Detail

Regular or to match existing

2.2.1.8 Finish

Factory-applied standard finish.

2.2.1.9 Minimum CAC

35

2.3 SUSPENSION SYSTEM

Provide standard width flange suspension system conforming to JIS A 1445 for intermediate-duty systems. Provide surfaces exposed to view of aluminum or galvanized steel with a factory-applied white baked-enamel finish. Provide wall molding having a flange of not less than 23 mm. Provide standard mitered corners. Suspended ceiling framing system must have the capability to support the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Provide a suspension system with a maximum deflection of 1/360 of the span length. Conform seismic details to the guidance in UFC 3-310-04 and ASTM E580/E580M or JIS A 1445, and in accordance with "Practical Guide on the Technical Standards concerning Measures to Prevent the Fall of Buildings" based on MLIT Notification No. 771 of the Ministry of Land, Infrastructure, Transport and Tourism.

2.4 HANGERS

Provide hangers and attachment capable of supporting a minimum 1330 N ultimate vertical load without failure of supporting material or attachment.

2.4.1 Wires

Conform wires to JIS G 3537, JIS G 4309 condition annealed stainless steel, 2.0 mm in diameter.

2.4.2 Straps

Provide straps of 25 by 5 mm galvanized steel with a light commercial zinc coating or JIS G 5111 with an electrodeposited zinc coating conforming to JIS H 8610.

2.4.3 Rods

Provide 5 mm diameter threaded steel rods, zinc or cadmium coated.

2.4.4 Eyebolts

Provide eyebolts of weldless, forged-carbon-steel, with a straight-shank in accordance with JIS B 1168. Eyebolt size must be a minimum 7 mm, zinc coated.

2.4.5 Anchorage Devices

Comply with JIS A 1445 for anchorage devices for eyebolts. Where aluminum is in contact with concrete, coat aluminum with bituminous paint or where exposed, with a chromatic primer and 2-coats of enamel paint.

2.5 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

2.6 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system components as indicated.

2.7 ACOUSTICAL SEALANT

Conform acoustical sealant to JIS A 5758, nonstaining.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering and concrete work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

3.1.1 Suspension System

Install suspension system in accordance with JIS A 1445 and in accordance with "Practical Guide on the Technical Standards concerning Measures to Prevent the Fall of Buildings" based on MLIT Notification No. 771 of the Ministry of Land, Infrastructure, Transport and Tourism and as specified herein. The suspension system shall conform to allowable stress performance per the static pressurization test designed by the Japan Ministry of Construction. Do not suspend hanger wires or other loads from underside of steel decking.

3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than 150 mm from each corner of each fixture.

3.1.1.2 Splayed Hangers

Where hangers must be splayed (sloped or slanted) around obstructions, offset the resulting horizontal force by bracing, countersplaying, or other acceptable means.

3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than 75 mm from ends of each length and not more than 400 mm on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

3.1.3 Acoustical Units

Install acoustical units in accordance with the approved installation instructions of the manufacturer. Ensure that edges of acoustical units are in close contact with metal supports, with each other, and in true alignment. Arrange acoustical units so that units less than one-half width are minimized. Hold units in exposed-grid system in place with manufacturer's standard hold-down clips, if units weigh less than 5 kg/square meter or if required for fire resistance rating.

3.2 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

3.3 RECLAMATION PROCEDURES

Neatly stack ceiling tile, designated for recycling by the Contracting Officer, on 1220 by 1220 mm pallets not higher than 1220 mm. Panels must be completely dry. Shrink wrap and symmetrically stack pallets on top of each other without falling over.

-- End of Section --

SECTION 09 65 00

RESILIENT FLOORING
08/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------|--|
| ASTM C920 | (2018) Standard Specification for Elastomeric Joint Sealants |
| ASTM E648 | (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source |

JAPANESE STANDARDS ASSOCIATION (JSA)

- | | |
|------------|--|
| JIS A 5536 | (2015) Adhesives for Resilient Textile or Laminate Floor Coverings |
| JIS A 5705 | (2010) Polyvinyl Chloride Floorcoverings |

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

- | | |
|--------------------|--|
| MLIT SS Chapter 19 | (2019) Public Building Construction Standard Specifications Chapter 19: Interior Finishing Works |
|--------------------|--|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Resilient Flooring and Accessories; G

SD-10 Operation and Maintenance Data

Resilient Flooring and Accessories; G

1.3 JAPAN PUBLIC BUILDING CONSTRUCTION STANDARDS

Follow MLIT SS Chapter 19 for interior finishing works.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in original unopened containers bearing the manufacturer's name, style name, pattern color name and

number, production run, project identification, and handling instructions. Store materials in a clean, dry, secure, and well-ventilated area free from strong contaminant sources and residues with ambient air temperature maintained above 20 degrees C and below 30 degrees C, stacked according to manufacturer's recommendations. Remove resilient flooring products from packaging to allow ventilation prior to installation. Protect materials from the direct flow of heat from hot-air registers, radiators and other heating fixtures and appliances. Observe ventilation and safety procedures specified in the MSDS. Do not store rubber surface products with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions. Do not store exposed rubber surface materials in occupied spaces. Do not store near materials that may off gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives.

1.5 ENVIRONMENTAL REQUIREMENTS

Maintain areas to receive resilient flooring at a temperature above 20 degrees C and below 30 degrees C for 3 days before application, during application and 2 days after application, unless otherwise directed by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature of 13 degrees C thereafter. Provide adequate ventilation to remove moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

1.6 SCHEDULING

Schedule resilient flooring application after the completion of other work which would damage the finished surface of the flooring.

1.7 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a one year period.

1.8 EXTRA MATERIALS

Provide extra flooring material of each color and pattern at the rate of 5 tiles for each 1000 tiles. Provide extra wall base material composed of 6 m of each type, color and pattern. Package all extra materials in original properly marked containers bearing the manufacturer's name, brand name, pattern color name and number, production run, and handling instructions. Provide extra materials from the same lot as those installed. Leave extra stock at the site in location assigned by Contracting Officer.

PART 2 PRODUCTS

2.1 VINYL COMPOSITION TILE

Conform to JIS A 5705, asbestos-free, 300 mm square and 3.2 mm thick. Provide color and pattern uniformly distributed throughout the thickness of the tile.

2.2 WALL BASE

Provide 100 mm high and a minimum 3.175 mm thick wall base. Provide preformed corners in matching height, shape, and color.

2.3 MOULDING

Provide tapered mouldings of clear anodized aluminum and types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified. Provide vertical lip on moulding of maximum 6 mm. Provide bevel change in level between 6 and 13 mm with a slope no greater than 1:2.

2.4 ADHESIVES

Provide adhesives for flooring, base and accessories as recommended by the manufacturer or per JIS A 5536 and comply with local indoor air quality standards. Submit manufacturer's descriptive data, documentation stating physical characteristics, and mildew and germicidal characteristics.

2.5 SURFACE PREPARATION MATERIALS

Provide surface preparation materials, such as panel type underlayment, lining felt, and floor crack fillers as recommended by the flooring manufacturer for the subfloor conditions. Use one of the following substrates:

- a. Concrete floor leveling compound.

2.6 POLISH/FINISH

Provide polish finish as recommended by the manufacturer.

2.7 SEALANTS

Provide ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Color(s) of sealant to be selected from manufacturer's full range of color options.

2.8 MANUFACTURER'S COLOR, PATTERN AND TEXTURE

Provide color, pattern and texture for resilient flooring and accessories as indicated on the Drawings. Provide flooring in any one continuous area or replacement of damaged flooring in continuous area from same production run with same shade and pattern. Submit scaled drawings indicating patterns (including location of patterns and colors) and dimensions. Submit manufacturer's descriptive data and three samples of each indicated color and type of flooring, base, mouldings, and accessories sized a minimum 60 by 100 mm. Submit Data Package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

2.9 FIRE RESISTANCE TESTING REQUIREMENTS

Provide a minimum average critical radiant flux of 0.45 watts per square centimeter for flooring in corridors and exits when tested in accordance with ASTM E648.

PART 3 EXECUTION

3.1 EXAMINATION

Examine and verify that site conditions are in agreement with the design package. Report all conditions that will prevent a proper installation. Do not take any corrective action without written permission from the Government. Work will proceed only when conditions have been corrected and accepted by the installer. Submit manufacturer's printed installation instructions for all flooring materials and accessories, including preparation of substrate, seaming techniques, and recommended adhesives.

3.2 SURFACE PREPARATION

Provide a smooth, true, level plane for surface preparation of the flooring, except where indicated as sloped. Floor to be flat to within 4.75 in 3048 mm. Prepare subfloor in accordance with flooring manufacturer's recommended instructions. Prepare the surfaces of lightweight concrete slabs (as defined by the flooring manufacturer) as recommended by the flooring manufacturer. Floor fills or toppings may be required as recommended by the flooring manufacturer. Install underlayments, when required by the flooring manufacturer, in accordance with manufacturer's recommended printed installation instructions. Before any work under this section is begun, correct all defects such as rough or scaling concrete, chalk and dust, cracks, low spots, high spots, and uneven surfaces. Repair all damaged portions of concrete slabs as recommended by the flooring manufacturer. Remove concrete curing and sealer compounds from the slabs, other than the type that does not adversely affect adhesion. Remove paint, varnish, oils, release agents, sealers, waxes, and adhesives, as required by the flooring product in accordance with manufacturer's printed installation instructions.

3.3 MOISTURE, ALKALINITY AND BOND TESTS

Determine the suitability of the concrete subfloor for receiving the resilient flooring with regard to moisture content and pH level by moisture and alkalinity tests. Conduct moisture testing recommended by the flooring manufacturer. Conduct alkalinity testing as recommended by the flooring manufacturer. Determine the compatibility of the resilient flooring adhesives to the concrete floors by a bond test in accordance with the flooring manufacturer's recommendations. Submit copy of test reports for moisture and alkalinity content of concrete slab, and bond test stating date of test, person conducting the test, and the area tested.

3.4 GENERAL INSTALLATION

Do not install building construction materials that show visual evidence of biological growth. Installation shall comply with MLIT SS Chapter 19.

3.5 PLACING VINYL COMPOSITION TILES

Install tile flooring and accessories in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's directions. Keep tile lines and joints square, symmetrical, tight, and even. Keep each floor in true, level plane, except where slope is indicated. Vary edge width as necessary to maintain full-size tiles in the field, no edge tile to be less than one-half the field tile size, except where irregular shaped rooms make it impossible. Cut flooring to fit around all permanent fixtures, built-in

furniture and cabinets, pipes, and outlets. Cut, fit, and scribe edge tile to walls and partitions after field flooring has been applied.

3.6 PLACING MOULDING

Provide moulding where flooring termination is higher than the adjacent finished flooring and at transitions between different flooring materials. When required, locate moulding under door centerline. Moulding is not required at doorways where thresholds are provided. Anchor aluminum moulding to floor surfaces as recommended by the manufacturer.

3.7 PLACING WALL BASE

Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent resilient flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

3.8 CLEANING

Immediately upon completion of installation of flooring in a room or an area, dry and clean the flooring and adjacent surfaces to remove all surplus adhesive. Clean flooring as recommended in accordance with manufacturer's printed maintenance instructions and within the recommended time frame. As required by the manufacturer, apply the recommended number of coats and type of polish and finish in accordance with manufacturer's written instructions.

3.9 PROTECTION

From the time of installation until acceptance, protect flooring from damage as recommended by the flooring manufacturer. Remove and replace flooring which becomes damaged, loose, broken, or curled and wall base which is not tight to wall or securely adhered.

-- End of Section --

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SECTION 09 69 13

RIGID GRID ACCESS FLOORING
11/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A1264.1 (2017) Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings; Stairs and Guardrail/Handrail Systems

ASTM INTERNATIONAL (ASTM)

ASTM E648 (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3 (2005) Standard for High-Pressure Decorative Laminates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 253 (2011) Standard Method of Test for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detailed Installation Drawings; G

SD-04 Samples

Floor Panels

Floor Covering

Panel Support System

Accessories

SD-07 Certificates

Qualification of Manufacturer

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Lifting Device

1.3 QUALITY CONTROL

1.3.1 Qualification of Manufacturer

Access flooring manufacturer must have at least 5 years experience in manufacturing access flooring systems. Certify that the manufacturer of the access flooring system meets requirements specified under paragraph entitled QUALIFICATION OF MANUFACTURER.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery

Deliver materials to site in undamaged condition, in original containers or packages, complete with accessories and instructions. Label packages with manufacturer's name and brand designations. Package materials covered by specific references bearing specification number, type and class as applicable.

1.4.2 Storage

Store all materials in original protective packaging in a safe, dry, and clean location. Store panels at temperatures between 4 and 32 degrees C, and between 20 and 70 percent humidity. Replace defective or damaged materials.

1.4.3 Handling

Handle and protect materials in a manner to prevent damage during the entire construction period.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

- a. Provide for self-alignment of floor panels and readily removable floor panels covered as specified.
- b. Lateral stability of floor support system must be independent of panels. Provide a finished assembly that is rigid and free of vibration, noises, and rocking panels. Existing raised access flooring system consist of laminate panels, stringers and pedestals.
- c. Submit manufacturer's product data for floor panels consisting of descriptive data, catalog cuts, and installation instructions. Include cleaning and maintenance instructions. Systems which contain

zinc electroplated anti-corrosion coatings are prohibited.

2.1.1 Design Requirements

Panels to match existing panels and provide the same performance criteria as the existing raised access flooring system.

2.1.1.1 Existing System Information

Information from Owner provided as-builts dated 06-25-2023 as follows:

- a. The raised computer floor system has lateral bracing and is designed to resist lateral loads from seismic forces as defined by the International Building Codes (IBC), formerly Building Officials Code Administrators International (BOCA).
- b. The raised computer floor panel system is designed to meet or exceed IBC requirements: panel module size - 600mm by 600mm, design loading (uniform load) - 12.0 kN/m², design loading (concentrated load) - 4450 N, maximum allowable deflection 1/300 of span (= 2mm over a 600mm = 2mm), height above subfloor (to top of panel) - 600mm maximum/457.2 minimum, panel wear surface is plastic laminate or vinyl tile.

2.1.2 Bond Strength of Factory Installed Covering

Bond strength of floor covering must be sufficient to permit handling of the panels by use of the panel lifting device, and to withstand moving caster loads up to equal to the existing panels of the raised access flooring system, without separation of the covering from the panel.

2.2 FLOOR PANELS

2.2.1 Detailed Installation Drawings

Submit Detailed Installation Drawings that as a minimum indicate the following:

- a. Location of panels
- b. Layout of supports, panels, and cutout locations
- c. Lateral bracing
- d. Typical cutout details
- e. Gasketing and perforated panels.
- f. Floor finishes

2.2.2 Panel Construction

- a. Base access floor system on a 600 by 600 mm square module. Fabricate so accurate job cutting and fitting may be done using standard sizes for perimeters and around columns.
- b. Do not expose metal on finished top surface of panels. Provide cutouts and cutout closures to accommodate utility systems and equipment intercabling. Reinforce cutouts to meet design load requirements. Provide extra support pedestals at each corner of

cutout for cutout panels that do not meet specified design load requirements.

- c. Panel design must provide for convenient panel removal for underfloor servicing and for openings for new equipment. Use panels of uniform dimensions within specified tolerances. Permanently mark panels to indicate load rating and model number.
- d. Machine square floor panels to within plus or minus 0.38 mm with edge straightness plus or minus 0.064 mm. If plastic edging is applied to the panel, the tolerances apply to the panel before the plastic edging is applied.

2.2.2.1 Cementitious-Filled Formed Steel (Composite Panels)

- a. Provide composite panels of die-formed steel construction totally enclosing the panel, including the top surface. The void spaces between the top sheet and the formed steel bottom sheet must be completely filled with an incombustible cementitious or concrete material. Seal cut edges in accordance with manufacturer's recommendations. Gravity held panels on existing stringers and pedestals. Replace damaged stringers and fasten using manufacturer's standard screws in accordance with manufacturer's written instructions

2.2.3 Floor Covering

Surface floor panels with factory applied finish materials firmly bonded in place with waterproof adhesive. Provide finish flooring materials in corridors and exits with a critical radiant flux of not less than 0.45 watts per square centimeter (Class 1) when tested in accordance with ASTM E648 or NFPA 253. The electrical resistance must remain stable over the life expectancy of the floor covering. Any anti-static agent used in the manufacturing process must be an integral part of the material, not surface applied. Submit three separate samples of each specified floor covering finish and color.

2.2.3.1 High Pressure Laminate

Provide factory applied high pressure laminate surfacing conforming to ANSI/NEMA LD 3, High-Wear type, Grade HDM, 2 mm thickness. Finish material must consist of one piece to cover the face of the panel. Provide edge detail that is integral to the finish material. The total system electrical resistance from the wearing surface of the floor to the ground connection must be between 1,000,000 (1.0×10^6) ohms and 20,000,000,000 ohms (2.0×10^{10}).

2.2.4 Accessories

Provide the manufacturer's standard perforated panels. Provide perforated panels designed to support the same static loads as floor panels without structural failure, and capable of delivering the air volumes indicated. Submit three samples and colors of each accessory. Provide round perforations.

2.2.5 Lifting Device

At turn over provide one floor panel lifting device standard with the. Furnish a minimum of two devices.

2.3 PANEL SUPPORT SYSTEM

Design support system to allow for 360 degree clearance in laying out cable and cutouts for service to machines and so that panel and stringer together take up maximum of 50 mm. Submit one sample of suspension system proposed for use.

2.3.1 Stringers

Provide stringers of rolled steel or extruded aluminum, to interlock with the pedestal heads to prevent lateral movement. Provide stringers that can be added or removed after floor is in place.

Provide positive fastener connections between floor panels and supporting stringers in lieu of reliance on gravity alone. Provide and install stringers damaged due to removal; to match existing raised access flooring system.

2.3.2 Gaskets

Provide continuous gasketing at contact surfaces between panel and stringers to deaden sound and seal off the underfloor cavity from above for air tightness, and to maintain panel alignment.

2.4 PERFORATED AIR SUPPLY PANELS

Provide air supply floor panels that meet the design criteria specified for standard panels, to sustain point loads of 4448 newton without failure or permanent deformation, are fabricated of 2 mm perforated steel sheet welded to minimum 1.6 mm side channels, are covered with high pressure laminate to match standard panels, and have a uniform perforated pattern to allow even air distribution.

2.5 COLOR

Color must match existing raised access flooring system.

PART 3 EXECUTION

3.1 INSTALLATION

Install access flooring at the location and elevation and in the arrangement shown on the approved detailed installation drawings. The floor system must be of the rigid grid stringer type, complete with all supplemental items, and be the standard product of a manufacturer specializing in access flooring systems.

Install the floor system in accordance with the manufacturer's instructions.

3.1.1 Stringers

Interlock stringers with the pedestal caps to preclude lateral movement, spaced uniformly in parallel lines at the indicated elevation.

3.1.2 Panels

Interlock panels with supports in a manner that will preclude lateral

movement.

3.2 FIELD TESTS

3.2.1 Grounding

Ground the access flooring system for safety hazard and static suppression. Provide positive contact between components for safe, continuous electrical grounding of entire floor system.

3.3 CLEANING AND PROTECTION

3.3.1 Cleaning

Keep the space below the completed floor free of all debris. Before any traffic or other work on the completed raised floor is started, clean the completed floor in accordance with the floor covering manufacturer's instructions.

3.3.2 Protection

Protect traffic areas of raised floor systems with a covering of building paper, fiberboard, or other suitable material to prevent damage to the surface.

3.3.3 Surplus Material Removal

Clean surfaces of the work, and adjacent surfaces soiled as a result of the work. Remove all installation equipment, surplus materials, and rubbish from the work site.

3.4 OPERATION AND MAINTENANCE MANUALS

Submit maintenance instructions for proper care of the floor panel surface. When conductive flooring is specified, also submit maintenance instructions to identify special cleaning and maintenance requirements to maintain "conductivity" properties of the panel finish. Include requirements for maintenance of required dry and wet flooring slip resistance for level and ramped surfaces complying with ASSP A1264.1.

-- End of Section --

SECTION 09 90 00

PAINTS AND COATINGS
05/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5758 (2022) Sealants for Sealing and Glazing in Buildings

JIS A 6909 (2021) Coating Materials for Textured Finishes of Buildings

JIS K5600 (1999) Testing Methods for Paints

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT SS Chapter 18 (2019) Public Building Construction Standard Specification: Chapter 18 Painting Work

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval.

Samples of specified materials may be taken and tested for compliance with specification requirements.

SD-08 Manufacturer's Instructions

Application Instructions

Mixing

1.3 APPLICATOR'S QUALIFICATIONS

1.3.1 Contractor Qualification

Submit the name, address, telephone number, FAX number, and e-mail address of the contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on a minimum of three similar projects within the past three years. List information by individual and include the following:

a. Name of individual and proposed position for this work.

b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address, telephone number, and telex number (if non-US) of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

1.4 REGULATORY REQUIREMENTS

1.4.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, indoor coating materials to conform to formaldehyde emission class F****. Notify Contracting Officer of any paint specified herein which fails to conform.

1.4.2 Lead Content

Do not use coatings having a lead content over 0.009 percent by weight of nonvolatile content.

1.4.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

1.4.4 Asbestos Content

Provide asbestos-free materials.

1.4.5 Mercury Content

Provide materials free of mercury or mercury compounds.

1.4.6 Silica

Provide abrasive blast media containing no free crystalline silica.

1.4.7 Human Carcinogens

Provide materials that do not contain confirmed human carcinogens or suspected human carcinogens.

1.5 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 20 liters. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 4 to 35 degrees C.

1.6 SAFETY AND HEALTH

Apply coating materials using safety methods and equipment in accordance with the following:

Comply with applicable local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS and in Chapter 2 of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

1.6.1 Safety Methods Used During Coating Application

Comply with the requirements of MLIT SS Chapter 18.

1.6.2 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. Removal and disposal of coatings which contain asbestos materials is specified in the Hazardous Materials Report provided by the Owner. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials.

1.7 ENVIRONMENTAL CONDITIONS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation.

1.7.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Unsuitable for drying, including when the air temperature at a location for coating is below 5 degrees C, the humidity is 85 percent or more and condensation occurs due to inadequate ventilation. Where it is not possible to avoid coating, curing measures, such as warming

and ventilation, shall be performed.

- b. External coatings shall generally not be performed if rain is likely to occur or during strong winds.

1.8 COLOR SELECTION

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors approximate colors indicated and the product conforms to specified requirements.

Tint each coat progressively darker to enable confirmation of the number of coats.

Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes.

1.9 LOCATION AND SURFACE TYPE TO BE PAINTED

1.9.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

1.9.1.1 Exterior Painting

Includes existing coated surfaces, and existing uncoated surfaces, of the building and appurtenances. Also included are existing coated surfaces made bare by cleaning operations.

1.9.1.2 Interior Painting

Includes new surfaces, existing uncoated surfaces, and existing coated surfaces of the building and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

1.9.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

1.9.3 Mechanical and Electrical Painting

Includes field coating of interior and exterior new and existing surfaces.

- a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.
 - (1) Exposed piping, conduit, and ductwork;
 - (2) Supports, hangers, air grilles, and registers;
 - (3) Miscellaneous metalwork and insulation coverings.
- b. Do not paint the following, unless indicated otherwise:
 - (1) New zinc-coated, aluminum, and copper surfaces under insulation
 - (2) New aluminum jacket on piping
 - (3) New interior ferrous piping under insulation.

1.9.4 Definitions and Abbreviations

1.9.4.1 Coating

A film or thin layer applied to a base material called a substrate. A coating may be a metal, alloy, paint, or solid/liquid suspensions on various substrates (such as metals, plastics, wood, paper, leather, cloth). They may be applied by electrolysis, vapor deposition, vacuum, or mechanical means such as brushing, spraying, calendaring, and roller coating. A coating may be applied for aesthetic or protective purposes or both. The term "coating" as used herein includes emulsions, enamels, stains, varnishes, sealers, epoxies, and other coatings, whether used as primer, intermediate, or finish coat. The terms paint and coating are used interchangeably.

1.9.4.2 Dry Film Thickness

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

1.9.4.3 micron / microns

The metric measurement for 0.001 mm or one/one-thousandth of a millimeter.

1.9.4.4 mm

The metric measurement for millimeter, 0.001 meter or one/one-thousandth of a meter.

1.9.4.5 Gloss Levels

Gloss Level	Description	Units at 60 degrees	Units at 85 degrees
Gloss 30	Semi-Gloss	15	30
Gloss 50	Semi-Gloss	35	50
Gloss 70	Gloss	60	70
Glossy	High-Gloss	70+	80+

Gloss is tested in accordance with JIS K5600. Historically, the Government has used Flat(Matte), Gloss 30(Eggshell), Gloss 50 (Semi-Gloss), and Gloss 70 (Gloss).

1.9.4.6 Paint

See Coating definition.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit product data sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems.

Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, potlife, and curing and drying times between coats.

PART 3 EXECUTION

3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

3.2 RESEALING OF EXISTING EXTERIOR JOINTS

3.2.1 Surface Condition

Begin with surfaces that are clean, dry to the touch, and free from frost and moisture; remove grease, oil, wax, lacquer, paint, defective backstop, or other foreign matter that would prevent or impair adhesion. Where adequate grooves have not been provided, clean out to a depth of 13 mm and grind to a minimum width of 6 mm without damage to adjoining work. Grinding is not required on metal surfaces.

3.2.2 Backstops

In joints more than 13 mm deep, install glass fiber roving or neoprene, butyl, polyurethane, or polyethylene foams free of oil or other staining elements as recommended by sealant manufacturer. Provide backstop material compatible with sealant. Do not use oakum and other types of absorptive materials as backstops.

3.2.3 Primer and Bond Breaker

Install the type recommended by the sealant manufacturer.

3.2.4 Ambient Temperature

Between 4 degrees C and 35 degrees C when applying sealant.

3.2.5 Exterior Sealant

For joints, provide JIS A 5758. Color(s) will be selected by the Contracting Officer. Apply the sealant in accordance with the manufacturer's printed instructions. Force sealant into joints with sufficient pressure to fill the joints solidly. Apply sealant uniformly smooth and free of wrinkles.

3.2.6 Cleaning

Immediately remove fresh sealant from adjacent areas using a solvent recommended by the sealant manufacturer. Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean condition. Allow sealant time to cure, in accordance with manufacturer's recommendations, prior to coating.

3.3 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, disintegrated coatings, and other foreign matter and substances deleterious to coating performance per MLIT SS Chapter 18 for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas.

3.3.1 Additional Requirements for Preparation of Surfaces With Existing Coatings

Before application of coatings, perform the following on surfaces covered by soundly-adhered coatings, defined as those which cannot be removed with a putty knife:

- a. Test existing finishes for lead before sanding, scraping, or removing. If lead is present, refer to paragraph Toxic Materials.
- b. Wipe previously painted surfaces to receive solvent-based coatings, except stucco and similarly rough surfaces clean with a clean, dry cloth saturated with mineral spirits or per paint manufacturer's requirements. Allow surface to dry. Wipe immediately preceding the

application of the first coat of any coating, unless specified otherwise.

- c. Sand existing glossy surfaces to be painted to reduce gloss. Brush, and wipe clean with a damp cloth to remove dust.
- d. The requirements specified are minimum. Comply also with the application instructions of MLIT SS Chapter 18 and the paint manufacturer.
- e. Thoroughly clean previously painted surfaces specified to be repainted and damaged during construction of all grease, dirt, dust or other foreign matter.
- f. Remove blistering, cracking, flaking and peeling or otherwise deteriorated coatings.
- g. Roughen slick surfaces. Repair damaged areas such as, but not limited to, nail holes, cracks, chips, and spalls with suitable material to match adjacent undamaged areas.
- h. Feather and sand smooth edges of chipped paint.
- i. Clean rusty metal surfaces per MLIT SS Chapter 18 and per paint manufacturer's instructions. Use solvent, mechanical, or chemical cleaning methods to provide surfaces suitable for painting.
- j. Provide new, proposed coatings that are compatible with existing coatings.

3.3.2 Existing Coated Surfaces with Minor Defects

Sand, spackle, and treat minor defects to render them smooth. Minor defects are defined as scratches, nicks, cracks, gouges, spalls, alligatoring, chalking, and irregularities due to partial peeling of previous coatings. Remove chalking by sanding or blasting.

3.3.3 Removal of Existing Coatings

Remove existing coatings from the following surfaces:

- a. Surfaces containing large areas of minor defects;
- b. Surfaces containing more than 20 percent peeling area; and
- c. Surfaces designated by the Contracting Officer, such as surfaces where rust shows through existing coatings.

3.3.4 Substrate Repair

- a. Repair substrate surface damaged during coating removal;
- b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and
- c. Clean and prime the substrate as specified.

3.4 PREPARATION OF METAL SURFACES

3.4.1 Existing and New Ferrous Surfaces

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

3.5 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

3.5.1 Concrete and Masonry

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.
 - (1) Dirt, Chalking, Grease, and Oil: Wash new and existing uncoated surfaces with a solution per paint manufacturer's recommendations. Then rinse thoroughly with fresh water. Wash existing coated surfaces with a suitable detergent and rinse thoroughly. For large areas, water blasting may be used.
 - (2) Fungus and Mold: Wash, existing coated, and existing uncoated surfaces with a solution per paint manufacturer's recommendations. Rinse thoroughly with fresh water.
 - (3) Paint and Loose Particles: Remove by wire brushing.
 - (4) Efflorescence: Remove by scraping or wire brushing followed by washing with a solution per paint manufacturer's recommendations.
 - (5) Removal of Existing Coatings: For surfaces to receive textured coating, remove existing coatings including soundly adhered coatings if recommended by textured coating manufacturer.
- c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.
- d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical or horizontal surfaces as determined manufacturer's recommendations. In all cases follow manufacturers recommendations. Allow surfaces to cure a minimum of 30 days before painting.

3.5.2 Gypsum Board

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

- a. Surface Cleaning: Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A

damp cloth or sponge may be used if paint will be water-based.

- b. Repair of Minor Defects: Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.
- c. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by the paint manufacturer. Verify that new plaster to be coated does not exceed maximum moisture content per manufacturer's recommendations. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting or as recommended by paint manufacturer.

3.5.3 Existing Asbestos Cement Surfaces

Remove oily stains by solvent cleaning with mineral spirits, per manufacturer's recommendations. Remove loose dirt, dust, and other deleterious substances by brushing with a soft brush or rubbing with a dry cloth prior to application of the first coat material. Do not wire brush or clean using other abrasive methods. Verify surfaces are dry and clean prior to application of the coating.

3.6 PREPARATION OF WOOD AND PLYWOOD SURFACES

3.6.1 New, Existing Uncoated, and Existing Coated Plywood and Wood Surfaces, Except Floors:

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

- a. Clean wood surfaces of foreign matter.

Surface Cleaning: Verify that surfaces are free from dust and other deleterious substances and in a condition approved by the Contracting Officer prior to receiving paint or other finish. Do not use water to clean uncoated wood. Scrape to remove loose coatings. Lightly sand to roughen the entire area of previously enamel-coated wood surfaces.
- b. Removal of Fungus and Mold: Wash existing coated surfaces with a solution as recommended by paint manufacturer. Rinse thoroughly with fresh water.
- c. Do not exceed 12 percent moisture content of the wood as measured by a moisture meter in accordance with paint manufacturer's recommendations.
- d. Prime or touch up wood surfaces adjacent to surfaces to receive water-thinned paints before applying water-thinned paints.
- e. Cracks and Nailheads: Set and putty stop nailheads and putty cracks after the prime coat has dried.

3.7 APPLICATION

3.7.1 Coating Application

Application of paint per MLIT SS Chapter 18. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of

pigments during application.

Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.

Only apply paints, except water-thinned types to surfaces that are completely free of moisture as determined by sight or touch.

Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.

Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.

Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.

3.7.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

3.7.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

3.7.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in MLIT SS Chapter 18.
- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness as specified in MLIT SS Chapter 18. Coating thickness where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat surfaces which have not been specified, the same as surfaces having similar conditions of exposure.

- d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:
 - (1) One coat of primer.
 - (2) One coat of undercoat or intermediate coat.
 - (3) One topcoat to match adjacent surfaces.
- e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

3.8 COATING SYSTEMS FOR METAL

Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

- a. Apply specified ferrous metal primer on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.
- d. Surface Previously Coated with Epoxy or Urethane: Apply MPI 101, 0.038 mm DFT immediately prior to application of epoxy or urethane coatings.
- e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal per paint manufacturer's recommendations.

3.9 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

3.10 COATING SYSTEMS FOR WOOD AND PLYWOOD

Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

3.11 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

3.12 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers.

3.13 PAINT TABLES

3.13.1 Exterior Paint Tables

DIVISION 3: EXTERIOR CONCRETE PAINT TABLE

- A. Existing, previously painted concrete; vertical surfaces, including undersides of balconies, soffits, roofs, columns and beams but excluding tops of slabs:

1. Paint system shall be formaldehyde emission class F**** minimum, Weatherproof Class 3 Type. Coatings shall comply with JIS A 6909, Multi-Layer, Type E:

Spray Tile Finish System (4-Layer System):

- | | |
|--|--------------------------|
| (1) Base Coat 1: | Not less than 0.1 kg/sm |
| (2) Texture Base Coat 1: | Not less than 0.7 kg/sm |
| (3) Texture Top Coat: | Not less than 0.8 kg/sm |
| (4) Top Coat, Acrylic, Resin, Emulsion Type, High Gloss Finish, 2 times: | Not less than 0.25 kg/sm |

Primer as recommended by manufacturer. Coating system shall be applied by spray application in accordance with manufacturer's instructions.

DIVISION 5: EXTERIOR METAL, FERROUS AND NON-FERROUS PAINT TABLE

STEEL / FERROUS SURFACES

- A. New Steel that has been hand or power tool cleaned or per MLIT SS Chapter 18.

1. MLIT SS Chapter 18

- B. New Steel that has been blast-cleaned:

2. MLIT SS Chapter 18

- C. Existing steel that has been spot-blasted:

1. MLIT SS Chapter 18

EXTERIOR GALVANIZED SURFACES

- F. New Galvanized surfaces:

1. MLIT SS Chapter 18
3. Waterborne Primer / Waterborne Light Industrial Coating
MLIT SS Chapter 18
System DFT: 112 microns
- G. Galvanized surfaces with slight coating deterioration; little or no rusting:
 1. MLIT SS Chapter 18
- H. Galvanized surfaces with severely deteriorated coating or rusting:
 1. MLIT SS Chapter 18

3.13.2 Interior Paint Tables

DIVISION 3: INTERIOR CONCRETE PAINT TABLE

- A. Existing, previously painted Concrete, vertical surfaces, not specified otherwise:
 1. MLIT SS Chapter 18
 2. High Performance Architectural Latex
MLIT SS Chapter 18

DIVISION 5: INTERIOR METAL, FERROUS AND NON-FERROUS PAINT TABLE

INTERIOR STEEL / FERROUS SURFACES

- A. Metal, Mechanical, Electrical, Fire extinguishing sprinkler systems including valves, conduit, hangers, supports . Surfaces adjacent to painted surfaces (Match surrounding finish), exposed copper piping, and miscellaneous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment:
 1. Waterborne Light Industrial Coating
MLIT SS Chapter 18

DIVISION 6: INTERIOR WOOD PAINT TABLE

- A. New Wood and plywood not otherwise specified:
 1. High Performance Architectural Latex
MLIT SS Chapter 18
- B. Existing, previously painted Wood and plywood not otherwise specified:
 1. High Performance Architectural Latex
MLIT SS Chapter 18

DIVISION 9: INTERIOR GYPSUM BOARD PAINT TABLE

- A. New and Existing, previously painted Wallboard not otherwise specified:
 2. High Performance Architectural Latex - High Traffic Areas

MLIT SS Chapter 18

B. Existing, previously painted Wallboard in restrooms

2. Alkyd

MLIT SS Chapter 18

-- End of Section --

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SECTION 10 28 13

TOILET ACCESSORIES
08/17

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-04 Samples

Finishes; G

Accessory Items

1.2 DELIVERY, STORAGE, AND HANDLING

Wrap toilet accessories for shipment and storage, then deliver to the jobsite in manufacturer's original packaging, and store in a clean, dry area protected from construction damage and vandalism.

1.3 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a 1 year period.

PART 2 PRODUCTS

2.1 MANUFACTURED UNITS

Provide toilet accessories where indicated in accordance with paragraph SCHEDULE. Provide each accessory item complete with the necessary mounting plates of sturdy construction with corrosion resistant surface.

2.1.1 Anchors and Fasteners

Provide anchors and fasteners capable of developing a restraining force commensurate with the strength of the accessory to be mounted and suited for use with the supporting construction. Provide tamperproof design exposed fasteners with finish to match the accessory.

2.1.2 Finishes

Except where noted otherwise, provide the following finishes on metal:

Metal	Finish
Stainless steel	No. 4 satin finish

2.2 ACCESSORY ITEMS

Conform to the requirements for accessory items specified below. Submit fasteners proposed for use for each type of wall construction, mounting, operation, and cleaning instructions and one sample of each other accessory proposed for use. Incorporate approved samples into the finished work, provided they are identified and their locations noted. Submit certificate for each type of accessory specified, attesting that the items meet the specified requirements.

2.2.1 Grab Bar (GB-1)

Grab bar of 18 gauge, 32 mm OD Type 304 stainless steel. Provide form and length for grab bar as indicated. Provide grab with satin finish. Snap-on flange covers for concealed mounting are stainless steel and equipped with two screw holes for attachment to wall. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 900 pounds when properly installed. Clearance from wall to grab bar is 38 mm to meet ADA and ANSI codes.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Provide the same finish for the surfaces of fastening devices exposed after installation as the attached accessory. Install accessories at the location and height indicated. Protect exposed surfaces of accessories with strippable plastic or by other means until the installation is accepted. After acceptance of accessories, remove and dispose of strippable plastic protection. Coordinate accessory manufacturer's mounting details with other trades as their work progresses. After installation, thoroughly clean exposed surfaces and restore damaged work to its original condition or replace with new work.

3.1.1 Surface Mounted Accessories

Mount on concealed backplates, unless specified otherwise. Conceal fasteners on accessories without backplates. Install accessories with sheet metal screws or wood screws in lead-lined braided jute, PTFE or neoprene sleeves, or lead expansion shields, or with toggle bolts or other approved fasteners as required by the construction. Install backplates in the same manner, or provide with lugs or anchors set in mortar, as required by the construction. Fasten accessories mounted on gypsum board and plaster walls without solid backing into the metal or wood studs or to solid wood blocking secured between wood studs, or to metal backplates secured to metal studs.

3.2 CLEANING

Clean material in accordance with manufacturer's recommendations. Do not use alkaline or abrasive agents. Take precautions to avoid scratching or marring exposed surfaces.

-- End of Section --

SECTION 10 44 16

FIRE EXTINGUISHERS
11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E814 (2024) Standard Test Method for Fire Tests of Penetration Firestop Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1 (2021) Fire Code

NFPA 10 (2022; ERTA 1 2021) Standard for Portable Fire Extinguishers

NFPA 101 (2024) Life Safety Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.157 (2003) Portable Fire Extinguishers

UL SOLUTIONS (UL)

UL 299 (2024) Dry Chemical Fire Extinguishers

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fire Extinguishers; G

Accessories; G

Cabinets; G

Wall Brackets; G

Schedule; G

SD-03 Product Data

Fire Extinguishers; G

Accessories; G

Cabinets; G

Wall Brackets; G

Replacement Parts List; G

SD-07 Certificates

Fire Extinguishers Certifications; G

Manufacturer's Warranty with Inspection Tag; G

1.3 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

1.4 WARRANTY

Guarantee that Fire Extinguishers are free of defects in materials, fabrication, finish, and installation and that they will remain so for a period of not less than six years after completion.

Submit the manufacturer's warranty with inspection tag.

1.5 PROJECT SCHEDULE

For fire extinguishers. Coordinate final fire extinguisher schedule with fire protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: Fire Extinguishers; Accessories, cabinets, Wall Brackets.

2.1 SYSTEM DESCRIPTION

2.1.1 Types

Submit fire extinguishers certifications showing compliance with local codes and regulations.

Provide fire extinguishers conforming to NFPA 10. Provide quantity and placement in compliance with the applicable sections of NFPA 1, NFPA 101, and 29 CFR 1910.157.

Provide dry chemical type fire extinguishers compliant with UL 299.

2.1.2 Material

Provide corrosion-resistant steel extinguisher shell.

2.1.3 Size

13.6 kilogram extinguishers.

2.1.4 Accessories

- a. Forged brass valve
- b. Fusible plug
- c. Safety release
- d. Pressure gage

2.2 EQUIPMENT

2.2.1 Cabinets

2.2.1.1 Material

Provide enameled steel cabinets.

2.2.1.2 Type

Provide semi-recessed cabinet for a 150 or 100 mm wall as indicated on drawings.

Provide a fire rated cabinet, listed and labeled to comply with ASTM E814 for fire resistance wall rating.

2.2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguishers.

2.2.2 Wall Brackets

Provide wall-hook fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

2.2.2.1 Identification

Provide lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by the drawings.

- a. Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

(1) Orientation: Vertical.

PART 3 EXECUTION

3.1 INSTALLATION

Install Fire Extinguishers where indicated on the drawings. Verify exact locations prior to installation.

Provide extinguishers which are fully charged and ready for operation upon installation. Provide extinguishers complete with Manufacturer's Warranty with Inspection Tag attached.

Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.

Comply with the manufacturer's recommendations for all installations.

3.2 PROTECTION

3.2.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

Submit replacement parts list indicating specified items replacement part, replacement cost, and name, address and contact for replacement parts distributor.

3.2.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --

SECTION 13 48 73

SEISMIC CONTROL FOR NONSTRUCTURAL COMPONENTS
05/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

- ACI 318 (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)
- ACI 355.2 (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary
- ACI 355.4 (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN WELDING SOCIETY (AWS)

- AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

- ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel
- ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
- ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
- ASTM A500/A500M (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
- ASTM A563 (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts

ASTM A563M	(2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)
ASTM A603	(2019) Standard Specification for Zinc-Coated Steel Structural Wire Rope
ASTM A992/A992M	(2022) Standard Specification for Structural Steel Shapes
ASTM B695	(2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM C827/C827M	(2023) Standard Test Method for Change in Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM F436/F436M	(2024) Standard Specification for Hardened Steel Washers Inch and Metric Dimensions
ASTM F844	(2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC ES AC156	(2012) Acceptable Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components
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INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code
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METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

MFMA-4	(2004) Metal Framing Standards Publication
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U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 2, 2024) Structural Engineering
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UFC 4-010-01 (2018; with Change 3, 2024) DoD Minimum
Antiterrorism Standards for Buildings

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION
(VISCMA)

VISCMA 412 (2014) Installing Seismic Restraints for
Mechanical Equipment

VISCMA 413 (2014) Installing Seismic Restraints for
Electrical Equipment

1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

Design and provide seismic supports and attachments in accordance with ASCE 7-22 Chapter 13 and as modified by UFC 3-301-01. Design and provide supports and attachments in accordance with requirements in UFC 4-010-01.

Components and their supports must be attached or anchored to structure or foundation. Components, supports, and attachments must comply with following structural design criteria:

Risk Category:	{I}{II}{III}{IV}{V}
Seismic Design Category:	{C}{D}{E}{F}
Seismic Design Spectral Response Acceleration Parameter at Short Period (SDS):	{ }2.02g
Seismic Design Spectral Response Acceleration Parameter at period of 1 second (SD1):	{ }0.93g

Apply the seismic requirements described in this section and on the drawings, of the electrical, mechanical, and miscellaneous equipment and components listed in paragraphs ELECTRICAL EQUIPMENT, MECHANICAL EQUIPMENT, and MISCELLANEOUS EQUIPMENT AND COMPONENTS below, in accordance with UFC 3-301-01 and additional data furnished by the Contracting Officer. Provide seismic protection measures in addition to any other requirements called for in other sections of these specifications. Where there is a conflict between the specifications and the drawings, the specifications will take precedence. Accomplish resistance to lateral forces induced by earthquakes without consideration of friction resulting from gravity loads.

For new construction and existing buildings required to comply with the UFC 4-010-01, all overhead electrical, mechanical, and miscellaneous equipment and other fixtures weighing or more must be designed to resist forces of 0.5 times the equipment weight in any horizontal direction and 1.5 times the equipment weight in the downward direction. This requirement does not preclude the need to design equipment mountings for structural design criteria stated above.

1.2.2 Electrical Equipment

Provide seismic supports and attachments for electrical equipment listed in Section 26 05 48.00 10 SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT,

paragraph ELECTRICAL EQUIPMENT.

1.2.3 Mechanical Equipment

Provide seismic supports and attachments for the mechanical equipment listed in Section 23 05 48.19 SEISMIC BRACING FOR MECHANICAL SYSTEMS, paragraph MECHANICAL EQUIPMENT.

1.2.4 Miscellaneous Equipment and Components

Provide seismic supports and attachments for the following miscellaneous equipment and components in accordance with the requirements of this specification:

Equipment/Components with $I_p = 1.0$
Storage cabinets
Ornamentations
Storage Racks
Signs and Billboards
Shelving
Furnishings
Partitions
Stacks
Pole or frame supported equipment
Storage tanks for water and oil

Equipment/Components with $I_p = 1.5$ (Designated Seismic Systems)
Insert edited list here similar to one above for $I_p = 1.0$

1.2.5 Contractor Designed Supports and Attachments

Provide seismic supports and attachments for electrical, mechanical, and miscellaneous equipment and components.

Contractor must retain services of a Professional Engineer to design supports and attachments for electrical, mechanical, and miscellaneous equipment and components. Supports and attachments, and isolation and energy dissipation systems that induce torsion in structural members are not permitted.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Resilient Vibration Isolation Devices; G

SD-03 Product Data

Material Supports; G

Common Bolts; G

High-Strength Bolts; G

Anchors Certified Manufacturer's Mill Reports; G

Vibration Isolators; G

Snubbers; G

Welding Electrodes And Rods;

SD-05 Design Data

Design Calculations and Drawings; G

Details of Equipment Supports and Attachments; G

SD-06 Test Reports

Anchor Test Results; G

Shake Table Test; G

SD-07 Certificates

Certificate Of Completion; G

Welder Qualifications; G

PART 2 PRODUCTS

2.1 EQUIPMENT REQUIREMENTS

Equipment must be rugged enough to survive design seismic event. Equipment components designated as Designated Seismic Systems must remain operational as required in paragraph SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT.

Submit design calculations and drawings of supports and attachments for the items listed in paragraph ELECTRICAL EQUIPMENT, MECHANICAL EQUIPMENT, and MISCELLANEOUS EQUIPMENT AND COMPONENTS. Design calculations and drawings must be stamped and signed by Contractor's Professional Engineer.

A 1 Kyu Kenchiku Shi (1st Class Qualified Architect) is an acceptable equivalent to a U.S. Licensed Professional Structural Engineer. Design must be based on actual equipment and system layout. Design must include calculated applicable loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods.

Drawings must indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction.

2.1.1 Designated Seismic System Equipment

Include drawing for Designated Seismic System Equipment indicating the equipment location in the facility to be used for the installation. Equipment must be rigidly or flexibly mounted to building structure or

foundation. Roof mounted equipment both vibration isolated and nonisolated, must have support members designed and anchored to building structure

2.1.2 Rigidly (Base and Suspended) Mounted Equipment

The following specific items of miscellaneous equipment and components must be constructed and assembled to withstand the seismic forces specified in UFC 3-301-01.

- Storage cabinets
- Storage Racks
- Signs and Billboards
- Shelving
- Stacks
- Pole or frame supported equipment

Equipment furnished under this contract must be rigidly mounted using cast-in-place anchors or post-installed anchors that are qualified for earthquake loading in accordance with ACI 355.2 or ACI 355.4. Cast-in-place anchors must conform to ASTM F1554 and ACI 318 Chapter 17. For any rigid equipment which is rigidly anchored, entirely locate each item of equipment and rigidly attach on one side only of a building expansion joint; provide flexible joints for piping, electrical conduit, cabletray, busway, raceway, mechanical ducts, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging. Base-mounted equipment installed in Risk Category (RC) IV buildings needed for continued operation after an earthquake assigned to Seismic Design Category (SDC) D, E, or F must have two nuts provided on each anchor.

Roof and floor mounted equipment curbs, framing and attachment to equipment and structure must be designed and braced to withstand seismic loads. Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads utilizing appropriate load factors per ASCE 7-22 or ICC IBC to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchors for combined shear and overturning.

2.1.3 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 6 mm. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubbers in each horizontal direction, and vertical restraints must be provided where required to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer must be a member of VISCMA. Design force, F_p , must be doubled for vibration isolators if the nominal clearance (air gap) between the equipment support frame and restraint is

greater than in accordance with ASCE 7-22, Chapter 13. Housed springs must not be used for seismic restraint applications because they cannot resist uplift.

2.2 SUPPORTS AND ATTACHMENTS

2.2.1 Supports

Submit Material supports used for members listed in this section and on the drawings. Material supports must conform with the following:

- a. Angles, plates, and bars, ASTM A36/A36M
- b. Wide flange and WT shapes, ASTM A992/A992M
- c. Threaded rods, ASTM A307
- d. Wire rope, ASTM A603 pre-stretched, with Class B weight zinc-coated wires throughout rope. Connect rope at ends using ferrule or saddle-type wire rope clamp systems. Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 413. Saddle-type clamps must be used with minimum of two clamps at each end of wire rope.
- e. Tubes, ASTM A500/A500M, Grade B.
- f. Pipes, ASTM A53/A53M, Grade B.
- g. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at discrete points conforming to MFMA-4.

2.2.2 Fasteners

Common bolts, ASTM A307, Grade A, plain finish hot dipped zinc coating. The bolt heads and the nuts of the supplied fasteners must be marked with the manufacturer's identification mark, the strength grade and type specified by ASTM specifications.

High-strength bolts, ASTM F3125/F3125M, Grade A325, heavy hex head style, with nuts per ASTM A563 and washers per ASTM F436/F436M, hot dipped zinc coating. Provide ASTM B695, Class 55, Type 1 galvanizing. Submit product data for direct tension indicator washers.

2.2.3 Anchors

Submit Anchors Certified Manufacturer's Mill Reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied fasteners.

2.2.3.1 Cast-in-Place Anchors

- a. Anchor rods must be per ASTM F1554 Gr 36 55, Class 2A.
- b. Anchor nuts must be per ASTM A563M, Grade A, hex style.
- c. Anchor washers must be per ASTM F844.
- d. Anchor plate washers must be per ASTM A36/A36M.

- e. Non-shrink grout must be per ASTM C1107/C1107M, with no ASTM C827/C827M shrinkage. Grout must be nonmetallic. Submit product data for non-shrink grout.

2.2.4 Welding

AWS D1.1/D1.1M. Submit product data for welding electrodes and rods.

PART 3 EXECUTION

3.1 SUPPORTS AND ATTACHMENTS

Provide supports and attachments with continuous load path to distribute equipment/component seismic loads to structure conforming to UFC 3-301-01 requirements. Install vertical diagonal braces at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412 and VISCMA 413.

Provide bolted and welded connections for supports and attachments in accordance with UFC 3-301-01. Provide welding in accordance with AWS D1.1/D1.1M. Provide AWS D1.1/D1.1M qualified welders, welding operators, and tackers. Submit AWS welder qualifications for approval.

3.2 BUILDING DRIFT

Do not attach sway braces for equipment to two dissimilar structural elements of a building that may respond differentially during an earthquake unless a flexible joint is provided. Equipment/components supports and attachments must be capable of accommodating building story drifts, deflections, and relative displacements.

3.3 ANCHORS

3.3.1 General

Submit copies of anchor test results to verify the adequacy of the specific anchor and application, as specified.

Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for the restraint anchors. Identify position of reinforcing steel and other embedded items prior to drilling holes for post-installed anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or fill the gap with epoxy on equipment anchors where clearance between anchor and equipment support hole exceeds 0.125 inches.

3.3.2 Cast-In-Place Anchors

Use templates to locate cast-in-place anchors accurately and securely in formwork. Provide anchors with an embedded straight length with heavy hex nut and washer to provide required strength and ductility. Anchors that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or be increased in depth to accommodate the required anchor embedment depth.

3.4 RESILIENT VIBRATION ISOLATION DEVICES

Where the need for vibration isolation devices (e.g. multidirectional

seismic snubbers, spring vibration isolators) is determined, install these devices as indicated in drawings and in accordance with the manufacturer's written instructions. After completion of installation of all isolation materials and before startup of isolated equipment, all debris must be cleared from areas surrounding and from beneath all isolated equipment, leaving equipment free to move on the isolation supports. Ensure pipe, duct, and electrical connections to the isolated equipment do not reduce system flexibility. Ensure that pipe, conduit, and duct passing through walls and floors do not transmit vibrations. Check for vibration and noise transmission through connections, piping, ductwork, foundation, and walls. Adjust, repair, or replace isolators as required to reduce vibration and noise transmissions to specified levels.

3.5 EQUIPMENT SUPPORT REQUIREMENTS

3.5.1 Suspended Equipment

Provide equipment supports and attachments for components supported from overhead floor or roof structural systems. Provide braces that consist of angles, rods, wire rope, bars, channels (struts) or pipes arranged as shown in bracing submittals and secured at both ends with not less than 13 mm bolts. Provide sufficient supports and attachments for equipment to resist horizontal and vertical forces as specified in paragraph GENERAL REQUIREMENT without exceeding capacity of structural elements

Submit details of equipment supports and attachments for acceptance. In lieu of bracing with vertical supports, these items may be supported with hangers inclined at 45 degrees directed up and radially away from equipment and oriented symmetrically in 90-degree intervals on the horizontal plane, bisecting the angles of each corner of the equipment, provided that supporting members are properly sized to support operating weight of equipment when hangers are inclined. Where interference is present, the inclined hanger slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412 and VISCMA 413.

3.5.2 Floor or Pad Mounted Equipment

Equipment and components must be positively attached to floor or pad. Provide and install anchors as indicated in approved drawings and in accordance with paragraph ANCHORS.

3.6 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT

Manufactured equipment and components assigned as Designated Seismic Systems in buildings assigned to SDC C, D, E, or F and Risk Category IV equipment and components required to remain operational after an earthquake must be seismic qualified by shake table testing conforming to ICC ES AC156 Shake Table Test procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific equipment and components. Prequalified certifications are acceptable unless noted otherwise.

A certification for each equipment and component in accordance with the requirement must be maintained in a file identified as "Equipment Certification Documentation." This file must be a part of the Operations & Maintenance (O&M) Manual that is turned over to the Authority Having Jurisdiction (AHJ). The following statement must be included in the O&M Manual: "Replaced or modified components in this Manual must be qualified per the original qualification criteria."

Equipment and components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water resistant material. Nameplates must be mechanically attached to all nonstructural components and placed on the component for clear identification. The nameplate must not be less than with red letters 1 inch in height on a white background stating "Certified Equipment." The following statement must be on the nameplate: "This equipment/component is seismically certified. Modification or replacement must be approved in advance by a qualified professional engineer and documented in the "Equipment Seismic Certification Documentation" file within the O&M manuals." The nameplate must also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.

3.7 SPECIAL INSPECTION FOR COMPONENTS, SUPPORTS, AND EQUIPMENT

- a. Perform special inspections for designated seismic systems, equipment, components, supports, and attachments in accordance with ICC IBC Chapter 17, and UFC 3-301-01.
- b. Contractor must employ Special Inspector, who must be an independent Professional Engineer, to perform periodic walk-down inspections to ensure that the equipment/components satisfy life safety mounting requirements. The Special Inspector must be familiar with the construction and installation of mechanical and electrical components, and their vulnerabilities to earthquakes. The selection of the Special inspector is subject to the approval of the Structural Engineer of Record (SER) of the main building or their representative and per Section 01 45 35 SPECIAL INSPECTION.
- c. Special Inspector must (1) inspect and test items; (2) examine Designated Seismic Systems requiring seismic qualification in accordance with ASCE 7-22 and (3) verify the label, anchorage installation, and mounting conform to the certificate of compliance.
- d. A final walk-down inspection for Designated Seismic Systems will be performed by the SER of the main building or their representative. The final walk-down inspection must be documented in a report. The final report prepared by the SER needs to include the following:
 - (1) Record/observations of final site visit
 - (2) Documentation that all required inspections were performed in accordance with the Statement of Special Inspections.
 - (3) Documentation that the Designated Seismic Systems were installed in accordance with the construction documents and inspected in accordance with the requirements of ICC IBC Chapter 17, as modified by UFC 3-301-01.
- e. A certificate of completion is required for each Designated Seismic System equipment to certify that seismic supports and attachments have been installed and inspected in accordance with the requirements in this section and UFC 3-301-01. The certification of completion must be a part of the O&M Manual that is turned over to the AHJ.

-- End of Section --

SECTION 22 00 00

PLUMBING, GENERAL PURPOSE
11/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.22/CSA 4.4 (2015; R 2020) Relief Valves for Hot Water Supply Systems

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.14.1 (2003; R 2017; R 2022) Backwater Valves

ASME B1.20.1 (2013; R 2018) Pipe Threads, General Purpose (Inch)

ASME B16.3 (2021) Malleable Iron Threaded Fittings, Classes 150 and 300

ASME B16.15 (2024) Cast Copper Alloy Threaded Fittings Classes 125 and 250

ASME B16.18 (2021) Cast Copper Alloy Solder Joint Pressure Fittings

ASME B16.22 (2021) Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings

ASME B16.24 (2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500

ASME B16.39 (2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1003 (2020) Performance Requirements for Water Pressure Reducing Valves for Domestic Water Distribution Systems - (ANSI approved 2010)

ASSE 1010 (2021) Performance Requirements for Water Hammer Arresters

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M (1999; R 2022; E 2022) Standard

Specification for Ferritic Malleable Iron Castings

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM A733	(2016) Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples
ASTM B42	(2020) Standard Specification for Seamless Copper Pipe, Standard Sizes
ASTM B43	(2020) Standard Specification for Seamless Red Brass Pipe, Standard Sizes
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)
ASTM B584	(2014; R 2022) Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM B828	(2016) Standard Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings
ASTM D1785	(2015; E 2018) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120
ASTM D2241	(2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)
ASTM D2464	(2015) Standard Specification for Threaded Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
ASTM D2466	(2017) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40
ASTM D2467	(2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
ASTM D2672	(2014) Joints for IPS PVC Pipe Using Solvent Cement
ASTM D3035	(2015) Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside

Diameter

ASTM D2239	(2012) Standard Specification for Polyethylene (PE) Plastic Pipe (SIDR-PR) Based on Controlled Inside Diameter
ASTM D3261	(2016) Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing
ASTM D2683	(2020) Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing
ASTM D2737	(2012a) Polyethylene (PE) Plastic Tubing
ASTM D2846/D2846M	(2019) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Hot- and Cold-Water Distribution Systems
ASTM F441/F441M	(2020) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80
ASTM F442/F442M	(2020) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe (SDR-PR)
ASTM F437	(2021) Standard Specification for Threaded Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
ASTM F438	(2017) Standard Specification for Socket-Type Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 40
ASTM F439	(2019) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
ASTM F877	(2020) Standard Specification for Crosslinked Polyethylene (PEX) Plastic Hot- and Cold-Water Distribution Systems
ASTM F2389	(2021) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523	(1974; A 1999) Safe Drinking Water Act
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INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 COMM	(2017) Standard and Commentary Accessible
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and Usable Buildings and Facilities

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (2024) Standard for the Installation of
Air Conditioning and Ventilating Systems

PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)

PPFA Fire Man (2016) Firestopping: Plastic Pipe in Fire
Resistive Construction

JAPAN WATER WORKS ASSOCIATION (JWWA)

JWWA K 115 (2018) Tar Epoxy Resin Paint of Water
Works and Painting Method

JWWA K 120 (2008) Water Works Sodium Hypochlorite

JWWA S 101 (2019) Polyethylene Pipe Fitting for Water
Distribution

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5207 (2014) Sanitary Wares

JIS A 5758 (2022) Sealants for Sealing and Glazing in
Buildings

JIS A 6005 (2005) Asphalt Roofing Felts

JIS B 0203 (1999) Taper Pipe Threads

JIS B 2003 (2013) General Rules for Inspection of
Valves

JIS B 2004 (1994) General Rules for Marking on Valves

JIS B 2011 (2013) Bronze, Gate, Globe, Angle, and
Check Valves (Amendment 2)

JIS B 2031 (2015) Gray Cast Iron Valves (Amendment 1)

JIS B 2061 (2017) Faucets, Ball Taps and Flush Valves

JIS B 2071 (2000) Steel Valves

JIS B 2220 (2012) Steel Pipe Flanges

JIS B 2240 (2006) Copper Alloy Pipe Flanges

JIS B 2301 (2013) Screwed Type Malleable Cast Iron
Pipe Fittings

JIS B 2302 (2013) Screwed Type Steel Pipe Fittings

JIS B 2404 (2018) Dimensions of Gaskets for Use with
Pipe Flanges

JIS B 7414	(2018) Glass Thermometers
JIS B 7505-1	(2017) Aneroid Pressure Gauges-Part 1: Bourdon Tube Pressure Gauges
JIS B 8225	(2012) Safety Valves-Measuring Methods for Coefficient of Discharge
JIS B 8285	(2010) Welding Procedure Qualification Test for Pressure Vessels
JIS B 8372-1	(2003) Pneumatic Fluid Power-Compressed Air Pressure Regulators and Filter-Regulators-Part 1: Main Characteristics to be Included in Literature from Suppliers and Product-Making Requirements
JIS B 8410	(2015) Pressure Reducing Valves for Water Works
JIS B 8414	(2011) Relief Valves for Hot Water Appliances (Amendment 3)
JIS G 3103	(2012) Carbon Steel and Molybdenum Alloy Steel Plates for Boilers and Pressure Vessels
JIS G 3118	(2017) Carbon Steel Plates for Pressure Vessels for Intermediate and Moderate Temperature Services
JIS G 3201	(2008) Carbon Steel Forgings for General Use (Amendment 1)
JIS G 3452	(2016) Carbon Steel Pipes for Ordinary Piping (Amendment 1)
JIS G 4051	(2018) Carbon Steels for Machine Structural Use (Amendment 1)
JIS G 4107	(2015) Alloy Steel Bolting Materials for High Temperature Service (Amendment 1)
JIS G 4303	(2012) Stainless Steel Bars
JIS G 5502	(2007) Spheroidal Graphite Iron Castings
JIS G 5526	(2014) Ductile Iron Pipes
JIS G 5705	(2018) Malleable Iron Castings
JIS H 3100	(2018) Copper and Copper Alloy Sheets, Plates and Strips
JIS H 3250	(2015) Copper and Copper Alloy Rods and Bars
JIS H 3300	(2018) Copper and Copper Alloy Seamless

Pipes and Tubes

JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 5120	(2016) Copper and Copper Alloy Castings
JIS K 6353	(2011) Rubber Goods for Water Works
JIS K 6739	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Drain
JIS K 6741	(2016; R 2021) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS K 6743	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Water Supply
JIS K 7013	(2009)Fibre Reinforced Plastic Pipes (Amendment 1)
JIS R 3644	(1998) Glass Tubes
JIS S 3200-3	(1997) Equipment for Water Supply Service-Test Method of Water Hammer
JIS S 3201	(2017) Testing Methods for Household Water Purifiers
JIS Z 3284-1	(2014) Solder Paste-Part 1: Kinds and Quality Classification
JIS Z 3313	(2009) Flux Cored Wires for Gas Shielded and Self-Shielded Metal Arc Welding of Mild Steel, High Strength Steel and Low Temperature Service Steel
JIS Z 3621	(2014) Recommended Practice for Brazing
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique
JIS Z 9102	(1987) Identification Marking for Piping Systems

JAPAN VALVE MANUFACTURERS' ASSOCIATION (JVMA)

JV-5	(2008) Pipe end Anticorrosion Screwed Valve
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JAPAN PIPE FITTINGS ASSOCIATION (JPFA)

JPF-MP-006	(2019) Rool Grooving Dimension Fitting
JPF-DP-001	(2010) Screwed Cast Iron Fitting

JAPAN CAST IRON COVER & WASTE FITTING ASSOCIATION (JCW)

JCW 203	(2012) Floor Clean Out
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MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-44	(2019) Steel Pipeline Flanges
MSS SP-67	(2022) Butterfly Valves
MSS SP-70	(2011) Gray Iron Gate Valves, Flanged and Threaded Ends
MSS SP-71	(2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends
MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-78	(2011) Cast Iron Plug Valves, Flanged and Threaded Ends
MSS SP-80	(2019) Bronze Gate, Globe, Angle and Check Valves
MSS SP-83	(2014) Class 3000 Steel Pipe Unions Socket Welding and Threaded
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification
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NSF INTERNATIONAL (NSF)

NSF 372	(2016) Drinking Water System Components - Lead Content
NSF/ANSI 14	(2021) Plastics Piping System Components and Related Materials
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects

</REF>

PLUMBING AND DRAINAGE INSTITUTE (PDI)

PDI WH 201	(2010) Water Hammer Arresters Standard
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Plumbing System; G

Detail drawings consisting of schedules, performance charts, instructions, diagrams, and other information to illustrate the requirements and operations of systems that are not covered by the Plumbing Code. Detail drawings for the complete plumbing system including piping layouts and locations of connections; dimensions for roughing-in, foundation, and support points; schematic diagrams and wiring diagrams or connection and interconnection diagrams. Detail drawings shall indicate clearances required for maintenance and operation. Where piping and equipment are to be supported other than as indicated, details shall include loadings and proposed support methods. Mechanical drawing plans, elevations, views, and details, shall be drawn to scale.

SD-06 Test Reports

Tests, Flushing and Disinfection

Test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, completion and testing of the installed system. Each test report shall indicate the final position of controls.

SD-10 Operation and Maintenance Data

Plumbing System

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.3 STANDARD PRODUCTS

Specified materials and equipment shall be standard products of a manufacturer regularly engaged in the manufacture of such products. Specified equipment shall essentially duplicate equipment that has performed satisfactorily at least two years prior to bid opening. Standard products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.3.1 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.3.2 Service Support

The equipment items shall be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of

the equipment which includes their addresses and qualifications. These service organizations shall be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.3.3 Manufacturer's Nameplate

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.3.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.3.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions shall be considered mandatory, the word "should" shall be interpreted as "shall." Reference to the "code official" shall be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" shall be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" shall be interpreted to mean the "lessor." References to the "permit holder" shall be interpreted to mean the "Contractor."

1.3.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, shall be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.5 PERFORMANCE REQUIREMENTS

1.5.1 Welding

Piping shall be welded in accordance with qualified procedures using performance-qualified welders and welding operators. Procedures and welders shall be qualified in accordance with JIS Z 3801. Welding procedures qualified by others, and welders and welding operators qualified by another employer, may be accepted as permitted by JIS B 8285. The Contracting Officer shall be notified 24 hours in advance of tests,

and the tests shall be performed at the work site if practicable. Welders or welding operators shall apply their assigned symbols near each weld they make as a permanent record.

1.6 REGULATORY REQUIREMENTS

Unless otherwise required herein, plumbing work shall be in accordance with MLIT-M.

1.7 PROJECT/SITE CONDITIONS

The Contractor shall become familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.8 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work.

Instruction shall be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.9 ACCESSIBILITY OF EQUIPMENT

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

2.1 MATERIALS

Materials for various services shall be in accordance with TABLES I and II. Pipe schedules shall be selected based on service requirements. Pipe fittings shall be compatible with the applicable pipe materials. Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF listed for the service intended. Plastic pipe, fittings, and solvent cement used for potable hot and cold water service shall bear the NSF seal "NSF-PW." Polypropylene pipe and fittings shall conform to dimensional requirements of Schedule 40, Iron Pipe size and shall comply with NSF/ANSI 14, NSF/ANSI 61 and ASTM F2389. Polypropylene piping that will be exposed to UV light shall be provided with a Factory applied UV resistant coating. Pipe threads (except dry seal) shall conform to

ASME B1.20.1. Grooved pipe couplings and fittings shall be from the same manufacturer. Material or equipment containing a weighted average of greater than 0.25 percent lead shall not be used in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61, Annex G or NSF 372. In line devices such as water meters, building valves, check valves, meter stops, valves, fittings and back flow preventers shall comply with or PL 93-523 and NSF/ANSI 61, Section 8. End point devices such as drinking water fountains, lavatory faucets, kitchen and bar faucets, residential ice makers, supply stops and end point control valves used to dispense water for drinking must meet the requirements of NSF/ANSI 61. Hubless cast-iron soil pipe shall not be installed underground, under concrete floor slabs, or in crawl spaces below kitchen floors. Plastic pipe shall not be installed in air plenums. Plastic pipe shall not be installed in a pressure piping system in buildings greater than three stories including any basement levels.

The NSF requirements of this section and Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS for Mandatory US Tested Products are independent of the Japanese standards listed elsewhere in this section.

Thus where Japanese standards are listed for potable water equipment, in line devices, and end point devices, the products must meet both NSF and the Japanese standards. The purpose is to promote the use of Japanese products to the greatest extent possible. Japanese products that are not NSF certified will not be allowed for potable water systems. For the majority of material and equipment that must be NSF certified and where other standards are required both the Japanese and US Standards are referenced in this specification.

A searchable database of NSF certified products and materials is available online at: <https://www.nsf.org/certified-products-systems>.

2.1.1.1 Pipe Joint Materials

Grooved pipe and hubless cast-iron soil pipe shall not be used underground. Solder containing lead shall not be used with copper pipe.

- a. Coupling for Cast-Iron Pipe: for hub and spigot type JIS G 5526, JPF-MP-006. For hubless type: JIS G 5526.
- b. Coupling for Steel Pipe: JPF-MP-006.
- c. Couplings for Grooved Pipe: Malleable Iron JIS G 5705.
- d. Flange Gaskets: Gaskets shall be made of non-asbestos material in accordance with JIS B 2404. Gaskets shall be flat, 1.6 mm thick, and contain Aramid fibers bonded with Styrene Butadiene Rubber (SBR) or Nitro Butadiene Rubber (NBR). Gaskets shall be the full face or self centering flat ring type. Gaskets used for hydrocarbon service shall be bonded with NBR.
- e. Brazing Material: Brazing material shall conform to JIS Z 3621.
- f. Brazing Flux: Flux shall be in paste or liquid form appropriate for use with brazing material. Flux shall be as follows: lead-free; have a 100 percent flushable residue; contain slightly acidic reagents; contain potassium borides; and contain fluorides.
- g. Solder Material: Solder metal shall conform to JIS Z 3284-1.

- h. Solder Flux: Flux shall be liquid form, non-corrosive, and conform to JIS Z 3313.
- i. PTFE Tape: PTFE Tape, for use with Threaded Metal or Plastic Pipe.
- j. Rubber Gaskets for Cast-Iron Soil-Pipe and Fittings (hub and spigot type and hubless type): JIS K 6353.
- k. Rubber Gaskets for Grooved Pipe: JIS B 2404.
- l. Flexible Elastomeric Seals: JIS A 5758.
- m. Bolts and Nuts for Grooved Pipe Couplings: Heat-treated carbon steel, JIS G 4051.
- n. Solvent Cement for Transition Joints between ABS and PVC Nonpressure Piping Components: JWWA S 101.
- o. Plastic Solvent Cement for ABS Plastic Pipe: JWWA S 101.
- p. Plastic Solvent Cement for PVC Plastic Pipe: JWWA S 101 and JWWA S 101.
- q. Plastic Solvent Cement for CPVC Plastic Pipe: JWWA S 101.
- r. Flanged fittings including, but not limited to, flanges, bolts, nuts and bolt patterns shall be in accordance with JIS B 2220 and shall have the manufacturer's trademark affixed in accordance with JIS B 2004 and JIS Z 9102. Flange material shall conform to JIS G 3201. Blind flange material shall conform to JIS G 3118 and JIS G 3118 cold service and JIS G 3103 and JIS G 3118 for hot service. Bolts shall be high strength or intermediate strength with material conforming to JIS G 4107 and JIS G 4303.
- s. Plastic Solvent Cement for Styrene Rubber Plastic Pipe: JIS K 6743.
- t. Press fittings for Copper Pipe and Tube: Copper press fittings shall conform to the material and sizing requirements of JIS H 3401. Sealing elements shall be factory installed or an alternative supplied fitting manufacturer. Sealing element shall be selected based on manufacturer's approved application guidelines.
- u. Copper tubing shall conform to JIS H 3300 or ASTM B88M, Type K, L or M.

2.1.2 Miscellaneous Materials

Miscellaneous materials shall conform to the following:

- a. Water Hammer Arrestor: Performance standards of products conforming to the Japanese Water Supply Law or PDI WH 201. Water hammer arrester shall be piston type.
- b. Copper, Sheet and Strip for Building Construction: JIS H 3100.
- c. Asphalt Roof Cement: JIS A 6005.
- d. Hose Clamps: MLIT-M.
- e. Metallic Cleanouts: JCW 203.

- f. Plumbing Fixture Setting Compound: A preformed flexible ring seal molded from hydrocarbon wax material. The seal material shall be nonvolatile nonasphaltic and contain germicide and provide watertight, gastight, odorproof and verminproof properties.
- g. Coal-Tar Protective Coatings and Linings for Steel Water Pipelines: JWWA K 115.
- h. Hypochlorites: JWWA K 120.
- i. Liquid Chlorine: JWWA K 120.
- j. Gauges - Pressure and Vacuum Indicating Dial Type - Elastic Element: JIS B 7505-1.
- k. Thermometers: JIS B 7414. Mercury shall not be used in thermometers.

2.1.3 Pipe Insulation Material

Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.2 PIPE HANGERS, INSERTS, AND SUPPORTS

Pipe hangers, inserts, and supports shall conform to MLIT-M.

2.3 VALVES

Valves shall be provided on supplies to equipment and fixtures. Valves 65 mm and smaller shall be bronze with threaded bodies for pipe and solder-type connections for tubing. Valves 80 mm and larger shall have flanged iron bodies and bronze trim. Pressure ratings shall be based upon the application. Grooved end valves may be provided if the manufacturer certifies that the valves meet the performance requirements of applicable MSS standard. Valves shall conform to the following standards:

Description	Standard
Butterfly Valves: MSS SP-67 for potable water system. JPF-DP-001 or MSS SP-67 for other than potable water system.	JPF-DP-001 or MSS SP-67
Cast-Iron Gate Valves, Flanged and Threaded Ends: MSS SP-70 for potable water system. JIS B 2031 or MSS SP-70 for other than potable water system.	JIS B 2031 or MSS SP-70
Cast-Iron Swing Check Valves, Flanged and Threaded Ends: MSS SP-71 for potable water system. JIS B 2011 or MSS SP-71 for other than potable water system.	JIS B 2011 or MSS SP-71

Ball Valves with Flanged Butt-Welding Ends for General Service: <u>MSS SP-72 for potable water system. JV-5 or MSS SP-72 for other than potable water system.</u>	<u>JV-5 or MSS SP-72</u> <u>JV-5</u>
Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends: <u>MSS SP-110 for potable water system. JV-5 or MSS SP-110 for other than potable water system.</u>	<u>MSS SP-110</u> <u>JV-5</u> or <u>JV-5</u>
Bronze Gate, Globe, Angle, and Check Valves <u>MSS SP-80 for potable water system. JV-5 or MSS SP-80 for other than potable water system.</u>	<u>JV-5 or MSS SP-78</u>
Steel Valves, Socket Welding and Threaded Ends: <u>ASME B16.34 for potable water system. JIS B 2003 or ASME B16.34 for other than potable water system.</u>	<u>JIS B 2003</u> , or <u>MSS SP-80</u>
Cast-Iron Globe and Angle Valves, Flanged and Threaded Ends: <u>MSS SP-85 for potable water system. JIS B 2071 or MSS SP-80 for other than potable water system.</u>	<u>JIS B 2071 or ASME A112.14.1</u>
Vacuum Relief Valves: <u>"ANSI Z21.22/CSA 4.4 for potable water system. JIS B 8372-1 or ANSI Z21.22/CSA 4.4 for other than potable water system.</u>	<u>JIS B 8372-1 and JIS B 8225 or ANSI Z21.22/CSA 4.4</u>
Water Pressure Reducing Valves: <u>ASSE 1003 for potable water system. JIS B 8410 or ASSE 1003 for other than potable water system.</u>	<u>JIS B 8410 or ASSE 1003</u>
Water Heater Drain Valves: <u>ASME BPVC SEC IV, Part HLW-810; Requirements for Potable-Water Heaters Bottom Drain Valve for potable water system. JIS B 2011 or ASME BPVC SEC IV, Part HLW-810; Requirements for Potable-Water Heaters Bottom Drain Valve for other than potable water system.</u>	<u>JIS B 2011</u> or <u>ANSI Z21.22/CSA 4.4</u>
Trap Seal Primer Valves	<u>JIS B 2061</u>

Temperature and Pressure Relief Valves for Hot Water Supply Systems: <u>ANSI Z21.22/CSA 4.4 for potable water system.</u> <u>JIS B 8414 and JIS B 2011 or ANSI Z21.22/CSA 4.4 for other than potable water system.</u>	JIS B 8414 and JIS B 2011 or ANSI Z21.22/CSA 4.4
Temperature and Pressure Relief Valves for Automatically Fired Hot Water Boilers	JIS B 8414

2.3.1 Backwater Valves

Backwater valves shall be either separate from the floor drain or a combination floor drain, P-trap, and backwater valve, as shown. Valves shall have cast-iron bodies with cleanouts large enough to permit removal of interior parts. Valves shall be of the flap type, hinged or pivoted, with revolving disks. Hinge pivots, disks, and seats shall be nonferrous metal. Disks shall be slightly open in a no-flow no-backwater condition. Cleanouts shall extend to finished floor and be fitted with threaded countersunk plugs.

2.3.2 Wall Faucets

Wall faucets with vacuum-breaker backflow preventer shall be brass with 20 mm male inlet threads, hexagon shoulder, and 20 mm hose connection. Faucet handle shall be securely attached to stem.

2.3.3 Lawn Faucets

Lawn faucets shall be brass, with either straight or angle bodies, and shall be of the compression type. Body flange shall be provided with internal pipe thread to suit 20 mm pipe. Body shall be suitable for wrench grip. Faucet spout shall have 20 mm exposed hose threads. Faucet handle shall be securely attached to stem.

2.3.4 Yard Hydrants

Yard box or post hydrants shall have valve housings located below frost lines. Water from the casing shall be drained after valve is shut off. Hydrant shall be bronze with cast-iron box or casing guard. "T" handle key shall be provided.

2.3.5 Relief Valves

Water heaters and hot water storage tanks shall have a combination pressure and temperature (P&T) relief valve. The pressure relief element of a P&T relief valve shall have adequate capacity to prevent excessive pressure buildup in the system when the system is operating at the maximum rate of heat input. The temperature element of a P&T relief valve shall have a relieving capacity which is at least equal to the total input of the heaters when operating at their maximum capacity. Relief valves shall be rated according to JIS B 8225. Relief valves for systems where the maximum rate of heat input is less than 59 kW shall have 20 mm minimum inlets, and 20 mm outlets. Relief valves for systems where the maximum rate of heat input is greater than 59 kW shall have 25 mm minimum inlets, and 25 mm outlets. The discharge pipe from the relief valve shall be the

size of the valve outlet.

2.3.6 Thermostatic Mixing Valves

Provide thermostatic mixing valve for lavatory faucets. Mixing valves, thermostatic type, pressure-balanced or combination thermostatic and pressure-balanced shall be line size and shall be constructed with rough or finish bodies either with or without plating. Each valve shall be constructed to control the mixing of hot and cold water and to deliver water at a desired temperature regardless of pressure or input temperature changes. The control element shall be of an approved type. The body shall be of heavy cast bronze, and interior parts shall be brass, bronze, corrosion-resisting steel or copper. The valve shall be equipped with necessary stops, check valves, unions, and sediment strainers on the inlets. Mixing valves shall maintain water temperature within 2 degrees C of any setting.

2.4 FIXTURES

Water closet replacements in major renovations may have a flush valve of up to 6.1 LPF to accommodate existing plumbing capacity. Vitreous China, nonabsorbent, hard-burned, and vitrified throughout the body shall be provided. Porcelain enameled ware shall have specially selected, clear white, acid-resisting enamel coating evenly applied on surfaces. No fixture will be accepted that shows cracks, crazes, blisters, thin spots, or other flaws. Fixtures shall be equipped with appurtenances such as traps, faucets, stop valves, and drain fittings. Each fixture and piece of equipment requiring connections to the drainage system shall be equipped with a trap. Brass expansion or toggle bolts capped with acorn nuts shall be provided for supports, and polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Fixtures with the supply discharge below the rim shall be equipped with backflow preventers. Internal parts of flush valves and flushometer valves, pop-up stoppers of lavatory waste drains, and pop-up stoppers and overflow tees and shoes of bathtub waste drains may contain acetal resin, fluorocarbon, nylon, acrylonitrile-butadiene-styrene (ABS) or other plastic material, if the material has provided satisfactory service under actual commercial or industrial operating conditions for not less than 2 years. Plastic in contact with hot water shall be suitable for 82 degrees C water temperature.

2.4.1 Lavatories

Provide faucet with a maximum flow rate of 1.9 L/min at a flowing pressure of 414 kPa. Water volume must be limited to 1.0 L per metering cycle.

2.4.2 Automatic Controls

Flushing and faucet systems shall consist of solenoid-activated valves with light beam sensors. Flush valve for water closet shall include an override pushbutton. Flushing devices shall be provided as described in paragraph FIXTURES AND FIXTURE TRIMMINGS.

2.4.3 Flush Valve Water Closets

JIS A 5207, vitreous china, siphon jet, elongated bowl, wall mounted, wall outlet. Top of toilet seat height above floor shall be 360 to 380 mm, except 435 to 480 mm for wheelchair water closets. Provide wax bowl ring including plastic sleeve. Provide solid plastic elongated open-front seat.

Water flushing volume of the water closet and flush valve combination shall not exceed 4.85 liters per flush.

Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components shall be chromium-plated or polished stainless steel. Flush valves shall be nonhold-open type. Mount flush valves not less than 280 mm above the fixture. Mounted height of flush valve shall not interfere with the hand rail in ADA stalls.

2.4.4 Countertop Lavatories

JIS A 5207 white vitreous china, self-rimming, minimum dimensions of 480mm wide by 430 mm front to rear, with supply openings for use with top mounted centerset faucets. Furnish template and mounting kit by lavatory manufacturer. Provide aerator with faucet. Provide lavatory faucets and accessories meeting the flow rate and product requirements of the paragraph LAVATORIES. Mount counter with the top surface 865 mm above floor and with 737 mm minimum clearance from bottom of the counter face to floor. Provide top mounted washerless centerset lavatory faucets.

2.5 MISCELLANEOUS PIPING ITEMS

2.5.1 Escutcheon Plates

Provide one piece or split hinge metal plates for piping entering floors, walls, and ceilings in exposed spaces. Provide chromium-plated on copper alloy plates or polished stainless steel finish in finished spaces. Provide paint finish on plates in unfinished spaces.

2.5.2 Pipe Sleeves

Provide where piping passes entirely through walls, ceilings, roofs, and floors. Sleeves are not required where drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade, except where penetrating a membrane waterproof floor.

2.5.2.1 Sleeves in Masonry and Concrete

Provide steel pipe sleeves or schedule 40 PVC plastic pipe sleeves. Sleeves are not required where drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade. Do not core drill masonry or concrete.

2.5.2.2 Sleeves Not in Masonry and Concrete

Provide 26 gage galvanized steel sheet or PVC plastic pipe sleeves.

2.5.3 Pipe Hangers (Supports)

Provide Pipe hangers and supports in accordance with MLIT-M.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Piping located in air plenums shall conform to NFPA 90A. Piping located in shafts that constitute air ducts or that enclose air ducts shall be

noncombustible in accordance with NFPA 90A. Installation of plastic pipe where in compliance with NFPA may be installed in accordance with PPFA Fire Man. The plumbing system shall be installed complete with necessary fixtures, fittings, traps, valves, and accessories.

3.1.1 Water Pipe, Fittings, and Connections

3.1.1.1 Utilities

The piping shall be extended to fixtures, outlets, and equipment. The hot-water and cold-water piping system shall be arranged and installed to permit draining. The supply line to each item of equipment or fixture, except faucets, flush valves, or other control valves which are supplied with integral stops, shall be equipped with a shutoff valve to enable isolation of the item for repair and maintenance without interfering with operation of other equipment or fixtures. Supply piping to fixtures, faucets, hydrants, shower heads, and flushing devices shall be anchored to prevent movement.

3.1.1.2 Cutting and Repairing

The work shall be carefully laid out in advance, and unnecessary cutting of construction shall be avoided. Damage to building, piping, wiring, or equipment as a result of cutting shall be repaired by mechanics skilled in the trade involved.

3.1.1.3 Protection of Fixtures, Materials, and Equipment

Pipe openings shall be closed with caps or plugs during installation. Fixtures and equipment shall be tightly covered and protected against dirt, water, chemicals, and mechanical injury. Upon completion of the work, the fixtures, materials, and equipment shall be thoroughly cleaned, adjusted, and operated. Safety guards shall be provided for exposed rotating equipment.

3.1.1.4 Mains, Branches, and Runouts

Piping shall be installed as indicated. Pipe shall be accurately cut and worked into place without springing or forcing. Structural portions of the building shall not be weakened. Aboveground piping shall run parallel with the lines of the building, unless otherwise indicated. Branch pipes from service lines may be taken from top, bottom, or side of main, using crossover fittings required by structural or installation conditions. Supply pipes, valves, and fittings shall be kept a sufficient distance from other work and other services to permit not less than 12 mm between finished covering on the different services. Bare and insulated water lines shall not bear directly against building structural elements so as to transmit sound to the structure or to prevent flexible movement of the lines. Water pipe shall not be buried in or under floors unless specifically indicated or approved. Changes in pipe sizes shall be made with reducing fittings. Use of bushings will not be permitted except for use in situations in which standard factory fabricated components are furnished to accommodate specific accepted installation practice. Change in direction shall be made with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center-line radius of bends shall be not less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be acceptable.

3.1.1.5 Commercial-Type Water Hammer Arresters

Commercial-type water hammer arresters shall be provided on hot- and cold-water supplies and shall be located as generally indicated, with precise location and sizing to be in accordance with PDI WH 201 or the Performance Standards Compliant Product (PSCP) of the Water Supply Law. Water hammer arresters, where concealed, shall be accessible by means of access doors or removable panels. Commercial-type water hammer arresters shall conform to JIS S 3200-3 or ASSE 1010. Vertical capped pipe columns will not be permitted.

3.1.2 Joints

Installation of pipe and fittings shall be made in accordance with the manufacturer's recommendations. Mitering of joints for elbows and notching of straight runs of pipe for tees will not be permitted. Joints shall be made up with fittings of compatible material and made for the specific purpose intended.

3.1.2.1 Threaded

Threaded joints shall have American Standard taper pipe threads conforming to JIS B 0203. Only male pipe threads shall be coated with graphite or with an approved graphite compound, or with an inert filler and oil, or shall have a polytetrafluoroethylene tape applied.

3.1.2.2 Mechanical Couplings

Mechanical couplings may be used in conjunction with grooved pipe for aboveground, ferrous or non-ferrous, domestic hot and cold water systems, in lieu of unions, brazed, soldered, welded, flanged, or threaded joints.

Mechanical couplings are permitted in accessible locations including behind access plates. Flexible grooved joints will not be permitted, except as vibration isolators adjacent to mechanical equipment. Rigid grooved joints shall incorporate an angle bolt pad design which maintains metal-to-metal contact with equal amount of pad offset of housings upon installation to ensure positive rigid clamping of the pipe.

Designs which can only clamp on the bottom of the groove or which utilize gripping teeth or jaws, or which use misaligned housing bolt holes, or which require a torque wrench or torque specifications will not be permitted.

Grooved fittings and couplings, and grooving tools shall be provided from the same manufacturer. Segmentally welded elbows shall not be used. Grooves shall be prepared in accordance with the coupling manufacturer's latest published standards. Grooving shall be performed by qualified grooving operators having demonstrated proper grooving procedures in accordance with the tool manufacturer's recommendations.

The Contracting Officer shall be notified 24 hours in advance of test to demonstrate operator's capability, and the test shall be performed at the work site, if practical, or at a site agreed upon. The operator shall demonstrate the ability to properly adjust the grooving tool, groove the pipe, and to verify the groove dimensions in accordance with the coupling manufacturer's specifications.

3.1.2.3 Unions and Flanges

Unions, flanges and mechanical couplings shall not be concealed in walls, ceilings, or partitions. Unions shall be used on pipe sizes 65 mm and smaller; flanges shall be used on pipe sizes 80 mm and larger.

3.1.2.4 Grooved Mechanical Joints

Grooves shall be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools shall be products of the same manufacturer. Pipe and groove dimensions shall comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field shall be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe shall be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Grooved joints shall not be used in concealed locations.

3.1.2.5 Cast Iron Soil, Waste and Vent Pipe

Bell and spigot compression and hubless gasketed clamp joints for soil, waste and vent piping shall be installed per the manufacturer's recommendations.

3.1.2.6 Copper Tube and Pipe

- a. **Brazed.** Brazed joints shall be made in conformance with JIS Z 3621, JIS H 3300, and JIS H 3300 with flux and are acceptable for all pipe sizes. Copper to copper joints shall include the use of copper-phosphorus or copper-phosphorus-silver brazing metal without flux. Brazing of dissimilar metals (copper to bronze or brass) shall include the use of flux with either a copper-phosphorus, copper-phosphorus-silver or a silver brazing filler metal.
- b. **Soldered.** Soldered joints shall be made with flux and are only acceptable for piping 50 mm and smaller. Soldered joints shall conform to JIS H 3300 and JIS H 3300. Soldered joints shall not be used in compressed air piping between the air compressor and the receiver.
- c. **Copper Tube Extracted Joint.** Mechanically extracted joints shall be made in accordance with MLIT-M.
- d. **Press connection.** Copper press connections shall be made in **strict** accordance with the manufacturer's installation instructions for manufactured rated size. The joints shall be pressed using the tool(s) approved by the manufacturer **of that joint**. Minimum distance between fittings shall be in accordance with the manufacturer's requirements.

3.1.2.7 Plastic Pipe

Acrylonitrile-Butadiene-Styrene (ABS) pipe shall have joints made with solvent cement. PVC and CPVC pipe shall have joints made with solvent cement elastomeric, threading, (threading of Schedule 80 Pipe is allowed only where required for disconnection and inspection; threading of Schedule 40 Pipe is not allowed), or mated flanged.

3.1.3 Dissimilar Pipe Materials

Connections between ferrous and non-ferrous copper water pipe shall be made with dielectric unions or flange waterways. Dielectric waterways shall have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways shall have metal connections on both ends suited to match connecting piping. Dielectric waterways shall be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges shall meet the performance requirements described herein for dielectric waterways. Connecting joints between plastic and metallic pipe shall be made with transition fitting for the specific purpose.

3.1.4 Pipe Sleeves and Flashing

Pipe sleeves shall be furnished and set in their proper and permanent location.

3.1.4.1 Sleeve Requirements

Unless indicated otherwise, provide pipe sleeves meeting the following requirements:

Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, roofs, and floors.

A modular mechanical type sealing assembly may be installed in lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve. The seals shall consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and sleeve using galvanized steel bolts, nuts, and pressure plates. The links shall be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolt shall cause the rubber sealing elements to expand and provide a watertight seal between the pipe and the sleeve. Each seal assembly shall be sized as recommended by the manufacturer to fit the pipe and sleeve involved.

Sleeves shall not be installed in structural members, except where indicated or approved. Rectangular and square openings shall be as detailed. Each sleeve shall extend through its respective floor, or roof, and shall be cut flush with each surface, except for special circumstances. Pipe sleeves passing through floors in wet areas such as mechanical equipment rooms, lavatories, kitchens, and other plumbing fixture areas shall extend a minimum of 100 mm above the finished floor.

Unless otherwise indicated, sleeves shall be of a size to provide a minimum of 6 mm clearance between bare pipe or insulation and inside of sleeve or between insulation and inside of sleeve. Sleeves in bearing walls and concrete slab on grade floors shall be steel pipe or cast-iron pipe. Sleeves in nonbearing walls or ceilings may be steel pipe, cast-iron pipe, galvanized sheet metal with lock-type longitudinal seam, or plastic.

Except as otherwise specified, the annular space between pipe and sleeve, or between jacket over insulation and sleeve, shall be sealed as indicated

with sealants conforming to JIS A 5758 and with a primer, backstop material and surface preparation. The annular space between pipe and sleeve, between bare insulation and sleeve or between jacket over insulation and sleeve shall not be sealed for interior walls which are not designated as fire rated.

3.1.4.2 Pipe Penetrations of Slab on Grade Floors

Where pipes, fixture drains, floor drains, cleanouts or similar items penetrate slab on grade floors, except at penetrations of floors with waterproofing membrane as specified in paragraphs FLASHING REQUIREMENTS and WATERPROOFING, a groove 6 to 13 mm wide by 6 to 10 mm deep shall be formed around the pipe, fitting or drain.

3.1.4.3 Pipe Penetrations

Provide sealants for all pipe penetrations. All pipe penetrations shall be sealed to prevent infiltration of air, insects, and vermin.

3.1.5 Supports

3.1.5.1 Pipe Supports and Structural Bracing, Seismic Requirements

Piping and attached valves shall be supported and braced to resist seismic loads as specified in Section 23 05 48.19 SEISMIC BRACING FOR HVAC. Structural steel required for reinforcement to properly support piping, headers, and equipment, but not shown, shall be provided. Material used for supports shall be as specified in Structural drawings.

3.1.5.2 Pipe Hangers, Inserts, and Supports

Installation of pipe hangers, inserts and supports shall conform to MLIT-M except as modified herein.

- a. Horizontal pipe supports shall be spaced as specified in MLIT-M and a support shall be installed not over 300 mm from the pipe fitting joint at each change in direction of the piping. Pipe supports shall be spaced not over 1.5 m apart at valves. Operating temperatures in determining hanger spacing for PVC or CPVC pipe shall be 49 degrees C for PVC and 82 degrees C for CPVC. Horizontal pipe runs shall include allowances for expansion and contraction.
- b. Vertical pipe shall be supported at each floor, except at slab-on-grade, at intervals of not more than 4.5 m nor more than 2 m from end of risers, and at vent terminations. Vertical pipe risers shall include allowances for expansion and contraction.

3.1.5.3 Structural Attachments

Attachment to building structure concrete and masonry shall be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors shall be applied with a safety factor not less than 5. Supports shall not be attached to metal decking. Supports shall not be attached to the underside of concrete filled floor or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications shall be constructed of ferrous materials only.

3.1.6 Welded Installation

Plumbing pipe weldments shall be as indicated. Changes in direction of piping shall be made with welding fittings only; mitering or notching pipe to form elbows and tees or other similar type construction will not be permitted. Branch connection may be made with either welding tees or forged branch outlet fittings. Branch outlet fittings shall be forged, flared for improvement of flow where attached to the run, and reinforced against external strains. Beveling, alignment, heat treatment, and inspection of weld shall conform to JIS B 8285. Weld defects shall be removed and repairs made to the weld, or the weld joints shall be entirely removed and rewelded. After filler metal has been removed from its original package, it shall be protected or stored so that its characteristics or welding properties are not affected. Electrodes that have been wetted or that have lost any of their coating shall not be used.

3.2 FIXTURES AND FIXTURE TRIMMINGS

Polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Angle stops, straight stops, stops integral with the faucets, or concealed type of lock-shield, and loose-key pattern stops for supplies with threaded, sweat or solvent weld inlets shall be furnished and installed with fixtures. Where connections between copper tubing and faucets are made by rubber compression fittings, a beading tool shall be used to mechanically deform the tubing above the compression fitting. Exposed traps and supply pipes for fixtures and equipment shall be connected to the rough piping systems at the wall, unless otherwise specified under the item. Floor and wall escutcheons shall be as specified. Drain lines and hot water lines of fixtures for handicapped personnel shall be insulated and do not require polished chrome finish. Plumbing fixtures and accessories shall be installed within the space shown.

3.2.1 Fixture Connections

Where space limitations prohibit standard fittings in conjunction with the cast-iron floor flange, special short-radius fittings shall be provided. Connections between earthenware fixtures and flanges on soil pipe shall be made gastight and watertight with a closet-setting compound or neoprene gasket and seal. Use of natural rubber gaskets or putty will not be permitted. Fixtures with outlet flanges shall be set the proper distance from floor or wall to make a first-class joint with the closet-setting compound or gasket and fixture used.

3.2.2 Height of Fixture Rims Above Floor

Lavatories shall be mounted with rim 775 mm above finished floor. Wall-hung drinking fountains and water coolers shall be installed with rim 1020 mm above floor. Wall-hung service sinks shall be mounted with rim 700 mm above the floor. Installation of fixtures for use by the physically handicapped shall be in accordance with ICC A117.1 COMM.

3.2.3 Fixture Supports

Fixture supports for off-the-floor lavatories, urinals, water closets, and other fixtures of similar size, design, and use, shall be of the chair-carrier type. The carrier shall provide the necessary means of mounting the fixture, with a foot or feet to anchor the assembly to the floor slab. Adjustability shall be provided to locate the fixture at the

desired height and in proper relation to the wall. Support plates, in lieu of chair carrier, shall be fastened to the wall structure only where it is not possible to anchor a floor-mounted chair carrier to the floor slab.

3.2.3.1 Support for Concrete-Masonry Wall Construction

Chair carrier shall be anchored to floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate shall be fastened to the concrete wall using through bolts and a back-up plate.

3.2.3.2 Support for Steel Stud Frame Partitions

Chair carrier shall be used. The anchor feet and tubular uprights shall be of the heavy duty design; and feet (bases) shall be steel and welded to a square or rectangular steel tube upright. Wall plates, in lieu of floor-anchored chair carriers, shall be used only if adjoining steel partition studs are suitably reinforced to support a wall plate bolted to these studs.

3.2.3.3 Support for Wood Stud Construction

Where floor is a concrete slab, a floor-anchored chair carrier shall be used. Where entire construction is wood, wood crosspieces shall be installed. Fixture hanger plates, supports, brackets, or mounting lugs shall be fastened with not less than No. 10 wood screws, 6 mm thick minimum steel hanger, or toggle bolts with nut. The wood crosspieces shall extend the full width of the fixture and shall be securely supported.

3.2.3.4 Wall-Mounted Water Closet Gaskets

Where wall-mounted water closets are provided, reinforced wax, treated felt, or neoprene gaskets shall be provided. The type of gasket furnished shall be as recommended by the chair-carrier manufacturer.

3.3 VIBRATION-ABSORBING FEATURES

Mechanical equipment, including compressors and pumps, shall be isolated from the building structure by approved vibration-absorbing features, unless otherwise shown. Each foundation shall include an adequate number of standard isolation units. Each unit shall consist of machine and floor or foundation fastening, together with intermediate isolation material, and shall be a standard product with printed load rating. Piping connected to mechanical equipment shall be provided with flexible connectors. Isolation unit installation shall limit vibration to 90 percent of the lowest equipment rpm.

3.4 ESCUTCHEONS

Escutcheons shall be provided at finished surfaces where bare or insulated piping, exposed to view, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons shall be fastened securely to pipe or pipe covering and shall be satin-finish, corrosion-resisting steel, polished chromium-plated zinc alloy, or polished chromium-plated copper alloy. Escutcheons shall be either one-piece or split-pattern, held in place by internal spring tension or setscrew.

3.5 TESTS, FLUSHING AND DISINFECTION

3.5.1 Plumbing System

The following tests shall be performed on the plumbing system in accordance with MLIT-M, except that the drainage and vent system final test shall include the smoke test. The Contractor has the option to perform a peppermint test in lieu of the smoke test. If a peppermint test is chosen, the Contractor must submit a testing procedure and reasons for choosing this option in lieu of the smoke test to the Contracting Officer for approval.

- a. Drainage and Vent Systems Test. The final test shall include a smoke test.
- b. Building Sewers Tests.
- c. Water Supply Systems Tests.

3.5.2 Defective Work

If inspection or test shows defects, such defective work or material shall be replaced or repaired as necessary and inspection and tests shall be repeated. Repairs to piping shall be made with new materials. Caulking of screwed joints or holes will not be acceptable.

3.5.3 System Flushing

3.5.3.1 During Flushing

Before operational tests or disinfection, potable water piping system shall be flushed with hot potable water. Sufficient water shall be used to produce a water velocity that is capable of entraining and removing debris in all portions of the piping system. This requires simultaneous operation of all fixtures on a common branch or main in order to produce a flushing velocity of approximately 1.2 meters per second through all portions of the piping system. In the event that this is impossible due to size of system, the Contracting Officer (or the designated representative) shall specify the number of fixtures to be operated during flushing. Contractor shall provide adequate personnel to monitor the flushing operation and to ensure that drain lines are unobstructed in order to prevent flooding of the facility. Contractor shall be responsible for any flood damage resulting from flushing of the system. Flushing shall be continued until entrained dirt and other foreign materials have been removed and until discharge water shows no discoloration. All faucets and drinking water fountains, to include any device considered as an end point device by NSF/ANSI 61, Section 9, shall be flushed a minimum of 1 L per 24 hour period, ten times over a 14 day period.

3.5.3.2 After Flushing

System shall be drained at low points. Strainer screens shall be removed, cleaned, and replaced. After flushing and cleaning, systems shall be prepared for testing by immediately filling water piping with clean, fresh potable water. Any stoppage, discoloration, or other damage to the finish, furnishings, or parts of the building due to the Contractor's failure to properly clean the piping system shall be repaired by the Contractor. When the system flushing is complete, the hot-water system

shall be adjusted for uniform circulation. Flushing devices and automatic control systems shall be adjusted for proper operation according to manufacturer's instructions. Flow rates on fixtures must not exceed those stated in PART 2 of this Section. Unless more stringent local requirements exist, lead levels shall not exceed limits established by JIS S 3201. The water supply to the building shall be tested separately to ensure that any lead contamination found during potable water system testing is due to work being performed inside the building.

3.5.4 Operational Test

Upon completion of flushing and prior to disinfection procedures, the Contractor shall subject the plumbing system to operating tests to demonstrate satisfactory installation, connections, adjustments, and functional and operational efficiency. Such operating tests shall cover a period of not less than 8 hours for each system and shall include the following information in a report with conclusion as to the adequacy of the system:

- a. Time, date, and duration of test.
- b. Water pressures at the most remote and the highest fixtures.
- c. Operation of each fixture and fixture trim.
- d. Operation of each valve, hydrant, and faucet.

3.5.5 Disinfection

After all system components are provided and operational tests are complete, the entire domestic hot- and cold-water distribution system shall be disinfected. Before introducing disinfecting chlorination material, entire system shall be flushed with potable water until any entrained dirt and other foreign materials have been removed.

Water chlorination procedure shall be in accordance with Japanese Water Supply Law, Article 20 as modified and supplemented by this specification. The chlorinating material shall be hypochlorites or liquid chlorine. The chlorinating material shall be fed into the water piping system at a constant rate at a concentration of at least 50 parts per million (ppm). Feed a properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or inject liquid chlorine into the system through a solution-feed chlorinator and booster pump until the entire system is completely filled.

Test the chlorine residual level in the water at 6 hour intervals for a continuous period of 24 hours. If at the end of a 6 hour interval, the chlorine residual has dropped to less than 25 ppm, flush the piping including tanks with potable water, and repeat the above chlorination procedures. During the chlorination period, each valve and faucet shall be opened and closed several times.

After the second 24 hour period, verify that no less than 25 ppm chlorine residual remains in the treated system. The 24 hour chlorination procedure must be repeated until no less than 25 ppm chlorine residual remains in the treated system.

Upon the specified verification, the system including tanks shall then be flushed with potable water until the residual chlorine level is

reduced to less than one part per million. During the flushing period, each valve and faucet shall be opened and closed several times.

Take additional samples of water in disinfected containers, for bacterial examination, at locations specified by the Contracting Officer

Test these samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with MLIT-M. The testing method used shall be EPA approved for drinking water systems and shall comply with applicable local and state requirements.

Disinfection shall be repeated until bacterial tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

3.6 POSTED INSTRUCTIONS

Framed instructions under glass or in laminated plastic, including wiring and control diagrams showing the complete layout of the entire system, shall be posted where directed. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system shall be prepared in typed form, framed as specified above for the wiring and control diagrams and posted beside the diagrams. The framed instructions shall be posted before acceptance testing of the systems.

3.7 TABLES

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
1	Cast iron soil pipe and fittings, hub and spigot, JIS G 5526 with compression gaskets. Pipe and fittings shall be marked with the CISPI trademark.	X	X	X	X	X		

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
2	Cast iron soil pipe and fittings hubless, JIS G 5526 and JIS B 2302 and JIS B 2301. Pipe and fittings shall be marked with the CISPI trademark.		X	X	X	X		
3	Cast iron drainage fittings, threaded, JIS B 2302 and JIS B 2301 for use with Item 10	X		X	X			
4	Cast iron screwed fittings (threaded) JIS B 2302 and JIS B 2301 for use with Item 10				X	X		
5	Grooved pipe couplings, ferrous and non-ferrous pipe JIS G 5502 And JIS G 5705	X	X		X	X		
6	Ductile iron grooved joint fittings for ferrous pipe JIS G 5502 and JIS G 5705 for use with Item 5	X	X		X	X		

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
7	Bronze sand casting grooved joint pressure fittings for non-ferrous pipe JIS H 5120, for use with Item 5	X	X		X	X		
8	Wrought copper grooved joint pressure fittings for non-ferrous pipe JIS H 3250, Copper Rand Bores JIS H 3401 JIS H 3401 for use with Item 5	X	X					
9	Malleable-iron threaded fittings, galvanized JIS B 2301 for use with Item 10				X	X		
10	Steel pipe, seamless galvanized, JIS G 3452 Type S, Grade B	X			X	X		
11	Seamless red brass pipe, JIS H 3300				X	X		X
12	Bronzed flanged fittings, JIS B 2240 for use with Items 11 and 14				X	X		X

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
13	Cast copper alloy solder joint pressure fittings, JIS H 3401 for use with Item 14				X	X		X
14	Seamless copper pipe, JIS H 3300						X	X
15	Cast bronze threaded fittings, JIS H 3401				X	X		
16	Copper drainage tube, (DWV), JIS H 3300	X*	X	X*	X	X		X
17	Wrought copper and wrought alloy solder-joint drainage fittings. JIS H 3401	X	X	X	X	X		X
18	Cast copper alloy solder joint drainage fittings, DWV, JIS H 3401	X	X	X	X	X		X
19	Acrylonitrile-Butadiene (ABS) plastic drain, waste, and vent pipe and fittings JIS H 3401	X	X	X	X	X	X	

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
20	Polyvinyl Chloride plastic drain, waste and vent pipe and fittings, JIS K 6739 OR JIS K 6741 JIS K 6742, (Sch 40) JIS K 6742	X	X	X	X	X	X	X
21	Process glass pipe and fittings, JIS R 3644						X	
22	High-silicon content cast iron pipe and fittings (hub and spigot, and mechanical joint), JPF-DP-001		X			X	X	
23	Filament-wound reinforced thermosetting resin (RTRP) pipe, JIS K 7013						X	
SERVICE: A - Underground Building Soil, Waste and Storm Drain B - Aboveground Soil, Waste, Drain In Buildings C - Underground Vent D - Aboveground Vent E - Interior Rainwater Conductors Aboveground F - Corrosive Waste And Vent Above And Belowground G - Condensate Drain Aboveground * - Hard Temper								

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
1	Malleable-iron threaded fittings:				
	a. Galvanized, ASME B16.3 for use with Item 4a	X	X	X	X
	b. Same as "a" but not galvanized for use with Item 4b			X	
2	Grooved pipe couplings, ferrous pipe ASTM A536 and ASTM A47/A47M, non-ferrous pipe, ASTM A536 and ASTM A47/A47M	X	X	X	
3	Ductile iron grooved joint fittings for ferrous pipe ASTM A536 and ASTM A47/A47M, for use with Item 2	X	X	X	
4	Steel pipe:				
	a. Seamless, galvanized, ASTM A53/A53M, Type S, Grade B	X	X	X	X
	b. Seamless, black, ASTM A53/A53M, Type S, Grade B			X	
5	Seamless red brass pipe, ASTM B43	X	X		X
6	Bronze flanged fittings, ASME B16.24 for use with Items 5 and 7	X	X		X
7	Seamless copper pipe, ASTM B42	X	X		X
8	Seamless copper water tube, ASTM B88, ASTM B88M	X**	X**	X**	X***
9	Cast bronze threaded fittings, ASME B16.15 for use with Items 5 and 7	X	X		X

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
10	Wrought copper and bronze solder-joint pressure fittings, ASME B16.22 for use with Items 5, 7 and 8	X	X	X	X
11	Cast copper alloy solder-joint pressure fittings, ASME B16.18 for use with Item 8	X	X	X	X
12	Bronze and sand castings groovedjoint pressure fittings for non-ferrous pipe ASTM B584, for use with Item 2	X	X	X	
13	Polyethylene (PE) plastic pipe, Schedules 40 and 80, based on outside diameter	X			X
14	Polyethylene (PE) plastic pipe (SDR-PR), based on controlled outside diameter, ASTM D3035	X			X
15	Polyethylene (PE) plastic pipe (SIDR-PR), based on controlled inside diameter, ASTM D2239	X			X
16	Butt fusion polyethylene (PE) plastic pipe fittings, ASTM D3261 for use with Items 14, 15, and 16	X			X
17	Socket-type polyethylene fittings for outside diameter-controlled polyethylene pipe, ASTM D2683 for use with Item 15	X			X
18	Polyethylene (PE) plastic tubing, ASTM D2737	X			X
19	Chlorinated polyvinyl chloride (CPVC) plastic hot and cold water distribution system, ASTM D2846/D2846M	X	X		X

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
20	Chlorinated polyvinyl chloride (CPVC) plastic pipe, Schedule 40 and 80, ASTM F441/F441M	X	X		X
21	Chlorinated polyvinyl chloride (CPVC) plastic pipe (SDR-PR) ASTM F442/F442M	X	X		X
22	Threaded chlorinated polyvinyl chloride (chloride CPVC) plastic pipe fittings, Schedule 80, ASTM F437 for use with Items 20, and 21	X	X		X
23	Socket-type chlorinated polyvinyl chloride (CPVC) plastic pipe fittings, Schedule 40, ASTM F438 for use with Items 20, 21, and 22	X	X		X
24	Socket-type chlorinated polyvinyl chloride (CPVC) plastic pipe fittings Schedule 80, ASTM F439 for use with Items 20, 21, and 22	X	X		X
25	Polyvinyl chloride (PVC) plastic pipe, Schedules 40, 80, and 120, ASTM D1785	X			X
26	Polyvinyl chloride (PVC) pressure-rated pipe (SDR Series), ASTM D2241	X			X
27	Polyvinyl chloride (PVC) plastic pipe fittings, Schedule 40, ASTM D2466	X			X
28	Socket-type polyvinyl chloride (PVC) plastic pipe fittings, schedule 80, ASTM D2467 for use with Items 26 and 27	X			X

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
29	Threaded polyvinyl chloride (PVC) plastic pipe fittings, schedule 80, ASTM D2464	X			X
30	Joints for IPS PVC pipe using solvent cement, ASTM D2672	X			X
31	Polypropylene (PP) plastic pipe and fittings; ASTM F2389	X	X		X
32	Steel pipeline flanges, MSS SP-44	X	X		
33	Fittings: brass or bronze; ASME B16.15, and ASME B16.18 ASTM B828	X	X		
34	Carbon steel pipe unions, socket-welding and threaded, MSS SP-83	X	X	X	
35	Malleable-iron threaded pipe unions ASME B16.39	X	X		
36	Nipples, pipe threaded ASTM A733	X	X	X	
37	Crosslinked Polyethylene (PEX) Plastic Pipe ASTM F877	X	X		X
38	Press Fittings	X	X		

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
	SERVICE: A - Cold Water Service Aboveground B - Hot and Cold Water Distribution 180 degrees F Maximum Aboveground C - Compressed Air Lubricated D - Cold Water Service Belowground Indicated types are minimum wall thicknesses. ** - Type L - Hard *** - Type K - Hard temper with brazed joints only or type K-soft temper without joints in or under floors **** - In or under slab floors only brazed joints				

-- End of Section --

SECTION 23 01 30.41

HVAC SYSTEM CLEANING
05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 62.1 (2016) Ventilation for Acceptable Indoor
Air Quality

INSTITUTE OF INSPECTION, CLEANING, AND RESTORATION CERTIFICATION
(IICRC)

ANSI/IICRC S520 (2015) Standard and Reference Guide for
Professional Mold Remediation

NATIONAL AIR DUCT CLEANERS ASSOCIATION (NADCA)

NADCA (2005) Introduction to HVAC System
Cleaning Services

NADCA ACR (2013) Standard for Assessment, Cleaning,
and Restoration of HVAC Systems

NADCA ASCS (2013) Air Systems Cleaning Specialist to
the NADCA Standard ACR

NADCA HVAC Inspection Manual (2021) Procedures for Assessing the
Cleanliness of Commercial HVAC Systems

NORTH AMERICAN INSULATION MANUFACTURERS ASSOCIATION (NAIMA)

NAIMA AH112 (1993) Cleaning Fibrous Glass or Lined
Sheet Metal Ducts

NAIMA AH122 (2006) Cleaning Fibrous Insulated Duct
Systems - Recommended Practices

NAIMA AH127 (1999) Facts About the Impact of Duct
Cleaning on Internal Duct Insulation

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

SMACNA 1966 (2020) HVAC Duct Construction Standards
Metal and Flexible, 4th Edition

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 402-C-01-001	(2001) IAQ Building Education and Assessment Tool (I-BEAM)
EPA 402-F-91-102	(1991) Building Air Quality: A Guide for Building Owners and Facility Managers

UL SOLUTIONS (UL)

UL 181	(2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors
UL 181A	(2013; Reprint Dec 2021) Standard for Safety Closure Systems for Use with Rigid Air Ducts
UL 181B	(2013; Reprint Dec 2021) UL Standard for Safety Closure Systems for Use with Flexible Air Ducts and Air Connectors

1.2 DEFINITIONS

1.2.1 NADCA Standards

Perform the services specified here in accordance with the current published standards of the National Air Duct Cleaners Association (NADCA, NADCA ASCS, NADCA ACR and NADCA HVAC Inspection Manual).

- a. All terms in this specification are defined as stated in the NADCA Standards.
- b. Follow NADCA Standards without modification or deviation.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Record of Existing Conditions

Coordination Plan

NADCA Firm

NADCA Team Assistants

NADCA Air System Cleaning Specialist (ASCS)

NADCA Supervisor Qualifications

Records of Experience in the Field of HVAC System Cleaning

NADCA Work Execution Schedule

SD-03 Product Data

Safety Data Sheets (SDS)

SD-06 Test Reports

Testing Procedures Summary

Gravimetric Analysis

Post-Project Report

1.4 QUALITY CONTROL

1.4.1 NADCA Firm

Submit information certifying that the NADCA firm is a first tier subcontractor who is not affiliated with any other company participating in work on this contract, including furnishing equipment. Further, submit the following, for the firm, to Contracting Officer for approval:

a. Independent NADCA firm:

NADCA Firm: NADCA registration number and expiration date of current certification;

NADCA Supervisor Qualifications: Name and copy of NADCA supervisor certificate and expiration date of current certification.

NADCA Air System Cleaning Specialist (ASCS): Name and documented evidence that the team field leader has satisfactorily performed full-time supervision of HVAC cleaning work in the field for not less than three years immediately preceding this contract's bid opening date.

NADCA Team Assistants: Names and documented evidence that each field technician has satisfactorily assisted a NADCA team field leader in performance of HVAC cleaning work in the field for not less than one year immediately preceding this contract's bid opening date.

Current Certificates: Ensure registrations and certifications are current, and valid for the duration of this contract. Renew Certifications which expire prior to completion of the HVAC cleaning work, in a timely manner so that there is no lapse in registration or certification. NADCA agency or NADCA team personnel without a current registration or current certification are not to perform HVAC cleaning work on this contract.

b. TAB Team Members: NADCA team approved to accomplish work on this contract are full-time employees of the NADCA firm. No other personnel is allowed to do HVAC cleaning work on this contract.

c. Replacement of NADCA Team Members: Replacement of members may occur if each new member complies with the applicable personnel qualifications and each is approved by the Contracting Officer.

1.4.2 Experience

Submit records of experience in the field of HVAC system cleaning. Bids will only be considered from firms which are regularly engaged in HVAC system maintenance with an emphasis on HVAC system cleaning and decontamination.

1.4.3 Equipment, Materials and Labor

Possess and furnish all necessary equipment, materials and labor to adequately perform the specified services and comply with the applicable provisions of NADCA General Specifications for the Cleaning of Commercial HVAC Systems and ASHRAE 62.1.

- a. Assure that all employees have received safety equipment training, medical surveillance programs, individual health protection measures, and manufacturer's product and Safety Data Sheets (SDS) as required for the work by the U.S. Occupational Safety and Health Administration, and as described by this specification. For work performed in countries outside of the U.S.A., comply with applicable national safety codes and standards.
- b. Maintain a copy of all current SDS documentation and safety certifications at the site at all times, as well as comply with all other site documentation requirements of applicable OSHA programs and this specification.
- c. Submit all Safety Data Sheets (SDS) for all chemical products proposed used in the cleaning process, including all VOC ratings.

1.4.4 Licensing

Provide proof of maintaining the proper license(s), if any, as required to do work in Japan. Comply with all Federal, State and local rules, regulations, and licensing requirements.

1.4.5 Health And Safety

1.4.5.1 Safety Standards

Comply with all applicable Federal, State, and local requirements for protecting the safety of the contractors' employees, building occupants, and the environment. In particular, follow all applicable standards of the Occupational Safety and Health Administration (OSHA) when working in accordance with this specification.

1.4.5.2 Occupant Safety

Employ no processes or materials in such a manner that introduce additional hazards into occupied spaces.

1.4.5.3 Disposal of Debris

Dispose of all debris removed from the HVAC System in accordance with applicable Federal, State and local requirements.

1.5 PROJECT/SITE CONDITIONS

1.5.1 Mechanical Drawings

Obtain one copy of the following documents:

- a. Project drawings and specifications
- b. Approved construction revisions pertaining to the HVAC system
- c. Any existing indoor air quality (IAQ) assessments or environmental reports prepared for the facility.

Submit a NADCA Work Execution Schedule to the Contracting Officer within 10 working days of the contract award.

1.5.2 Site Conditions

The HVAC system includes any interior surface of the facility's air distribution system for conditioned spaces and/or occupied zones. This includes the entire heating, air-conditioning and ventilation system from the points where the air enters the system to the points where the air is discharged from the system. The return air grilles, return air ducts (except ceiling plenums and mechanical room) to the air handling unit (AHU), the interior surfaces of the AHU, mixing box, coil compartment, condensate drain pans, humidifiers and dehumidifiers, supply air ducts, fans, fan housing, fan blades, air wash systems, spray eliminators, turning vanes, filters, filter housings, reheat coils, and supply diffusers are all considered part of the HVAC system. The HVAC system may also include other components such as dedicated exhaust and ventilation components and make-up air systems.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Perform the services specified here in accordance with the current published standards of the National Air Duct Cleaners Association (NADCA, NADCA ASCS, NADCA ACR and NADCA HVAC Inspection Manual).

- a. All terms in this specification have their meaning defined as stated in the NADCA Standards.
- b. Follow NADCA Standards with no modifications or deviations being allowed. Remove visible surface contaminants and deposits from within the HVAC system in strict accordance with these specifications.

3.1 PREPARATION

3.1.1 HVAC System Inspections And Site Preparations

3.1.1.1 HVAC System Evaluation

Prior to the commencement of any cleaning work, perform a visual inspection of the HVAC system in the presence of the Contracting Officer to determine appropriate methods, tools, and equipment required to satisfactorily complete this project. Cleanliness inspection should

include air handling units and representative areas of HVAC system components and ductwork. In HVAC systems that include multiple air handling units, a representative sample of units should be inspected. Notify the Contracting Officer 10 days prior to the planned inspection. As part of evaluation, record photographs and videos of each inspection location to document the "as found" condition prior to cleaning.

Document damaged system components found during the inspection and submit to the Contracting Officer, clearly labeled "Record of Existing Conditions."

3.1.1.2 Site Evaluation and Preparations

Conduct without negatively impacting indoor environment through excessive disruption of settled dust, microbial amplification or other debris. In cases where contamination is suspected, and/or in sensitive environments where even small amounts of contaminant may be of concern, implement environmental engineering control measures.

Conduct a site evaluation, and establish a specific, coordination plan which details how each area of the building is protected during the various phases of the project.

3.2 APPLICATION

3.2.1 General HVAC System Cleaning Requirements

3.2.1.1 Containment

Collect debris removed during cleaning and take precautions to ensure that debris is not otherwise dispersed outside the HVAC system during the cleaning process.

3.2.1.2 Particulate Collection

Where the Particulate Collection Equipment (PCE) is exhausting inside the building, use HEPA filtration with 99.97 percent collection efficiency for 0.3-micron size (or greater). When the PCE is exhausting outside the building, undertake mechanical cleaning operations only with PCE, including adequate filtration to contain debris removed from the HVAC system. When the PCE is exhausting outside the building, take precautions to locate the equipment down wind and away from all air intakes and other points of entry into the building.

3.2.1.3 Controlling Odors

Take all reasonable measures to control offensive odors and/or mist vapors during the cleaning process.

3.2.1.4 Component Cleaning

Employ cleaning methods such that all HVAC system components are Visibly Clean as defined in applicable standards. Upon completion, return all components to those settings recorded just prior to cleaning operations.

3.2.1.5 Air-Volume Control Devices

Mark the position of dampers and any air-directional mechanical devices inside the HVAC system prior to cleaning and, upon completion, restore to their marked position.

3.2.1.6 Service Openings

Utilize service openings, as required for proper cleaning, at various points of the HVAC system for physical and mechanical entry, and inspection. Utilize the existing service openings already installed in the HVAC system where possible.

Create other openings where needed, created and resealed in conformance with NADCA Standard 05. Place closures so they do not significantly hinder, restrict, alter the air-flow within the system, or compromise the structural integrity of the system. Properly insulate closures to prevent heat loss/gain or condensation on surfaces within the system. Conform construction techniques used in the creation of openings to requirements of applicable building and fire codes, and applicable NFPA, SMACNA and NADCA Standards. Cutting service openings into flexible duct is not permitted. Disconnect flexible duct at the ends as needed for proper cleaning and inspection.

Reseal rigid fiber glass ductboard duct systems in accordance with NAIMA recommended practices; NAIMA AH112, NAIMA AH122, and NAIMA AH127. Only closure techniques which comply with UL 181, UL 181A, or UL 181B are suitable for fiber glass duct system closures.

Provide access doors for openings that need to be re-opened for future inspection or remediation. Refer to Section 23 30 00 HVAC AIR DISTRIBUTION for access doors. Clearly mark all service openings, capable of being re-opened for future inspection or remediation, and report their location in project report documents.

3.2.1.7 Ceiling Sections (Tile)

Carefully remove and reinstall ceiling sections to gain access to HVAC systems during the cleaning process. Replace any damaged ceiling sections caused by the removal at no cost to the Government.

3.2.1.8 Air Distribution Devices (Registers, Grilles and Diffusers)

Clean all air distribution devices.

3.2.1.9 Air Handling Units, Terminal Units, Blowers and Exhaust Fans

Ensure that supply, return, and exhaust fans and blowers are thoroughly cleaned. Areas for cleaning include blowers, fan housings, plenums (except ceiling supply and return plenums), scrolls, blades, or vanes, shafts, baffles, dampers and drive assemblies. Remove all visible surface contamination deposits in accordance with NADCA Standards.

- a. Clean all air handling unit (AHU) internal surfaces, components and condensate collectors and drains.
- b. Assure that a suitable operative drainage system is in place prior to beginning wash down procedures.
- c. Clean all coils and related components, including evaporator fins.

3.2.1.10 Duct Systems

- a. Create service openings in the system as necessary in order to

accommodate cleaning of otherwise inaccessible areas.

- b. Mechanically clean all duct systems to remove all visible contaminants, such that the systems are capable of passing NADCA Cleaning Verification Testings Standards.
- c. Any exposed edges in internal duct lining within duct, including but not limited to interfaces with externally insulated duct, must be fully protected, sealed and encapsulated to prevent future erosion.

3.2.2 Mechanical Cleaning Methodology

3.2.2.1 Source Removal Cleaning Methods

Clean the HVAC system using Source Removal mechanical cleaning methods designed to extract contaminants from within the HVAC system and safely remove contaminants from the facility. Select Source Removal methods which will render the HVAC System Visibly Clean and capable of passing NADCA cleaning verification methods Standards and other specified standards and tests, in accordance with all general requirements. Use no cleaning method, or combination of methods, which could potentially damage components of the HVAC system or negatively alter the integrity of the system.

Incorporate the use of vacuum collection devices that are operated continuously during cleaning for all methods used. Connect a vacuum device to the downstream end of the section being cleaned through a predetermined opening. Use a vacuum collection device of sufficient power to render all areas being cleaned under negative pressure, such that containment of debris and the protection of the indoor environment is assured.

Equip all vacuum devices exhausting air inside the building, including hand-held vacuums and wet-vacuums, with HEPA filters (minimum efficiency).

Equip all vacuum devices exhausting air outside the facility with Particulate Collection including adequate filtration to contain Debris removed from the HVAC system, in a manner that does not allow contaminants to re-enter the facility. Release of debris outdoors which violates any outdoor environmental standards, codes or regulations is not allowed.

All methods require mechanical agitation devices to dislodge debris adhered to interior HVAC system surfaces, such that debris may be safely conveyed to vacuum collection devices. Acceptable methods include those which will not potentially damage the integrity of the ductwork, nor damage porous surface materials such as liners inside the ductwork or system components.

3.2.2.2 Methods of Cleaning Fibrous Glass Insulated Components

Thoroughly clean glass thermal insulation elements present in any equipment or ductwork with HEPA vacuuming equipment. Clean while the HVAC system is under constant negative pressure, and not permitted to get wet in accordance with applicable NADCA and NAIMA standards and recommendations.

Do not use cleaning methods that cause damage to fibrous glass components or renders the system capable of passing Cleaning Verification Tests NADCA Standards.

Provide surface treatment for insulation for sections of internally-lined duct. Select and apply encapsulants, coatings, and insulation repair products to completely restore surface integrity of fibrous glass surfaces in accordance with applicable standards and manufacturer's installation instructions.

3.2.2.3 Damaged Fibrous Glass Material

If there is any evidence of damage, deterioration, delamination, friable material, mold or fungus growth, or moisture such that fibrous glass materials cannot be restored by cleaning or resurfacing with an acceptable insulation repair coating, identify them to the Contracting Officer for replacement.

When requested or specified, remediate exposed damaged insulation in air handlers and/or ductwork requiring replacement.

If insulation is damaged as a result of this work under this specification, notify Contracting Officer and initiate a meeting with same to determine options for repair or replacement of insulation.

3.2.2.4 Replacement Material

If replacement of fiber glass materials is required, conform all materials to applicable industry codes and standards, including those of UL and SMACNA 1966.

Replacement of damaged insulation is **not** covered by this specification. Refer to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.2.2.5 Cleaning of Coils

Use any cleaning method which renders the coil visibly clean and capable of passing NADCA Coil Cleaning Verification Standards. Coil drain pans are subject to Non-Porous Surfaces Cleaning Verification. Maintain operability of the drain for the condensate at all times. Do not damage, displace, inhibit heat transfer, or cause erosion of the coil surface or fins, and conform to coil manufacturer recommendations when available. Thoroughly rinse coils with clean water to remove any latent residues.

3.2.2.6 Antimicrobial Agents and Coatings

Perform application of antimicrobial agents used to control the growth of fungal or bacteriological contaminants after the removal of surface deposits and debris. Perform mold remediation in accordance with ANSI/IICRC S520.

Use only antimicrobial agents registered by the U.S. Environmental Protection Agency (EPA 402-F-91-102) (EPA 402-C-01-001) specifically for use within HVAC system.

Apply antimicrobial agents in strict accordance with manufacturer's instructions.

Use only antimicrobial coating products, for both porous and non-porous surfaces, which are EPA registered, water soluble solutions with supporting efficacy data and SDS records.

Apply antimicrobial coatings according to manufacturer's instructions. Spray coatings directly onto interior ductwork surfaces, rather than "fog" downstream onto surfaces. Achieve a continuous film on the surface treated by the coating application, and apply in strict accordance with manufacturer's minimum millage surface application rate standards for effectiveness.

3.3 FIELD QUALITY CONTROL

3.3.1 Cleanliness Verification

3.3.1.1 General

Verification of HVAC System cleanliness is determined after mechanical cleaning and before the application of any treatment or introduction of any treatment-related substance to the HVAC system, including antimicrobial agents and coatings.

3.3.1.2 Visual Inspection

Visually inspect the HVAC system to ensure that no visible contaminants are present.

If no contaminants are evident through visual inspection, consider the HVAC system clean; however, further verification of the system cleanliness through gravimetric or wipe testing analysis testing may be requested at the discretion of the Contracting Officer.

If visible contaminants are evident through visual inspection, re-clean those portions of the system where contaminants are visible, and subject to re-inspection for cleanliness.

As part of inspection, record photographs and videos of inspection locations to document post-cleaning condition.

3.3.1.3 Gravimetric Analysis

At the expense of the **Contractor**, test sections of the HVAC system for cleanliness using the NADCA Vacuum Test (gravimetric analysis) as specified in applicable NADCA Standards. Ensure levels of debris collected are equal to or less than acceptable levels defined in applicable NADCA Standards.

If gravimetric analysis determines that levels of debris are equal to or lower than those levels specified, the system is considered clean and to have passed cleanliness verification.

If gravimetric analysis determines that levels of debris exceed those specified in applicable NADCA standards, the system will not be considered clean, and re-cleaning of those sections of the system which failed cleanliness verification will be required at the expense of the HVAC system cleaning contractor.

Perform cleanliness verification immediately after mechanical cleaning and before the HVAC system is restored to normal operation.

3.3.1.4 Verification of Coil Cleaning

Cleaning is to restore the coil pressure drop to within 10 percent of the

pressure drop measured when the coil was first installed. If the original pressure drop is not known, the coil will be considered clean only if the coil is free of foreign matter and chemical residue, based on a thorough visual inspection (see NADCA HVAC Inspection Manual Standards).

3.3.2 Post-Project Report

At the conclusion of the project, provide a Testing Procedures Summary and Post-Project Report indicating the following:

- a. Success of the cleaning project, as verified through visual inspection and gravimetric analysis.
- b. Areas of the system found to be damaged and/or in need of repair.

-- End of Section --

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SECTION 23 05 15

COMMON PIPING FOR HVAC
02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 0203	(2014) Concrete Terminology
JIS A 1108	(2018) Method of Test for Compressive Strength of Concrete
JIS A 1110	(2006) Methods of Test for Density and Water Absorption of Coarse Aggregates
JIS A 5758	(2022) Sealants for Sealing and Glazing in Buildings
JIS A 9504	(2021) Man Made Mineral Fibre Thermal Insulation Materials
JIS B 1112	(2019) Cross-Recessed Head Wood Screws
JIS B 1181	(2014) Hexagon Nuts and Hexagon Thin Nuts
JIS B 1220	(2015) Set of Anchor Bolt for Structures
JIS B 2031	(2015) Gray Cast Iron Valves (Amendment 1)
JIS B 2032	(2013) Wafer Type Rubber-Seated Butterfly Valves
JIS B 2061	(2017) Faucets, Ball Taps and Flush Valves
JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2239	(2013) Cast Iron Pipe Flanges
JIS B 2240	(2006) Copper Alloy Pipe Flanges
JIS B 2301	(2013) Screwed Type Malleable Cast Iron Pipe Fittings
JIS B 2311	(2015) Steel Butt-Welding Pipe Fittings for Ordinary Use
JIS B 2312	(2015) Steel Butt-Welding Pipe Fittings
JIS B 7410	(1997) Liquid-In-Glass Thermometers for Testing of Petroleum Product

JIS B 7505-1	(2017) Aneroid Pressure Gauges-Part 1: Bourdon Tube Pressure Gauges
JIS B 8267	(2015) Construction of Pressure Vessel
JIS B 8285	(2010) Welding Procedure Qualification Test for Pressure Vessels
JIS F 0602	(1995) Shipbuilding-Non-Asbestos Gaskets to Cargo Piping System-Application Standard
JIS G 3138	(2021) Rolled Steel Bars for Building Structure
JIS G 3454	(2019) Carbon Steel Pipes for Pressure Service (Amendment 1)
JIS G 3455	(2016) Carbon Steel Pipes for High Pressure Service
JIS G 3456	(2019) Carbon Steel Pipes for High Temperature Service
JIS G 3459	(2017) Stainless Steel Pipes (Amendment 1)
JIS G 4051	(2018) Carbon Steels for Machine Structural Use (Amendment 1)
JIS G 4053	(2018) Low-Alloyed Steels for Machine Structural Use (Amendment 1)
JIS G 4107	(2015) Alloy Steel Bolting Materials for High Temperature Service (Amendment 1)
JIS G 4305	(2015) Cold-Rolled Stainless Steel Plate, Sheet and Strip (Amendment 1)
JIS G 5501	(2020) Grey Iron Castings
JIS H 3100	(2018) Copper and Copper Alloy Sheets, Plates and Strips
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS K 7311	(1995) Testing Methods for Thermoplastic Polyurethane Elastomers
JIS HB 40-1	(2019) Ferrous Materials & Metallurgy I
JIS HB 40-2	(2019) Ferrous Materials & Metallurgy II

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 834-1	(1999) Fire-Resistance Tests-Elements of
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Building Construction-Part 1: General
Requirements

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M

(2019) Public Building Construction
Standard Specification

1.2 GENERAL REQUIREMENTS

Section 23 30 00 AIR SUPPLY, DISTRIBUTION, VENTILATION, AND EXHAUST SYSTEMS applies to work specified in this section

Submit Material, Equipment, and Fixture Lists for pipes, valves and specialties including manufacturer's style or catalog numbers, specification and drawing reference numbers, and warranty information.

Submit Record Drawings for pipes, valves and accessories providing current factual information including deviations and amendments to the drawings, and concealed and visible changes in the work.

Submit Coordination Drawings for pipes, valves and specialties showing coordination of work between different trades and with the structural and architectural elements of work. Detail all drawings sufficiently to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate on drawings where conflicts or clearance problems exist between various trades.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists G

SD-02 Shop Drawings

Record Drawings; G

Coordination Drawings G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals

1.4 QUALITY ASSURANCE

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Provide standard products in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use includes applications of equipment and materials under similar circumstances and of similar size. Ensure the

product has been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.4.2 Alternative Qualifications

Products having less than a two-year field service record are acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.4.3 Service Support

Ensure the equipment items are supported by service organizations located in Japan. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. Select service organizations that are reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.4.4 Manufacturer's Nameplate

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable.

1.4.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer.

1.4.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions are considered mandatory, the word "should" is interpreted as "shall." Reference to the "code official" is interpreted to mean the "Contracting Officer."

1.4.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, are applied as appropriate by the Contracting Officer and as authorized by his administrative cognizance and the FAR.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Provide instructors thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work.

Give instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished is as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

2.1 PIPE AND FITTINGS

Submit equipment and performance data for pipe and fittings consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.1.1 Type BCS, Black Carbon Steel

Ensure pipe DN6 through DN300 is Schedule 40 black carbon steel, conforming to JIS G 3454.

Ensure pipe DN6 through DN250 is Schedule 40 seamless or electric-resistance welded black carbon steel, conforming to JIS G 3454.

Ensure pipe DN300 through DN610 is 9.52 millimeter wall seamless black carbon steel, conforming to JIS G 3454.

Ensure fittings DN50 and under are 1034 kilopascal working steam pressure (wsp) banded black malleable iron screwed, conforming to JIS G 3454 and JIS B 2301.

Ensure unions DN50 and under are 1724 kilopascal female, screwed, black

malleable iron with brass-to-iron seat, and ground joint, conforming to JIS B 2301.

Ensure fittings DN65 and over are Steel butt weld, conforming to JIS B 2312 to match pipe wall thickness.

Ensure flanges DN65 and over are 1034 kilopascal forged-steel conforming to JIS B 2220, welding neck to match pipe wall thickness.

2.1.2 Type BCS-125, 862 kilopascal Service

Ensure pipe DN6 through DN40 is Schedule 40 steam, Schedule 80 condensate, furnace butt weld, black carbon steel, conforming to JIS G 3456.

Ensure pipe DN50 through DN250 is Schedule 40 steam, Schedule 80 condensate, seamless or electric-resistance welded black carbon steel, conforming to Grade B (electric-resistance welded) or Type S (seamless) and JIS G 3456.

Ensure pipe DN300 through DN610 is 9.52 millimeter wall, seamless or electric-resistance welded black carbon steel, conforming to JIS G 3454.

Ensure fittings DN50 and under are 862 kilopascal wsp, cast iron, screwed end, conforming to JIS G 5501 and JIS G 3454 and JIS B 2301.

Ensure fittings DN50 and under are 1034 kilopascal wsp banded black malleable iron screwed, conforming to JIS G 3455 and JIS B 2301.

Ensure fittings DN25 through DN50 are 14 or 21 megapascal water, oil, or gas (wog) to match pipe wall, forged carbon steel socket weld, conforming to JIS G 3455 and JIS B 2301.

Ensure fittings DN50 and under are 862 kilopascal wsp, cast iron, screwed end, conforming to JIS G 3454 and JIS B 2301.

Ensure fittings DN65 and over are wall thickness to match pipe, long radius butt weld, black carbon steel, conforming to JIS G 4051 and JIS B 2311.

Ensure couplings DN50 and under are commercial standard weight for Schedule 40 pipe and commercial extra heavy weight for Schedule 80 pipe, black carbon steel where threaded, and 14 or 21 megapascal wog forged carbon steel, conforming to JIS G 3455 and JIS B 2301, where welded.

Ensure flanges DN65 and over are 1035 kilopascal, forged carbon-steel welding neck, with raised face or flat face and concentric serrated finish, conforming to JIS G 3455 and JIS B 2301.

Conform grooved pipe couplings and fittings in accordance with paragraph GROOVED PIPE COUPLINGS AND FITTINGS.

2.1.3 Type CPR, Copper

2.1.3.1 Type CPR-A, Copper Above Ground

Ensure tubing DN50 and under is seamless copper tubing, conforming to JIS H 3300, Type L (hard-drawn for all horizontal and all exposed vertical lines, annealed for concealed vertical lines).

Ensure fittings DN50 and under are 1034 kilopascal wsp wrought-copper solder joint fittings conforming to JIS H 3401.

Ensure unions DN50 and under are 1034 kilopascal wsp wrought-copper solder joint, conforming to JIS H 3401.

2.2 PIPING SPECIALTIES

Submit equipment and performance data for piping specialties consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.2.1 Air Separator

Air separated from converter discharge water is ejected by a reduced-velocity device vented to the compression tank.

Provide a commercially constructed separator, designed and certified to separate not less than 80 percent of entrained air on the first passage of water and not less than 80 percent of residual on each successive pass. Provide shop drawings detailing all piping connections proposed for this work.

Ensure the air separator is carbon steel, designed, fabricated, tested, and stamped in conformance with JIS B 8267 for service pressures not less than 862 kilopascal.

2.2.2 Air Vents

Provide manual air vents using 10 millimeter globe valves.

Provide automatic air vents on pumps, mains, and where indicated using ball-float construction. Ensure the vent inlet is not less than DN20 and the outlet not less than 8 millimeter. Orifice size is 3 millimeter. Provide corrosion-resistant steel trim conforming to JIS G 4305. Fit vent with try-cock. Ensure vent discharges air at any pressure up to 1034 kilopascal. Ensure outlet is copper tube routed.

2.2.3 Dielectric Connections

Electrically insulate dissimilar pipe metals from each other by couplings, unions, or flanges commercially manufactured for that purpose and rated for the service pressure and temperature.

2.2.4 Expansion Vibration Isolation Joints

Construct single or multiple arch-flanged expansion vibration isolation joints of steel-ring reinforced chloroprene-impregnated cloth materials. Design joint to absorb the movement of the pipe sections in which installed with no detrimental effect on the pipe or connected equipment. Back flanges with ferrous-metal backing rings. Provide control rod assemblies to restrict joint movement. Coat all nonmetallic exterior surfaces of the joint with chlorosulphinated polyethylene. Provide grommets in limit bolt hole to absorb noise transmitted through the bolts.

Ensure joints are suitable for continuous-duty working temperature of at least 121 degrees C.

Fill arches with soft chloroprene.

Ensure joint, single-arch, movement limitations and size-related, pressure characteristics conform to MLIT-M.

2.2.5 Flexible Pipe

Construct flexible pipe vibration and pipe-noise eliminators of wire-reinforced, rubber-impregnated cloth and cord materials and be flanged. Back the flanges with ferrous-metal backing rings. Ensure service pressure-rating is a minimum 1.5 times actual service, with surge pressure at 82 degrees C.

Construct flexible pipe vibration and pipe noise eliminators of wire-reinforced chloroprene-impregnated cloth and cord materials. Ensure the pipe is flanged. Provide all flanges backed with ferrous-metal backing rings. Coat nonmetallic exterior surfaces of the flexible pipe with an acid- and oxidation-resistant chlorosulphinated polyethylene. Rate the flexible pipe for continuous duty at 896 kilopascal and 121 degrees C.

Ensure unit pipe lengths, face-to-face, are not less than the following:

<u>INSIDE DIAMETER (DN)</u>	<u>UNIT PIPE LENGTH</u>
To 65, inclusive	305 millimeter
80 to 100, inclusive	450 millimeter

2.2.6 Flexible Metallic Pipe

Ensure flexible pipe is the bellows-type with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of MLIT-M.

Minimum working pressure rating is 345 kilopascal at 149 degrees C.

Ensure minimum burst pressure is four times working pressure at 149 degrees C. Bellows material is JIS G 3459 corrosion-resistant steel. Ensure braid is JIS G 4053 corrosion-resistant steel wire.

Ensure welded end connections are Schedule 80 carbon steel pipe, conforming to JIS G 3456.

Provide threaded end connections; hex-collared Schedule 40, JIS G 3459 corrosion-resistant steel, conforming to JIS G 3459.

Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

2.2.7 Metallic Expansion Joints

Provide metallic-bellows expansion joints conforming to MLIT-M.

2.2.8 Pressure Gages

Ensure pressure gages conform to JIS B 7505-1 and to requirements specified herein. Pressure-gage size is 90 millimeter nominal diameter. Ensure case is corrosion-resistant steel, conforming to any of JIS G 4053 series of JIS G 3138. Equip gages with adjustable red marking pointer and damper-screw adjustment in inlet connection. Align service-pressure reading at midpoint of gage range.

Fit steam gages with black steel syphons and steam service pressure-rated gage cocks or valves.

2.2.9 Sight-Flow Indicators

Construct sight-flow indicators for pressure service on 80 millimeter and smaller of bronze with specially treated single- or double-glass sight windows and have a bronze, nylon, or tetrafluoroethylene rotating flow indicator mounted on SUS 304, or SUS 316 corrosion-resistant steel shaft. Body may have screwed or flanged end. Provide pressure- and temperature-rated assembly for the applied service. Flapper flow-type indicators are not acceptable.

2.2.10 Sleeve Couplings

Sleeve couplings for plain-end pipe consist of one steel middle ring, two steel followers, two chloroprene or Buna-N elastomer gaskets, and the necessary steel bolts and nuts.

2.2.11 Thermometers

Ensure thermometers conform to JIS B 7410, except for being filled with a red organic liquid. Provide an industrial pattern armored glass thermometer, (well-threaded and seal-welded). Ensure thermometers installed 1800 millimeter or higher above the floor have an adjustable angle body. Ensure scale is not less than 180 millimeter long and the case face is manufactured from manufacturer's standard polished aluminum JIS G 4053 series polished corrosion-resistant steel. Provide thermometers with nonferrous separable wells. Provide lagging extension to accommodate insulation thickness.

2.2.12 Pump Suction Strainers

Provide a cast iron strainer body, rated for not less than 172 kilopascal at 38 degrees C, with flanges conforming to JIS B 2239. Strainer construction is such that there is a machined surface joint between body and basket that is normal to the centerline of the basket.

Ensure minimum ratio of open area of each basket to pipe area is 3 to 1. Provide a basket with JIS G 4053 series corrosion-resistant steel wire mesh with perforated backing.

Ensure mesh is capable of retaining all particles larger than 1,000 micrometer, with a pressure drop across the strainer body of not more than 5 kilopascal when the basket is two-thirds dirty at maximum system flow rate. Provide reducing fittings from strainer-flange size to pipe size.

Provide a differential-pressure gage fitted with a two-way brass cock across the strainer.

Provide manual air vent cocks in cap of each strainer.

2.2.13 Line Strainers, Water Service

Install Y-type strainers with removable basket. Ensure strainers in sizes DN50 and smaller have screwed ends; in sizes DN65 and larger, strainers have flanged ends. Ensure body working-pressure rating exceeds maximum service pressure of installed system by at least 50 percent. Ensure body has cast-in arrows to indicate direction of flow. Ensure all strainer bodies fitted with screwed screen retainers have straight threads and gasketed with nonferrous metal. For strainer bodies DN65 and larger, fitted with bolted-on screen retainers, provide offset blowdown holes. Fit all strainers larger than DN65 with manufacturer's standard ball-type blowdown valve. Ensure body material is cast bronze conforming to MLIT-M. Where system material is nonferrous, use nonferrous metal for the metal strainer body material.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 1.14 millimeter. Ensure strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is SUS 304, or SUS 316 corrosion-resistant steel.

2.3 VALVES

Submit equipment and performance data for valves consisting of corrosion resistance and life expectancy. Submit design analysis and calculations consisting of rates of flow, head losses, inlet and outlet design, and pressure calculations. Also include in data, pipe dimensions, as well as temperature ratings, vibration and thrust limitations, minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.3.1 Ball and Butterfly Valves

Ensure ball valves conform to JIS B 2061. For valve bodies in sizes DN50 and smaller, use screwed-end connection-type constructed of copper alloy. For valve bodies in sizes DN50 DN65 and larger, use flanged-end connection type, constructed of material. Balls and stems of valves DN50 and smaller are manufacturer's standard with hard chrome plating finish. Balls and stems of valves DN65 and larger are manufacturer's standard corrosion-resistant steel alloy with hard chrome plating. Balls of valves DN150 and larger may be Brinell hard chrome plating. Ensure valves are suitable for flow from either direction and seal equally tight in either direction. Valves with ball seals held in place by spring washers are not acceptable. Ensure all valves have adjustable packing glands. Seats and seals are fabricated from tetrafluoroethylene.

Ensure butterfly valves conform to JIS B 2032 and are the wafer type for mounting between specified flanges. Ensure valves are rated for 1034 kilopascal shutoff and nonshock working pressure. Select bodies of cast ferrous metal conforming to JIS B 2239 for body wall thickness. Seats and seals are fabricated from resilient elastomer designed for field removal and replacement.

2.3.2 Drain, Vent, and Gage Cocks

Provide T-head drain, vent, and gage cocks, ground key type, with washer and screw, constructed of polished JIS B 2240 and rated 862 kilopascal

wsp. Ensure end connections are rated for specified service pressure.

Ensure pump vent cocks, and where spray control is required, constructed of manufacturer's standard polished brass. Ensure cocks are 15 millimeter male, end threaded, and rated at not less than 862 kilopascal at 107 degrees C.

2.3.3 Standard Check Valves (SCV)

Ensure standard check valves in sizes DN50 and smaller are 862 kilopascal swing check valves except as otherwise specified. Provide lift checks where indicated. Ensure swing-check pins are nonferrous and suitably hard for the service. Select composition type discs. Ensure the swing-check angle of closure is manufacturer's standard unless a specific angle is needed.

Use cast iron, bronze trim, swing type check valves in sizes DN65 and larger. Ensure valve bodies are cast iron, conforming to JIS B 2031 and valve ends are flanged in conformance with JIS B 2239. Swing-check pin is approved equal corrosion-resistant steel. Angle of closure is manufacturer's standard unless a specific angle is needed. Ensure valves have bolted and gasketed covers.

2.3.4 Nonslam Check Valves (NSV)

Provide check valves at pump discharges in sizes DN50 and larger with nonslam or silent-check operation conforming to JIS B 2031. Select a valve disc or plate that closes before line flow can reverse to eliminate slam and water-hammer due to check-valve closure. Ensure valve is Class 125 rated for 1379 kilopascal maximum, nonshock pressure at 66 degrees C in sizes to DN300. Use valves that are wafer type to fit between flanges conforming to JIS B 2239. Valve body may be cast iron, or equivalent strength ductile iron. Select disks using manufacturer's standard bronze, aluminum bronze, or corrosion-resistant steel. Ensure pins, springs, and miscellaneous trim are manufacturer's standard corrosion-resistant steel.

2.4 MISCELLANEOUS MATERIALS

Submit equipment and performance data for miscellaneous materials consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

2.4.1 Bituminous Coating

Ensure the bituminous coating is a solvent cutback, heavy-bodied material to produce not less than a 0.30 millimeter dry-film thickness in one coat, and is recommended by the manufacturer to be compatible with factory-applied coating and rubber joints.

For previously coal-tar coated and uncoated ferrous surfaces underground, use bituminous coating solvent cutback coal-tar type.

2.4.2 Bolting

Ensure flange and general purpose bolting is hex-head and conforms to JIS G 4107, above (bolts, for flanged joints in piping systems where one or both flanges are cast iron). Heavy hex-nuts conform to JIS B 1181. Square-head bolts and nuts are not acceptable. Ensure threads are coarse-thread series.

2.4.3 Elastomer Caulk

Use two-component polysulfide- or polyurethane-base elastomer caulking material, conforming to JIS K 7311.

2.4.4 Escutcheons

Manufacture escutcheons from nonferrous metals and chrome-plated except when JIS G 4053 series corrosion-resistant steel is provided. Ensure metals and finish conforms to Japanese standard.

Use one-piece escutcheons where mounted on chrome-plated pipe or tubing, and one-piece of split-pattern type elsewhere. Ensure all escutcheons have provisions consisting of internal spring-tension devices or setscrews for maintaining a fixed position against a surface.

2.4.5 Flashing

Ensure sheetlead conforms to Japanese standard

Ensure sheet copper conforms to JIS H 3100 and be not less than 4.88 kilogram per square meter weight.

2.4.6 Flange Gaskets

Provide compressed non-asbestos sheets, conforming to JIS F 0602, coated on both sides with graphite or similar lubricant, with nitrile composition, binder rated to 399 degrees C.

2.4.7 Grout

Provide shrink-resistant grout as a premixed and packaged metallic-aggregate, mortar-grouting compound conforming to JIS A 0203.

Ensure shrink-resistant grout is a combination of pre-measured and packaged epoxy polyamide or amine resins and selected aggregate mortar grouting compound conforming to the following requirements:

Tensile strength	13.100 Megapascal, minimum
Compressive strength	96.527 Megapascal, minimum JIS A 1108
Shrinkage, linear	0.003 mm per millimeter, maximum
Water absorption	0.1 percent, maximum JIS A 1110
Bond strength to	6.895 Megapascal, minimum steel in shear minimum

2.4.8 Pipe Thread Compounds

Use polytetrafluoroethylene tape not less than 0.05 to 0.08 millimeter thick in potable and process water and in chemical systems for pipe sizes to and including DN25. Use polytetrafluoroethylene dispersions and other suitable compounds for all other applications upon approval by the Contracting Officer; however, do not use lead-containing compounds in

potable water systems.

2.5 SUPPORTING ELEMENTS

Submit equipment and performance data for the supporting elements consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

Provide all necessary piping systems and equipment supporting elements, including but not limited to: building structure attachments; supplementary steel; hanger rods, stanchions, and fixtures; vertical pipe attachments; horizontal pipe attachments; anchors; guides; and spring-cushion, variable, or constant supports. Ensure supporting elements are suitable for stresses imposed by systems pressures and temperatures and natural and other external forces normal to this facility without damage to supporting element system or to work being supported.

Ensure supporting elements conform to requirements of MLIT-M, except as noted.

Ensure attachments welded to pipe are made of materials identical to that of pipe or materials accepted as permissible raw materials by referenced code or standard specification.

Ensure supporting elements exposed to weather are hot-dip galvanized or stainless steel. Select materials of such a nature that their apparent and latent-strength characteristics are not reduced due to galvanizing process. Electroplate supporting elements in contact with copper tubing with copper.

Ensure masonry anchor group-, type-, and style-combination designations are in accordance with JIS A 5758 and JIS B 1112. Provide support elements, except for supplementary steel, that are cataloged, load rated, commercially manufactured products.

2.5.1 Building Structure Attachments

2.5.1.1 Anchor Devices, Concrete and Masonry

Ensure anchor devices conform to JIS A 5758 and JIS B 1220.

For cast-in, floor mounted, equipment anchor devices, provide adjustable positions.

2.5.1.2 Beam Clamps

Ensure beam clamps are center-loading conforming to MLIT-M.

When it is not possible to use center-loading beam clamps, eccentric-loading beam clamps, conforming to MLIT-M may be used for piping sizes DN50 and less and for piping sizes DN50 through DN250 provided two counterbalancing clamps are used per point of pipe support. Where more than one rod is used per point of pipe support, determine rod diameter in accordance with referenced standards.

2.5.1.3 C-Clamps

Do not use C-clamps.

2.5.1.4 Inserts, Concrete

Use concrete conforming to MLIT-M inserts. When applied to piping in sizes DN50 and larger and where otherwise required by imposed loads, insert and wire a 305 millimeter length of 13 millimeter reinforcing rod through wing slots. Submit proprietary-type continuous inserts for approval.

2.5.2 Horizontal Pipe Attachments

2.5.2.1 Single Pipes

Support piping in sizes to and including DN50 by conforming to MLIT-M solid malleable iron pipe rings, except that, use split-band-type rings in sizes up to DN25.

Support piping in sizes through DN200 inclusive by conforming to MLIT-M.

Use MLIT-M assemblies on vapor-sealed insulated piping and have an inside diameter larger than pipe being supported to provide adequate clearance during pipe movement.

Where thermal movement of a point in a piping system DN100 and larger would cause a hanger rod to deflect more than 4 degrees from the vertical or where a horizontal point movement exceeds 13 millimeter, use conforming to MLIT-M.

Support piping in sizes larger than DN200 with conforming to MLIT-M.

Use conforming to MLIT-M shields on all insulated piping. Ensure area of the supporting surface is such that compression deformation of insulated surfaces does not occur. Roll away longitudinal and transverse shield edges from the insulation.

Provide insulated piping without vapor barrier on roll supports with conforming to MLIT-M saddles.

Provide spring supports as indicated.

2.5.2.2 Parallel Pipes

Use trapeze hangers fabricated from structural steel shapes, with U-bolts, in congested areas and where multiple pipe runs occur. Ensure structural steel shapes be of commercially available, proprietary design, rolled steel.

2.5.3 Vertical Pipe Attachments

Ensure vertical pipe attachments are conforming to MLIT-M.

Include complete fabrication and attachment details of any spring supports in shop drawings.

2.5.4 Hanger Rods and Fixtures

Use only circular cross section rod hangers to connect building structure attachments to pipe support devices. Use pipe, straps, or bars of equivalent strength for hangers only where approved by the Contracting Officer.

Provide turnbuckles, swing eyes, and clevises as required by support

system to accommodate temperature change, pipe accessibility, and adjustment for load and pitch. Rod couplings are not acceptable.

2.5.5 Supplementary Steel

Where it is necessary to frame structural members between existing members or where structural members are used in lieu of commercially rated supports, design and fabricate such supplementary steel in accordance with MLIT-M.

PART 3 EXECUTION

3.1 PIPE INSTALLATION

Submit certificates for pipes, valves and specialties showing conformance with test requirements as contained in the reference standards contained in this section. Provide certificates verifying Surface Resistance, Shear and Tensile Strengths, Temperature Ratings, Bending Tests, Flattening Tests and Transverse Guided Weld Bend Tests.

Fabricate and install piping systems in accordance with JIS HB 40-1 and JIS HB 40-2.

Ensure connections between steel piping and copper piping are electrically isolated from each other with dielectric couplings (or unions) rated for the service.

Make final connections to equipment with flanges provided every 30480 millimeter of straight run. Provide unions in the line downstream of screwed- and welded-end valves.

Ream all pipe ends before joint connections are made.

Make screwed joints with specified joint compound with not more than three threads showing after joint is made up.

Apply joint compounds to the male thread only and exercise care to prevent compound from reaching the unthreaded interior of the pipe.

Provide screwed unions, welded unions, or bolted flanges wherever required to permit convenient removal of equipment, valves, and piping accessories from the piping system for maintenance.

Securely support piping systems with due allowance for thrust forces, thermal expansion and contraction. Do not subject the system to mechanical, chemical, vibrational or other damage as specified in MLIT-M.

Ensure field welded joints conform to the requirements of the JIS HB 40-1 and JIS HB 40-2 and JIS B 8285.

Accomplish preheat and postheat treatment of welds in accordance with JIS B 8285.

Take all necessary precautions during installation of flexible pipe and hose including flushing and purging with water, steam, and compressed air to preclude bellows failure due to pipe line debris lodged in bellows. Ensure installation conforms to manufacturer's instructions.

3.2 VALVES

Provide valves in piping mains and all branches and at equipment where indicated and as specified.

Provide valves to permit isolation of branch piping and each equipment item from the balance of the system.

Provide riser and downcomer drains above piping shutoff valves in piping DN65 and larger. Tap and fit shutoff valve body with a DN15 plugged globe valve.

Provide valves unavoidably located in furred or other normally inaccessible places with access panels adequately sized for the location and located so that concealed items may be serviced, maintained, or replaced.

3.3 SUPPORTING ELEMENTS INSTALLATION

Provide supporting elements in accordance with the referenced codes and standards.

Support piping from building structure. Do not support piping from roof deck or from other pipe.

Run piping parallel with the lines of the building. Space and install piping and components so that a threaded pipe fitting may be removed between adjacent pipes and so that there is no less than DN15 of clear space between the finished surface and other work and between the finished surface of parallel adjacent piping. Arrange hangars on different adjacent service lines running parallel with each other in line with each other and parallel to the lines of the building.

Install piping support elements at intervals specified hereinafter, at locations not more than 900 millimeter from the ends of each runout, and not over 300 millimeter from each change in direction of piping.

Base load rating for all pipe-hanger supports on insulated weight of lines filled with water and forces imposed. Deflection per span is not exceed slope gradient of pipe. Ensure supports are in accordance with the following minimum rod size and maximum allowable hanger spacing for specified pipe. For concentrated loads such as valves, reduce the allowable span proportionately:

<u>PIPE SIZE (DN)</u> <u>MILLIMETER</u>	<u>ROD SIZE</u> <u>MILLIMETER</u>	<u>STEEL PIPE</u> <u>MILLIMETER</u>	<u>COPPER PIPE</u> <u>MILLIMETER</u>
25 and smaller	10	2500	1850
32 to 40	10	3050	2500
50	10	3050	3050
65 to 90	13	3700	3700
100 to 125	16	5000	4300

Support vertical risers independently of connected horizontal piping, whenever practicable, with fixed or spring supports at the base and at

intervals to accommodate system range of thermal conditions. Ensure risers have guides for lateral stability. For risers subject to expansion, provide only one rigid support at a point approximately one-third down from the top. Place clamps under fittings unless otherwise specified. Support carbon-steel pipe at each floor and at not more than 4572 millimeter intervals for pipe DN50 and smaller and at not more than 6096 millimeter intervals for pipe DN65 and larger.

3.4 PENETRATIONS

Provide effective sound stopping and adequate operating clearance to prevent structure contact where piping penetrates walls, floors, or ceilings into occupied spaces adjacent to equipment rooms; where similar penetrations occur between occupied spaces; and where penetrations occur from pipe chases into occupied spaces. Occupied spaces include space above ceilings where no special acoustic treatment of ceiling is provided. Finish penetrations to be compatible with surface being penetrated.

Accomplish sound stopping and vapor-barrier sealing of pipe shafts and large floor and wall openings by packing to high density with properly supported fibrous-glass insulation or, where ambient or surface temperatures do not exceed 49 degrees C, by foaming-in-place with self-extinguishing, 0.9 kilogram density polyurethane foam to a depth not less than 152 millimeter. Finish foam with a rasp. Ensure vapor barrier is not less than 3 millimeter thick vinyl coating applied to visible and accessible surfaces. Where high temperatures and fire stopping are a consideration, use only mineral wool with openings covered by 1.6 millimeter sheet metal.

3.5 SLEEVES

Provide sleeves where piping passes through roofs, masonry, concrete walls and floors.

Continuously weld sleeves passing through steel decks to the deck.

Ensure sleeves that extend through floors, roofs, load bearing walls, and fire barriers are continuous and fabricated from Schedule 40 steel pipe, with welded anchor lugs. Form all other sleeves by molded linear polyethylene liners or similar materials that are removable. Ensure diameter of sleeves is large enough to accommodate pipe, insulation, and jacketing without touching the sleeve and provides a minimum 10 millimeter clearance. Install a sleeve size to accommodate mechanical and thermal motion of pipe precluding transmission of vibration to walls and the generation of noise.

Pack the space between a pipe, bare or insulated, and the inside of a pipe sleeve or a construction surface penetration solid with a mineral fiber conforming to JIS A 9504. Provide this packing wherever the piping passes through firewalls, equipment room walls, floors, and ceilings connected to occupied spaces, and other locations where sleeves or construction-surface penetrations occur between occupied spaces. Where sleeves or construction surface penetrations occur between conditioned and unconditioned spaces, fill the space between a pipe, bare or insulated, and the inside of a pipe sleeve or construction surface penetration with an elastomer caulk to a depth of 13 millimeter. Ensure all caulked surfaces are oil- and grease-free.

Ensure through-penetration fire stop materials and methods are in

accordance with ISO 834-1.

Caulk exterior wall sleeves watertight with lead and oakum or mechanically expandable chloroprene inserts with mastic-sealed metal components.

Ensure sleeve height above roof surface is a minimum of 305 and a maximum of 457 millimeter.

3.6 ESCUTCHEONS

Provide escutcheons at all penetrations of piping into finished areas. Where finished areas are separated by partitions through which piping passes, provide escutcheons on both sides of the partition. Where suspended ceilings are installed, provide plates at the underside only of such ceilings. For insulated pipes, select plates large enough to fit around the insulation. Use chrome-plated escutcheons in all occupied spaces and of size sufficient to effectively conceal openings in building construction. Firmly attach escutcheons with setscrews.

3.7 FLASHINGS

Provide flashings at penetrations of building boundaries by mechanical systems and related work.

3.8 DISINFECTION

Disinfect water piping, including all valves, fittings, and other devices, with a solution of chlorine and water. Ensure the solution contains not less than 50 parts per million (ppm) of available chlorine. Hold solution for a period of not less than 8 hours, after which the solution contains not less than 10 ppm of available chlorine or redisinfect the piping. After successful sterilization, thoroughly flush the piping before placing into service. Flushing is complete when the flush water contains less than 0.5 ppm of available chlorine. Water for disinfected will be furnished by the Government. Approve disposal of contaminated flush water in accordance with written instructions received from the Environmental authority having jurisdiction through the Contracting Officer and all local, State and Federal Regulations.

3.9 OPERATION AND MAINTENANCE

Provide Operation and Maintenance Manuals consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions. Submit test data that is clear and readily legible.

3.10 PAINTING OF NEW EQUIPMENT

Factory or shop apply new equipment painting, as specified herein, and provided under each individual section.

3.10.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied withstands 125 hours in a salt-spray fog test, except that equipment located outdoors withstand 500 hours in a salt-spray fog test. Conduct salt-spray fog test is in accordance with Japanese Industry Standards, and for that test the acceptance criteria is as follows: immediately after completion of the

test, the inspected paint shows no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shows no signs of rust creepage beyond 3 mm on either side of the scratch mark.

Ensure the film thickness of the factory painting system applied on the equipment is not less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, design the factory painting system for the temperature service.

3.10.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal, surfaces subject to temperatures in excess of 50 degrees C.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Selected color of finish coat is aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C receives one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.
- b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C Receives two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm.
- c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C receives two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

-- End of Section --

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SECTION 23 05 48.19

SEISMIC BRACING FOR HVAC
05/18, CHG 2: 08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

- ACI 355.2 (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary
- ACI 355.4 (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

- AISC 325 (2017) Steel Construction Manual

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN WATER WORKS ASSOCIATION (AWWA)

- AWWA C105/A21.5 (2018) Polyethylene Encasement for Ductile-Iron Pipe Systems
- AWWA C116/A21.16 (2015) Protective Fusion-Bonded Coatings for the Interior and Exterior Surfaces of Ductile-Iron and Gray Iron Fittings
- AWWA C153/A21.53 (2019) Ductile-Iron Compact Fittings for Water Service
- AWWA C213 (2015) Fusion-Bonded Epoxy Coating for the Interior and Exterior of Steel Water Pipelines

ASTM INTERNATIONAL (ASTM)

- ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel
- ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A325	(2014) Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength
ASTM A490	(2014a) Standard Specification for Structural Bolts, Alloy Steel, Heat Treated, 150 ksi Minimum Tensile Strength
ASTM A500/A500M	(2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM A603	(2019) Standard Specification for Zinc-Coated Steel Structural Wire Rope
ASTM D1785	(2015; E 2018) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120
ASTM D2665	(2014) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings
ASTM E488/E488M	(2015) Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements
ASTM F891	(2016) Standard Specification for Coextruded Poly (Vinyl Chloride) (PVC) Plastic Pipe with a Cellular Core
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)

FEMA P-414	(January 2004) Installing Seismic Restraints for Duct and Pipe
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ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC ES AC193	(2012) Acceptance Criteria for Mechanical Anchors in Concrete Elements
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JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 1186	(2013) Sets of High Strength Hexagon Bolt, Hexagon Nut and Plain Washers for Friction
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Grip Joints

JIS G 3101	(2020) Rolled Steels for General Structure
JIS G 3106	(2020) Rolled Steels for Welded Structure
JIS G 3114	(2022) Hot-Rolled Atmospheric Corrosion Resisting Steels for Welded Structure
JIS G 3136	(2022) Rolled Steels for Building Structure
JIS G 3444	(2021) Carbon Steel Tubes for General Structure
JIS G 3466	(2021) Carbon Steel Square and Rectangular Tubes for General Structure
JIS G 3475	(2021) Carbon Steel Tubes for Building Structure

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

MFMA-4	(2004) Metal Framing Standards Publication
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NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
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SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1981	(2008) Seismic Restraint Manual Guidelines for Mechanical Systems, 3rd Edition
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U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 2, 2024) Structural Engineering
UFC 4-010-01	(2018; with Change 3, 2024) DoD Minimum Antiterrorism Standards for Buildings

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION (VISCMA)

VISCMA 412	(2014) Installing Seismic Restraints for Mechanical Equipment
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1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

Apply the requirements for seismic protection measures described in this section and on the drawings to the mechanical equipment and mechanical systems both inside and outside of the building along with exterior utilities and systems listed below. Where there is a conflict between the specifications and the drawings, the specifications will take precedence. Accomplish resistance to lateral forces induced by earthquakes without

consideration of friction resulting from gravity loads.

1.2.2 Mechanical Equipment

Mechanical equipment to be seismically protected must include the following items to the extent required on the drawings or in other sections of these specifications:

Equipment/Components with $I_p = 1.0$

Expansion Air Separator Tanks	Valves and Fittings for Piping
Water Chiller Units	Thermal Storage Units
Computer Room Air Conditioners	Air Handling Units
Pumps with Motors	Lab Scrubbers
Air Terminal Units	Unitary HVAC Systems

1.2.3 Mechanical Systems

Mechanical systems to be seismically protected must include the following items to the extent required on the drawings or in this or other sections of these specifications:

Mechanical systems with $I_p = 1.0$

- a. All Piping and Ducts Inside the Building Except as Specifically Stated Below Under "Items Not Covered By This Section".
- b. Chilled Water Distribution Systems Outside of Buildings.
- c. Water Piping Outside of Buildings.
- d. All Water Supply Systems Outside of Buildings.
- f. All Process Piping Outside of Buildings.

1.2.4 Contractor Designed Bracing

Submit copies of the design calculations with the drawings. Calculations must be approved, certified, stamped and signed by a registered Professional Structural Engineer. A non-U.S. citizen may qualify with a 1 Kyu Kenchikushi (1st Class Qualified Architect), in lieu of U.S. licensing qualifications. Calculations must verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design the bracing in accordance with UFC 3-301-01, , UFC 4-010-01 and additional data furnished by the Contracting Officer. Resistance to lateral forces induced by earthquakes must be accomplished without consideration of friction resulting from gravity loads. UFC 3-301-01 uses parameters for the building, not for the equipment in the building; therefore, corresponding adjustments to the formulas must be required. Loadings determined using UFC 3-301-01 are based on strength design; therefore, AISC 325 Specifications must be used for the design. The bracing for the mechanical equipment designated in paragraph 1.2.2 and systems designated in paragraph 1.2.3 must be developed by the Contractor.

1.2.5 Items Not Covered By This Section

1.2.5.1 Items Requiring No Seismic Restraints

Seismic restraints are not required for the following items:

- b. Piping in mechanical equipment rooms less than 32 mm nominal pipe size.
- c. All other piping equal to or less than 80 mm nominal pipe size.
- d. Rectangular air handling ducts less than 0.56 square meters in cross sectional area.
- e. Round air handling ducts less than 711 mm in diameter.
- f. Piping suspended by individual hangers 300 mm or less in length from the top of pipe to the bottom of the supporting structural member where the hanger is attached, except as noted below.
- g. Ducts suspended by hangers 300 mm or less in length from the top of the duct to the bottom of the supporting structural member, except as noted below.

In exemptions f. and g. all hangers must meet the length requirements. If the length requirement is exceeded by one hanger in the run, brace the entire run. Seismically protect interior piping and ducts not listed above in accordance with the provisions of this specification.

Non-critical items may require seismic restraints if adjacent to critical equipment or systems that must remain operational after an earthquake and could be compromised by impact with non-critical adjacent components.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Coupling and Bracing

Flexible Couplings or Joints

Equipment Restraint

Contractor Designed Bracing; G

SD-03 Product Data

Coupling and Bracing; G

Flexible Couplings Or Joints; G

Equipment Restraint; G

Contractor Designed Bracing; G

Anchor Bolts

Vibration Isolators

SD-05 Design Data

Design Calculations

SD-06 Test Reports

Anchor Bolts; G

PART 2 PRODUCTS

2.1 GENERAL DESIGN REQUIREMENTS

Submit detailed seismic restraint drawings for mechanical equipment, duct systems, piping systems and any other mechanical systems along with calculations, catalog cuts, templates, and erection and installation details, as appropriate, for the items listed below. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction. Calculations must be stamped, by a registered structural engineer, and verify the capability of structural members to which bracing is attached for carrying the load from the brace. Include drawing for Mission Critical Equipment indicating the equipment location in the facility sufficient to be used for the installation. Design must be based on actual equipment and system layout. Design must include calculated dead loads, static seismic loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods.

2.2 EQUIPMENT RESTRAINT

Equipment must be rigidly or flexibly mounted as indicated in the specifications and/or drawings depending on vibration isolation requirements as follows below.

Roof mounted equipment such as cooling towers and condensers, both vibration isolated and nonisolated, must have support members designed and anchored to building structural steel or concrete as required for seismic restraint and wind loads.

2.2.1 Rigidly (Base and Suspended) Mounted Equipment

HVAC equipment furnished under this contract must be rigidly mounted that are qualified for earthquake loading in accordance with ACI 355.2 and ACI 355.4. Anchor bolts must conform to ASTM F1554 or JIS B 1186. For any rigid equipment which is rigidly anchored, provide flexible joints for piping, ductwork, electrical conduit, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging. Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and overturning.

Roof mounted HVAC equipment roof curbs, framing and attachment to equipment and structure must be designed and braced to withstand seismic loads. Designated Seismic Systems (DSS) assigned to Seismic Design Category (SDC) C, D, E, or F and Risk Category IV components needed for continued operation after an earthquake must have two nuts provided on each anchor bolt.

2.2.2 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 6 mm. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where required to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer's will be a member of VISCMA. Design force, F_p , must be doubled for vibration isolators with an air gap greater than 0.25 inches as specified in ASCE 7-16, Chapter 13. Housed springs must not be used for seismic restraint applications because they cannot resist uplift.

2.3 BOLTS AND NUTS

Hex head bolts, and heavy hexagon nuts must be ASTM A325 or JIS B 1186, or ASTM A490 or JIS B 1186 bolts and ASTM A563 or JIS B 1186 nuts. Provide bolts and nuts galvanized in accordance with ASTM A153/A153M when used underground or exposed to weather.

2.4 FLEXIBLE JOINTS

Flexible joints must have same pressure and temperature ratings as adjoining pipe. Braided hoses must not be used where there is torsional or axial movement unless manufacturer allows it.

2.4.1 Braided Hose Expansion Joint

Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops are not acceptable. Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops must not be acceptable. Braided hose in a 60 degree flexible V loop arrangement must be used for small diameter pipe connections to coils in variable-air-volume (VAV) terminal units and fan coil units installed in suspended ductwork whether braced or unbraced.

All braided hose expansion joints must be manufactured in accordance with the documented manufacturers weld procedure specifications. The procedure qualification record must be used to document the execution of this procedure and must follow the general "guidelines" of ASME Section IX. Each individual welder must conform to the in-house procedure qualification record and be qualified prior to each production lot. The testing of each individual welder must be documented in a welding procedure qualification record.

2.4.1.1 Corrugated Hose

Corrugated hose must be Type 304 stainless steel. Braid must be Type 304 stainless steel for any series 300 stainless steel hose. Fittings materials of construction and end fitting type must be consistent with pipe material and equipment/ pipe connection fittings. Copper fittings must not be attached to stainless steel hose.

2.4.1.2 Flexible Hose Expansion Loops

Flexible hose expansion loops must have a factory supplied, hanger / support lug located at the bottom of the 180deg return. Flexible hose expansion loop(s) must be rated with an operating pressure which is the same as the adjoining pipe. The operating pressure must be based on burst pressure with a 4 to 1 safety factor.

2.4.2 Double Ball Flexible Expansion Joint

Install flexible expansion joints manufactured of ductile iron conforming to the material requirements of ASTM A536 and AWWA C153/A21.53 in the locations indicated on the drawings. Provide foundry certification of material upon request. Each flexible expansion joint must be pressure tested prior to shipment against its own restraint to a minimum of 350 psi (250 psi for flexible expansion joints 2 inch and 30 inches diameter and larger.) A minimum 2:1 safety factor, determined from the published pressure rating, must apply. Factory Mutual Approval for the 3 inch through 12 inch sizes is required. Each flexible expansion joint must consist of an expansion joint designed and cast as an integral part of a ball and socket type flexible joint, having a minimum per ball deflection of: 20°, 2" - 12"; 15°, 14" - 36"; 12°, 42"-48" and 4-inches minimum expansion. Additional expansion sleeves must be available and easily added or removed at the factory or in the field. Both standardized mechanical joint and flange end connections must be available.

2.4.2.1 Internal Surfaces

Line all internal surfaces (wetted parts) with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C213. Sealing gaskets must be constructed of EPDM. The coating must meet NSF/ANSI 61.

2.4.2.2 Exterior Surfaces

Coat exterior surfaces with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C116/A21.16. Include appropriately sized polyethylene sleeves, meeting AWWA C105/A21.5, for direct buried applications.

2.4.3 Double Ball Flexible Expansion Joint Gravity Drain (Non-Pressurized)

Flexible expansion joints gravity drain must be installed in the locations indicated on the drawings and must be manufactured of pvc. All connections whether solvent weld or mechanical must be restrained to allow movement to be transferred to expansion joint. Each ball must allow up to 15 degrees deflection.

End connection outside diameters must be compatible with ASTM D1785, ASTM D2665 and ASTM F891 PVC pipe and are to be solvent welded.

2.5 SWAY BRACING MATERIALS

Material used for members listed in this section and on the drawings, must be structural steel conforming with the following:

- a. Plates, rods, and rolled shapes, ASTM A36/A36M or JIS G 3101, JIS G 3106, JIS G 3114, JIS G 3136.
- b. Wire rope, ASTM A603 pre-stretched. Class B galv coating Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 412. A minimum of two clamps are required on each end of wire rope.
- c. Tubes, ASTM A500/A500M, Grade B. or JIS G 3444, JIS G 3475
- d. Pipes, ASTM A53/A53M, Grade B or JIS G 3444, JIS G 3466.
- e. Angles, ASTM A36/A36M or JIS G 3101, JIS G 3106, JIS G 3114, JIS G 3136.
- f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at random points conforming to MFMA-4

PART 3 EXECUTION

3.1 COUPLING AND BRACING

- a. Submit detail drawings, as specified here and throughout this specification, along with catalog cuts, templates, and erection and installation details, as appropriate, for the items listed. Submittals must be complete in detail; must indicate thickness, type, grade, class of metal, and dimensions; and must show construction details, reinforcement, anchorage, and installation with relation to the building construction.
- b. Provide coupling installation conforming to the details shown on the drawings. Provisions of this paragraph apply to all piping within a 1.5 m line around outside of building unless buried in the ground. Piping grouped for support on trapeze-type hangers must be braced at the most frequent interval as determined by applying the requirements of this specification to each piping run on the common support.
- c. Size bracing components as required for the total load carried by the common supports. Bracing rigidly attached to pipe flanges, or similar, must not be used where it would interfere with thermal expansion of piping.
- d. Adjust isolators and restraints after piping systems has been filled and equipment is at its operating weight, following the manufacturer's

written instructions.

- e. Install cables at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412.

3.2 FLEXIBLE COUPLINGS OR JOINTS

3.2.1 Building Piping

Provide flexible couplings or joints in building piping at bottom of all pipe risers for pipe larger than 90 mm in diameter. Laterally brace flexible couplings or joints without interfering with the action of the flexible coupling or joint. Cast iron waste and vent piping need only comply with these provisions when caulked joints are used. Flexible bell and spigot pipe joints using rubber gaskets may be used at each branch adjacent to tees and elbows for underground waste piping inside of building to satisfy these requirements.

3.3 PIPE SLEEVES

Size pipe sleeves in interior non-fire rated walls as indicated on the drawings to provide clearances that will permit differential movement of piping without the piping striking the pipe sleeve.

3.4 SPREADERS

Provide spreaders between adjacent piping runs to prevent contact during seismic activity whenever pipe or insulated pipe surfaces are less than 100 mm apart. Apply spreaders at same interval as sway braces at an equal distance between the sway braces. If rack type hangers are used where the pipes are restrained from contact by mounting to the rack, spreaders are not required for pipes mounted in the rack. Apply spreaders to surface of bare pipe and over insulation on insulated pipes utilizing high-density inserts and pipe protection shields in accordance with the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.5 SWAY BRACES FOR PIPING

Provide sway braces to prevent movement of the pipes under seismic loading. Provide braces in both the longitudinal and transverse directions, relative to the axis of the pipe. Provide sufficient braces for equipment to resist a horizontal force as specified in UFC 3-301-01 without exceeding safe working stress of bracing components. Provide bracing that does not interfere with thermal expansion requirements for the pipes as described in other sections of these specifications. For seismic analysis of horizontal pipes, the equivalent static force should be considered to act concurrently with the full dead load of the pipe, including contents.

3.5.1 Transverse Sway Bracing

Provide transverse sway bracing for steel and copper pipe at intervals not to exceed those shown on the drawings. All runs (length of pipe between end joints) must have a minimum of transverse bracing at each end. Provide transverse sway bracing for pipes of materials other than steel and copper at intervals not to exceed the hanger spacing as specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.5.2 Longitudinal Sway Bracing

Provide longitudinal sway bracing at 12 m intervals unless otherwise indicated. All runs (length of pipe between end joints) must have one longitudinal brace minimum. Construct sway braces in accordance with the drawings. Do not use branch lines, walls, or floors as sway braces.

3.5.3 Vertical Runs

Run is defined as length of pipe between end joints. Do not brace vertical runs of piping no more than 3 m vertical intervals. Braces for vertical runs must be above the center of gravity of the segment being braced. Flexible couplings should be provided at the bottoms of risers for pipes larger than 3.5 in. (89 mm) in diameter. Flexible couplings and expansion joints should be braced laterally and longitudinally unless such bracing would interfere with the action of the couplings or joints. When pipes enter buildings, flexible couplings should be provided to allow for relative movement between the soil and building. Construct all sway braces in accordance with the drawings. Attach sway braces to the structural system. Do not connect to branch lines, walls, or floors.

3.5.4 Clamps and Hangers

Apply clamps or hangers on uninsulated pipes directly to pipe. Insulated piping must have clamps or hangers applied over insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

Hanger rod stiffener angle or strut bracing must be securely attached by a series of attachment clamps manufactured from a one piece metal stamping and must include all require attachment hardware and locking nuts. Attachment clamps made from aluminum or cast iron must not be used in seismic applications. Do not weld vertical braces to hanger rods.

3.6 SWAY BRACES FOR DUCTS

3.6.1 Braced Ducts

Provide bracing details and spacing for rectangular and round ducts in accordance with SMACNA 1981. However, the design seismic loadings for these items must not be less than loadings obtained using the procedures in UFC 3-301-01. Bracing must not attach to duct joints. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct.

3.6.2 Unbraced Ducts

Attach hangers for unbraced ducts to the duct within 50 mm of the top of the duct with a minimum of two #10 sheet metal screws in accordance with FEMA P-414. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct. Install unbraced ducts with a 150 mm minimum clearance to vertical ceiling hanger wires.

3.7 EQUIPMENT

3.7.1 General

Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel

and other embedded items prior to drilling holes for anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or till the gap with epoxy on equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

3.7.2 Controls

Ensure that controls for critical equipment that must remain operational after an earthquake are certified per paragraph 3.11 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT and are served by emergency power as required.

3.8 ANCHOR BOLTS

3.8.1 Drilled-In Anchor Bolts

Drill holes with rotary impact hammer drills Drill bits must be of diameters as specified by the anchor manufacturer. Unless otherwise shown on the Drawings, all holes must be drilled perpendicular to the concrete surface. Where anchors are permitted to be installed in cored holes, use core bits with matched tolerances as specified by the manufacturer. Properly clean cored hole per manufacturer's instructions. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Exercise care in coring or drilling to avoid damaging existing reinforcing or embedded items. Notify the COR if reinforcing steel or other embedded items are encountered during drilling. Take precautions as necessary to avoid damaging prestressing tendons, electrical and telecommunications conduit, and gas lines. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Perform anchor installation in accordance with manufacturer instructions.

3.8.1.1 Wedge Anchors, Heavy-Duty Sleeve Anchors, and Undercut Anchors

Protect threads from damage during anchor installation. Heavy-duty sleeve anchors must be installed with sleeve fully engaged in part to be fastened. Set anchors to manufacturer's recommended torque, using a torque wrench. Following attainment of 10% of the specified torque, 100% of the specified torque must be reached within 7 or fewer complete turns of the nut. If the specified torque is not achieved within the required number of turns, the anchor must be removed and replaced unless otherwise directed by the Engineer.

3.8.1.2 Cartridge Injection Adhesive Anchors

Where approved for seismic application, clean all holes per manufacturer instructions to remove loose material and drilling dust prior to installation of adhesive. Inject adhesive into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive. Follow manufacturer recommendations to ensure proper mixing of adhesive components. Sufficient adhesive must be injected in the hole to ensure that the annular gap is filled to the surface. Remove excess adhesive from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

3.8.1.3 Capsule Anchors

Where approved for seismic application, perform drilling and setting operations in accordance with manufacturer instructions. Clean all holes to remove loose material and drilling dust prior to installation of adhesive. Remove water from drilled holes in such a manner as to achieve a surface dry condition. Capsule anchors must be installed with equipment conforming to manufacturer recommendations. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

Observe manufacturer recommendations with respect to installation temperatures for cartridge injection adhesive anchors and capsule anchors.

3.9 ANCHOR BOLT TESTING

Test in place expansion and chemically bonded anchors not more than 24 hours after installation of the anchor, conducted by an independent testing agency; testing must be performed on random anchor bolts as described below.

3.9.1 Torque Wrench Testing

Perform torque wrench testing on not less than 50 percent of the total installed applied torque expansion anchors and at least one anchor for every piece of equipment containing more than two anchors. The test torque must equal the minimum required installation torque as required by the bolt manufacturer. Calibrate torque wrenches at the beginning of each day the torque tests are performed. Recalibrate torque wrenches for each bolt diameter whenever tests are run on bolts of various diameters. Apply torque between 20 and 100 percent of wrench capacity. Reach the test torque within one half turn of the nut, except for 9 mm sleeve anchors which must reach their torque by one quarter turn of the nut. If any anchor fails the test, test similar anchors not previously tested until 20 consecutive anchors pass. Failed anchors must be retightened and retested to the specified torque; if the anchor still fails the test it must be replaced.

3.9.2 Pullout Testing

Test expansion and chemically bonded anchors by applying a pullout load using a hydraulic ram attached to the anchor bolt. Testing must be in accordance with ASTM E488/E488M or ICC ES AC193. At least 10 percent of each type and size of anchors, but not less than 3 per day must be tested. Apply the load to the anchor without removing the nut; when that is not possible, the nut must be removed and a threaded coupler must be installed of the same tightness as the original nut. Check the test setup to verify that the anchor is not restrained from withdrawing by the baseplate, the test fixture, or any other fixtures. The support for the testing apparatus must be at least 1.5 times the embedment length away from the bolt being tested. Load each tested anchor to 1 times the design tension value for the anchor. The anchor must have no observable movement at the test load. If any anchor fails the test, similar type and size anchors not previously tested must be tested until 10 percent of those type consecutive anchors pass. Remove and replace failed anchors. Fill empty anchor holes and patch failed anchor locations with high-strength non-shrink, nonmetallic grout.

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SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC
11/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 203 (1990; R 2011) Field Performance
Measurements of Fan Systems

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 62.1 (2016) Ventilation for Acceptable Indoor
Air Quality

ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1 (2002; 6th ed) National Standards for
Total System Balance

AABC MN-4 (1996) Test and Balance Procedures

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB MASV (2006) Procedural Standards for
Measurements and Assessment of Sound and
Vibration

NEBB PROCEDURAL STANDARDS (2015) Procedural Standards for TAB
(Testing, Adjusting and Balancing)
Environmental Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

SMACNA 1780 (2023) HVAC Systems - Testing, Adjusting
and Balancing, 3rd Edition

SMACNA 1858 (2004) HVAC Sound And Vibration Manual -
First Edition

1.2 DEFINITIONS

- a. AABC: Associated Air Balance Council
- b. COTR: Contracting Officer's Technical Representative
- c. HVAC: Heating, ventilating, and air conditioning; or heating,
ventilating, and cooling

- d. NEBB: National Environmental Balancing Bureau
- e. Out-of-tolerance data: Pertains only to field acceptance testing of Final TAB report. When applied to TAB work this phase means "a measurement taken during TAB field acceptance testing which does not fall within the range of plus 5 to minus 5 percent of the original measurement reported on the TAB Report for a specific parameter."
- f. Season of maximum heating load: The time of year when the outdoor temperature at the project site remains within plus or minus 17.5 degrees Celsius of the project site's winter outdoor design temperature, throughout the period of TAB data recording.
- g. Season of maximum cooling load: The time of year when the outdoor temperature at the project site remains within plus or minus 3 degrees Celsius of the project site's summer outdoor design temperature, throughout the period of TAB data recording.
- h. Season 1, Season 2: Depending upon when the project HVAC is completed and ready for TAB, Season 1 is defined, thereby defining Season 2. Season 1 could be the season of maximum heating load, or the season of maximum cooling load.
- i. Sound measurements terminology: Defined in AABC MN-1, NEBB MASV, or SMACNA 1858 (TABB).
- j. TAB: Testing, adjusting, and balancing (of HVAC systems)
- k. TAB'd: HVAC Testing/Adjusting/Balancing procedures performed
- l. TAB Agency: TAB Firm
- m. TAB team field leader: TAB team field leader
- n. TAB team supervisor: TAB team engineer
- o. TAB team technicians: TAB team assistants
- p. TABB: Testing Adjusting and Balancing Bureau

1.2.1 Similar Terms

In some instances, terminology differs between the Contract and the TAB Standard primarily because the intent of this Section is to use the industry standards specified, along with additional requirements listed herein to produce optimal results.

The following table of similar terms is provided for clarification only. Contract requirements take precedent over the corresponding AABC, NEBB, or TABB requirements where differences exist.

SIMILAR TERMS			
Contract Term	AABC Term	NEBB Term	TABB Term
TAB Standard	National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems	Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems	International Standards for Environmental Systems Balance
TAB Specialist	TAB Engineer	TAB Supervisor	TAB Supervisor
Systems Readiness Check	Construction Phase Inspection	Field Readiness Check & Preliminary Field Procedures	Field Readiness Check & Prelim. Field Procedures

1.3 WORK DESCRIPTION

The work includes testing, adjusting, and balancing (TAB) of new heating, ventilating, and cooling (HVAC) air and water distribution systems including equipment and performance data, ducts, and piping which are located within, on, under, between, and adjacent to buildings, including records of existing conditions.

Perform TAB in accordance with the requirements of the TAB procedural standard recommended by the TAB trade association that approved the TAB Firm's qualifications. Comply with requirements of AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 (TABB) as supplemented and modified by this specification section. All recommendations and suggested practices contained in the TAB procedural standards are considered mandatory.

Conduct TAB of the indicated existing systems and equipment and submit the specified TAB reports for approval. Conduct TAB work in accordance with the requirements of this section.

1.3.1 Air Distribution Systems

Test, adjust, and balance systems (TAB) in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to exterior of air distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

1.3.2 Water Distribution Systems

TAB systems in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to water distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. At Contractor's option and with Contracting Officer's written approval, the piping systems may be insulated before systems are TAB'd.

Terminate piping insulation immediately adjacent to each flow control valve, automatic control valve, or device. Seal the ends of pipe insulation and the space between ends of pipe insulation and piping, with

waterproof vapor barrier coating.

After completion of work under this section, insulate the flow control valves and devices as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

1.3.3 TAB Schematic Drawings

Show the following information on TAB Schematic Drawings:

- (1) A unique number or mark for each piece of equipment or terminal.
- (2) Air quantities at air terminals.
- (3) Air quantities and temperatures in air handling unit schedules.
- (4) Water quantities and temperatures in thermal energy transfer equipment schedules.
- (5) Water quantities and heads in pump schedules.
- (6) Water flow measurement fittings and balancing fittings.

The Testing, Adjusting, and Balancing (TAB) Specialist must review the Contract Plans and Specifications and advise the Contracting Officer of any deficiencies that would prevent the effective and accurate TAB of the system, including records of existing conditions, and systems readiness check. The TAB Specialist must provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for proper system operation.

Submit three copies of the TAB Schematic Drawings and Report Forms to the Contracting Officer, no later than 21 days prior to the start of TAB field measurements.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Records of Existing Conditions; G

TAB Firm; G

Designation of TAB Team Assistants; G

Designation of TAB Team Engineer; G

TAB Specialist; G

Designation of TAB Team Field Leader; G

SD-02 Shop Drawings

TAB Schematic Drawings and Report Forms; G

SD-03 Product Data

Equipment and Performance Data; G

TAB Related HVAC Submittals; G

A list of the TAB Related HVAC Submittals, no later than 7 days after the approval of the TAB team engineer.

TAB Procedures; G

Proposed procedures for TAB, submitted with the TAB Schematic Drawings and Report Forms.

Calibration; G

Systems Readiness Check; G

TAB Execution; G

SD-06 Test Reports

TAB Design Review Report; G

TAB Report for Season 1; G

SD-07 Certificates

Independent TAB Agency and Personnel Qualifications; G

TAB Submittal and Work Schedule; G

TAB Pre-Field Engineering Report; G

TAB Firm; G

Design Review Report; G

Advanced Notice for Season 1 TAB Field Work; G

Prerequisite HVAC Work Check Out List For Season 1; G

Advanced Notice for Season 2 TAB Field Work; G

Prerequisite HVAC Work Check Out List For Season 2; G

1.5 QUALITY ASSURANCE

1.5.1 Independent TAB Agency and Personnel Qualifications

To secure approval for the proposed agency, submit information certifying that the TAB agency is a first tier subcontractor who is not affiliated with any other company participating in work on this contract, including design, furnishing equipment, or construction. Further, submit the following, for the agency, to Contracting Officer for approval:

- a. Independent AABC or NEBB or TABB TAB agency:

- (1) TAB agency: AABC registration number and expiration date of current certification; or NEBB certification number and expiration date of current certification; or TABB certification number and expiration date of current certification.
 - (2) TAB team supervisor: Name and copy of AABC or NEBB or TABB TAB supervisor certificate and expiration date of current certification.
 - (3) TAB team field leader: Name and documented evidence that the team field leader has satisfactorily performed full-time supervision of TAB work in the field for not less than 3 years immediately preceding this contract's bid opening date.
 - (4) TAB team field technicians: Names and documented evidence that each field technician has satisfactorily assisted a TAB team field leader in performance of TAB work in the field for not less than one year immediately preceding this contract's bid opening date.
 - (5) Current certificates: Registrations and certifications are current, and valid for the duration of this contract. Renew Certifications which expire prior to completion of the TAB work, in a timely manner so that there is no lapse in registration or certification. TAB agency or TAB team personnel without a current registration or current certification are not to perform TAB work on this contract.
- b. TAB Team Members: TAB team approved to accomplish work on this contract are full-time employees of the TAB agency. No other personnel is allowed to do TAB work on this contract.
- c. Replacement of TAB team members: Replacement of members may occur if each new member complies with the applicable personnel qualifications and each is approved by the Contracting Officer.

1.5.2 TAB Standard

Perform TAB in accordance with the requirements of the standard under which the TAB Firm's qualifications are approved, i.e., AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 unless otherwise specified herein. All recommendations and suggested practices contained in the TAB Standard are considered mandatory. Use the provisions of the TAB Standard, including checklists, report forms, etc., as nearly as practical, to satisfy the Contract requirements. Use the TAB Standard for all aspects of TAB, including qualifications for the TAB Firm and Specialist and calibration of TAB instruments. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the TAB Standard, adhere to the manufacturer's recommendations.

All quality assurance provisions of the TAB Standard such as performance guarantees are part of this contract. For systems or system components not covered in the TAB Standard, TAB procedures must be developed by the TAB Specialist. Where new procedures, requirements, etc., applicable to the Contract requirements have been published or adopted by the body responsible for the TAB Standard used (AABC, NEBB, or TABB), the requirements and recommendations contained in these procedures and requirements are considered mandatory, including the latest requirements

of ASHRAE 62.1.

1.5.3 Qualifications

1.5.3.1 TAB Firm

The TAB Firm must be either a member of AABC or certified by the NEBB or the TABB and certified in all categories and functions where measurements or performance are specified on the plans and specifications, including building systems commissioning and the measuring of sound and vibration in environmental systems.

Certification must be maintained for the entire duration of duties specified herein. If, for any reason, the firm loses subject certification during this period, the Contractor must immediately notify the Contracting Officer and submit another TAB Firm for approval. Any firm that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not be eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections to be performed by the TAB Firm will be considered invalid if the TAB Firm loses its certification prior to Contract completion and must be performed by an approved successor.

These TAB services are to assist the prime Contractor in performing the quality oversight for which it is responsible. The TAB Firm must be a prime subcontractor of the Contractor and be financially and corporately independent of the mechanical subcontractor, reporting directly to and paid by the Contractor.

1.5.3.2 TAB Specialist

The TAB Specialist must be either a member of AABC, an experienced technician of the Firm certified by the NEBB, or a Supervisor certified by the TABB. The certification must be maintained for the entire duration of duties specified herein. If, for any reason, the Specialist loses subject certification during this period, immediately notify the Contracting Officer and submit another TAB Specialist for approval. Any individual that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections performed by the TAB Specialist will be considered invalid if the TAB Specialist loses its certification prior to Contract completion and must be performed by the approved successor.

1.5.3.3 TAB Specialist Responsibilities

TAB Specialist responsibilities include all TAB work specified herein and in related sections under his direct guidance. The TAB specialist is required to be onsite on a daily basis to direct TAB efforts. The TAB Specialist must participate in the commissioning process specified in Section 01 91 00.15 BUILDING COMMISSIONING.

1.5.3.4 TAB Related HVAC Submittals

The TAB Specialist must prepare a list of the submittals from the Contract Submittal Register that relate to the successful accomplishment of all HVAC TAB. Accompany the submittals identified on this list with a letter

of approval signed and dated by the TAB Specialist when submitted to the Government. Ensure that the location and details of ports, terminals, connections, etc., necessary to perform TAB are identified on the submittals.

1.5.4 Responsibilities

The Contractor is responsible for ensuring compliance with the requirements of this section. The following delineation of specific work responsibilities is specified to facilitate TAB execution of the various work efforts by personnel from separate organizations. This breakdown of specific duties is specified to facilitate adherence to the schedule listed in the paragraph TAB SUBMITTAL AND WORK SCHEDULE.

1.5.4.1 Contractor

- a. TAB personnel: Ensure that the TAB work is accomplished by a group meeting the requirements specified in the paragraph TAB PERSONNEL QUALIFICATION REQUIREMENTS.
- b. Pre-TAB meeting: Attend the meeting with the TAB Supervisor, and ensure that a representative is present for the sheetmetal contractor, mechanical contractor, electrical contractor, and automatic temperature controls contractor.
- c. HVAC documentation: Furnish one complete set of the following HVAC-related documentation to the TAB agency:
 - (1) Contract drawings and specifications
 - (2) Approved submittal data for equipment
 - (3) Construction work schedule
 - (4) Up-to-date revisions and change orders for the previously listed items
- d. Submittal and work schedules: Ensure that the schedule for submittals and work required by this section and specified in the paragraph TAB SUBMITTAL AND WORK SCHEDULE is met.
- e. Coordination of supporting personnel:
- f. Provide the technical personnel, such as factory representatives or HVAC controls installer required by the TAB field team to support the TAB field measurement work.
- g. Provide equipment mechanics to operate HVAC equipment and ductwork mechanics to provide the field designated test ports to enable TAB field team to accomplish the TAB field measurement work. Ensure these support personnel are present at the times required by the TAB team, and cause no delay in the TAB field work.
- h. Conversely, ensure that the HVAC controls installer has required support from the TAB team field leader to complete the controls check out.
- i. Deficiencies: Ensure that the TAB Agency supervisor submits all Design/Construction deficiency notifications directly to the

Contracting officer within 3 days after the deficiency is encountered. Further, ensure that all such notification submittals are complete with explanation, including documentation, detailing deficiencies.

- j. Prerequisite HVAC work: Complete check out and debugging of HVAC equipment, ducts, and controls prior to the TAB engineer arriving at the project site to begin the TAB work. Debugging includes searching for and eliminating malfunctioning elements in the HVAC system installations, and verifying all adjustable devices are functioning as designed. Include as prerequisite work items, the deficiencies pointed out by the TAB team supervisor in the design review report.
- k. Prior to the TAB field team's arrival, ensure completion of the applicable inspections and work items listed in the TAB team supervisor's pre-field engineering report. Do not allow the TAB team to commence TAB field work until all of the following are completed.
 - (1) HVAC system installations are fully complete.
 - (2) HVAC prerequisite checkout work lists specified in the paragraph PRE-FIELD TAB ENGINEERING REPORT are completed, submitted, and approved. Ensure that the TAB Agency gets a copy of the approved prerequisite HVAC work checklist.
 - (3) HVAC system filters are clean for both Season 1 and Season 2 TAB field work.
- l. Advance notice: Furnish to the Contracting Officer with advance written notice for the commencement of the TAB field work.
- m. Insulation work: Ensure that openings in duct and machinery insulation coverings for TAB test ports are marked, closed and sealed.

1.5.4.2 TAB Agency

Provide the services of a TAB team which complies with the requirements of the paragraph INDEPENDENT TAB AGENCY PERSONNEL QUALIFICATIONS. The work to be performed by the TAB agency is limited to testing, adjusting, and balancing of HVAC air and water systems to satisfy the requirements of this specification section.

1.5.4.3 TAB Team Supervisor

- a. Overall management: Supervise and manage the overall TAB team work effort, including preliminary and technical TAB procedures and TAB team field work.
- b. Pre-TAB meeting: Attend meeting with Contractor.
- c. Design review report: Review project specifications and accompanying drawings to verify that the air systems and water systems are designed in such a way that the TAB engineer can accomplish the work in compliance with the requirements of this section. Verify the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, circuit setters, balancing valves, and manual volume dampers.
- d. Support required: Specify the technical support personnel required

from the Contractor other than the TAB agency; such as factory representatives for temperature controls or for complex equipment. Inform the Contractor in writing of the support personnel needed and when they are needed. Furnish the notice as soon as the need is anticipated, either with the design review report, or the pre-field engineering report, the during the TAB field work.

- e. Pre-field engineering report: Utilizing the following HVAC-related documentation; contract drawings and specifications, approved submittal data for equipment, up-to-date revisions and change orders; prepare this report.
- f. Prerequisite HVAC work checklist: Ensure the Contractor gets a copy of this checklist at the same time as the pre-field engineering report is submitted.
- g. Technical Assistance for TAB Work: Provide immediate technical assistance to the TAB field team for the TAB work.
 - (1) TAB field visit: At the midpoint of the Season 1 and Season 2 TAB field work effort, visit the contract site to inspect the HVAC installation and the progress of the TAB field work. Conduct site visit full-time for a minimum of one 8 hour workday duration.
 - (2) TAB field visit: Near the end of the TAB field work effort, visit the contract site to inspect the HVAC installation and the progress of the TAB field work. Conduct site visit full-time for a minimum of one 8 hour workday duration. Review the TAB final report data and certify the TAB final report.
 - (3) TAB field visit: Near the end of the TAB field work effort, visit the contract site to inspect the HVAC installation and the progress of the TAB field work. Conduct site visit full-time for a minimum of one 8 hour workday duration. Review the TAB final report data and certify the TAB final report.
- h. Certified TAB report: Certify the TAB report. This certification includes the following work:
 - (1) Review: Review the TAB field data report. From this field report, prepare the certified TAB report.
 - (2) Verification: Verify adherence, by the TAB field team, to the TAB plan prescribed by the pre-field engineering report and verify adherence to the procedures specified in this section.
- i. Design/Construction deficiencies: Within 3 working days after the TAB Agency has encountered any design or construction deficiencies, the TAB Supervisor must submit written notification directly to the Contracting Officer, with a separate copy to the Contractor, of all such deficiencies. Provide in this submittal a complete explanation, including supporting documentation, detailing deficiencies. Where deficiencies are encountered that are believed to adversely impact successful completion of TAB, the TAB Agency must issue notice and request direction in the notification submittal.
- j. TAB Field Check: The TAB team supervisor must attend and supervise Season 1 TAB field check.

1.5.4.4 TAB Team Field Leader

- a. Field manager: Manage, in the field, the accomplishment of the work specified in Part 3, EXECUTION.
- b. Full time: Be present at the contract site when TAB field work is being performed by the TAB team; ensure day-to-day TAB team work accomplishments are in compliance with this section.
- c. Prerequisite HVAC work: Do not bring the TAB team to the contract site until a copy of the prerequisite HVAC Checklist, with all work items certified by the Contractor to be working as designed, reaches the office of the TAB Agency.

1.5.5 Test Reports

1.5.5.1 Certified TAB Reports

Submit: TAB Report for Season 1 in the following manner:

- a. Report format: Submit the completed pre-field data forms approved in the pre-field TAB Engineering Report completed by TAB field team, reviewed and certified by the TAB supervisor. Bind the report with a waterproof front and back cover. Include a table of contents identifying by page number the location of each report. Report forms and report data must be typewritten. Handwritten report forms or report data are not acceptable.
- b. Temperatures: On each TAB report form reporting TAB work accomplished on HVAC thermal energy transfer equipment, include the indoor and outdoor dry bulb temperature range and indoor and outdoor wet bulb temperature range within which the TAB data was recorded. Include in the TAB report continuous time versus temperature recording data of wet and dry bulb temperatures for the rooms, or zones, as designated in the following list:
 - (1) Measure and compile data on a continuous basis for the period in which TAB work affecting those rooms is being done.
 - (2) Measure and record data only after the HVAC systems installations are complete, the systems fully balanced and the HVAC systems controls operating in fully automatic mode.
 - (3) Data may be compiled using direct digital controls trend logging where available. Otherwise, temporarily install calibrated time versus temperature/humidity recorders for this purpose. The HVAC systems and controls must be fully operational a minimum of 24 hours in advance of commencing data compilation. Include the specified data in the Season I TAB Report.
- c. System Diagrams: Provide updated diagrams with final installed locations of all terminals and devices, any numbering changes, and actual test locations. Use a key numbering system on the diagram which identifies each outlet contained in the outlet airflow report sheets.
 - (1) Report static pressure drop across air filters, acoustic silencers, moisture eliminators, air flow straighteners, air flow measuring stations or other pressure drop producing specialty

items installed in unit cabinetry, or in the system ductwork. Examples of these specialty items are smoke detectors, white sound generators, RF shielding, wave guides, security bars, blast valves, small pipes passing through ductwork, and duct mounted humidifiers.

- (2) Do not report static pressure drop across duct fittings provided for the sole purpose of conveying air, such as elbows, transitions, offsets, plenums, manual dampers, and branch takes-offs.
- (3) Report static pressure drop across outside air and relief/exhaust air louvers.
- (4) Report static pressure readings of supply air, return air, exhaust/relief air, and outside air in duct at the point where these ducts connect to each air moving unit and also at the following locations:

Main Duct: Take readings at four locations along the full length of the main duct 50 percent and 100 percent of the total duct length.

VAV Terminals: Take readings at inlet static pressure at VAV terminal box primary air branch ducts.

VAV Terminals, Fan Powered: Take readings at fan discharge and inlet static pressures for series and parallel fan powered VAV terminal boxes.

- d. Duct Traverses: Report duct traverses for main supply, return, exhaust, relief and outside air ducts. This includes all ducts, including those which lack 7 1/2 duct diameters upstream and 2 1/2 duct diameters downstream of straight duct unobstructed by duct fittings/offsets/elbows. The TAB Agency must evaluate and report findings on the duct traverses taken. Evaluate the suitability of the duct traverse measurement based on satisfying the qualifications for a pilot traverse plane as defined by AMCA 203, "Field Measurements", Section 8, paragraph 8.3, "Location of Traverse Plane."
- e. Instruments: List the types of instruments actually used to measure the tab data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date.
- f. Instrumentation, used for taking wet bulb temperature readings must provide accuracy of plus or minus 5 percent at the measured face velocities. Submit instrument manufacturer's literature to document instrument accuracy performance is in compliance with that specified.
- g. Certification: Include the typed name of the TAB supervisor and the dated signature of the TAB supervisor.
- h. Performance Curves: The TAB Supervisor must include, in the TAB Reports, factory pump curves and fan curves for pumps and fans TAB'd on the job.
- i. Calibration Curves: The TAB Supervisor must include, in the TAB Reports, a factory calibration curve for installed flow control

balancing valves, flow venturi's and flow orifices TAB'd on the job.

1.6 PROJECT/SITE CONDITIONS

1.6.1 TAB Services to Obtain Existing Conditions

Conduct TAB of the indicated existing systems and equipment and submit the specified TAB reports for approval. Conduct this TAB work in accordance with the requirements of this section.

1.7 SEQUENCING AND SCHEDULING

1.7.1 Projects with Phased Construction

This specification section is structured as though the HVAC construction, and thereby the TAB work, will be completed in a single phase. When the construction is completed in phases, the TAB work must be planned, completed, and accepted for each construction phase.

1.7.1.1 Phasing of Work

This specification section is structured as though the HVAC construction, and thereby the TAB work, is going to be completed in a single phase. All elements of the TAB work are addressed on this premise. When a contract is to be completed in construction phases, including the TAB work, the TAB work must be planned for, completed and approved by the Contracting Officer with each phase. An example of this case would be one contract that requires the rehabilitation of the HVAC in each of several separated buildings. At the completion of the final phase, compile all approved reports and submit as one document.

1.7.2 TAB Submittal and Work Schedule

Submit this schedule, and TAB Schematic Drawings, adapted for this particular contract, to the Contracting Officer (CO) for review and approval. Include with the submittal the planned calendar dates for each submittal or work item. Resubmit an updated version for CO approval every 90 calendar days. Compliance with the following schedule is the Contractor's responsibility.

Qualify TAB Personnel: Within 45 calendar days after date of contract award, submit TAB agency and personnel qualifications.

Pre-TAB Meeting: Within 30 calendar days after the date of approval of the TAB agency and personnel, meet with the COTR.

Design Review Report: Within 60 calendar days after the date of the TAB agency personnel qualifications approval, submit design review report.

Pre-Field TAB Engineering Report: Within 30 calendar days after approval of the TAB agency Personnel Qualifications, submit the Pre-Field TAB Engineering Report.

Prerequisite HVAC Work Check Out List For Season 1 and Advanced Notice For Season 1 TAB Field Work: At a minimum of 115 calendar days prior to CCD, submit Season 1 prerequisite HVAC work check out list certified as complete, and submit advance notice of commencement of Season 1 TAB field work.

Season 1 TAB Field Work: At a minimum of 90 calendar days prior to CCD, and when the ambient temperature is within Season 1 limits, accomplish Season 1 TAB field work; submit Season 1 TAB report; and conduct Season 1 field check.

Complete Season 1 TAB Work: Prior to CCD, complete all TAB work except Season 2 TAB work.

Prerequisite HVAC Work Check Out List For Season 2 and Advanced Notice For Season 2 TAB Field Work: Within 150 calendar days after date of the commencement of the Season 1 TAB field work, submit the Season 2 prerequisite HVAC work check out list certified as complete and submit advance notice of commencement of Season 2 TAB field work.

Season 2 TAB Field Work: Within 180 calendar days after date of commencement of the Season 1 TAB field work and when the ambient temperature is within Season 2 limits, accomplish Season 2 TAB field work; submit Season 2 TAB report; and conduct Season 2 field check.

Complete Season 2 TAB Work: Within 15 calendar days after the completion of Season 2 field data check, complete TAB work.

1.7.2.1 TAB Design Review Report

Submit typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the duct leakage testing work and the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. State that no deficiencies are evident if that is the case.

1.7.2.2 TAB Pre-Field Engineering Report

Submit report containing the following information:

a. Step-by-step TAB procedure:

- (1) Strategy: Describe the method of approach to the TAB field work from start to finish. Include in this description a complete methodology for accomplishing each seasonal TAB field work session.
- (2) Air System Diagrams: Use the contract drawings and duct fabrication drawings if available to provide air system diagrams in the report showing the location of all terminal outlet supply, return, exhaust and transfer registers, grilles and diffusers. Use a key numbering system on the diagrams which identifies each outlet contained in the outlet airflow report sheets. Show intended locations of all traverses and static pressure readings.
- (3) Procedural steps: Delineate fully the intended procedural steps to be taken by the TAB field team to accomplish the required TAB work of each air distribution system and each water distribution system. Include intended procedural steps for TAB work for subsystems and system components.

b. Pre-field data: Submit AABC or NEBB or SMACNA 1780 data report forms

with the following pre-field information filled in:

- (1) Design data obtained from system drawings, specifications, and approved submittals.
 - (2) Notations detailing additional data to be obtained from the contract site by the TAB field team.
 - (3) Designate the actual data to be measured in the TAB field work.
 - (4) Provide a list of the types of instruments, and the measuring range of each, which are anticipated to be used for measuring in the TAB field work. By means of a keying scheme, specify on each TAB data report form submitted, which instruments will be used for measuring each item of TAB data. If the selection of which instrument to use, is to be made in the field, specify from which instruments the choice will be made. Place the instrument key number in the blank space where the measured data would be entered.
- c. Prerequisite HVAC work checkout list: Provide a list of inspections and work items which are to be completed by the Contractor. This list must be acted upon and completed by the Contractor and then submitted and approved by the Contracting Officer prior to the TAB team coming to the contract site.
- d. At a minimum, a list of the applicable inspections and work items listed in the NEBB PROCEDURAL STANDARDS, Section III, "Preliminary TAB Procedures" under paragraphs titled, "Air Distribution System Inspection" and "Hydronic Distribution System Inspection" must be provided for each separate system to be TAB'd.
- 1.8 WARRANTY

Furnish workmanship and performance warranty for the TAB system work performed for a period not less than 1 years from the date of Government acceptance of the work; issued directly to the Government. Include provisions that if within the warranty period the system shows evidence of major performance deterioration, or is significantly out of tolerance, resulting from defective TAB workmanship, the corrective repair or replacement of the defective materials and correction of the defective workmanship is the responsibility of the TAB firm. Perform corrective action that becomes necessary because of defective materials and workmanship while system TAB is under warranty 7 days after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within the specified period of time constitutes grounds for having the corrective action and repairs performed by others and the cost billed to the TAB firm. The Contractor must also provide a 1 year contractor installation warranty.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 WORK DESCRIPTIONS OF PARTICIPANTS

Comply with requirements of this section.

3.2 PRETAB MEETING

Meet with the Contracting Officer's technical representative (COTR) to develop a mutual understanding relative to the details of the TAB work requirements. Ensure that the TAB supervisor is present at this meeting. Requirements to be discussed include required submittals, work schedule, and field quality control.

3.3 TAB PROCEDURES

3.3.1 TAB Field Work

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents.

That is, comply with the requirements of AABC MN-1 or SMACNA 1780 (TABB) and SMACNA 1858 (TABB), except as supplemented and modified by this section.

Provide instruments and consumables required to accomplish the TAB work. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents. Conduct TAB work, including measurement accuracy, and sound measurement work in conformance with the AABC MN-1 and AABC MN-4, or NEBB TABES and NEBB MASV, or SMACNA 1780 (used by TABB) and SMACNA 1858 sound measurement procedures, except as supplemented and modified by this section.

3.3.2 Preliminary Procedures

Use the approved pre-field engineering report as instructions and procedures for accomplishing TAB field work. TAB engineer is to locate, in the field, test ports required for testing. It is the responsibility of the sheet metal contractor to provide and install test ports as required by the TAB engineer.

3.3.3 TAB Air Distribution Systems

3.3.3.1 Units With Coils

Report heating and cooling performance capacity tests for hot water, chilled water, DX and steam coils for the purpose of verifying that the coils meet the indicated design capacity. Submit the following data and calculations with the coil test reports:

- a. For air handlers with capacities greater than 26,370 Watts cooling, such as factory manufactured units, central built-up units and rooftop units, conduct capacity tests in accordance with AABC MN-4, procedure 3.5, "Coil Capacity Testing."
- b. Do not determine entering and leaving wet and dry bulb temperatures by single point measurement, but by the average of multiple readings in compliance with paragraph 3.5-5, "Procedures", (in subparagraph d.) of AABC MN-4, Procedure 3.5, "Coil Capacity Testing."

- c. Submit part-load coil performance data from the coil manufacturer converting test conditions to design conditions; use the data for the purpose of verifying that the coils meet the indicated design capacity in compliance with AABC MN-4, Procedure 3.5, "Coil Capacity Testing," paragraph 3.5.7, "Actual Capacity Vs. Design Capacity" (in subparagraph c.).
- d. For units with capacities of 26370 Watts or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, such as through-the-wall heat pumps:
- e. Determine the apparent coil capacity by calculations using single point measurement of entering and leaving wet and dry bulb temperatures; submit the calculations with the coil reports.

3.3.3.2 Air Handling Units

Air handling unit systems including fans (air handling unit fans, exhaust fans and winter ventilation fans), coils, ducts, plenums, mixing boxes, terminal units, variable air volume boxes, and air distribution devices for supply air, return air, outside air, mixed air relief air, and makeup air.

3.3.3.3 Fan Coils

Fan coil unit systems including fans, coils, ducts, plenums, and air distribution devices for supply air, return air, and outside air.

3.3.3.4 Exhaust Fans

Exhaust fan systems including fans, ducts, plenums, grilles, and hoods for exhaust air.

3.3.4 TAB Water Distribution Systems

3.3.4.1 Chilled Water

Chilled water systems including chillers, condensers, cooling towers, pumps, coils, system balance valves and flow measuring devices.

For water chillers, report data as required by AABC, NEBB and TABB standard procedures, including refrigeration operational data.

3.3.5 TAB Work on Performance Tests Without Seasonal Limitations

3.3.5.1 Performance Tests

In addition to the TAB proportionate balancing work on the air distribution systems and the water distribution systems, accomplish TAB work on the HVAC systems which directly transfer thermal energy. TAB the operational performance of the cooling systems.

3.3.5.2 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data

taking.

3.3.5.3 Water Chillers

For water chillers, report data as required by NEBB Form TAB 15-83, NEBB PROCEDURAL STANDARDS, including refrigeration operational data.

3.3.5.4 Refrigeration Units

For refrigeration compressors/condensers/condensing units, report data as required by NEBB Form TAB 15-83, NEBB PROCEDURAL STANDARDS, including refrigeration operational data.

3.3.5.5 Coils

Report heating and cooling performance capacity tests for , chilled water, DX for the purpose of verifying that the coils meet the indicated design capacity. Submit the following data and calculations with the coil test reports:

- a. For Central station air handlers with capacities greater than 26,370 Watts cooling, such as factory manufactured units, central built-up units and rooftop units, conduct capacity tests in accordance with AABC MN-4, procedure 3.5, "Coil Capacity Testing".
- b. Entering and leaving wet and dry bulb temperatures are not determined by single point measurement, but the average of multiple readings in compliance with paragraph 3.5-5, "Procedures", (in subparagraph d.) of AABC MN-4, Procedure 3.5, "Coil Capacity Testing."
- c. Submit part-load coil performance data from the coil manufacturer converting test conditions to design conditions; use the data for the purpose of verifying that the coils meet the indicated design capacity in compliance with AABC MN-4, Procedure 3.5, "Coil Capacity Testing," paragraph 3.5.7, "Actual Capacity Vs. Design Capacity" (in subparagraph c.).
- d. For units with capacities of 26370 Watts or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, such as through-the-wall heat pumps:
- e. Determine the apparent coil capacity by calculations using single point measurement of entering and leaving wet and dry bulb temperatures; submit the calculations with the coil reports.

3.3.6 Workmanship

Conduct TAB work on the HVAC systems until measured flow rates are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents. This TAB work includes adjustment of balancing valves, balancing dampers, and sheaves. Further, this TAB work includes changing out fan sheaves and pump impellers if required to obtain air and water flow rates specified or indicated. If, with these adjustments and equipment changes, the specified or indicated design flow rates cannot be attained, contact the Contracting Officer for direction.

3.3.7 Deficiencies

Strive to meet the intent of this section to maximize the performance of

the equipment as designed and installed. However, if deficiencies in equipment design or installation prevent TAB work from being accomplished within the range of design values specified in the paragraph WORKMANSHIP, provide written notice as soon as possible to the Contractor and the Contracting Officer describing the deficiency and recommended correction.

Responsibility for correction of installation deficiencies is the Contractor's. If a deficiency is in equipment design, call the TAB team supervisor for technical assistance. Responsibility for reporting design deficiencies to Contractor is the TAB team supervisor's.

3.3.8 TAB Reports

After completion of the TAB work, prepare a pre-final TAB report using the reporting forms approved in the pre-field engineering report. Data required by those approved data report forms is to be furnished by the TAB team. Except as approved otherwise in writing by the Contracting Officer, the TAB work and the TAB report is considered incomplete until the TAB work is accomplished to within the accuracy range specified in the paragraph WORKMANSHIP of this section.

Prepare the report neatly and legibly; the pre-final TAB report is the final TAB report minus the TAB supervisor's review and certification. Obtain, at the contract site, the TAB supervisor's review and certification of the TAB report.

Verbally notify the COTR that the field check of the TAB report data can commence; give this verbal notice 48 hours in advance of field check commencement. Do not schedule field check of the TAB report until the specified workmanship requirements have been met or written approval of the deviations from the requirements have been received from the Contracting Officer.

3.3.9 Quality Assurance - COTR TAB Field Acceptance Testing

3.3.9.1 TAB Field Acceptance Testing

During the field acceptance testing, verify, in the presence of the COTR, random selections of data (water, air quantities, air motion, recorded in the TAB Report. Points and areas for field acceptance testing are to be selected by the COTR. Measurement and test procedures are the same as approved for TAB work for the TAB Report.

Field acceptance testing includes verification of TAB Report data recorded for the following equipment groups:

Group 1: All chillers, computer room units, and air handling units.

Group 2: 25 percent of the VAV terminal boxes and associated diffusers and registers.

Group 3: 25 percent of the supply diffusers, registers, grilles associated with constant volume air handling units.

Group 4: 25 percent of the return grilles, return registers, exhaust grilles and exhaust registers.

Group 5: 25 percent of the supply fans, exhaust fans, and pumps.

Further, if any data on the TAB Report for Groups 2 through 5 is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, additional group data verification is required in the presence of the COTR. Verify TAB Report data for one additional piece of equipment in that group. Continue this additional group data verification until out-of-tolerance data ceases to be found.

3.3.9.2 Additional COTR TAB Field Acceptance Testing

If any of the acceptance testing measurements for a given equipment group is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, terminate data verification for all affected data for that group. The affected data for the given group will be disapproved. Make the necessary corrections and prepare a revised TAB Report. Reschedule acceptance testing of the revised report data with the COTR.

Further, if any data on the TAB Report for a given field acceptance test group is out-of-tolerance, then field test data for one additional field test group as specified herein. Continue this increase field test work until out-of-tolerance data ceases to be found. This additional field testing is up and above the original 25 percent of the of reported data entries to be field tested.

If there are no more similar field test groups from which to choose, additional field testing from another, but different, type of field testing group must be tested.

3.3.9.3 Prerequisite for Approval

Compliance with the field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer approval of the TAB Report submitted.

3.4 MARKING OF SETTINGS

Upon the final TAB work approval, permanently mark the settings of HVAC adjustment devices including valves, gauges, splitters, and dampers so that adjustment can be restored if disturbed at any time. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the submitted TAB report.

3.5 MARKING OF TEST PORTS

The TAB team is to permanently and legibly mark and identify the location points of the duct test ports. If the ducts have exterior insulation, make these markings on the exterior side of the duct insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

-- End of Section --

SECTION 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS
02/13

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. At the discretion of the Government, the manufacturer of any material supplied will be required to furnish test reports pertaining to any of the tests necessary to assure compliance with the standard or standards referenced in this specification.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM C647	(2008; R 2013) Properties and Tests of Mastics and Coating Finishes for Thermal Insulation
ASTM C921	(2010) Standard Practice for Determining the Properties of Jacketing Materials for Thermal Insulation
ASTM C1136	(2021) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
ASTM D882	(2012) Tensile Properties of Thin Plastic Sheeting
ASTM D5590	(2000; R 2010; E 2012) Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E2231	(2018) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide
<https://www.approvalguide.com/>

GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2758 (2014) Paper - Determination of Bursting Strength

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 90B (2018) Standard for the Installation of Warm Air Heating and Air Conditioning Systems

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2017) Adhesive and Sealant Applications

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-24179 (1969; Rev A; Am 2 1980; Notice 1 1987) Adhesive, Flexible Unicellular-Plastic Thermal Insulation

MIL-A-3316 (1987; Rev C; Am 2 1990) Adhesives, Fire-Resistant, Thermal Insulation

MIL-PRF-19565 (1988; Rev C) Coating Compounds, Thermal Insulation, Fire- and Water-Resistant, Vapor-Barrier

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5538 (2018) Adhesives for Wall and Ceiling Boards

JIS A 9504 (2021) Man Made Mineral Fibre Thermal Insulation Materials

JIS G 4304 (2021) Hot-Rolled Stainless Steel Plate, Sheet and Strip

JIS H 4000 (2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M (2019) Public Building Construction Standard Specification

UL SOLUTIONS (UL)

UL 723	(2020) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
UL 2818	(2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168	(2017) Adhesive and Sealant Applications
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1.2 SYSTEM DESCRIPTION

1.2.1 General

Provide field-applied insulation and accessories on mechanical systems as specified herein; factory-applied insulation is specified under the piping, duct or equipment to be insulated. Field applied insulation materials required for use on Government-furnished items as listed in the SPECIAL CONTRACT REQUIREMENTS shall be furnished and installed by the Contractor.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submit the three SD types, SD-02 Shop Drawings, SD-03 Product Data, and SD-08 Manufacturer's Instructions at the same time for each system.

SD-02 Shop Drawings

Pipe Insulation Systems and Associated Accessories

Duct Insulation Systems and Associated Accessories

Equipment Insulation Systems and Associated Accessories

1.4 CERTIFICATIONS

1.4.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein. Include VOC content requirements of SCAQMD Rule 1168, GS-36, or F 4-Star.

1.5 QUALITY ASSURANCE

1.5.1 Installer Qualification

Qualified installers shall have successfully completed three or more similar type jobs within the last 5 years.

1.6 DELIVERY, STORAGE, AND HANDLING

Materials shall be delivered in the manufacturer's unopened containers. Materials delivered and placed in storage shall be provided with protection from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval shall have manufacturer's stamp or label attached giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable). Insulation packages and containers shall be asbestos free.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. The product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation shall be included. The product data must be copyrighted, have an identifying or publication number, and shall have been published prior to the issuance date of this solicitation. Materials furnished under this section shall be submitted together in a booklet.

2.1.1 Insulation System

Provide insulation systems in accordance with MLIT-M. Provide field-applied insulation for heating, ventilating, and cooling (HVAC) air distribution systems and piping systems that are located within, on, under, and adjacent to buildings; and for plumbing systems.

2.1.2 Surface Burning Characteristics

Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flame spread, and smoke developed indexes, shall be determined by ASTM E84 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ASTM E2231.

2.2 MATERIALS

Insulation exterior shall be cleanable, grease resistant, non-flaking and non-peeling. Materials shall be compatible and shall not contribute to corrosion, soften, or otherwise attack surfaces to which applied in either

wet or dry state.

2.2.1 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants must meet limit requirements of "Other" category within SCAQMD Rule 1168 sealants table). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

2.2.1.1 Acoustical Lining Insulation Adhesive

Adhesive shall be a nonflammable, fire-resistant adhesive conforming to JIS A 5538.

2.2.1.2 Mineral Fiber Insulation Cement

Cement shall be in accordance with JIS A 9504.

2.2.1.3 Lagging Adhesive

Lagging is the material used for thermal insulation, especially around a cylindrical object. This may include the insulation as well as the cloth/material covering the insulation. To resist mold/mildew, lagging adhesive shall meet ASTM D5590 with 0 growth rating. Lagging adhesives shall be nonflammable and fire-resistant and shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Adhesive shall be MIL-A-3316, Class 1, pigmented white and be suitable for bonding fibrous glass cloth to faced and unfaced fibrous glass insulation board; for bonding cotton brattice cloth to faced and unfaced fibrous glass insulation board; for sealing edges of and bonding glass tape to joints of fibrous glass board; for bonding lagging cloth to thermal insulation; or Class 2 for attaching fibrous glass insulation to metal surfaces. Lagging adhesives shall be applied in strict accordance with the manufacturer's recommendations for pipe and duct insulation.

2.2.1.4 Contact Adhesive

Adhesives may be any of, but not limited to, the neoprene based, rubber based, or elastomeric type that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. The adhesive shall not adversely affect, initially or in service, the insulation to which it is applied, nor shall it cause any corrosive effect on metal to which it is applied. Any solvent dispersing medium or volatile component of the adhesive shall have no objectionable odor and shall not contain any benzene or carbon tetrachloride. The dried adhesive shall not emit nauseous, irritating, or toxic volatile matters or aerosols when the adhesive is heated to any temperature up to 100 degrees C. The dried adhesive shall be nonflammable and fire resistant. Flexible Elastomeric Adhesive: Comply with MIL-A-24179, Type II, Class I. Provide product listed in FM APP GUIDE.

2.2.2 Caulking

Caulking shall be in accordance with MLIT-M.

2.2.3 Corner Angles

Nominal 0.406 mm aluminum 25 by 25 mm with factory applied kraft backing. Aluminum shall be JIS H 4000.

2.2.4 Finishing Cement

MLIT-M: Mineral fiber hydraulic-setting thermal insulating and finishing cement. All cements that may come in contact with Austenitic stainless steel must comply with MLIT-M.

2.2.5 Fibrous Glass Cloth and Glass Tape

Fibrous glass cloth, with 20X20 maximum mesh size, and glass tape shall be have maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Tape shall be 100 mm wide rolls. Class 3 tape shall be 0.15 kg/square m. Elastomeric Foam Tape: Black vapor-retarder foam tape with acrylic adhesive containing an anti-microbial additive.

2.2.6 Staples

Outward clinching type JIS G 4304, Type 304 or 316 stainless steel.

2.2.7 Jackets

2.2.7.1 Stainless Steel Jackets

ASTM A240/A240M; Type 304, minimum thickness of 33 gauge (0.25mm), smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 13 mm.

2.2.7.2 Vapor Barrier/Weatherproofing Jacket

Vapor barrier/weatherproofing jacket shall be in accordance with MLIT-M, Table 2.3.1 and 2.3.2.

2.2.8 Vapor Retarder Required

ASTM C921, Type I, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork, where a minimum puncture resistance of 25 Beach units is acceptable. Minimum tensile strength, 6.1 N/mm width. ASTM C921, Type II, minimum puncture resistance 25 Beach units, tensile strength minimum 3.5 N/mm width. Jackets used on insulation exposed in finished areas shall have white finish suitable for painting without sizing. Based on the application, insulation materials that require manufacturer or fabricator applied pipe insulation jackets are cellular glass, when all joints are sealed with a vapor barrier mastic, and mineral fiber. All non-metallic jackets shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible elastomerics require (in addition to vapor barrier skin) vapor retarder jacketing for high relative humidity and below ambient temperature applications.

2.2.8.1 Vapor Retarder/Vapor Barrier Mastic Coatings

2.2.8.1.1 Vapor Barrier

The vapor barrier shall be self adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) greater than 3 plies standard grade, silver, white, black and embossed white jacket for use on hot/cold pipes. Permeability shall be less than 0.02 when tested in accordance with ASTM E96/E96M. Products shall meet UL 723 or ASTM E84. Flame and smoke requirements and shall be UV resistant.

2.2.8.1.2 Vapor Retarder

The vapor retarder coating shall be fire and water resistant and appropriately selected for either outdoor or indoor service. Color shall be white. The water vapor permeance of the compound shall be 0.013 perms or less at 1 mm dry film thickness as determined according to procedure B of ASTM E96/E96M or utilizing apparatus described in ASTM E96/E96M. The coating shall be nonflammable, fire resistant type. To resist mold/mildew, coating shall meet ASTM D5590 with 0 growth rating. Coating shall meet MIL-PRF-19565 Type II (if selected for indoor service) and be Qualified Products Database listed. All other application and service properties shall be in accordance with ASTM C647.

2.2.8.2 Laminated Film Vapor Retarder

ASTM C1136, Type I, maximum moisture vapor transmission 0.02 perms, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork; where Type II, maximum moisture vapor transmission 0.02 perms, a minimum puncture resistance of 25 Beach units is acceptable. Vapor retarder shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible Elastomeric exterior foam with factory applied UV Jacket. Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

2.2.8.3 Polyvinylidene Chloride (PVDC) Film Vapor Retarder

The PVDC film vapor retarder shall have a maximum moisture vapor transmission of 0.02 perms, minimum puncture resistance of 150 Beach units, a minimum tensile strength in any direction of 5.3 kN/m when tested in accordance with ASTM D882, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

2.2.8.4 Vapor Barrier/Weather Barrier

The vapor barrier shall be greater than 3 ply self adhesive laminate -white vapor barrier jacket- superior performance (less than 0.0000 permeability when tested in accordance with ASTM E96/E96M). Vapor barrier shall meet UL 723 or ASTM E84. 25 flame and 50 smoke requirements; and UV resistant. Minimum burst strength 1.3 MPa in accordance with ISO 2758. Tensile strength 0.12 kg/m width (PSTC-1000). Tape shall be as specified for laminated film vapor barrier above.

2.2.9 Vapor Retarder Not Required

ASTM C921, Type II, Class D, minimum puncture resistance 50 Beach units on all surfaces except ductwork, where maximum moisture vapor transmission 0.10, a minimum puncture resistance of 25 Beach units is acceptable.

Jacket shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

2.2.10 Wire

Wire shall be in accordance with MLIT-M.

2.2.11 Sealants

Sealants shall be chosen from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Sealants shall have a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

2.3 PIPE INSULATION SYSTEMS

Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet. Limit pipe insulation materials to those listed herein and meeting the following requirements:

2.3.1 Aboveground Cold Pipeline (-34 to 16 deg. C)

Insulation for outdoor, indoor, exposed or concealed applications, shall be as follows:

2.3.1.1 Mineral Fiber Insulation with Integral Wicking Material (MFIWM)

JIS A 9504. Install in accordance with manufacturer's instructions. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.

2.3.2 Aboveground Hot Pipeline (Above 16 deg. C)

Insulation for outdoor, indoor, exposed or concealed applications shall meet the following requirements. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

2.3.2.1 Mineral Fiber

JIS A 9504, supply the insulation with manufacturer's recommended factory-applied jacket.

2.4 DUCT INSULATION SYSTEMS

2.4.1 Field Applied Insulation

Provide field applied insulation according to manufacturer's installation guidelines. Provide manufacturer's standard reinforced fire-retardant vapor barrier, with identification of installed thermal resistance (R) value and out-of-package R values.

2.4.2 Duct Insulation Jackets

2.4.2.1 All-Purpose Jacket

Provide insulation with insulation manufacturer's standard reinforced fire-retardant jacket with or without integral vapor barrier as required by the service. In exposed locations, provide jacket with a white surface

suitable for field painting.

2.4.2.2 Metal Jackets

2.4.2.2.1 Aluminum Jackets

JIS H 4000, with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide smooth surface jackets for jacket outside dimension 200 mm and larger. Provide corrugated surface jackets for jacket outside dimension 200 mm and larger. Provide stainless steel bands, minimum width of 13 mm.

2.4.2.2.2 Stainless Steel Jackets

JIS G 4304; Type 304, smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 13 mm.

2.5 EQUIPMENT INSULATION SYSTEMS

Insulate equipment and accessories as specified in accordance with MLIT-M, Chapter 3.

PART 3 EXECUTION

3.1 APPLICATION - GENERAL

Insulation shall only be applied to unheated and uncooled piping and equipment. Flexible elastomeric cellular insulation shall not be compressed at joists, studs, columns, ducts, hangers, etc. The insulation shall not pull apart after a one hour period; any insulation found to pull apart after one hour, shall be replaced.

3.1.1 Display Samples

Submit and display, after approval of materials, actual sections of installed systems, properly insulated in accordance with the specification requirements. Such actual sections must remain accessible to inspection throughout the job and will be reviewed from time to time for controlling the quality of the work throughout the construction site. Each material used shall be identified, by indicating on an attached sheet the specification requirement for the material and the material by each manufacturer intended to meet the requirement. The Contracting Officer will inspect display sample sections at the jobsite. Approved display sample sections shall remain on display at the jobsite during the construction period. Upon completion of construction, the display sample sections will be closed and sealed.

3.1.1.1 Pipe Insulation Display Sections

Display sample sections shall include as a minimum an elbow or tee, a valve, dielectric waterways and flanges, a hanger with protection shield and insulation insert, or dowel as required, at support point, method of fastening and sealing insulation at longitudinal lap, circumferential lap, butt joints at fittings and on pipe runs, and terminating points for each type of pipe insulation used on the job, and for hot pipelines and cold pipelines, both interior and exterior, even when the same type of insulation is used for these services.

3.1.1.2 Duct Insulation Display Sections

Display sample sections for rigid and flexible duct insulation used on the job. Use a temporary covering to enclose and protect display sections for duct insulation exposed to weather

3.1.2 Installation

Except as otherwise specified, material shall be installed in accordance with the manufacturer's written instructions. Insulation materials shall not be applied until tests specified in other sections of this specification are completed. Material such as rust, scale, dirt and moisture shall be removed from surfaces to receive insulation. Insulation shall be kept clean and dry. Insulation shall not be removed from its shipping containers until the day it is ready to use and shall be returned to like containers or equally protected from dirt and moisture at the end of each workday. Insulation that becomes dirty shall be thoroughly cleaned prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, the insulation will be rejected, and shall be immediately removed from the jobsite. Joints shall be staggered on multi layer insulation. Mineral fiber thermal insulating cement shall be mixed with demineralized water when used on stainless steel surfaces. Insulation, jacketing and accessories shall be installed in accordance with plates except where modified herein or on the drawings.

3.1.3 Firestopping

Where pipes and ducts pass through fire walls, fire partitions, above grade floors, and fire rated chase walls, the penetration shall be sealed with fire stopping materials. The protection of ducts at point of passage through firewalls must be in accordance with NFPA 90A and/or NFPA 90B. All other penetrations, such as piping, conduit, and wiring, through firewalls must be protected with a material or system of the same hourly rating that is listed by MLIT-M.

3.1.4 Painting and Finishing

Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.1.5 Welding

No welding shall be done on piping, duct or equipment without written approval of the Contracting Officer. The capacitor discharge welding process may be used for securing metal fasteners to duct.

3.1.6 Pipes/Ducts/Equipment That Require Insulation

Insulation is required on all pipes, ducts, or equipment, except for omitted items as specified.

3.2 PIPE INSULATION SYSTEMS INSTALLATION

Install pipe insulation systems in accordance with the approved plates as supplemented by the manufacturer's published installation instructions.

3.2.1 Pipe Insulation

3.2.1.1 General

Pipe insulation shall be installed on aboveground hot and cold pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise. Installation shall be with full length units of insulation and using a single cut piece to complete a run. Cut pieces or scraps abutting each other shall not be used. Pipe insulation shall be omitted on the following:

- a. Pipe used solely for fire protection.
- b. Chromium plated pipe to plumbing fixtures. However, fixtures for use by the physically handicapped shall have the hot water supply and drain, including the trap, insulated where exposed.
- c. Access plates of fan housings.

3.2.1.2 Pipe Insulation Material and Thickness

Pipe insulation materials must be as listed in Table 1.

TABLE 1					
Insulation Material for Piping					
Service					
	Material	Specification	Type	Class	VR/VB Req'd
Chilled Water (Supply & Return, Dual Temperature Piping, 40 F nominal)					
	Mineral Fiber with Wicking Material. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.	JIS A 9504	I		Yes
Heating Hot Water Supply & Return, Heated Oil (Max 250 F)					
	Mineral Fiber	JIS A 9504			
Cold Domestic Water Piping, Makeup Water & Drinking Fountain Drain Piping					
	Mineral Fiber	JIS A 9504			
Hot Domestic Water Supply & Recirculating Piping (Max 200 F)					
	Mineral Fiber	JIS A 9504			
Refrigerant Suction Piping (35 degrees F nominal)					
	Shall be pre-insulated by manufacturer				
Compressed Air Discharge, Steam and Condensate Return (201 to 250 Degrees F)					
	Mineral Fiber	JIS A 9504			

TABLE 1					
Insulation Material for Piping					
Service					
	Material	Specification	Type	Class	VR/VB Req'd
Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel					
	Mineral Fiber	JIS A 9504			
Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)					
	Mineral Fiber	JIS A 9504			
Condensate Drain Located Inside Building					
	Mineral Fiber	JIS A 9504			
Medium Temperature Hot Water, Steam and Condensate (251 to 350 Degrees F)					
	Mineral Fiber	JIS A 9504			
High Temperature Hot Water & Steam (351 to 700 Degrees F)					
	Mineral Fiber	JIS A 9504			
Brine Systems Cryogenics (-30 to 0 Degrees F)					
	Mineral Fiber	JIS A 9504			
Brine Systems Cryogenics (0 to 34 Degrees F)					
	Mineral Fiber	JIS A 9504			
Note: VR/VB = Vapor Retarder/Vapor Barrier					

TABLE 2						
Piping Insulation Thickness (inch) Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4.						
Service						
	Material	Tube And Pipe Size (inch)				
		<1	1-<1.5	1.5-<4	4-<8	> or = >8
{Chilled Water (Supply & Return, Dual Temperature Piping, 40 Degrees F nominal)}						
	Mineral Fiber with Wicking Material	1	1.5	1.5	2	2
{Chilled Water (Supply & Return, Dual Temperature Piping, 40 Degrees F nominal)}						
	Mineral Fiber with Wicking Material	1	1.5	1.5	2	2
Heating Hot Water Supply & Return, Heated Oil (Max 250 F)						
	Mineral Fiber	1.5	1.5	2	2	2
Cold Domestic Water Piping, Makeup Water & Drinking Fountain Drain Piping						
	Mineral Fiber	1.5	1.5	50	1.5	1.5
	Flexible Elastomeric Cellular	1	1	1	N/A	N/A
Hot Domestic Water Supply & Recirculating Piping (Max 200 F)						
	Mineral Fiber	1	1	1	1.5	1.5
Refrigerant Suction Piping (35 degrees F nominal)						
	Shall be pre-insulated by manufacturer					
Compressed Air Discharge, Steam and Condensate Return (201 to 250 Degrees F)						
	Mineral Fiber	1.5	1.5	2	2	2
		1.5*	2*	2.5*	3*	3.5*
Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel						
	Mineral Fiber	40	40	0.5	0.5	0.5
Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)						

TABLE 2						
Piping Insulation Thickness (inch) Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4.						
Service						
	Material	Tube And Pipe Size (inch)				
		<1	1-<1.5	1.5-<4	4-<8	> or = >8
	Mineral Fiber	1.5	1.5	1.5	1.5	1.5
Condensate Drain Located Inside Building						
	Mineral Fiber	1.5	1.5	1.5	1.5	50
Medium Temperature Hot Water, Steam and Condensate (251 to 350 Degrees F)						
	Mineral Fiber	1.5	3	3	4	4
		2.5*	*	3.5*		
High Temperature Hot Water & Steam (351 to 700 Degrees F)						
	Mineral Fiber	2.5	3	3	4	4
Brine Systems Cryogenics (-30 to 0 Degrees F)						
	Mineral Fiber	40	40	50	50	50
Brine Systems Cryogenics (0 to 34 Degrees F)						
	Mineral Fiber	40	40	2	50	50

3.2.2 Aboveground Cold Pipelines

The following cold pipelines for minus 34 to plus 16 degrees C, shall be insulated in accordance with Table 2 except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted. This includes but is not limited to the following:

- Refrigerant suction lines.
- Chilled water.
- Air conditioner condensate drains.

- d. Exposed lavatory drains and domestic water lines serving plumbing fixtures for handicap persons.

3.2.2.1 Insulation Material and Thickness

Insulation thickness for cold pipelines shall be determined using Table 2.

3.2.3 Piping Exposed to Weather

Piping exposed to weather shall be insulated and jacketed in accordance with MLIT-M.

3.3 DUCT INSULATION SYSTEMS INSTALLATION

Install duct insulation systems in accordance with the approved plates as supplemented by the manufacturer's published installation instructions. Duct insulation minimum thickness and insulation level must be as listed in Table 4.

3.3.1 Duct Insulation Minimum Thickness

Duct insulation minimum thickness in accordance with Table 3.

Table 43 - Minimum Duct Insulation (inches)	
Cold Air Ducts	2.0
Relief Ducts	1.5
Fresh Air Intake Ducts	1.5
Warm Air Ducts	2.0
Relief Ducts	1.5
Fresh Air Intake Ducts	1.5

3.3.2 Insulation and Vapor Retarder/Vapor Barrier for Cold Air Duct

Insulation and vapor retarder/vapor barrier shall be provided for the following cold air ducts and associated equipment.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief ducts.
- d. Flexible run-outs (field-insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil headers and return bends.
- h. Coil casings.

- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes (field-insulated).
- l. Supply fans (field-insulated).
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.
- o. Combustion air intake ducts.

3.3.3 Insulation for Warm Air Duct

Insulation and vapor barrier shall be provided for the following warm air ducts and associated equipment:.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief air ducts
- d. Flexible run-outs (field insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil-headers and return bends.
- h. Coil casings.
- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes.
- l. Supply fans.
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.

3.3.4 Ducts Handling Air for Dual Purpose

For air handling ducts for dual purpose below and above 16 degrees C, ducts shall be insulated as specified for cold air duct.

3.3.5 Duct Test Holes

After duct systems have been tested, adjusted, and balanced, breaks in the insulation and jacket shall be repaired in accordance with the applicable section of this specification for the type of duct insulation to be repaired.

3.3.6 Duct Exposed to Weather

3.3.6.1 Installation

Ducts exposed to weather shall be insulated and finished as specified in accordance with MLIT-M.

3.4 EQUIPMENT INSULATION SYSTEMS INSTALLATION

Install equipment insulation systems in accordance with MLIT-M.

3.4.1 Insulation for Cold Equipment

Cold equipment below 16 degrees C: Insulation shall be furnished on equipment handling media below 16 degrees C including the following:

- a. Pumps.
- b. Refrigeration equipment parts that are not factory insulated.
- c. Drip pans under chilled equipment.
- d. Cold and chilled water pumps.
- e. Air handling equipment parts that are not factory insulated.
- f. Expansion and air separation tanks.

3.4.1.1 Insulation Type

Insulation shall be suitable for the temperature encountered. Material and thicknesses shall be as shown in Table 5:

TABLE 5		
Insulation Thickness for Cold Equipment (inches)		
Equipment handling media at indicated temperature		
	Material	Thickness (inches)
	Mineral Fiber	MLIT-M

3.4.2 Equipment Handling Dual Temperature Media

Below and above 16 degrees C: equipment handling dual temperature media shall be insulated as specified for cold equipment.

3.4.3 Equipment Exposed to Weather

3.4.3.1 Installation

Equipment exposed to weather shall be insulated and finished in accordance with the requirements for ducts exposed to weather in paragraph DUCT INSULATION INSTALLATION.

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SECTION 23 08 00

COMMISSIONING OF MECHANICAL SYSTEMS
05/23, CHG 1: 08/24

PART 1 GENERAL

Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users. Comply with specification Section 01 91 00.15 BUILDING COMMISSIONING.

1.1 SEQUENCING AND SCHEDULING

Complete the following prior to starting Functional Performance Tests of mechanical systems:

- a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications
- b. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- c. Performance Verification Tests of the controls systems have been completed and the Performance Verification Test Report has been submitted and approved in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- d. The Certificate of Readiness submitted and approved in accordance with Section 01 91 00.15 BUILDING COMMISSIONING
- e. Tests, Flushing, and Disinfection in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G

SD-06 Test Reports

Pipe Flushing, Testing, And Water Treatment Reports; G

Completed Pre-Functional Checklists; G

Seasonal Test Report; G

Full-Load Test Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate Of Readiness; G

1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

1.4 COORDINATION

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the Lead Commissioning Specialist in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the Lead Commissioning Specialist. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the Lead Commissioning Specialist as specified in the commissioning plan, as it is completed.

1.5 PIPE FLUSHING, TESTING, AND WATER TREATMENT REPORTS

Test requirements are specified in Division 23 piping Sections. Prepare a pipe system cleaning, flushing, and hydrostatic testing log. Provide cleaning, flushing, testing, and water treatment log and final reports.

Include the following in the pipe system cleaning, flushing, and hydrostatic testing log:

- a. Minimum flushing water velocity.
- b. Water treatment reports.
- c. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

1.6 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports; Performance Verification Test Reports;. Do not schedule Functional

Performance Tests for the system until the Certificate of Readiness for that system receives approval by the Government. The CxC, Quality Control Manager, and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.3 degrees C and a resolution of plus or minus 0.1 degrees C. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

PART 3 EXECUTION

3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the Lead Commissioning Specialist to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Download example Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists for Section 01 91 00.15 BUILDING COMMISSIONING at the following location: <https://www.wbdg.org/dod/ufgs/ufgs-01-91-00-15>. The checklists submitted in the Interim and Final Construction Phase Commissioning Plans must contain the same level of detail shown in the examples. The submitted checklists are not required to match the format of the examples.

3.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS. Pre-functional checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

3.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract and Owner's Project Requirements Document. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

- a. Notable system features including information about controls to facilitate understanding of system operation and maintenance
- b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements and specifications and Owner's Project Requirements Document. Recommendation must clearly indicate that the system should or should not be approved by the Government.
- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions
- d. Attendees
- e. Identification of the equipment involved in the test
- f. **System feature identification**
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station
- h. Actuator operation observations demonstrating actuator responses to commands from the control system

- i. As-found condition of the system operation
- j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- k. Space for comments for each test item.

3.2.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract and Owner's Project Requirements Document. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections in the Integrated Systems Test Checklists:

- a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements and specifications and the Owner's Project Requirements Document. Recommendation must clearly indicate that the systems should or should not be approved by the Government
- c. Test conditions including date and beginning and ending time
- d. Identification of the equipment and systems involved in the test
- e. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- f. Space for comments for each test item.

3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Validation of each item by each team member must occur immediately following completion of their respective duties related to the item. Commissioning Specialist validation of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents and the Owner's Project Requirements Document. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, Quality Control Manager, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB

representative (for items impacting TAB). CxC will witness 100 percent of the Pre-Functional Checklist for critical equipment and a minimum of a single sample of each type of non-critical equipment. Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklists items for each system.

3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

- a. Startup tests and factory testing reports.
- b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.
- c. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.
- d. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing

3.5 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

3.5.1 Testing, Adjusting, and Balancing (TAB) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The CxC witness verification of the Testing, Adjusting, and Balancing shall be a 100 percent sample strategy with any failed results requiring retest. All deficiencies must be resolved prior to TAB Report approval.

3.5.2 Testing, Adjusting, and Balancing (TAB) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The CxC witness verification of the Testing, Adjusting, and Balancing shall be a 100 percent sample strategy with any failed results requiring retest. All deficiencies must be resolved prior to TAB Report approval.

3.6 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

The Mechanical System Technical Commissioning Specialist must review the Start-Up Testing Report, PVT Procedures and PVT Reports, including endurance testing trend submittals, required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The Mechanical System Technical Commissioning Specialist must review each submittal and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All deficiencies must be resolved prior to final acceptance.

3.7 FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING

Functional Performance and Integrated Systems Testing are a type of Commissioning Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents and the Owner's Project Requirements Document.

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

Integrated System Test (IST) must be accomplished without impacting mission equipment power. "Blackout" test shall only impact the power to the HVAC system. Fire system shunt trip may be used for IST if applicable.

3.7.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests and Integrated Systems Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

3.7.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance and integrated systems testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

3.7.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their

pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

3.7.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

- a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).
- b. Dirty filter pressure drops can be simulated by partially blocking filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer, as noted above, for special cases.
- g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 12 degrees C, when the outside air temperature is above 12 degrees C, temporarily change the lockout set point to be minus 17

degrees C above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.

- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

3.7.5 Manufacturer's Representative

Provide a factory trained representative authorized by the equipment manufacturer to perform Functional Performance Testing for the following equipment:

Chillers
Packaged Direct-Expansion Refrigeration Equipment, Fan Coil Units
Condensing Units
Air Handling Units
Exhaust Fans
Pumps
Packaged Computer Room Air Conditioners (CRAC)

Ensure the test representative reviews, approves, and signs the completed field test report. Include person's name with signatures.

3.7.6 Integrated Systems Tests

Follow the Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan. Integrated Systems Tests must be performed for the interactive operation between HVAC systems and other systems such as fire protection systems, smoke control systems, or back-up electrical supply. Verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation or correct operation of uninterruptible power supplies or energy generators and connected systems.

3.7.7 Sample Strategy

Perform Functional Performance Tests using the following sample strategy. Complete a Functional Performance Test Checklist for each item of equipment or system to be tested. No equipment or system to be tested for sample sizes less than 100 percent. Equipment Identifiers are as indicated on the design drawings:

Equipment Identifier	Sample Size (Percent)
AHU	100
Chiller	100

CRAC	100
Condensing Unit	100
DX Fan Coil Units	100
Exhaust Fan	100
Pumps	100

Perform Integrated Systems Tests and complete an Integrated Systems Test Checklist for for all (100 percent) systems and equipment having interactive operation.

3.7.7.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.7.7.2 Less than 100 Percent Sample Procedures

No equipment will have a sample rate of less than 100 percent.

3.8 INTEGRATED SYSTEMS TEST

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

3.9 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

3.10 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

3.11 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

3.12 FULL-LOAD TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover. In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems under full-load conditions. Develop and implement means of artificial loading to demonstrate the ability of the process cooling systems to handle peak process loads. Schedule Full-Load Functional Performance Tests in coordination with the Contracting Officer. Submit Full-Load Test Report within 14 days of test completion.

Execute full-load functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any full load testing.

3.13 TRAINING

The Mechanical Systems Technical Commissioning Specialist must review the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

3.14 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance and Integrated Systems Test Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

3.15 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph ENDURANCE TEST in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required. Use the Temporary Trending Hardware, if necessary, in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract, and the Owner's Project Requirements Document. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation, recommendations for correction, and includes a

graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist. Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

-- End of Section --

DRAFT

SECTION 23 09 00

INSTRUMENTATION AND CONTROL FOR HVAC
08/24

PART 1 GENERAL

1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as indicated and shown and in accordance with Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM, and other referenced Sections.

1.1.1 Proprietary Systems

1.1.1.1 Proprietary Systems Exempted From Open Protocol Requirements

The following systems are specifically exempted from the open protocol requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS:

- a. A simple split (DX) system consisting of a single indoor unit and a single outdoor unit from the same manufacturer.
- b. Systems in Table I (previously approved by the designer in accordance with UFC 3-410-02).

TABLE I: Systems Approved to Use Proprietary Communications		
System	Type (Multi-Split/VRF or Chiller/Boiler Plant)	Proprietary Multi-Split Engineering Tool Software Required (for Multi-Split/VRF only)

- c. A system (not already shown Table I) of multiple boilers or multiple chillers communicating with a proprietary network for which an approved request has been obtained and for which: all units are from the same manufacturer, they are all co-located in the same room, the network connecting them is fully contained in that room, and the units are operating using a common "plant" sequence of operation which stages the units in a manner that requires operational parameters be shared between them and which cannot be accomplished with a single lead-lag command from a third-party controller.

1.1.1.2 Implementation of Proprietary Systems

For proprietary systems exempted from open protocol requirements, a proprietary network and DDC hardware communicating via a proprietary protocol are permitted. For these systems a building control network meeting the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS must also be provided,

along with a gateway or interface to connect the proprietary system to the open building control network.

The proprietary system gateway or interface must provide the required functionality as shown on the points schedule. Scheduling, alarming, trending, overrides, network inputs, network outputs and other protocol related requirements must be met on the open protocol control system as specified in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

1.1.1.3 Proprietary Multi-Split Engineering Tool Software

For each permitted proprietary systems in Table 1 shown as requiring Proprietary Multi-Split Engineering Tool Software, provide the software needed to replace a unit and configure the replacement. Submit hard copies of the software user manuals with the software submittal.

Submit Proprietary Multi-Split Engineering Tool Software on CD-ROM as a Technical Data Package. Submit 3 hard copies of the software user manual for each piece of software.

1.1.2 System Requirements

Provide systems meeting the requirements this Section and other Sections referenced by this Section, and which have the following characteristics:

- a. The system implements the control sequences of operation shown in the Contract Drawings using DDC hardware to control mechanical and electrical equipment
- b. The system meet the requirements of this specification as a stand-alone system and does not require connection to any other system.
- c. Control sequences reside in DDC hardware in the building. The building control network is not dependent upon connection to a Utility Monitoring and Control System (UMCS) Front End or to any other system for performance of control sequences. To the greatest extent practical, the hardware performs control sequences without reliance on the building network.
- d. The hardware is installed such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- e. All necessary documentation, configuration information, programming tools, programs, drivers, and other software are licensed to and otherwise remain with the Government such that the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- f. Sufficient documentation and data, including rights to documentation and data, are provided such that the Government or their agents can execute work to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- g. Hardware is installed and configured such that the Government or their agents are able to perform repair, replacement, and upgrades of

individual hardware without further interaction with the Contractor, Vendor or Manufacturer.

- h. All Niagara Framework components have an unrestricted interoperability license with a Niagara Compatibility Statement (NiCS) following the Tridium Open NiCS Specification and have a value of "ALL" for "Station Compatibility In", "Station Compatibility Out", "Tool Compatibility In" and "Tool Compatibility Out". Note that this will result in the following entries in the license file:

```
accept.station.in="*"
accept.station.out="*"
accept.wb.in="*"
accept.wb.out=""
```

1.1.3 End to End Accuracy

Select products, install and configure the system such that the maximum error of a measured value as read from the DDC Hardware over the network is less than the maximum allowable error specified for the sensor or instrumentation.

1.1.4 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.1.5 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

1.2 RELATED SECTIONS

Related work specified elsewhere:

- a. Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet systems with or without Niagara Framework.
- b. Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
- c. Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM
- d. Section 23 81 00 DECENTRALIZED UNITARY HVAC EQUIPMENT
- e. Section 01 91 00.15 BUILDING COMMISSIONING

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

ASHRAE FUN SI (2021) Fundamentals Handbook, SI Edition

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

TRIDIUM, INC (TRIDIUM)

Niagara Framework (2021) NiagaraAX User's Guide

Tridium Open NiCS (2021) Understanding the NiagaraAX Compatibility Statement (NiCS)

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-410-02 (2018; with Change 2, 2021) Direct Digital Control for HVAC and Other Building Control Systems

UL SOLUTIONS (UL)

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers

1.4 DEFINITIONS

The following list of definitions includes terms used in Sections referenced by this Section and are included here for completeness. The definitions contained in this Section may disagree with how terms are defined or used in other documents, including documents referenced by this Section. The definitions included here are the authoritative definitions for this Section and all Sections referenced by this Section.

After each term the protocol related to that term is included in parenthesis.

1.4.1 Alarm Generation (All protocols)

Alarm Generation is the monitoring of a value, comparison of the value to alarm conditions and the creation of an alarm when the conditions set for the alarm are met. Note that this does NOT include delivery of the alarm to the final destination (such as a user interface).

1.4.2 Building Automation and Control Network (BACnet) (BACnet)

The term BACnet is used in two ways. First meaning the BACnet Protocol Standard - the communication requirements as defined by ASHRAE 135 including all annexes and addenda. The second to refer to the overall technology related to the ASHRAE 135 protocol.

1.4.3 BACnet Advanced Application Controller (B-AAC) (BACnet)

A hardware device BTL Listed as a B-AAC, which is required to support BACnet Interoperability Building Blocks (BIBBs) for scheduling and alarming, but is not required to support as many BIBBs as a B-BC.

1.4.4 BACnet Application Specific Controller (B-ASC) (BACnet)

A hardware device BTL Listed as a B-ASC, with fewer BIBB requirements than a B-AAC. It is intended for use in a specific application.

1.4.5 BACnet Building Controller (B-BC) (BACnet)

A hardware device BTL Listed as a B-BC. A general-purpose, field-programmable device capable of carrying out a variety of building automation and control tasks including control and monitoring via direct digital control (DDC) of specific systems and data storage for trend information, time schedules, and alarm data. Like the other BTL Listed controller types (B-AAC, B-ASC etc.) a B-BC device is required to support the server ("B") side of the ReadProperty and WriteProperty services, but unlike the other controller types it is also required to support the client ("A") side of these services. Communication between controllers requires that one of them support the client side and the other support the server side, so a B-BC is often used when communication between controllers is needed.

1.4.6 BACnet Broadcast Management Device (BBMD) (BACnet)

A communications device, typically combined with a BACnet router. A BBMD forwards BACnet broadcast messages to BACnet/IP devices and other BBMDs connected to the same BACnet/IP network. Each IP subnet that is part of a BACnet/IP network must have at least one BBMD. Note there are additional restrictions when multiple BBMDs share an IP subnet.

1.4.7 BACnet/IP (BACnet)

An extension of BACnet, Annex J, defines the use of a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP subnets that share the same BACnet network number. See also paragraph BACNET BROADCAST MANAGEMENT DEVICE.

1.4.8 BACnet Internetwork (BACnet)

Two or more BACnet networks, connected with BACnet routers. In a BACnet Internetwork, there exists only one message path between devices.

1.4.9 BACnet Interoperability Building Blocks (BIBBs) (BACnet)

A BIBB is a collection of one or more ASHRAE 135 Services intended to define a higher level of interoperability. BIBBs are combined to build the BACnet functional requirements for a device in a specification. Some BIBBs define additional requirements (beyond requiring support for specific services) in order to achieve a level of interoperability. For example, the BIBB DS-V-A (Data Sharing-View-A), which would typically be used by a front-end, not only requires the client to support the ReadProperty Service, but also provides a list of data types (Object / Properties) which the client must be able to interpret and display for the user.

In the BIBB shorthand notation, -A is the client side and -B is the server side.

The following is a list of some BIBBs used by this or referenced Sections:	
DS-COV-A	Data Sharing-Change of Value (A side)
DS-COV-B	Data Sharing-Change of Value (B side)
NM-RC-B	Network Management-Router Configuration (B side)
DS-RP-A	Data Sharing-Read Property (A side)
DS-RP-B	Data Sharing-Read Property (B side)
DS-RPM-A	Data Sharing-Read Property Multiple (A Side)
DS-RPM-B	Data Sharing-Read Property Multiple (B Side)
DS-WP-A	Data Sharing-Write Property (A Side)
DM-TS-B	Device Management-Time Synchronization (B Side)
DM-UTC-B	Device Management-UTC Time Synchronization (B Side)
DS-WP-B	Data Sharing-Write Property (B side)
SCHED-E-B	Scheduling-External (B side)
DM-OCD-B	Device Management-Object Creation and Deletion (B side)
AE-N-I-B	Alarm and Event-Notification Internal (B Side)
AE-N-E-B	Alarm and Event-Notification External (B Side)
T-VMT-I-B	Trending-Viewing and Modifying Trends Internal (B Side)

The following is a list of some BIBBs used by this or referenced Sections:	
T-VMT-E-B	Trending-Viewing and Modifying Trends External (B Side)

1.4.10 BACnet Network (BACnet)

In BACnet, a portion of the control Internetwork consisting of one or more segments connected by repeaters. Networks are separated by routers.

1.4.11 BACnet Operator Display (B-OD) (BACnet)

A basic operator interface with limited capabilities relative to a B-OWS. It is not intended to perform direct digital control. A B-OD profile could be used for LCD devices, displays affixed to BACnet devices, handheld terminals or other very simple user interfaces.

1.4.12 BACnet Segment (BACnet)

One or more physical segments interconnected by repeaters (ASHRAE 135).

1.4.13 BACnet Smart Actuator (B-SA) (BACnet)

A simple actuator device with limited resources intended for specific applications.

1.4.14 BACnet Smart Sensor (B-SS) (BACnet)

A simple sensing device with limited resources.

1.4.15 BACnet Testing Laboratories (BTL) (BACnet)

Established by BACnet International to support compliance testing and interoperability testing activities and consists of BTL Manager and the BTL Working Group (BTL-WG). BTL also publishes Implementation Guidelines.

1.4.16 BACnet Testing Laboratories (BTL) Listed (BACnet)

A device that has been listed by BACnet Testing Laboratory. Devices may be certified to a specific device profile, in which case the listing indicates that the device supports the required capabilities for that profile, or may be listed as "other".

1.4.17 Binary (All protocols)

A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level. 'Digital' is sometimes used interchangeably with 'binary'.

1.4.18 Broadcast (BACnet)

Unlike most messages, which are intended for a specific recipient device, a broadcast message is intended for all devices on the network.

1.4.19 Building Control Network (BCN) (All protocols)

The network connecting all DDC Hardware within a building (or specific group of buildings).

1.4.20 Building Point of Connection (BPOC) (All protocols)

A FPOC for a Building Control System. (This term is being phased out of use in preference for FPOC but is still used in some specifications and criteria. When it was used, it typically referred to a piece of control hardware. The current FPOC definition typically refers instead to IT hardware.)

1.4.21 Commandable (All protocols)

See Overridable.

1.4.22 Commandable Objects (BACnet)

Commandable Objects have a Commandable Property, Priority_Array, and Relinquish_Default Property as defined in ASHRAE 135, Clause 19.2, Command Prioritization.

1.4.23 Configurable (All protocols)

A property, setting, or value is configurable if it can be changed via hardware settings on the device, via the use of engineering software or over the control network from the front end, and is retained through (after) loss of power.

- a. In a non-Niagara Framework BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:
 - (1) via BACnet services (including proprietary BACnet services)
 - (2) via hardware settings on the device
- b. In a Niagara Framework BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:
 - (1) via BACnet services (including proprietary BACnet services)
 - (2) via hardware settings on the device
 - (3) via the Niagara Framework

Note this is more stringent than the ASHRAE 135 definition.

1.4.24 Control Logic Diagram (All protocols)

A graphical representation of control logic for multiple processes that make up a system.

1.4.25 Device (BACnet)

A Digital Controller that contains a BACnet Device Object and uses BACnet to communicate with other devices.

1.4.26 Device Object (BACnet)

Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet Internetwork.

This number is often referred to as the device instance or device ID.

1.4.27 Device Profile (BACnet)

A collection of BIBBs determining minimum BACnet capabilities of a device, defined in ASHRAE 135. Standard device profiles include BACnet Advanced Workstations (B-AWS), BACnet Building Controllers (B-BC), BACnet Advanced Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS).

1.4.28 Digital Controller (All protocols)

An electronic controller, usually with internal programming logic and digital and analog input/output capability, which performs control functions.

1.4.29 Direct Digital Control (DDC) (All protocols)

Digital controllers performing control logic. Usually the controller directly senses physical values, makes control decisions with internal programs, and outputs control signals to directly operate switches, valves, dampers, and motor controllers.

1.4.30 Field Point of Connection (FPOC) (All protocols)

The FPOC is the point of connection between the UMCS IP Network and the field control network (either an IP network, a non-IP network, or a combination of both). The hardware at this location which provides the connection is generally an IT device such as a switch, IP router, or firewall.

In general, the term "FPOC Location" means the place where this connection occurs, and "FPOC Hardware" means the device that provides the connection. Sometimes the term "FPOC" is used to mean either and its actual meaning (i.e. location or hardware) is determined by the context in which it is used.

1.4.31 Fox Protocol (Niagara Framework)

The protocol used for communication between components in the Niagara Framework. By default, Fox uses TCP port 1911.

1.4.32 Gateway (All protocols)

A device that translates from one protocol application data format to another. Devices that change only the transport mechanism of the protocol - "translating" from TP/FT-10 to Ethernet/IP or from BACnet MS/TP to BACnet over IP for example - are not gateways as the underlying data format does not change. Gateways are also called Communications Bridges or Protocol Translators.

A Niagara Framework Supervisory Gateway is one type of Gateway.

1.4.33 IEEE 802.3 Ethernet (All protocols)

A family of local-area-network technologies providing high-speed networking features over various media, typically Cat 5, 5e or Cat 6 twisted pair copper or fiber optic cable.

1.4.34 Internet Protocol (IP, TCP/IP, UDP/IP) (All protocols)

A communication method, the most common use is the World Wide Web. At the lowest level, it is based on Internet Protocol (IP), a method for conveying and routing packets of information over various LAN media. Two common protocols using IP are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP). UDP conveys information to well-known "sockets" without confirmation of receipt. TCP establishes connections, also known as "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

1.4.35 Input/Output (I/O) (All protocols)

Physical inputs and outputs to and from a device, although the term sometimes describes network or "virtual" inputs or outputs. See also "Points".

1.4.36 I/O Expansion Unit (All protocols)

An I/O expansion unit provides additional point capacity to a digital controller

1.4.37 IP subnet (All protocols)

A group of devices which share a defined range IP addresses. Devices on a common IP subnet can share data (including broadcasts) directly without the need for the traffic to traverse an IP router.

1.4.38 JACE (Niagara Framework)

Java Application Control Engine. See paragraph NIAGARA FRAMEWORK SUPERVISORY GATEWAY

1.4.39 Local-Area Network (LAN) (All protocols)

A communication network that spans a limited geographic area and uses the same basic communication technology throughout.

1.4.40 Local Display Panels (LDPs) (All protocols)

A DDC Hardware with a display and navigation buttons, and must provide display and adjustment of points as shown on the Points Schedule and as indicated.

1.4.41 MAC Address (All protocols)

Media Access Control address. The physical device address that identifies a device on a Local Area Network.

1.4.42 Master-Slave/Token-Passing (MS/TP) (BACnet)

Data link protocol as defined by the BACnet standard. Multiple speeds (data rates) are permitted by the BACnet MS/TP standard.

1.4.43 Monitoring and Control (M&C) Software (All protocols)

The UMCS 'front end' software which performs supervisory functions such as alarm handling, scheduling and data logging and provides a user interface for monitoring the system and configuring these functions.

1.4.44 Network Number (BACnet)

A site-specific number assigned to each network. This network number must be unique throughout the BACnet Internetwork.

1.4.45 Niagara Framework (Niagara Framework)

A set of hardware and software specifications for building and utility control owned by Tridium Inc. and licensed to multiple vendors. The Framework consists of front end (M&C) software, web based clients, field level control hardware, and engineering tools. While the Niagara Framework is not adopted by a recognized standards body and does not use an open licensing model, it is sufficiently well-supported by multiple HVAC vendors to be considered a de-facto Open Standard.

1.4.46 Niagara Framework Supervisory Gateway (Niagara Framework)

DDC Hardware component of the Niagara Framework. A typical Niagara architecture has Niagara specific supervisory gateways at the IP level and other (non-Niagara specific) controllers on field networks (TP/FT-10, MS/TP, etc.) beneath the Niagara supervisory gateways. The Niagara specific controllers function as a gateway between the Niagara framework protocol (Fox) and the field network beneath. These supervisory gateways may also be used as general purpose controllers and also have the capability to provide a web-based user interface.

Note that different vendors refer to this component by different names. The most common name is "JACE"; other names include (but are not limited to) "EC-BOS", "FX-40", "TMN", "SLX" and "UNC".

1.4.47 Object (BACnet)

An ASHRAE 135 Object. The concept of organizing BACnet information into standard components with various associated Properties. Examples include Analog Input objects and Binary Output objects.

1.4.48 Object Identifier (BACnet)

A grouping of two Object properties: Object Type (e.g. Analog Value, Schedule, etc.) and Object Instance (in this case, a number). Object Identifiers must be unique within a device.

1.4.49 Object Instance (BACnet)

See paragraph OBJECT IDENTIFIER

1.4.50 Object Properties (BACnet)

Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.

1.4.51 Operator Configurable (All protocols)

Operator configurable values are values that can be changed from a single common front end user interface across multiple vendor systems.

For Niagara Framework Systems, a property, setting, or value is Operator Configurable when it is configurable from a Niagara Framework Front End.

For non Niagara-based BACnet systems, a property, setting, or value in a device is Operator Configurable when it is Configurable and is either:

- a. a Writable Property of a Standard BACnet Object; or
- b. a Property of a Standard BACnet Object that is Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable.

1.4.52 Override (All protocols)

Changing the value of a point outside of the normal sequence of operation where the change has priority over the sequence and where there is a mechanism for releasing the change such that the point returns to the normal value. Overrides persist until released or overridden at the same or higher priority but are not required to persist through a loss of power. Overrides are often used by operators to change values, and generally originate at a user interface (workstation or local display panel).

1.4.53 Packaged Equipment (All protocols)

Packaged equipment is a single piece of equipment provided by a manufacturer in a substantially complete and operable condition, where the controls (DDC Hardware) are factory installed, and the equipment is sold and shipped from the manufacturer as a single entity. Disassembly and reassembly of a large piece of equipment for shipping does not prevent it from being packaged equipment. Package units may require field installation of remote sensors. Packaged equipment is also called a "packaged unit".

Note industry may use the term "Packaged System" to mean a collection of equipment that is designed to work together where each piece of equipment is packaged equipment and there is a network that connects the equipment together. A "packaged system" of this type is NOT packaged equipment; it is a collection of packaged equipment, and each piece of equipment must individually meet specification requirements.

1.4.54 Packaged Unit (All protocols)

See packaged equipment.

1.4.55 Performance Verification Test (PVT) (All protocols)

The procedure for determining if the installed BAS meets design criteria prior to final acceptance. The PVT is performed after installation, testing, and balancing of mechanical systems. Typically the PVT is performed by the Contractor in the presence of the Government.

1.4.56 Physical Segment (BACnet)

A single contiguous medium to which BACnet devices are attached (ASHRAE 135).

1.4.57 Polling (All protocols)

A device periodically requesting data from another device.

1.4.58 Points (All protocols)

Physical and virtual inputs and outputs. See also paragraph INPUT/OUTPUT (I/O).

1.4.59 Proportional, Integral, and Derivative (PID) Control Loop (All protocols)

Three parameters used to control modulating equipment to maintain a setpoint. Derivative control is often not required for HVAC systems (leaving "PI" control).

1.4.60 Proprietary (BACnet)

Within the context of BACnet, any extension of or addition to object types, properties, PrivateTransfer services, or enumerations specified in ASHRAE 135. Objects with Object_Type values of 128 and above are Proprietary Objects. Properties with Property_Identifier of 512 and above are proprietary Properties.

1.4.61 Protocol Implementation Conformance Statement (PICS) (BACnet)

A document, created by the manufacturer of a device, which describes which portions of the BACnet standard may be implemented by a given device. ASHRAE 135 requires that all ASHRAE 135 devices have a PICS, and also defines a minimum set of information that must be in it. A device as installed for a specific project may not implement everything in its PICS.

1.4.62 Repeater (All protocols)

A device that connects two control network segments and retransmits all information received on one side onto the other.

1.4.63 Router (All protocols)

A device that connects two ASHRAE 135 networks and controls traffic between the two by retransmitting signals received from one side onto the other based on the signal destination. Routers are used to subdivide a BACnet internetwork and to limit network traffic.

1.4.64 Segment (All protocols)

A 'single' section of a control network that contains no repeaters or routers. There is generally a limit on the number of devices on a segment, and this limit is dependent on the topology/media and device type.

1.4.65 Standard BACnet Objects (BACnet)

Objects with Object_Type values below 128 and specifically enumerated in Clause 21 of ASHRAE 135. Objects which are not proprietary. See paragraph PROPRIETARY.

1.4.66 Standard BACnet Properties (BACnet)

Properties with Property_Identifier values below 512 and specifically enumerated in Clause 21 of ASHRAE 135. Properties which are not proprietary. See Proprietary.

1.4.67 Standard BACnet Services (BACnet)

ASHRAE 135 services other than ConfirmedPrivateTransfer or UnconfirmedPrivateTransfer. See paragraph PROPRIETARY.

1.4.68 UMCS (All protocols)

UMCS stands for Utility Monitoring and Control System. The term refers to all components by which a project site monitors, manages, and controls real-time operation of HVAC and other building systems. These components include the UMCS "front-end" and all field building control systems connected to the front-end. The front-end consists of Monitoring and Control Software (user interface software), browser-based user interfaces and network infrastructure.

The network infrastructure (the "UMCS Network"), is an IP network connecting multiple building or facility control networks to the Monitoring and Control Software.

1.4.69 UMCS Network (All protocols)

The UMCS Network connects multiple building or facility control networks to the Monitoring and Control Software.

1.4.70 Writable Property (BACnet)

A Property is Writable when it can be changed through the use of one or more of the WriteProperty services defined in ASHRAE 135, Clause 15 regardless of the value of any other Property. Note that in the ASHRAE 135 standard, some Properties may be writable when the Out of Service Property is TRUE; for purposes of this Section, Properties that are only writable when the Out of Service Property is TRUE are not considered to be Writable.

1.5 PROJECT SEQUENCING

TABLE II: PROJECT SEQUENCING lists the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column). TABLE II does not specify overall project milestone and completion dates; these dates are specified in the contract documents.

- a. Sequencing for Submittals: The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon resubmittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.
- b. Sequencing for Activities: The sequencing specified for activities

indicates the earliest the activity may begin.

- c. Abbreviations: In TABLE II the abbreviation AAO is used for 'after approval of' and 'ACO' is used for 'after completion of'.

TABLE II. PROJECT SEQUENCING			
ITEM #	TYPE	DESCRIPTION	SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL)
1	S	Existing Conditions Report	
2	S	DDC Contractor Design Drawings	
3	S	Manufacturer's Product Data	
4	S	Pre-construction QC Checklist	
5	E	Install Building Control System	AAO #1 thru #4
6	E	Start-Up and Start-Up Testing	ACO #5
7	S	Post-Construction QC Checklist	{{[] days }ACO #6
8	S	Programming Software Configuration Software Niagara Framework Engineering Tool Niagara Framework Wizards	{{[] days }ACO #6
9	S	Draft As-Built Drawings	{{[] days }ACO #6
10	S	Start-Up Testing Report	{{[] days }ACO #6
11	S	PVT Procedures	{{[] days }before schedule start of #12 and AAO #10
12	E	Execute PVT	AAO #9 and #11
13	S	PVT Report	{{[] days }ACO #12
14	S	Controller Application Programs Controller Configuration Settings Niagara Framework Supervisory Gateway Backups	{{[] days }AAO #13
15	S	Final As-Built Drawings	{{[] days }AAO #13

TABLE II. PROJECT SEQUENCING			
ITEM #	TYPE	DESCRIPTION	SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL)
16	S	O&M Instructions	AAO #15
17	S	Training Documentation	AAO #10 and [[] days] before scheduled start of #18
18	E	Training	AAO #16 and #17
19	S	Closeout QC Checklist	ACO #18

1.6 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

DDC Contractor Design Drawings; G

Draft As-Built Drawings; G

Final As-Built Drawings; G

SD-03 Product Data

Programming Software; G

Controller Application Programs; G

Configuration Software; G

Controller Configuration Settings; G

Proprietary Multi-Split Engineering Tool Software; G

Manufacturer's Product Data; G

Niagara Framework Supervisory Gateway Backups; G

Niagara Framework Engineering Tool; G

SD-05 Design Data

Boiler Or Chiller Plant Gateway Request

SD-06 Test Reports

Existing Conditions Report

Pre-Construction Quality Control (QC) Checklist; G

Post-Construction Quality Control (QC) Checklist; G

Start-Up Testing Report; G

PVT Procedures; G

PVT Report; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Training Documentation; G

SD-11 Closeout Submittals

Enclosure Keys; G

Password Summary Report; G

Closeout Quality Control (QC) Checklist; G

1.7 DATA PACKAGE AND SUBMITTAL REQUIREMENTS

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

1.8 SOFTWARE FOR DDC HARDWARE AND GATEWAYS

Provide all software related to the programming and configuration of DDC Hardware and Gateways as indicated. License all Software to the project site. The term "controller" as used in these requirements means both DDC Hardware and Gateways.

1.8.1 Configuration Software

For each type of controller, provide the configuration tool software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of the software user manuals for each software with the software submittal.

Submit Configuration Software on CD-ROM as a Technical Data Package. Submit 2 hard copies of the software user manual for each piece of software.

1.8.2 Controller Configuration Settings

For each controller, provide copies of the installed configuration settings as source code compatible with the configuration tool software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Configuration Settings on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which files are associated with each device. Submit 2 copies of the Controller Configuration Settings CD-ROM.

1.8.3 Programming Software

For each type of programmable controller, provide the programming software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of software user manuals for each software with the software submittal.

Submit Programming Software on CD-ROM as a Technical Data Package. Submit 2 hard copies of the software user manual for each piece of software.

1.8.4 Controller Application Programs

For each programmable controller, provide copies of the application program as source code compatible with the programming software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Application Programs on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which application program is associated with each device. Submit 2 copies of the Controller Application Programs CD-ROM.

1.8.5 Niagara Framework Supervisory Gateway Backups

For each Niagara Framework Supervisory Gateway, provide a backup of all software within the Niagara Framework Supervisory Gateway, including configuration settings. This backup must be sufficient to allow the restoration of the Niagara Framework Supervisory Gateway or the replacement of the Niagara Framework Supervisory Gateway.

Submit backups for each Niagara Framework Supervisory Gateway on CD-ROM as a Technical Data Package. Mark each backup indicating clearly the source Niagara Framework Supervisory Gateway.

1.8.6 Niagara Framework Engineering Tool(for all Niagara Framework system)

Provide a Niagara Framework Engineering Tool in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit software user manuals with the Niagara Framework Engineering Tool submittal.

Submit the Niagara Framework Engineering Tool on CD-ROM as a Technical Data Package. Submit 2 hard copies of the software user manual for the Niagara Framework Engineering Tool.

1.9 BOILER OR CHILLER PLANT GATEWAY REQUEST

If requesting the use of a gateway to a boiler or chiller plant as indicated in paragraph Proprietary Systems Exempted From Open Protocol Requirements, submit a Boiler or Chiller Plant Gateway Request describing the configuration of the boilers or chillers including model numbers for equipment and controllers, the sequence of operation for the units, and a justification for the need to operate the units on a shared non-BACnet network.

1.10 QUALITY CONTROL CHECKLISTS

The QC Checklist for BACnet Systems at the end of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Checklist for Niagara Framework Based BACnet Systems at the end of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Representative must verify each item indicated and initial in the space provided to indicate that the requirement has been met. The QC Representative must sign and date the Checklist prior to submission to the Government.

1.10.1 Pre-Construction Quality Control (QC) Checklist

Complete items indicated as Pre-Construction QC Checklist items in the QC Checklist. Submit four copies of the Pre-Construction QC Checklist.

1.10.2 Post-Construction Quality Control (QC) Checklist

Complete items indicated as Post-Construction QC Checklist items in the QC Checklist. Submit four copies of the Post-Construction QC Checklist.

1.10.3 Closeout Quality Control (QC) Checklist

Complete items indicated as Closeout QC Checklist items in the QC Checklist. Submit four copies of the Closeout QC Checklist.

PART 2 PRODUCTS

Provide products meeting the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, other referenced Sections, and this Section.

2.1 GENERAL PRODUCT REQUIREMENTS

Units of the same type of equipment must be products of a single manufacturer. Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of these and similar products. The standard products must have been in a satisfactory commercial or industrial use for two years prior to use on this project. The two year use must include applications of equipment and materials under similar circumstances and of similar size. DDC Hardware not meeting

the two-year field service requirement is acceptable provided it has been successfully used by the Contractor in a minimum of two previous projects. The equipment items must be supported by a service organization. Items of the same type and purpose must be identical, including equipment, assemblies, parts and components.

2.2 PRODUCT DATA

Provide manufacturer's product data sheets documenting compliance with product specifications for each product provided under Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, or this Section. Provide product data for all products in a single indexed compendium, organized by product type.

For all BACnet hardware: for each manufacturer, model and version (revision) of DDC Hardware provide the Protocol Implementation Conformance Statement (PICS) in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Manufacturer's Product Data on CD-ROM.

2.3 OPERATION ENVIRONMENT

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

- a. Pressure: Pressure conditions normally encountered in the installed location.
- b. Vibration: Vibration conditions normally encountered in the installed location.
- c. Temperature:
 - (1) Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.
 - (2) Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -37 to +66 degrees C and temperature conditions outside this range normally encountered at the installed location.
- d. Humidity: 10 to 95 percent relative humidity, noncondensing and humidity conditions outside this range normally encountered at the installed location.

2.4 WIRELESS CAPABILITY

Wireless capabilities must comply with Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM

2.5 ENCLOSURES

Enclosures supplied as an integral (pre-packaged) part of another product are acceptable. Provide two Enclosure Keys for each lockable enclosure on a single ring per enclosure with a tag identifying the enclosure the keys operate. Provide enclosures meeting the following minimum requirements:

2.5.1 Outdoors

For enclosures located outdoors, provide enclosures meeting NEMA 250 Type 4 requirements.

2.5.2 Mechanical and Electrical Rooms

For enclosures located in mechanical or electrical rooms, provide enclosures meeting NEMA 250 Type 4 requirements.

2.5.3 Other Locations

For enclosures in other locations including but not limited to occupied spaces, above ceilings, and in plenum returns, provide enclosures meeting NEMA 250 Type 1 requirements.

2.6 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

2.6.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

2.6.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

2.6.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 50 mm (2 inch) lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation
- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

2.6.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or

thicker stranded copper wire rated for 600-volt service.

2.6.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

3.1.1 Existing Conditions Survey

Perform a field survey, including testing and inspection of the equipment to be controlled and submit an Existing Conditions Report documenting the current status and its impact on the Contractor's ability to meet this specification. For those items considered nonfunctional, document the deficiency in the report including explanation of the deficiencies and estimated costs to correct the deficiencies. As part of the report, define the scheduled need date for connection to existing equipment. Make written requests and obtain Government approval prior to disconnecting any controls and obtaining equipment downtime.

Submit four copies of the Existing Conditions Report.

3.1.2 Existing Equipment Downtime

Make written requests and obtain Government approval prior to disconnecting any controls and obtaining equipment downtime.

3.1.3 Existing Control System Devices

Inspect, calibrate, and adjust as necessary to place in proper working order all existing devices which are to be reused.

3.2 INSTALLATION

Fully install and test the control system in accordance Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, and this Section.

3.2.1 Dielectric Isolation

Provide dielectric isolation where dissimilar metals are used for connection and support. Install control system in a manner that provides clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Install control system such that it does not interfere with the clearance requirements for mechanical and electrical system maintenance.

3.2.2 Penetrations in Building Exterior

Make all penetrations through and mounting holes in the building exterior watertight.

3.2.3 Device Mounting Criteria

Install devices in accordance with the manufacturer's recommendations and as indicated and shown. Provide a weathershield for all devices installed outdoors. Provide clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Provide clearance for mechanical and electrical system maintenance; do not not interfere with the clearance requirements for mechanical and electrical system maintenance.

3.2.4 Labels and Tags

Key all labels and tags to the unique identifiers shown on the As-Built drawings. For labels exterior to protective enclosures provide engraved plastic labels mechanically attached to the enclosure or DDC Hardware. Labels inside protective enclosures may be attached using adhesive, but must not be hand written. For tags, provide plastic or metal tags mechanically attached directly to each device or attached by a metal chain or wire.

- a. Label all Enclosures and DDC Hardware.
- b. Tag Airflow measurement arrays (AFMA) with flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient.
- c. Tag duct static pressure taps at the location of the pressure tap

3.2.5 Surge Protection

3.2.5.1 Power-Line Surge Protection

Protect equipment connected to AC circuits to withstand power-line surges in accordance with IEEE C62.41. Do not use fuses for surge protection.

3.2.5.2 Surge Protection for Transmitter and Control Wiring

Protect DDC hardware against or provided DDC hardware capable of withstanding surges induced on control and transmitter wiring installed outdoors and as shown. Protect equipment against the following two waveforms:

- a. A waveform with a 10-microsecond rise time, a 1000-microsecond decay time and a peak current of 60 amps.
- b. A waveform with an 8-microsecond rise time, a 20-microsecond decay time and a peak current of 500 amperes.

3.2.6 Basic Cybersecurity Requirements

3.2.6.1 Passwords

Password development and the Password Summary Report submittal must comply with Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM.

3.2.6.2 Wireless Capability

Wireless capabilities must comply with Section 25 05 11.23 CYBERSECURITY

FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM.

3.2.6.3 IP Network Physical Security

IP network physical security must comply with Section 25 05 11.23
CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL
SYSTEM.

3.3 DRAWINGS AND CALCULATIONS

Provide drawings in the form and arrangement indicated and shown. Use the same abbreviations, symbols, nomenclature and identifiers shown. Assign a unique identifier as shown to each control system element on a drawing. When packaging drawings, group schedules by system. When space allows, it is permissible to include multiple schedules for the same system on a single sheet. Except for drawings covering all systems, do not put information for different systems on the same sheet.

Submit hardcopy drawings on ISO A1 841 by 594 mm or A3 420 by 297 mm sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

- a. Submit DDC Contractor Design Drawings consisting of each drawing indicated with pre-construction information depicting the intended control system design and plans. Submit DDC Contractor Design Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.
- b. Submit Draft As-Built Drawings consisting of each drawing indicated updated with as-built data for the system prior to PVT. Submit Draft As-Built Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.
- c. Submit Final As-Built Drawings consisting of each drawing indicated updated with all final as-built data. Final As-Built Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.

3.3.1 Sample Drawings

Sample drawings in electronic format are available at the Whole Building Design Guide page for this section:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-23-09-00>

These drawings may prove useful in demonstrating expected drawing formatting and example content and are provided for illustrative purposes only. Note that these drawings do not meet the content requirements of this Section and must be completed to meet project requirements.

3.3.2 Drawing Index and Legend

Provide an HVAC Control System Drawing Index showing the name and number of the building, military site, State or other similar designation, and Country. In the Drawing Index, list all Contractor Design Drawings, including the drawing number, sheet number, drawing title, and computer filename when used. In the Design Drawing Legend, show and describe all symbols, abbreviations and acronyms used on the Design Drawings. Provide a single Index and Legend for the entire drawing package.

3.3.3 Thermostat and Occupancy Sensor Schedule

Provide a thermostat and occupancy sensor schedule containing each thermostat's unique identifier, room identifier and control features and functions as shown. Provide a single thermostat and occupancy sensor schedule for the entire project.

3.3.4 Valve Schedule

Provide a valve schedule containing each valve's unique identifier, size, flow coefficient Kv (Cv), pressure drop at specified flow rate, spring range, positive positioner range, actuator size, close-off pressure to torque data, dimensions, and access and clearance requirements data. In the valve schedule include actuator selection data supported by calculations of the force required to move and seal the valve, access and clearance requirements. Provide a single valve schedule for the entire project.

3.3.5 Damper Schedule

Provide a damper schedule containing each damper's unique identifier, type (opposed or parallel blade), nominal and actual sizes, orientation of axis and frame, direction of blade rotation, actuator size and spring ranges, operation rate, positive positioner range, location of actuators and damper end switches, arrangement of sections in multi-section dampers, and methods of connecting dampers, actuators, and linkages. Include the AMCA 511 maximum leakage rate at the operating static-pressure differential for each damper in the Damper Schedule. Provide a single damper schedule for the entire project.

3.3.6 Project Summary Equipment Schedule

Provide a project summary equipment schedule containing the manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a single project equipment schedule for the entire project.

3.3.7 Equipment Schedule

Provide system equipment schedules containing the unique identifier, manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a separate equipment schedule for each HVAC system.

3.3.8 Occupancy Schedule

Provide an occupancy schedule drawing containing the same fields as the occupancy schedule Contract Drawing with Contractor updated information. Provide a single occupancy schedule for the entire project.

3.3.9 DDC Hardware Schedule

Provide a single DDC Hardware Schedule for the entire project and including following information for each device.

3.3.9.1 DDC Hardware Identifier

The Unique DDC Hardware Identifier for the device.

3.3.9.2 HVAC System

The system "name" used to identify a specific system (the name used on the system schematic drawing for that system).

3.3.9.3 BACnet Device Information

3.3.9.3.1 Device Object Identifier

The Device Object Identifier: The Object_Identifier of the Device Object

3.3.9.3.2 Network Number

The Network Number for the device.

3.3.9.3.3 MAC Address

The MAC Address for the device

3.3.9.3.4 BTL Listing

The BTL Listing of the device. If the device is listed under multiple BTL Profiles, indicate the profile that matches the use and configuration of the device as installed.

3.3.9.3.5 Proprietary Services Information

If the device uses non-standard ASHRAE 135 services as defined and permitted in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, indicate that the device uses non-standard services and include a description of all non-standard services used. Describe usage and content such that a device from another vendor can interoperate with the device using the non-standard service. Provide descriptions with sufficient detail to allow a device from a different manufacturer to be programmed to both read and write the non-standard service request:

- a. read: interpret the data contained in the non-standard service and;
- b. write: given similar data, generate the appropriate non-standard service request.

3.3.9.3.6 Alarming Information

Indicate whether the device is used for alarm generation, and which types of alarm generation the device implements: intrinsic, local algorithmic, remote algorithmic.

3.3.9.3.7 Scheduling Information

Indicate whether the device is used for scheduling.

3.3.9.3.8 Trending Information

Indicate whether the device is used for trending, and indicate if the device is used to trend local values, remote values, or both.

3.3.9.4 Niagara Station ID

The Niagara Station ID for each Niagara Framework Supervisory Gateway

3.3.10 Points Schedule

Provide a Points Schedule in tabular form for each HVAC system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets ("<" and ">"), such as <____> for a required entry or <value> for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval.

Points Schedule Columns must include:

3.3.10.1 Point Name

The abbreviated name for the point using the indicated naming convention.

3.3.10.2 Description

A brief functional description of the point such as "Supply Air Temperature".

3.3.10.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

3.3.10.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

3.3.10.5 Range

The range of values, including units, associated with the point, including but not limited to a zone temperature setpoint adjustment range, a sensor measurement range, occupancy values for an occupancy input, or the status of a safety.

3.3.10.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output
- c. BI: The value comes from a hardware (physical) Binary Input

- d. B0: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.
- g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

3.3.10.7 Object and Property Information

The Object Type and Instance Number for the Object associated with the point. If the value of the point is not in the Present_Value Property, then also provide the Property ID for the Property containing the value of the point. Any point that is displayed at the front end or on an LDP, is trended, is used by another device on the network, or has an alarm condition must be documented here.

3.3.10.8 Niagara Station ID

The Niagara Station ID of the Niagara Framework Supervisory Gateway the point is mapped into.

3.3.10.9 Network Data Exchange Information (Gets Data From, Sends Data To)

Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

3.3.10.10 Override Information (Object Type and Instance Number)

For each point requiring an Override and not residing in a Niagara Framework Supervisory Gateway, indicate if the Object for the point is Commandable or, if the use of a separate Object was specifically approved by the Contracting Officer, provide the Object Type and Instance Number of the Object to be used in overriding the point.

3.3.10.11 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote, the trend Object type and the trend Object instance number. For remote trends provide the DDC Hardware Identifier for the device containing the trend Object in the Points Schedule notes.

3.3.10.12 Alarm Information

Indicate the Alarm Generation Type, Event Enrollment Object Instance Number, and Notification Class Object Instance Number for each point

requiring an alarm. (Note that not all alarms will have Event Enrollment Objects.)

For Niagara BACnet systems: Indicate the Alarm Generation Type and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have a Notification Class Object.)

3.3.10.13 Configuration Information

Indicate the means of configuration associated with each point. For points in a Niagara Framework Supervisory Gateway, indicate the point within the Niagara Framework Supervisory Gateway used to configure the value. For other points:

- a. For Operator Configurable Points indicate BACnet Object and Property information (Name, Type, Identifiers) containing the configurable value. Indicate whether the property is writable always, or only when Out_Of_Service is TRUE.
- b. For Configurable Points indicate the BACnet Object and Property information as for Operator Configurable points, or identification of the configurable settings from within the engineering software for the device or identification of the hardware settings on the device.

3.3.11 Riser Diagram

The Riser Diagram of the Building Control Network may be in tabular form, and must show all DDC Hardware and all Network Hardware, including network terminators. For each item, provide the unique identifier, common descriptive name, physical sequential order (previous and next device on the network), room identifier and location within room. A single riser diagram must be submitted for the entire system.

3.3.12 Control System Schematics

Provide control system schematics in the same form as the control system schematic Contract Drawing with Contractor updated information. Provide a control system schematic for each HVAC system.

3.3.13 Sequences of Operation Including Control Logic Diagrams

Provide HVAC control system sequence of operation and control logic diagrams in the same format as the Contract Drawings. Within these drawings, refer to devices by their unique identifiers. Submit sequences of operation and control logic diagrams for each HVAC system

3.3.14 Controller, Motor Starter and Relay Wiring Diagram

Provide controller wiring diagrams as functional wiring diagrams which show the interconnection of conductors and cables to each controller and to the identified terminals of input and output devices, starters and package equipment. Show necessary jumpers and ground connections and the labels of all conductors. Identify sources of power required for control systems and for packaged equipment control systems back to the panel board circuit breaker number, controller enclosures, magnetic starter, or packaged equipment control circuit. Show each power supply and transformer not integral to a controller, starter, or packaged equipment. Show the connected volt-ampere load and the power supply volt-ampere rating. Provide wiring diagrams for each HVAC system.

3.4 CONTROLLER TUNING

Tune each controller in a manner consistent with that described in the ASHRAE FUN SI and in the manufacturer's instruction manual. Tuning must consist of adjustment of the proportional, integral, and where applicable, the derivative (PID) settings to provide stable closed-loop control. Each loop must be tuned while the system or plant is operating at a high gain (worst case) condition, where high gain can generally be defined as a low-flow or low-load condition. Upon final adjustment of the PID settings, in response to a change in controller setpoint, the controlled variable must settle out at the new setpoint with no more than two (2) oscillations above and below setpoint. Upon settling out at the new setpoint the controller output must be steady. With the exception of naturally slow processes such as zone temperature control, the controller must settle out at the new setpoint within five (5) minutes. Set the controller to its correct setpoint and record and submit the final PID configuration settings with the O&M Instructions and on the associated Points Schedule.

3.5 START-UP

3.5.1 Start-Up Test

Perform the following startup tests for each control system to ensure that the described control system components are installed and functioning per this specification.

Adjust, calibrate, measure, program, configure, set the time schedules, and otherwise perform all necessary actions to ensure that the systems function as indicated and shown in the sequence of operation and other contract documents.

3.5.1.1 Systems Check

An item-by-item check must be performed for each HVAC system

3.5.1.1.1 Step 1 - System Inspection

With the system in unoccupied mode and with fan hand-off-auto switches in the OFF position, verify that power and main air are available where required and that all output devices are in their failsafe and normal positions. Inspect each local display panel to verify that all displays indicate shutdown conditions.

3.5.1.1.2 Step 2 - Calibration Accuracy Check

Perform a two-point accuracy check of the calibration of each HVAC control system sensing element and transmitter by comparing the value from the test instrument to the network value provided by the DDC Hardware. Use digital indicating test instruments, such as digital thermometers, motor-driven psychrometers, and tachometers. Use test instruments with accuracy at least twice as accurate as the specified sensor accuracy and with calibration traceable to National Institute of Standards and Technology standards. Check one the first check point in the bottom one-third of the sensor range, and the second in the top one-third of the sensor range. Verify that the sensing element-to-DDC readout accuracies at two points are within the specified product accuracy tolerances, and if not recalibrate or replace the device and repeat the calibration check.

3.5.1.1.3 Step 3 - Actuator Range Check

With the system running, apply a signal to each actuator through the DDC Hardware controller. Verify proper operation of the actuators and positioners for all actuated devices and record the signal levels for the extreme positions of each device. Vary the signal over its full range, and verify that the actuators travel from zero stroke to full stroke within the signal range. Where applicable, verify that all sequenced actuators move from zero stroke to full stroke in the proper direction, and move the connected device in the proper direction from one extreme position to the other. For valve actuators and damper actuators, perform the actuator range check under normal system pressures.

3.5.1.2 Weather Dependent Test

Perform weather dependent test procedures in the appropriate climatic season.

3.5.2 Start-Up Testing Report

Submit 4 copies of the Start-Up Testing Report. The report may be submitted as a Technical Data Package documenting the results of the tests performed and certifying that the system is installed and functioning per this specification, and is ready for the Performance Verification Test (PVT).

3.6 PERFORMANCE VERIFICATION TEST (PVT)

3.6.1 PVT Procedures

Prepare PVT Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system performs in accordance with the sequences of operation, and other contract documents. Submit 4 copies of the PVT Procedures. The PVT Procedures may be submitted as a Technical Data Package.

3.6.1.1 Sensor Accuracy Checks

Include a one-point accuracy check of each sensor in the PVT procedures.

3.6.1.2 Endurance Test

Include a one-week endurance test as part of the PVT during which the system is operated continuously.

Use the building control system BACnet Trend Log or Trend Log Multiple Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.

Use the building control system Niagara Trend Log Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.

3.6.1.3 PVT Equipment List

Include in the PVT procedures a control system performance verification test equipment list that lists the equipment to be used during performance verification testing. For each piece of equipment, include manufacturer name, model number, equipment function, the date of the latest calibration, and the results of the latest calibration

3.6.2 PVT Execution

Demonstrate compliance of the control system with the contract documents. Using test plans and procedures approved by the Government, software capable of reading and writing COV Notification Subscriptions, Notification Class Recipient List Properties, event enrollments, demonstrate all physical and functional requirements of the project. Show, step-by-step, the actions and results demonstrating that the control systems perform in accordance with the sequences of operation. Do not start the performance verification test until after receipt of written permission by the Government, based on Government approval of the PVT Plan and Draft As-Built and completion of balancing. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. Do not conduct tests during scheduled seasonal off periods of base heating and cooling systems. If the system experiences any failures during the endurance test portion of the PVT, repair the system repeat the endurance test portion of the PVT until the system operates continuously and without failure for the specified endurance test period.

3.6.3 PVT Report

Prepare and submit a PVT report documenting all tests performed during the PVT and their results. Include all tests in the PVT procedures and any additional tests performed during PVT. Document test failures and repairs conducted with the test results.

Submit four copies of the PVT Report. The PVT Report may be submitted as a Technical Data Package.

3.7 OPERATION AND MAINTENANCE (O&M) INSTRUCTIONS

Provide HVAC control System Operation and Maintenance Instructions which include:

- a. "Data Package 3" as indicated in Section 01 78 23 OPERATION AND MAINTENANCE DATA for each piece of control equipment.
- b. "Data Package 4" as described in Section 01 78 23 OPERATION AND MAINTENANCE DATA for all air compressors.
- c. HVAC control system sequences of operation formatted as indicated.
- d. Procedures for the HVAC system start-up, operation and shut-down including the manufacturer's supplied procedures for each piece of equipment, and procedures for the overall HVAC system.
- e. As-built HVAC control system detail drawings formatted as indicated.
- f. Routine maintenance checklist. Provide the routine maintenance checklist arranged in a columnar format, where the first column lists

all installed devices, the second column states the maintenance activity or that no maintenance required, the third column states the frequency of the maintenance activity, and the fourth column is used for additional comments or reference.

- g. Qualified service organization list, including at a minimum company name, contact name and phone number.
- h. Start-Up Testing Report.
- i. Performance Verification Test (PVT) Procedures and Report.

Submit 2 copies of the Operation and Maintenance Instructions, indexed and in booklet form. The Operation and Maintenance Instructions may be submitted as a Technical Data Package.

3.8 MAINTENANCE AND SERVICE

Provide services, materials and equipment as necessary to maintain the entire system in an operational state as indicated for a period of one year from the date of final acceptance of the project. Minimize impacts on facility operations.

- a. The integration of the system specified in this section into a Utility Monitoring and Control System must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period. Integration into a UMCS includes but is not limited to establishing communication between devices in the control system and the front end or devices in another system.
- b. The changing of configuration properties must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period.

3.8.1 Description of Work

Provide adjustment and repair of the system including the manufacturer's required sensor and actuator (including transducer) calibration, span and range adjustment.

3.8.2 Personnel

Use only service personnel qualified to accomplish work promptly and satisfactorily. Advise the Government in writing of the name of the designated service representative, and of any changes in personnel.

3.8.3 Scheduled Inspections

Perform two inspections at six-month intervals and provide work required. Perform inspections in June and December. During each inspection perform the indicated tasks:

- a. Perform visual checks and operational tests of equipment.
- b. Clean control system equipment including interior and exterior surfaces.
- c. Check and calibrate each field device. Check and calibrate 50 percent of the total analog inputs and outputs during the first inspection.

Check and calibrate the remaining 50 percent of the analog inputs and outputs during the second major inspection. Certify analog test instrumentation accuracy to be twice the specified accuracy of the device being calibrated. Randomly check at least 25 percent of all binary inputs and outputs for proper operation during the first inspection. Randomly check at least 25 percent of the remaining binary inputs and outputs during the second inspection. If more than 20 percent of checked inputs or outputs failed the calibration check during any inspection, check and recalibrate all inputs and outputs during that inspection.

- d. Run system software diagnostics and correct diagnosed problems.
- e. Resolve any previous outstanding problems.

3.8.4 Scheduled Work

This work must be performed during regular working hours, Monday through Friday, excluding Federal holidays.

3.8.5 Emergency Service

The Government will initiate service calls when the system is not functioning properly. Qualified personnel must be available to provide service to the system. A telephone number where the service supervisor can be reached at all times must be provided. Service personnel must be at the site within 24 hours after receiving a request for service. The control system must be restored to proper operating condition as required per Section 01 78 00 CLOSEOUT SUBMITTALS.

3.8.6 Operation

After performing scheduled adjustments and repairs, verify control system operation as demonstrated by the applicable tests of the performance verification test.

3.8.7 Records and Logs

Keep dated records and logs of each task, with cumulative records for each major component, and for the complete system chronologically. Maintain a continuous log for all devices, including initial analog span and zero calibration values and digital points. Keep complete logs and provide logs for inspection onsite, demonstrating that planned and systematic adjustments and repairs have been accomplished for the control system.

3.8.8 Work Requests

Record each service call request as received and include its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion. Submit a record of the work performed within 5 days after work is accomplished.

3.8.9 System Modifications

Submit recommendations for system modification in writing. Do not make system modifications, including operating parameters and control settings, without prior approval of the Government.

3.9 TRAINING

Conduct a training course for 3 operating staff members designated by the Government in the maintenance and operation of the system, including specified hardware and software. Conduct 32 hours of training at the project site within 30 days after successful completion of the performance verification test. The Government reserves the right to make audio and visual recordings (using Government supplied equipment) of the training sessions for later use. Provide audiovisual equipment and other training materials and supplies required to conduct training. A training day is defined as 8 hours of classroom instruction, including two 15 minute breaks and excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility.

3.9.1 Training Documentation

Prepare training documentation consisting of:

- a. Course Attendee List: Develop the list of course attendees in coordination with and signed by the Controls AND HVAC shop supervisor.
- b. Training Manuals: Provide training manuals which include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. When presenting portions of the course material by audiovisuals, deliver copies of those audiovisuals as a part of the printed training manuals.

3.9.2 Training Course Content

For guidance in planning the required instruction, assume that attendees will have a high school education, and are familiar with HVAC systems. During the training course, cover all of the material contained in the Operating and Maintenance Instructions, the layout and location of each controller enclosure, the layout of one of each type of equipment and the locations of each, the location of each control device external to the panels, the location of the compressed air station, preventive maintenance, troubleshooting, diagnostics, calibration, adjustment, commissioning, tuning, and repair procedures. Typical systems and similar systems may be treated as a group, with instruction on the physical layout of one such system. Present the results of the performance verification test and the Start-Up Testing Report as benchmarks of HVAC control system performance by which to measure operation and maintenance effectiveness.

3.9.3 Training Documentation Submittal Requirements

Submit hardcopy training manuals and all training materials on CD-ROM. Provide one hardcopy manual for each trainee on the Course Attendee List and 2 additional copies for archive at the project site. Provide 2 copies of the Course Attendee List with the archival copies. Training Documentation may be submitted as a Technical Data Package.

<u>QC CHECKLIST FOR BACNET SYSTEMS</u>		
This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such. Instructions: Initial each item in the space provided (____) verifying that the requirement has been met.		
This checklist is for (circle one:) Pre-Construction QC Checklist Submittal Post-Construction QC Checklist Submittal Close-out QC Checklist Submittal		
Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:		
1	All DDC Hardware is numbered on Control System Schematic Drawings.	____
2	Signal lines on Control System Schematic are labeled with the signal type.	____
3	Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.	____
Items verified for Post-Construction and Closeout QC Checklist Submittals:		
4	All sequences are performed as specified using DDC Hardware.	____
5	Training schedule and course attendee list has been developed and coordinated with shops and submitted.	____
Items verified for Closeout QC Checklist Submittal:		
6	Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.	____
7	Programming software has been submitted for all programmable controllers.	____
8	All software has been licensed to the Government.	____
9	O&M Instructions have been completed and submitted.	____
10	Training course has been completed.	____
11	All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.	____

<u>QC CHECKLIST FOR BACNET SYSTEMS</u>		
12	All DDC Hardware is BTL listed.	____
13	Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule.	____
14	Scheduling, Alarming, and Trending have been implemented using the standard BACnet Objects for these functions.	____
15	All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated	____
(QC Representative Signature) (Date)		

<u>QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS</u>		
This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such. Instructions: Initial each item in the space provided (____) verifying that the requirement has been met.		
This checklist is for (circle one:) Pre-Construction QC Checklist Submittal Post-Construction QC Checklist Submittal Close-out QC Checklist Submittal		
Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:		
1	All DDC Hardware is numbered on Control System Schematic Drawings.	____
2	Signal lines on Control System Schematic are labeled with the signal type.	____
3	Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.	____
Items verified for Post-Construction and Closeout QC Checklist Submittals:		

<u>QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS</u>		
4	All sequences are performed as specified using DDC Hardware.	____
5	Training schedule and course attendee list has been developed and coordinated with shops and submitted.	____
Items verified for Closeout QC Checklist Submittal:		
6	Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.	____
7	Programming software has been submitted for all programmable controllers.	____
8	All software has been licensed to the Government.	____
9	O&M Instructions have been completed and submitted.	____
10	Training course has been completed.	____
11	All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.	____
12	All DDC Hardware is BTL listed.	____
13	Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule.	____
14	Scheduling, Alarming, and Trending have been implemented using Niagara Framework objects and services, and BACnet Intrinsic Alarming as indicated.	____
15	All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated	____
(QC Representative Signature) (Date)		

-- End of Section --

SECTION 23 09 13

INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
11/15, CHG 2: 05/21

PART 1 GENERAL

1.1 SUMMARY

This section provides for the instrumentation control system components excluding direct digital controllers, network controllers, gateways etc. that are necessary for a completely functional automatic control system. When combined with a Direct Digital Control (DDC) system, the Instrumentation and Control Devices covered under this section must be a complete system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and indicated.

- a. Install hardware to perform the control sequences as specified and indicated and to provide control of the equipment as specified and indicated.
- b. Install hardware such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- c. Install and configure hardware such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the installing Contractor.

1.1.1 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.1.2 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

1.2 RELATED SECTIONS

Related work specified elsewhere.

Section 23 30 00 HVAC AIR DISTRIBUTION

Section 23 05 15 COMMON PIPING FOR HVAC

Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

- AMCA 500-D (2018) Laboratory Methods of Testing
Dampers for Rating
- AMCA 511 (2010; R 2016) Certified Ratings Program
for Air Control Devices

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

- ANSI C12.1 (2014; Errata 2016) Electric Meters - Code
for Electricity Metering

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- ASME B16.15 (2024) Cast Copper Alloy Threaded Fittings
Classes 125 and 250
- ASME B16.34 (2021) Valves - Flanged, Threaded and
Welding End
- ASME B40.100 (2022) Pressure Gauges and Gauge
Attachments

ASTM INTERNATIONAL (ASTM)

- ASTM A536 (2024) Standard Specification for Ductile
Iron Castings

FLUID CONTROLS INSTITUTE (FCI)

- FCI 70-2 (2021) Control Valve Seat Leakage

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 142 (2007; Errata 2014) Recommended Practice
for Grounding of Industrial and Commercial
Power Systems - IEEE Green Book

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- ANSI C12.20 (2015; E 2018) Electricity Meters - 0.1,
0.2, and 0.5 Accuracy Classes
- NEMA 250 (2020) Enclosures for Electrical Equipment
(1000 Volts Maximum)
- NEMA/ANSI C12.10 (2011; R 2021) Physical Aspects of
Watthour Meters - Safety Standard

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National

Electrical Code

NFPA 90A (2024) Standard for the Installation of
Air Conditioning and Ventilating Systems

UL SOLUTIONS (UL)

UL 555 (2006; Reprint Aug 2016) UL Standard for
Safety Fire Dampers

UL 555S (2014; Reprint Oct 2020) UL Standard for
Safety Smoke Dampers

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for
Safety Low Voltage Transformers - Part 3:
Class 2 and Class 3 Transformers

1.4 SUBMITTALS

Submittal requirements are specified in Section 23 09 00 INSTRUMENTATION
AND CONTROL FOR HVAC.

1.5 DELIVERY AND STORAGE

Store and protect products from the weather, humidity, and temperature
variations, dirt and dust, and other contaminants, within the storage
condition limits published by the equipment manufacturer.

1.6 INPUT MEASUREMENT ACCURACY

Select, install and configure sensors, transmitters and DDC Hardware such
that the maximum error of the measured value at the input of the DDC
hardware is less than the maximum allowable error specified for the sensor
or instrumentation.

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 General Requirements

All products used to meet this specification must meet the indicated
requirements, but not all products specified here will be required by
every project. All products must meet the requirements both Section
23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section.

2.1.2 Operation Environment Requirements

Unless otherwise specified, provide products rated for continuous
operation under the following conditions:

2.1.2.1 Pressure

Pressure conditions normally encountered in the installed location.

2.1.2.2 Vibration

Vibration conditions normally encountered in the installed location.

2.1.2.3 Temperature

- a. Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.
- b. Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -37 to +66 degrees C and temperature conditions outside this range normally encountered at the installed location.

2.1.2.4 Humidity

10 to 95 percent relative humidity, non-condensing and also humidity conditions outside this range normally encountered at the installed location.

2.2 WEATHERSHIELDS

Provide weathershields constructed of galvanized steel painted white, unpainted aluminum, aluminum painted white, or white PVC.

2.3 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

2.3.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

2.3.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

2.3.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 50 mm (2 inch) lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation
- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

2.3.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

2.3.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

2.4 AUTOMATIC CONTROL VALVES

Provide valves with stainless-steel stems and stuffing boxes with extended necks to clear the piping insulation. Provide valves with bodies meeting ASME B16.34 or ASME B16.15 pressure and temperature class ratings based on the design operating temperature and 150 percent of the system design operating pressure. Unless otherwise specified or indicated, provide valves meeting FCI 70-2 Class III leakage rating. Provide valves rated for modulating or two-position service as indicated, which close against a differential pressure indicated as the Close-Off pressure and which are Normally-Open, Normally-Closed, or Fail-In-Last-Position as indicated.

2.4.1 Valve Type

2.4.1.1 Liquid Service 150 Degrees F or Less

Use either globe valves or ball valves except that butterfly valves may be used for sizes 100 mm and larger.

2.4.2 Valve Flow Coefficient and Flow Characteristic

2.4.2.1 Two-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide equal-percentage flow characteristic for liquid service except for butterfly valves. Provide linear flow characteristic for steam service except for butterfly valves.

2.4.2.2 Three-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide linear flow characteristic with constant total flow throughout full plug travel.

2.4.3 Two-Position Valves

Use full line size full port valves with maximum available (Kv).

2.4.4 Globe Valves

2.4.4.1 Liquid Service Not Exceeding 66 Degrees C

a. Valve body and body connections:

- (1) valves 38 mm and smaller: brass or bronze body, with threaded or union ends

- (2) valves from 51 mm to 76 mm inclusive: brass, bronze, or iron bodies. 51 millimeters valves with threaded connections; 63 to 76 mm valves with flanged connections

- b. Internal valve trim: Brass or bronze.
- c. Stems: Stainless steel.
- d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

2.4.4.2 Liquid Service Not Exceeding 121 Degrees C

- a. Valve body and body connections:
 - (a) 1) valves 38 mm and smaller: brass or bronze body, with threaded or union ends
 - (b) 2) valves from 51 mm to 76 mm inclusive: brass, bronze, or iron bodies. 51 millimeters valves with threaded connections; 63 to 76 mm valves with flanged connections
- b. Internal trim: Type 316 stainless steel including seats, seat rings, modulation plugs, valve stems, and springs.
- c. Provide valves with non-metallic parts suitable for a minimum continuous operating temperature of 121 degrees C or 28 degrees C above the system design temperature, whichever is higher.
- d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol

2.4.5 Ball Valves

2.4.5.1 Liquid Service Not Exceeding 66 Degrees C

- a. Valve body and connections:
 - (a) 1) valves 38 mm and smaller: bodies of brass or bronze, with threaded or union ends
 - (b) 2) valves from 51 mm to 76 mm inclusive: bodies of brass, bronze, or iron. 50 mm valves with threaded connections; valves from 63 to 76 mm with flanged connections.
- b. Ball: Stainless steel or nickel-plated brass or chrome-plated brass.
- c. Seals: Reinforced Teflon seals and EPDM O-rings.
- d. Stem: Stainless steel, blow-out proof.
- e. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

2.4.6 Butterfly Valves

Provide butterfly valves which are threaded lug type suitable for dead-end service and modulation to the fully-closed position, with carbon-steel bodies or with ductile iron bodies in accordance with ASTM A536. Provide

butterfly valves with non-corrosive discs, stainless steel shafts supported by bearings, and EPDM seats suitable for temperatures from -28.9 to +121.1 degrees C. Provide valves with rated Kv of the Kv at 70 percent (60 degrees) open position. Provide valves meeting FCI 70-2 Class VI leakage rating.

2.4.7 Pressure Independent Control Valves (PICV)

Provide pressure independent control valves which include a regulator valve which maintains the differential pressure across a flow control valve. Pressure independent control valves must accurately control the flow from 0-100 percent full rated flow regardless of changes in the piping pressure and not vary the flow more than plus or minus 5 percent at any given flow control valve position when the PICV differential pressure lies between the manufacturer's stated minimum and maximum. The rated minimum differential pressure for steady flow must not exceed 34.5 kPa across the PICV. Provide either globe or ball type valves meeting the indicated requirements for globe and ball valves. Provide valves with a flow tag listing full rated flow and minimum required pressure drop. Provide valves with factory installed Pressure/Temperature ports ("Pete's Plugs") to measure the pressure drop to determine the valve flow rate.

2.5 DAMPERS

2.5.1 Damper Assembly

Provide single damper sections with blades no longer than 1.2 m and which are no higher than 1.8 m and damper blade width of 203 mm or less. When larger sizes are required, combine damper sections. Provide dampers made of steel, or other materials where indicated and with assembly frames constructed of 2.8 mm minimum thickness stainless steel channels with mitered and welded corners. Steel channel frames constructed of 1.5 mm minimum thickness are acceptable provided the corners are reinforced.

- a. Flat blades must be made rigid by folding the edges. Blade-operating linkages must be within the frame so that blade-connecting devices within the same damper section must not be located directly in the air stream.
- b. Damper axles must be 13 mm minimum, plated steel rods supported in the damper frame by stainless steel or bronze bearings. Blades mounted vertically must be supported by thrust bearings.
- c. Provide dampers which do not exceed a pressure drop through the damper of 10 Pa at 5 m/s in the wide-open position. Provide dampers with frames not less than 50 mm in width. Provide dampers which have been tested in accordance with AMCA 500-D.

2.5.2 Operating Linkages

For operating links external to dampers, such as crank arms, connecting rods, and line shafting for transmitting motion from damper actuators to dampers, provide links able to withstand a load equal to at least 300 percent of the maximum required damper-operating force without deforming. Rod lengths must be adjustable. Links must be brass, bronze, zinc-coated steel, or stainless steel. Working parts of joints and clevises must be brass, bronze, or stainless steel. Adjustments of crank arms must control the open and closed positions of dampers.

2.5.3 Damper Types

2.5.3.1 Flow Control Dampers

Provide parallel-blade or opposed blade type dampers for outside air, return air, relief air, exhaust, face and bypass dampers as indicated on the Damper Schedule. Blades must have interlocking edges. The channel frames of the dampers must be provided with jamb seals to minimize air leakage. Unless otherwise indicated, dampers must meet AMCA 511 Class 1A requirements. Outside air damper seals must be suitable for an operating temperature range of -40 to +75 degrees C. Dampers must be rated at not less than 10 m/s air velocity.

2.5.3.2 Mechanical Rooms and Other Utility Space Ventilation Dampers

Provide utility space ventilation dampers as indicated. Unless otherwise indicated provide AMCA 511 class 3 dampers. Provide dampers rated at not less than 7.6 m/s air velocity.

2.5.3.3 Smoke Dampers

Provide smoke-damper and actuator assemblies which meet the current requirements of NFPA 90A, UL 555, and UL 555S. For combination fire and smoke dampers provide dampers rated for 121 degrees C Class II leakage per UL 555S.

2.6 SENSORS AND INSTRUMENTATION

Unless otherwise specified, provide sensors and instrumentation which incorporate an integral transmitter. Sensors and instrumentation, including their transmitters, must meet the specified accuracy and drift requirements at the input of the connected DDC Hardware's analog-to-digital conversion.

2.6.1 Analog and Binary Transmitters

Provide transmitters which match the characteristics of the sensor. Transmitters providing analog values must produce a linear 4-20 mAdc, 0-10 Vdc signal corresponding to the required operating range and must have zero and span adjustment. Transmitters providing binary values must have dry contacts rated at 1A at 24 Volts AC.

2.6.2 Network Transmitters

Sensors and Instrumentation incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS when used in a BACnet network.

2.6.3 Temperature Sensors

Provide the same sensor type throughout the project. Temperature sensors may be provided without transmitters. Where transmitters are used, the range must be the smallest available from the manufacturer and suitable for the application such that the range encompasses the expected range of temperatures to be measured. The end to end accuracy includes the combined effect of sensitivity, hysteresis, linearity and repeatability between the measured variable and the end user interface (graphic presentation) including transmitters if used.

2.6.3.1 Sensor Accuracy and Stability of Control

2.6.3.1.1 Conditioned Space Temperature

Plus or minus 0.3 degrees C over the operating range.

2.6.3.1.2 Unconditioned Space Temperature

- a. Plus or minus 0.6 degrees C over the range of -1 to +55 degrees C AND
- b. Plus or minus 2 degrees C over the rest of the operating range.

2.6.3.1.3 Duct Temperature

Plus or minus 0.3 degrees C

2.6.3.1.4 Outside Air Temperature

- a. Plus or minus 1 degree C over the range of -35 to +55 degrees C AND
- b. Plus or minus 0.6 degrees C over the range of -1 to +40 degrees C.

2.6.3.1.5 Chilled Water

Plus or minus 0.4 degrees C over the range of 2 to 18 degrees C.

2.6.3.2 Transmitter Drift

The maximum allowable transmitter drift: 0.1 degrees C per year.

2.6.3.3 Point Temperature Sensors

Point Sensors must be encapsulated in epoxy, series 300 stainless steel, anodized aluminum, or copper.

2.6.3.4 Temperature Sensor Details

2.6.3.4.1 Room Type

Provide the sensing element components within a decorative protective cover suitable for surrounding decor.

2.6.3.4.2 Duct Probe Type

Ensure the probe is long enough to properly sense the air stream temperature.

2.6.3.4.3 Duct Averaging Type

Continuous averaging sensors must be one foot in length for each 0.1 square m of duct cross-sectional area, and a minimum length of 1.5 m.

2.6.3.4.4 Pipe Immersion Type

For pipes with larger than 7.6 cm diameter, provide minimum 7.6 cm immersion. For pipes with less than 7.6 cm diameter, provide immersion at least half the diameter of the pipe. Provide each sensor with a corresponding pipe-mounted sensor well, unless indicated otherwise.

Sensor wells must be stainless steel when used in steel piping, and brass when used in copper piping.

2.6.3.4.5 Outside Air Type

Provide the sensing element rated for outdoor use

2.6.4 Relative Humidity Sensor

Relative humidity sensors must use bulk polymer resistive or thin film capacitive type non-saturating sensing elements capable of withstanding a saturated condition without permanently affecting calibration or sustaining damage. The sensors must include removable protective membrane filters. Where required for exterior installation, sensors must be capable of surviving below freezing temperatures and direct contact with moisture without affecting sensor calibration. When used indoors, the sensor must be capable of being exposed to a condensing air stream (100 percent relative humidity) with no adverse effect to the sensor's calibration or other harm to the instrument. The sensor must be of the wall-mounted or duct-mounted type, as required by the application, and must be provided with any required accessories. Sensors used in duct high-limit applications must have a bulk polymer resistive sensing element. Duct-mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. Relative humidity (RH) sensors must measure relative humidity over a range of 0 percent to 100 percent with an accuracy of plus or minus 3 percent. RH sensors must function over a temperature range of 4.4 to 57.2 degrees C and must not drift more than 1 percent per year.

2.6.5 Carbon Dioxide (CO2) Sensors

Provide photometric type CO2 sensors with integral transducers and linear output. Carbon dioxide (CO2) sensors must measure CO2 concentrations between 0 to 2000 parts per million (ppm) using non-dispersible infrared (NDIR) technology with an accuracy of plus or minus 50 ppm and a maximum response time of 1 minute. The sensor must be rated for operation at ambient air temperatures within the range of 0 to 50 degrees C and relative humidity within the range of 20 to 95 percent (non-condensing). The sensor must have a maximum drift of 2 percent per year. The sensor chamber must be manufactured with a non-corrosive material that does not affect carbon dioxide sample concentration. Duct mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. The sensor must have a calibration interval no less than 5 years.

2.6.6 Differential Pressure Instrumentation

2.6.6.1 Differential Pressure Sensors

Provide Differential Pressure Sensors with ranges as indicated or as required for the application. Pressure sensor ranges must not exceed the high end range indicated on the Points Schedule by more than 50 percent. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The accuracy must be plus or minus 1 percent of full scale. The sensor must have a maximum drift of 2 percent per year

2.6.6.2 Differential Pressure Switch

Provide differential pressure switches with a user-adjustable setpoint which are sized for the application such that the setpoint is between 25 percent and 75 percent of the full range. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The switch must have two sets of contacts and each contact must have a rating greater than it's connected load. Contacts must open or close upon rise of pressure above the setpoint or drop of pressure below the setpoint as indicated.

2.6.7 Flow Sensors

2.6.7.1 Airflow Measurement Array (AFMA)

2.6.7.1.1 Airflow Straightener

Provide AFMAs which contain an airflow straightener if required by the AFMA manufacturer's published installation instructions. The straightener must be contained inside a flanged sheet metal casing, with the AFMA located as specified according to the published recommendation of the AFMA manufacturer. In the absence of published documentation, provide airflow straighteners if there is any duct obstruction within 5 duct diameters upstream of the AFMA. Air-flow straighteners, where required, must be constructed of 3 mm aluminum honeycomb and the depth of the straightener must not be less than 40 mm.

2.6.7.1.2 Resistance to Airflow

The resistance to air flow through the AFMA, including the airflow straightener must not exceed 20 Pa at an airflow of 10 m/s. AFMA construction must be suitable for operation at airflows of up to 25 m/s over a temperature range of 4 to 49 degrees C.

2.6.7.1.3 Outside Air Temperature

In outside air measurement or in low-temperature air delivery applications, provide an AFMA certified by the manufacturer to be accurate as specified over a temperature range of -29 to +49 degrees C.

2.6.7.1.4 Pitot Tube AFMA

Each Pitot Tube AFMA must contain an array of velocity sensing elements. The velocity sensing elements must be of the multiple pitot tube type with averaging manifolds. The sensing elements must be distributed across the duct cross section in the quantity and pattern specified or recommended by the published installation instructions of the AFMA manufacturer.

- a. Pitot Tube AFMAs for use in airflows over 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 2.5 to 12.5 m/s.
- b. Pitot Tube AFMAs for use in airflows under 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s.

2.6.7.1.5 Electronic AFMA

Each electronic AFMA must consist of an array of velocity sensing elements of the resistance temperature detector (RTD) or thermistor type. The sensing elements must be distributed across the duct cross section in the

quantity and pattern specified or recommended by the published application data of the AFMA manufacturer. Electronic AFMAs must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s and the output must be temperature compensated over a range of 0 to 100 degrees C.

2.6.7.1.6 Fan Inlet Measurement Devices

Fan inlet measurement devices cannot be used unless indicated on the drawings or schedules.

2.6.7.2 Orifice Plate

Orifice plate must be made of an austenitic stainless steel sheet of 3. mm nominal thickness with an accuracy of plus or minus 1 percent of full flow. The orifice plate must be flat within 0.1 mm. The orifice surface roughness must not exceed 0.5 μm . The thickness of the cylindrical face of the orifice must not exceed 2 percent of the pipe inside diameter or 12.5 percent of the orifice diameter, whichever is smaller. The upstream edge of the orifice must be square and sharp. Where orifice plates are used, concentric orifice plates must be used in all applications except steam flow measurement in horizontal pipelines.

2.6.7.3 Flow Nozzle

Flow nozzle must be made of austenitic stainless steel with an accuracy of plus or minus 1 percent of full flow. The inlet nozzle form must be elliptical and the nozzle throat must be the quadrant of an ellipse. The thickness of the nozzle wall and flange must be such that distortion of the nozzle throat from strains caused by the pipeline temperature and pressure, flange bolting, or other methods of installing the nozzle in the pipeline must not cause the accuracy to degrade beyond the specified limit. The outside diameter of the nozzle flange or the design of the flange facing must be such that the nozzle throat must be centered accurately in the pipe.

2.6.7.4 Venturi Tube

Venturi tube must be made of cast iron or cast steel and must have an accuracy of plus or minus 1 percent of full flow. The throat section must be lined with austenitic stainless steel. Thermal expansion characteristics of the lining must be the same as that of the throat casting material. The surface of the throat lining must be machined to a plus or minus 1.2 μm finish, including the short curvature leading from the converging entrance section into the throat.

2.6.7.5 Annular Pitot Tube

Annular pitot tube must be made of austenitic stainless steel with an accuracy of plus or minus 2 percent of full flow and a repeatability of plus or minus 0.5 percent of measured value. The unit must have at least one static port and no less than four total head pressure ports with an averaging manifold.

2.6.7.6 Insertion Turbine Flowmeter

Provide dual axial turbine flowmeter with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Insertion Turbine Flowmeter accuracy

must be plus or minus 0.5 percent of rate at calibrated velocity., within plus or minus of rate over a 10:1 turndown and within plus or minus 2 percent of rate over a 50:1 turndown. Repeatability must be plus or minus 0.25 percent of reading. The meter flow sensing element must operate over a range suitable for the installed location with a pressure loss limited to 1 percent of operating pressure at maximum flow rate. The flowmeter ,must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs. The turbine rotor assembly must be constructed of Series 300 stainless steel and use Teflon seals.

2.6.7.7 Vortex Shedding Flowmeter

Vortex Shedding Flowmeter accuracy must be within plus or minus 0.8 percent of the actual reading over the range of the meter. Steam meters must contain density compensation by direct measurement of temperature. Mass flow inferred from specified steam pressure are not acceptable. The flow meter body must be made of austenitic stainless steel and include a weather tight NEMA 4X electronics enclosure. The vortex shedding flowmeter body must not require removal from the piping in order to replace the shedding sensor.

2.6.7.8 Ultrasonic Flow Meter

Provide Ultrasonic Flow Meters complete with matched transducers, self aligning installation hardware and transducer cables. Ultrasonic transducers must be optimized for the specific pipe and process conditions for the application. The flow meter accuracy must plus or minus 1 percent of rate from 0.3 to 12 meters/sec. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc output.

2.6.7.9 Insertion Magnetic Flow Meter

Provide insertion type magnetic flowmeters with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Flowmeter accuracy must be no greater than plus or minus 1 percent of rate from 0.6 to 6 meters/sec. Wetted material parts must be 300 series stainless steel. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs.

2.6.7.10 Positive Displacement Flow Meter

The flow meter must be a direct reading, gerotor, nutating disc or vane type displacement device rated for liquid service as indicated. A counter must be mounted on top of the meter, and must consist of a non-resettable mechanical totalizer for local reading, and a pulse transmitter for remote reading. The totalizer must have a six digit register to indicate the volume passed through the meter in liters, and a sweep-hand dial to indicate down to 1 L. The pulse transmitter must have a hermetically sealed reed switch which is activated by magnets fixed on gears of the counter. The meter must have a bronze body with threaded or flanged connections as required for the application. Output accuracy must be plus or minus 2 percent of the flow range. The maximum pressure drop at full flow must be 34 kPa.

2.6.7.11 Flow Meters, Paddle Type

Sensor must be non-magnetic, with forward curved impeller blades designed

for water containing debris. Sensor accuracy must be plus or minus 1 percent of rate of flow, minimum operating flow velocity must be 0.3 meters/second. Sensor repeatability and linearity must be plus or minus 1 percent. Materials which will be wetted must be made from non-corrosive materials and must not contaminate water. The sensor must be rated for installation in pipes of 76 mm to 1 m diameters. The transmitter housing must be a NEMA 250 Type 4 enclosure.

2.6.7.12 Flow Switch

Flow switch must have a repetitive accuracy of plus or minus 10 percent of actual flow setting. Switch actuation must be adjustable over the operating flow range, and must be sized for the application such that the setpoint is between 25 percent and 75 percent of the full range.. The switch must have Form C snap-action contacts, rated for the application. The flow switch must have non flexible paddle with magnetically actuated contacts and be rated for service at a pressure greater than the installed conditions. Flow switch for use in sewage system must be rated for use in corrosive environments encountered.

2.6.7.13 Gas Flow Meter

Gas flow meter must be diaphragm or bellows type (gas positive displacement meters) for flows up to 19.7 L/sec and axial flow turbine type for flows above 19.7 L/sec, designed specifically for natural gas supply metering, and rated for the pressure, temperature, and flow rates of the installation. Meter must have a minimum turndown ratio of 10 to 1 with an accuracy of plus or minus 1 percent of actual flow rate. The meter index must include a direct reading mechanical totalizing register and electrical impulse dry contact output for remote monitoring. The electrical impulse dry contact output must not require field adjustment or calibration. The electrical impulse dry contact output must have a minimum resolution of 3 cubic meters of gas per pulse and must not exceed 15 pulses per second at the design flow.

2.6.8 Electrical Instruments

Provide Electrical Instruments with an input range as indicated or sized for the application. Unless otherwise specified, AC instrumentation must be suitable for 60 Hz operation.

2.6.8.1 Current Transducers

Current transducers must accept an AC current input and must have an accuracy of plus or minus 0.5 percent of full scale. The device must have a means for calibration. Current transducers for variable frequency applications must be rated for variable frequency operation.

2.6.8.2 Current Sensing Relays (CSRs)

Current sensing relays (CSRs) must provide a normally-open contact with a voltage and amperage rating greater than its connected load. Current sensing relays must be of split-core design. The CSR must be rated for operation at 200 percent of the connected load. Voltage isolation must be a minimum of 600 volts. The CSR must auto-calibrate to the connected load or be adjustable and field calibrated. Current sensors for variable frequency applications must be rated for variable frequency operation.

2.6.8.3 Voltage Transducers

Voltage transducers must accept an AC voltage input and have an accuracy of plus or minus 0.25 percent of full scale. The device must have a means for calibration. Line side fuses for transducer protection must be provided.

2.6.8.4 Energy Metering

2.6.8.4.1 Watt or Watthour Transducers

Watt transducers must measure voltage and current and must output kW or kWh or both kW and kWh as indicated. kW outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1. kWh outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1.

2.6.8.4.2 Watthour Revenue Meter (with and without Demand Register)

All Watthour revenue meters must measure voltage and current and must be in accordance with ANSI C12.1 with an ANSI C12.20 Accuracy class of 0.5 and must have pulse initiators for remote monitoring of Watthour consumption. Pulse initiators must consist of form C contacts with a current rating not to exceed two amperes and voltage not to exceed 500 V, with combinations of VA not to exceed 100 VA, and a life rating of one billion operations. Meter sockets must be in accordance with NEMA/ANSI C12.10. Watthour revenue meters with demand registers must output instantaneous demand in addition to the pulse initiators.

2.6.8.4.3 Hydronic BTU Meters

The BTU meter is to be supplied with wall mount hardware and be capable of being installed remote from the flow meter. The BTU meter must include an LCD display for local indication of energy rate and for display of parameters and settings during configuration. Each BTU meter must be factory configured for its specific application and be completely field configurable by the user via a front panel keypad (no special interface device or computer required). The unit must output Energy Rate, Energy Total, Flow Rate, Supply Temperature, and Return Temperature. An integral transmitter is to provide a linear analog or configurable pulse output signal representing the energy rate; and the signal must be compatible with building automation system DDC Hardware to which the output is connected.

2.6.9 pH Sensor

The sensor must be suitable for applications and chemicals encountered in water treatment systems of boilers, chillers and condenser water systems. Construction, wiring, fittings and accessories must be corrosion and chemical resistant with fittings for tank or suspension installation. Housing must be polyvinylidene fluoride with O-rings made of chemical resistant materials which do not corrode or deteriorate with extended exposure to chemicals. The sensor must be encapsulated. Periodic replacement must not be required for continued sensor operation. Sensors must use a ceramic junction and pH sensitive glass membrane capable of withstanding a pressure of 689 kPa at 66 degrees C. The reference cell must be double junction configuration. Sensor range must be 0 to 12 pH, stability 0.05, sensitivity 0.02, and repeatability of plus or minus 0.05 pH value, response of 90 percent of full scale in one second and a

linearity of 99 percent of theoretical electrode output measured at 24 degrees C.

2.6.10 Oxygen Analyzer

Oxygen analyzer must consist of a zirconium oxide sensor for continuous sampling and an air-powered aspirator to draw flue gas samples. The analyzer must be equipped with filters to remove flue air particles. Sensor probe temperature rating must be 435 degrees C. The sensor assembly must be equipped for flue flange mounting.

2.6.11 Occupancy Sensors

Occupancy sensors must have occupancy-sensing sensitivity adjustment and an adjustable off-delay timer with a setpoint of 15 minutes. Adjustments accessible from the face of the unit are preferred. Occupancy sensors must be rated for operation in ambient air temperatures ranging from 5 to 35 degrees C or temperatures normally encountered in the installed location. Sensors integral to wall mount on-off light switches must have an auto-off switch. Wall switch sensors must be decorator style and must fit behind a standard decorator type wall plate. All occupancy sensors, power packs, and slave packs must be UL listed. In addition to any outputs required for lighting control, the occupancy sensor must provide an output for the HVAC control system.

2.6.11.1 Passive Infrared (PIR) Occupancy Sensors

PIR occupancy sensors must have a multi-level, multi-segmented viewing lens and a conical field of view with a viewing angle of 180 degrees and a detection of at least 6 m unless otherwise indicated or specified. PIR Sensors must provide field-adjustable background light-level adjustment with an adjustment range suitable to the light level in the sensed area, room or space. PIR sensors must be immune to false triggering from RFI and EMI.

2.6.11.2 Ultrasonic Occupancy Sensors

Ultrasonic sensors must operate at a minimum frequency 32 kHz and must be designed to not interfere with hearing aids.

2.6.11.3 Dual-Technology Occupancy Sensor (PIR and Ultrasonic)

Dual-Technology Occupancy Sensors must meet the requirements of both PIR and Ultrasonic Occupancy Sensors.

2.6.12 Vibration Switch

Vibration switch must be solid state, enclosed in a NEMA 250 Type 4 or Type 4X housing with sealed wire entry. Unit must have two independent sets of Form C switch contacts with one set to shutdown equipment upon excessive vibration and a second set for monitoring alarm level vibration. The vibration sensing range must be a true rms reading, suitable for the application. The unit must include either displacement response for low speed or velocity response for high speed application. The frequency range must be at least 3 Hz to 500 Hz. Contact time delay must be 3 seconds. The unit must have independent start-up and running delay on each switch contact. Alarm limits must be adjustable and setpoint accuracy must be plus or minus 10 percent of setting with repeatability of plus or minus 2 percent.

2.6.13 Conductivity Sensor

Sensor must include local indicating meter and must be suitable for measurement of conductivity of water in boilers, chilled water systems, condenser water systems, distillation systems, or potable water systems as indicated. Sensor must sense from 0 to 10 microSeimens per centimeter ($\mu\text{S}/\text{cm}$) for distillation systems, 0 to 100 $\mu\text{S}/\text{cm}$ for boiler, chilled water, and potable water systems and 0 to 1000 $\mu\text{S}/\text{cm}$ for condenser water systems. Contractor must field verify the ranges for particular applications and adjust the range as required. The output must be temperature compensated over a range of 0 to 100 degrees C. The accuracy must be plus or minus 2 percent of the full scale reading. Sensor must have automatic zeroing and must require no periodic maintenance or recalibration.

2.6.14 Compressed Air Dew Point Sensor

Sensor must be suitable for measurement of dew point from -40 +27 degrees C over a pressure range of 0 to 1 MPa. The transmitter must provide both dry bulb and dew point temperatures on separate outputs. The end to end accuracy of the dew point must be plus or minus 2.8 degrees C and the dry bulb must be plus or minus 0.6 degrees C. Sensor must be automatic zeroing and must require no normal maintenance or periodic recalibration.

2.6.15 Turbidity Sensor

Sensor must include a local indicating meter and must be suitable for measurement of turbidity of water. Sensor must sense from 0 to 1000 Nephelometric Turbidity Units (NTU). Range must be field-verified for the particular application and adjusted as required. The output must be temperature compensated over a range of 0 to 100 degrees C. The accuracy must be plus or minus 5 percent of full scale reading. Sensor must have automatic zeroing and must not require periodic maintenance or recalibration.

2.6.16 Chlorine Detector

The detector must measure concentrations of chlorine in water in the range 0 to 20 ppm with a repeatability of plus or minus 1 percent of full scale and an accuracy of plus or minus 2 percent of full scale. The Chlorine Detector transmitter must be housed in a non-corrosive NEMA 250 Type 4X enclosure. Detector must include a local panel with adjustable alarm trip level, local audio and visual alarm with silence function.

2.6.17 Floor Mounted Leak Detector

Leak detectors must use electrodes mounted at slab level with a minimum built-in-vertical adjustment of 3 mm. Detector must have a binary output. The indicator must be manual reset type.

2.6.18 Temperature Switch

2.6.18.1 Duct Mount Temperature Low Limit Safety Switch (Freezestat)

Duct mount temperature low limit switches (Freezestats) must be manual reset, low temperature safety switches at least 3 meters long per square meter of coverage which must respond to the coldest 450 mm segment with an accuracy of plus or minus 2 degrees C. The switch must have a

field-adjustable setpoint with a range of at least $-1 +10$ degrees C. The switch must have two sets of contacts, and each contact must have a rating greater than its connected load. Contacts must open or close upon drop of temperature below setpoint as indicated and must remain in this state until reset.

2.6.18.2 Pipe Mount Temperature Limit Switch (Aquastat)

Pipe mount temperature limit switches (aquastats) must have a field adjustable setpoint between 15 and 32 degrees C, an accuracy of plus or minus 2 degrees C and a 5 degrees C fixed deadband. The switch must have two sets of contacts, and each contact must have a rating greater than its connected load. Contacts must open or close upon change of temperature above or below setpoint as indicated.

2.6.19 Damper End Switches

Each end switch must be a hermetically sealed switch with a trip lever and over-travel mechanism. The switch enclosure must be suitable for mounting on the duct exterior and must permit setting the position of the trip lever that actuates the switch. The trip lever must be aligned with the damper blade.

End switches integral to an electric damper actuator are allowed as long as at least one is adjustable over the travel of the actuator.

2.6.20 Air Quality Sensors

Provide full spectrum air quality sensors using a hot wire element based on the Taguchi principle. The sensor must monitor a wide range of gaseous volatile organic components common in indoor air contaminants like paint fumes, solvents, cigarette smoke, and vehicle exhaust. The sensor must automatically compensate for temperature and humidity, have span and calibration potentiometers, operate on 24 VDC power with output of 0-10 VDC, and have a service rating of 0 to 60 degrees C and 5 to 95 percent relative humidity.

2.7 INDICATING DEVICES

All indicating devices must display readings in metric (SI) units.

2.7.1 Thermometers

Provide bi-metal type thermometers at locations indicated. Thermometers must have either 230 mm long scales or 90 mm diameter dials, with insertion, immersion, or averaging elements. Provide matching thermowells for pipe-mounted installations. Select scale ranges suitable for the intended service, with the normal operating temperature near the scale's midpoint. The thermometer's accuracy must be plus or minus 2 percent of the scale range.

2.7.1.1 Piping System Thermometers

Piping system thermometers must have brass, malleable iron or aluminum alloy case and frame, clear protective face, permanently stabilized glass tube with indicating-fluid column, white face, black numbers, and a 230 mm scale. Piping system thermometers must have an accuracy of plus or minus 1 percent of scale range. Thermometers for piping systems must have rigid stems with straight, angular, or inclined pattern. Thermometer stems must

have expansion heads as required to prevent breakage at extreme temperatures. On rigid-stem thermometers, the space between bulb and stem must be filled with a heat-transfer medium.

2.7.1.2 Air-Duct Thermometers

Air-duct thermometers must have perforated stem guards and 45-degree adjustable duct flanges with locking mechanism.

2.7.2 Pressure Gauges

Provide pipe-mounted pressure gauges at the locations indicated. Gauges must conform to ASME B40.100 and have a 100 mm diameter dial and shutoff cock. Select scale ranges suitable for the intended service, with the normal operating pressure near the scale's midpoint. The gauge's accuracy must be plus or minus 2 percent of the scale range.

Gauges must be suitable for field or panel mounting as required, must have black legend on white background, and must have a pointer traveling through a 270-degree arc. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus 3 percent of scale range. Gauges must meet requirements of ASME B40.100.

2.7.3 Low Differential Pressure Gauges

Gauges for low differential pressure measurements must be a minimum of 90 mm (nominal) size with two sets of pressure taps, and must have a diaphragm-actuated pointer, white dial with black figures, and pointer zero adjustment. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus two percent of scale range.

2.8 OUTPUT DEVICES

2.8.1 Actuators

Actuators must be electric (electronic). All actuators must be normally open (NO), normally closed (NC) or fail-in-last-position (FILP) as indicated. Normally open and normally closed actuators must be of mechanical spring return type. Electric actuators must have an electronic cut off or other means to provide burnout protection if stalled. Actuators must have a visible position indicator. Electric actuators must provide position feedback to the controller as indicated. Actuators must smoothly and fully open or close the devices to which they are applied. Electric actuators must have a full stroke response time in both directions of 90 seconds or less at rated load. Electric actuators must be of the foot-mounted type with an oil-immersed gear train or the direct-coupled type. Where multiple electric actuators operate from a common signal, the actuators must provide an output signal identical to its input signal to the additional devices. All actuators must be rated for their operating environment. Actuators used outdoors must be designed and rated for outdoor use. Actuators under continuous exposure to water, such as those used in sumps, must be submersible.

Actuators incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

2.8.1.1 Valve Actuators

Valve actuators must provide shutoff pressures and torques as indicated on the Valve Schedule.

2.8.1.2 Damper Actuators

Damper actuators must provide the torque necessary per damper manufacturer's instructions to modulate the dampers smoothly over its full range of operation and torque must be at least 7.3 Nm/square m of damper area for opposed blade dampers and 10.9 Nm/square m of damper area for parallel blade dampers.

2.8.1.3 Electric Actuators

Each actuator must have distinct markings indicating the full-open and full-closed position. Each actuator must deliver the torque required for continuous uniform motion and must have internal end switches to limit the travel, or be capable of withstanding continuous stalling without damage. Actuators must function properly within 85 to 110 percent of rated line voltage. Provide actuators with hardened steel running shafts and gears of steel or copper alloy. Fiber or reinforced nylon gears may be used for torques less than 16 inch-pounds..

- a. Two-position actuators must be single direction, spring return, or reversing type. Two position actuator signals may either be the control power voltage or line voltage as needed for torque or appropriate interlock circuits.
- b. Modulating actuators must be capable of stopping at any point in the cycle, and starting in either direction from any point. Actuators must be equipped with a switch for reversing direction, and a button to disengage the clutch to allow manual adjustments. Provide the actuator with a hand crank for manual adjustments, as applicable. Modulating actuator input signals can either be a 4 to 20 mAdc or a 0-10 VDC signal.
- c. Floating or pulse width modulation actuators are acceptable for non-fail safe applications unless indicated otherwise provided that the floating point control (timed actuation) must have a scheduled re-calibration of span and position no more than once a day and no less than once a week. The schedule for the re-calibration should not affect occupied conditions and be staggered between equipment to prevent falsely loading or unloading central plant equipment.

2.8.2 Solenoid-Operated Electric to Pneumatic Switch (EPS)

Solenoid-Operated Electric to Pneumatic Switches (EPS) must accept a voltage input to actuate its air valve. Each valve must have three-port operation: common, normally open, and normally closed. Each valve must have an outer cast aluminum body and internal parts of brass, bronze, or stainless steel. The air connection must be a 10 mm NPT threaded connection. Valves must be rated for 345 kPa.

2.8.3 Relays

Relays must have contacts rated for the intended application, indicator light, and dust proof enclosure. The indicator light must be lit when the coil is energized and off when coil is not energized.

Control relay contacts must have utilization category and ratings selected for the application. Each set of contacts must incorporate a normally open (NO), normally closed (NC) and common contact. Relays must be rated for a minimum life of one million operations.

2.9 USER INPUT DEVICES

User Input Devices, including potentiometers, switches and momentary contact push-buttons. Potentiometers must be of the thumb wheel or sliding bar type. Momentary Contact Push-Buttons may include an adjustable timer for their output. User input devices must be labeled for their function.

2.10 MULTIFUNCTION DEVICES

Multifunction devices are products which combine the functions of multiple sensor, user input or output devices into a single product. Unless otherwise specified, the multifunction device must meet all requirements of each component device. Where the requirements for the component devices conflict, the multifunction device must meet the most stringent of the requirements.

2.10.1 Current Sensing Relay Command Switch

The Current Sensing Relay portion must meet all requirements of the Current Sensing Relay input device. The Command Switch portion must meet all requirements of the Relay output device except that it must have at least one normally-open (NO) contact.

Current Sensing Relays used for Variable Frequency Drives must be rated for Variable Frequency applications unless installed on the source side of the drive. If used in this situation, the threshold for showing status must be set to allow for the VFD's control power when the drive is not enabled and provide indication of operation when the drive is enabled at minimum speed.

2.10.2 Space Sensor Module

Space Sensor Modules must be multifunction devices incorporating a temperature sensor and one or more of the following as specified and indicated on the Space Sensor Module Schedule:

- a. A temperature indicating device.
- b. A User Input Device which must adjust a temperature setpoint output.
- c. A User Input Momentary Contact Button and an output to the control system indicating zone occupancy.
- d. A three position User Input Switch labeled to indicate heating, cooling and off positions ('HEAT-COOL-OFF' switch) and providing corresponding outputs to the control system.
- e. A two position User Input Switch labeled with 'AUTO' and 'ON' positions and providing corresponding output to the control system..
- f. A multi-position User Input Switch with 'OFF' and at least two fan speed positions and providing corresponding outputs to the control

system.

Space Sensor Modules cannot contain mercury (Hg).

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 General Installation Requirements

Perform the installation under the supervision of competent technicians regularly employed in the installation of DDC systems.

3.1.1.1 Device Mounting Criteria

All devices must be installed in accordance with manufacturer's recommendations and as specified and indicated. Control devices to be installed in piping and ductwork must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Strap-on temperature sensing elements must not be used except as specified. Spare thermowells must be installed adjacent to each thermowell containing a sensor and as indicated. Devices located outdoors must have a weathershield.

3.1.1.2 Labels and Tags

Match labels and tags to the unique identifiers indicated on the As-Built drawings. Label all enclosures and instrumentation. Tag all sensors and actuators in mechanical rooms. Tag airflow measurement arrays to show flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient. Tag duct static pressure taps at the location of the pressure tap. Provide plastic or metal tags, mechanically attached directly to each device or attached by a metal chain or wire. Labels exterior to protective enclosures must be engraved plastic and mechanically attached to the enclosure or instrumentation. Labels inside protective enclosures may be attached using adhesive, but must not be hand written.

3.1.2 Weathershield

Provide weathershields for sensors located outdoors. Install weathershields such that they prevent the sun from directly striking the sensor and prevent rain from directly striking or dripping onto the sensor. Install weather shields with adequate ventilation so that the sensing element responds to the ambient conditions of the surroundings. When installing weathershields near outside air intake ducts, install them such that normal outside air flow does not cause rainwater to strike the sensor.

3.1.3 Room Instrument Mounting

Mount room instruments, including but not limited to wall mounted non-adjustable space sensor modules and sensors located in occupied spaces, 1.5 meters above the floor unless otherwise indicated. Install adjustable devices to be ADA compliant unless otherwise indicated on the Room Sensor Schedule:

- a. Space Sensor Modules for Fan Coil Units may be either unit or wall

mounted but not mounted on an exterior wall.

b. Wall mount all other Space Sensor Modules.

3.1.4 Indication Devices Installed in Piping and Liquid Systems

Provide snubbers for gauges in piping systems subject to pulsation. For gauges for steam service use pigtail fittings with cock. Install thermometers and temperature sensing elements in liquid systems in thermowells. Provide spare Pressure/Temperature Ports (Pete's Plug) for all temperature and pressure sensing elements installed in liquid systems for calibration/testing.

3.1.5 Occupancy Sensors

Provide a sufficient quantity of occupancy sensors to provide complete coverage of the area (room or space). Occupancy sensors are to be ceiling mounted. Install occupancy sensors in accordance with NFPA 70 requirements and the manufacturer's instructions. Do not locate occupancy sensors within 2 m of HVAC outlets or heating ducts, or where they can "see" beyond any doorway. Installation above doorway(s) is preferred. Do not use ultrasonic sensors in spaces containing ceiling fans. Install sensors to detect motion to within 600 mm of all room entrances and to not trigger due to motion outside the room. Set the off-delay timer to 15 minutes unless otherwise indicated. Adjust sensors prior to beneficial occupancy, but after installation of furniture systems, shelving, partitions, etc. For each controlled area, provide one hundred percent coverage capable of detecting small hand-motion movements, accommodating all occupancy habits of single or multiple occupants at any location within the controlled room.

3.1.6 Switches

3.1.6.1 Temperature Limit Switch

Provide a temperature limit switch (freezestat) to sense the temperature at the location indicated. Provide a sufficient number of temperature limit switches (freezestats) to provide complete coverage of the duct section but no less than 3 m in length per square meter of cross sectional area. Install manual reset limit switches in approved, accessible locations where they can be reset easily. Install temperature limit switch (freezestat) sensing elements in a side-to-side (not top-to-bottom) serpentine pattern with the relay section at the highest point and in accordance with the manufacturer's installation instructions.

3.1.6.2 Hand-Off Auto Switches

Wire safety controls such as smoke detectors and freeze protection thermostats to protect the equipment during both hand and auto operation.

3.1.7 Temperature Sensors

Install temperature sensors in locations that are accessible and provide a good representation of sensed media. Installations in dead spaces are not acceptable. Calibrate and install sensors according to manufacturer's instructions. Select sensors only for intended application as designated or recommended by manufacturer.

3.1.7.1 Room Temperature Sensors

Mount the sensors on interior walls to sense the average room temperature at the locations indicated. Avoid locations near heat sources such as copy machines or locations by supply air outlet drafts. Mount the center of all user-adjustable sensors 1220 mm above the floor to meet ADA requirements. Non user-adjustable sensors can be mounted as indicated in paragraph ROOM INSTRUMENT MOUNTING.

3.1.7.2 Duct Temperature Sensors

3.1.7.2.1 Probe Type

Place tip of the sensor in the middle of the airstream or in accordance with manufacturer's recommendations or instructions. Provide a gasket between the sensor housing and the duct wall. Seal the duct penetration air tight. When installed in insulated duct, provide enclosure or stand off fitting to accommodate the thickness of duct insulation to allow for maintenance or replacement of the sensor and wiring terminations. Seal the duct insulation penetration vapor tight.

3.1.7.2.2 Averaging Type

Weave the sensing element in a serpentine fashion from side to side perpendicular to the flow, across the duct or air handler cross-section, using durable non-metal supports in accordance with manufacturer's installation instructions. Avoid tight radius bends or kinking of the sensing element. Prevent contact between the sensing element and the duct or air handler internals. Provide a duct access door at the sensor location. The access door must be hinged on the side, factory insulated, have cam type locks, and be as large as the duct will permit, maximum 18 by 18 inches. For sensors inside air handlers, the sensors must be fully accessible through the air handler's access doors without removing any of the air handler's internals.

3.1.7.3 Immersion Temperature Sensors

Provide thermowells for sensors measuring piping, tank, or pressure vessel temperatures. Locate wells to sense continuous flow conditions. Do not install wells using extension couplings. When installed on insulated piping, provide stand enclosure or stand off fitting to accommodate the thickness of the pipe insulation and allow for maintenance or replacement of the sensor or wiring terminations. Where piping diameters are smaller than the length of the wells, provide wells in piping at elbows to sense flow across entire area of well. Wells must not restrict flow area to less than 70 percent of pipe area. Increase piping size as required to avoid restriction. Provide the sensor well with a heat-sensitive transfer agent between the sensor and the well interior ensuring contact between the sensor and the well.

3.1.7.4 Outside Air Temperature Sensors

Provide outside air temperature sensors on the building's north side with a protective weather shade that does not inhibit free air flow across the sensing element, and protects the sensor from snow, ice, and rain. Location must not be near exhaust hoods and other areas such that it is not influenced by radiation or convection sources which may affect the reading. Provide a shield to shade the sensor from direct sunlight.

3.1.8 Air Flow Measurement Arrays (AFMA)

Locate Outside Air AFMAs downstream from the Outside Air filters.

Install AFMAs with the manufacturer's recommended minimum distances between upstream and downstream disturbances. Airflow straighteners may be used to reduce minimum distances as recommended by the AFMA manufacturer.

3.1.9 Duct Static Pressure Sensors

Locate the duct static pressure sensing tap at 75 percent of the distance between the first and last air terminal units. If the transmitter output is a 0-10Vdc signal, locate the transmitter in the same enclosure as the air handling unit (AHU) controller for the AHU serving the terminal units. If a remote duct static pressure sensor is to be used, run the signal wire back to the controller for the air handling unit.

3.1.10 Relative Humidity Sensors

Install relative humidity sensors in supply air ducts at least 3 m downstream of humidity injection elements.

3.1.11 Meters

3.1.11.1 Flowmeters

Install flowmeters to ensure minimum straight unobstructed piping for at least 10 pipe diameters upstream and at least 5 pipe diameters downstream of the flowmeter, and in accordance with the manufacturer's installation instructions.

3.1.11.2 Energy Meters

Locate energy meters as indicated. Connect each meter output to the DDC system, to measure both instantaneous demand/energy and other variables as indicated.

3.1.12 Dampers

3.1.12.1 Damper Actuators

Provide spring return actuators which fail to a position that protects the served equipment and space on all control dampers related to freeze protection or force protection. For all outside, makeup and relief dampers provide dampers which fail closed. Terminal fan coil units, terminal VAV units, convectors, and unit heaters may be non-spring return unless indicated otherwise. Do not mount actuators in the air stream. Do not connect multiple actuators to a common drive shaft. Install actuators so that their action seal the damper to the extent required to maintain leakage at or below the specified rate and so that they move the blades smoothly throughout the full range of motion.

3.1.12.2 Damper Installation

Install dampers straight and true, level in all planes, and square in all dimensions. Dampers must move freely without undue stress due to twisting, racking (parallelogramming), bowing, or other installation error. External linkages must operate smoothly over the entire range of

motion, without deformation or slipping of any connecting rods, joints or brackets that will prevent a return to it's normal position. Blades must close completely and leakage must not exceed that specified at the rated static pressure. Provide structural support for multi-section dampers. Acceptable methods of structural support include but are not limited to U-channel, angle iron, corner angles and bolts, bent galvanized steel stiffeners, sleeve attachments, braces, and building structure. Where multi-section dampers are installed in ducts or sleeves, they must not sag due to lack of support. Do not use jackshafts to link more than three damper sections. Do not use blade to blade linkages. Install outside and return air dampers such that their blades direct their respective air streams towards each other to provide for maximum mixing of air streams.

3.1.13 Valves

Install the valves in accordance with the manufacturer's instructions.

3.1.13.1 Valve Actuators

Provide spring return actuators on all control valves where freeze protection is required. Spring return actuators for terminal fan coil units, terminal VAV units, convectors, and unit heaters are not required unless indicated otherwise.

3.1.14 Thermometers and Gauges

3.1.14.1 Thermometers

Mount devices to allow reading while standing on the floor or ground, as applicable.

3.1.15 Wire and Cable

Provide complete electrical wiring for the Control System, including wiring to transformer primaries. Wire and Cable must be installed without splices between control devices and in accordance with NFPA 70 and NFPA 90A. Instrumentation grounding must be installed per the device manufacturer's instructions and as necessary to prevent ground loops, noise, and surges from adversely affecting operation of the system. Test installed ground rods as specified in IEEE 142. Cables and conductor wires must be tagged at both ends, with the identifier indicated on the shop drawings. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as indicated. Wiring external to enclosures must be run in raceways, except low-voltage control and low-voltage network wiring may be installed as follows:

- a. plenum rated cable in suspended ceilings over occupied spaces may be run without raceways
- b. nonmetallic-sheathed cables or metallic-armored cables may be installed as permitted by NFPA 70.

Install control circuit wiring not in raceways in a neat and safe manner. Wiring must not use the suspended ceiling system (including tiles, frames or hangers) for support. Where conduit or raceways are required, control circuit wiring must not run in the same conduit/raceway as power wiring over 50 volts. Run all circuits over 50 volts in conduit, metallic tubing, covered metal raceways, or armored cable.

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SECTION 23 09 23.02

BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS
08/24

PART 1 GENERAL

1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and shown and in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

1.1.1 System Requirements

Provide a system meeting the requirements of both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section and with the following characteristics:

- a. Except for Gateways, the control system must be an open implementation of BACnet technology using ASHRAE 135 and Fox as the communications protocols. The system must use standard ASHRAE 135 Objects and Properties and the Niagara Framework. The system must use standard ASHRAE 135 Services and the Niagara Framework exclusively for communication over the network. Gateways to packaged units must communicate with other DDC hardware using ASHRAE 135 or the Fox protocol exclusively and may communicate with packaged equipment using other protocols. The control system must be installed such that any two ASHRAE 135 devices on the Internetwork can communicate using standard ASHRAE 135 Services.
- b. Install and configure control hardware to provide ASHRAE 135 Objects and Properties or Niagara Framework Objects as indicated and as needed to meet the requirements of this specification.
- c. Use Niagara Framework hardware and software exclusively for scheduling, trending, and communication with a front end (UMCS). Use Niagara Framework or standard BACnet Objects and services for alarming. Use the Fox protocol for all communication between Niagara Framework Supervisory Gateways; use the ASHRAE 135 protocol for all other building communication.
- d. Use Niagara Framework either AX or Version 4.0 or later.

1.1.2 Verification of Specification Requirements

Review all specifications related to the control system installation and advise the Contracting Officer of any discrepancies before performing any work. If Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC or any other Section referenced in this specification is not included in the project specifications advise the Contracting Officer and either obtain the missing Section or obtain Contracting Officer approval before performing any work.

1.2 REFERENCES

The publications listed below form a part of this specification to the

extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

BACNET TESTING LABORATORIES (BTL)

BTL Guide (v.50; 2022) BACnet Testing Laboratory Implementation Guidelines

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3 (2022) Ethernet

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485 (1998a; R 2012) Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems

TRIDIUM, INC (TRIDIUM)

Niagara Framework (2021) NiagaraAX User's Guide

Tridium Open NiCS (2021) Understanding the NiagaraAX Compatibility Statement (NiCS)

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15 Radio Frequency Devices (47 CFR 15)

UL SOLUTIONS (UL)

UL 916 (2015; Reprint Oct 2021) UL Standard for Safety Energy Management Equipment

1.3 DEFINITIONS

For definitions related to this section, see Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

1.4 SUBMITTALS

Submittal requirements related to this Section are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

PART 2 PRODUCTS

All products used to meet this specification must meet the indicated requirements, but not all products specified here will be required by

every project. All products must meet the requirements of Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC, Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM and this Section.

2.1 NETWORK HARDWARE

2.1.1 BACnet Router

All BACnet Routers must be BACnet/IP Routers and must perform layer 3 routing of ASHRAE 135 packets over an IP network in accordance with ASHRAE 135 Annex J and Clause 6. The router must provide the appropriate connection to the IP network and connections to one or more ASHRAE 135 MS/TP networks. Devices used as BACnet Routers must meet the requirements for DDC Hardware, and except for Niagara Framework Supervisory Gateways, devices used as BACnet routers must support the NM-RC-B BIBB.

2.1.2 BACnet Gateways

In addition to the requirements for DDC Hardware, the BACnet Gateway must be a Niagara Framework Supervisory Gateway or must meet the following requirements:

- a. It must perform bi-directional protocol translation from one non-ASHRAE 135 protocol to ASHRAE 135. BACnet Gateways must incorporate a network connection to an ASHRAE 135 network (either BACnet over IP in accordance with Annex J or MS/TP) and a separate connection appropriate for the non-ASHRAE 135 protocol and media.
- b. It must retain its configuration after a power loss of an indefinite time, and must automatically return to their pre-power loss state once power is restored.
- c. It must allow bi-directional mapping of data between the non-ASHRAE 135 protocol and Standard Objects as defined in ASHRAE 135. It must support the DS-RP-B BIBB for Objects requiring read access and the DS-WP-B BIBB for Objects requiring write access.
- d. It must support the DS-COV-B BIBB.

Although Gateways must meet DDC Hardware requirements, except for Niagara Framework Supervisory Gateways, they are not DDC Hardware and must not be used when DDC Hardware is required. (Niagara Framework Supervisory Gateways are both Gateways and DDC Hardware.)

2.1.3 Ethernet Switch

Ethernet Switches must be as specified in Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM.

2.2 CONTROL NETWORK WIRING

- a. BACnet MS/TP communications wiring must be in accordance with ASHRAE 135. The wiring must use shielded, three wire (twisted-pair with reference) cable with characteristic impedance between 100 and 120 ohms. Distributed capacitance between conductors must be less than 100 pF per meter.

- b. Building Control Network Backbone IP Network must use Ethernet media. Ethernet cables must be CAT-5e at a minimum and meet all requirements of IEEE 802.3.

2.3 DIRECT DIGITAL CONTROL (DDC) HARDWARE

2.3.1 General Requirements

All DDC Hardware must meet the following requirements:

- a. It must be locally powered and must incorporate a light to indicate the device is receiving power.
- b. It must conform to the BTL Guide
- c. It must be BACnet Testing Laboratory (BTL) Listed.
- d. The Manufacturer's Product Data submittal for each piece of DDC Hardware must include the Protocol Implementation Conformance Statement (PICS) for that hardware as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- e. It must communicate and be interoperable in accordance with ASHRAE 135 and have connections for BACnet IP or MS/TP control network wiring.
- f. Other than devices controlling terminal units or functioning solely as a BACnet Router, it must support DS-COV-B, DS-RPM-A and DS-RPM-B BIBBs.
- g. Devices supporting the DS-RP-A BIBB must also support the DS-COV-A BIBB.
- h. Application programs, configuration settings and communication information must be stored in a manner such that they persist through loss of power:
 - (1) Application programs must persist regardless of the length of time power is lost.
 - (2) Configured settings must persist for any loss of power less than 2,500 hours.
 - (3) Communication information, including but not limited to COV subscriptions, event reporting destinations, Notification Class Object settings, and internal communication settings, must persist for any loss of power less than 2,500 hours.
- i. Internal Clocks:
 - (1) Clocks in DDC Hardware incorporating a Clock must continue to function for 120 hours upon loss of power to the DDC Hardware.
 - (2) DDC Hardware incorporating a Clock must support the DM-TS-B or DM-UTC-B BIBB.
- j. It must have all functionality indicated and required to support the application (Sequence of Operation or portion thereof) in which it is used, including but not limited to providing Objects or Niagara Framework Points as specified and as indicated on the Points Schedule.

- k. In addition to these general requirements and the DDC Hardware Input-Output (I/O) Function requirements, all DDC Hardware must also meet any additional requirements for the application in which it is used (e.g. scheduling, alarming, trending, etc.).
- l. It must meet FCC Part 15 requirements and have UL 916 or equivalent safety listing.
- m. Except for Niagara Framework Supervisory Gateways, Device must support Commandable Objects to support Override requirements as detailed in PART 3 EXECUTION
- n. User interfaces which allow for modification of Properties or settings must be password-protected.
- o. Devices communicating BACnet MS/TP must meet the following requirements:
 - (1) Must have a configurable Max_Master Property.
 - (2) DDC Hardware other than hardware controlling a single terminal unit must have a configurable Max_Info_Frames Property.
 - (3) Must respond to any valid request within 50 msec with either the appropriate response or with a response of "Reply Postponed".
 - (4) Must use twisted pair with reference and shield (3-wire media) wiring.
- p. Devices communicating BACnet/IP must use UDP Port 0xBAC0. Devices with configurable UDP Ports must default to 0xBAC0.
- q. All Device IDs, Network Numbers, and BACnet MAC addresses of devices must be fully configurable without limitation, except MS/TP MAC addresses may be limited by ASHRAE 135 requirements.
- r. Except for Niagara Framework Supervisory Gateways, DDC Hardware controlling a single terminal unit must have:
 - (1) Objects (including the Device Object) with an Object Name Property of at least 8 characters in length.
 - (2) A configurable Device Object Name.
 - (3) A configurable Device Object Description Property at least 16 characters in length.
- s. Except for Objects in either Niagara Framework Supervisory Gateways or DDC Hardware controlling a single terminal unit, all Objects (including Device Objects) must:
 - (1) Have a configurable Object Name Property of at least 12 characters in length.
 - (2) Have a configurable Object Description Property of at least 24 characters in length.
- t. For programmable DDC Hardware, provide and license to the project site

all programming software required to program the Hardware in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

- u. For programmable DDC Hardware, provide copies of the installed application programs (all software that is not common to every controller of the same manufacturer and model) as source code compatible with the supplied programming software in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The submitted application program must be the complete application necessary for controller to function as installed and be sufficient to allow replacement of the installed controller with another controller of the same type.

2.3.2 Hardware Input-Output (I/O) Functions

DDC Hardware incorporating hardware input-output (I/O) functions must meet the following requirements:

2.3.2.1 Analog Inputs

DC Hardware analog inputs (AIs) must be implemented using ASHRAE 135 Analog Input Objects and perform analog to digital (A-to-D) conversion with a minimum resolution of 8 bits plus sign or better as needed to meet the accuracy requirements specified in Section 23 09 00. Signal conditioning including transient rejection must be provided for each analog input. Analog inputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. The AI must incorporate common mode noise rejection of at least 50 dB from 0 to 100 Hz for differential inputs, and normal mode noise rejection of at least 20 dB at 60 Hz from a source impedance of 10,000 ohms.

2.3.2.2 Analog Outputs

DDC Hardware analog outputs (AOs) must be implemented using ASHRAE 135 Analog Output Objects and perform digital to analog (D-to-A) conversion with a minimum resolution of 8 bits plus sign, and output a signal with a range of 4-20 mA_{dc} or 0-10 V_{dc}. Analog outputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. DDC Hardware with Hand-Off-Auto (H-O-A) switches for analog outputs must provide for overriding the output to 0 percent and to 100 percent.

2.3.2.3 Binary Inputs

DDC Hardware binary inputs (BIs) must be implemented using ASHRAE 135 Binary Input Objects and accept contact closures and must ignore transients of less than 5 milli-second duration. Protection against a transient 50VAC must be provided.

2.3.2.4 Binary Outputs

DDC Hardware binary outputs (BOs) must be implemented using ASHRAE 135 Binary Output Objects and provide relay contact closures or triac outputs for momentary and maintained operation of output devices. DDC Hardware with H-O-A switches for binary outputs must provide for overriding the output open or closed.

2.3.2.4.1 Relay Contact Closures

Closures must have a minimum duration of 0.1 second. Relays must provide at least 180V of isolation. Electromagnetic interference suppression must be provided on all output lines to limit transients to 50 Vac. Minimum contact rating must be 0.5 amperes at 24 Vac.

2.3.2.4.2 Triac Outputs

Triac outputs must provide at least 180 V of isolation. Minimum contact rating must be 0.5 amperes at 24 Vac.

2.3.2.5 Pulse Accumulator

DDC Hardware pulse accumulators must be implemented using either an ASHRAE 135 Accumulator Object or an ASHRAE 135 Analog Value Object where the Present_Value is the totalized pulse count. Pulse accumulators must accept contact closures, ignore transients less than 5 msec duration, protect against transients of 50 VAC, and accept rates of at least 20 pulses per second.

2.3.2.6 ASHRAE 135 Objects for Hardware Inputs and Outputs

The requirements for use of ASHRAE 135 objects for hardware input and outputs includes devices where the hardware sensor or actuator is integral to the controller (e.g. a VAV box with integral damper actuator, a smart sensor, a VFD, etc.)

2.3.2.7 Integrated H-O-A Switches

Where integrated H-O-A switches are provided on hardware outputs, controller must provide means of monitoring position or status of H-O-A switch. This feedback may be provided via the Niagara Framework or via any valid BACnet method, including the use of proprietary Objects, Properties, or Services.

2.3.3 Local Display Panel (LDP)

The Local Display Panels (LDPs) must be DDC Hardware with a display and navigation buttons or a touch screen display, and must provide display and adjustment of Niagara Framework points or ASHRAE 135 Properties as indicated on the Points Schedule and as specified. LDPs must be either BTL Listed as a B-OD, B-OWS, B-AWS, or be an integral part of another piece of DDC Hardware listed as a B-BC. For LDPs listed as B-OWS or B-AWS, the hardware must be BTL listed and the product must come factory installed with all applications necessary for the device to function as an LDP.

The adjustment of values using display and navigation buttons must be password protected.

2.3.4 Expansion Modules and Tethered Hardware

A single piece of DDC Hardware may consist of a base unit and also:

- a. An unlimited number of hardware expansion modules, where the individual hardware expansion modules are designed to directly connect, both mechanically and electrically, to the base unit hardware. The expansion modules must be commercially available as an

optional add-on to the base unit.

- b. A single piece of hardware connected (tethered) to a base unit by a single cable where the cable carries a proprietary protocol between the base unit and tethered hardware. The tethered hardware must not contain control logic and be commercially available as an optional add-on to the base unit as a single package.

Note that this restriction on tethered hardware does not apply to sensors or actuators using standard binary or analog signals (not a communications protocol); sensors or actuators using standard binary or analog signals are not considered part of the DDC Hardware.

Hardware capable of being installed stand-alone, or without a separate base unit, is DDC Hardware and must not be used as expansion modules or tethered hardware.

2.3.5 Supervisory Control Requirements

2.3.5.1 Alarm Generation Hardware

Non-Niagara Framework DDC Hardware used for alarm generation must meet the following requirements:

- a. Device must support the AE-N-I-B BIBB
- b. The Recipient_List Property must be Writable for all Notification Class Objects used for alarm generation.
- c. For all Objects implementing Intrinsic Alarming, the following Properties must be Writable:
 - (1) Time_Delay
 - (2) High_Limit
 - (3) Low_Limit
 - (4) Deadband
 - (5) Event_Enable
 - (6) If the issue date of this project specification is after 1 January 2016, Time_Delay_Normal must be writable.
- d. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Notification Class Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Notification Class Objects.

2.3.6 Niagara Framework Supervisory Gateway

Any device implementing the Niagara Framework is a Niagara Framework Supervisory Gateway and must meet these requirements. In addition to the general requirements for all DDC Hardware, Niagara Framework Supervisory Gateway Hardware must:

- a. Be direct digital control hardware.

- b. Have an unrestricted interoperability license and its Niagara Compatibility Statement (NiCS) must follow the Tridium Open NiCS Specification.
- c. Manage communications between a field control network and the Niagara Framework Monitoring and Control Software, and between itself and other Niagara Framework Supervisory Gateways. Niagara Framework Supervisory Gateway Hardware must use Fox protocol for communication with other Niagara Framework Components, regardless of the manufacturer of the other components.
- d. Be fully programmable using the Niagara Framework Engineering Tool and must support the following:
 - (1) Time synchronization, Calendar, and Scheduling using Niagara Scheduling Objects
 - (2) Alarm generation and routing using the Niagara Alarm Service
 - (3) Trending using the Niagara History Service and Niagara Trend Log Objects
 - (4) Integration of field control networks using the Niagara Framework Engineering Tool
 - (5) Configuration of integrated field control system using the Niagara Framework Engineering Tool when supported by the field control system
- e. Meet the following minimum hardware requirements:
 - (1) One 10/100 Mbps Ethernet Port
 - (2) One or more MS/TP ports.
- f. Provide access to field control network data and supervisory functions via web interface and support a minimum of 16 simultaneous users.
Note: implementation of this capability may not be required on all projects.
- g. Submit a backup of each Niagara Framework Supervisory Gateway as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The backup must be sufficient to restore a Niagara Framework Supervisory Gateway to the final as-built condition such that a new Niagara Framework Supervisory Gateway loaded with the backup is indistinguishable in functionality from the original.

2.4 NIAGARA FRAMEWORK ENGINEERING TOOL

The Niagara Framework Engineering Tool must be Niagara Workbench or an equivalent Niagara Framework engineering tool software must:

- a. Have an unrestricted interoperability license and its Niagara Compatibility Statement (NiCS) must follow the Tridium Open NiCS Specification.
- b. Be capable of performing network configuration for Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.

- c. Be capable of programming and configuring of Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.
- d. Be capable of discovery of Niagara Framework Supervisory Gateways and all points mapped into each Niagara Framework Supervisory Gateway and making these points accessible to Niagara Framework Monitoring and Control Software.

PART 3 EXECUTION

3.1 CONTROL SYSTEM INSTALLATION

3.1.1 Niagara Framework Engineering Tool

Provide a Niagara Framework Engineering Tool.

3.1.2 Building Control Network (BCN)

Install the Building Control Network (BCN) as a single BACnet Internetwork consisting of a single IP network as the BCN Backbone and zero or more BACnet MS/TP networks. Note that in some cases there may only be a single device on the BCN Backbone.

Except for the IP Network and as permitted for the non-BACnet side of Gateways, use exclusively ASHRAE 135 networks.

3.1.2.1 Building Control Network IP Backbone

Install IP Network Cabling in conduit. Install Ethernet Switches in lockable enclosures. Install the Building Control Network (BCN) IP Backbone such that it is available at the Facility Point of Connection (FPOC) location. When the FPOC location is a room number, provide sufficient additional media to ensure that the Building Control Network (BCN) IP Backbone can be extended to any location in the room.

Use UDP port 0xBAC0 for all BACnet traffic on the IP network. (Note that in a Niagara Framework system there may not be BACnet traffic on the IP Network)

3.1.2.2 BACnet MS/TP Networks

When using MS/TP, provide MS/TP networks in accordance with ASHRAE 135 and in accordance with the ASHRAE 135 figure "Mixed Devices on 3-Conductor Cable with Shield" (Figure 9-1.4 in the 2020 version of ASHRAE 135). Ground the shield at the BACnet Router and at no other point. Ground the reference wire at the BACnet Router through a 100 ohm resistor and do not ground it at any other point. In addition:

- a. Provide each segment in a doubly terminated bus topology in accordance with TIA-485.
- b. Provide each segment with 2 sets of network bias resistors in accordance with ASHRAE 135, with one set of resistors at each end of the MS/TP network.
- c. Use 3 wire (twisted pair and reference) with shield media for all MS/TP media installed inside. Use fiber optic isolation in accordance

with ASHRAE 135 for all MS/TP media installed outside buildings, or between multiple buildings.

- d. For 18 AWG cable, use segments with a maximum length of 1200 m. When using greater distances or different wire gauges comply with the electrical specifications of TIA-485.
- e. For each controller that does not use the reference wire provide transient suppression at the network connection of the controller if the controller itself does not incorporate transient suppression.
- f. Install no more than 32 devices on each MS/TP segment. Do not use MS/TP to MS/TP routers.
- g. Connect each MS/TP network to the BCN backbone via a Niagara Framework Supervisory Gateway configured as a BACnet Router.
- h. For BACnet Routers, configure the MS/TP MAC address to 0. Assign MAC Addresses to other devices consecutively beginning at 1, with no gaps.
- i. Configure the Max_Master Property of all devices to be 31.

3.1.2.3 Building Control Network (BCN) Installation

Provide a building control network meeting the following requirements:

- a. Install all DDC Hardware connected to the Building Control Network.
- b. Where multiple pieces of DDC Hardware are used to execute one sequence, install all DDC Hardware executing that sequence on a single MS/TP network dedicated to that sequence.
- c. Traffic between BACnet networks must be exclusively via BACnet routers.
- d. Use the Fox protocol for all traffic both originating and terminating at Niagara Framework components. Use the Fox protocol for all traffic originating or terminating at a Niagara Framework UMCS (including traffic to or from a future UMCS). All other traffic, including traffic between ASHRAE 135 devices and traffic between Niagara Framework Supervisory Gateways and ASHRAE 135 devices must be in accordance with ASHRAE 135.

3.1.3 DDC Hardware

Install all DDC Hardware that connects to an IP network in lockable enclosure. Install other DDC Hardware that is not in suspended ceilings in lockable enclosures. For all DDC hardware with a user interface, coordinate with site to determine proper passwords and configure passwords into device.

- a. Except for zone sensors (thermostats), install all Tethered Hardware within 2 m of its base unit.
- b. Install and configure all BTL-Listed devices in a manner consistent with their BTL Listing such that the device as provided still meets all requirements necessary for its BTL Listing.
- c. Install and configure all BTL-Listed devices in a manner consistent with the BTL Device Implementation Guidelines such that the device as

provided meets all those Guidelines.

3.1.3.1 Device Identifiers, Network Addresses, and IP addresses

- a. Do not use any Device Identifier or Network Number already used by another BACnet system at the project site. Coordinate Device IDs and Network Numbers with the installation.
- b. Coordinate device IP addresses with installation.

3.1.3.2 ASHRAE 135 Object Name Property and Object Description Property

Configure the Object_Names and Object_Descriptions properties of all ASHRAE 135 Objects (including Device Objects) as indicated on the Points Schedule (Point Name and Point Description) and as specified. At a minimum:

- a. Except for DDC Hardware controlling a single terminal unit, configure the Object_Name and Object_Description properties of all Objects (including Device Objects) as indicated on the Points Schedule and as specified.
- b. In DDC Hardware controlling a single terminal unit, configure the Device Object_Name and Device Object_Description as indicated on the Points Schedule and as specified.

When Points Schedule entries exceed the length limitations in the device, notify Controls Supervisor and provide recommended alternatives for approval.

3.1.3.3 Niagara Framework Point Names and Descriptions

Configure the names and descriptions of all Points in Niagara Framework Supervisory Gateways as indicated on the Points Schedule and as specified.

3.1.3.4 Niagara Station IDs

Ensure that Niagara Station IDs of new Niagara Framework Supervisory Gateways are maintained as unique within UMCS front-end, including ensuring they do not conflict with any existing Niagara Station ID.

3.1.3.5 Hand-Off-Auto (H-O-A) Switches

Provide Hand-Off-Auto (H-O-A) switches as specified and as indicated on the Points Schedule. Provide H-O-A switches that are integral to the controller hardware, an external device co-located with (in the same enclosure as) the controller, integral to the controlled equipment, or an external device co-located with (in the same enclosure as) the controlled equipment.

- a. For H-O-A switches integral to DDC Hardware, meet the requirements specified in paragraph DIRECT DIGITAL CONTROL (DDC) HARDWARE.
- b. For external H-O-A switches used for binary outputs, provide for overriding the output open or closed.
- c. For eternal H-O-A switches used for analog outputs, provide for overriding to 0 percent or 100 percent.

3.1.3.6 Local Display Panels

Provide LDPs to display and override values of points in a Niagara Framework Supervisory Gateway or ASHRAE 135 Object Properties as indicated on the Points Schedule. Install LDPs displaying points for anything other than a terminal unit in the same room as the equipment. Install LDPs displaying points for only terminal units in a mechanical room central to the group of terminal units it serves. For LDPs using WriteProperty to commandable objects to implement an override, write values with priority 9.

3.1.3.7 MS/TP Slave Devices

Configure all MS/TP devices as Master devices. Do not configure any devices to act as slave devices.

3.1.3.8 Change of Value (COV) and Read Property

- a. To the greatest extent possible, configure all devices to support the SubscribeCOV service (the DS-COV-B BIBB). At a minimum, all devices supporting the DS-RP-B BIBB, other than devices controlling only a single terminal unit, must be configured to support the DS-COV-B BIBB.
- b. Whenever supported by the server side, configure client devices to use the DS-COV-A BIBB.

3.1.3.9 Engineering Units

Configure devices to use SI (Metric) units as follows:

- a. Temperature in degrees C
- b. Air or natural gas flows in Liters per Second (LPS)
- c. Water flow in Liters per Second (LPS)
- d. Steam flow in kilograms per second (kg/s)
- e. Differential Air pressures in Pascals (Pa)
- f. Water, steam and natural gas pressures in kiloPascals (kPa)
- g. Enthalpy in kiloJoules per kilogram (kJ/kg)
- h. Heating and Cooling Energy in kilowatt-hours (kWh)
- i. Heating and Cooling load in kilowatts (kW)
- j. Electrical Power: kilowatts (kW)
- k. Electrical Energy: kilowatt-hours (kWh)

3.1.3.10 Occupancy Modes

Use the following correspondence between value and occupancy mode whenever an occupancy state or value is required:

- a. OCCUPIED mode: a value of one
- b. UNOCCUPIED mode: a value of two

- c. WARM-UP/COOL-DOWN (PRE-OCCUPANCY) mode: a value of three

Note that elsewhere in this Section the Schedule Object is required to also support a value of four, which is reserved for future use. Also note that the behavior of a system in each of these occupancy modes is indicated in the sequence of operation for the system.

3.1.3.11 Use of BACnet Objects

Except as specifically indicated for Niagara Framework Objects, use only standard non-proprietary ASHRAE 135 Objects and services to accomplish the project scope of work as follows:

- a. Use Analog Input or Analog Output Objects for all analog hardware I/O. Do not use Analog Value Object for analog hardware I/O) .
- b. Use Binary Input or Binary Output Objects for all binary hardware I/O. Do not use Binary Value Objects for binary hardware I/O.
- c. Use Analog Value Objects for analog setpoints.
- d. Use Accumulator Objects or Analog Value Objects for pulse inputs.
- e. For occupancy modes, use Multistate Value Objects and the correspondence between value and occupancy mode specified in paragraph OCCUPANCY MODES.
- f. Use a combination of Niagara Framework Alarm Extensions and Alarm Services, Intrinsic Alarming, and Notification Class Objects for alarm generation.
- g. For all other points shown on the Points Schedule as requiring an ASHRAE 135 Object, use the Object type shown on the Points Schedule or, if no Object Type is shown, use a standard Object appropriate to the point.

3.1.3.11.1 Niagara Framework Objects

Points in the Niagara Framework Supervisory Gateway, even if used in a sequence or are shown on the Points Schedule, are not required to be exposed as BACnet Objects unless they are required to be available on the network by another device or sequence of operation (i.e. there is some other reason they are needed).

Use a Niagara Framework Supervisory Gateway as specified for all scheduling and trending. Use a Niagara Framework Supervisory Gateway as specified for all alarming except for intrinsic alarming.

3.1.3.12 Use of Standard BACnet Services

Except as noted in this paragraph, for all DDC Hardware (including Niagara Frameworks Supervisory Gateways when communicating with non-Niagara Framework DDC Hardware) use Standard BACnet Services as defined in this specification (which excludes some ASHRAE 135 services) exclusively for application control functionality and communication.

DDC Hardware that cannot meet this requirement may use non-standard services provided they can provide identical functionality using Standard

BACnet Services when communicating with BACnet devices from a different vendor. When implementing non-standard services, document all non-standard services in the DDC Hardware Schedule as specified and as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

3.1.3.13 Device Application Configuration

- a. For every property, setting or value shown on the Points Schedule or otherwise indicated as Configurable, provide a value that is retained through loss of power and can be changed via one or more of:
 - (1) BACnet services (including proprietary services)
 - (2) Hardware settings on the device
 - (3) The Niagara Framework
- b. For every property, setting or value in non-Niagara Framework Hardware shown on the Points Schedule or otherwise indicated as Operator Configurable, provide a value that is retained through loss of power and can be changed via one or more of:
 - (1) A Writable Property of a standard BACnet Object
 - (2) A Property of a standard BACnet Object that is Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable.
 - (3) Using some other method supported by a Niagara Framework Supervisory Gateway
- c. Configure Niagara Framework Supervisory Gateways such that the property, setting or value is configurable from a Niagara Framework Front End.
- d. For every property, setting or value in a Niagara Framework Supervisory Gateway which is shown on the Points Schedule or otherwise indicated as Operator Configurable, configure the value to be configurable from within the Niagara Framework such that it can be configured from a system graphic page at a Niagara Framework Front End.

3.1.3.14 Niagara Framework Engineering Tool

Use the Niagara Framework Engineering Tool to fully discover the field control system and make all field control system information available to the Niagara Framework Supervisory Gateway. Ensure that all points on the points schedule are available to the front end via the Fox protocol.

3.1.3.15 Graphics and Web Pages

Configure Niagara Framework Supervisory Gateways to use web pages to provide a graphical user interface including System Displays using the project site sample displays, including overrides, as indicated on the Points Schedule and as specified. Label all points on displays with the point name as indicated on the Points Schedule and the point description as indicated on the Points Schedule. Configure user permissions for access to and executions of action using graphic pages. Coordinate user permissions with the Controls and HVAC shop supervisor. Configure the web server to use HTTPS based on the Transport Layer Security (TLS) protocol in accordance with RFC 5246 using a Government furnished certificate.

3.1.4 Scheduling, Alarming, Trending, and Overrides

3.1.4.1 Scheduling

Configure schedules in Niagara Framework Supervisory Gateway using Niagara Schedule Objects as indicated on the Points Schedule and as specified. When the schedule is controlling occupancy modes in DDC Hardware other than a Niagara Framework Supervisory Gateway use the indicated correspondence between value and occupancy mode.

Provide a separate schedule for each AHU including it's associated Terminal Units and for each stand-alone Terminal Unit (those not dependent upon AHU service).

3.1.4.2 Alarm Configuration

Configure alarm generation and management as indicated on the Points Schedule and as specified. Configure alarm generation in Niagara Framework Supervisory Gateways using Niagara Framework Alarm Extensions and Alarm Services or in other DDC Hardware (not Niagara Framework Supervisory Gateways) using ASHRAE 135 Intrinsic Alarming. Configure alarm management and routing for all alarms, including those generated via intrinsic alarming in other devices, in the Niagara Framework Supervisory Gateway such that the alarms are able to be accessed from the Niagara Framework Front End.

Where Intrinsic Alarming is used, configure intrinsic alarming as specified in paragraph "Configuration of ASHRAE 135 Intrinsic Alarm Generation". Configure a Niagara Framework Supervisory Gateway to provide a means to configure the intrinsic alarm parameters such that the Intrinsic Alarm is configurable from the front end via the Niagara Framework.

3.1.4.3 Configuration of ASHRAE 135 Intrinsic Alarm Generation

Intrinsic alarm generation must meet the following requirements:

Configure alarm generation as indicated on the Points Schedule and as specified using Intrinsic Alarming in accordance with ASHRAE 135 or Algorithmic Alarming in accordance with ASHRAE 135. Alarm generation must meet the following requirements:

- a. Send alarm events as Alarms (not Events).
- b. Use the ConfirmedNotification Service for alarm events.
- c. For alarm generation, support two priority levels for alarms: critical and non-critical. Configure the Priority of Notification Class Objects to use Priority 112 for critical and 224 for non-critical alarms.
- d. Number of Notification Class Objects for Alarm Generation:
 - (1) If the device implements non-critical alarms, or if any Object in the device supports Intrinsic Alarms, then provide a single Notification Class Object specifically for (shared by) all non-critical alarms.

- (2) If the device implements critical alarms, provide a single Notification Class Object specifically for (shared by) all critical alarms.
 - (3) If the device implements both critical and non-critical alarms, provide both Notification Class Objects (one for critical, one for non-critical).
 - (4) If the device controls equipment other than a single terminal unit, provide both Notification Class Objects (one for critical, one for non-critical) even if no alarm generation is required at time of installation.
- e. For all intrinsic alarms configure the Limit_Enable Property to set both HighLimitEnable and LowLimitEnable to TRUE. If the specified alarm conditions are for a single-sided alarm (only High_Limit used or only Low_Limit used) assign a value to the unused limit such that the unused alarm condition will not occur.
- f. For all objects supporting intrinsic alarming, even if no alarm generation is required during installation, configure the following Properties as follows:
- (1) Notification_Class to point to the non-Critical Notification Class Object in that device.
 - (2) Limit_Enable to enable both the HighLimitEnable and LowLimitEnable
 - (3) Notify_Type to Alarm
- g. Configure the Recipient_List Property of the Notification Class Object to point to the Niagara Framework Supervisory Gateway managing the alarm.

3.1.4.4 Trending

Perform all trending using a Niagara Framework Supervisory Gateway using Niagara Framework History Extensions and Niagara Framework History Service exclusively.

3.1.4.5 Overrides

Provide an override for each point shown on the Points Schedule as requiring an override. Use the Niagara Framework for all overrides to points in Niagara Framework Supervisory Gateways. For overrides to other points, provide an override to a point in a Niagara Framework Supervisory Gateway via the Niagara Framework where the Niagara Framework Supervisory Gateway overrides the other point as specified.

Unless otherwise approved, provide Commandable Objects to support all Overrides in non-Niagara Framework Supervisory Gateway DDC Hardware. With specific approval from the Contracting Officer, Overrides for points which are not hardware outputs and which are in DDC hardware controlling a single terminal unit may support overrides via an additional Object provided for the override. No other means of implementing Overrides may be used.

- a. Where Commandable Objects are used, ensure that WriteProperty service requests with a Priority of 10 or less take precedence over the

SEQUENCE VALUE and that WriteProperty service request with a priority of 11 or more have a lower precedence than the SEQUENCE VALUE.

- b. For devices implementing overrides via additional Objects, provide Objects which are NOT Written to as part of the normal Sequence of Operations and are Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable. Use this point as an Override of the normal value when Out_Of_Service is TRUE and the normal value otherwise. Note these Objects may be modified as part of the sequence via local processes, but must not be modified by local processes when Out_Of_Service is TRUE.

3.1.5 BACnet Gateways

The requirements in this paragraph do not themselves permit the installation of hardware not meeting the other requirements of this section. Except for proprietary systems specifically indicated in Section 23 09 00, all control hardware installed under this project must meet the requirements of this specification, including the control hardware providing the network interface for a package unit or split system specified under another section. Only use gateways to connect to pre-existing control devices, and to proprietary systems specifically permitted by Section 23 09 00.

3.1.5.1 General Gateway Requirements

Provide BACnet Gateways to connect non-BACnet control hardware in accordance with the following:

- a. Configure gateways to map writable data points in the controlled equipment to Writable Properties of Standard Objects, or to Niagara Framework points, as indicated in the Points Schedule and as specified.
- b. Configure gateway to map readable data points in the controlled equipment to Readable Properties of Standard Objects, or to Niagara Framework points, as indicated in the Points Schedule and as specified.
- c. Configure gateway to support the DS-COV-B BIBB for all points mapped to BACnet Objects.
- d. Do not use non-BACnet control hardware for controlling built-up units or any other equipment that was not furnished with factory-installed controls. (Note: A Niagara Framework Supervisory Gateway is BACnet control hardware.)
- e. Do not use non-BACnet control hardware for system scheduling functions.
- f. Each gateway must communicate with and perform protocol translation for non-BACnet control hardware controlling one and only one package unit or a single non-BACnet system specifically permitted by Section 23 09 00.
- g. Connect one network port on the gateway to the Building Control Backbone IP Network or to a BACnet MS/TP network and the other port to the single piece of controlled equipment or the non-BACnet system specifically permitted by Section 23 09 00..
- h. For gateways to existing package units or simple split systems, non-BACnet network wiring connecting the gateway to the package unit

must not exceed 3 meters in length and must connect to exactly two devices: the controlled equipment (packaged unit) or split system interface and the gateway.

-- End of Section --

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SECTION 23 21 23

HYDRONIC PUMPS
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 (2024) Unified Inch Screw Threads (UN, UNR, and UNJ Thread Form)

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

HYDRAULIC INSTITUTE (HI)

HI 1.1-1.2 (2014) Rotodynamic (Centrifugal) Pump for Nomenclature and Definitions

HI 9.6.4 (2009) Rotodynamic Pumps for Vibration Analysis and Allowable Values

HI ANSI/HI 2.1-2.2 (2014) Rotodynamic Vertical Pumps of Radial, Mixed, and Axial Flow Types for Nomenclature and Definitions

HI ANSI/HI 9.6.3 (2024) Rotodynamic Pumps - Guideline for Operating Regions - B120

HI ANSI/HI 14.6 (2011) Rotodynamic Pumps for Hydraulic Performance Acceptance Tests - A136

INTERNATIONAL CODE COUNCIL (ICC)

ICC IgCC (2018) International Green Construction Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

NEMA Z535.4 (2023) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National
Electrical Code

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 21 (1982; E 2004) White or Colored Silicone
Alkyd Paint (Type I, High Gloss and Type
II, Medium Gloss)

SSPC Paint 25 (1997; E 2004) Zinc Oxide, Alkyd, Linseed
Oil Primer for Use Over Hand Cleaned
Steel, Type I and Type II

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.219 Mechanical Power Transmission Apparatus

UL SOLUTIONS (UL)

UL 778 (2016; Reprint Jul 2024) UL Standard for
Safety Motor-Operated Water Pumps

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

System Coordination; G

SD-03 Product Data

Instructions; G

Equipment Data; G

Training Period; G

SD-06 Test Reports

Field Quality Control

SD-07 Certificates

Manufacturer's Representative

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Training; G

1.3 QUALITY ASSURANCE

1.3.1 Manufacturer Services

Provide the services of a manufacturer's representative experienced in the installation, adjustment, and operation of the equipment specified. The representative must supervise the installation, adjustment, testing of the equipment, and conduct training.

Submit the names and qualifications of the manufacturer's representative and training engineers and written certification from the manufacturer that the representative and trainers are technically qualified.

1.3.2 Standard Products

Provide material and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate equipment that has been in satisfactory HVAC operation at least 2 years prior to issuance of this solicitation. Support equipment with a service organization that is reasonably convenient to the jobsite. Pumps of the same types must each be the product of one manufacturer.

1.3.3 Conformance with Agency Requirements

Where materials or equipment are specified to be an approved type, attach the seal or label of approval from a nationally recognized testing agency, adequately equipped and competent to perform such services. A written certificate from the testing agency must accompany the materials or equipment and be submitted stating that the items have been tested and that they conform to the applicable requirements of the specifications and to the standards listed herein. The certificate must indicate the methods of testing used by the testing agency. In lieu of a certificate from a testing agency, published catalog specification data, accompanied by the manufacturer's certified statement to the effect that the items are in accordance with the applicable requirements of the specifications and the referenced standards, will be considered and may be acceptable as evidence that the items conform with agency requirements.

1.4 DELIVERY, STORAGE, AND HANDLING

Protect equipment, delivered and designated for storage, from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Hydronic pumps used for heating and air conditioning applications are defined by the type of impeller, number of impellers, type of casing, method of connection to the driver, and mounting position. Provide centrifugal water pumps of the types indicated and specified. Use an electric motor driving unit for each pump as indicated and specified.

2.1.1 Selection Criteria

Select pumps at a point within the maximum efficiency for a given impeller casing combination. Deviations within 3 percent of maximum efficiency are permissible, provided the lesser efficiency is not less than the scheduled

efficiency in the construction design documents. Pumps having impeller diameters larger or smaller than manufacturer's published maximum and minimum impeller diameters for a given impeller casing combination will be rejected. Pump performance data, as shown in performance curves, must be based on factory tests using precision instrumentation and exacting procedures as detailed in HI ANSI/HI 14.6.

2.1.2 System Coordination

Submit drawings containing complete wiring and piping schematic diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Show the proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for maintenance and operation. Provide a complete listing of equipment, materials and miscellaneous components including mechanical seals, bearings, and couplings.

2.1.3 Safety Requirements

Fully enclose or guard couplings, projecting set-screws, keys, and other rotating parts, that pose an entangling hazards..

2.2 MATERIALS AND EQUIPMENT

2.2.1 Nameplates

Securely affix a standard nameplate to pumps and motors in a conspicuous place showing the manufacturer's name, address, type or style, model, serial number, and catalog number. In addition, for each pump show the capacity in L/second at rated speed in rpm and total head in mm of water. For each electric motor show at least the minimum information required by NEMA MG 1. Show such other information as the manufacturer may consider necessary to complete identification on the nameplate. Pumps must be listed and labeled by UL, and comply with UL 778 for pumps not using universal motors rated more than 250 volts such as circulating pumps.

2.2.2 Framed Instructions

Submit proposed diagrams, instructions, and other sheets, prior to posting. Post approved wiring and control diagrams showing the complete layout of the entire system, including equipment, piping valves, and control sequence, framed under glass or in approved laminated plastic, where directed. Provide condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system, framed as specified above for the wiring and control diagrams, and posted beside the diagrams. Post the framed instructions before acceptance testing of the systems.

2.2.3 Pump Characteristic

Construct hydronic water pumps in accordance with HI 1.1-1.2 and HI ANSI/HI 2.1-2.2. The pumps must be capable of discharging quantities at total discharge heads measured at the discharge flange, between the following limits:

Operate pumps at optimum efficiencies to produce the most economical pumping system under the conditions encountered. Pumps must furnish not

less than 150 percent of rated capacity at a total discharge head of not less than 65 percent of total rated head. Operate pumps at specified system fluid temperatures without vapor binding and cavitation. Operate pumps to HI ANSI/HI 9.6.3 standard for Preferred Operating Region (POR).

2.2.4 Pump Drivers

Provide electric motors as indicated for each pump and in compliance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.2.5 Equipment Data

Submit manufacturer's descriptive data and technical literature, performance charts and curves for all impeller sizes for a given casing, catalog cuts, and installation instructions. Provide spare parts data for each different item of material and equipment specified, after approval of the detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include a complete list of parts and supplies, with current unit prices and local source of supply with contact information.

Submit catalog information, certified pumps curves, rated capacities, final impeller dimensions, and accessories provided for the product indicated. Indicate operating point of each pump on curves. Furnish pump curves for each pump and combination of pumps designed to operate in parallel. The pump curve must show as a minimum; bhp, flow, total dynamic head, efficiency, NPSH, impeller diameter and system curve (individually and in combination for each pump operating in a parallel application). Select pumps operating in parallel operation to cross the system curve when operating individually.

2.3 HYDRONIC PUMPS

Provide centrifugal, single-stage type, designed for HVAC service in the following configurations:

Configuration	Pump No.
Base-Mounted, Close Coupled, End Suction	CWP-1,2
Automatic Cooling Coil Condensate Pump Units	

2.3.1 Base-Mounted, Close Coupled, End Suction

Provide pumps with capacities as indicated. Pump must be base mounted, close coupled, single stage, end suction design capable of being serviced without disturbing piping connections.

2.3.1.1 Casing

Provide pump volute of Class 30 cast iron suitable for 1204 kPa working pressure. Include vent, drain and gauge tapings.

2.3.1.2 Seal Assembly

Seal off the liquid cavity at the motor shaft by an internally flushed mechanical seal or a positive pressure external seal flushing line with ceramic seal seat and carbon seal ring, suitable for continuous operation at 107 degrees C. A replaceable shaft sleeve of bronze alloy must

completely cover the wetted area under the seal.

2.3.1.3 Impeller

Provide cast bronze or 304 stainless steel impeller, enclosed type, hydraulically and dynamically balanced to HI 9.6.4 balance grade G6.3, keyed to shaft and secured by a locking capscrew.

2.3.1.4 Motor

Electric Motors must comply with NEMA MG 1 and be the wattage, and voltage indicated. Motor enclosure must be totally enclosed fan cooled. provide with heavy duty grease lubricated ball bearings completely adequate for the maximum load for which the motor and pump impeller is designed. Motor must be non-overloading at any point on the pump curve and premium efficiency. Provide motor efficiencies as shown in the ICC IgCC standard. Totally enclosed fan cooled motor efficiencies must be as shown in NEMA MG 1.

2.3.2 Cooling Coil Condensate Pump Units

Provide pumps with capacities as indicated. Cooling Coil Condensate Pump Unit must be a packaged unit including a corrosion-resistant pump, plastic tank with cover, and automatic controls. Include factory installed check valve and a 1800 mm minimum, electrical power cord with plug for 120V/1PH/60HZ electrical service.

2.3.2.1 Motor

Electric motor must comply with NEMA MG 1 and be the size, voltage and enclosure indicated. Provide heavy duty grease lubricated ball bearings completely adequate for the maximum load for which the motor is designed.

2.4 ELECTRICAL WORK

Provide electrical motor driven equipment specified herein complete with motors, motor starters, and controls. Provide electric equipment and wiring in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical characteristics must be as indicated. Provide motor starters complete with properly sized thermal overload protection in each phase and other appurtenances necessary for the motor control specified. Each motor must be of sufficient capacity to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor when operating at proper electrical system voltage and frequency. Manual or automatic control and protective or signal devices required for the operation herein specified and any control wiring required for controls and devices but not indicated must be provided under this section of the specifications.

2.5 ELECTRICAL EQUIPMENT

Provide electrical equipment in conformance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical motor driven equipment herein specified complete with motors, motor starters, and controls. Motor controls, equipment, and wiring must be in accordance with NFPA 70.

2.5.1 Electric Motors

Drive each electric motor-driven pump by a continuous-duty electric motor

with enclosure type for specific service as defined in paragraph HYDRONIC PUMPS. Motor must have a 1.5 service factor. Provide motors having normal-starting-torque and low-starting-current characteristics, and of sufficient size so that the nameplate wattage rating will not be exceeded throughout the entire published pump characteristic curve. Integral size motors must be the premium efficiency type in accordance with NEMA MG 1. Pump electric motor efficiencies must meet or exceed the requirements of the ICC IgCC standard. Motor bearings must provide smooth operations under the conditions encountered for the life of the motor. Provide adequate thrust bearing in the motor to carry the weight of all rotating parts plus the hydraulic thrust and be capable of withstanding upthrust imposed during pump starting and under variable pumping head conditions specified. Provide motors in conformance with NEMA MG 1.

2.5.2 Control Equipment

Automatically controlled pumps must have three-position "MANUAL-OFF-AUTOMATIC" selector switch in cover. Provide additional controls or protective devices as indicated.

2.5.3 Variable Speed Control

The variable speed motor controllers must meet the requirements of UFGS 26 29 23 ADJUSTABLE SPEED DRIVE SYSTEMS UNDER 600 VOLTS.

2.6 EQUIPMENT APPURTENANCES

2.6.1 Attachments

Furnish all necessary bolts, nuts, washers, bolt sleeves, and other types of attachments with the equipment for the installation of the equipment. Bolts conform to the requirements of ASTM A307 and hexagonal nuts of the same quality as the bolts used. Threads must be clean-cut and conform to ASME B1.1. Bolts, nuts, and washers specified to be galvanized or not otherwise indicated or specified, must be zinc coated after being threaded, by the hot-dip process conforming to ASTM A123/A123M as appropriate. Bolts, nuts, and washers specified or indicated to be stainless steel must be Type 316.

2.6.2 Equipment Guards

Provide equipment driven by open shafts, belts, chains, or gears with all-metal guards enclosing the drive mechanism. Secure guards in position with steel braces or straps that permit easy removal for servicing the equipment. Coupler guards must comply with current national safety standards including 29 CFR 1910.219 and NEMA Z535.4. Provide guards with gaps no greater than 6 mm, safety orange in color, and have an NEMA Z535.4 compliant warning label.

2.6.3 Tools

Furnish a complete set of all special tools which may be necessary for the adjustment, operation, maintenance, and disassembly of all equipment. Special tools are considered to be those tools which because of their limited use are not normally available, but which are necessary for the particular equipment. Special tools must be high-grade, smooth, forged, alloy, tool steel. Furnish one pressure grease gun for each type of grease required. Deliver all tools at the same time as the equipment to which they pertain. Properly store and safeguard such tools until

completion of the work, at which time deliver them to the Contracting Officer.

2.7 FINISHES

All motors, pump casings, and similar parts of equipment must be thoroughly cleaned, primed, and given two finish coats of paint at the factory in accordance with the recommendations of the manufacturer. Give ferrous surfaces not to be painted a shop coat of grease or other suitable rust-resistant coating.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field and advise the Contracting Officer of any discrepancy before performing the work.

3.2 INSTALLATION

Install each pump and motor in accordance with the written instructions of the manufacturer. Provide access space around the device for servicing no less than the minimum recommended by the manufacturer.

3.3 FIELD QUALITY CONTROL

After installation of the pumping units and appurtenances, including coupling guard, is complete, carry out operating tests to assure that the pumping installation operates properly. Give each pumping unit a running field test in the presence of the Contracting Officer for a minimum of 2 hours. Operate each pumping unit at its rated capacity or such other point on its head-capacity curve selected by the Contracting Officer. Provide an accurate and acceptable method of measuring the discharge flow. Tests must assure that the units and appurtenances have been installed correctly, that there is no objectionable heating, vibration, or noise from any parts, and that all manual and automatic controls function properly. If any deficiencies are revealed during any tests, correct such deficiencies and reconduct the tests.

Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of the installed system. Each test report must indicate the final position of controls.

3.4 FIELD PAINTING

Do not paint stainless steel, galvanized steel, and nonferrous surfaces.

3.4.1 Touch-up painting

Factory painted items requiring touching up in the field must be thoroughly cleaned of all foreign material, and primed and topcoated with the manufacturer's standard factory finish.

3.4.2 Exposed Ferrous Surfaces

Paint exposed ferrous surfaces with two coats of enamel paint conforming

to SSPC Paint 21. Solvent clean factory primed surfaces before painting. Surfaces that have not been factory primed must be prepared and primed with one coat of SSPC Paint 25 or in accordance with the enamel paint manufacturer's recommendations.

3.5 CLOSEOUT ACTIVITIES

3.5.1 Operation and Maintenance Manuals

Submit one complete set at the time the tests procedure is submitted; remaining sets before the contract is completed. Permanently bind each in a hard cover. Inscribe the following identification on the covers: the words "OPERATING AND MAINTENANCE INSTRUCTIONS," name and location of the building, name of the Contractor, and contract number. Place flysheets before instructions covering each subject. Use 216 by 279 mm paper for instruction sheets, with large sheets of drawings folded in.

Include, but do not limit to, the following in the Instructions:

- a. System layout showing piping, valves, and controls.
- b. Approved wiring and control diagrams including variable frequency drives.
- c. A control sequence describing startup, operation, and shutdown.
- d. Operating and maintenance instructions for each piece of equipment, including task list for routine maintenance, routine inspections, intermediate inspections, and annual inspections; lubrication instructions; and troubleshooting guide.
- e. Manufacturer's bulletins, cuts, and descriptive data; and parts list and recommended spare parts.

3.5.2 Training

Upon completion of the work, and at a time designated by the Contracting Officer, provide the services of one or more competent engineers for a training period of not less than 4 hours to instruct a representative of the Government in the contents of the operation and maintenance manuals for the equipment furnished under these specifications. These field instructions must cover all the items contained in the bound instructions. Submit the training course curriculum and training instructions 14 days prior to the start of training.

-- End of Section --

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SECTION 23 23 00

REFRIGERANT PIPING
08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 711	(2009) Performance Rating of Liquid-Line Driers
AHRI 720	(2002) Refrigerant Access Valves and Hose Connectors
AHRI 751 SI	(2016) Performance Rating of Thermostatic Refrigerant Expansion Valves
AHRI 760	(2014) Performance Rating of Solenoid Valves for Use With Volatile Refrigerants
AHRI 1371 SI	(2017) Performance Rating of Electronic Expansion Valves

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 15 & 34	(2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016
ASHRAE 17	(2022) Method of Testing Capacity of Thermostatic Refrigerant Expansion Valves
ASHRAE 90.1 - SI	(2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.2M	(2006; R 2011) Pipe Threads, 60 Deg. General Purpose (Metric)
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Butt welding Fittings

ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
ASME B16.26	(2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes
ASME B31.1	(2024) Power Piping
ASME B31.5	(2022) Refrigeration Piping and Heat Transfer Components
ASME B31.9	(2020) Building Services Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding
AWS A5.31/A5.31M	(2012) Specification for Fluxes for Brazing and Braze Welding
AWS BRH	(2007; 5th Ed) Brazing Handbook
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS Z49.1	(2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A193/A193M	(2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A334/A334M	(2004a; R 2021) Standard Specification for Seamless and Welded Carbon and Alloy-Steel Tubes for Low-Temperature Service
ASTM A653/A653M	(2023) Standard Specification for Steel

Sheet, Zinc-Coated (Galvanized) or
Zinc-Iron Alloy-Coated (Galvannealed) by
the Hot-Dip Process

ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B75/B75M	(2020) Standard Specification for Seamless Copper Tube
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B280	(2023) Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service
ASTM B813	(2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
ASTM D520	(2000; R 2011) Zinc Dust Pigment
ASTM D3308	(2012; R 2017) Standard Specification for PTFE Resin Skived Tape
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)	
MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
U.S. DEPARTMENT OF DEFENSE (DOD)	
UFC 3-301-01	(2023; with Change 2, 2024) Structural Engineering

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Refrigerant Piping System; G

SD-03 Product Data

Refrigerant Piping System

Spare Parts

Qualifications

Refrigerant Piping Tests

Verification of Dimensions

SD-06 Test Reports

Refrigerant Piping Tests

SD-07 Certificates

Service Organization

SD-10 Operation and Maintenance Data

Maintenance; G

Operation and Maintenance Manuals; G

Demonstrations; G

1.3 QUALITY ASSURANCE

1.3.1 Qualifications

Submit 2 copies of qualified procedures, and list of names and identification symbols of qualified welders and welding operators, prior to non-factory welding operations. Weld piping in accordance with the qualified procedures using performance qualified welders and welding operators. Procedures and welders must be qualified in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ASME B31.1. Notify the Contracting Officer 24 hours in advance of tests to be performed at the work site, if practical. The welder or welding operator must apply the personally assigned symbol near each weld made, as a permanent record. Weld structural members in accordance with Structural drawings.

1.3.2 Contract Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

1.4 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation is the Contractor's responsibility. Replace any materials found to be damaged at the Contractor's expense. During installation, cap piping and similar

openings to keep out dirt and other foreign matter.

1.5 MAINTENANCE

1.5.1 General

Submit Data Package 2 plus operation and maintenance data complying with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

1.5.2 Extra Materials

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis in the data.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for 2 years prior to bid opening.
- b. The 2 year use must include applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown.
- c. Products must be supported by a service organization. System components must be environmentally suitable for the indicated locations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.
- d. Exposed equipment moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.
- e. Provide the manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component. Highlight the data to show information such as, but not limited to,

material, size, options, performance charts, and curves in adequate detail to demonstrate compliance with contract requirements. Include the manufacturer's recommended installation instructions and procedures in the data provided. Provide data for the following components as a minimum:

- (1) Piping and Fittings
- (2) Valves
- (3) Piping Accessories
- (4) Pipe Hangers, Inserts, and Supports

2.2 ELECTRICAL WORK

Electrical equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Field wiring must be in accordance with manufacturer's instructions. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

2.3 REFRIGERANT PIPING SYSTEM

Provide refrigerant piping, valves, fittings, and accessories in accordance with ASHRAE 15 & 34 and ASME B31.5, except as specified herein. Refrigerant piping, valves, fittings, and accessories must be compatible with the fluids used and capable of withstanding the pressures and temperatures of the service. Refrigerant piping, valves, and accessories used for refrigerant service must be cleaned, dehydrated, and sealed (capped or plugged) prior to shipment from the manufacturer's plant. Submit drawings, at least 5 weeks prior to beginning construction, provided in adequate detail to demonstrate compliance with contract requirements. Drawings must consist of:

- a. Piping layouts which identify all valves and fittings.
- b. Plans and elevations which identify clearances required for maintenance and operation.

2.4 PIPE, FITTINGS AND END CONNECTIONS (JOINTS)

2.4.1 Steel Pipe

Steel pipe for refrigerant service must conform to ASTM A53/A53M, Schedule 40, Type E or S, Grades A or B. Do not use Type F pipe.

2.4.1.1 Welded Fittings and Connections

Butt-welded fittings must conform to ASME B16.9. Socket-welded fittings must conform to ASME B16.11. Identify welded fittings with the appropriate grade and marking symbol. Welded valves and pipe connections (both butt-welds and socket-welds types) must conform to ASME B31.9.

2.4.1.2 Threaded Fittings and Connections

Threaded fitting must conform to ASME B16.3. Threaded valves and pipe connections must conform to ASME B1.20.2M.

2.4.1.3 Flanged Fittings and Connections

Flanges must conform to ASME B16.5, Class 150. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1.59 mm thickness, full face or self-centering flat ring type. Gaskets must contain aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR). Bolts, nuts, and bolt patterns must conform to ASME B16.5. Bolts must be high or intermediate strength material conforming to ASTM A193/A193M.

2.4.2 Steel Tubing

Tubing must be cold-rolled, electric-forged, welded-steel in accordance with ASTM A334/A334M, Grade 1. Joints and fittings must be socket type provided by the steel tubing manufacturer.

2.4.3 Copper Tubing

Provide copper tubing conforming to ASTM B280 annealed or hard drawn as required. Copper tubing must bear the product identification markings in accordance with ASTM B280, "ACR" must be present on copper tubing. Copper tubing must be soft annealed where bending is required and hard drawn where no bending is required. Soft annealed copper tubing must not be used in sizes larger than 35 mm. Joints must be brazed except that joints on lines 22 mm and smaller may be flared. Cast copper alloy fittings for flared copper tube must conform to ASME B16.26 and ASTM B62. Wrought copper and bronze solder-joint pressure fittings must conform to ASME B16.22 and ASTM B75/B75M. Joints and fittings for brazed joint must be wrought-copper or forged-brass sweat fittings. Cast sweat-type joints and fittings are not allowed for brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

2.4.4 Solder

Solder must conform to ASTM B32, grade Sb5, tin-antimony alloy for service pressures up to 1034 kPa. Solder flux must be liquid or paste form, non-corrosive and conform to ASTM B813.

2.4.5 Brazing Filler Metal

Filler metal must conform to AWS A5.8/A5.8M, Type BAg-5 with AWS Type FB3-A or Type FB3-C flux, except Type BCuP-3, BCuP-4, or BCuP-5 may be used for brazing copper-to-copper joints. BAlSi-4 with AWS Type FB1-A flux may be used when joining copper piping to aluminum components.

2.4.6 Brazing Flux

Brazing flux must conform to AWS A5.31/A5.31M, Type FB3-A or Type FB3-C when using Type BAg-5 filler metal. Type FB1-A is to be used with Type BAlSi-4 filler metal.

2.4.7 Press Fittings

Press fittings are not acceptable for use in refrigerant piping systems.

2.5 VALVES

Valves must be designed, manufactured, and tested specifically for refrigerant service. The valve material and all internal components must be compatible with the specific refrigerant and lubricant used. Valve bodies must be of brass, bronze, steel, or ductile iron construction. Valves 25 mm and smaller must have brazed or socket welded connections. Valves larger than 25 mm must have tongue-and-groove flanged end connections. Do not use threaded end connections, except in pilot pressure or gauge lines where maintenance disassembly is required and welded flanges cannot be used. Internal parts must be removable for inspection or replacement without applying heat or breaking pipe connections. Valve stems exposed to the atmosphere must be stainless steel or corrosion resistant metal plated carbon steel. Direction of flow must be legibly and permanently indicated on the valve body. Control valve inlets must be fitted with integral or adapted strainer or filter where recommended or required by the manufacturer. Purge, charge and receiver valves must be of manufacturer's standard configuration.

2.5.1 Refrigerant Stop Valves

Valve must be the globe or full-port ball type with a back-seating stem especially packed for refrigerant service. Valve packing must be replaceable under line pressure. Provide valve with a handwheel or wrench operator and a seal cap. Valve must be the straight or angle pattern design as indicated.

2.5.2 Check Valves

Valve must be the swing or lift type as required to provide positive shutoff at the differential pressure indicated. Valve must be provided with resilient seat.

2.5.3 Liquid Solenoid Valves

Provide valves that comply with AHRI 760 and are suitable for continuous duty with applied voltages 15 percent under and 5 percent over nominal rated voltage at maximum and minimum encountered pressure and temperature service conditions. Valves must be direct-acting or pilot-operating type, packless, except that packed stem, seal capped, manual lifting provisions must be furnished. Provide solenoid coils that are moisture-proof, UL approved, totally encapsulated or encapsulated and metal jacketed as required. Valves must have safe working pressure of 4206 kPa and a maximum operating pressure differential of at least 1375 kPa at 85 percent rated voltage. Valves must have an operating pressure differential suitable for the refrigerant used.

2.5.4 Expansion Valves

Provide valve conforming to AHRI 751 SI and ASHRAE 17. Valve must be the diaphragm and spring-loaded type with internal or external equalizers, and bulb and capillary tubing. Provide valve with an external superheat adjustment along with a seal cap. Internal equalizers may be utilized where flowing refrigerant pressure drop between outlet of the valve and inlet to the evaporator coil is negligible and pressure drop across the evaporator is less than the pressure difference corresponding to 1 degree C of saturated suction temperature at evaporator conditions. Bulb charge must be determined by the manufacturer for the application and such that liquid will remain in the bulb at all operating conditions. Do not use

gas limited liquid charged valves and other valve devices for limiting evaporator pressure without a distributor or discharge tube or effective means to prevent loss of control when bulb becomes warmer than valve body. Pilot-operated valves must have a characterized plug to provide required modulating control. A de-energized solenoid valve may be used in the pilot line to close the main valve in lieu of a solenoid valve in the main liquid line. Provide an isolatable pressure gauge in the pilot line, at the main valve. Automatic pressure reducing or constant pressure regulating expansion valves may be used only where indicated or for constant evaporator loads.

2.5.5 Electronic Expansion Valves

Valve must conform to AHRI 1371 SI and ASHRAE 17. The valve must prevent the return of liquid to the compressor in the event of power loss or low superheat.

2.5.6 Safety Relief Valves

Valve must be the two-way type, unless indicated otherwise. Valve must bear the ASME code symbol. Valve capacity must be certified by the National Board of Boiler and Pressure Vessel Inspectors. Valve must be of an automatically reseating design after activation.

2.5.7 Evaporator Pressure Regulators, Direct-Acting

Valve must include a diaphragm/spring assembly, external pressure adjustment with seal cap, and pressure gauge port. Valve must maintain a constant inlet pressure by balancing inlet pressure on diaphragm against an adjustable spring load. Pressure drop at system design load must not exceed the pressure difference corresponding to a 1 degree C change in saturated refrigerant temperature at evaporator operating suction temperature. Spring must be selected for indicated maximum allowable suction pressure range.

2.5.8 Refrigerant Access Valves

Provide refrigerant access valves and hose connections in accordance with AHRI 720.

2.6 PIPING ACCESSORIES

2.6.1 Filter Driers

Driers must conform to AHRI 711. Sizes 15 mm and larger must be the full flow, replaceable core type. Sizes 13 mm and smaller must be the sealed type. Cores must be of suitable desiccant that will not plug, cake, dust, channel, or break down, and must remove water, acid, and foreign material from the refrigerant. Construct filter driers so that none of the desiccant will pass into the refrigerant lines. Minimum bursting pressure must be 10.3 MPa.

2.6.2 Sight Glass and Liquid Level Indicator

2.6.2.1 Assembly and Components

Assembly must be pressure- and temperature-rated and constructed of materials suitable for the service. Glass must be borosilicate type. Ferrous components subject to condensation must be electro-galvanized.

2.6.2.2 Gauge Glass

Gauge glass must include top and bottom isolation valves fitted with automatic checks, and packing followers; red-line or green-line gauge glass; elastomer or polymer packing to suit the service; and gauge glass guard.

2.6.2.3 Bull's-Eye and Inline Sight Glass Reflex Lens

Provide bull's-eye and inline sight glass reflex lens for dead-end liquid service. For pipe line mounting, provide two plain lenses in one body suitable for backlighting viewing.

2.6.2.4 Moisture Indicator

Indicator must be a self-reversible action, moisture reactive, color changing media. Indicator must be furnished with full-color-printing tag containing color, moisture, and temperature criteria. Unless otherwise indicated, the moisture indicator must be an integral part of each corresponding sight glass.

2.6.3 Vibration Dampeners

Dampeners must be of the all-metallic bellows and woven-wire type.

2.6.4 Flexible Pipe Connectors

Connector must be a composite of interior corrugated phosphor bronze or Type 300 Series stainless steel, as required for fluid service, with exterior reinforcement of bronze, stainless steel or monel wire braid. Assembly must be constructed with a safety factor of not less than 4 at 150 degrees C. Unless otherwise indicated, the length of a flexible connector must be as recommended by the manufacturer for the service intended.

2.6.5 Strainers

Strainers used in refrigerant service must have brass or cast-iron body, Y-or angle-pattern, cleanable, not less than 60-mesh noncorroding screen of an area to provide net free area not less than ten times the pipe diameter with pressure rating compatible with the refrigerant service. Screens must be stainless steel or monel and reinforced spring-loaded where necessary for bypass-proof construction.

2.6.6 Pressure and Vacuum Gauges

Provide gauges conforming to ASME B40.100 with throttling type needle valve or a pulsation dampener and shut-off valve. Gauge must be a minimum of 85 mm in diameter with a range from 0 kPa to approximately 1.5 times the maximum system working pressure. Select each gauge range so that at normal operating pressure, the needle is within the middle-third of the range.

2.6.7 Temperature Gauges

Provide industrial duty type temperature gauges for the required temperature range. Gauges must have Celsius scale in 1 degree graduations scale (black numbers) on a white face. The pointer must be adjustable.

Provide rigid stem type temperature gauges in thermowells located within 1.5 m of the finished floor. Provide universal adjustable angle type or remote element type temperature gauges in thermowells located 1.5 to 2.1 m above the finished floor. Provide remote element type temperature gauges in thermowells located 2.1 m above the finished floor.

2.6.7.1 Stem Cased-Glass

Provide stem cased-glass case composed of polished stainless steel or cast aluminum, 229 mm long, with clear acrylic lens, and non-mercury filled glass tube with indicating-fluid column.

2.6.7.2 Bimetallic Dial

Provide bimetallic dial type case that is greater than 89 mm, stainless steel, and hermetically sealed with clear acrylic lens. Bimetallic element must be silicone dampened and unit fitted with external calibrator adjustment. Accuracy must be one percent of dial range.

2.6.7.3 Liquid-, Solid-, and Vapor-Filled Dial

Provide liquid-, solid-, and vapor-filled dial type cases that are greater than 89 mm, stainless steel or cast aluminum with clear acrylic lens. Fill must be nonmercury, suitable for encountered cross-ambients, and connecting capillary tubing must be double-braided bronze.

2.6.7.4 Thermowell

Thermowell must be identical size, 13 or 19 mm NPT connection, brass or stainless steel. Where test wells are indicated, provide captive plug-fitted type 13 mm NPT connection suitable for use with either engraved stem or standard separable socket thermometer or thermostat. Mercury must not be used in thermometers. Extended neck thermowells must be of sufficient length to clear insulation thickness by 25 mm.

2.6.8 Pipe Hangers, Inserts, and Supports

Provide pipe hangers, inserts, guides, and supports conforming to MSS SP-58.

2.6.9 Escutcheons

Escutcheons must be chromium-plated iron or chromium-plated brass, either one piece or split pattern, held in place by internal spring tension or set screws.

2.7 FABRICATION

2.7.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type

I.

2.7.2 Factory Applied Insulation

Factory installed insulation must be in accordance with ASHRAE 90.1 - SI. Refrigerant suction lines between the cooler and each compressor must be insulated with not less than 13 mm thick unicellular plastic foam. Factory insulated items installed outdoors are not required to be fire-rated. As a minimum, factory insulated items installed indoors must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Factory insulated items (no jacket) installed indoors and which are located in air plenums, in ceiling spaces, and in attic spaces must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Flame spread and smoke developed indexes must be determined by ASTM E84. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Test material supplied by a manufacturer with a jacket as a composite material. Provide jackets, facings, and adhesives that have a flame spread index less than 25 and a smoke developed index less than 50 when tested in accordance with ASTM E84.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, perform a verification of dimensions in the field. Submit a letter, at least 2 weeks prior to beginning construction, including the date the site was visited, conformation of existing conditions, and any discrepancies found before performing any work.

3.2 INSTALLATION

Pipe and fitting installation must conform to the requirements of ASME B31.1. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove by reaming, and permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

3.2.1 Directional Changes

Make changes in direction with fittings, except that bending of pipe 100 mm and smaller is permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends must not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted.

3.2.2 Functional Requirements

Install piping 4 mm/m of pipe in the direction of flow to ensure adequate oil drainage. Properly cap or plug open ends of refrigerant lines or equipment during installation to keep moisture, dirt, or other foreign material out of the system. Piping must remain capped until installation. Equipment piping must be in accordance with the equipment manufacturer's recommendations and the contract drawings. Equipment and

pipng arrangements must fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.

3.2.3 Fittings and End Connections

3.2.3.1 Threaded Connections

Make threaded connections with tapered threads and make tight with PTFE tape complying with ASTM D3308 or equivalent thread-joint compound applied to the male threads only. Do not show more than three threads after the joint is made.

3.2.3.2 Brazed Connections

Perform brazing in accordance with AWS BRH, except as modified herein. During brazing, fill the pipe and fittings with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux on copper-to-copper connections. Remove surplus brazing material at all joints. Make steel tubing joints in accordance with the manufacturer's recommendations. Paint joints in steel tubing with the same material as the baked-on coating within 8 hours after joints are made. Protect tubing against oxidation during brazing by continuous purging of the inside of the piping using nitrogen. Support piping prior to brazing and do not spring or force.

3.2.3.3 Welded Connections

Fusion-weld joints in steel refrigerant piping. Make branch connections with welding tees or forged welding branch outlets. Thoroughly clean pipe of all scale and foreign matter before the piping is assembled. During welding, fill the pipe and fittings with an inert gas, such as nitrogen, to prevent the formation of scale. Beveling, alignment, heat treatment, and inspection of weld must conform to ASME B31.1. Remove and reweld weld defects at no additional cost to the Government. Store and dry electrodes in accordance with AWS D1.1/D1.1M or as recommended by the manufacturer. Do not use electrodes that have been wetted or that have lost any of their coating

3.2.3.4 Flared Connections

When flared connections are used, use a suitable lubricant between the back of the flare and the nut in order to avoid tearing the flare while tightening the nut.

3.2.3.5 Flanged Connections

When steel refrigerant piping is used, provide union or flange joints in each line immediately preceding the connection to each piece of equipment requiring maintenance, such as compressors, coils, chillers, control valves, and other similar items. Flanged joints must be assembled square end tight with matched flanges, gaskets, and bolts. Provide gaskets that are suitable for use with the refrigerants to be handled.

3.2.4 Valves

3.2.4.1 General

Install refrigerant stop valves on each side of each piece of equipment such as compressors condensers, evaporators, receivers, and other similar items in multiple-unit installation, to provide partial system isolation as required for maintenance or repair. Install stop valves with stems horizontal unless otherwise indicated. Install ball valves must be installed with stems positioned to facilitate operation and maintenance. Isolating valves for pressure gauges and switches must be external to thermal insulation. Safety switches must not be fitted with isolation valves. Filter dryers having access ports may be considered a point of isolation. Purge valves must be provided at all points of systems where accumulated non-condensable gases would prevent proper system operation. Valves must be furnished to match line size, unless otherwise indicated or approved.

3.2.4.2 Expansion Valves

Install expansion valves with the thermostatic expansion valve bulb located on top of the suction line when the suction line is less than 54 mm in diameter and at the 4 o'clock or 8 o'clock position on lines larger than 54 mm. Fasten the bulb securely with two clamps. Insulate the bulb. Install the bulb in a horizontal portion of the suction line, if possible, with the pigtail on the bottom. If the bulb must be installed in a vertical line, the bulb tubing must be facing up.

3.2.4.3 Valve Identification

Tag each system valve, including those which are part of a factory assembly. Tags must be in alphanumeric sequence, progressing in direction of fluid flow. Tags must be embossed, engraved, or stamped plastic or nonferrous metal of various shapes, sized approximately 34 mm diameter, or equivalent dimension, substantially attached to a component or immediately adjacent thereto. Attach tags with nonferrous, heavy duty, bead or link chain, 14 gauge annealed wire, nylon cable bands or as approved. Reference tag numbers in Operation and Maintenance Manuals and system diagrams.

3.2.5 Vibration Dampers

Provide vibration damper in the suction and discharge lines on spring mounted compressors. Install vibration dampers parallel with the shaft of the compressor and anchor firmly at the upstream end on the suction line and the downstream end in the discharge line.

3.2.6 Strainers

Provide strainers immediately ahead of solenoid valves and expansion devices. Strainers may be an integral part of an expansion valve.

3.2.7 Filter Dryer

Provide a liquid line filter dryer on each refrigerant circuit located such that all liquid refrigerant passes through a filter dryer. Size dryers in accordance with the manufacturer's recommendations for the system in which it is installed. Install dryers such that it can be isolated from the system, the isolated portion of the system evacuated,

and the filter dryer replaced. Install dryers in the horizontal position except replaceable core filter dryers may be installed in the vertical position with the access flange on the bottom.

3.2.8 Sight Glass

Install a moisture indicating sight glass in all refrigerant circuits downstream of all filter dryers and where indicated. Provide full line size sight glasses.

3.2.9 Discharge Line Oil Separator

Provide discharge line oil separator in the discharge line from each compressor. Connect the oil return line to the compressor as recommended by the compressor manufacturer.

3.2.10 Accumulator

Provide accumulators in the suction line to each compressor.

3.2.11 Flexible Pipe Connectors

Install connectors perpendicular to line of motion being isolated. Fit piping for equipment with bidirectional motion with two flexible connectors, in perpendicular planes. Install reinforced elastomer flexible connectors in accordance with manufacturer's instructions. Provide piping guides and restraints related to flexible connectors as required.

3.2.12 Temperature Gauges

Locate temperature gauges specifically on, but not limited to the following: the liquid line leaving a receiver and the suction line at each evaporator or liquid cooler. Thermowells for insertion thermometers and thermostats must extend beyond thermal insulation surface not less than 25 mm.

3.2.13 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports must conform to MSS SP-58, except as modified herein. Do not use pipe hanger types 5, 12, and 26. Fabricate hangers used to support piping 50 mm and larger to permit adequate adjustment after erection while still supporting the load. Support piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, by variable spring hangers and supports or by constant support hangers.

3.2.13.1 Hangers

Do not use Type 3 on insulated piping. Type 24 may be used only on trapeze hanger systems or on fabricated frames.

3.2.13.2 Inserts

Secure Type 18 inserts to concrete forms before concrete is placed. Continuous inserts which allow more adjustments may be used if they otherwise meet the requirements for Type 18 inserts.

3.2.13.3 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with MSS SP-58 and have both locknuts and retaining devices, furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

3.2.13.4 Angle Attachments

Furnish Type 20 attachments used on angles and channels with an added malleable-iron heel plate or adapter.

3.2.13.5 Saddles and Shields

Where Type 39 saddle or Type 40 shield are permitted for a particular pipe attachment application, the Type 39 saddle, connected to the pipe, must be used on all pipe 100 mm and larger when the temperature of the medium is 16 degrees C or higher. Use Type 40 shields on all piping less than 100 mm and all piping 100 mm and larger carrying medium less than 16 degrees C. Use a high-density insulation insert of cellular glass under the Type 40 shield for piping 50 mm and larger.

3.2.13.6 Horizontal Pipe Supports

Space horizontal pipe supports as specified in MSS SP-58 and install a support no more than 300 mm from the pipe fitting joint at each change in direction of the piping. Space pipe supports no more than 1.5 m apart at valves.

3.2.13.7 Vertical Pipe Supports

Support vertical pipe at each floor, except at slab-on-grade, and at intervals of not more than 4.5 m not more than 2.4 m from end of risers, and at vent terminations.

3.2.13.8 Pipe Guides

Provide Type 35 guides using, steel, reinforced polytetrafluoroethylene (PTFE) or graphite slides where required to allow longitudinal pipe movement. Provide lateral restraints as required. Provide slide materials that are suitable for the system operating temperatures, atmospheric conditions, and bearing loads encountered.

3.2.13.9 Steel Slides

Where steel slides do not require provisions for restraint of lateral movement, an alternate guide method may be used. On piping 100 mm and larger, use a Type 39 saddle. On piping under 100 mm, a Type 40 protection shield may be attached to the pipe or insulation and freely rest on a steel slide plate.

3.2.13.10 High Temperature Guides with Cradles

Where there are high system temperatures and welding to piping is not desirable, the Type 35 guide must include a pipe cradle, welded to the guide structure and strapped securely to the pipe. Separate the pipe from the slide material by at least 100 mm, or by an amount adequate for the insulation, whichever is greater.

3.2.13.11 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, use a clip or clamp where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

3.2.13.12 Seismic Requirements

Support and brace piping and attached valves to resist seismic loads as specified under UFC 3-301-01 and Section 23 05 48.19 SEISMIC BRACING FOR HVAC. Provide structural steel required for reinforcement to properly support piping, headers, and equipment but not shown under this section.

3.2.13.13 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Provide structural steel brackets required to support piping, headers, and equipment, but not shown, under this section.

3.2.14 Pipe Alignment Guides

Provide pipe alignment guides where indicated for expansion loops, offsets, and bends and as recommended by the manufacturer for expansion joints, not to exceed 1.5 m on each side of each expansion joint, and in lines 100 mm or smaller not more than 600 mm on each side of the joint.

3.2.15 Pipe Anchors

Provide anchors wherever necessary or indicated to localize expansion or to prevent undue strain on piping. Provide anchors consisting of heavy steel collars with lugs and bolts for clamping and attaching anchor braces, unless otherwise indicated. Install anchor braces in the most effective manner to secure the desired results using turnbuckles where required. Do not attach supports, anchors, or stays where they will injure the structure or adjacent construction during installation or by the weight of expansion of the pipeline. Where pipe and conduit penetrations of vapor barrier sealed surfaces occur, immediately anchor these items adjacent to each penetrated surface, to provide essentially zero movement within penetration seal. Submit detailed drawings of pipe anchors for approval before installation.

3.2.16 Building Surface Penetrations

Do not install sleeves in structural members except where indicated or approved. Provide galvanized sheet metal sleeves in non-load bearing surfaces conforming to ASTM A653/A653M, Coating Class G-90, 1.0 mm (20 gauge). Provide uncoated carbon steel pipe sleeves in load bearing surfaces conforming to ASTM A53/A53M, Standard weight. Apply sealants to moisture and oil-free surfaces and elastomers to not less than 13 mm depth. Do not install sleeves in structural members.

3.2.16.1 Refrigerated Space

Fit refrigerated space building surface penetrations with sleeves

fabricated from hand-lay-up or helically wound, fibrous glass reinforced polyester or epoxy resin with a minimum thickness equal to equivalent size Schedule 40 steel pipe. Construct sleeves with integral collar or fit cold side with a bonded slip-on flange or extended collar. In the case of masonry penetrations where sleeve is not cast-in, fill voids with latex mixed mortar cast to shape of sleeve and assemble flange/external collar type sleeve with butyl elastomer vapor barrier sealant through penetration to cold side surface vapor barrier overlap and fastened to surface with masonry anchors. Flash integral cast-in collar type sleeve with not less than 100 mm of cold side vapor barrier overlap of sleeve surface. Normally seal noninsulated penetrating round surfaces to sleeve bore with mechanically expandable seals in vapor tight manner and insulate remaining warm and cold side sleeve depth with not less than 100 mm of foamed-in-place rigid polyurethane or foamed-in-place silicone elastomer. Apply vapor barrier sealant to finish warm side insulation surface. Insulate warm side of penetrating surface beyond vapor barrier sealed sleeve insulation for a distance which prevents condensation. Seal wires in refrigerated space surface penetrating conduit with vapor barrier plugs or compound to prevent moisture migration through conduit and condensation therein.

3.2.16.2 General Service Areas

Extend each sleeve through its respective wall, floor, or roof, and cut flush with each surface. Provide pipes passing through concrete or masonry wall or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Provide sleeves that allow a minimum of 6 mm all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, seal the annular space between pipe and sleeve or between jacket over-insulation and sleeve.

3.2.16.3 Waterproof Penetrations

Install pipes passing through roof or floor waterproofing membrane through a 5.17 kg/sq. m copper sleeve, or a 0.81 mm thick aluminum sleeve, each within an integral skirt or flange. Form flashing sleeve, and extend skirt or flange greater than 200 mm from the pipe and set over the roof or floor membrane in a troweled coating of bituminous cement. Extend the flashing sleeve up the pipe a minimum of 50 mm above the roof or floor penetration. Seal the annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation as indicated. Seal penetrations by either one of the following methods.

3.2.16.3.1 Waterproofing Clamping Flange

Pipes up to and including 250 mm in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Clamp waterproofing membrane into place and place sealant in the caulking recess.

3.2.16.3.2 Modular Mechanical Type Sealing Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular mechanical type sealing assembly may be installed. Provide seals consisting of interlocking synthetic rubber links shaped to continuously

fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Loosely assemble links with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tighten the bolt to cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Size each seal assembly as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals must provide sleeves of the proper diameters.

3.2.16.4 Escutcheons

Provide escutcheons for finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Where sleeves project slightly from floors, use special deep-type escutcheons. Secure escutcheon to pipe or pipe covering.

3.2.17 Access Panels

Provide access panels for all concealed valves, vents, controls, and items requiring inspection or maintenance. Provide access panels of sufficient size and locate so that the concealed items may be serviced and maintained or completely removed and replaced.

3.2.18 Field Applied Insulation

Field installed insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

3.2.19 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

3.2.19.1 Color Coding

Color coding for piping identification is specified in Section 09 90 00 PAINTS AND COATINGS.

3.2.19.2 Color Coding Scheme

Provide a color coding scheme for locating hidden piping in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.2.20 Identification Tags

Provide identification tags made of brass, engraved laminated plastic or engraved anodized aluminum indicating service and item number on all valves and dampers. Tags must be 35 mm minimum diameter and marking must be stamped or engraved. Indentations must be black for reading clarity. Attach tags to valves with No. 12 AWG copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

3.3 CLEANING AND ADJUSTING

Clean uncontaminated system(s) by evacuation and purging procedures

currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant, or any foreign matter are considered contaminated systems. Restore contaminated systems to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging, using currently approved refrigerant and refrigeration manufacturer's procedures. Restore contaminated systems at no additional cost to the Government as determined by the Contracting Officer. Do not use water in any procedure or test.

3.4 TRAINING COURSE

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training. Conduct a training course for 3 members of the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.
- b. Cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations in the field posted instructions..
- c. Submit 6 complete copies of an operation manual in bound 216 by 279 booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. Include the manufacturer's name, model number, and parts list in the booklets. Include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features in the manuals.
- d. Submit 6 complete copies of maintenance manual in bound 216 by 279 booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. Include piping layouts and simplified wiring and control diagrams of the system as installed in the manuals.

3.5 REFRIGERANT PIPING TESTS

After all components of the refrigerant system have been installed and connected, subject the entire refrigeration system to pneumatic, evacuation, and startup tests as described herein. Submit a schedule, at least 2 weeks prior to the start of related testing, for each test. Identify the proposed date, time, and location for each test. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test. Provide the services of a qualified technician, as required, to perform all tests and procedures indicated herein. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the tests report in bound 216 by 279 mm booklets documenting all phases of the tests performed. Include initial test summaries, all repairs/adjustments made, and the final test results in the report.

3.5.1 Preliminary Procedures

Prior to pneumatic testing, isolate equipment which has been factory tested and refrigerant charged as well as equipment which could be damaged or cause personnel injury by imposed test pressure, positive or negative, from the test pressure, or remove from the system. Remove safety relief valves and rupture discs that are not part of factory sealed systems, and cap or plug openings.

3.5.2 Pneumatic Test

Provide pressure control and excess pressure protection at the source of test pressure. Valves must be wide open, except those leading to the atmosphere. Test gas must be dry nitrogen, with minus 55 degrees C dewpoint and less than 5 ppm oil. Apply test pressure in two stages before any refrigerant pipe is insulated or covered. In accordance with ASME B31.5, a preliminary test not to exceed 170 kPa must be applied as a means of locating major leaks. Every joint being tested must be coated with a thick soap or color indicating solution. The second stage test pressure must be at least 110 percent of the design pressure, but cannot exceed 130 percent of the design pressure of any component in the system. For large systems that are not completely visible, the pressure in the system must be gradually increased to one-half of the test pressure after which the pressure must be increased in steps of one-tenth of the test pressure, until the required test pressure has been reached. The test pressure must be continuously maintained for at least 24 hours, after which it can be reduced to the leak test pressure. A correction factor of 2 kPa will be allowed for each degree C change between test space initial and final ambient temperature, plus for increase and minus for a decrease. The leak test pressure must be the design pressure, or a pressure specified in the engineering design. To repair leaks, the joint must be taken apart, thoroughly cleaned, and reconstructed as a new joint. Joints repaired by caulking, re-melting, or back-welding/brazing are not acceptable. Following repair, the entire system must be retested using the pneumatic tests described above. Reassemble the entire system once the pneumatic tests are satisfactorily completed.

3.5.3 Evacuation Test

Following satisfactory completion of the pneumatic tests, relieve the pressure and evacuate the entire system to an absolute pressure of 300 micrometers. During evacuation of the system, the ambient temperature must be higher than 2 degrees C. Do not evacuate no more than one system at one time by one vacuum pump. Once the desired vacuum has been reached, close the vacuum line and allow the system to stand for 1 hour. If the pressure rises over 500 micrometers after the 1 hour period, evacuate the system again down to 300 micrometers and let set for another 1 hour period. Do not charge the system until a vacuum of at least 500 micrometers is maintained for a period of 1 hour without the assistance of a vacuum line. If during the testing the pressure rises above 500 micrometers, continue to repeat the evacuation procedures until all residual moisture has been removed. During evacuation, record pressures by a thermocouple-type, electronic-type, or a calibrated-micrometer type gauge.

3.5.4 System Charging and Startup Test

Following satisfactory completion of the evacuation tests, charge the system with the required amount of refrigerant by raising pressure to

normal operating pressure and in accordance with manufacturer's procedures. Following charging, the system must operate with high-side and low-side pressures and corresponding refrigerant temperatures, at design or improved values. Test the entire system tested for leaks. Test fluorocarbon systems with halide torch or electronic leak detectors.

3.5.5 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must be immediately isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

3.5.6 Contractor's Responsibility

At all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time will the allowable leak rate exceed the leak rates allowed in Section 608 of the Clean Air Act: 30 percent of the full charge per year for industrial refrigeration, 20 percent of the full charge per year for commercial refrigeration, and 10 percent of the full charge per year for comfort cooling. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

-- End of Section --

SECTION 23 25 00

CHEMICAL TREATMENT OF WATER FOR MECHANICAL SYSTEMS
05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B40.100 (2022) Pressure Gauges and Gauge Attachments

ASTM INTERNATIONAL (ASTM)

ASTM D596 (2001; R 2018) Standard Guide for Reporting Results of Analysis of Water

ASTM D2688 (2015; E 2016) Standard Test Method for Corrosivity of Water in the Absence of Heat Transfer (Weight Loss Methods)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 200-1-13 (2016) Environmental Quality -- Minimizing the Risk of Legionellosis Associated with Building Water Systems on Army Installation

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Water Analysis; G

Spare Parts

Field Instructions

Tests; G

Training Course; G

1.3 MAINTENANCE MATERIAL SUBMITTALS

Submit spare parts data for each different item of material and equipment specified, after approval of the detail drawings, not later than 2 months prior to the date of beneficial occupancy. Include a complete list of parts and supplies, with source of supply, with the data.

1.4 QUALITY CONTROL

1.4.1 Safety

Ensure exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel are insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired.

1.4.2 Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Summary

This section covers the provisions and installation procedures necessary for a complete and totally functional water system(s) chemical treatment. Provide and install the system with all necessary System Components, Accessories, Piping Components, and Supplemental Components/Services. Minimize to risk of Legionellosis by following the guidance in EM 200-1-13.

2.1.2 Standard Products

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for two years' prior to bid opening.
- b. Include in the two-year use all applications of equipment and materials under similar circumstances and of similar size. Ensure the two years' experience has been satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a two-year field service record are acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory

tests, can be shown.

- c. All products are required to be supported by a service organization. Submit a certified list of qualified permanent service organizations for support of the equipment, including their addresses and qualifications. These service organizations are required to be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.
- d. The selected service organization provides the chemicals required, the concentrations required, and the water treatment equipment sizes and flow rates required. The company provides all chemicals required for the chilled water systems and fills the systems with chemicals to the levels specified. The chemical is required to meet the requirements of this specification as well as the recommendations from the manufacturers of the condenser and cooling tower. Acid treatment chemicals are not allowed to be used.

2.1.3 Water Analysis

Conditions of make-up water to be supplied to the boilers, cooling towers and chilled water systems reported in accordance with ASTM D596 are as follows:

Date of Sample	_____ {_____}
Temperature	_____ {_____} degrees C
Silica (SiO ₂)	_____ {_____} ppm (mg/L)
Insoluble	_____ {_____} ppm (mg/L)
Iron, total (Fe)	_____ {_____} ppm (mg/L)
Aluminum (Al)	_____ {_____} ppm (mg/L)
Calcium (Ca)	_____ {_____} ppm (mg/L)
Magnesium (Mg)	_____ {_____} ppm (mg/L)
Carbonate (HCO ₃)	_____ {_____} ppm (mg/L)
Sulfate (SO ₄)	_____ {_____} ppm (mg/L)
Chloride (Cl)	_____ {_____} ppm (mg/L)
Nitrate (NO ₃)	_____ {_____} ppm (mg/L)
Turbidity	_____ {_____} ntu
pH	_____ {_____}
Residual Chlorine	_____ {_____} ppm (mg/L)
Total Alkalinity	_____ {_____} ppm (mg/L)
Non-Carbonate Hardness	_____ {_____} ppm (mg/L)

Total Hardness	_____f_____} ppm (mg/L)
Dissolved Solids	_____f_____} ppm (mg/L)
Conductivity	_____f_____} Micromho/cm

2.2 EQUIPMENT

2.2.1 Nameplates

Provide a nameplate for each major component of equipment that includes the manufacturer's name, address, type or style, and catalog or serial number securely attached to the item of equipment. Provide nameplates for:

- a. Pump(s)
- b. Pump Motor(s)
- c. Water Treatment Controller(s)

2.2.2 Electrical Work

Ensure all electrical equipment, motors, motor efficiencies, and wiring complies with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical motor driven equipment specified complete with motors, motor starters, and controls. Provide electrical characteristics and enclosure types as shown, and unless otherwise indicated, provide all motors of 745 W and above with open, drip-proof, or totally enclosed fan cooled enclosures, high efficiency type. Perform field wiring in accordance with manufacturer's instructions. Each motor is required to conform to NEMA MG 1

and be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Provide continuous duty motors with the enclosure specified. Provide motor starters complete with thermal overload protection and other appurtenances necessary for the motor control indicated. Furnish motors with a magnetic across-the-line or reduced voltage type starter as required by the manufacturer. Furnish motor starters with NEMA 3R enclosures. Provide manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown.

2.2.3 Gauges

Provide gauges that conform to ASME B40.100, Class 1, 2, or 3, Style X, Type I or III as required, 115 mm in diameter with phenolic or metal case.

2.3 COMPONENTS

2.3.1 Chilled Water System

Provide a 7.57 L shot feeder on the chilled water piping as indicated. Furnish the feeder with an air vent, gauge glass, funnel, valves, fittings, and piping.

2.3.1.1 Chilled Water Treatment

Treat chilled water with either a borax/nitrite type treatment or a molybdate type treatment. Both types of treatment are acceptable for use with glycol. Maintain borax/nitrite treatment at the limits of 600-1000 ppm nitrite, 40-50 ppm copper corrosion inhibitor (TT or MBT), and pH of 8.5 to 9.5. Maintain molybdate treatment at the limits of 100-125 ppm molybdate, 40-50 ppm copper corrosion inhibitor (TT or MBT), and pH of 8.0 to 9.0.

2.3.1.2 Chilled Water Test Kits

Provide one test kit of each type required to determine the water quality as outlined within the operation and maintenance manuals (e.g. pH and nitrite or molybdate).

2.3.2 Test Kit

Provide one test kit of each type required to determine the water quality as outlined within the operation and maintenance manuals (e.g. pH, hardness and sulfite).

2.3.3 Supplemental Components/Services

Ensure drain and makeup water piping complies with the requirements of Section 22 00 00 PLUMBING, GENERAL PURPOSE. Connect drains to sanitary sewer systems by means of an indirect waste connection.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy, before performing any work.

3.2 INSTALLATION

Provide all chemicals, equipment and labor necessary to bring all system waters in conformance with the specified requirements. Perform all work in accordance with the manufacturer's published diagrams, recommendations, and equipment warranty requirements.

3.2.1 Piping

Fabricate all connections between dissimilar metals using dielectric unions.

3.3 FIELD QUALITY CONTROL

3.3.1 Tests

If the waters of the mechanical systems are not in conformance with the specified requirements or in accordance with manufacturer's recommendations, the contractor is required to direct the water treatment company to take corrective action to achieve compliance. Perform daily operational tests in the directed frequencies to maintain required control to prevent corrosion, scaling and damage to equipment during operation. Submit test schedules, at least 2 weeks prior to the start of related

testing, for the condenser/chilled/boiler/condensate/feedwater water quality tests. Identify the date, time, frequency and collection location for each test within the schedules.

3.3.2 Chilled Water Testing (monthly)

Perform the following tests on chilled water on a monthly basis.

pH	_____ {_____}
Nitrite or Molybdate	_____ {_____} ppm (mg/L)
Conductivity	_____ {_____} micromho/cm

3.3.3 Quality Assurance Testing

Conduct QA testing periodically by an independent water treatment lab/consultant to verify to managers that the mechanical and water treatment systems are being maintained properly. Provide the QA evaluation reports to the government COR.

3.3.3.1 Chilled Water Quality Assurance Testing (quarterly)

Perform the following tests quarterly on chilled water.

pH	_____ {_____}
Nitrite or Molybdate	_____ {_____} ppm (mg/L)
Conductivity	_____ {_____} micromho/cm
Iron (total, as Fe(2)O(3))	_____ {_____} ppm (mg/L)
Written evaluation summary	

3.3.4 Corrosion Testers

Install corrosion coupon and rack systems to verify corrosion control in the systems. Install testers or coupons in flowing system water through a sidestream or rack system. Test both mild steel and copper metal samples in the corrosion testers in accordance with ASTM D2688. Replace and analyze samples every 3 months. Rates of corrosion less than 3 mpy for steel and 0.2 mpy for copper are acceptable. Install corrosion testers on the piping systems for chilled water system.

3.4 CLOSEOUT ACTIVITIES

3.4.1 Training Course

Submit a schedule, at least 2 weeks prior to the date of the proposed training course that identifies the date, time, and location for the training. Conduct a training course for the operating staff as designated by the Contracting Officer. Conduct the training to include a total of 4

hours of normal working time and start after the system is functionally completed but prior to final acceptance tests. Submit field instructions, at least 2 weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. Include within the condensed operation instructions all preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. Frame the posted instructions under glass or laminated plastic and post where indicated by the Contracting Officer. Ensure the field instructions cover all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations.

3.5 INSPECTIONS

3.5.1 Inspection General Requirements

Thirty days after project completion, inspect the cooling tower and condenser for problems due to corrosion, scale, and biological growth. If the cooling tower and condenser are found not to conform to the manufacturer's recommended conditions, and the water treatment company recommendations have been followed; instruct the water treatment company to provide all chemicals and labor for cleaning or repairing the equipment as required by the manufacturer's recommendations.

-- End of Section --

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SECTION 23 30 00

HVAC AIR DISTRIBUTION
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (2024) Standard for the Installation of
Air Conditioning and Ventilating Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

SMACNA 1819 (2002) Fire, Smoke and Radiation Damper
Installation Guide for HVAC Systems, 5th
Edition

UL SOLUTIONS (UL)

UL 555 (2006; Reprint Aug 2016) UL Standard for
Safety Fire Dampers

UL Electrical Construction (2012) Electrical Construction Equipment
Directory

UL Fire Resistance (2014) Fire Resistance Directory

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 4009 (2017) Components of Air Duct

JIS A 9504 (2021) Man Made Mineral Fibre Thermal
Insulation Materials

JIS A 9511 (2021) Preformed Cellular Plastics Thermal
Insulation Materials

JIS B 1518 (2013) Rolling Bearings-Dynamic Load
Ratings and Rating Life

JIS B 1521 (2012) Rolling Bearings-Deep Groove Ball
Bearings

JIS B 1534 (2013) Rolling Bearings-Tapered Roller
Bearings

JIS B 8330 (2000) Testing Methods for Turbo-Fans

JIS B 8616 (2015) Package Air Conditioners

JIS B 9908-1	(2019) Test Method of Air Filter Units for Ventilation and Electric Air Cleaners for Ventilation-Part 1: Technical Specifications, Requirements and Classification System Based Upon Particulate Matter Efficiency
JIS C 4203	(2010) Single Phase Induction Motors for General Purpose (Amendment 1)
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 9603	(1988) Ventilating Fans
JIS G 3302	(2022) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3452	(2016) Carbon Steel Pipes for Ordinary Piping (Amendment 1)
JIS G 3553	(2011) Crimped Wire Cloth (Amendment 1)
JIS G 4305	(2015) Cold-Rolled Stainless Steel Plate, Sheet and Strip (Amendment 1)
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 4000	(2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)
JIS H 8610	(1999) Electroplated-Coatings of Zinc on Iron or Steel
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS HB 71	(2019) Electrical Safety
JIS K 5600-5-5	(1999) Testing Methods for Paints- Part 5 : Mechanical Property of Film Section 5 : Scratch Hardness (Stylus Method)
JIS K 5600-5-6	(1999) Testing Methods for Paints - Part 5 : Mechanical Property of Film-Section 6 : Adhesion Test (Cross-Cut Test)
JIS K 5600-7-9	(2006) Testing Methods for Paints Part 7 : Determination Of Resistance to Cyclic Corrosion Conditions Section 9 : Salt Fog/Dry/Humidity
JIS Z 2330	(2012) Non-Destructive Testing-Selection of Leak Testing Method
JIS Z 2371	(2015) Methods of Salt Spray Testing
JIS Z 4812	(1995) HEPA Filters for Radioactive

Aerosols

THE JAPAN REFRIGERATION AND AIR CONDITIONING INDUSTRY ASSOCIATION
(JRAIA)

JRA 4036

(2014) Air Handling Unit

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M

(2019) Public Building Construction
Standard Specification

JBE-S

(2014) Japanese Building Equipment
Seismic Standard Design

1.2 SYSTEM DESCRIPTION

Furnish ductwork, piping offsets, fittings, and accessories as required to provide a complete installation. Coordinate the work of the different trades to avoid interference between piping, equipment, structural, and electrical work. Provide complete, in place, all necessary offsets in piping and ductwork, and all fittings, and other components, required to install the work as indicated and specified.

1.2.1 Mechanical Equipment Identification

The number of charts and diagrams must be equal to or greater than the number of mechanical equipment rooms. Where more than one chart or diagram per space is required, mount these in edge pivoted, swinging leaf, extruded aluminum frame holders which open to 170 degrees.

1.2.1.1 Charts

Provide chart listing of equipment by designation numbers and capacities such as flow rates, pressure and temperature differences, heating and cooling capacities, horsepower, pipe sizes, and voltage and current characteristics.

1.2.1.2 Diagrams

Submit proposed diagrams, at least 2 weeks prior to start of related testing. provide neat mechanical drawings provided with extruded aluminum frame under 3 mm glass or laminated plastic, system diagrams that show the layout of equipment, piping, and ductwork, and typed condensed operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system. After approval, post these items where directed.

1.2.2 Service Labeling

Label equipment, including fans, air handlers, terminal units, etc. with labels made of self-sticking, plastic film designed for permanent installation. Provide labels in accordance with the typical examples below:

SERVICE	LABEL AND TAG DESIGNATION
Air handling unit Number	AHU - _____
Control and instrument air	CONTROL AND INSTR.
Exhaust Fan Number	EF - _____
VAV Box Number	VAV - _____
Fan Coil Unit Number	FC - _____
Terminal Box Number	TB - _____

Identify similar services with different temperatures or pressures. Where pressures could exceed 860 kilopascal, include the maximum system pressure in the label. Label and arrow piping in accordance with the following:

- Each point of entry and exit of pipe passing through walls.
- Each change in direction, i.e., elbows, tees.
- In congested or hidden areas and at all access panels at each point required to clarify service or indicated hazard.
- In long straight runs, locate labels at distances within eyesight of each other not to exceed 22 meter. All labels must be visible and legible from the primary service and operating area.

For Bare or Insulated Pipes	
for Outside Diameters of	Lettering
1/2 thru 1-3/8 inch	1/2 inch
1-1/2 thru 2-3/8 inch	3/4 inch
2-1/2 inch and larger	1-1/4 inch

1.2.3 Color Coding

Color coding of all piping systems must be in accordance with MLIT-M or the base standard.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G

SD-03 Product Data

Fire Dampers

Air Handling Units; G

Variable Volume, Single Duct Terminal Units; G

Test ProceduresDiagrams; G

SD-06 Test Reports

Performance Tests

Damper Acceptance Test; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals

Fire Dampers

Manual Balancing Dampers

Centrifugal Fans

In-Line Centrifugal Fans

Air Handling Units

Variable Volume, Single Duct Terminal Units

1.4 QUALITY ASSURANCE

Except as otherwise specified, approval of materials and equipment is based on manufacturer's published data.

- a. Submit a written certificate from any recognized testing agency, adequately equipped and competent to perform such services, stating that the items have been tested and that the units conform to the specified requirements. Outline methods of testing used by the specified agencies.
- b. Where materials or equipment are specified to be constructed or tested, or both, in accordance with the standards of the JIS a manufacturer's certificate of compliance of each item is acceptable as proof of compliance.
- c. Conformance to such agency requirements does not relieve the item from compliance with other requirements of these specifications.

1.4.1 Prevention of Corrosion

Protect metallic materials against corrosion. Provide rust-inhibiting treatment and standard finish for the equipment enclosures. Do not use aluminum in contact with earth, and where connected to dissimilar metal. Protect aluminum by approved fittings, barrier material, or treatment. Provide hot-dip galvanized ferrous parts such as anchors, bolts, braces, boxes, bodies, clamps, fittings, guards, nuts, pins, rods, shims,

thimbles, washers, and miscellaneous parts not of corrosion-resistant steel or nonferrous materials in accordance with JIS H 8641 for exterior locations and cadmium-plated in conformance with JIS H 8610 for interior locations. Provide written certification from the bolt manufacturer that the bolts furnished comply with the requirements of this specification. Include illustrations of product markings, and the number of each type of bolt to be furnished in the certification.

1.4.2 Asbestos Prohibition

Do not use asbestos and asbestos-containing products.

1.4.3 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be Refrigerant Handling Technician (Reibai-Furontou-Toriatsukai-Gijutsusha). Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4.4 Detail Drawings

Submit detail drawings showing equipment layout, including assembly and installation details and electrical connection diagrams; ductwork layout showing the location of all supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications. Include any information required to demonstrate that the system has been coordinated and functions properly as a unit on the drawings and show equipment relationship to other parts of the work, including clearances required for operation and maintenance. Submit drawings showing bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Submit function designation of the equipment and any other requirements specified throughout this Section with the shop drawings.

1.4.5 Test Procedures

Submit proposed test procedures and test schedules for the performance tests of systems, at least 2 weeks prior to the start of related testing.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored equipment at the jobsite from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all pipes until installed.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

Provide components and equipment that are "standard products" of a manufacturer regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. "Standard products" is defined as being in satisfactory commercial or industrial use for 2 years before bid opening, including applications of components and equipment under similar circumstances and of similar size, satisfactorily completed by a product that is sold on the commercial market through advertisements, manufacturers' catalogs, or brochures. Products having less than a 2-year

field service record are acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Provide equipment items that are supported by a service organization located in Japan.

2.2 IDENTIFICATION PLATES

In addition to standard manufacturer's identification plates, provide engraved laminated phenolic identification plates for each piece of mechanical equipment. Identification plates are to designate the function of the equipment. Submit designation with the shop drawings. Provide identification plates that are layers, black-white-black, engraved to show white letters on black background. Letters must be upper case. Identification plates 40 mm that are high and smaller must be 1.6 mm thick, with engraved lettering 3 mm high; identification plates larger than 40 mm high must be 3 mm thick, with engraved lettering of suitable height. Identification plates 40 mm high and larger must have beveled edges. Install identification plates using a compatible adhesive.

2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover with insulation of a type specified, high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard.

2.4 ELECTRICAL WORK

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.
- b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with JIS C 4203.
- c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with JIS C 4212.
- d. Provide motors in accordance with JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Provide motors rated for continuous duty with the enclosure specified. Provide motor duty that allows for maximum frequency start-stop operation and minimum encountered interval between start and stop. Provide motor torque capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload

protection and other necessary appurtenances. Fit motor bearings with grease supply fittings and grease relief to outside of the enclosure.

- e. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers are allowed to accomplish the same function. Use solid-state variable-speed controllers for motors rated 7.45 kW or less and adjustable frequency drives for larger motors. Provide variable frequency drives for motors as specified in Section 26 29 23 VARIABLE FREQUENCY DRIVE SYSTEMS UNDER 600 VOLTS.

2.5 ANCHOR BOLTS

Provide anchor bolts for equipment placed on concrete equipment pads or on concrete slabs. Bolts to be of the size and number recommended by the equipment manufacturer and located by means of suitable templates. Installation of anchor bolts must not degrade the surrounding concrete.

2.6 SEISMIC ANCHORAGE

Anchor equipment in accordance with applicable seismic criteria for the area and as defined in JBE-S

2.7 PAINTING

Paint equipment units in accordance with approved equipment manufacturer's standards unless specified otherwise. Field retouch only if approved. Otherwise, return equipment to the factory for refinishing.

2.8 DUCT SYSTEMS

2.8.1 Metal Ductwork

Provide metal ductwork construction, including all fittings and components, that complies with MLIT-M.

2.8.1.1 Metallic Flexible Duct

- a. Provide duct that conforms to MLIT-M with factory-applied insulation, vapor barrier, and end connections. Provide ducts designed for working pressures of 497 Pa and 373 Pa. Provide flexible round duct length that does not exceed 1525 mm. Secure connections by applying adhesive for 51 mm over rigid duct, apply flexible duct 51 mm over rigid duct, apply metal clamp, and provide minimum of three No. 8 sheet metal screws through clamp and rigid duct.
- b. Inner duct core: Provide interlocking spiral or helically corrugated flexible core constructed of zinc-coated steel, aluminum, or stainless steel; or constructed of inner liner of continuous galvanized spring steel wire helix fused to continuous, fire-retardant, flexible vapor barrier film, inner duct core.
- c. Insulation: Provide inner duct core that is insulated with mineral fiber blanket type flexible insulation, minimum of 25 mm thick. Provide insulation covered on exterior with manufacturer's standard fire retardant vapor barrier jacket for flexible round duct.

2.8.1.2 Insulated Nonmetallic Flexible Duct Runouts

Use flexible duct runouts only where indicated. Runout length is

indicated on the drawings, and is not to exceed 1.5 m. Provide runouts that are preinsulated, factory fabricated. Provide either field or factory applied vapor barrier. Provide not less than 0.60 L glass fabric duct connectors coated on both sides with neoprene. Where coil induction or high velocity units are supplied with vertical air inlets, use a streamlined, vaned and mitered elbow transition piece for connection to the flexible duct or hose. Provide a die-stamped elbow and not a flexible connector as the last elbow to these units other than the vertical air inlet type. Insulated flexible connectors are allowed as runouts. Provide insulated material and vapor barrier that conform to the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Do not expose the insulation material surface to the air stream.

2.8.1.3 General Service Duct Connectors

Provide a flexible duct connector approximately 150 mm in width where sheet metal connections are made to fans or where ducts of dissimilar metals are connected. For round/oval ducts, secure the flexible material by stainless steel or zinc-coated, iron clinch-type draw bands. For rectangular ducts, install the flexible material locked to metal collars using normal duct construction methods.

2.8.1.4 Aluminum Ducts

JIS H 4000, alloy 3003-H14 for aluminum sheet and alloy 6061-T6 or equivalent strength for aluminum connectors and bar stock.

2.8.1.5 Corrosion Resisting (Stainless) Steel Sheets

JIS G 4305

2.8.2 Duct Access Doors

Provide hinged access doors conforming to MLIT-M in ductwork and plenums where indicated and at all air flow measuring primaries, automatic dampers, fire dampers, coils, thermostats, and other apparatus requiring service and inspection in the duct system. Provide access doors upstream and downstream of air flow measuring primaries and heating and cooling coils. Provide doors that are a minimum 375 by 450 mm, unless otherwise shown. Where duct size does not accommodate this size door, make the doors as large as practicable. Equip doors 600 by 600 mm or larger with fasteners operable from inside and outside the duct. Use insulated type doors in insulated ducts.

2.8.3 Fire Dampers

Use 1.5 hour rated fire dampers unless otherwise indicated. Provide fire dampers that conform to the requirements of NFPA 90A and UL 555. Perform the fire damper test as outlined in NFPA 90A. Provide a pressure relief door upstream of the fire damper. If the ductwork connected to the fire damper is to be insulated then provide a factory installed pressure relief damper. Provide automatic operating fire dampers with a dynamic rating suitable for the maximum air velocity and pressure differential to which it is subjected. Provide fire dampers approved for the specific application, and install according to their listing. Equip fire dampers with a steel sleeve or adequately sized frame installed in such a manner that disruption of the attached ductwork, if any, does not impair the operation of the damper. Equip sleeves or frames with perimeter mounting angles attached on both sides of the wall or floor opening. Construct

ductwork in fire-rated floor-ceiling or roof-ceiling assembly systems with air ducts that pierce the ceiling of the assemblies in conformance with UL Fire Resistance. Provide curtain type with damper blades out of the air stream fire dampers. Install dampers that do not reduce the duct or the air transfer opening cross-sectional area. Install dampers so that the centerline of the damper depth or thickness is located in the centerline of the wall, partition or floor slab depth or thickness. Unless otherwise indicated, comply with the installation details given in SMACNA 1819 and in manufacturer's instructions for fire dampers. Perform acceptance testing of fire dampers according to paragraph Fire Damper Acceptance Test and NFPA 90A.

2.8.4 Manual Balancing Dampers

Furnish manual balancing dampers with accessible operating mechanisms. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building. Provide manual volume control dampers that are operated by locking-type quadrant operators. Install dampers that are 2 gauges heavier than the duct in which installed. Unless otherwise indicated, provide opposed blade type multileaf dampers with maximum blade width of 300 mm. Provide access doors or panels for all concealed damper operators and locking setscrews. Provide stand-off mounting brackets, bases, or adapters not less than the thickness of the insulation when the locking-type quadrant operators for dampers are installed on ducts to be thermally insulated, to provide clearance between the duct surface and the operator. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5 Air Supply And Exhaust Air Dampers

Provide outdoor air supply and exhaust air dampers that have a maximum leakage rate when tested in accordance with JIS A 4009 as required by MLIT-M, including maximum Damper Leakage for:

- a. Climate Zones 1,2,6,7,8 the maximum damper leakage at 250 Pa for motorized dampers is 20 L/s per square m of damper area and non-motorized dampers are not allowed.
- b. All other Climate Zones the maximum damper leakage at 250 Pa is 50 L/s per square m and for non-motorized dampers is 100 L/s per square m of damper area.

Dampers smaller than 600 mm in either direction may have leakage of 200 L/s per square m.

2.8.6 Plenums and Casings for Field-Fabricated Units

2.8.6.1 Plenum and Casings

Fabricate and erect plenums and casings as shown in MLIT-M, as applicable. Construct system casing of not less than 1.6 mm galvanized sheet steel. Furnish cooling coil drain pans with 25 mm threaded outlet to collect condensation from the cooling coils. Fabricate drain pans from not lighter than 1.6 mm steel, galvanized after fabrication or of 1.3 mm corrosion-resisting sheet steel conforming to JIS G 4305, welded and stiffened. Thermally insulate drain pans exposed to the atmosphere to prevent condensation. Coat insulation with a flame resistant waterproofing material. Provide separate drain pans for each vertical coil section, and a separate drain line for each pan. Size pans to ensure

capture of entrained moisture on the downstream-air side of the coil. Seal openings in the casing, such as for piping connections, to prevent air leakage. Size the water seal for the drain to maintain a pressure of at least 500 Pa greater than the maximum negative pressure in the coil space.

2.8.6.2 Casing

Terminate casings at the curb line and bolt each to the curb using galvanized angle, as indicated in MLIT-M.

2.8.6.3 Access Doors

Provide access doors in each section of the casing. Weld doorframes in place, gasket each door with neoprene, hinge with minimum of two brass hinges, and fasten with a minimum of two brass tension fasteners operable from inside and outside of the casing. Where possible, make doors 900 by 450 mm and locate them 450 mm above the floor. Where the space available does not accommodate doors of this size, use doors as large as the space accommodates. Swing doors so that fan suction or pressure holds doors in closed position, airtight. Provide a push-button station, located inside the casing, to stop the supply.

2.8.6.4 Factory-Fabricated Insulated Sheet Metal Panels

Factory-fabricated components are allowed for field-assembled units, provided all requirements specified for field-fabricated plenums and casings are met. Provide panels of modular design, pretested for structural strength, thermal control, condensation control, and acoustical control. Seal and insulate panel joints. Provide and gasket access doors to prevent air leakage. Provide panel construction that is not less than one mm galvanized sheet steel, assembled with fasteners treated against corrosion. Provide standard length panels that deflect not more than 13 mm under operation. Construct details, including joint sealing, not specifically covered, as indicated in MLIT-M. Construct the plenums and casings to withstand the specified internal pressure of the air systems.

2.8.6.5 Duct Liner

Unless otherwise specified, duct liner is not permitted.

2.8.7 Sound Attenuation Equipment

2.8.7.1 Systems with total pressure above 1 kPa

Provide sound attenuators on the discharge duct of each fan operating at a total pressure above 1 kPa, and, when indicated, at the intake of each fan system. Provide sound attenuators elsewhere as indicated. Provide factory fabricated sound attenuators, tested by an independent laboratory for sound and performance characteristics. Provide a net sound reduction as indicated. Maximum permissible pressure drop is not to exceed 157 Pa. Construct traps to be airtight when operating under an internal static pressure of 2.5 kPa. Provide air-side surface capable of withstanding air velocity of 50 m/s. Certify that the equipment can obtain the sound reduction values specified after the equipment is installed in the system and coordinated with the sound information of the system fan to be provided. Provide sound absorbing material conforming to JIS A 9511. Provide sound absorbing material that meets the fire hazard rating requirements for insulation specified in Section 23 07 00 THERMAL

INSULATION FOR MECHANICAL SYSTEMS. For connection to ductwork, provide a duct transition section. Factory fabricated double-walled internally insulated spiral lock seam and round duct and fittings designed for high pressure air system can be provided if complying with requirements specified for factory fabricated sound attenuators, in lieu of factory fabricated sound attenuators. Construct the double-walled duct and fittings from an outer metal pressure shell of zinc-coated steel sheet, 25 mm thick acoustical blanket insulation, and an internal perforated zinc-coated metal liner. Provide a sufficient length of run to obtain the noise reduction coefficient specified. Certify that the sound reduction value specified can be obtained within the length of duct run provided. Provide welded or spiral lock seams on the outer sheet metal of the double-walled duct to prevent water vapor penetration. Provide duct and fittings with an outer sheet that conforms to the metal thickness of high-pressure spiral and round ducts and fittings shown in MLIT-M. Provide acoustical insulation with a thermal conductivity "k" of not more than 0.0389 W/m-K at 24 degrees C mean temperature. Provide an internal perforated zinc-coated metal liner that is not less than 0.7 mm with perforations not larger than 6.35 mm in diameter providing a net open area not less than 10 percent of the surface.

2.8.7.2 System with total pressure of 1 kPa and Lower

Use sound attenuators only where indicated. Provide factory fabricated sound attenuators that are constructed of galvanized steel sheets. Provide attenuator with outer casing that is not less than 0.85 mm. Provide fibrous glass acoustical fill. Provide net sound reduction indicated. Obtain values on a test unit not less than 600 by 600 mm outside dimensions made by a certified nationally recognized independent acoustical laboratory. Provide air flow capacity as indicated or required. Provide pressure drop through the attenuator that does not exceed the value indicated, or that is not in excess of 15 percent of the total external static pressure of the air handling system, whichever is less. Acoustically test attenuators with metal duct inlet and outlet sections while under the rated air flow conditions. Include with the noise reduction data the effects of flanking paths and vibration transmission. Construct sound attenuators to be airtight when operating at the internal static pressure indicated or specified for the duct system, but in no case less than 500 Pa.

2.8.8 Diffusers, Registers, and Grilles

Provide factory-fabricated units of corrosion-resistant steel or aluminum that distribute the specified quantity of air evenly over space intended without causing noticeable drafts, air movement faster than 0.25 m/s in occupied zone, or dead spots anywhere in the conditioned area. Provide outlets for diffusion, spread, throw, and noise level as required for specified performance. Certify performance according to Japanese Industry Standards (JIS). Provide sound rated and certified inlets and outlets according to MLIT-M. Provide sound power level as indicated. Provide diffusers and registers with volume damper with accessible operator, unless otherwise indicated; or if standard with the manufacturer, an automatically controlled device is acceptable. Provide opposed blade type volume dampers for all diffusers and registers, except linear slot diffusers. Provide linear slot diffusers with round or elliptical balancing dampers. Where the inlet and outlet openings are located less than 2 m above the floor, protect them by a grille or screen.

2.8.8.1 Diffusers

Provide diffuser types indicated. Furnish ceiling mounted units with anti-smudge devices, unless the diffuser unit minimizes ceiling smudging through design features. Provide diffusers with air deflectors of the type indicated. Provide air handling troffers or combination light and ceiling diffusers conforming to the requirements of UL Electrical Construction for the interchangeable use as cooled or heated air supply diffusers or return air units. Install ceiling mounted units with rims tight against ceiling. Provide sponge rubber gaskets between ceiling and surface mounted diffusers for air leakage control. Provide suitable trim for flush mounted diffusers. For connecting the duct to diffuser, provide duct collar that is airtight and does not interfere with volume controller. Provide return or exhaust units that are similar to supply diffusers.

2.8.8.2 Registers and Grilles

Provide units that are four-way directional-control type, except provide return and exhaust registers that are fixed horizontal or vertical louver type similar in appearance to the supply register face. Furnish registers with sponge-rubber gasket between flanges and wall or ceiling. Install wall supply registers at least 150 mm below the ceiling unless otherwise indicated. Locate return and exhaust registers 150 mm above the floor unless otherwise indicated. Achieve four-way directional control by a grille face which can be rotated in 4 positions or by adjustment of horizontal and vertical vanes. Provide grilles as specified for registers, without volume control damper.

2.8.9 Louvers

Provide louvers for installation in exterior walls that are associated with the air supply and distribution system.

2.8.10 Air Vents, Penthouses, and Goosenecks

Fabricate air vents, penthouses, and goosenecks from galvanized steel or aluminum sheets with galvanized or aluminum structural shapes. Provide sheet metal thickness, reinforcement, and fabrication that conform to MLIT-M. Accurately fit and secure louver blades to frames. Fold or bead edges of louver blades for rigidity and baffle these edges to exclude driving rain. Provide air vents, penthouses, and goosenecks with bird screen.

2.8.11 Bird Screens and Frames

Provide bird screens that conform to JIS G 3553, No. 2 mesh, aluminum or stainless steel. Provide "medium-light" rated aluminum screens. Provide "light" rated stainless steel screens. Provide removable type frames fabricated from either stainless steel or extruded aluminum.

2.9 AIR SYSTEMS EQUIPMENT

2.9.1 Fans

Test and rate fans according to JIS B 8330. Calculate system effect on air moving devices in accordance with JIS B 8330 where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine

the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving device. The sound power level of the fans must not exceed 85 dBA when tested according to JIS B 8330 and rated in accordance with JIS B 8330. Connect fans to the motors either directly or indirectly with V-belt drive. Use V-belt drives designed for not less than 150 or 120 percent of the connected driving capacity. Provide variable pitch motor sheaves for 11 kW and below, and fixed pitch as defined by MLIT-M the fan shaft and the motor shaft. This is a non-adjustable speed. Select variable pitch sheaves to drive the fan at a speed which can produce the specified capacity when set at the approximate midpoint of the sheave adjustment. When fixed pitch sheaves are furnished, provide a replaceable sheave when needed to achieve system air balance. Provide motors for V-belt drives with adjustable rails or bases. Provide removable metal guards for all exposed V-belt drives, and provide speed-test openings at the center of all rotating shafts. Provide fans with personnel screens or guards on both suction and supply ends, except that the screens need not be provided, unless otherwise indicated, where ducts are connected to the fan. Provide fan and motor assemblies with vibration-isolation supports or mountings as indicated. Use vibration-isolation units that are standard products with published loading ratings. Select each fan to produce the capacity required at the fan static pressure indicated. Provide sound power level as indicated. Obtain the sound power level values according to JIS B 8330. Provide standard JIS C 9603, arrangement, rotation, and discharge as indicated.

2.9.1.1 Centrifugal Fans

Provide fully enclosed, single-width single-inlet, or double-width double-inlet centrifugal fans, with JIS B 8330 as required or indicated for the design system pressure. Provide impeller wheels that are rigidly constructed and accurately balanced both statically and dynamically. Provide forward curved or backward-inclined airfoil design fan blades in wheel sizes up to 750 mm. Provide backward-inclined airfoil design fan blades for wheels over 750 mm in diameter. Provide fan wheels over 900 mm in diameter with overhung pulleys and a bearing on each side of the wheel. Provide fan wheels 900 mm or less in diameter that have one or more extra long bearings between the fan wheel and the drive. Provide sleeve type, self-aligning and self-oiling bearings with oil reservoirs, or precision self-aligning roller or ball-type with accessible grease fittings or permanently lubricated type. Connect grease fittings to tubing for serviceability from a single accessible point. Provide L50 rated bearing life at not less than the requirement of JIS B 1518 as defined by JIS B 1521 and JIS B 1534. Provide steel, accurately finished fan shafts, with key seats and keys for impeller hubs and fan pulleys. Provide fan outlets of ample proportions, designed for the attachment of angles and bolts for attaching flexible connections. Unless otherwise indicated, provide motors that do not exceed 1800 rpm and have totally enclosed enclosures. Provide reduced-voltage-start type motor starters with weather-resistant enclosure.

2.9.1.2 In-Line Centrifugal Fans

Provide in-line fans with centrifugal backward inclined blades, stationary discharge conversion vanes, internal and external belt guards, and adjustable motor mounts. Mount fans in a welded tubular casing. Provide a fan that axially flows the air in and out. Streamline inlets with conversion vanes to eliminate turbulence and provide smooth discharge air flow. Enclose and isolate fan bearings and drive shafts from the air

stream. Provide precision, self aligning ball or roller type fan bearings that are sealed against dust and dirt and are permanently lubricated. Provide L50 rated bearing life at not less than the requirement of JIS B 1518 as defined by JIS B 1521 and JIS B 1534. Provide motors with totally enclosed enclosure.

2.9.2 Coils

Provide fin-and-tube type coils constructed of seamless copper tubes and copper fins mechanically bonded or soldered to the tubes. Provide copper tube wall thickness that is a minimum of 0.406 mm. Provide copper fins that are 0.114 mm minimum thickness. Provide casing and tube support sheets that are not lighter than 1.6 mm galvanized steel, formed to provide structural strength. When required, provide multiple tube supports to prevent tube sag. Mount coils for counterflow service. Rate and certify coils to meet the requirements of JIS B 8616.

2.9.2.1 Direct-Expansion Coils

Provide suitable direct-expansion coils for the refrigerant involved. Provide refrigerant piping that conforms to JIS H 3300 and clean, dehydrate and seal. Provide seamless copper tubing suction headers or seamless or resistance welded steel tube suction headers with copper connections. Provide supply headers that consist of a distributor which distributes the refrigerant through seamless copper tubing equally to all circuits in the coil. Provide circuited tubes to ensure minimum pressure drop and maximum heat transfer. Provide circuiting that permits refrigerant flow from inlet to suction outlet without causing oil slugging or restricting refrigerant flow in coil. Provide field installed coils which are completely dehydrated and sealed at the factory upon completion of pressure tests. Pressure test coils in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

2.9.2.2 Water Coils

Install water coils with a pitch of not less than 10 mm/m of the tube length toward the drain end. Use headers constructed of cast iron, welded steel or copper. Furnish each coil with a plugged vent and drain connection extending through the unit casing. Provide removable water coils with drain pans. Pressure test coils in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

2.9.3 Air Filters

List air filters according to requirements of JIS B 9908-1, except list high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test method under the Label Service to meet the requirements of JIS B 9908-1.

2.9.3.1 Extended Surface Pleated Panel Filters

Provide 50 mm depth, sectional, disposable type filters of the size indicated with a MERV of 8 when tested according to JIS B 9908-1. Provide initial resistance at 2.54 m/s that does not exceed 0.09 kPa. Provide UL Class 2 filters, and nonwoven cotton and synthetic fiber mat media. Attach a wire support grid bonded to the media to a moisture resistant fiberboard frame. Bond all four edges of the filter media to the inside of the frame to prevent air bypass and increase rigidity.

2.9.3.2 Cartridge Type Filters

Provide 305 mm depth, sectional, replaceable dry media type filters of the size indicated when tested according to JIS B 9908-1. Provide initial resistance at 2.54 m/s that does not exceed 0.14 kPa. Provide JIS Z 4812, and pleated microglass paper media with corrugated aluminum separators, sealed inside the filter cell to form a totally rigid filter assembly. Fluctuations in filter face velocity or turbulent airflow have no effect on filter integrity or performance. Install each filter with an extended surface pleated media panel filter as a prefilter in a factory preassembled side access housing, or a factory-made sectional frame bank, as indicated.

2.9.3.3 Replaceable Media Filters

Provide the dry-media type replaceable media filters, of the size required to suit the application. Provide filtering media that is not less than 50 mm thick fibrous glass media pad supported by a structural wire grid or woven wire mesh. Enclose pad in a holding frame of not less than 1.6 mm galvanized steel, equipped with quick-opening mechanism for changing filter media. Base the air flow capacity of the filter on net filter face velocity not exceeding 1.5 m/s, with initial resistance of 32 Pa.

2.9.3.4 Holding Frames

Fabricate frames from not lighter than 1.6 mm sheet steel with rust-inhibitor coating. Equip each holding frame with suitable filter holding devices. Provide gasketed holding frame seats. Make all joints airtight.

2.9.3.5 Filter Gauges

Provide dial type filter gauges, diaphragm actuated draft for all filter stations, including those filters which are furnished as integral parts of factory fabricated air handling units. Provide gauges that are at least 98 mm in diameter, with white dials with black figures, and graduations with a minimum range of 0.25 kPa beyond the specified final resistance for the filter bank on which each gauge is applied. Provide each gauge with a screw operated zero adjustment and two static pressure taps with integral compression fittings, two molded plastic vent valves, two 1.5 m minimum lengths of 6.35 mm diameter aluminum tubing, and all hardware and accessories for gauge mounting.

2.10 AIR HANDLING UNITS

2.10.1 Factory-Fabricated Air Handling Units

Provide single-zone draw-through type units as indicated. Units must include fans, coils, airtight insulated casing, prefilters, secondary filter sections, adjustable V-belt drives, belt guards for externally mounted motors, access sections where indicated, mixing box vibration-isolators, and appurtenances required for specified operation. Provide vibration isolators as indicated. Physical dimensions of each air handling unit must be suitable to fit space allotted to the unit with the capacity indicated. Provide air handling unit that is rated in accordance with JRA 4036.

2.10.1.1 Casings

Provide the following:

- a. Casing sections 50 mm double wall type, constructed of a minimum 1.3 mm galvanized steel, or 1.3 mm corrosion-resisting sheet steel conforming to JIS G 4305. Inner casing of double-wall units that are a minimum one mm solid galvanized steel or corrosion-resisting sheet steel conforming to JIS G 4305. Design and construct casing with an integral insulated structural galvanized steel frame such that exterior panels are non-load bearing.
- b. Individually removable exterior panels with standard tools. Removal must not affect the structural integrity of the unit. Furnish casings with access sections, according to paragraph AIR HANDLING UNITS, inspection doors, and access doors, all capable of opening a minimum of 90 degrees, as indicated.
- c. Insulated, fully gasketed, double-wall type inspection and access doors, of a minimum 1.3 mm outer and one mm inner panels made of either galvanized steel or corrosion-resisting sheet steel conforming to JIS G 4305. Provide rigid doors with heavy duty hinges and latches. Inspection doors must be a minimum 300 mm wide by 300 mm high. Access doors must be a minimum 600 mm wide, the full height of the unit casing or a minimum of 1800 mm, whichever is less.
- d. Double-wall insulated type drain pan (thickness equal to exterior casing) constructed of 1.4 mm corrosion resisting sheet steel conforming to JIS G 4305. Construct drain pans water tight, treated to prevent corrosion, and designed for positive condensate drainage. When 2 or more cooling coils are used, with one stacked above the other, condensate from the upper coils must not flow across the face of lower coils. Provide intermediate drain pans or condensate collection channels and downspouts, as required to carry condensate to the unit drain pan out of the air stream and without moisture carryover. Construct drain pan to allow for easy visual inspection, including underneath the coil without removal of the coil and to allow complete and easy physical cleaning of the pan underneath the coil without removal of the coil. Provide coils that are individually removable from the casing.
- e. Insulate single-wall casing sections handling conditioned air with not less than 25 mm thick, 24 kg/cubic meter density coated fibrous glass material having a thermal conductivity not greater than 0.033 W/m-K. Insulate double-wall casing sections handling conditioned air with not less than 50 mm of the same insulation specified for single-wall casings. Foil-faced insulation is not an acceptable substitute for use with double wall casing. Seal double wall insulation completely by inner and outer panels.
- f. Factory applied fibrous glass insulation that conforms to JIS A 9511, except that the minimum thickness and density requirements do not apply. Make air handling unit casing insulation uniform over the entire casing. Foil-faced insulation is not an acceptable substitute for use on double-wall access doors and inspections doors.
- g. Duct liner material, coating, and adhesive that conforms to fire-hazard requirements specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Protect exposed insulation edges

and joints where insulation panels are butted with a metal nosing strip or coat to meet erosion resistance requirements of JIS A 9511.

- h. A latched and hinged inspection door, in the fan and coil sections. Plus additional inspection doors, access doors and access sections where indicated.

2.10.1.2 Heating and Cooling Coils

Provide coils as specified in paragraph AIR SYSTEMS EQUIPMENT.

2.10.1.3 Air Filters

Provide air filters as specified in paragraph AIR SYSTEMS EQUIPMENT for types and thickness indicated.

2.10.1.4 Fans

Provide the following:

- a. Fans that are double-inlet, centrifugal type with each fan in a separate scroll. Dynamically balance fans and shafts prior to installation into air handling unit, then after it has been installed in the air handling unit, statically and dynamically balance the entire fan assembly. Mount fans on steel shafts, accurately ground and finished.
- b. Fan bearings that are sealed against dust and dirt and are precision self-aligning ball or roller type, with L50 rated bearing life at not less than 200,000 hours as defined by JIS B 1521 and JIS B 1534. Provide bearings that are permanently lubricated or lubricated type with lubrication fittings readily accessible at the drive side of the unit. Support bearings by structural shapes, or die formed sheet structural members, or support plates securely attached to the unit casing. Do not fasten bearings directly to the unit sheet metal casing. Furnish fans and scrolls with coating indicated.
- c. Fans that are driven by a unit-mounted, or a floor-mounted motor connected to fans by V-belt drive complete with belt guard for externally mounted motors. Furnish belt guards that are the three-sided enclosed type with solid or expanded metal face. Design belt drives for not less than a 1.3 service factor based on motor nameplate rating.
- d. Where fixed sheaves are required, the use of variable pitch sheaves is allowed during air balance, but replace them with an appropriate fixed sheave after air balance is completed. Select variable pitch sheaves to drive the fan at a speed that produces the specified capacity when set at the approximate midpoint of the sheave adjustment. Furnish motors for V-belt drives with adjustable bases, and with totally enclosed enclosures.
- e. Motor starters of reduced-voltage-start type with general-purpose enclosure. Select unit fan or fans to produce the required capacity at the fan static pressure with sound power level as indicated. Obtain the sound power level values according to JIS B 8330, JIS B 8330.

2.10.1.5 Access Sections and Filter/Mixing Boxes

Provide access sections where indicated and furnish with access doors as shown. Construct access sections and filter/mixing boxes in a manner identical to the remainder of the unit casing and equip with access doors. Design mixing boxes to minimize air stratification and to promote thorough mixing of the air streams.

2.11 TERMINAL UNITS

2.11.1 Variable Air Volume (VAV) and Dual Duct Terminal Units

- a. Provide VAV and dual duct terminal units that are the type, size, and capacity shown, mounted in the ceiling or wall cavity, plus units that are suitable for single or dual duct system applications. Provide actuators and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.
- b. Provide unit enclosures that are constructed of galvanized steel not lighter than 0.85 mm or aluminum sheet not lighter than 1.3 mm. Provide single or multiple discharge outlets as required. Units with flow limiters are not acceptable. Provide unit air volume that is factory preset and readily field adjustable without special tools. Provide reheat coils as indicated.
- c. Attach a flow chart to each unit. Base acoustic performance of the terminal units upon units tested according to JRA 4036 with the calculations prepared in accordance with JIS B 8616. Provide sound power level as indicated. Show discharge sound power for minimum and 375 Pa inlet static pressure.

2.11.1.1 Variable Volume, Single Duct Terminal Units

Provide variable volume, single duct, terminal units with a calibrated air volume sensing device, air valve or damper, actuator, and accessory relays. Provide units that control air volume to within plus or minus 5 percent of each air set point volume as determined by the thermostat with variations in inlet pressures from 200 to 1500 Pa. Provide units with an internal resistance not exceeding 100 Pa at maximum flow range. Provide external differential pressure taps separate from the control pressure taps for air flow measurement with a 0 to 250 Pa range.

2.12 FACTORY PAINTING

Factory paint new equipment, which are not of galvanized construction. Paint with a corrosion resisting paint finish according to JIS H 8641 or JIS G 3302. Clean, phosphatize and coat internal and external ferrous metal surfaces with a paint finish which has been tested according to JIS Z 2371 JIS K 5600-7-9, and JIS K 5600-5-5 or JIS K 5600-5-6. Submit evidence of satisfactory paint performance for a minimum of 125 hours for units to be installed indoors and 500 hours for units to be installed outdoors. Provide rating of failure at the scribe mark that is not less than 6, average creepage not greater than 3 mm. Provide rating of the inscribed area that is not less than 10, no failure. On units constructed of galvanized steel that have been welded, provide a final shop docket of zinc-rich protective paint on exterior surfaces of welds or welds that have burned through from the interior according to JIS H 8641.

Field paint factory painting that has been damaged prior to acceptance by

the Contracting Officer in compliance with the requirements of paragraph
FIELD PAINTING OF MECHANICAL EQUIPMENT.

2.13 SUPPLEMENTAL COMPONENTS/SERVICES

2.13.1 Chilled, Condenser, or Dual Service Water Piping

The requirements for chilled, condenser, or dual service water piping and accessories are specified in Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS

2.13.2 Refrigerant Piping

The requirements for refrigerant piping are specified in Section 23 23 00 REFRIGERANT PIPING.

2.13.3 Condensate Drain Lines

Provide and install condensate drainage for each item of equipment that generates condensate in accordance with Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS except as modified herein.

2.13.4 Insulation

The requirements for shop and field applied insulation are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.13.5 Controls

The requirements for controls are specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS and Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

3.2 INSTALLATION

- a. Install materials and equipment in accordance with the requirements of the contract drawings and approved manufacturer's installation instructions. Accomplish installation by workers skilled in this type of work. Perform installation so that there is no degradation of the designed fire ratings of walls, partitions, ceilings, and floors.
- b. No installation is permitted to block or otherwise impede access to any existing machine or system. Install all hinged doors to swing open a minimum of 120 degrees. Provide an area in front of all access doors that clears a minimum of 910 mm. In front of all access doors to electrical circuits, clear the area the minimum distance to energized circuits JIS HB 71 (Electrical-Safety Related work practices) and an additional 910 mm.
- c. Except as otherwise indicated, install emergency switches and alarms in conspicuous locations. Mount all indicators, to include gauges,

meters, and alarms in order to be easily visible by people in the area.

3.2.1 Condensate Drain Lines

Provide water seals in the condensate drain from all units. Provide a depth of each seal of 50 mm plus 0.1 mm for each Pa, of the total static pressure rating of the unit to which the drain is connected. Provide water seals that are constructed of 2 tees and an appropriate U-bend with the open end of each tee plugged. Provide pipe cap or plug cleanouts where indicated. Connect drains indicated to connect to the sanitary waste system using an indirect waste fitting. Insulate air conditioner drain lines as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.2.2 Equipment and Installation

Provide frames and supports for tanks, compressors, pumps, valves, air handling units, fans, coils, dampers, and other similar items requiring supports. Floor mount or ceiling hang air handling units as indicated. Anchor and fasten as detailed. Set floor-mounted equipment on not less than 150 mm concrete pads or curbs doweled in place unless otherwise indicated. Make concrete foundations heavy enough to minimize the intensity of the vibrations transmitted to the piping, duct work and the surrounding structure, as recommended in writing by the equipment manufacturer. In lieu of a concrete pad foundation, build a concrete pedestal block with isolators placed between the pedestal block and the floor. Make the concrete foundation or concrete pedestal block a mass not less than three times the weight of the components to be supported. Provide the lines connected to the pump mounted on pedestal blocks with flexible connectors. Submit foundation drawings as specified in paragraph DETAIL DRAWINGS. Provide concrete for foundations as specified in Structural drawings.

3.2.3 Access Panels

Install access panels for concealed valves, vents, controls, dampers, and items requiring inspection or maintenance of sufficient size, and locate them so that the concealed items are easily serviced and maintained or completely removed and replaced. Provide access panels as specified in Structural Drawings.

3.2.4 Flexible Duct

Install pre-insulated flexible duct in accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Provide hangers, when required to suspend the duct, of the type recommended by the duct manufacturer and set at the intervals recommended.

3.2.5 Metal Ductwork

Install according to MLIT-M unless otherwise indicated. Install duct supports for sheet metal ductwork according to MLIT-M, unless otherwise specified. Do not use friction beam clamps indicated in MLIT-M. Anchor risers on high velocity ducts in the center of the vertical run to allow ends of riser to move due to thermal expansion. Erect supports on the risers that allow free vertical movement of the duct. Attach supports only to structural framing members and concrete slabs. Do not anchor supports to metal decking unless a means is provided and approved for

preventing the anchor from puncturing the metal decking. Where supports are required between structural framing members, provide suitable intermediate metal framing. Where C-clamps are used, provide retainer clips.

3.2.6 Dust Control

To prevent the accumulation of dust, debris and foreign material during construction, perform temporary dust control protection. Protect the distribution system (supply and return) with temporary seal-offs at all inlets and outlets at the end of each day's work. Keep temporary protection in place until system is ready for startup.

3.2.7 Insulation

Provide thickness and application of insulation materials for ductwork, piping, and equipment according to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Externally insulate outdoor air intake ducts and plenums up to the point where the outdoor air reaches the conditioning unit or up to the point where the outdoor air mixes with the return air stream.

3.2.8 Duct Test Holes

Provide holes with closures or threaded holes with plugs in ducts and plenums as indicated or where necessary for the use of pitot tube in balancing the air system. Plug insulated duct at the duct surface, patched over with insulation and then marked to indicate location of test hole if needed for future use.

3.2.9 Power Transmission Components Adjustment

Test V-belts and sheaves for proper alignment and tension prior to operation and after 72 hours of operation at final speed. Uniformly load belts on drive side to prevent bouncing. Make alignment of direct driven couplings to within 50 percent of manufacturer's maximum allowable range of misalignment.

3.3 EQUIPMENT PADS

Provide equipment pads to the dimensions shown or, if not shown, to conform to the shape of each piece of equipment served with a minimum 75 mm margin around the equipment and supports. Allow equipment bases and foundations, when constructed of concrete or grout, to cure a minimum of 28 calendar days before being loaded.

3.4 CUTTING AND PATCHING

Install work in such a manner and at such time that a minimum of cutting and patching of the building structure is required. Make holes in exposed locations, in or through existing floors, by drilling and smooth by sanding. Use of a jackhammer is permitted only where specifically approved. Make holes through masonry walls to accommodate sleeves with an iron pipe masonry core saw.

3.5 CLEANING

Thoroughly clean surfaces of piping and equipment that have become covered with dirt, plaster, or other material during handling and construction before such surfaces are prepared for final finish painting or are

enclosed within the building structure. Before final acceptance, clean mechanical equipment, including piping, ducting, and fixtures, and free from dirt, grease, and finger marks. When the work area is in an occupied space such as office, laboratory or warehouse protect all furniture and equipment from dirt and debris. Incorporate housekeeping for field construction work which leaves all furniture and equipment in the affected area free of construction generated dust and debris; and, all floor surfaces vacuum-swept clean.

3.6 PENETRATIONS

Provide sleeves and prepared openings for duct mains, branches, and other penetrating items, and install during the construction of the surface to be penetrated. Cut sleeves flush with each surface. Place sleeves for round duct 380 mm and smaller. Build framed, prepared openings for round duct larger than 380 mm and square, rectangular or oval ducts. Sleeves and framed openings are also required where grilles, registers, and diffusers are installed at the openings. Provide 25 mm clearance between penetrating and penetrated surfaces except at grilles, registers, and diffusers. Pack spaces between sleeve or opening and duct or duct insulation with mineral fiber conforming with JIS A 9504, Type 1, Class B-2.

3.6.1 Sleeves

Fabricate sleeves, except as otherwise specified or indicated, from 1 mm thick mill galvanized sheet metal. Where sleeves are installed in bearing walls or partitions, provide black steel pipe conforming with JIS G 3452, Schedule 20.

3.6.2 Framed Prepared Openings

Fabricate framed prepared openings from 1 mm galvanized steel, unless otherwise indicated.

3.6.3 Insulation

Provide duct insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS continuous through sleeves and prepared openings except firewall penetrations. Terminate duct insulation at fire dampers and flexible connections. For duct handling air at or below 16 degrees C, provide insulation continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air.

3.6.4 Closure Collars

Provide closure collars of a minimum 100 mm wide, unless otherwise indicated, for exposed ducts and items on each side of penetrated surface, except where equipment is installed. Install collar tight against the surface and fit snugly around the duct or insulation. Grind sharp edges smooth to prevent damage to penetrating surface. Fabricate collars for round ducts 380 mm in diameter or less from 1 mm galvanized steel. Fabricate collars for square and rectangular ducts, or round ducts with minimum dimension over 380 mm from 1.40 mm galvanized steel. Fabricate collars for square and rectangular ducts with a maximum side of 380 mm or less from 1 mm galvanized steel. Install collars with fasteners a maximum of 150 mm on center. Attach to collars a minimum of 4 fasteners where the opening is 300 mm in diameter or less, and a minimum of 8 fasteners where the opening is 500 mm in diameter or less.

3.6.5 Firestopping

Where ducts pass through fire-rated walls, fire partitions, and fire rated chase walls, seal the penetration with fire stopping materials.

3.7 FIELD PAINTING OF MECHANICAL EQUIPMENT

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal on metal surfaces subject to temperatures in excess of 50 degrees C. Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Provide aluminum or light gray finish coat.

3.7.1 Temperatures less than 50 degrees C

Immediately after cleaning, apply one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat to metal surfaces subject to temperatures less than 50 degrees C.

3.7.2 Finish Painting

The requirements for finish painting of items only primed at the factory, and surfaces not specifically noted otherwise, are specified in Section 09 90 00 PAINTS AND COATINGS.

3.7.3 Color Coding Scheme for Locating Hidden Utility Components

Use scheme in buildings having suspended grid ceilings. Provide color coding scheme that identifies points of access for maintenance and operation of components and equipment that are not visible from the finished space and are accessible from the ceiling grid, consisting of a color code board and colored metal disks. Make each colored metal disk approximately 13 mm diameter and secure to removable ceiling panels with fasteners. Insert each fastener into the ceiling panel so as to be concealed from view. Provide fasteners that are manually removable without the use of tools and that do not separate from the ceiling panels when the panels are dropped from ceiling height. Make installation of colored metal disks follow completion of the finished surface on which the disks are to be fastened. Provide color code board that is approximately 1 m wide, 750 mm high, and 13 mm thick. Make the board of wood fiberboard and frame under glass or 1.6 mm transparent plastic cover. Make the color code symbols approximately 19 mm in diameter and the related lettering in 13 mm high capital letters. Mount the color code board in the mechanical or equipment room.

3.8 IDENTIFICATION SYSTEMS

Provide identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and item number on all valves and dampers. Provide tags that are 35 mm minimum diameter with stamped or engraved markings. Make indentations black for reading clarity. Attach tags to valves with No. 12 AWG 2 mm diameter

corrosion-resistant steel wire, copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

3.9 DUCTWORK LEAK TEST

Perform ductwork leak test for the entire air distribution and exhaust system, including fans, coils, filters, etc. Provide test procedure, apparatus, and report that conform to JIS Z 2330. The maximum allowable leakage rate is as calculated using ductwork surface area. Complete ductwork leak test with satisfactory results prior to applying insulation to ductwork exterior or concealing ductwork.

3.10 DAMPER ACCEPTANCE TEST

Submit the proposed schedule, at least 2 weeks prior to the start of test. Operate all fire dampers and smoke dampers under normal operating conditions, prior to the occupancy of a building to determine that they function properly. Test each fire damper equipped with fusible link by having the fusible link cut in place. Test dynamic fire dampers with the air handling and distribution system running. Reset all fire dampers with the fusible links replaced after acceptance testing. To ensure optimum operation and performance, install the damper so it is square and free from racking.

3.11 TESTING, ADJUSTING, AND BALANCING

The requirements for testing, adjusting, and balancing are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC. Begin testing, adjusting, and balancing only when the air supply and distribution, including controls, has been completed, with the exception of performance tests.

3.12 PERFORMANCE TESTS

After testing, adjusting, and balancing is complete as specified, test each system as a whole to see that all items perform as integral parts of the system and temperatures and conditions are evenly controlled throughout the building. Record the testing during the applicable season. Make corrections and adjustments as necessary to produce the conditions indicated or specified. Conduct capacity tests and general operating tests by an experienced engineer. Provide tests that cover a period of not less than 2 days for each system and demonstrate that the entire system is functioning according to the specifications. Make coincidental chart recordings at points indicated on the drawings for the duration of the time period and record the temperature at space thermostats or space sensors, the humidity at space humidistats or space sensors and the ambient temperature and humidity in a shaded and weather protected area.

Submit test reports for the ductwork leak test, and performance tests in booklet form, upon completion of testing. Document phases of tests performed including initial test summary, repairs/adjustments made, and final test results in the reports.

3.13 CLEANING AND ADJUSTING

Provide a temporary bypass for water coils to prevent flushing water from passing through coils. Inside of air terminal units, thoroughly clean ducts, plenums, and casing of debris and blow free of small particles of

rubbish and dust and then vacuum clean before installing outlet faces. Wipe equipment clean, with no traces of oil, dust, dirt, or paint spots. Provide temporary filters prior to startup of all fans that are operated during construction, and provide new filters after all construction dirt has been removed from the building, and the ducts, plenums, casings, and other items specified have been vacuum cleaned. Perform and document that proper "Indoor Air Quality During Construction" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. Maintain system in this clean condition until final acceptance. Properly lubricate bearings with oil or grease as recommended by the manufacturer. Tighten belts to proper tension. Adjust control valves and other miscellaneous equipment requiring adjustment to setting indicated or directed. Adjust fans to the speed indicated by the manufacturer to meet specified conditions. Maintain all equipment installed under the contract until close out documentation is received, the project is completed and the building has been documented as beneficially occupied.

3.14 OPERATION AND MAINTENANCE

3.14.1 Operation and Maintenance Manuals

Submit six manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

3.14.2 Operation And Maintenance Training

Conduct a training course for the members of the operating staff as designated by the Contracting Officer. Make the training period consist of a total of 8 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course.

-- End of Section --

SECTION 23 64 10

WATER CHILLERS, VAPOR COMPRESSION TYPE
11/16, CHG 2: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 480	(2007) Refrigerant-Cooled Liquid Coolers, Remote Type
AHRI 550/590 I-P	(2020) Performance Rating Of Water-Chilling and Heat Pump Water-Heating Packages Using the Vapor Compression Cycle
AHRI 575	(2017) Method of Measuring Machinery Sound Within an Equipment Space
ANSI/AHRI 460	(2005) Performance Rating of Remote Mechanical-Draft Air-Cooled Refrigerant Condensers

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 9	(2015) Load Ratings and Fatigue Life for Ball Bearings
ABMA 11	(2014) Load Ratings and Fatigue Life for Roller Bearings

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ANSI/ASHRAE 15 & 34	(2022) ANSI/ASHRAE Standard 15-Safety Standard for Refrigeration Systems and ANSI/ASHRAE Standard 34-Designation and Safety Classification of Refrigerants
ASHRAE 90.1 - SI	(2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1
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AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1	(2021) Safety in Welding and Cutting and Allied Processes
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ASTM INTERNATIONAL (ASTM)

ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D520	(2000; R 2011) Zinc Dust Pigment
ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM F104	(2011; R 2020) Standard Classification System for Nonmetallic Gasket Materials

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1	(2021) Motors and Generators
NEMA MG 11	(1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 82	Protection of Stratospheric Ozone
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Water Chiller; G

Posted Instructions

Verification of Dimensions

Factory Tests

System Performance Tests

Demonstrations

Refrigerant

Water Chiller - Field Acceptance Test Plan

SD-06 Test Reports

Field Acceptance Testing

Water Chiller - Field Acceptance Test Report

Factory Tests

System Performance Tests

SD-07 Certificates

Refrigeration System; G

Ozone Depleting Substances Technician Certification

SD-08 Manufacturer's Instructions

Water Chiller - Installation Instructions; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

1.3 CERTIFICATIONS

1.3.1 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4 SAFETY REQUIREMENTS

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.

1.5 DELIVERY, STORAGE, AND HANDLING

Stored items must be protected from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation will be the Contractor's responsibility. Any materials found to be damaged must be replaced at the Contractor's expense. During installation, piping and similar openings must be capped to keep out dirt and other foreign matter.

1.6 PROJECT REQUIREMENTS

1.6.1 Verification of Dimensions

The Contractor must become familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

Materials and equipment will be standard Commercial cataloged products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. These products must have a two year record of satisfactory field service prior to bid opening. The two year record of service must include applications of equipment and materials under similar circumstances and of similar size. Products having less than a two year record of satisfactory field service will be acceptable if a certified record of satisfactory field service for not less than 6000 hours can be shown. The 6000 hour service record must not include any manufacturer's prototype or factory testing. Satisfactory field service must have been completed by a product that has been, and presently is being sold or offered for sale on the commercial market through the following copyrighted means: advertisements, manufacturer's catalogs, or brochures.

2.2 MANUFACTURER'S STANDARD NAMEPLATES

Major equipment including chillers, compressors, compressor drivers, condensers, water coolers, receivers, refrigerant leak detectors, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life. Plates must be fixed in prominent locations with nonferrous screws or bolts.

2.3 ELECTRICAL WORK

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown, must be provided. For packaged equipment, the manufacturer must provide controllers including the required monitors and timed restart.
- b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with NEMA MG 11.
- c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with NEMA MG 1.
- d. Provide motors in accordance with NEMA MG 1 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Motors must be rated for continuous duty with the enclosure specified. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque must be capable of

accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Motor bearings must be fitted with grease supply fittings and grease relief to outside of the enclosure. Motor enclosure type may be either TEAO or TEFC.

- e. Where two-speed motors are indicated, variable-speed controllers may be provided to accomplish the same function. Provide variable frequency drives for motors as specified in Section 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS.
- f. Provide inverter duty premium efficiency motors for use with variable frequency drives.

2.4 SPLIT-SYSTEM WATER CHILLER, VAPOR COMPRESSION TYPE

Total chiller system must be certified for performance per AHRI 550/590 I-P. If chiller is not in scope of AHRI 550/590 I-P then chiller must be rated in accordance with AHRI 550/590 I-P. Individual chiller components must be constructed and rated in accordance with the applicable AHRI standards. Chiller system must conform to ANSI/ASHRAE 15 & 34. The chiller must be ASHRAE 90.1 - SI compliant and meet 10 CFR Part 433, 434 and 435 efficiency performance standards for federal construction. The manufacturer must provide certification of compliance. Chiller must be assembled, leak-tested, charged (refrigerant and oil), and adjusted at the job site in strict accordance with manufacturer's recommendations. Unit components delivered separately must be sealed and charged with a nitrogen holding charge. Unit assembly must be completed in strict accordance with manufacturer's recommendations. Chiller must operate within capacity range and speed recommended by the manufacturer. Parts weighing 23 kg or more which must be removed for inspection, cleaning, or repair, must have lifting eyes or lugs. Chiller must include all customary auxiliaries deemed necessary by the manufacturer for safe, controlled, automatic operation of the equipment. Chiller's water cooler must be provided with standard water boxes with flanged connections. Chillers must operate at partial load conditions without increased vibration over normal vibration at full load, and must be capable of continuous operation down to minimum capacity. As a minimum, chiller must include the following components as defined in paragraph CHILLER COMPONENTS.

- a. Refrigerant and oil
- b. Structural base
- c. Chiller refrigerant circuit
- d. Controls package
- e. Tools

2.4.1 Compressor-Chiller Unit

As a minimum, the compressor-chiller unit must include the following components as defined in paragraph CHILLER COMPONENTS.

- a. Scroll, reciprocating, or rotary screw compressor
- b. Compressor driver, electric motor

- c. Compressor driver connection
- d. Water cooler (evaporator)

2.4.2 Condensing Unit

As a minimum, the condensing unit must include the following components as defined in paragraph CHILLER COMPONENTS.

- a. Scroll, reciprocating, or rotary screw compressor
- b. Compressor driver, electric motor
- c. Compressor driver connection
- d. Air or water cooled condenser

2.4.3 Remote Water Cooler (Evaporator)

2.4.3.1 Shell and Tube Type

Cooler must be constructed and rated in accordance with AHRI 480. Cooler must be of the shell-and-coil or shell-and-tube type design. Cooler's refrigerant side must be designed and factory pressure tested to comply with ANSI/ASHRAE 15 & 34. Cooler's water side must be designed and factory pressure tested for not less than 1,000 kPa. Cooler shell must be constructed of seamless or welded steel. Tubes must be seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable. Tubes must be installed into carbon mild steel tube sheets by rolling. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Cooler must be skid-mounted. Refrigerant circuit must be complete with liquid solenoid valve and expansion device capable of modulating to the minimum step of capacity unloading. For the water side of water cooler, performance must be based on a fluid velocity not less than 0.91 m/s and not more than 3.7 m/s and a fouling factor per AHRI 550/590 I-P.

2.4.4 Remote Air-Cooled Condenser

Condenser must be a factory-fabricated and assembled unit, consisting of coils, fans, and condenser fan motors. Condenser must be rated in accordance with ANSI/AHRI 460. Unless the condenser coil is completely protected through inherent design, louvered panel coil guards must be provided by the manufacturer to prevent physical damage to the coil. Manufacturer must certify that the condenser and associated equipment are designed for the submitted condensing temperature. For design conditions, if matched combination catalog ratings matching remote condensers to compressors are not available, the Contractor must furnish a crossplotting of the gross heat rejection of the condenser against the gross heat rejection of the compressor, for the design conditions to show the compatibility of the equipment furnished.

2.4.4.1 Condenser Casing

Condenser casing must be aluminum not less than 1.016 mm or hot-dip galvanized steel not lighter than 18 gauge 1.311 mm.

2.4.4.2 Coil

Condenser coil must be of the extended-surface fin-and-tube type and must be constructed of seamless copper tubes with compatible copper fins. Fins must be soldered or mechanically bonded to the tubes and installed in a metal casing. Coils must be circuited and sized for a minimum of 3 degrees C subcooling and full pumpdown capacity. Coil must be factory leak and pressure tested after assembly in accordance with ANSI/ASHRAE 15 & 34.

Coil must be entirely coated in accordance with the requirements of paragraph COIL CORROSION PROTECTION.

2.4.4.3 Fans

Provide centrifugal or propeller type fans as best suited for the application. Fans must be direct driven. Fans must be statically and dynamically balanced.

2.4.4.4 Condenser Sizing

Size condensers for full capacity at 16.67 degrees C temperature difference between entering outside air and condensing refrigerant. Subcooling must not be considered in determining compressor and condenser capacities. For design conditions, submit a cross-plot of net refrigeration effect of compressor to establish net refrigeration effect and compatibility of equipment furnished.

2.4.4.5 High Ambient Unloading

Provide unloading capability to allow operation in high ambient conditions above design conditions.

2.5 CHILLER COMPONENTS

2.5.1 Refrigerant and Oil

Refrigerants must be one of the fluorocarbon gases. Refrigerants must have number designations and safety classifications in accordance with ANSI/ASHRAE 15 & 34. CFC-based refrigerants are prohibited. Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0, with the exception of R-123. Provide SDS sheets for all refrigerants.

2.5.2 Structural Base

Chiller and individual chiller components must be provided with a factory-mounted structural steel base (welded or bolted) or support legs. Chiller and individual chiller components must be isolated from the building structure by means of molded neoprene isolation pads.

2.5.3 Chiller Refrigerant Circuit

Chiller refrigerant circuit must be completely piped and factory leak tested in accordance with ANSI/ASHRAE 15 & 34. Circuit must include as a minimum a combination filter and drier, combination sight glass and moisture indicator, an electronic or thermostatic expansion valve with external equalizer or float valve, charging ports, compressor service valves for field-serviceable compressors, and superheat adjustment.

2.5.4 Controls Package

Provide chillers with a complete factory-mounted, microprocessor based operating and safety control system. Controls package must contain as a minimum a digital display, an on-auto-off switch, variable frequency motor controller, disconnect switches, power wiring, and control wiring. Controls package must provide operating controls, monitoring capabilities, programmable setpoints, safety controls, and UMCS interfaces as defined below.

2.5.4.1 Operating Controls

Chiller must be provided with the following adjustable operating controls as a minimum.

- a. Leaving chilled water temperature control
- b. Adjustable timer or automated controls to prevent a compressor from short cycling
- c. Automatic lead/lag controls (adjustable) for multi-compressor units
- d. Load limiting
- e. System capacity control to adjust the unit capacity in accordance with the system load and the programmable setpoints. Controls must automatically re-cycle the chiller on power interruption.
- f. Startup and head pressure controls to allow system operation at all ambient temperatures down to 5 degrees C.
- g. Fan sequencing for air-cooled condenser

2.5.4.2 Monitoring Capabilities

During normal operations, the control system must be capable of monitoring and displaying the following operating parameters. Access and operation of display must not require opening or removing any panels or doors.

- a. Entering and leaving chilled water temperatures
- b. Entering and leaving chilled water pressure. Chilled water flow
- c. Self diagnostic
- d. Operation status
- e. Operating hours
- f. Number of starts
- g. Compressor status (on or off)
- h. Compressor load (percent)
- i. Refrigerant discharge and suction pressures
- j. Magnetic bearing levitation status (if applicable)

- k. Magnetic bearing temperatures (if applicable)
- l. Oil pressure
- m. Number of purge cycles over the last 7 days

2.5.4.3 Configurable Setpoints

The control system must be capable of being configured directly at the unit's interface panel. No parameters may be capable of being changed without first entering a security access code. The programmable setpoints must include the following as a minimum:

- a. Leaving Chilled Water Temperature
- b. Time Clock/Calendar Date

2.5.4.4 Safety Controls with Manual Reset

Chiller must be provided with the following safety controls which automatically shutdown the chiller and which require manual reset.

- a. Low chilled water temperature protection
- b. High condenser refrigerant discharge pressure protection
- c. Low evaporator pressure protection
- d. Chilled water flow detection
- e. High motor winding temperature protection
- f. Low oil flow protection if applicable
- g. Magnetic bearing controller (MBC), Internal fault (if applicable)
- h. MBC, High bearing temperature (if applicable)
- i. MBC, Communication fault (if applicable)
- j. MBC, Power supply fault (if applicable)
- k. Motor current overload and phase loss protection

2.5.4.5 Safety Controls with Automatic Reset

Chiller must be provided with the following safety controls which automatically shutdown the chiller and which provide automatic reset.

- a. Over/under voltage protection
- b. Chilled water flow interlock
- c. MBC, Vibration (if applicable)
- d. MBC, No levitation (if applicable)

2.5.4.6 Remote Alarm

During the initiation of a safety shutdown, a chiller's control system must be capable of activating a remote alarm bell. In coordination with the chiller, the Contractor must provide an alarm circuit (including transformer if applicable) and a minimum 100 mm diameter alarm bell. Alarm circuit must activate bell in the event of machine shutdown due to the chiller's monitoring of safety controls. The alarm bell must not sound for a chiller that uses low-pressure cutout as an operating control.

2.5.4.7 Utility Monitoring and Control System Interface

Provide a Utility Monitoring and Control System (UMCS) interface meeting the requirements of Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. The interface must provide all system operating conditions, capacity controls, and safety shutdown conditions as network points. In addition, the following points must be overridable via the network interface:

- a. Unit Start/Stop
- b. Leaving Chilled Water Temperature Setpoint

2.5.5 Compressor(s)

2.5.5.1 Rotary Screw Compressor(s)

Compressors must operate stably for indefinite time periods to at least 25 percent capacity reduction without gas bypass external to the compressor. Provision must be made to insure proper lubrication of bearings and shaft seals on shutdown with or without electric power supply. Rotary screw compressors must include:

- a. An open or hermetic, positive displacement, oil-injected design directly driven by the compressor driver. Allow access to internal compressor components for repairs, inspection, and replacement of parts.
- b. Rotors must be solid steel, possessing sufficient rigidity for proper operation.
- c. A maximum rotor operating speed no greater than 3600 RPM. Provide cast iron rotor housing.
- d. Casings of cast iron, precision machined for minimal clearance about periphery of rotors with minimal clearance at rotor tops and rotor ends.
- e. A lubrication system of the forced-feed type that provides oil at the proper pressure to all parts requiring lubrication.
- f. Bearing housing must be conservatively loaded and rated for an L(10) life of not less than 200,000 hours. Shaft main bearings of the sleeve type with heavy duty bushings or rolling element type in accordance with ABMA 9 or ABMA 11.
- g. A differential oil pressure or flow cutout to allow the compressor to operate only when the required oil pressure or flow is provided to the

bearings.

- h. Use a Variable Frequency Drive (VFD) to modulate capacity modulation from 100 percent to 15 percent.
- i. An oil separator and oil return system to remove oil entrained in the refrigerant gas and automatically return the oil to the compressor.
- j. Crankcase oil heaters must be provided.

2.5.6 Compressor Driver, Electric Motor

Components such as motors, variable speed drives and wiring must be in accordance with paragraph ELECTRICAL WORK. Variable frequency drive must be unit mounted, remote mounted as indicated with variable frequency drive type, wiring, and accessories coordinated with the chiller manufacturer.

2.5.7 Compressor Driver Connections

Each compressor must be driven by a V-belt drive or direct connected through a flexible coupling, except that flexible coupling is not required on hermetic units. V-belt drives must be designed for not less than 150 percent of the driving motor capacity. Flexible couplings must be of the type that does not require lubrication.

2.5.8 Water Cooler (Evaporator)

Cooler must be of the shell-and-coil or shell-and-tube type design. Cooler shell must be constructed of seamless or welded steel. Coil bundles must be totally removable and arranged to drain completely. Tubes must be seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable. Tubes must be installed into carbon mild steel tube sheets by rolling. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Performance must be based on a water velocity not less than 0.91 m/s nor more than 3.7 m/s and a fouling factor per AHRI 550/590 I-P.

Brazed plate heat exchanger must be constructed of 304 or 316 stainless steel, designed to a refrigerant-side working pressure of 3,000 kPa and a waterside working pressure of 1,000 kPa. Evaporator must be factory tested at 1.1 times maximum allowable refrigerant side working pressure and 1.5 times maximum allowable water side working pressure. Provide cooler with factory-installed flow switches. All water connections must use either flanged or grooved-pipe connections. Factory insulate all cold surfaces.

2.5.9 Air-Cooled Condenser Coil

Condenser coil must be of the extended-surface fin-and-tube type and must be constructed of seamless copper tubes with compatible copper fins. Fins must be soldered or mechanically bonded to the tubes and installed in a metal casing. Coils must be circuited and sized for a minimum of 3 degrees C subcooling and full pumpdown capacity. Coil must be factory leak and pressure tested after assembly in accordance with ANSI/ASHRAE 15 & 34.

Coil must be entirely coated in accordance with the requirements of paragraph COIL CORROSION PROTECTION.

2.5.10 Chiller Purge System

Chillers which operate at pressures below atmospheric pressure must be provided with a purge system. Purge system must automatically remove air, water vapor, and non-condensable gases from the chiller's refrigerant while keeping refrigerant emissions below requirements of ASHRAE Std 147. Purge units must be certified per AHRI 580. Purge system must condense, separate, and return all refrigerant back to the chiller. An oil separator must be provided with the purge system if required by the manufacturer. Purge system must not discharge to occupied areas, or create a potential hazard to personnel. Purge system must include a purge pressure gauge, number of starts counter, and an elapsed time meter. Purge system must include lights or an alarm which indicate excessive purge or an abnormal air leakage into chiller.

2.5.11 Tools

One complete set of special tools, as recommended by the manufacturer for field maintenance of the system, must be provided. Tools must be mounted on a tool board in the equipment room or contained in a toolbox as directed by the Contracting Officer.

2.6 ACCESSORIES

2.6.1 Refrigerant Leak Detector

Detector must be the continuously-operating, halogen-specific type. Detector must be appropriate for the refrigerant in use. Detector must be specifically designed for area monitoring and must include a single sampling point installed where indicated. Detector design and construction must be compatible with the temperature, humidity, barometric pressure and voltage fluctuations of the operating area. Detector must have an adjustable sensitivity such that it can detect refrigerant at or above 3 parts per million (ppm). Detector must be supplied factory-calibrated for the appropriate refrigerant(s). Detector must be provided with an alarm relay output which energizes when the detector detects a refrigerant level at or above the TLV-TWA (or toxicity measurement consistent therewith) for the refrigerant(s) in use. The detector's relay must be capable of initiating corresponding alarms and ventilation systems as indicated on the drawings. Detector must be provided with a failure relay output that energizes when the monitor detects a fault in its operation. Detector must be compatible with the facility's Building Control Network (BCN). The BCN must be capable of generating an electronic log of the refrigerant level in the operating area, monitoring for detector malfunctions, and monitoring for any refrigerant alarm conditions.

2.6.2 Refrigerant Relief Valve/Rupture Disc Assembly

The assembly must be a combination pressure relief valve and rupture disc designed for refrigerant usage. The assembly must be in accordance with ASME BPVC SEC VIII D1 and ANSI/ASHRAE 15 & 34. The assembly must be provided with a pressure gauge assembly which will provide local indication if a rupture disc is broken. Rupture disc must be the non-fragmenting type.

2.6.3 Refrigerant Signs

Refrigerant signs must be a medium-weight aluminum type with a baked enamel finish. Signs must be suitable for indoor or outdoor service. Signs must have a white background with red letters not less than 13 mm in height.

2.6.3.1 Installation Identification

Each new refrigerating system must be provided with a refrigerant sign which indicates the following as a minimum:

- a. Contractor's name.
- b. Refrigerant number and amount of refrigerant.
- c. The lubricant identity and amount.
- d. Field test pressure applied.

2.6.3.2 Controls and Piping Identification

Refrigerant systems containing more than 50 kg of refrigerant must be provided with refrigerant signs which designate the following as a minimum:

- a. Valves or switches for controlling the refrigerant flow and the refrigerant compressor(s).
- b. Pressure limiting device(s).

2.6.4 Automatic Tube Brush Cleaning System

2.6.4.1 Brush and Basket Sets

One brush and basket set (one brush and two baskets) must be furnished for each condenser tube. Brushes must be made of nylon bristles, with titanium wire. Baskets must be polypropylene.

2.6.4.2 Flow-Diverter Valve

Each system must be equipped with one flow-diverter valve specifically designed for the automatic tube brush cleaning system and have parallel flow connections. The flow-diverter valve must be designed for a working pressure of 1,000 kPa. End connections must be flanged. Each valve must be provided with an electrically operated air solenoid valve and position indicator.

2.6.4.3 Control Panel

The control panel must provide signals to the diverter valve at a preset time interval to reverse water flow to drive the tube brushes down the tubes and then signal the valve to reverse the water flow to drive the brushes back down the tubes to their original position. The controller must have the following features as a minimum:

- a. Timer to initiate the on-load cleaning cycle.
- b. Manual override of preset cleaning cycle.

- c. Power-on indicator.
- d. Diverter-position indicator.
- e. Cleaning-cycle-time adjustment
- f. Flow-switch bypass.

2.6.5 Gaskets

Gaskets must conform to ASTM F104 - classification for compressed sheet with nitrile binder and acrylic fibers for maximum 371 degrees C service.

2.6.6 Bolts and Nuts

Bolts and nuts, except as required for piping applications, must be in accordance with ASTM A307. The bolt head must be marked to identify the manufacturer and the standard with which the bolt complies in accordance with ASTM A307.

2.7 FABRICATION

2.7.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 10,000 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type I.

2.7.2 Factory Applied Insulation

Chiller must be provided with factory installed insulation on surfaces subject to sweating including the water cooler, suction line piping, economizer, and cooling lines. Insulation on heads of coolers may be field applied, however it must be installed to provide easy removal and replacement of heads without damage to the insulation. Where motors are the gas-cooled type, factory installed insulation must be provided on the cold-gas inlet connection to the motor per manufacturer's standard practice. Factory insulated items installed outdoors are not required to be fire-rated. As a minimum, factory insulated items installed indoors must have a flame spread index no higher than 75 and a smoke developed index no higher than 150. Factory insulated items (no jacket) installed indoors and which are located in air plenums, in ceiling spaces, and in attic spaces must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Flame spread and smoke developed indexes must be determined by ASTM E84. Insulation must be tested in the same density and installed thickness as the material to be used in the actual construction. Material supplied by a manufacturer with a jacket must be tested as a composite material. Jackets, facings, and adhesives must have a flame spread index no higher than 25 and a smoke developed index no higher than 50 when tested in accordance with ASTM E84.

2.7.3 Coil Corrosion Protection

Provide coil with a uniformly applied epoxy electrodeposition type coating to all coil surface areas without material bridging between fins. Submit product data on the type coating selected, the coating thickness, the application process used, the estimated heat transfer loss of the coil, and verification of conformance with the salt spray test requirement. Coating must be applied at either the coil or coating manufacturer's factory. Coating process must ensure complete coil encapsulation. Coating must be capable of withstanding a minimum 10,000 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution.

2.8 FACTORY TESTS

2.8.1 Chiller Performance Test

The Contractor and proposed chiller manufacturer must be responsible for performing the chiller factory test to validate the specified full load capacity, full load EER, and IPLV in accordance with AHRI 550/590 I-P except as indicated. The Contractor and chiller manufacturer must provide to the Government a certified chiller factory test report in accordance with AHRI 550/590 I-P to confirm that the chiller performs as specified. Tests must be conducted in an AHRI certified test facility in conformance with AHRI 550/590 I-P procedures and tolerances, except as indicated. At a minimum, chiller capacity must be validated to meet the scheduled requirements indicated on the drawings. Tolerance or deviation must be in strict accordance with AHRI 550/590 I-P. Stable operation at minimum load of 10 percent of total capacity must be demonstrated during the factory test.

2.8.1.1 Temperature Adjustments

Temperature adjustments must adhere to AHRI 550/590 I-P to adjust from the design fouling factor to the clean tube condition. Test temperature adjustments must be verified prior to testing by the manufacturer. There must be no exceptions to conducting the test with clean tubes with the temperature adjustments per AHRI 550/590 I-P. The manufacturer must clean the tubes prior to testing to obtain a test fouling factor of 0.0000.

2.8.1.2 Test Instrumentation

The factory test instrumentation must be per AHRI 550/590 I-P and the calibration must be traceable to the National Institute of Standards and Technology.

2.8.1.3 Equipment Adjustments

If the equipment fails to perform within allowable tolerances, the manufacturer must be allowed to make necessary revisions to his equipment and retest as required. The manufacturer must assume all expenses incurred by the Government to witness the retest.

2.8.2 Chiller Sound Test

Chillers must be sound tested at the factory prior to shipment to confirm the sound pressure level specified herein. Tests and data must be conducted and measured in strict accordance with AHRI 575 at the full load system operating conditions. The chiller sound pressure level, in

decibels (dB), with a reference pressure of 20 micropascals, must not exceed 90 dB, A weighted. Ratings must be in accordance with AHRI 575. No reduction of entering condenser water temperature or raising of leaving chilled water temperature will be allowed. A minimum of 75 percent of the sound data points must be taken along the length of the machine, and established as the minimum percentage of total possible points used to determine sound levels. In the event that the chiller does not meet the dBA sound pressure level, the manufacturer must, at his expense, provide sufficient attenuation to the machine to meet the specified value. This attenuation must be applied in such a manner that it does not hinder the operation or routine maintenance procedures of the chiller. The attenuation material, adhesives, coatings, and other accessories must have surface burning characteristics as determined by ASTM E84.

2.9 SUPPLEMENTAL COMPONENTS/SERVICES

2.9.1 Chilled and Condenser Water Piping and Accessories

Chilled and condenser water piping and accessories must be provided and installed in accordance with Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS.

2.9.2 Refrigerant Piping

Refrigerant piping for split-system water chillers must be provided and installed in accordance with Section 23 23 00 REFRIGERANT PIPING.

2.9.3 Temperature Controls

Chiller control packages must be fully coordinated with and integrated into the temperature control system indicated in Section 23 30 00 HVAC AIR DISTRIBUTION and Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

PART 3 EXECUTION

3.1 INSTALLATION

Installation of water chiller systems including materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with the manufacturer's written installation instructions, including the following:

(1) Water chiller - installation instructions

3.1.1 Installation Instructions

Provide manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component, highlighted to show features such as materials, dimensions, options, performance and efficiency. Data must include manufacturer's recommended installation instructions and procedures. Data must be adequate to demonstrate compliance with contract requirements.

3.1.2 Vibration Isolation

If vibration isolation is specified for a unit, vibration isolator literature must be included containing catalog cuts and certification that

the isolation characteristics of the isolators provided meet the manufacturer's recommendations.

3.1.3 Posted Instructions

Provide posted instructions, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

3.1.4 Verification of Dimensions

Provide a letter including the date the site was visited, conformation of existing conditions, and any discrepancies found.

3.1.5 System Performance Test Schedules

Provide a schedule, at least 2 weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test.

3.1.6 Certificates

Where the system, components, or equipment are specified to comply with requirements of AGA, NFPA, ARI, ASHRAE, ASME, or UL, proof of such compliance must be provided. The label or listing of the specified agency must be acceptable evidence. In lieu of the label or listing, a written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of the specified agency may be submitted. When performance requirements of this project's drawings and specifications vary from standard ARI rating conditions, computer printouts, catalog, or other application data certified by ARI or a nationally recognized laboratory as described above must be included. If ARI does not have a current certification program that encompasses such application data, the manufacturer may self certify that his application data complies with project performance requirements in accordance with the specified test standards.

3.1.7 Operation and Maintenance Manuals

Provide Six complete copies of an operation manual in bound 216 by 279 mm booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features. Six complete copies of maintenance manual in bound 216 by 279 mm booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

3.1.8 Connections to Existing Systems

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Furnish materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Furnish labor and tools for making actual connections to existing systems.

3.1.9 Refrigeration System

3.1.9.1 Equipment

Refrigeration equipment and the installation thereof must conform to ANSI/ASHRAE 15 & 34. Necessary supports must be provided for all equipment, appurtenances, and pipe as required, including frames or supports for compressors, pumps, cooling towers, condensers, water coolers, and similar items. Compressors must be isolated from the building structure. If mechanical vibration isolators are not provided, vibration absorbing foundations must be provided. Each foundation must include isolation units consisting of machine and floor or foundation fastenings, together with intermediate isolation material. Other floor-mounted equipment must be set on not less than a 150 mm concrete pad doweled in place. Concrete foundations for floor mounted pumps must have a mass equivalent to three times the weight of the components, pump, base plate, and motor to be supported. In lieu of concrete pad foundation, concrete pedestal block with isolators placed between the pedestal block and the floor may be provided. Concrete pedestal block must be of mass not less than three times the combined pump, motor, and base weights. Isolators must be selected and sized based on load-bearing requirements and the lowest frequency of vibration to be isolated. Isolators must limit vibration to 10 percent at lowest equipment rpm. Lines connected to pumps mounted on pedestal blocks must be provided with flexible connectors. Foundation drawings, bolt-setting information, and foundation bolts must be furnished prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Concrete for foundations must be as specified in Structural drawings. Equipment must be properly leveled, aligned, and secured in place in accordance with manufacturer's instructions.

3.1.9.2 Field Refrigerant Charging

- a. Initial Charge: Upon completion of all the refrigerant pipe tests, the vacuum on the system must be broken by adding the required charge of dry refrigerant for which the system is designed, in accordance with the manufacturer's recommendations. Contractor must provide the complete charge of refrigerant in accordance with manufacturer's recommendations. Upon satisfactory completion of the system performance tests, any refrigerant that has been lost from the system must be replaced. After the system is fully operational, service valve seal caps and blanks over gauge points must be installed and tightened.
- b. Refrigerant Leakage: If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must immediately be isolated from the remainder of the system and the refrigerant must be pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

- c. Contractor's Responsibility: The Contractor must, at all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time must more than 85 g of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year must be repaired in accordance with the specified requirements including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

3.1.9.3 Oil Charging

Except for factory sealed units, two complete charges of lubricating oil for each compressor crankcase must be furnished. One charge must be used during the performance testing period, and upon the satisfactory completion of the tests, the oil must be drained and replaced with the second charge.

3.1.10 Mechanical Room Ventilation

Mechanical ventilation systems must be in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

3.1.11 Field Applied Insulation

Field installed insulation must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

3.1.12 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

3.2 FACTORY TEST SCHEDULING AND REPORTS

Provide schedules which identify the date, time, and location for each test. Schedules must be submitted for the Chiller Performance Tests.

Six copies of the certified test report must be forwarded to the Government for approval prior to project acceptance. Calibration curves and information sheets for all instrumentation must be included. Provide copies in bound 216 by 279 mm booklets. Reports must certify the compliance with performance requirements and follow the format of the required testing standard for the Chiller Performance Tests. Test report must include certified calibration report of all test instrumentation. Calibration report must include certification that all test instrumentation has been calibrated within 6 months prior to the test date, identification of all instrumentation, and certification that all instrumentation complies with requirements of the test standard. Test report must be submitted 1 week after completion of the factory test.

3.3 MANUFACTURER'S FIELD SERVICE

The services of a factory-trained representative must be provided for 14

days. The representative must advise on the following:

a. Hermetic machines:

- (1) Testing hermetic water-chilling unit under pressure for refrigerant leaks; evacuation and dehydration of machine to an absolute pressure of not over 300 micrometers.
- (2) Charging the machine with refrigerant.
- (3) Starting the machine.

b. Open Machines:

- (1) Erection, alignment, testing, and dehydrating.
- (2) Charging the machine with refrigerant.
- (3) Starting the machine.

3.4 CLEANING AND ADJUSTING

Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Provide temporary filters for all fans that are operated during construction. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions. At least one week before the official equipment warranty start date, all condenser coils on air-cooled water chillers and split-system water chillers must be cleaned in accordance with the chiller manufacturer's instructions. This work covers two coil cleanings. The condenser coils must be cleaned with an approved coil cleaner by a service technician, factory trained by the chiller manufacturer. The condenser coil cleaner must not have any detrimental affect on the materials or protective coatings on the condenser coils. Testing, adjusting, and balancing must be as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

3.5 FIELD ACCEPTANCE TESTING

3.5.1 Test Plans

- a. Manufacturer's Test Plans: Within 120 calendar days after contract award, submit the following plans:

(1) Water chiller - Field Acceptance Test Plan

Field acceptance test plans must be developed by the chiller manufacturer detailing recommended field test procedures for that particular type and size of equipment. Field acceptance test plans developed by the installing Contractor, or the equipment sales agency furnishing the equipment, will not be acceptable.

The Contracting Officer will review and approve the field acceptance test plan for each of the listed equipment prior to commencement of field testing of the equipment. The approved field acceptance tests of the chiller and subsequent test reporting.

- b. Coordinated testing: Indicate in each field acceptance test plan when work required by this section requires coordination with test work required by other specification sections. Furnish test procedures for the simultaneous or integrated testing of tower system controls which interlock and interface with controls for the equipment provided under Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.
- c. Prerequisite testing: Chillers for which performance testing is dependent upon the completion of the work covered by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must have that work completed as a prerequisite to testing work under this section. Indicate in each field acceptance test plan when such prerequisite work is required.
- d. Test procedure: Indicate in each field acceptance test plan each equipment manufacturers published installation, start-up, and field acceptance test procedures. Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives. Procedures must be structured to test the controls through all modes of control to confirm that the controls are performing with the intended sequence of control.

Controller must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

- e. Performance variables: Each test plan must list performance variables that are required to be measured or tested as part of the field test.

Include in the listed variables performance requirements indicated on the equipment schedules on the design drawings. Chiller manufacturer must furnish with each test procedure a description of acceptable results that have been verified.

Chiller manufacturer must identify the acceptable limits or tolerance within which each tested performance variable must acceptably operate.

- f. Job specific: Each test plan must be job specific and must address the particular cooling towers and particular conditions which exist in this contract. Generic or general preprinted test procedures are not acceptable.
- g. Specialized components: Each test plan must include procedures for field testing and field adjusting specialized components, such as hot gas bypass control valves, or pressure valves.

3.5.2 Testing

- a. Each water chiller system must be field acceptance tested in compliance with its approved field acceptance test plan and the resulting following field acceptance test report submitted for approval:

(1) Water chiller - Field Acceptance Test Report
- b. Manufacturer's recommended testing: Conduct the manufacturer's recommended field testing in compliance with the approved test plan. Furnish a factory trained field representative authorized by and to represent the equipment manufacturer at the complete execution of the field acceptance testing.
- c. Operational test: Conduct a continuous 24 hour operational test for each item of equipment. Equipment shutdown before the test period is completed must result in the test period being started again and run for the required duration. For the duration of the test period, compile an operational log of each item of equipment. Log required entries every two hours. Use the test report forms for logging the operational variables.
- d. Notice of tests: Conduct the manufacturer's recommended tests and the operational tests; record the required data using the approved reporting forms. Notify the Contracting Officer in writing at least 15 calendar days prior to the testing. Within 30 calendar days after acceptable completion of testing, submit each test report for review and approval.
- e. Report forms: Type data entries and writing on the test report forms. Completed test report forms for each item of equipment must be reviewed, approved, and signed by the Contractor's test director. The manufacturer's field test representative must review, approve, and sign the report of the manufacturer's recommended test. Signatures must be accompanied by the person's name typed.
- f. Deficiency resolution: The test requirements acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations and corrections retested in order to verify compliance.

3.6 SYSTEM PERFORMANCE TESTS

Six copies of the report must be provided in bound 216 by 279 mm booklets.

3.6.1 General Requirements

Before each refrigeration system is accepted, tests to demonstrate the general operating characteristics of all equipment must be conducted by the manufacturer's approved start-up representative experienced in system start-up and testing, at such times as directed. Tests must cover a period of not less than 48 hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications. Corrections and adjustments must be made as necessary and tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, service valve seal caps and blanks over gauge points must be installed and tightened. Any refrigerant lost during the system startup must be replaced. If tests do

not demonstrate satisfactory system performance, deficiencies must be corrected and the system must be retested. Tests must be conducted in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Any material, equipment, instruments, and personnel required for the test must be provided by the Contractor. Field tests must be coordinated with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

3.6.2 Test Report

The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. The report must also include the following information and must be taken at least three different times at outside dry-bulb temperatures that are at least 3 degrees C apart:

- a. Date and outside weather conditions.
- b. The load on the system based on the following:
 - (1) The refrigerant used in the system.
 - (2) Condensing temperature and pressure.
 - (3) Suction temperature and pressure.
 - (4) Running current, voltage and proper phase sequence for each phase of all motors.
 - (5) The actual on-site setting of all operating and safety controls.
 - (6) Chilled water pressure, flow and temperature in and out of the chiller.
 - (7) The position of the capacity-reduction gear at machine off, one-third loaded, one-half loaded, two-thirds loaded, and fully loaded.

3.7 DEMONSTRATIONS

Contractor must conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 16 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests. The training course must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations.

Provide a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.

-- End of Section --

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SECTION 23 64 26

CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS
08/09

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- | | |
|----------|--|
| NFPA 70 | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code |
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems |

JAPANESE STANDARDS ASSOCIATION (JSA)

- | | |
|--------------|---|
| JIS B 1180 | (2014) Hexagon Head Bolts and Hexagon Screws |
| JIS B 2011 | (2013) Bronze, Gate, Globe, Angle, and Check Valves (Amendment 2) |
| JIS B 2031 | (2015) Gray Cast Iron Valves (Amendment 1) |
| JIS B 2032 | (2013) Wafer Type Rubber-Seated Butterfly Valves |
| JIS B 2220 | (2012) Steel Pipe Flanges |
| JIS B 2301 | (2013) Screwed Type Malleable Cast Iron Pipe Fittings |
| JIS B 2311 | (2015) Steel Butt-Welding Pipe Fittings for Ordinary Use |
| JIS B 2316 | (2017) Steel Socket-Welding Pipe Fittings |
| JIS B 2352 | (2013) Bellows Type Expansion Joints |
| JIS B 2404 | (2018) Dimensions of Gaskets for Use with Pipe Flanges |
| JIS B 6801 | (2003) Manual Blowpipes for Welding, Cutting and Heating |
| JIS B 7505-1 | (2017) Aneroid Pressure Gauges-Part 1: Bourdon Tube Pressure Gauges |
| JIS B 8285 | (2010) Welding Procedure Qualification Test for Pressure Vessels |

JIS B 8301	(2009) Technical Specifications for Centrifugal Pumps-Class II
JIS B 8308	(2009) Technical Specifications for Centrifugal Pumps-Class II
JIS B 8410	(2015) Pressure Reducing Valves for Water Works
JIS B 8414	(2011) Relief Valves for Hot Water Appliances (Amendment 3)
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS G 3106	(2020) Rolled Steels for Welded Structure
JIS G 3302	(2022) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3454	(2019) Carbon Steel Pipes for Pressure Service (Amendment 1)
JIS G 5502	(2007) Spheroidal Graphite Iron Castings
JIS G 5705	(2018) Malleable Iron Castings
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 5120	(2016) Copper and Copper Alloy Castings
JIS K 0102	(2019) Testing Methods for Industrial Wastewater (Amendment 1)
JIS Z 2371	(2015) Methods of Salt Spray Testing
JIS Z 3197	(2012) Test Methods for Soldering Fluxes
JIS Z 3202	(2007) Copper and Copper Alloy Gas Welding Rods
JIS Z 3284-1	(2014) Solder Paste-Part 1: Kinds and Quality Classification
JIS Z 3422-1	(2003) Specification and Approval of Welding Procedures for Metallic Materials-Welding Procedure Tests-Arc and Gas Welding of Steels and Arc Welding of Nickel and Nickel Alloys
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique

JIS Z 8802 (2011) Methods for Determination of PH of Aqueous Solutions

THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN (SHASE)

SHASE-S003 (2012) Sleeve Type Expansion Pipe Joints

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M (2019) Public Building Construction Standard Specification

MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)

MHLW-177 (2018) WATER SUPPLY LAW

JAPAN VALVE MANUFACTURERS' ASSOCIATION (JVMA)

JV-5 (2008) Pipe end Anticorrosion Screwed Valve

1.2 SYSTEM DESCRIPTION

Provide the water systems having the minimum service (design) temperature-pressure rating indicated. Provision of the piping systems, including materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing shall be in accordance with the required and advisory provisions of MLIT-M except as modified or supplemented by this specification section or design drawings. This specification section covers the water systems piping which is located within, on, and adjacent to building(s) within the building(s) 1.66 meter line.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Pumps; G

SD-06 Test Reports

Piping Welds NDE Report

Pressure Tests Reports; G

Report shall be provided in bound 216 by 279 mm booklets. In the reports, document all phases of the tests performed. Include initial test summaries, all repairs/adjustments made, and the final test results.

SD-10 Operation and Maintenance Data

Requirements for data packages are specified Section 01 78 23 OPERATION AND MAINTENANCE DATA, except as supplemented and modified by this specification section.

Submit spare parts data for each different item of equipment specified, with operation and maintenance data packages. Include a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

Submit a list of qualified permanent service organizations with operation and maintenance data packages. Include service organization addresses and service area or expertise. The service organizations shall be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

Water Treatment Systems

An operation manual in bound 216 by 279 mm booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown. Include testing procedures used in determining water quality.

A maintenance manual in bound 216 by 279 mm booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide.

Calibrated Balancing Valves, Data Package 3;

Automatic Flow Control Valves, Data Package 3;

Pump Discharge Valve, Data Package 2;

Water Temperature Mixing Valve, Data Package 3;

Water Temperature Regulating Valves, Data Package 3;

Water Pressure Reducing Valve, Data Package 3;

Pressure Relief Valve, Data Package 2;

Combination Pressure and Temperature Relief Valves, Data Package 2;

Expansion Joints, Data Package 2;

Pumps, Data Package 3; G

Combination Strainer and Pump Suction Diffuser, Data Package 2;

Expansion Tanks, Data Package 2; G

Air Separator Tanks, Data Package 2; G

1.4 MODIFICATIONS TO REFERENCES

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in

these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions shall be considered mandatory, the word "should" shall be interpreted as "shall." Reference to the "code official" shall be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" shall be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" shall be interpreted to mean the "lessor." References to the "permit holder" shall be interpreted to mean the "Contractor."

1.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, shall be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.5 SAFETY REQUIREMENTS

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel shall be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices shall be installed so that proper operation of equipment is not impaired.

1.6 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation shall be the Contractor's responsibility. Any materials found to be damaged shall be replaced at the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter. Any porous materials found to be contaminated with mold or mildew will be replaced at the Contractor's expense. Non-porous materials found to be contaminated with mold or mildew will be disinfected and cleaned prior to installation.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Verification of Dimensions

The Contractor shall become familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.7.2 Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. The Contractor shall carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to

be performed and shall arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

1.7.3 Accessibility

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

Materials and equipment shall be standard products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. The standard products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening.

The two year use shall include applications of equipment and materials under similar circumstances and of similar size. The 2 years experience shall be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures.

Products having less than a 2 year field service record shall be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. System components shall be environmentally suitable for the indicated locations.

The equipment items shall be supported by service organizations. These service organizations shall be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

2.2 STEEL PIPING

Water piping shall be steel pipe. Provide steel piping with a Japanese Industrial Standards (JIS) Class 125 service rating, which for 66 degrees C the pressure rating is 1207 kPa.

2.2.1 Pipe

Steel pipe, conform to JIS G 3454, Schedule 40 and MLIT-M.

2.2.2 Fittings and End Connections (Joints)

Piping and fittings 25 mm and smaller shall have threaded connections. Piping and fittings larger than 25 mm and smaller than 80 mm shall have either threaded, or welded connections. Piping and fittings 80 mm and larger shall have welded, or flanged connections. The manufacturer of each fitting shall be permanently identified on the body of the fitting.

2.2.2.1 Threaded Connections

Use threaded fitting conforming to JIS B 2301. Use threaded unions conforming to JIS B 2301. Use threaded pipe nipples conforming to MLIT-M.

2.2.2.2 Flanged Connections

Flanges shall conform to JIS B 2220. Gaskets shall be nonasbestos compressed material in accordance with JIS B 2404, 1.59 mm thickness, full face or self-centering flat ring type. These gaskets shall contain aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR). Bolts, nuts, and bolt patterns shall conform to JIS B 1180.

2.2.2.3 Welded Connections

Welded valves and pipe connections (both butt-welds and socket-welds types) shall conform to JIS B 2301. Butt-welded fittings shall conform to JIS B 2311. Socket-welded fittings shall conform to JIS B 2316. Welded fittings shall be identified with the appropriate grade and marking symbol.

2.2.2.4 Grooved Mechanical Connections For Steel

Rigid grooved mechanical connections may only be used in serviceable aboveground locations where the temperature of the circulating medium does not exceed 110 degrees C. Flexible grooved connections shall be used only as a flexible connector with grooved pipe system. Unless otherwise specified, grooved piping components shall meet the corresponding criteria specified for the similar welded, flanged, or threaded component specified herein.

Each grooved mechanical joint shall be a system, including coupling housing, gasket, fasteners, all furnished by the same manufacturer. Joint installation shall be in compliance with joint manufacturer's written instructions.

Use fitting and coupling houses of malleable iron conforming to JIS G 5705; steel conforming JIS G 3454. Use gaskets of molded synthetic rubber with central cavity, pressure responsive configuration and conforming to MLIT-M. Grooved mechanical connections shall conform to MLIT-M. Coupling nuts and bolts shall be steel and shall conform to MLIT-M. Pipe connections and fittings shall be the product of the same manufacturer. Provide joint installation be in compliance with joint manufacturer's written instructions.

2.2.2.5 Dielectric Waterways and Flanges

Provide dielectric waterways with a water impervious insulation barrier capable of limiting galvanic current to 1 percent of short circuit current in a corresponding bimetallic joint. When dry, insulation barrier shall be able to withstand a 600-volt breakdown test. Provide dielectric waterways constructed of galvanized steel and have threaded end connections to match connecting piping. Dielectric waterways shall be suitable for the required operating pressures and temperatures. Provide dielectric flanges with the same pressure ratings as standard flanges and provide complete electrical isolation between connecting pipe and/or equipment as described herein for dielectric waterways.

2.3 COPPER TUBING

Provide copper tubing and fittings with a JIS H 3300 Class 125 service rating, which for 66 degrees C, the pressure rating is 1207 kPa.

2.3.1 Tube

Use copper tube conforming to JIS H 3300, Type L or M for aboveground tubing, and Type K for buried tubing.

2.3.2 Fittings and End Connections (Solder and Flared Joints)

Wrought copper and bronze solder joint pressure fittings, including unions and flanges, shall conform to JIS H 3401. Provide adapters as required. Cast copper alloy solder-joint pressure fittings, including unions and flanges, shall conform to JIS H 3401. Cast copper alloy fittings for flared copper tube shall conform to JIS H 3401 and JIS H 5120. JIS H 3300 copper pipe nipples with threaded end connections shall conform to JIS H 3300.

Copper tubing of sizes larger than 100 mm shall have brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

Extracted brazed tee joints may be used if produced with an acceptable tool and installed in accordance with tool manufacturer's written procedures.

2.3.3 Grooved Mechanical Connections For Copper

Rigid grooved mechanical connections may only be used in serviceable aboveground locations where the temperature of the circulating medium does not exceed 110 degrees C. Flexible grooved connections shall be used only as a flexible connector with grooved pipe system. Unless otherwise specified, grooved piping components shall meet the corresponding criteria specified for the similar welded, flanged, or threaded component specified herein.

Each grooved mechanical joint shall be a system, including coupling housing, gasket, fasteners, all furnished by the same manufacturer. Joint installation shall be in compliance with joint manufacturer's written instructions.

Grooved fitting and mechanical coupling housing shall be ductile iron conforming to JIS G 5502. Provide gaskets for use in grooved joints shall constructed of molded synthetic polymer of pressure responsive design and shall conform to MLIT-M for circulating medium up to 110 degrees C. Provide grooved joints in conformance with MLIT-M.

2.3.4 Solder

Provide solder in conformance with JIS Z 3284-1, grade Sb5, tin-antimony alloy. Solder flux shall be liquid or paste form, non-corrosive and conform to JIS Z 3197.

2.3.5 Brazing Filler Metal

Filler metal shall conform to JIS Z 3202, Type BAg-5 with AWS Type 3 flux, except Type BCuP-5 or BCuP-6 may be used for brazing copper-to-copper joints.

2.4 VALVES

Provide valves with a Japanese standard service rating.

Valves in sizes larger than 25 mm and used on steel pipe systems, may be provided with rigid grooved mechanical joint ends. Such grooved end valves shall be subject to the same requirements as rigid grooved mechanical joints and fittings and, shall be furnished by the same manufacturer as the grooved pipe joint and fitting system.

2.4.1 Gate Valve

Gate valves 65 mm and smaller shall conform to JIS B 2011 Class 125 and shall be bronze with wedge disc, rising stem and threaded, soldered, or flanged ends. Gate valves 80 mm and larger shall conform to JIS B 2031, Class 125, cast iron with bronze trim, outside screw and yoke, and flanged or threaded ends.

2.4.2 Globe and Angle Valve

Globe and angle valves 65 mm and smaller shall conform to JIS B 2011, Class 125. Globe and angle valves 80 mm and larger shall conform to JIS B 2031, Class 125.

2.4.3 Check Valve

Check valves 65 mm and smaller shall conform to JIS B 2011. Check valves 80 mm and larger shall conform to JIS B 2031, Class 125.

2.4.4 Butterfly Valve

Butterfly valves shall conform to JIS B 2032 and JIS B 2011, Type 1 and shall be either the wafer or lug type. Valves smaller than 200 mm shall have throttling handles with a minimum of two locking positions. Valves 200 mm and larger shall have totally enclosed manual gear operators with adjustable balance return stops and position indicators.

2.4.5 Plug Valve

Plug valves 50 mm and larger shall conform to JIS B 2031, have flanged or threaded ends, and have cast iron bodies with bronze trim. Valves 50 mm and smaller shall be bronze with NPT connections for black steel pipe and brazed connections for copper tubing. Valve shall be lubricated, non-lubricated, or tetrafluoroethylene resin-coated type. Valve shall be resilient, double seated, trunnion mounted with tapered lift plug capable of 2-way shutoff. Valve shall operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves 200 mm or larger shall be provided with manual gear operators with position indicators.

2.4.6 Ball Valve

Full port design. Ball valves 15 mm and larger shall conform to JV-5 and shall be cast iron or bronze with threaded, soldered, or flanged ends. Valves 200 mm or larger shall be provided with manual gear operators with position indicators. Ball valves may be provided in lieu of gate valves.

2.4.7 Square Head Cocks

Provide copper alloy or cast-iron body with copper alloy plugs, suitable for 125 psig water working pressure.

2.4.8 Calibrated Balancing Valves

Copper alloy or cast iron body, copper alloy or stainless internal working parts. Provide valve calibrated so that flow can be determined when the temperature and pressure differential across valve is known. Valve shall have an integral pointer which registers the degree of valve opening. Valve shall function as a service valve when in fully closed position. Valve shall be constructed with internal seals to prevent leakage and shall be supplied with preformed insulation.

Provide valve bodies with tapped openings and pipe extensions with positive shutoff valves outside of pipe insulation. The pipe extensions shall be provided with quick connecting hose fittings for a portable differential pressure meter connections to verify the pressure differential. Provide metal tag on each valve showing the liters per second flow for each differential pressure reading.

2.4.9 Automatic Flow Control Valves

Valve shall automatically maintain the constant flow indicated on the design drawings. Valve shall modulate by sensing the pressure differential across the valve body. Valve shall be selected for the flow required and provided with a permanent nameplate or tag carrying a permanent record of the factory-determined flow rate and flow control pressure levels. Provide valve that controls the flow within 5 percent of the tag rating. Valve materials shall be the same as specified for the ball or plug valves.

Provide valve that are electric type as indicated. Valve shall be capable of positive shutoff against the system pump head, valve bodies shall be provided with tapped openings and pipe extensions with shutoff valves outside of pipe insulation. The pipe extensions shall be provided with quick connecting hose fittings and differential meter, suitable for the operating pressure specified. Provide the meter complete with hoses, vent, integral metering connections, and carrying case as recommended by the valve manufacturer.

2.4.10 Pump Discharge Valve

Valve shall perform the functions of a nonslam check valve, a manual balancing valve, and a shutoff. Valve shall be of cast iron or ductile iron construction with bronze and/or stainless steel accessories. Provide an integral pointer on the valve which registers the degree of valve opening. Flow through the valve shall be manually adjustable from bubble tight shutoff to full flow. Valves smaller than 50 mm shall have NPT connections. Valves 50 mm and larger shall have flanged or grooved end connections. Valve design shall allow the back seat for the stem to be replaced in the field under full line pressure.

2.4.11 Water Temperature Mixing Valve

Provide water temperature mixing valve that meets performance standards of the water supply Law (MHLW-177).

2.4.12 Water Temperature Regulating Valves

Provide copper alloy body, direct acting, pilot operated, for the intended service.

2.4.13 Water Pressure Reducing Valve

Valve, JIS B 8410 for water service, copper alloy body.

2.4.14 Pressure Relief Valve

Valve shall prevent excessive pressure in the piping system when the piping system reaches its maximum heat buildup. Valve, JIS B 8414 and shall have cast iron bodies with corrosion resistant internal working parts. The discharge pipe from the relief valve shall be the size of the valve outlet unless otherwise indicated.

2.4.15 Combination Pressure and Temperature Relief Valves

JIS B 8414, copper alloy body, automatic re-seating, test lever, and discharge capacity based on AGA temperature steam rating.

2.4.16 Float Valve

Valve bodies 80 mm nominal pipe size and smaller shall be bronze. Valve bodies larger than 80 mm shall be cast iron or bronze. Steel parts shall be corrosion resistant. Where float rods are extended for tank applications, extension shall be properly supported and guided to avoid bending of float rod or stressing of valve pilot linkage.

2.4.17 Drain Valves

Valves, JIS B 2011 gate valves. Valve shall be manually-operated, 20 mm pipe size and above with a threaded end connection. Provide valve with a water hose nipple adapter.

2.4.18 Air Venting Valves

Automatic type air venting shall be the ball-float type with brass/bronze or brass bodies, 300 series corrosion-resistant steel float, linkage and removable seat. Air venting valves on water coils shall have not less than 3 mm threaded end connections. Air venting valves on water mains shall have not less than 20 mm threaded end connections. Air venting valves on all other applications shall have not less than 15 mm threaded end connections.

2.4.19 Vacuum Relief Valves

JIS B 8414

2.5 PIPING ACCESSORIES

2.5.1 Strainer

Strainer, MLIT-M, except as modified and supplemented in this specification. Strainer shall be the cleanable, basket or "Y" type, the same size as the pipeline. Strainer bodies shall be fabricated of cast iron with bottoms drilled, and tapped. Provide blowoff outlet with pipe nipple, gate valve, and discharge pipe nipple. The bodies shall have arrows clearly cast on the sides indicating the direction of flow.

Provide strainer with removable cover and sediment screen. The screen shall be made of minimum 0.8 mm corrosion-resistant steel, with small perforations numbering not less than 60 per square centimeter to provide a

net free area through the basket of at least 3.30 times that of the entering pipe. The flow shall be into the screen and out through the perforations.

2.5.2 Combination Strainer and Pump Suction Diffuser

Angle type body with removable strainer basket and internal straightening vanes, a suction pipe support, and a blowdown outlet and plug. Strainer shall be in accordance with MLIT-M, except as modified and supplemented by this specification. Unit body shall have arrows clearly cast on the sides indicating the direction of flow.

Strainer screen shall be made of minimum 0.8 mm corrosion-resistant steel, with small perforations numbering not less than 60 per square centimeter to provide a net free area through the basket of at least 3.30 times that of the entering pipe. Flow shall be into the screen and out through the perforations. Provide an auxiliary disposable fine mesh strainer which shall be removed 30 days after start-up. Provide warning tag for operator indicating scheduled date for removal.

Casing shall have connection sizes to match pump suction and pipe sizes, and be provided with adjustable support foot or support foot boss to relieve piping strains at pump suction. Provide unit casing with blowdown port and plug. Provide a magnetic insert to remove debris from system.

2.5.3 Flexible Pipe Connectors

Provide flexible bronze or stainless steel piping connectors with single braid. Equip flanged assemblies with limit bolts to restrict maximum travel to the manufacturer's standard limits. Unless otherwise indicated, the length of the flexible connectors shall be as recommended by the manufacturer for the service intended. Internal sleeves or liners, compatible with circulating medium, shall be provided when recommended by the manufacturer. Provide covers to protect the bellows where indicated.

2.5.4 Pressure and Vacuum Gauges

Gauges, JIS B 7505-1 with throttling type needle valve or a pulsation dampener and shut-off valve. Provide gauges with 115 mm dial, brass or aluminum case, bronze tube, and siphon. Gauge shall have a range from 0 kPa to approximately 1.5 times the maximum system working pressure. Each gauge range shall be selected so that at normal operating pressure, the needle is within the middle-third of the range.

2.5.5 Temperature Gauges

Temperature gauges, shall be the industrial duty type and be provided for the required temperature range. Provide gauges with fixed thread connection, dial face gasketed within the case; and an accuracy within 2 percent of scale range. Gauges shall have Celsius scale in 1 degree graduations scale (black numbers) on a white face. The pointer shall be adjustable. Rigid stem type temperature gauges shall be provided in thermal wells located within 1.5 m of the finished floor. Universal adjustable angle type or remote element type temperature gauges shall be provided in thermal wells located 1.5 to 2.1 m above the finished floor or in locations indicated. Remote element type temperature gauges shall be provided in thermal wells located 2.1 m above the finished floor or in locations indicated.

2.5.5.1 Stem Cased-Glass

Stem cased-glass case shall be polished stainless steel or cast aluminum, 229 mm long, with clear acrylic lens, and non-mercury filled glass tube with indicating-fluid column.

2.5.5.2 Bimetallic Dial

Bimetallic dial type case shall be not less than 89 mm stainless steel, and shall be hermetically sealed with clear acrylic lens. Bimetallic element shall be silicone dampened and unit fitted with external calibrator adjustment.

2.5.5.3 Liquid-, Solid-, and Vapor-Filled Dial

Liquid-, solid-, and vapor-filled dial type cases shall be not less than 89 mm, stainless steel or cast aluminum with clear acrylic lens. Fill shall be nonmercury, suitable for encountered cross-ambients, and connecting capillary tubing shall be double-braided bronze.

2.5.5.4 Thermal Well

Thermal well shall be identical size, 15 or 20 mm NPT connection, brass or stainless steel. Where test wells are indicated, provide captive plug-fitted type 15 mm NPT connection suitable for use with either engraved stem or standard separable socket thermometer or thermostat. Mercury shall not be used in thermometers. Extended neck thermal wells shall be of sufficient length to clear insulation thickness by 25 mm.

2.5.6 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, guides, and supports shall be in accordance with MLIT-M.

2.5.7 Escutcheons

Provide one piece or split hinge metal plates for piping entering floors, walls, and ceilings in exposed spaces. Secure plates in place by internal spring tension or set screws. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

2.5.8 Expansion Joints

2.5.8.1 Slip-Tube Type

Slip-tube expansion joints, SHASE-S003. Joints shall be provided with internally-externally alignment guides, injected semi-plastic packing, and service outlets. End connections shall be flanged or beveled for welding as indicated. Initial settings shall be made in accordance with the manufacturer's recommendations to compensate for ambient temperature at time of installation. Pipe alignment guides shall be installed as recommended by the joint manufacturer.

2.5.8.2 Flexible Ball Type

Flexible ball expansion joints shall be capable of 360 degrees rotation plus 15 degrees angular flex movement. Joints shall be constructed of carbon steel with the exterior spherical surface of carbon steel balls

plated with a minimum 0.12 mm of hard chrome in accordance with MLIT-M. Joint end connections shall be threaded for piping 50 mm or smaller. Joint end connections larger than 50 mm shall be grooved, flanged, or beveled for welding. Provide joint with pressure-molded composition gaskets suitable for continuous operation at twice design temperature.

2.5.8.3 Bellows Type

Bellows expansion type joints, JIS B 2352 with Type 304 stainless steel corrugated bellows, reinforced with equalizing rings, internal sleeves, and external protective covers. Joint end connections shall be grooved, flanged, or beveled for welding. Guiding of piping on both sides of expansion joint shall be in accordance with the published recommendations of the manufacturer of the expansion joint.

2.6 PUMPS

Pumps shall be the electrically driven, non-overloading, centrifugal type which conform to JIS B 8308 and JIS B 8301. Pumps shall be selected at or within 5 percent of peak efficiency. Pump curve shall rise continuously from maximum capacity to shutoff. Pump motor shall conform to JIS C 4212, be totally enclosed, and have sufficient wattage for the service required. Pump motor shall have the required capacity to prevent overloading with pump operating at any point on its characteristic curve. Pump speed shall not exceed 3,600 rpm, except where the pump head is less than 180 kPa, the pump speed shall not exceed 1,750 rpm. Pump motor shall be equipped with an across-the-line magnetic controller in a industrial use, dust tight enclosure with "START-STOP" switch in the cover.

2.6.1 Construction

Each pump casing shall be designed to withstand the discharge head specified plus the static head on system plus 50 percent of the total, but not less than 862 kPa. Pump casing and bearing housing shall be close grained cast iron. High points in the casing shall be provided with manual air vents; low points shall be provided with drain plugs. Provide threaded suction and discharge pressure gage tapping with square-head plugs.

Impeller shall be statically and dynamically balanced. Impeller, impeller wearing rings, glands, casing wear rings, and shaft sleeve shall be bronze. Shaft shall be carbon or alloy steel, turned and ground. Bearings shall be ball-bearings, roller-bearings, or oil-lubricated bronze-sleeve type bearings, and be efficiently sealed or isolated to prevent loss of oil or entrance of dirt or water.

Close-coupled pump shall be provided with integrally cast or fabricated steel feet with bolt holes for securing feet to supporting surface. Close-coupled pumps shall be provided with drip pockets and tapped openings. Pump shall be accessible for servicing without disturbing piping connections. Shaft seals shall be mechanical-seals.

2.6.2 Mechanical Shaft Seals

Seals shall be single, inside mounted, end-face-elastomer bellows type with stainless steel spring, brass or stainless steel seal head, carbon rotating face, and tungsten carbide or ceramic sealing face. Glands shall be bronze and of the water-flush design to provide lubrication flush across the face of the seal. Bypass line from pump discharge to flush

connection in gland shall be provided, with filter or cyclone particle separator in line.

2.7 EXPANSION TANKS

Tank shall be welded steel, constructed for, and tested to pressure-temperature rating of 862 kPa at 66 degrees C. Provide tanks precharged to the minimum operating pressure. Tank shall have a replaceable polypropylene or butyl lined diaphragm which keeps the air charge separated from the water; shall be the captive air type.

Tanks shall accommodate expanded water of the system generated within the normal operating temperature range, limiting this pressure increase at all components in the system to the maximum allowable pressure at those components. Each tank air chamber shall be fitted with a drain, fill, an air charging valve, and system connections. Tank shall be supported by steel legs or bases for vertical installation or steel saddles for horizontal installations. The only air in the system shall be the permanent sealed-in air cushion contained within the expansion tank.

2.8 AIR SEPARATOR TANKS

Design to separate air from water and to direct released air to automatic air vent. Unit shall be of one piece cast-iron construction with internal baffles and two air chambers at top of unit; one air chamber shall have outlet to expansion tank and other air chamber shall be provided with automatic air release device. Tank shall be steel, constructed for, and tested to pressure-temperature rating in accordance with Ordinance on Safety of Boilers and Pressure Vessels (MHLW-177).

2.9 WATER TREATMENT SYSTEMS

When water treatment is specified, the use of chemical-treatment products containing equivalent chromium (CPR) is prohibited.

2.9.1 Water Analysis

Conditions of make-up water to be supplied to the condenser and chilled water systems were reported in accordance with JIS K 0102 and JIS Z 8802 and are as follows:

Date of Sample	_____ {_____}
Temperature	_____ {_____} degrees F
Silica (Sino 2)	_____ {_____} pp (mg/1)
Insoluble	_____ {_____} pp (mg/1)
Iron and Aluminum Oxides	_____ {_____} pp (mg/1)
Calcium (Ca)	_____ {_____} pp (mg/1)
Magnesium (Mg)	_____ {_____} pp (mg/1)

Sodium and Potassium (Nan and AK)	_____ {_____} pp (mg/1)
Carbonate (HO 3)	_____ {_____} pp (mg/1)
Sulfate (SO 4)	_____ {_____} pp (mg/1)
Chloride (JCL)	_____ {_____} pp (mg/1)
Nitrate (NO 3)	_____ {_____} pp (mg/1)
Turbidity	_____ {_____} unit
pH	_____ {_____}
Residual Chlorine	_____ {_____} pp (mg/1)
Total Alkalinity	_____ {_____} PM (me/1)
Non-Carbonate Hardness	_____ {_____} PM (me/1)
Total Hardness	_____ {_____} PM (me/1)
Dissolved Solids	_____ {_____} pp (mg/1)
Fluorine	_____ {_____} pp (mg/1)
Conductivity	_____ {_____} McMahon/cm

2.9.2 Chilled and Condenser Water

Water to be used in the chilled and condenser water systems shall be treated to maintain the conditions recommended by this specification as well as the recommendations from the manufacturers of the condenser and evaporator coils. Chemicals shall meet all required local and base environmental regulations for the treatment of evaporator coils and direct discharge to the sanitary sewer.

2.9.3 Water Treatment Services

The services of a company regularly engaged in the treatment of chilled water systems shall be used to determine the correct chemicals required, the concentrations required, and the water treatment equipment sizes and flow rates required. The company shall maintain the chemical treatment and provide all chemicals required for the chilled water systems for a period of 1 year from the date of occupancy. The chemical treatment and services provided over the 1 year period shall meet the requirements of this specification as well as the recommendations from the manufacturers of the condenser and evaporator coils. Acid treatment and proprietary chemicals shall not be used.

2.9.4 Chilled Water System

A shot feeder shall be provided on the chilled water piping as indicated. Size and capacity of feeder shall be based on local requirements and water analysis. The feeder shall be furnished with an air vent, gauge glass, funnel, valves, fittings, and piping.

2.10 ELECTRICAL WORK

Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown, shall be provided. For packaged equipment, the manufacturer shall provide controllers including the required monitors and timed restart.

Provide high efficiency type, single-phase, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with JIS C 4212.

Provide polyphase, squirrel-cage medium induction motors, including motors that are part of a system, that meet the efficiency ratings for premium efficiency motors in accordance with JIS C 4212. Provide motors in accordance with JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor.

Motors shall be rated for continuous duty with the enclosure specified. Motor duty requirements shall allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque shall be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Motor bearings shall be fitted with grease supply fittings and grease relief to outside of the enclosure.

Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers may be provided to accomplish the same function. Use solid-state variable-speed controllers for motors rated 7.45 kW or less and adjustable frequency drives for larger motors. Provide variable frequency drives for motors as specified in Section 26 29 23 VARIABLE FREQUENCY DRIVE SYSTEMS UNDER 600 VOLTS.

2.11 PAINTING OF NEW EQUIPMENT

New equipment painting shall be factory applied or shop applied, and shall be as specified herein, and provided under each individual section.

2.11.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided. The factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors shall withstand 500 hours in a salt-spray fog test.

Salt-spray fog test shall be in accordance with JIS Z 2371, and for that test, the acceptance criteria shall be as follows: immediately after completion of the test, the paint shall show no signs of blistering, wrinkling, or cracking, and no loss of 3 mm on either side of the scratch mark. The film thickness of the factory painting system applied on the equipment shall not be less than the film thickness used on the test specimen.

If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, the factory painting system shall be designed for the temperature service.

2.11.2 Shop Painting Systems for Metal Surfaces

Shop painting for metal surface shall be in accordance with MLIT-M.

2.12 NAMEPLATES

Major equipment including pumps, pump motors, expansion tanks, and air separator tanks shall have the manufacturer's name, type or style, model or serial number on a plate secured to the item of equipment. The nameplate of the distributing agent will not be acceptable. Plates shall be durable and legible throughout equipment life and made of stainless steel. Plates shall be fixed in prominent locations with nonferrous screws or bolts.

2.13 RELATED COMPONENTS/SERVICES

2.13.1 Drain and Make-Up Water Piping

Requirements for drain and make-up water piping and backflow preventer is specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

2.13.2 Field Applied Insulation

Requirements for field applied insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.13.3 Field Applied Insulation

Requirements for field installed insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as supplemented and modified by this specification section.

2.13.4 Field Painting

Requirements for painting of surfaces not otherwise specified, and finish painting of items only primed at the factory, are specified in Section 09 90 00 PAINTS AND COATINGS.

2.13.4.1 Color Coding

Requirements for color coding for piping identification are specified in Section 09 90 00 PAINTS AND COATINGS.

2.13.4.2 Color Coding For Hidden Piping

A color coding scheme for locating hidden piping shall be in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

PART 3 EXECUTION

3.1 INSTALLATION

Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows,

doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove burrs by reaming, and fashion to permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Furnish materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Furnish labor and tools for making actual connections to existing systems.

3.1.1 Welding

Provide welding work specified this section for piping systems in conformance with JIS B 2301, as modified and supplemented by this specification section and the accompanying drawings. The welding work includes: qualification of welding procedures, welders, welding operators, brazers, brazing operators, and nondestructive examination personnel; maintenance of welding records, and examination methods for welds.

3.1.1.1 Employer's Record Documents (For Welding)

Submit for review and approval the following documentation. This documentation and the subject qualifications shall be in compliance with JIS Z 3801.

- a. List of qualified welding procedures that is proposed to be used to provide the work specified in this specification section.
- b. List of qualified welders, brazers, welding operators, and brazing operators that are proposed to be used to provide the work specified in this specification section.
- c. List of qualified weld examination personnel that are proposed to be used to provide the work specified in this specification section.

3.1.1.2 Welding Procedures and Qualifications

- a. Specifications and Test Results: Submit copies of the welding procedures specifications and procedure qualification test results for each type of welding required. Approval of any procedure does not relieve the Contractor of the responsibility for producing acceptable welds. Submit this information on the forms printed in JIS B 8285 or their equivalent.
- b. Certification: Before assigning welders or welding operators to the work, submit a list of qualified welders, together with data and certification that each individual is performance qualified as specified. Do not start welding work prior to submitting welder, and welding operator qualifications. The certification shall state the type of welding and positions for which each is qualified, the code and procedure under which each is qualified, date qualified, and the firm and individual certifying the qualification tests.

3.1.1.3 Examination of Piping Welds

Conduct non-destructive examinations (NDE) on piping welds and brazing and

verify the work meets the acceptance criteria specified in JIS Z 3801. NDE on piping welds covered by JIS B 2301 is visual inspection only. Submit a piping welds NDE report meeting the requirements specified in JIS B 2301.

3.1.1.4 Welding Safety

Welding and cutting safety requirements shall be in accordance with JIS Z 3422-1 and JIS B 6801.

3.1.2 Directional Changes

Make changes in direction with fittings, except that bending of pipe 100 mm and smaller is permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends shall not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations is not acceptable.

3.1.3 Functional Requirements

Pitch horizontal supply mains down in the direction of flow as indicated. The grade shall not be less than 2 mm in 1 m. Reducing fittings shall be used for changes in pipe sizes. Cap or plug open ends of pipelines and equipment during installation to keep dirt or other foreign materials out of the system.

Pipe not otherwise specified shall be uncoated. Connections to appliances shall be made with malleable iron unions for steel pipe 65 mm or less in diameter, and with flanges for pipe 80 mm and above in diameter. Connections between ferrous and copper piping shall be electrically isolated from each other with dielectric waterways or flanges.

Piping located in air plenums shall conform to NFPA 90A requirements. Pipe and fittings installed in inaccessible conduits or trenches under concrete floor slabs shall be welded. Equipment and piping arrangements shall fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance. Electric isolation fittings shall be provided between dissimilar metals.

3.1.4 Fittings and End Connections

3.1.4.1 Threaded Connections

Threaded connecting shall be in accordance with MLIT-M.

3.1.4.2 Brazed Connections

Brazing, MLIT-M, except as modified herein. During brazing, the pipe and fittings shall be filled with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, both the outside of the tube and the inside of the fitting shall be cleaned with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux. Surplus brazing material shall be removed at all joints. Steel tubing joints shall be made in accordance with the manufacturer's recommendations. Piping shall be supported prior to brazing and not be sprung or forced.

3.1.4.3 Welded Connections

Branch connections shall be made with welding tees or forged welding branch outlets. Pipe shall be thoroughly cleaned of all scale and foreign matter before the piping is assembled. During welding, the pipe and fittings shall be filled with an inert gas, such as nitrogen, to prevent the formation of scale. Beveling, alignment, heat treatment, and inspection of weld shall conform to JIS Z 3801. Weld defects shall be removed and rewelded at no additional cost to the Government. Electrodes shall be stored and dried in accordance with JIS G 3106 or as recommended by the manufacturer. Electrodes that have been wetted or that have lost any of their coating shall not be used.

3.1.4.4 Grooved Mechanical Connections

Prepare grooves in accordance with the coupling manufacturer's instructions. Pipe and groove dimensions shall comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field shall be measured using a "go/no-go" gauge, vernier or dial caliper, or narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe shall be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Grooved joints shall not be used in concealed locations, such as behind solid walls or ceilings, unless an access panel is shown on the drawings for servicing or adjusting the joint.

3.1.4.5 Flared Connections

When flared connections are used, a suitable lubricant shall be used between the back of the flare and the nut in order to avoid tearing the flare while tightening the nut.

3.1.4.6 Flanges and Unions

Except where copper tubing is used, union or flanged joints shall be provided in each line immediately preceding the connection to each piece of equipment or material requiring maintenance such as coils, pumps, control valves, and other similar items. Flanged joints shall be assembled square end tight with matched flanges, gaskets, and bolts. Gaskets shall be suitable for the intended application.

3.1.5 Valves

Isolation gate or ball valves shall be installed on each side of each piece of equipment, at the midpoint of all looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Each valve except check valves shall be identified. Valves in horizontal lines shall be installed with stems horizontal or above.

3.1.6 Air Vents

Air vents shall be provided at all high points, on all water coils, and where indicated to ensure adequate venting of the piping system.

3.1.7 Drains

Drains shall be provided at all low points and where indicated to ensure complete drainage of the piping. Drains shall be accessible, and shall

consist of nipples and caps or plugged tees unless otherwise indicated.

3.1.8 Flexible Pipe Connectors

Connectors shall be attached to components in strict accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Hangers, when required to suspend the connectors, shall be of the type recommended by the flexible pipe connector manufacturer and shall be provided at the intervals recommended.

3.1.9 Temperature Gauges

Temperature gauges shall be located on coolant supply and return piping at each heat exchanger, on condenser water piping entering and leaving a condenser, at each automatic temperature control device without an integral thermometer, and where indicated or required for proper operation of equipment. Thermal wells for insertion thermometers and thermostats shall extend beyond thermal insulation surface not less than 25 mm.

3.1.10 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports shall conform to MLIT-M, except as supplemented and modified in this specification section. Hangers used to support piping 50 mm and larger shall be fabricated to permit adequate adjustment after erection while still supporting the load. Piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, shall be supported by variable spring hangers and supports or by constant support hangers.

3.1.10.1 Horizontal Pipe Supports

Horizontal pipe supports shall be spaced as specified in equivalent Japanese Standard and a support shall be installed not over 300 mm from the pipe fitting joint at each change in direction of the piping. Pipe supports shall be spaced not over 1.5 m apart at valves. Pipe hanger loads suspended from steel joist with hanger loads between panel points in excess of 23 kg shall have the excess hanger loads suspended from panel points.

3.1.10.2 Vertical Pipe Supports

Vertical pipe shall be supported at each floor, except at slab-on-grade, and at intervals of not more than 4.5 m, not more than 2.4 m from end of risers, and at vent terminations.

3.1.10.3 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, a clip or clamp shall be used where each pipe crosses the base support member. Spacing of the base support members shall not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

3.1.10.4 Seismic Requirements

Piping and attached valves shall be supported and braced to resist seismic loads as specified under Section 23 05 48.19 SEISMIC BRACING FOR HVAC Structural steel required for reinforcement to properly support piping, headers, and equipment but not shown shall be provided under this section. Material used for support shall be as specified in Structural

drawings.

3.1.10.5 Structural Attachments

Attachment to building structure concrete and masonry shall be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors shall be applied with a safety factor not less than 5. Supports shall not be attached to metal decking. Supports shall not be attached to the underside of concrete filled floors or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications shall be constructed of ferrous materials only. Structural steel brackets required to support piping, headers, and equipment, but not shown, shall be provided under this section. Material used for support shall be as specified in Structural drawings.

3.1.11 Pipe Alignment Guides

Pipe alignment guides shall be provided where indicated for expansion loops, offsets, and bends and as recommended by the manufacturer for expansion joints, not to exceed 1.5 m on each side of each expansion joint, and in lines 100 mm or smaller not more than 600 mm on each side of the joint.

3.1.12 Pipe Anchors

Anchors shall be provided where indicated. Unless indicated otherwise, anchors shall comply with the requirements specified. Anchors shall consist of heavy steel collars with lugs and bolts for clamping and attaching anchor braces, unless otherwise indicated. Anchor braces shall be installed in the most effective manner to secure the desired results using turnbuckles where required.

Supports, anchors, or stays shall not be attached where they will injure the structure or adjacent construction during installation or by the weight of expansion of the pipeline. Where pipe and conduit penetrations of vapor barrier sealed surfaces occur, these items shall be anchored immediately adjacent to each penetrated surface, to provide essentially zero movement within penetration seal.

3.1.13 Building Surface Penetrations

Sleeves shall not be installed in structural members except where indicated or approved. Except as indicated otherwise piping sleeves shall comply with requirements specified. Sleeves in nonload bearing surfaces shall be galvanized sheet metal, conforming to JIS G 3302. Sleeves in load bearing surfaces shall be uncoated carbon steel pipe, conforming to JIS G 3454 and JIS B 2301, Standard weight. Sealants shall be applied to moisture and oil-free surfaces and elastomers to not less than 13 mm depth. Sleeves shall not be installed in structural members.

3.1.13.1 General Service Areas

Each sleeve shall extend through its respective wall, floor, or roof, and shall be cut flush with each surface. Pipes passing through concrete or masonry wall or concrete floors or roofs shall be provided with pipe sleeves fitted into place at the time of construction. Sleeves shall be of such size as to provide a minimum of 6.35 mm all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, the annular space between pipe

and sleeve or between jacket over-insulation and sleeve shall be sealed.

3.1.13.2 Waterproof Penetrations

Pipes passing through roof or floor waterproofing membrane shall be installed through a 5.17 kg/sq. m copper sleeve, or a 0.81 mm thick aluminum sleeve, each within an integral skirt or flange.

Flashing sleeve shall be suitably formed, and skirt or flange shall extend not less than 200 mm from the pipe and be set over the roof or floor membrane in a troweled coating of bituminous cement. The flashing sleeve shall extend up the pipe a minimum of 50 mm above the roof or floor penetration. The annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation shall be sealed as indicated. Penetrations shall be sealed by either one of the following methods.

- a. Waterproofing Clamping Flange: Pipes up to and including 250 mm in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Waterproofing membrane shall be clamped into place and sealant shall be placed in the caulking recess.
- b. Modular Mechanical Type Sealing Assembly: In lieu of a waterproofing clamping flange, a modular mechanical type sealing assembly may be installed. Seals shall consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Links shall be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut.

After the seal assembly is properly positioned in the sleeve, tightening of the bolt shall cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Each seal assembly shall be sized as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals shall provide sleeves of the proper diameters.

3.1.13.3 Fire-Rated Penetrations

Penetration of fire-rated walls, partitions, and floors shall be sealed.

3.1.13.4 Escutcheons

Finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms, shall be provided with escutcheons. Where sleeves project slightly from floors, special deep-type escutcheons shall be used. Escutcheon shall be secured to pipe or pipe covering.

3.1.14 Access Panels

Access panels shall be provided where indicated for all concealed valves, vents, controls, and additionally for items requiring inspection or maintenance. Access panels shall be of sufficient size and located so

that the concealed items may be serviced and maintained or completely removed and replaced. Access panels shall be as specified in Structural drawings.

3.2 ELECTRICAL INSTALLATION

Install electrical equipment in accordance with NFPA 70 and manufacturers instructions.

3.3 CLEANING AND ADJUSTING

Pipes shall be cleaned free of scale and thoroughly flushed of all foreign matter. A temporary bypass shall be provided for all water coils to prevent flushing water from passing through coils. Strainers and valves shall be thoroughly cleaned. Prior to testing and balancing, air shall be removed from all water systems by operating the air vents. Temporary measures, such as piping the overflow from vents to a collecting vessel shall be taken to avoid water damage during the venting process. Air vents shall be plugged or capped after the system has been vented. Control valves and other miscellaneous equipment requiring adjustment shall be adjusted to setting indicated or directed.

3.4 FIELD TESTS

Field tests shall be conducted in the presence of the QC Manager or his designated representative to verify systems compliance with specifications. Any material, equipment, instruments, and personnel required for the test shall be provided by the Contractor.

3.4.1 Equipment and Component Isolation

Prior to testing, equipment and components that cannot withstand the tests shall be properly isolated.

3.4.2 Pressure Tests

Each piping system shall be hydrostatically tested at a pressure not less than 1297 kPa (gage) for period of time sufficient to inspect every joint in the system and in no case less than 2 hours. Test pressure shall be monitored by a currently calibrated test pressure gauge. Leaks shall be repaired and piping retested until test requirements are met. No leakage or reduction in gage pressure shall be allowed.

Leaks shall be repaired by rewelding or replacing pipe or fittings. Caulking of joints will not be permitted. Concealed and insulated piping shall be tested in place before concealing.

Submit for approval pressure tests reports covering the above specified piping pressure tests; describe the systems tested, test results, defects found and repaired, and signature of the pressure tests' director. Obtain approval from the QC Manager before concealing piping or applying insulation to tested and accepted piping.

3.4.3 Related Field Inspections and Testing

3.4.3.1 Piping Welds

Examination of Piping Welds is specified in the paragraph EXAMINATION OF PIPING WELDS (above).

3.4.3.2 HVAC TAB

Requirements for testing, adjusting, and balancing (TAB) of HVAC water piping, and associated equipment is specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Coordinate with the TAB team, and provide support personnel and equipment as specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC to assist TAB team to meet the TAB work requirements.

3.5 INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the chilled water. Instructors shall be thoroughly familiar with all parts of the installation and shall be instructed in operating theory as well as practical operation and maintenance work. Submit a lesson plan for the instruction course for approval. The lesson plan and instruction course shall be based on the approved operation and maintenance data and maintenance manuals.

Conduct a training course for the operating staff and maintenance staff selected by the Contracting Officer. Give the instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be one man-day. Use approximately half of the time for classroom instruction and the other time for instruction at the location of equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

-- End of Section --

SECTION 23 81 00

DECENTRALIZED UNITARY HVAC EQUIPMENT
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 8607	(2008) Flare Type and Brazing Type Fittings for Refrigerants
JIS B 8615-1	(2013) Non-ducted Air Conditioners and Heat Pumps-Testing and Rating for Performance
JIS B 8616	(2015) Package Air Conditioners
JIS B 9908-1	(2019) Test Method of Air Filter Units for Ventilation and Electric Air Cleaners for Ventilation-Part 1: Technical Specifications, Requirements and Classification System Based Upon Particulate Matter Efficiency
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 9730-1	(2019) Automatic Electrical Controls-Part 1: General Requirements
JIS C 9730-2-9	(2010) Automatic Electrical Controls for Household and Similar Use-Part 2-9: Particular Requirements for Temperature Sensing Controls
JIS G 3302	(2022) Hot Dip Zinc Coated Steel Sheet and Strip
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS Z 2371	(2015) Methods of Salt Spray Testing

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M

(2019) Public Building Construction
Standard Specification

THE JAPAN REFRIGERATION AND AIR CONDITIONING INDUSTRY ASSOCIATION
(JRAIA)

JRA Standard

Japan Refrigeration and Air Conditioning

THE JAPAN ELECTRICAL MANUFACTURERS' ASSOCIATION (JEMA)

JEM 1038

AC Electromagnetic Contactor

JEM 1167

High Voltage Electromagnetic Contactor

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval or for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Field-Assembled Refrigerant Piping

Control System Wiring Diagrams

SD-03 Product Data

Room Air Conditioners

Packaged Terminal Units

Air Conditioners

Training; G

Posted Instructions

Spare Parts

SD-06 Test Reports

Start-Up and Initial Operational Tests

SD-07 Certificates

Service Organizations

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Room Air Conditioners, Data Package 3

Packaged Terminal Units, Data Package 3

Air Conditioners, Data Package 3

Filters, Data Package 2

Thermostats, Data Package 2

Submit in accordance with Section 01 78 23 OPERATION AND
MAINTENANCE DATA.

1.3 QUALITY ASSURANCE

1.3.1 Modification of References

Accomplish work in accordance with the referenced publications, except as modified by this section. Consider the advisory or recommended provisions to be mandatory, as though the word "shall" had been substituted for the words "should" or "could" or "may," wherever they appear. Interpret reference to "the Authority having jurisdiction," "the Administrative Authority," "the Owner," or "the Design Engineer" to mean the Contracting Officer.

1.3.2 Detail Drawing

For refrigerant piping, submit piping, including pipe sizes. Submit control system wiring diagrams.

1.3.3 Safety

Design, manufacture, and installation of unitary air conditioning equipment shall conform to Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

1.3.4 Sizing

Size equipment based on MLIT-M; do not oversize.

1.3.5 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be Refrigerant Handling Technician (Reibai-Furontou-Toriatsukai-Gijutsusha) under Japan Refrigeration and Air-Conditioning Industry Association. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4 REFRIGERANTS

Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0, with the exception of R-123. CFC-based refrigerants are prohibited. HCFCs and Halons shall not be permitted.

1.5 ENVIRONMENTAL REQUIREMENTS

For proper Indoor Environmental Quality, maintain positive pressure within the building. Meet or exceed filter media efficiency as tested in accordance with JIS B 9908-1.

PART 2 PRODUCTS

2.1 ROOM AIR CONDITIONERS

JIS B 8616. Minimum energy efficiency ratio (EER) shall be in accordance with Room Air Conditioners: Provide units removable from inside the building for servicing without removing the outside cabinet. Construct outside cabinets, including metal grilles to protect condenser coils, of zinc-coated steel or aluminum. Steel and zinc-coated surfaces shall receive at least one coat of primer and manufacturer's standard factory-applied finish. Insulate cabinets to prevent condensation and run off of moisture. Provide mounting hardware made of corrosion-resistant material or protected by a corrosion-resistant finish. Provide air filters of the throw-away or permanent washable type removable without the use of tools and arranged to filter both room and ventilating air. Remove condensate by means of a drain or by evaporation and diffusion. Provide with metal or plastic mounting flanges on each side, top, and bottom of unit. For thru-the-wall installations provide aluminum or shop painted zinc-coated steel flanged telescopic wall sleeves. Design wall sleeves to restrict driving rain. For window mounted units provide shop-painted metal mounting brackets, braces, and sill plates. Mount compressors on vibration isolators. Minimum cooling capacity shall be not less than that indicated.

2.1.1 Units for Operation on 100 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords shall have a 15- or 20-amp, 3-pole, 125-volt ground type plug to match receptacle.

2.1.2 Units for Operation on 200 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords shall have a 15-, 20-, or 30-amp, 3-pole, 250-volt or manufacturer's standard ground type plug to match receptacle.

2.1.3 Controls

Mount controls in cabinet. Manual controls shall permit operation of either the fan or the fan and refrigerating equipment. Fan control shall provide two fan speed settings. Automatic controls shall include a thermostat for controlling air temperature. Thermostat shall have an adjustable range, including 18 to 28 degrees C and shall automatically turn the refrigeration system on or off to maintain the preselected temperature within plus or minus 20 degrees C.

2.2 PACKAGED TERMINAL UNITS

2.2.1 Air Conditioners

JIS B 8616, air-cooled, split type. Provide units with cooling only section with indicated capacity. Minimum seasonal energy efficiency ratio (SEER) shall be 13.4 SEER.

2.2.2 Indoor Noise Rating

Rate in accordance with Japanese manufacture's standard ratings. Indoor

rating shall not exceed 85 bels while entire unit is operating at any fan or compressor speed.

2.2.3 Wall Sleeves and Mounts

Provide manufacturer's standard wall sleeves and mounts. Wall sleeves shall have seals designed to restrict driving rain and wind. Provide subbase with 24-volt remote control circuitry and wall mounted thermostat.

2.2.4 Refrigeration Sections

Completely self-contained, slide-in assembly or removable chassis with welded, hermetically sealed, air-cooled refrigeration system, outdoor fan, indoor fan, control box, and ventilation damper. Provide refrigeration sections capable of installation or removal without the use of tools. Refrigeration sections shall include refrigeration circuit tubing, wiring, and safety controls, and shall operate down to 2 degrees C outdoor temperature and 21 degrees C indoor temperature, without compressor short cycling while delivering not less than 100 percent of rated cooling capacity. Units shall have drains to the building exterior to eliminate excess driving rain. Condensate shall not drain onto building exterior or interior.

- a. Compressors: Hermetic type with vibration isolation devices.
- b. Coils: Constructed of seamless copper tubing with copper or aluminum fins bonded to tubes. Coat outdoor air coils with factory applied corrosion resistant treatment. Coils to be coated shall be part of manufacturer's standard product for capacities and ratings indicated and specified. Provide plate type fins.
- c. Outdoor Fans: Direct connected centrifugal type with aluminum or plastic wheel and forward curved blades or direct connected aluminum propeller type. Design fans so that condensate will evaporate without drip, splash, or spray on building exterior.
- d. Indoor Fans: Direct connected centrifugal type with aluminum, galvanized steel, or plastic wheel and forward curved blades. Provide minimum two-speed motor with built-in overload protection.

2.2.5 Air Filters

Removable without use of tools, and shall filter both recirculated and ventilating air.

2.2.6 Controls

Provide controls including, an adjustable thermostat, and switches, to regulate room air temperature through control of refrigerant compressors or heating elements. Controls shall at least have positions for off, high or low fan speed for heating and cooling, and fan only operation. Provide remote mounted night set-back thermostat.

2.3 AIR CONDITIONERS

2.3.1 Split-System Type

Provide separate assemblies designed to be used together. Base ratings on the use of matched assemblies. Units shall have a minimum SEER of 14.1

when tested in accordance with JIS B 8616 or JIS B 8615-1 as applicable. Provide capacity, electrical characteristics and operating conditions as indicated. Condensers shall provide not less than 10 degrees F liquid subcooling at standard ratings.

2.3.2 Single Zone Units

Provide single zone type units arranged to draw through coil sections. Air may be blown or drawn through heating section.

2.3.3 Multizone Units

Provide multizone type units arranged to draw through the cooling and heating sections.

2.3.4 Heaters

Provide as an integral part of the evaporator-blower unit. Provide electric fin tubular heaters.

2.3.5 Compressors

For compressors over 70 kw, compressor speed shall not exceed 3450 rpm. For systems over 35 kw provide automatic capacity reduction of at least 50 percent of rated capacity. Capacity reduction may be accomplished by cylinder unloading, use of multi- or variable speed compressors, use of multiple, but not more than four compressors, or a combination of the two methods. Units with cylinder unloading shall start with capacity reduction devices in the unloaded position. Units with multiple compressors shall have means to sequence starting of compressors. Provide compressors with devices to prevent short cycling when shut down by safety controls. Device shall delay operation of compressor motor for at least 3 minutes but not more than 6 minutes. Provide a pumpdown cycle for units 70 kw and over. Provide reciprocating compressors with crankcase heaters in accordance with the manufacturer's recommendations. If compressors are paralleled, provide not less than two independent circuits.

2.3.6 Coils

On coils with all-aluminum construction, provide tubes of aluminum alloy; provide fins of aluminum alloy and provide tube sheets of aluminum alloy. Provide a separate air cooled condenser circuit for each compressor or parallel compressor installation. Provide a coating on condenser and evaporator coils and fins as specified in the paragraph COATINGS FOR FINNED TUBE COILS. Coils to be coated shall be part of manufacturer's standard product for capacities and ratings indicated and specified. Provide plate type fins.

2.3.7 Condenser Controls

Provide start-up and head pressure controls to allow for system operation at ambient temperatures down to 4 degrees C.

2.3.8 Fans

Provide belt-driven evaporator fans with adjustable pitch pulleys; except for units less than 17 1/2 kw capacity, direct drive with at least two speed taps may be used. Select pulleys at approximately midpoint of the adjustable range.

2.3.9 Filters

Provide filters of the type specified in this section.

2.3.10 Filter Boxes

Provide when filters are not included integral with air conditioning units. Construct of not less than No. 20 US gage steel with track, hinged access doors with latches, and gaskets between frame and filters. Arrange filters to filter outside and return air. Provide removable filter assemblies, replaceable without the use of tools.

2.3.11 Mixing Boxes

Provide of the physical size to match the basic unit and include equal sized flanged openings, sized to individually handle full air flow. Arrange openings as indicated. Provide openings with dampers of parallel or opposed blade type. Provide opposed blade type for modulating dampers and parallel type for two-position dampers. Connect damper shafts together by one continuous linkage bar. Arrange dampers for automatic operation so that when one starts to close from its opened position, the other starts to open from its closed position.

2.3.12 Thermostats

Provide adjustable type that conforms to applicable requirements of JIS C 9730-1 and JIS C 9730-2-9. Provide combination heating-cooling type with contacts hermetically sealed against moisture, corrosion, lint, dust, and foreign material. Design to operate on not more than 0.83 degrees C differential and of suitable range calibrated in degrees C. Provide adjustable heat anticipation and fixed cooling anticipation. Provide two independent temperature sensing elements electrically connected to control the compressor and heating equipment, respectively. Accomplish manual switching for system changeover from heating to cooling or cooling to heating and fan operation through the use of a thermostat subbase. Provide system selector switches to provide "COOL" and "OFF" and "HEAT" and fan selector switches to provide "AUTOMATIC" and "ON." Provide relays, contactors, and transformers located in a panel or panels for replacement and service.

2.3.12.1 Cooling

- a. When thermostat is in "COOL" position with fan selector switch in "AUTO" position, compressor, evaporator fan, and condenser fan shall cycle together.
- b. When thermostat is in "COOL" position with fan selector switch in "ON" position, compressor, and condenser fan shall cycle together and evaporator fan shall run continuously.

2.3.12.2 Heating

- a. When thermostat is in "HEAT" position with fan selector switch in "AUTO" position, heater and supply air fan shall cycle together. Provide a separate thermostat to keep the fan running until the heater cools.
- b. When thermostat is in "HEAT" position with fan selector switch in "ON"

position, heater shall cycle and supply air fan shall run continuously.

2.3.12.3 Supply Air Fan

- a. When fan selector switch is in "AUTO" position with thermostat in "OFF" position, fan shall not run.
- b. When fan selector switch is in "ON" position, fan shall run continuously.

2.4 FILTERS

Provide filters to filter outside air and return air and locate inside combination air filter mixing box. Provide replaceable (throw-away) type. Filters shall conform to JIS B 9908-1. Polyurethane filters shall not be used on units with multiframe filters.

2.4.1 Replaceable Type Filters

Throw-away frames and media, standard dust holding capacity, 1.79 m/s maximum face velocity, and 25 mm thick. Filters shall be in accordance with JIS B 9908-1.

2.4.2 Manometers

Provide inclined-type manometers for filter stations of 944 L/s capacity or larger including filters furnished as integral parts of air-handling units and filters installed separately. Provide sufficient length to read at least 250 Pa with 10 major graduations, and equipped with spirit level. Equip manometers with overpressure safety traps to prevent loss of fluid, and two three-way vent valves for checking zero setting. Mercury shall not be used as the operating fluid.

2.5 COATINGS FOR FINNED TUBE COILS

Where stipulated in equipment specifications of this section, coat finned tube coils of the affected equipment as specified below. Apply coating at the premises of a company specializing in such work. Degrease and prepare for coating in accordance with the coating applicator's procedures for the type of metals involved. Completed coating shall show no evidence of softening, blistering, cracking, crazing, flaking, loss of adhesion, or "bridging" between the fins.

2.5.1 Phenolic Coating

Provide a resin base thermosetting phenolic coating. Apply coating by immersion dipping of the entire coil. Provide a minimum of two coats. Bake or heat dry coils following immersions. After final immersion and prior to final baking, spray entire coil with particular emphasis given to building up coating on sheared edges. Total dry film thickness shall be 0.064 to 0.076 mm.

2.5.2 Chemical Conversion Coating with Polyelastomer Finish Coat

Dip coils in a chemical conversion solution to molecularly deposit a corrosion resistant coating by electrolysis action. Cure conversion coating at a temperature of 43 to 60 degrees C for a minimum of 3 hours. Coat coil surfaces with a complex polymer primer with a dry film thickness of 0.025 mm. Cure primer coat for a minimum of 1 hour. Using dip tank

method, provide three coats of a complex polyelastomer finish coat. After each of the first two finish coats, cure the coils for 1 hour. Following the third coat, spray a fog coat of an inert sealer on the coil surfaces. Total dry film thickness shall be 0.064 to 0.076 mm. Cure finish coat for a minimum of 3 hours. Coating materials shall have 300 percent flexibility, operate in temperatures of minus 46 to plus 104 degrees C, and protect against atmospheres of a pH range of 1 to 14.

2.5.3 Vinyl Coating

Apply coating using an airless fog nozzle. For each coat, make at least two passes with the nozzle. Materials to be applied are as follows:

Total dry film thickness, 0.165 mm maximum.

Vinyl Primer, 24 percent solids by volume: One coat 0.051 mm thick

Vinyl Copolymer, 30 percent solids by volume: One coat 0.114 mm thick.

2.6 MOTORS AND STARTERS

JIS C 4212 or JEM 1038, and JEM 1167. Variable speed. Motors less than 3/4 kw shall meet JIS C 4212 requirements. Motors 3/4 kw and larger shall meet JIS C 4212 requirements. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters. Provide motors to operate at full capacity with a voltage variation of plus or minus 10 percent of the motor voltage rating. Motor size shall be sufficient for the duty to be performed and shall not exceed its full load nameplate current rating when driven equipment is operated at specified capacity under the most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, the Contractor shall make the necessary adjustments to the wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide reduced voltage type motor starters. Provide watertight type starter enclosures.

2.7 REFRIGERANT PIPING AND ACCESSORIES

Provide accessories as specified in JIS B 8615-1 and this section. Provide suction line accumulators as recommended by equipment manufacturer's installation instructions. Provide a filter-drier in the liquid line.

2.7.1 Factory Charged Tubing

Provide extra soft, deoxidized, bright annealed copper tubing conforming to JIS H 3300, factory dehydrated and furnished with a balanced charge of refrigerant recommended by manufacturer of equipment being connected. Factory insulate suction line tubing with 9.52 mm minimum thickness of closed cell, foamed plastic conforming to manufacturer's standard close cell foamed plastic material with a permeance rating not to exceed 1.0. Provide quick-connectors with caps or plugs to protect couplings. Include couplings for suction and liquid line connections of the indoor and outdoor sections.

2.7.2 Field-Assembled Refrigerant Piping

Material and dimensional requirements for field-assembled refrigerant piping, valves, fittings, and accessories shall conform to in accordance

with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese product and JIS H 3300 and JIS B 8607, except as herein specified. Factory clean, dehydrate, and seal piping before delivery to the project location. Provide seamless copper tubing, hard drawn, Type K or L, conforming to JIS H 3300, except that tubing with outside diameters of 6.35 mm and 9.52 mm shall have nominal wall thickness of not less than 7.62 mm and 0.81 mm, respectively. Soft annealed copper tubing conforming to JIS H 3300 may be used where flare connections to equipment are required only in nominal sizes less than one inch outside diameter.

2.7.3 Fittings

JIS H 3401 for solder-joint fittings. Recommended by manufacturer for flared tube fittings.

2.7.4 Pipe Hangers and Supports

Pipe hangers and supports shall be in accordance with MLIT-M.

2.7.5 Pipe Sleeves

Provide sleeves where piping passes through walls, floors, roofs, and partitions. Secure sleeves in proper position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, floors, roofs, and partitions. Provide not less than 6.35 mm space between exterior of piping or pipe insulation and interior of sleeve. Firmly pack space with insulation and caulk at both ends of the sleeve with plastic waterproof cement which will dry to a firm but pliable mass, or provide a segmented elastomeric seal.

2.7.5.1 Sleeves in Masonry and Concrete Walls, Floors, and Roofs

Provide Schedule 40 or Standard Weight zinc-coated steel pipe sleeves. Extend sleeves in floor slabs 80 mm above finished floor.

2.7.5.2 Sleeves in Partitions and Non-Masonry Structures

Provide zinc-coated steel sheet sleeves having a nominal weight of not less than 4.39 kg per square meter, in partitions and other than masonry and concrete walls, floors, and roofs.

2.8 FINISHES

Provide steel surfaces of equipment including packaged terminal units, heat pumps, and air conditioners, that do not have a zinc coating conforming to JIS H 8641, JIS G 3302 or a duplex coating of zinc and paint, with a factory applied coating or paint system. Provide a coating or paint system on actual equipment identical to that on salt-spray test specimens with respect to materials, conditions of application, and dry-film thickness.

2.9 SOURCE QUALITY CONTROL

2.9.1 Salt-Spray Tests

Salt-spray test the factory-applied coating or paint system of equipment including packaged terminal units, heat pumps, and air conditioners in accordance with JIS Z 2371.

2.10 MANUFACTURER'S STANDARD NAMEPLATES

Major equipment including compressors, condensers, receivers, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life. Plates must be fixed in prominent locations.

PART 3 EXECUTION

3.1 EQUIPMENT INSTALLATION

Install equipment and components in a manner to ensure proper and sequential operation of equipment and equipment controls. Install equipment not covered in this section, or in manufacturer's instructions, as recommended by manufacturer's representative. Provide proper foundations for mounting of equipment, accessories, appurtenances, piping and controls including, but not limited to, supports, vibration isolators, stands, guides, anchors, clamps and brackets. Foundations for equipment shall conform to equipment manufacturer's recommendation, unless otherwise indicated. Set anchor bolts and sleeves using templates. Provide anchor bolts of adequate length, and provide with welded-on plates on the head end embedded in the concrete. Level equipment bases, using jacks or steel wedges, and neatly grout-in with a nonshrinking type of grouting mortar. Locate equipment to allow working space for servicing including shaft removal, disassembling compressor cylinders and pistons, replacing or adjusting drives, motors, or shaft seals, access to water heads and valves of shell and tube equipment, tube cleaning or replacement, access to automatic controls, refrigerant charging, lubrication, oil draining and working clearance under overhead lines. Provide electric isolation between dissimilar metals for the purpose of minimizing galvanic corrosion.

3.1.1 Packaged Terminal Air Conditioners and Heat Pumps

Wall sleeve installation shall provide a positive weathertight and airtight seal.

3.1.2 Unitary Air Conditioning System

Install as indicated, in accordance with requirements in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese product, and the manufacturer's installation and operational instructions.

3.1.3 Room Air Conditioners

Install units in accordance with manufacturer's instructions. Provide structural mountings, closures, and seals for weathertight assembly. Pitch unit as recommended by manufacturer to ensure condensate drain to drain pan without overflow.

3.2 PIPING

Brazing, bending, forming and assembly of refrigerant piping shall conform to JIS H 3300 and JIS B 8607.

3.2.1 Pipe Hangers and Supports

Design and fabrication of pipe hangers, supports, and welding attachments shall conform to MLIT-M. Installation of hanger types and supports for bare and covered pipes shall conform to MLIT-M. Unless otherwise indicated, horizontal and vertical piping attachments shall conform to MLIT-M.

3.2.2 Refrigerant Piping

Cut pipe to measurements established at the site and work into place without springing or forcing. Install piping with sufficient flexibility to provide for expansion and contraction due to temperature fluctuation. Where pipe passes through building structure pipe joints shall not be concealed, but shall be located where they may be readily inspected. Install piping to be insulated with sufficient clearance to permit application of insulation. Install piping as indicated and detailed, to avoid interference with other piping, conduit, or equipment. Except where specifically indicated otherwise, run piping plumb and straight and parallel to walls and ceilings. Trapping of lines will not be permitted except where indicated. Provide sleeves of suitable size for lines passing through building structure. Braze refrigerant piping with silver solder. Inside of tubing and fittings shall be free of flux. Clean parts to be jointed with emery cloth and keep hot until solder has penetrated full depth of fitting and extra flux has been expelled. Cool joints in air and remove flame marks and traces of flux. During brazing operation, prevent oxide film from forming on inside of tubing by slowly flowing dry nitrogen through tubing to expel air. Make provisions to automatically return oil on halocarbon systems. Installation of piping shall comply with JIS H 3300 and JIS B 8607.

3.2.3 Returning Oil From Refrigerant System

Install refrigerant lines so that gas velocity in the evaporator suction line is sufficient to move oil along with gas to the compressor. Where equipment location requires vertical risers, line shall be sized to maintain sufficient velocity to lift oil at minimum system loading and corresponding reduction of gas volume. Install a double riser when excess velocity and pressure drop would result from full system loading. Larger riser shall have a trap, of minimum volume, obtained by use of 90- and 45-degree ells. Arrange small riser with inlet close to bottom of horizontal line, and connect to top of upper horizontal line. Do not install valves in risers.

3.2.4 Refrigerant Driers, Sight Glass Indicators, and Strainers

Provide refrigerant driers, sight glass liquid indicators, and strainers in refrigerant piping in accordance with this section when not furnished by the manufacturer as part of the equipment. Install driers in liquid line with service valves and valved bypass line the same size as liquid line in which dryer is installed. Size of driers shall be determined by piping and installation of the unit on location. Install dryers of 820 mL and larger vertically with the cover for removing cartridge at the bottom. Install moisture indicators in the liquid line downstream of the drier. Indicator connections shall be the same size as the liquid line in which it is installed. These devices shall be provided as optional accessories.

3.2.5 Strainer Locations and Installation

Locate strainers close to equipment they are to protect. Provide a strainer in common refrigerant liquid supply to two or more thermal valves in parallel when each thermal valve has a built-in strainer. Install strainers with screen down and in direction of flow as indicated on strainer's body.

3.2.6 Solenoid Valve Installation

Install solenoid valves in horizontal lines with stem vertical and with flow in direction indicated on valve. If not incorporated as integral part of the valve, provide a strainer upstream of the solenoid valve. Provide service valves upstream of the solenoid valve, upstream of the strainer, and downstream of the solenoid valve. Remove the internal parts of the solenoid valve when brazing the valve.

3.3 AUXILIARY DRAIN PANS, DRAIN CONNECTIONS, AND DRAIN LINES

Provide auxiliary drain pans under units located above finished ceilings or over mechanical or electrical equipment where condensate overflow will cause damage to ceilings, piping, and equipment below. Provide separate drain lines for the unit drain and auxiliary drain pans. Trap drain pans from the bottom to ensure complete pan drainage. Provide drain lines full size of drain opening. Traps and piping to drainage disposal points shall conform to Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.4 ACCESS PANELS

Provide access panels for concealed valves, controls, dampers, and other fittings requiring inspection and maintenance.

3.5 AIR FILTERS

Allow access space for servicing filters. Install filters with suitable sealing to prevent bypassing of air. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed.

3.6 IDENTIFICATION TAGS AND PLATES

Provide equipment, gages, thermometers, valves, and controllers with tags numbered and stamped for their use. Provide plates and tags of brass or suitable nonferrous material, securely mounted or attached. Provide minimum letter and numeral size of 3.18 mm high.

3.7 FIELD QUALITY CONTROL

3.7.1 Leak Testing

Upon completion of installation of air conditioning equipment, test factory- and field-installed refrigerant piping with an electronic-type leak detector. Use same type of refrigerant to be provided in the system for leak testing. When nitrogen is used to boost system pressure for testing, ensure that it is eliminated from the system before charging. Minimum refrigerant leak field test pressure shall be as specified in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. If leaks are detected at

time of installation or during warranty period, remove the entire refrigerant charge from the system, correct leaks, and retest system.

3.7.2 Evacuation, Dehydration, and Charging

After field charged refrigerant system is found to be without leaks or after leaks have been repaired on field-charged and factory-charged systems, evacuate the system using a reliable gage and a vacuum pump capable of pulling a vacuum of at least 133 Pa absolute. Evacuate system in accordance with the triple-evacuation and blotter method or in accordance with equipment manufacturer's printed instructions and recharge system.

3.7.3 Start-Up and Initial Operational Tests

Test the air conditioning systems and systems components for proper operation. Adjust safety and automatic control instruments as necessary to ensure proper operation and sequence. Conduct operational tests for not less than 8 hours.

3.7.4 Performance Tests

Upon completion of evacuation, charging, startup, final leak testing, and proper adjustment of controls, test the systems to demonstrate compliance with performance and capacity requirements. Test systems for not less than 8 hours, record readings hourly. At the end of the test period, average the readings, and the average shall be considered to be the system performance.

3.8 TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.
- b. Submit the field posted instructions, at least 2 weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.
- c. The posted instructions must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations. Submit 6 complete copies of an operation manual in bound 216 by 279 booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief

description of all equipment and their basic operating features.

- d. Submit 6 complete copies of maintenance manual in bound 216 by 279 mm booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

3.9 MAINTENANCE

3.9.1 Extra Materials

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

3.9.2 Maintenance Service

Submit a certified list of qualified permanent service organizations, which includes their addresses and qualifications, for support of the equipment. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

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SECTION 23 81 23

COMPUTER ROOM AIR CONDITIONING UNITS
11/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 410 (2001; Addendum 1 2002; Addendum 2 2005; Addendum 3 2011) Forced-Circulation Air-Cooling and Air-Heating Coils

AHRI 1360 (2017) Performance Rating of Computer and Data Processing Room Air Conditioners

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASHRAE 90.1 - SI (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 127 (2012) Method of Testing for Rating Computer and Data Processing Room Unitary Air-Conditioners

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1 (2024) Power Piping

ASME B31.5 (2022) Refrigeration Piping and Heat Transfer Components

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C1071 (2019) Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)

ASTM C1338 (2019) Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings

ASTM D5864 (2011) Standard Test Method for Determining Aerobic Aquatic Biodegradation

of Lubricants or Their Components

ASTM D6081 (1998; R 2014) Aquatic Toxicity Testing of Lubricants: Sample Preparation and Results Interpretation

ASTM G21 (2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

ETL TESTING LABORATORIES (ETL)

ETL DLP (updated continuously) ETL Listed Mark Directory

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-1424 (2016) Engineering and Design -- Lubricants and Hydraulic Fluids

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 82 Protection of Stratospheric Ozone

UL SOLUTIONS (UL)

UL 181 (2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors

UL Elec Equip Dir (2011) Electrical Appliance and Utilization Equipment Directory

1.2 DEFINITIONS

Computer Room Air Conditioner (CRAC): A single, self-contained unit or split-system unit designed and manufactured specifically for temperature and humidity control of data processing environments.

Cold Aisle: The aisle between or adjacent to rows of racks from which the computing equipment draws cool air.

Hot Aisle: The aisle between or adjacent to rows of racks to which the computing equipment ejects hot air.

Rack: Telecommunications support frame that can consist of post-and-frame or full cabinet construction. Racks are provided under Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Computer Room Air Conditioner; G
Space Temperature Control System Drawings; G
Filters
Refrigerants; S
Leak Detection; G

SD-06 Test Reports

Manufacturer's Factory Test Plans; G
Factory Test Reports; G
Field Test Schedule; G
Manufacturer's Field Test Plans; G
Field Test Reports; G

SD-07 Certificates

Credentials of the Manufacturer's Field Test Representative; G
Ozone Depleting Substances Technician Certification
Certified List Of Qualified Permanent Service Organizations
Seismic Certification; G

SD-08 Manufacturer's Instructions

Installation Manual for Each Type of CRAC

SD-10 Operation and Maintenance Data

Computer Room Air Conditioner Operation and Maintenance Data, Data Package 4; G

SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

1.4 REFRIGERANTS

Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0. CFC-based refrigerants are prohibited. HCFCs and Halons are not

permitted. Provide SDS sheets for all refrigerants.

1.5 QUALIFICATIONS

1.5.1 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.6 QUALIFICATIONS

1.6.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design, and workmanship. Standard products must have been in satisfactory commercial or industrial use for two years prior to bid opening. The two-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the two-year period.

1.6.2 Alternative Equipment Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.6.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.6.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.6.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

1.6.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.7 PROJECT REQUIREMENTS

1.7.1 Verification of Dimensions

Become familiar with the details of the work, verify all dimensions in the field, and provide adequate clearance for all connections and service access. Notify the Contracting Officer of any discrepancy before performing any work.

1.7.2 Energy Efficiency

Provide equipment with minimum efficiencies as required by ASHRAE 90.1 - SI.

1.8 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

PART 2 PRODUCTS

2.1 COMPUTER ROOM AIR CONDITIONER (CRAC)

Provide complete working CRACs, designed, factory assembled, and factory tested. Equipment must be listed in UL Elec Equip Dir or ETL DLP for computer room application. CRACs must have a minimum sensible coefficient of performance of 2.73 in accordance with ASHRAE 127. CRACs must include room cabinet and frame, floor stand, fan section, filter section, cooling coil, controls, and, interconnecting piping internal to the CRAC. Provide units rated in accordance with AHRI 1360.

2.1.1 Unit Airflow Configuration

2.1.1.1 Downflow Units

The CRAC must draw return air in at the top of the cabinet and discharge

supply air at the bottom of the cabinet.

2.1.2 Cabinet and Frame

2.1.2.1 Unit Frame

Unit frame must be manufactured of welded steel tubes and must be mill-galvanized or coated with an epoxy finish.

2.1.2.2 Unit Cabinet

Exterior panels must be steel sheet, minimum of 1.0 mm, mill-galvanized or coated with a corrosion-inhibiting epoxy finish in manufacturer's standard color. Mill galvanized sheet metal must be coated with not less than 380 gram of zinc per square meter of two-sided surface. Mill rolled structural steel must be hot-dip galvanized or primed and painted. Cut edges, burns and scratches in hot-dip galvanized surfaces must be coated with galvanizing repair coating. Manufacturer's standard cabinet materials and finishes will be acceptable if equivalent to the above requirements and approved by the Contracting Officer.

Provide removable panel for access to controls without interrupting airflow. Panels must be gasketed to prevent air leakage under system operating pressure and must be removable for service access without the use of special tools.

2.1.2.3 Cabinet Interiors Sound Attenuation

CRAC cabinet panels interior must be provided with 25 millimeters of 24 kilogram per cubic meter fiber glass insulation on interior of cabinet panels. Insulation must be applied to the cabinet panels with 100 percent adhesive coverage and both the insulation and the adhesive must conform to NFPA 90A. Insulation must be rated for 6000 fpm per UL 181 and ASTM C1071. Insulation must resist the growth of microorganisms per ASTM C1338 and ASTM G21.

CRAC cabinet panels interior must be provided with minimum 50 millimeters thick acoustical sound absorbing foam with a minimum Noise Reduction Coefficient (NRC) of 0.85.

Fans and compressors located in the CRAC interior cabinet must be provided with vibration isolators between their respective support frames and the cabinet framing.

CRAC manufacturer's standard interior cabinet sound attenuation materials and finishes will be acceptable if equivalent to the above requirements and approved by the Contracting Officer.

2.1.3 Fan Section

Provide fan(s) and fan motor(s) as integral, factory installed components of the CRAC.

2.1.3.1 Fan Wheel

The supply air fan must be AMCA certified. Provide steel or aluminum, backward curved, plenum/plug type fan wheel. The fan must be statically and dynamically balanced. The fan must have self-aligning, permanently lubricated ball bearings with a minimum life span of 100,000 hours.

Assess potential effects of lubricant on aquatic organisms in accordance with ASTM D6081 and submit aquatic toxicity reports. Assess biodegradation in accordance with ASTM D5864. In accordance with EM 1110-2-1424 Chapter 8, aquatic toxicity must exceed 1,000 ppm at LL50 and biodegradation must exceed 60 percent conversion of carbon to carbon dioxide in 28 days.

2.1.3.2 Motor and Drive

Provide fan wheel directly coupled to motor shaft.

Provide electronically commutated motor with integrated electronic control board and direct microprocessor control signaling for speed control.

Provide variable frequency drive(s) in accordance with Section 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS.

2.1.4 Cooling Coil

Provide AHRI 410 coil and slope for drainage. Coil must be manufactured of seamless copper tubes with plate aluminum fins. Each coil, in the production process, must be individually tested at 2200 kPa with compressed air under water and verified to be air tight. Factory dehydrate and seal each coil after testing and prior to evaluation and charging. Provide hydronic coils complete with drain and vent connections.

Provide condensate drain pan of minimum 1.0 millimeter Type 304 stainless steel with nonferrous connections, internal trap, and a condensate pump system complete with integral pump discharge check valve, integral float switch, reservoir, and pump and motor assembly.

2.1.5 Filters

Provide UL listed 50 mm thick deep pleated fiberglass throwaway type filters. Provide filtration media with a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2. Provide one complete spare filter bank set per unit for installation prior to final acceptance testing covered in Part 3 of this section.

2.1.6 Floorstand

Provide a 600 mm high seismic rated floorstand for each CRAC for freestanding installation on the main building structural floor. Floorstand must elevate the unit to the height of the raised computer floor and must allow for leveling and locking at the desired height. Floorstand must be retractable, or removable, for installing the unit directly on the raised floor. Unit must be fully gasketed (rubber or neoprene) to prevent air leakage at the raised floor penetration. Provide radiused turning vane integral to floorstand.

For units requiring seismic certification, the floorstand must be included in the unit certification.

2.2 INSTRUMENTATION AND CONTROLS

All controls provided under this section must comply with the requirements of Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM.

2.2.1 Unit Level Controls

Provide factory installed components and wiring to control a unit's basic functions and space ambient conditions at one factory installed and tested station. Controller modules must provide automatic centralized control of computer room critical equipment, simplifying emergency switching and unit testing. When the module recognizes an alarm condition, it must automatically switch to a stand-by device. User must be able to program a switching delay to allow time to correct emergency conditions. Provide modules with capability to balance the runtime of all connected air units. Provide clear, simplified instructions for programming and configuration of controllers, minimizing the chances of operator error. Provide an electronic temperature and humidity recorder, integral or external to the unit, readable to specified control accuracy, complete with supplies required for one year of operation. Controls must include a control system interface to an HVAC control system. The control system interface must meet DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Unit controls must comply with the requirements of Section 25 05 11.23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM.

2.2.1.1 Display Panel

Provide color LED touchscreen display with graphical menu navigation. Display panel must include the following minimum data: power on, power off, unit in alarm, description of alarm, filter status, room temperature, room relative humidity, event log, service contact information, and unit run hours. Display must have capability to set up password protection.

Provide the following minimum externally accessible controls at the unit: start and stop total system functions, silence audible alarm, main power disconnect.

2.2.1.2 Alarms

Display alarms on unit display panel. Alarm for the following: high and low space temperature, high and low space humidity, dirty filters, loss of airflow, loss of water flow, compressor high head pressure, custom alarms as indicated on the controls drawings, and leak detection. Provide field accessible local audible alarm with silence pushbutton. Provide push-to-test lamps or all-lamp test pushbutton. CRACs must have local devices which provide signals for remote audible and visual alarming capability for the above specified alarm conditions.

2.2.1.3 Leak Detection

Provide rope moisture detection system for each computer room. Leak detection must be designed for installation on the subfloor below the raised floor of the computer room. Leak detection system must interface with the associated CRAC control panel to alarm upon detection of moisture on the subfloor.

2.2.1.4 Factory Wired Components

Provide factory installed and wired chilled water valve. Valve must meet the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC.

Provide CRAC manufacturer's remote room temperature sensor and room humidity sensor. Sensors must meet the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC.

Provide factory wired discharge air temperature sensor. Sensors must meet the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC.

2.2.2 Supervisory CRAC Controls

In addition to stand alone controls, provide device to network together all CRACs in each computer room. The network device must integrate all data for each CRAC, as required under stand alone controls, and display it on any connected CRAC's display panel. The network device must balance runtime across all connected units. The network device must automatically switch to a standby unit upon detection of failure of a duty unit. Provide all control wiring among CRACs and network devices as required to meet this specification.

2.2.3 Integration to HVAC control system and Basewide Utility Monitoring and Control System (UMCS)

Integrate CRAC control into the HVAC control system defined in Section 23 09 23.02 BACNET DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Refer to controls drawings for minimum points required to interface with the HVAC control system and UMCS.

2.3 FACTORY PAINTING SYSTEMS

Provide manufacturer's standard factory painting. Certify that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 3 mm on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. The factory painting system must be designed for the anticipated temperature service.

2.4 ELECTRICAL

Provide an integral electrical panel of similar construction to the unit cabinet. Within the electrical panel, provide a single point power connection terminal block and fused disconnect switch. The electrical panel must provide at least 65,000 amp Short Circuit Current Rating (SCCR).

2.4.1 Electrical Motors, Controllers, Contactors, and Disconnects

Provide motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide internal wiring

for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors must have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls provided. When motors and equipment provided are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.4.2 Electrical Control Wiring

Provide control wiring under Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Provide Space temperature control system drawings which include point-to-point electrical wiring diagrams.

2.5 HVAC WATER PIPING

Requirements for HVAC water piping and metal ductwork are specified in Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS.

2.6 FIRE PROTECTION DEVICES

The requirements for duct smoke detectors are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

2.7 SOURCE QUALITY CONTROL

Provide factory test plan, factory test schedule, factory test and factory test report on each of the CRACs.

2.7.1 Manufacturer's Factory Test Plans

For each CRAC, submit a factory test plan which when followed during factory testing must verify that the performance scheduled on the drawings is met by the produced CRAC models.

The manufacturer must perform factory tests on the actual CRACs produced for this project. The test reports must document the performance tests conducted on the factory assembled computer room air conditioning units. Performance testing on the individual computer room air conditioning unit components, not factory assembled, is not acceptable.

Submit the required test plans for review and approval to the Contracting Officer at least 90 calendar days before scheduled factory test date.

2.7.1.1 Test Procedure

Indicate in each test plan the factory acceptance test procedures. Procedures must be structured to test all modes of operation to confirm that the controls are performing in accordance with the intended sequence of control.

Controllers must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

2.7.1.2 Performance Variables

Each test plan must list performance variables that are required to be measured or tested as part of the field test. Include in the performance variables list the performance indicated on the equipment schedules on the contract design drawings.

Manufacturer must provide with each test procedure a description of acceptable performance results that must be verified. Manufacturer must identify the acceptable limits or tolerances within which each tested performance variable must acceptably operate.

2.7.1.3 Test Configuration

Plans must indicate that tests are to be performed for a minimum of four continuous hours. If test period is interrupted, the four hour test period must be started over. Each test plan must be job specific and must address the particular CRACs and particular conditions which exist with this contract. Generic or general preprinted test procedures are not acceptable.

2.7.1.4 Tested Variables

Plans must provide for air side testing which includes verification of the airflow, total static pressure; fan drive motor KW, amperage and RPM; and fan RPM. Provide entering air temperatures equal to those indicated on the CRAC schedules.

2.7.1.5 Thermal Testing

Plans must provide thermal testing utilizing chilled water with temperatures equal to those indicated on the CRAC schedules. Thermal testing must verify CRAC heating, sensible cooling, total cooling, and humidifying performance scheduled on the contract drawings.

2.7.1.6 Specialized Components

Include procedures for field testing and field adjusting specialized components, such as hot gas bypass control valves, or pressure valves.

2.7.1.7 Factory Test For Sound Pressure Level

Determine the A-weighted sound pressure level for the indoor portion of each of the CRACs.

Each unit must be mounted on a raised floor duplicating of the installation configuration indicated on the contract drawings. Unit must be located at least 1.5 meters 150 mm from test room walls. No other equipment must be operating in the test room during sound level testing of subject unit. Background sound levels must be at least 10 dB below lowest sound pressure level measured on subject unit. Testing must be conducted by using an ANSI Type 1 or 2 sound level meter located 1.0 meter from the unit under test and 1.0 meter above raised floor. Measure and record A-weighted sound pressure level on all four sides of unit.

2.7.1.8 Factory Tests Reporting Forms

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives. Submit factory test reports, referencing each tested CRAC serial number, and receive approval before delivery of CRAC to the project site.

2.7.2 Factory Tests

Conduct the factory testing in compliance with the Contracting Officer approved manufacturer's field test plan, and in accordance with additional field testing requirements specified herein. Record the required data using the test reporting forms approved of the approved field test plan. Conduct the test for each CRAC for the continuous test period in the approved test plan. A CRAC shutdown before the continuous test period is completed must result in the test period being started again and run for the required duration.

2.7.3 Deficiency Resolution

The test requirements must be acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations and corrections tested as specified in the paragraph MANUFACTURER'S FACTORY TEST PLANS.

2.7.4 Factory Test Reports

Use the test reporting forms approved in the factory test plan. Final test report forms must be typed including data entries and remarks. Completed test report forms for each CRAC must be reviewed, approved, and signed by the Manufacturer's test director.

2.8 SEISMIC REQUIREMENTS

CRAC units must be seismically certified in accordance with the requirements in Section 23 05 48.19 SEISMIC BRACING FOR HVAC. Provide seismic bracing in accordance with Section 23 05 48.19 SEISMIC BRACING FOR HVAC.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 CRAC System

Installation of each CRAC system including equipment, materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing, must be in accordance with ASME B31.1, ASME B31.5, NFPA 70, as modified and supplemented by the requirements of this section and the CRAC manufacturer's written installation instructions.

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

3.1.2 Installation Instructions

Provide a manufacturer's installation manual for each type of CRAC.

3.1.3 Operation and Maintenance Data

Submit Computer Room Air Conditioner Operation and Maintenance Data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

3.1.4 Connections to Existing Systems

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Provide materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Provide labor and tools for making actual connections to existing systems.

3.2 FIELD QUALITY CONTROL

Upon completion and before final acceptance of work, test each CRAC subsystem in service to demonstrate compliance with the contract requirements, including field testing specified below. Adjust controls and balance systems prior to final acceptance of completed systems. Test controls through every cycle of operation. Test safety controls to demonstrate performance of required function. Correct defects in work provided and repeat tests. Provide steam, fuel, water, electricity, instruments, connecting devices, and personnel for tests. Flush and clean piping before placing in operation. Clean equipment, piping, strainers, and ducts. Prior to commencement of field testing, remove all filters and provide new filters. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided.

3.3 FIELD TESTING

Provide field test plan, field test schedule, field test and field test report on each of the CRACs. Field test each CRAC for Contracting Officer acceptance in accordance with the CRAC manufacturer's approved field test plan.

3.3.1 Manufacturer's Field Test Plans

Submit field test plans developed by the manufacturer for each CRAC; submit the field test plans at least 90 calendar days prior to planned date of the field test. Field test plans developed by the installing Contractor, or the equipment sales agency furnishing the CRAC, will not be acceptable.

The Contracting Officer will review and approve the field test plan for each of the listed CRACs prior to commencement of field testing of the equipment. The approved field test plans must be followed for the field tests of the CRAC and test reporting.

3.3.1.1 Coordinated Testing

Indicate in each field test plan when work required by this section requires coordination with test work required by other specification sections. Provide test procedures for the simultaneous or integrated testing of: CRAC controls which interlock and interface with controls factory prewired; and external controls for the CRAC provided under

Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

3.3.1.2 Prerequisite Testing

Each CRAC for which performance testing is dependent upon the completion of the work covered by Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC must have that work completed as a prerequisite to testing work under this section. Indicate in each field test plan when such prerequisite work is required.

3.3.1.3 Test Procedure

Indicate in each field test plan the CRAC manufacturer's published start-up, and field acceptance test procedures. Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

Procedures must be structured to test the controls through all modes of control to confirm that the controls are performing with the intended sequence of control.

Controllers must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

3.3.1.4 Performance Variables

Each test plan must list performance variables that are required to be measured or tested as part of the field test.

Include, in the listed performance variables, requirements indicated on the CRAC schedules on the design drawings. Manufacturer must provide, with each test procedure, a description of acceptable results that have been verified.

Manufacturer must identify the acceptable limits or tolerances within which each tested performance variable must acceptably operate.

3.3.1.5 Test Configuration

Plans must indicate that tests are to be performed for a minimum of four continuous hours in a wet coil condition. If test period is interrupted, the four hour test period must be started over. Each test plan must be job specific and must address the particular CRACs and particular conditions which exist with this contract. Generic or general preprinted test procedures are not acceptable. Tests must include a pressurized raised floor discharge configuration at the specified or indicated height above the floor, with or without the air discharge elbows;

3.3.1.6 Tested Variables

Plans must provide for air side testing which includes verification of the airflow, total static pressure; fan drive motor KW, amperage and RPM; and fan RPM. Provide entering air temperatures equal to those indicated on the CRAC schedules.

3.3.1.7 Thermal Testing

Plans must provide thermal testing utilizing chilled water with temperatures equal to those indicated on the CRAC schedules. Thermal

testing must verify CRAC heating, sensible cooling, total cooling, and humidifying performance scheduled on the contract drawings.

3.3.1.8 Specialized Components

Include procedures for field testing and field adjusting specialized components, such as hot gas bypass control valves, or pressure valves.

3.3.1.9 Field Test Reporting Forms

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives.

3.3.2 Field Test Schedule

Notify the Contracting Officer in writing at least 30 calendar days prior to the testing. Within 30 calendar days after acceptable completion of testing, submit each test report for the review and approval of the Contracting Officer.

3.3.3 Manufacturer's Test Representative

Provide a factory trained field test representative authorized by the CRAC manufacturer to oversee the complete execution of the field testing. This test representative must also review, approve, and sign the completed field test report. Signatures must be accompanied by the person's name typed.

Submit credentials of the manufacturer's field test representative proposed, including current telephone number, to the Contracting Officer for review and approval. Submit these credentials with the written advance notice of the field tests.

3.3.4 Field Tests

Conduct the field testing in compliance with the Contracting Officer approved manufacturer's field test plan, and in accordance with additional field testing requirements specified herein. Record the required data using the test reporting forms approved of the approved field test plan. Conduct the test for each CRAC for a continuous 24-hour test period. A CRAC shutdown before the continuous 24-hour test period is completed must result in the 24-hour test period being started again and run for the required duration.

3.3.5 Deficiency Resolution

The test requirements must be acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations. Corrections must be tested again in compliance with the requirements specified in the paragraph MANUFACTURER'S FIELD TEST PLANS.

3.3.6 Field Test Reports

Use the test reporting forms approved in the field test plan. Final test report forms must be typed, including data entries and remarks. Completed test report forms for each CRAC must be reviewed, approved, and signed by the Contractor's test director and the QC manager.

3.4 INSTRUCTION TO GOVERNMENT PERSONNEL

Provide the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. Provide 4 hours of training for each type of CRAC specified.

-- End of Section --

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SECTION 25 05 11.21

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - FIRE AND LIFE SAFETY
11/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

DTM 08-060

(2008) Policy on Use of Department of
Defense (DoD) Information Systems -
Standard Consent Banner and User Agreement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Wireless Communication Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

SD-02 Shop Drawings

User Interface Banner Schedule; G

Network Communication Report; G

Control System Inventory Report; G

Cybersecurity Interconnection Schedule; G

SD-03 Product Data

Control System Cybersecurity; G

SD-11 Closeout Submittals

Password Summary Report; G

Device Audit Record Upload Software; G

1.3 CYBERSECURITY DOCUMENTATION

1.3.1 Cybersecurity Interconnection Schedule

Provide a completed Cybersecurity Interconnection Schedule documenting connections between the installed system and other systems. See Attachment 25 05 11.21-A.

1.3.2 Network Communication Report

Provide a network communication report. See Attachment 25 05 11.21-B.

1.3.3 Control System Inventory Report

Provide a Control System Inventory report. See Attachment 25 05 11.21-C.

1.4 SOFTWARE UPDATE LICENSING

In addition to all other licensing requirements, all software licensing must include licensing of the following software updates for a period of no less than 5 years:

- a. Security and bug-fix patches issued by the software manufacturer.
- b. Security patches to address any vulnerability identified in the National Vulnerability Database at <http://nvd.nist.gov> with a Common Vulnerability Scoring System (CVSS) severity rating of MEDIUM or higher.

1.5 CYBERSECURITY DURING CONSTRUCTION

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.5.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.5.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.5.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.5.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a

minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers that will be connected to network controllers during construction. See Attachment 25 05 11.21-D.

1.5.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.5.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.5.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification

1.5.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.5.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. If no temporary IP networks will be used, provide a single copy of the Statement indicating this. See Attachment 25 05 11.21 -E.

1.6 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

(NOT USED)

PART 3 EXECUTION

3.1 ACCESS CONTROL REQUIREMENTS

3.1.1 User Accounts

3.1.1.1 Default Requirements for Control System Devices

For control system devices where User Account requirements are not otherwise indicated in this Section:

- a. Devices with web interfaces must support user accounts (such as "admin", "user 1", "user 2") or have their web interface disabled.

3.1.2 Unsuccessful Logon Attempts

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts.

3.1.2.1 Devices Supporting Accounts

Devices which MINIMALLY support accounts are not required to lock based on unsuccessful logon attempts.

3.1.2.2 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.1.3 System Use Notification

Web interfaces must display a warning banner meeting the requirements of DTM 08-060.

Devices which are connected to a network and have a user interface must display a warning banner meeting the requirements of DTM 08-060 if capable of doing so. Devices which are connected to a network and have a user interface but are not capable of displaying a banner must have a permanently affixed label displaying an approved banner from DTM 08-060. Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.1.3.1 User Interface Banner Schedule

Provide a User Interface Schedule using the format indicated showing each user interface provided and how the information banner requirement has been implemented for each user interface.

User Interface Schedule Format (with sample entries)			
User Interface Description	User Interface Location	Type of User Interface	Banner Implementation
Sample 1	Room 1	Remote	DTM 08-060 Banner "A" Displayed at Logon
Sample 2	Room 2	Limited Local	DTM 08-060 Banner "B" on Affixed Label
Sample 3	Room 3	Full Local	DTM 08-060 Banner "B" Displayed on Screen

3.1.4 Permitted Actions Without Identification or Authentication

The control system must require identification and authentication before allowing any actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.1.5 Wireless Access

Unless explicitly authorized by the Government, do not use any wireless communication. Any device with wireless communication capability is considered to be using wireless communication, regardless of whether or not the device is actively communicating wirelessly, except when wireless communication has been physically permanently disabled (such as through the removal of the wireless transceiver).

3.1.5.1 Wireless IP Communications

Do not install wireless IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

When explicitly authorized by the Government, wireless IP communication may be used to communicate with an existing wireless network.

3.1.5.2 Non-IP Wireless Communication

When non-IP wireless communication is explicitly authorized by the Government, use the maximum level of encryption supported by the specific protocol employed and select signal strength and radiated power to the minimum necessary for reliable communication.

3.1.5.3 Wireless Communication Request

Provide a report documenting the proposed use of wireless communication prior to beginning construction using the Wireless Communication Request Schedule. See Attachment 25 05 11.21-F.

3.2 CYBERSECURITY AUDITING

3.2.1 Audit Events, Content of Audit Records, and Audit Generation

3.2.1.1 Default Requirements for Control System Devices

For control system devices where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.2.1.1.1 Devices Which Do Not FULLY Support Accounts

For each Device which does not FULLY support accounts configure the device to audit all device shutdown and startup events and to record for each event the type of event and when the event occurred.

3.2.2 Audit Storage Capacity and Audit Upload

{For Reference Only: This subpart (and its subparts) relates to AU-4; CCI-001848, CCI-001849}

- a. For non-computer control system devices capable of generating audit records, provide 60 days worth of secure local storage, assuming 10 auditable events per day.

3.2.2.1 Device Audit Record Upload Software

For each non-computer device required to audit events, provide, and license to the Government, software implementing a secure mechanism of uploading audit records from the device to a computer and of exporting the uploaded audit records as a Microsoft Excel file. Where different devices use different software, provide software of each type required to upload audit logs from all devices.

Submit copies of device audit record upload software. If there are no non-computer devices requiring auditing, provide a document stating this in lieu of this submittal.

3.2.3 Response to Audit Processing Failures

{For Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490}.

In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

3.2.4 Time Stamps

3.2.4.1 Default Requirements for Control System Devices

For control system devices where Time Stamps requirements are not otherwise indicated in this Section: Devices generating audit records must have internal clocks capable of providing time with a resolution of 1 second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device generating audit records maintains accurate time to within 1 second.

3.3 REQUIREMENTS FOR LEAST FUNCTIONALITY

Do not provide devices with user interfaces where one was not required.

Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

3.3.1 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol, or use any protocol on ports other than those specified.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.4 SAFE MODE AND FAIL SAFE OPERATION

For all control system components configure the component to implement fail safe procedures to include standard control system design of sequences and alarm requirements.

3.5 IDENTIFICATION AND AUTHENTICATION

3.5.1 User Identification and Authentication

3.5.1.1 Default Requirements for Control System Devices

For control system devices where User Identification and Authentication requirements are not otherwise indicated in this Section, User Identification and Authentication for access to privileged accounts must be implemented by inheriting identification and authentication from the system.

3.5.2 Authenticator Management

3.5.2.1 Authentication Type

3.5.2.1.1 Default Requirements for Control System Devices

For control system devices where Authentication Type requirements are not otherwise indicated in this Section:

- a. Devices MINIMALLY supporting accounts must use password-based authentication.

3.5.2.2 Password-Based Authentication Requirements

3.5.2.2.1 Passwords for Devices Minimally Supporting Accounts

Devices minimally supporting accounts must support passwords with a minimum length of four characters.

3.5.2.2.2 Password Configuration and Reporting

For all devices with a password, change the password from the default password. Coordinate selection of passwords with the Contracting Officer. Do not use the same password for more than one device unless specifically instructed to do so. Provide a Password Summary Report documenting the password for each device and describing the procedure to change the password for each device.

Do not provide the Password Summary Report in electronic format. Provide two hardcopies of the Password Summary Report, each copy in its own sealed envelope.

3.5.3 Authenticator Feedback

{For Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable.

3.6 EMERGENCY POWER

Emergency power is specified in the control system and equipment specifications.

3.7 DURABILITY TO VULNERABILITY SCANNING

3.7.1 Default Requirements for Control System Devices

Non-computer control system devices where Durability to Vulnerability Scanning requirements are not otherwise indicated in this Section are not required to respond to scans.

3.8 SYSTEM AND COMMUNICATION PROTECTION

3.8.1 Denial of Service Protection, Process Isolation and Boundary Protection

To the greatest extent practical, implement control logic in non-computer hardware and without reliance on the network.

3.9 FIELD QUALITY CONTROL

3.9.1 Tests

In addition to testing and testing support required by other Sections, provide a minimum of 16 hours of technical support for cybersecurity testing of control systems.

-- End of Section --

Cybersecurity Interconnection Schedule Template

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Network Communication Report

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Contractor Computer Cybersecurity Compliance Statement

Number of contractor-owned computers used during construction: _____

For each contractor-owned computer, list the make and model of the device, the device serial number, the operating system version, and the anti-malware software version. Attach additional sheets if required to document all computers.

Number of additional sheets attached: _____

	Make/Model	Serial Number(s)	Operating System Type and Version	Anti-Malware Software Type and Version
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

I hereby certify that:

- The information provided above is accurate as of the date this document is signed
- All computers listed will be provided any and all patches and updates released during the period of construction
- The computers listed above will be scanned by Anti-Malware software at least once per day during the period of construction.
- Passwords for computers have been changed from defaults

Signature: _____

Date: _____

Completed By:

Name: _____

Position Title: _____

Company: _____

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Contractor Temporary Network Cybersecurity Compliance Statement

Check ONE: Temporary IP networks () WILL or () WILL NOT be used

Check ONE: Temporary Wireless IP networks () WILL or () WILL NOT be used

I hereby certify that:

- Temporary IP networks will not connect to any other IP network.
- Temporary IP networks will not extend outside the project site.
- Temporary Wireless IP networks will use WPA2 encryption.
- There will be NO off-site access of any kind to temporary networks.
- Passwords for network hardware and network access have been changed from defaults.

Signature: _____

Date: _____

Completed By:

Name: _____

Position Title: _____

Company: _____

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Wireless Network Request form (during construction)

Device ID	Device Description	Location	Receiving Device IDs	Protocol	Encryption (type / bits)	RF Frequency	Radiated Power (dBm)	Free-space range	As-installed range	Approved (Y/N)	Testing Required (Y/N)	Network connectivity at required points (Y/N)?				
												Location 1	Location 2	Location 3	...	Location <n>

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SECTION 25 05 11.23

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING CONTROL SYSTEM
11/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x (2010) Local and Metropolitan Area Networks - Port Based Network Access Control

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01 (2014) Ports, Protocols, and Services Management (PPSM)

DTM 08-060 (2008) Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Device Account Lock Exception Request; G

Multiple IP Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

SD-02 Shop Drawings

User Interface Banner Schedule; G

Network Communication Report; G

Cybersecurity Riser Diagram; G

Control System Inventory Report; G

Cybersecurity Interconnection Schedule; G

SD-03 Product Data

Control System Cybersecurity; G

SD-11 Closeout Submittals

Password Summary Report; G

Device Audit Record Upload Software; G

1.3 CYBERSECURITY DOCUMENTATION

1.3.1 Cybersecurity Interconnection Schedule

Provide a completed Cybersecurity Interconnection Schedule documenting connections between the installed system and other systems. See Attachment 25 05 11.23-A.

1.3.2 Network Communication Report

Provide a network communication report. See Attachment 25 05 11.23-B.

1.3.3 Control System Inventory Report

Provide a Control System Inventory report. See Attachment 25 05 11.23-C.

1.3.4 Cybersecurity Riser Diagram

Provide a Cybersecurity Riser Diagram.

1.4 SOFTWARE UPDATE LICENSING

In addition to all other licensing requirements, all software licensing must include licensing of the following software updates for a period of no less than 5 years:

- a. Security and bug-fix patches issued by the software manufacturer.
- b. Security patches to address any vulnerability identified in the National Vulnerability Database at <http://nvd.nist.gov> with a Common Vulnerability Scoring System (CVSS) severity rating of MEDIUM or higher.

1.5 CYBERSECURITY DURING CONSTRUCTION

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.5.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.5.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.5.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.5.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers that will be connected to network controllers during construction. See Attachment 25 05 11.23-D.

1.5.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.5.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.5.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification

1.5.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless

network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.5.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. If no temporary IP networks will be used, provide a single copy of the Statement indicating this. See Attachment 25 05 11.23 -E.

1.6 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

(NOT USED)

PART 3 EXECUTION

3.1 ACCESS CONTROL REQUIREMENTS

3.1.1 User Accounts

3.1.1.1 Default Requirements for Control System Devices

For control system devices where User Account requirements are not otherwise indicated in this Section:

- a. Devices with web interfaces must support user accounts (such as "admin", "user 1", "user 2") or have their web interface disabled.

3.1.2 Unsuccessful Logon Attempts

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts.

3.1.2.1 Devices Supporting Accounts

Devices which MINIMALLY support accounts are not required to lock based on unsuccessful logon attempts.

3.1.2.2 Devices

Devices which FULLY support accounts must meet the following requirements. If a device cannot meet these requirements, document device capabilities to protect from subsequent unsuccessful logon attempts and propose alternate protections in a Device Account Lock Exception Request

submittal. Do not implement alternate protection measures without explicit permission from the Government.

- a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.
- b. Once an account is locked, the account must stay locked until unlocked by an administrator.
- c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.1.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.1.3 System Use Notification

Web interfaces must display a warning banner meeting the requirements of DTM 08-060.

Devices which are connected to a network and have a user interface must display a warning banner meeting the requirements of DTM 08-060 if capable of doing so. Devices which are connected to a network and have a user interface but are not capable of displaying a banner must have a permanently affixed label displaying an approved banner from DTM 08-060. Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.1.3.1 User Interface Banner Schedule

Provide a User Interface Schedule using the format indicated showing each user interface provided and how the information banner requirement has been implemented for each user interface.

User Interface Schedule Format (with sample entries)			
User Interface Description	User Interface Location	Type of User Interface	Banner Implementation
Sample 1	Room 1	Remote	DTM 08-060 Banner "A" Displayed at Logon
Sample 2	Room 2	Limited Local	DTM 08-060 Banner "B" on Affixed Label
Sample 3	Room 3	Full Local	DTM 08-060 Banner "B" Displayed on Screen

3.1.4 Permitted Actions Without Identification or Authentication

The control system must require identification and authentication before

allowing any actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.1.5 Wireless Access

Unless explicitly authorized by the Government, do not use any wireless communication. Any device with wireless communication capability is considered to be using wireless communication, regardless of whether or not the device is actively communicating wirelessly, except when wireless communication has been physically permanently disabled (such as through the removal of the wireless transceiver).

3.1.5.1 Wireless IP Communications

Do not install wireless IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

When explicitly authorized by the Government, wireless IP communication may be used to communicate with an existing wireless network.

3.1.5.2 Non-IP Wireless Communication

When non-IP wireless communication is explicitly authorized by the Government, use the maximum level of encryption supported by the specific protocol employed and select signal strength and radiated power to the minimum necessary for reliable communication.

3.2 CYBERSECURITY AUDITING

3.2.1 Audit Events, Content of Audit Records, and Audit Generation

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

3.2.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event

3.2.1.1.1 Audited Events

Configure each computer to audit the following events:

- a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)
- a. Successful and unsuccessful logon attempts
- b. Privileged activities or other system level access
- c. Starting and ending time for user access to the system
- d. Concurrent logons from different workstations
- e. Successful and unsuccessful accesses to objects

- f. All program initiations
- g. All direct access to the information system
- h. All account creations, modifications, disabling, and terminations
- i. All kernel module load, unload, and restart

3.2.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

- a. What type of event occurred
- b. When the event occurred
- c. Where the event occurred
- d. The source of the event
- e. The outcome of the event
- f. The identity of any individuals or subjects associated with the event

3.2.1.2 For HVAC Control System Devices

3.2.1.2.1 HVAC Control System Devices FULLY Supporting User Accounts

For devices FULLY supporting accounts, provide the capability to select audited events, and the contents of audit logs. Configure devices to audit the following events:

- a. Successful and unsuccessful logon attempts to the device
- b. Starting and ending time for user access to the device
- c. All account creations, modifications, disabling, and terminations
- d. All device shutdown and startup

Configure the device to record for each event the following information (as applicable): the type of event, when the event occurred and the identity of any individuals or subjects associated with the event

3.2.1.2.2 Other HVAC Control System Devices

There are no requirements to perform auditing at HVAC field devices that do not FULLY support accounts.

3.2.1.3 Default Requirements for Control System Devices

For control system devices where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.2.1.3.1 Devices Which FULLY Support Accounts

For each device which FULLY supports accounts, provide the capability to

select audited events and the content of audit logs. Configure devices to audit the indicated events, and to record the indicated information for each auditable event

3.2.1.3.1.1 Audited Events

Configure each device to audit the following events:

- a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)
- a. Successful and unsuccessful logon attempts
- b. Privileged activities or other system level access
- c. Starting and ending time for user access to the system
- d. Concurrent logons from different workstations
- e. All account creations, modifications, disabling, and terminations
- f. All kernel module load, unload, and restart

3.2.1.3.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

- a. what type of event occurred
- b. when the event occurred
- c. where the event occurred
- d. the source of the event
- e. the outcome of the event
- f. the identity of any individuals or subjects associated with the event

3.2.1.3.2 Devices Which Do Not FULLY Support Accounts

For each Device which does not FULLY support accounts configure the device to audit all device shutdown and startup events and to record for each event the type of event and when the event occurred.

3.2.2 Audit Storage Capacity and Audit Upload

{For Reference Only: This subpart (and its subparts) relates to AU-4; CCI-001848, CCI-001849}

- a. For devices that have STIG/SRGs related to audit storage capacity (CCI-001848 or CCI-001849) comply with the requirements of those STIG/SRGs.
- b. For non-computer control system devices capable of generating audit records, provide 60 days worth of secure local storage, assuming 10 auditable events per day.

c. For computers, provide storage for at least 20 mb audit records.

3.2.2.1 Device Audit Record Upload Software

For each non-computer device required to audit events, provide, and license to the Government, software implementing a secure mechanism of uploading audit records from the device to a computer and of exporting the uploaded audit records as a Microsoft Excel file. Where different devices use different software, provide software of each type required to upload audit logs from all devices.

Submit copies of device audit record upload software. If there are no non-computer devices requiring auditing, provide a document stating this in lieu of this submittal.

3.2.3 Response to Audit Processing Failures

{For Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490}.

Front end computers associated with auditing must, in the case of a failure in the auditing system, notify ISSM via e-mail. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

3.2.4 Time Stamps

3.2.4.1 Computers

Computers generating audit records must have internal clocks capable of providing time with a resolution of 1 second. Clocks must not drift more than 10 seconds per day.

Configure the system so that each computer generating audit records maintains accurate time to within 1 second.

3.2.4.2 For HVAC Control System Devices

Time stamp requirements for HVAC Control Systems are as indicated in the HVAC Control System specifications. Devices generating audit records must have internal clocks capable of providing time with a resolution of 1 second. Clocks cannot drift more than 10 seconds per day. Configure the system so that each device generating audit records maintains accurate time to within 1 second.

3.2.4.3 Default Requirements for Control System Devices

For control system devices where Time Stamps requirements are not otherwise indicated in this Section: Devices generating audit records must have internal clocks capable of providing time with a resolution of 1 second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device generating audit records maintains accurate time to within 1 second.

3.3 REQUIREMENTS FOR LEAST FUNCTIONALITY

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O

access for least functionality), install and configure the device in accordance with that STIG or SRGs.

For HVAC Control Systems: Do not provide devices with user interfaces where one was not required. Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

For Other Control Systems: Do not provide devices with user interfaces where one was not required. Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

3.3.1 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol, or use any protocol on ports other than those specified.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.3.2 IP Control Networks

Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

3.4 SAFE MODE AND FAIL SAFE OPERATION

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs to implement fail safe procedures to include standard control system design of sequences and alarm requirements.

3.5 IDENTIFICATION AND AUTHENTICATION

3.5.1 User Identification and Authentication

{For Reference Only: This subpart (and its subparts) relates to IA-2,(1),(12); CCI-000764, CCI-000765, CCI-001953, CCI-001954}

- a. Devices that FULLY support accounts must uniquely identify and authenticate organizational users.
- b. Devices which allow network access to privileged accounts must implement multifactor authentication for network access to privileged accounts.

3.5.1.1 HVAC Control Systems Devices

Identification and Authentication for network access to privileged accounts must be implemented by either accepting and electronically verify Personal Identity Verification (PIV) credentials or inheriting identification and authentication from the operating system.

3.5.1.2 Default Requirements for Control System Devices

For control system devices where User Identification and Authentication requirements are not otherwise indicated in this Section, User Identification and Authentication for network access to privileged accounts must be implemented by accepting and electronically verify Personal Identity Verification (PIV) credentials or inheriting identification and authentication from the operating system.

3.5.2 Authenticator Management

3.5.2.1 Authentication Type

3.5.2.1.1 For HVAC Control System Devices

Unless otherwise indicated:

- a. Software which FULLY supports accounts and which runs on a computer must use password-based authentication or hardware token-based authentication.
- b. Other devices which FULLY support accounts must use password-based authentication.
- c. Devices MINIMALLY supporting accounts must use password-based authentication.

3.5.2.1.2 Default Requirements for Control System Devices

For control system devices where Authentication Type requirements are not otherwise indicated in this Section:

- a. Software which FULLY supports accounts and which runs on a computer must use password-based authentication or hardware token-based authentication.
- b. Other devices which FULLY support accounts must use either password-based authentication or hardware token-based authentication.
- c. Devices MINIMALLY supporting accounts must use password-based authentication.

3.5.2.2 Password-Based Authentication Requirements

3.5.2.2.1 Passwords for Computers

All computers supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of 12 characters
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character.
- f. Password must have a minimum lifetime of 24 hours.

- g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.
- i. Passwords must be cryptographically protected during storage and transmission.

3.5.2.2.2 Passwords for Non-Computer Devices FULLY Supporting Accounts

All non-computer devices FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of twelve (12) characters
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character.
- f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- g. Password must differ from previous five (5) passwords, where differ is defined as changing at least fifty percent of the characters.
- h. Passwords must be cryptographically protected during storage and transmission.

3.5.2.2.3 Passwords for Web Interfaces

Passwords for connecting to a web interface supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of 12 characters
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character.
- f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.
- h. Passwords must be cryptographically protected during storage and transmission.

3.5.2.2.4 Passwords for Devices Minimally Supporting Accounts

Devices minimally supporting accounts must support passwords with a minimum length of four characters.

3.5.2.2.5 Password Configuration and Reporting

For all devices with a password, change the password from the default password. Coordinate selection of passwords with the Contracting Officer. Do not use the same password for more than one device unless specifically instructed to do so. Provide a Password Summary Report documenting the password for each device and describing the procedure to change the password for each device.

Do not provide the Password Summary Report in electronic format. Provide two hardcopies of the Password Summary Report, each copy in its own sealed envelope.

3.5.2.3 Hardware Token-Based Authentication Requirements

Devices supporting hardware token-based authentication must use Personal Identity Verification (PIV) credentials for the hardware token.

3.5.3 Authenticator Feedback

{For Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

3.5.4 Device Identification and Authentication

All computers must use IEEE 802.1x for authentication to the network. All web servers running on computers must use HTTPS and must implement HTTPS using web server certificates obtained from the Contracting Officer.

3.5.4.1 For HVAC Control System Devices

Devices using Fox Protocol must use HTTPS using a web server certificate obtained from the Contracting Officer. Devices using Fox Protocol must support IEEE 802.1x. Devices using Ethernet must support IEEE 802.1x. Devices using BACnet must support Network Security as specified in Clause 24 of ASHRAE 135.

3.5.4.2 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using Ethernet must support IEEE 802.1x. Devices using HTTP as a control protocol must use HTTPS using a web server certificate obtained from the Contracting Officer instead.

3.5.5 Cryptographic Module Authentication

{For Reference Only: This subpart (and its subparts) relates to IA-7;

CCI-000803}

For devices that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.6 EMERGENCY POWER

Emergency power is specified in the control system and equipment specifications.

3.7 DURABILITY TO VULNERABILITY SCANNING

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

For control system devices other than computers:

3.7.1 HVAC Control System Devices Other Than Computers

HVAC control system devices other than computers are not required to respond to scans.

3.7.2 Default Requirements for Control System Devices

Non-computer control system devices where Durability to Vulnerability Scanning requirements are not otherwise indicated in this Section are not required to respond to scans.

3.8 DEVICES WITH CONNECTION TO MULTIPLE IP NETWORKS

Except for Ethernet switches, do not use more than one physical connection to IP networks on the same device unless doing so is both required by the project specifications and the specific application is approved. If a device with multiple IP connections is required, provide a Multiple IP Connection Device Request using the Multiple IP Connection Device Request Schedule to request approval for each device.

3.9 SYSTEM AND COMMUNICATION PROTECTION

3.9.1 Denial of Service Protection, Process Isolation and Boundary Protection

To the greatest extent practical, implement control logic in non-computer hardware and without reliance on the network.

3.9.2 Cryptographic Protection

For devices that have STIG/SRGs related to cryptographic protection (CCI-002450), comply with the requirements of those STIG/SRGs. Ensure that IP network traffic is encrypted using NSA-approved cryptography; provision of digital signatures and hashing, and FIPS-validated cryptography.

3.10 SYSTEM AND INTEGRATION INTEGRITY

3.10.1 Malicious Code Protection

For all computers installed under this project, install and configure malware protection software in accordance with the relevant STIGs.

3.11 FIELD QUALITY CONTROL

3.11.1 Tests

In addition to testing and testing support required by other Sections, provide a minimum of 40 hours of technical support for cybersecurity testing of control systems.

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Cybersecurity Interconnection Schedule Template

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Network Communication Report

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Contractor Computer Cybersecurity Compliance Statement

Number of contractor-owned computers used during construction: _____

For each contractor-owned computer, list the make and model of the device, the device serial number, the operating system version, and the anti-malware software version. Attach additional sheets if required to document all computers.

Number of additional sheets attached: _____

	Make/Model	Serial Number(s)	Operating System Type and Version	Anti-Malware Software Type and Version
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

I hereby certify that:

- The information provided above is accurate as of the date this document is signed
- All computers listed will be provided any and all patches and updates released during the period of construction
- The computers listed above will be scanned by Anti-Malware software at least once per day during the period of construction.
- Passwords for computers have been changed from defaults

Signature: _____

Date: _____

Completed By:

Name: _____

Position Title: _____

Company: _____

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Contractor Temporary Network Cybersecurity Compliance Statement

Check ONE: Temporary IP networks () WILL or () WILL NOT be used

Check ONE: Temporary Wireless IP networks () WILL or () WILL NOT be used

I hereby certify that:

- Temporary IP networks will not connect to any other IP network.
- Temporary IP networks will not extend outside the project site.
- Temporary Wireless IP networks will use WPA2 encryption.
- There will be NO off-site access of any kind to temporary networks.
- Passwords for network hardware and network access have been changed from defaults.

Signature: _____

Date: _____

Completed By:

Name: _____

Position Title: _____

Company: _____

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SECTION 26 05 48.00 10

SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT
10/07

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

ARCHITECTURAL INSTITUTE OF JAPAN (AIJ)

JASS 6 (2015) Structural Steelwork Specification
for Building Construction

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 8105-2 Series Luminaires: Particular Requirements for
Safety

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 2, 2024) Structural
Engineering

UL SOLUTIONS (UL)

UL 1598 (2021; Reprint Jun 2021) Luminaires

1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

The requirements for seismic protection measures described in this section shall be applied to the electrical equipment and systems listed below.

1.2.2 Electrical Equipment

Electrical equipment shall include the following items to the extent required on the drawings or in other sections of these specifications:

Control Panels	Air Handling Units
Pumps with Motors	Switchgear Light Fixtures

1.2.3 Electrical Systems

The electrical systems shall be installed as required on the drawings and other sections of these specifications and shall be seismically protected

in accordance with this specification.

1.2.4 Contractor Designed Bracing

Submit copies of the Design Calculations with the Drawings. Calculations shall be approved, certified, stamped and signed by a Registered Professional Engineer. Calculations shall verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design the bracing in accordance with UFC 3-301-01 and additional data furnished by the Contracting Officer. Resistance to lateral forces induced by earthquakes shall be accomplished without consideration of friction resulting from gravity loads. UFC 3-301-01 uses parameters for the building, not for the equipment in the building; therefore, corresponding adjustments to the formulas shall be required. Loadings determined using UFC 3-301-01 are based on strength design; therefore, AISC 325 or JASS 6 shall be used for the design.

1.2.5 Conduits Requiring No Special Seismic Restraints

Seismic restraints may be omitted from electrical conduit less than 64 mm trade size. All other interior conduit, shall be seismically protected as specified.

1.3 EQUIPMENT REQUIREMENTS

Submit detail drawings along with catalog cuts, templates, and erection and installation details, as appropriate, for the items listed. Submittals shall be complete in detail, indicating thickness, type, grade, class of metal, and dimensions; and shall show construction details, reinforcement, anchorage, and installation with relation to the building construction. Submit copies of the design calculations with the detail drawings. Calculations shall be stamped by a registered engineer and shall verify the capability of structural members to which bracing is attached for carrying the load from the brace.

1.3.1 Rigidly Mounted Equipment

Each item of rigid electrical equipment shall be entirely located and rigidly attached on one side only of a building expansion joint. Piping, electrical conduit, etc., which cross the expansion joint shall be provided with flexible joints that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Lighting Fixtures in Buildings

Equipment Requirements

SD-03 Product Data

Lighting Fixtures in Buildings; G

Equipment Requirements; G

Contractor Designed Bracing; G

PART 2 PRODUCTS

2.1 LIGHTING FIXTURE SUPPORTS

Lighting fixtures and supports shall conform to UL 1598 or JIS C 8105-2.

PART 3 EXECUTION

3.1 LIGHTING FIXTURES IN BUILDINGS

Lighting fixtures and supports shall conform to the following:

3.1.1 Pendant Fixtures

Pendant fixtures shall conform to the requirements of UFC 3-301-01.

3.1.2 Ceiling Attached Fixtures

3.1.2.1 Recessed Fixtures

Seismic protection for the fixtures shall conform to the requirements of UFC 3-301-01. Recessed lighting fixtures not over 25 kg in weight may be supported by and attached directly to the ceiling system runners using screws or bolts, number and size as required by the seismic design. Fixture accessories, including louvers, diffusers, and lenses shall have lock or screw attachments.

3.1.2.2 Surface-Mounted Fixtures

Seismic protection for the fixtures shall conform to the requirements of UFC 3-301-01.

3.1.3 Assembly Mounted on Outlet Box

A supporting assembly, that is intended to be mounted on an outlet box, shall be designed to accommodate mounting features on 100 mm boxes, plaster rings, and fixture studs.

3.1.4 Wall-Mounted Emergency Light Unit

Attachments for wall-mounted emergency light units shall be designed and secured for the worst expected seismic disturbance at the site.

-- End of Section --

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SECTION 26 05 73

POWER SYSTEM STUDIES
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1584	(2018; E 2019) Guide for Performing Arc-Flash Hazard Calculations
IEEE 1584.1	(2022) Guide for the Specification of Scope and Deliverable Requirements for an Arc-Flash Hazard Calculation Study
IEEE 3002.2	(2018) Recommended Practice for Conducting Load-Flow Studies and Analysis of Industrial and Commercial Power Systems
IEEE 3002.3	(2018) Recommended Practice for Conducting Short-Circuit Studies and Analysis of Industrial and Commercial Power Systems
IEEE C2	(2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
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1.2 SYSTEM DESCRIPTION

The power system covered by this specification consists of: Equipment affected by this project as shown on drawings, extend study to breakers immediately ahead of affected equipment. Assume infinite bus on primary side of service transformer.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Field Examination Plan; G

Arc Flash Label Formats; G

SD-06 Test Reports

Field Examination

SD-07 Certificates

System Analyzer; G

SD-11 Closeout Submittals

Model Files; G

Load Flow Study; G

Fault Current Study; G

System Coordination Study; G

Arc Flash Hazard Study; G

1.4 QUALITY ASSURANCE

1.4.1 System Analyzer

The System Analyzer must perform the power system studies. The System Analyzer must be a registered professional electrical engineer with a minimum of 3 years of experience with power system studies. Include a list of three comparable jobs performed by the System Analyzer with specific names and telephone numbers for reference. Include the license number and state of the registered Professional Engineer.

1.4.2 System Verifier

The System Verifier may inspect and record settings, markings, and otherwise document the existing equipment. The System Verifier may also place appropriate hazard labels on equipment. The System Verifier must not make any calibrations or adjustments or place equipment into service. When working with energized equipment, the system verifier must be a Qualified Person per NFPA 70E or working under the direct supervision of a Qualified Person.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 FIELD EXAMINATION

For each pre- and post-study inspection, submit a field examination plan identifying which facilities must be examined to complete the required work. Include a complete information verification procedure, where Qualified Personnel will be required, PPE requirements for "live" equipment examinations, and equipment to be used within the limited approach boundary of the equipment.

3.1.1 General

Perform field inspections of the site and equipment before the first study and after the last study to determine the state and settings of the equipment and to verify the final settings agree with the studies. Perform the post-study inspection(s) after changing settings, after performing acceptance tests, and before placing equipment in service.

Schedule the field examination by the System Verifier with the Contracting Officer at least 10 business days before conducting examination. Furnish all materials, labor, and equipment necessary to conduct the examination. Maintain a written record of the all equipment examined, equipment location, equipment ratings, settings, personnel involved, and the date the examination was performed. Submit the written record of each field examination.

3.1.2 Safety

Where examination of the equipment requires energized equipment to be opened, provide a Qualified Person to directly supervising any non-Qualified Person within the hazard area. Use PPE, protective barriers, danger signs and other safety devices to protect and warn personnel in the vicinity of "live" equipment being examined.

3.2 POWER SYSTEM STUDIES

Perform power system studies to demonstrate that the equipment selected and system constructed meet the contract requirements for fault current and interrupt ratings, coordination, protection, and Arc Flash Hazard. Submit reports of the studies along with protective device equipment submittals. Apply Arc Flash Hazard labels to equipment after the studies are approved. Update and resubmit the studies after any changes to the equipment or systems which may affect the study results, and re-apply Arc Flash Hazard labels to equipment after the resubmitted studies are approved. The Government is not responsible for any changes to equipment, device ratings, settings, or additional labor for installation of equipment or devices ordered and/or procured before approval of the study.

3.2.1 Scope of Studies

The scope of the studies must begin at the nearest upstream device in the existing source system and extend through the downstream devices at the load end.

The "source bus" is the source of energy for system being analyzed. This may be the energy feed from a utility, the first bus upstream of the work, generators within the work or upstream of the work, or any other source capable of contributing significant energy into the system being analyzed.

In the systems model for the studies, incorporate all existing and new equipment within the scope of the studies. Incorporate any additional sources or load equipment necessary to accurately model the system's performance.

Where multiple sources of power may be in service in various combinations or where electrical loops are in the system, incorporate into the studies all the modes of operation to evaluate the impact of the modes of operation on the system.

3.2.2 Determination of Facts

Determine and document the time-current characteristics, features, ratings, ampacities, and nameplate data for each existing protective device, electrical equipment, and feeder cables.

Utilize field-verified data in the power system studies. Document the field-verified data in the report.

3.2.3 Single Line Diagram

Provide a single line diagram showing the electrical system buses, devices, transformation points, and all sources of load current and fault current, including generator and motor contributions. Provide a diagram from the system model. Each bus, device or transformation point must have a unique identifier. Show the location of switches, breakers, and circuit interrupting devices on the diagram together with available fault data, and the device interrupting rating.

The naming of existing components within the system model and single line diagram must match existing installed equipment names. The naming of new components within the system model and single line diagram must use unique identifiers and be coordinated with the Government.

3.2.4 Load Flow Study

Perform the load flow study in accordance with IEEE 3002.2 to identify initial steady-state conditions for the fault current study. For each operating scenario, provide load flow results on the diagram or in the report. Show or describe the loading factors and assumptions used in each operating mode.

3.2.5 Fault Current Study

Use the results of the load flow study to perform the fault current study in accordance with IEEE 3002.3. Provide balanced three-phase fault, bolted line-to-line fault, and single line-to-ground fault current values at each voltage transformation point and at each power distribution bus. For each location, show in tabular form on the diagram or in the report the maximum and minimum available fault currents of all modes of operation for that location.

Where the available fault current at the source bus is not available, describe how the fault contribution from the source bus was determined and why this method is reasonable for the study.

3.2.6 System Coordination Study

Use the results of the load flow study and fault current study. For normal modes of operation, demonstrate that selectivity has been obtained between the devices within the scope of the project. Demonstrate the equipment, machines, and conductors are protected from damage from overloads and fault conditions. Include a description of the coordination of the protective devices in this project. Provide a written narrative describing which devices may operate in the event of a fault at each bus; the logic used to arrive at device ratings and settings; situations where system coordination is not achievable due to device limitations; coordination between upstream and downstream devices; and relay settings.

Provide recommendations to improve or enhance system reliability, and detail where such changes would involve additions or modifications to the contract and cost damages (addition or reduction). Provide composite coordination plots on a log-log scale. Where recommendations are provided to improve or enhance system reliability, provide separate coordination plots on log-log scale showing the system coordination before and after the implementation of the recommendations. Where coordination has been adjusted to lower arc flash values, include a discussion of the change and the impact on the system within the report.

3.2.7 Arc Flash Hazard Study

Perform the arc flash hazard study in accordance with IEEE 1584.1. Utilize the data from the fault current study to determine the worst case incident energy per IEEE 1584, IEEE C2, and OSHA 29 CFR 1910.269 Appendix E. Use identified modes of operation to determine the worst case arc flash energy. If not included in another study, include a description of the devices and device settings for the operating modes that provided the highest arc flash energy. Where the adjusting the coordination of devices to obtain lower arc flash values, include a discussion of the change and the impact on the system within the report.

Where the AC system voltage is 208 volts or less with an available short-circuit current less than 2000 A, an arc flash hazard study is not required.

3.2.8 Study report(s)

- a. Include a narrative describing the studies performed; the bases and methods used; and the desired method of coordinated protection of the power system.
- b. Include descriptive and technical data for existing devices and new protective devices proposed. Include manufacturers published data, nameplate data, and definition of the fixed or adjustable features of the existing or new protective devices. For existing devices, included statements on the condition of the equipment based upon field inspections and owner's statements and reports.
- c. Identify all assumptions about the protection devices, equipment, and system where data was not available.
- d. For each bus in the system, provide fully coordinated composite time-current characteristics (TCC) curves as required to ensure coordinated power system protection between protective devices and equipment. In a tabular format, provide existing and recommended ratings and settings of all protective devices.
- e. Provide an arc flash study report in accordance with IEEE 1584.1.
- f. Provide the calculations performed for the studies, including computer programs utilized. Provide the name of the software package, developer, and version number.
- g. The System Analyzer must stamp the study reports.

3.2.9 Arc Flash Labels

Provide arc flash warning labels on electrical equipment likely to require

examination, servicing, or maintenance while energized. Typical types of equipment include pad-mounted transformers, switchgear, switchboards, panelboards, disconnect switches, industrial control panels, meter socket enclosures, and motor control centers that are in other than dwelling occupancies. The arc flash label naming must match the naming used in the system modeling and the single line diagram.

Comply with the label requirements specified in NFPA 70E. Obtain approval of arc flash label formats before printing.

- a. Provide a thermal transfer type label of high adhesion polyester for each location device analyzed. The label must remain in place and be legible for at least 5 years in the installed environment.
- b. Labels must be machine printed with no field markings. Provide arc flash labels in the following manner. All labels must be based on implemented overcurrent devices and settings.
 1. Provide at least one arc flash label for each 480 volt and 208 volt panelboards.
 2. Provide at least one arc flash label for each low voltage (less than 1000 V) switchboard.
 3. Where the equipment construction has different hazards levels in different areas, provide a single label representing the worst case hazard for all branch circuits and a separate unique label for the compartment containing the main protective device. On the worst case label, indicate which branch circuit compartments have the worst case hazard.
- c. Use the worst case hazard of all operating scenarios unless mitigation procedures are used. If mitigation procedures are used, explain the procedures on the label.

3.3 MODELING

Develop a software model of the electrical system identified in the scope of the studies. Use the latest version of SKM PowerTools(TM). Develop the model with accurate, verified information. Model existing electrical equipment, machines, devices, and conductors directly connected to, altered by, or otherwise affected by the work. This includes, but is not limited to generators, transformers, switchgear, switchboards, panelboards, motor control centers, motors, voltage regulators, tap changers, protective relays, circuit breakers, switches, fuses, conductors, capacitors, reactors, grounding system, and control equipment.

Provide the final model files in their native editable formats for future use by the Government. Include device information for devices not in the software vendor's standard distribution.

-- End of Section --

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM
02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Ministry of Land, Infrastructure and Transport (MLIT), Japan Electrical Safety Inspection Associations, Japanese Electrotechnical Committee (JEC) or Japan Wire Industry Association (JCS) standards for non-Japanese standards, as approved by the Contracting Officer's representative.

JAPANESE ELECTROTECHNICAL COMMITTEE (JEC)

JEC 2200 (2015) Transformer

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 0365 (2007) Protection Against Electric Shock -
Common Aspects for Installation and
Equipment

JIS C 0920 (2003) Degrees Of Protection Provided By
Enclosures (IP Code)

JIS C 2336 (2012) Pressure-sensitive polyvinyl
chloride tapes for electrical purposes

JIS C 2338 (2012) Polyester adhesive tape for
electrical insulation

JIS C 2805 (2010) Crimp terminal for copper wire

JIS C 2806 (2003; R 2018) Bare crimping sleeve for
copper wire

JIS C 2810 (1995; R 2021) General rules on
non-separable type wire connectors for
interior wiring

JIS C 3101 (1994; R 2021) Hard-drawn copper wires for
electrical purposes

JIS C 3105 (1994; R 2021) Hard-drawn copper stranded
conductors

JIS C 3401 (2002; R 2022) Control Cables

JIS C 3605 (2022) 600 V Polyethylene Insulated
Cables, Type CV

JIS C 3612 (2002; R 2022) 600V Flame Retardant

Polyethylene Insulated Wires

JIS C 5381-11	(2014; R 2019) Low-voltage surge protective devices -- Part 11: Surge protective devices connected to low-voltage power systems -- Requirements and test methods
JIS C 5381-12	(2021) Low-voltage surge protective devices -- Part 12: Surge protective devices connected to low-voltage power distribution systems -- Selection and application principles
JIS C 60364-6	(2010) Low-voltage Electrical Installations - Part 6: Verification
JIS C 6436	(1995; R 2019) Power transformer for electronic equipment
JIS C 8201-3	(2009; R 2018) Low-voltage switchgear and control units-Part 3: Switches, Disconnector, disconnecting switch and fuse assembly unit
JIS C 8201-5-1	(2022) Low-Voltage Switchgear And Control Gear - Part 5-1: Control Circuit Devices And Switching Elements - Electromechanical Control Circuit Devices
JIS C 8201-4-1	(2023) Low-voltage switchgear and controlgear -- Part 4-1: Contactors and motor-starters: Electromechanical contactors and motor-starters
JIS C 8201-4-2	(2010; 2020) Low-voltage switchgear and controlgear -- Part 4-2: Contactors and motor-starters -- AC semiconductor motor controllers and starters
JIS C 8201-7-1	(2016) Low-Voltage Switchgear And Control Gear -- Part 7-1: Ancillary Equipment -- Terminal Blocks For Copper Conductors
JIS C 8281-1	(2019) Plugs and socket-outlets for household and similar purposes -- Part 1: General requirements
JIS C 8303	(2007; R 2022) Plugs And Receptacles For Domestic And Similar General Use
JIS C 8304	(2009; R 2019) Small switches for indoor use
JIS C 8305	(2019) Rigid Steel Conduits
JIS C 8309	(2019) Pliable metal conduits
JIS C 8330	(1999; R 2019) Fittings for rigid metal

conduits

JIS C 8340	(1999; R 2019) Boxes And Box Covers For Rigid Metal Conduits
JIS C 8350	(1999; R 2014) Fittings for pliable metal conduits
JIS C 8380	(2009; R 2019) Plastic coated steel pipes for cable-ways
JIS C 8430	(2019) Unplasticized polyvinyl chloride (PVC-U) conduits
JIS C 8432	(2019) Fittings of unplasticized polyvinyl chloride (PVC-U) conduits
JIS C 8435	(2022) Boxes And Box Covers Of Plastic Conduits
JIS C 8462-1	(2021) Boxes and enclosures for electrical accessories for household and similar fixed electrical installations -- Part 1: General requirements
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 61558-1	(2019) Safety of transformers, reactors, power supply units and combinations thereof -- Part 1: General requirements and tests
JIS K 6743	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Water Supply
JIS K 6911	(2006; R 2021) Thermosetting plastic general test method
JIS K 6741	(2016; R 2021) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS Z 9101	(2018) Graphical symbols -- Safety colours and safety signs -- Part 1: Design principles for safety signs and safety markings
JIS Z 9290-1	(2014; R 2019) Protection Against Lightning - Part 1: General Principles

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 1226	(2003) Soft Stranded Wire
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MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT ESS

(2019) MLIT Electrical Standard
Specification (ESS)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E

(2024) Standard for Electrical Safety in
the Workplace

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-02 Shop Drawings

Wireways; G

Marking strips drawings; G

SD-03 Product Data

Receptacles;

Switches; G

Motor controllers; G

Manual motor starters; G

Surge protective devices; G

Include performance and characteristic curves.

SD-06 Test Reports

600-volt wiring test

Ground-fault receptacle test

1.4 QUALITY ASSURANCE

1.4.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" or "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory

and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.4.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.2.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.2.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.5 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6 SEISMIC REQUIREMENTS

Provide seismic details as indicated.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL or Japanese standards, where UL or applicable Japanese codes and standards are established for those items for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit or Type G Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit or Type G

JIS C 8305, Type G. Diameter of conduit shall be as indicated.

2.2.2 Rigid Nonmetallic Conduit or Unplasticized Polyvinyl Chloride

JIS C 8430 for Type VE or HIVE conduit of diameter of less than 100mm and
JIS K 6741 for Type VP or HIVE conduit diameter of 100mm and larger.

2.2.3 Electrical, Zinc-Coated Steel Metallic Tubing (EMT) or Type E Metallic Conduit

JIS C 8305, Type E.

2.2.4 Plastic-Coated Rigid Steel and IMC Conduit or Type G, C or E; LL or LT

JIS C 8380.

2.2.5 Flexible Metal Conduit

JIS C 8309.

2.2.5.1 Liquid-Tight Flexible Metal Conduit, Steel

JIS C 8309.

2.2.6 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

JIS C 8330 and JIS C 8350. Ferrous fittings: cadmium- or zinc-coated in
accordance with JIS C 8330 and JIS C 8350.

2.2.6.1 Fittings for Rigid Metal Conduit and IMC or Type G and Type C

Threaded-type. Split couplings unacceptable.

2.2.6.2 Fittings for EMT or Type E Conduit

Steel type.

2.2.7 Fittings for Rigid Nonmetallic Conduit or Unplasticized Polyvinyl Chloride

JIS C 8432 for diameters less than 100mm and JIS K 6743 for diameters
100mm or larger.

2.3 OUTLET BOXES AND COVERS

JIS C 8340, cadmium- or zinc-coated, if ferrous metal. JIS C 8435, if
nonmetallic. Rated IP65 for wet locations.

2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES

Volume greater than 1640 mL, JIS C 8340, hot-dip, zinc-coated, if sheet
steel. Rated IP65 for wet locations.

2.5 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of Japanese standards for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 12 months prior to date of delivery to site.

2.5.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.
- b. Conductors 5.5 sqmm and larger cross sectional area: stranded.
- c. Conductors 2.0 mm and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper.

2.5.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

- a. Branch circuits: 2.0 mm.
- b. Class 1 remote-control and signal circuits: 1.6 mm.
- c. Class 2 low-energy, remote-control and signal circuits: 1.2 mm.
- d. Class 3 low-energy, remote-control, alarm and signal circuits: 0.65 mm.

2.5.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.5.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White or Gray.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.5.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 480/277/208/120 volt, three-phase, four-wire

- (1) Phase A - red
- (2) Phase B - black
- (3) Phase C - blue

b. 480 volt/208 volt, three-phase, three-wire

- (1) Phase A - red
- (2) Phase B - black
- (3) Phase C - blue

2.5.3 Insulation

Unless specified or indicated otherwise, provide power and lighting wires rated for 600-volts, Type EM-CE, EM-CET or EM-ECEQ conforming to JIS C 3605, except that grounding wire may be type TW conforming to JIS C 3612, Type EM-IE; remote-control and signal circuits: Type TW or TF, conforming to JIS C 3401, Type CEV. Where lighting fixtures require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.5.4 Bonding Conductors

JIS C 3101, solid bare copper wire for sizes 8 sqmm and smaller diameter; JIS C 3105 and JCS 1226, Class B, stranded bare copper wire for sizes 14 sqmm and larger diameter, JIS C 3102 Annealed Copper Wire.

2.5.5 Polyethylene or Cross-Linked Polyethylene Cable

Type CV conforming to JIS C 3605 for 600V.

2.6 SPLICES AND TERMINATION COMPONENTS

JIS C 2805, JIS C 2806 and JIS C 2810 for wire connectors and JIS C 2336, JIS C 2338 for insulating tapes. Connectors for 5.5 sqmm and smaller diameter wires: insulated, pressure-type in accordance with JIS C 2805, JIS C 2806 and JIS C 2810 or JIS C 2806 (twist-on splicing connector). Provide solderless terminal lugs on stranded conductors. Resin splices, where made, must be made of a rigid tube used to inject polyurethane insulating and sealing resin during a splice installation. Splices shall be performed in accordance with the manufacturer's instructions. The installation must not require heat or any additional materials such as covering.

2.7 DEVICE PLATES

Provide the following:

- a. UL listed, one-piece device plates for outlets to suit the devices installed.
- b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.
- c. For nonmetallic boxes and fittings, other suitable plates may be

provided.

- d. Plates on finished walls: nylon or lexan, minimum 0.792 mm wall thickness and same color as receptacle or toggle switch with which they are mounted.
- e. Screws: machine-type with countersunk heads in color to match finish of plate.
- f. Sectional type device plates are not be permitted.
- g. Plates installed in wet locations: gasketed and UL listed or JIS listed for "wet locations."

2.8 SWITCHES

2.8.1 Toggle Switches

JIS C 8304, single pole, double pole, three-way, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

- a. Handles: white thermoplastic.
- b. Wiring terminals: screw-type, side-wired or of the solderless pressure type having suitable conductor-release arrangement.
- c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.
- d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

2.8.2 Disconnect Switches

JIS C 8201-3. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in IP rated enclosure as indicated per JIS C 8462-1.

2.9 RECEPTACLES

Provide the following:

- a. JIS C 8303, JIS C 8281-1, general purpose specification grade (also designated heavy-duty), grounding-type.
- b. Ratings and configurations: as indicated.
- c. Bodies: white.
- d. Face and body: thermoplastic supported on a metal mounting strap.
- e. Dimensional requirements: per JIS C 8462-1.
- f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.

- g. Grounding pole connected to mounting strap.
- h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.

2.9.1 Switched Duplex Receptacles

Provide separate terminals for each ungrounded pole. Top receptacle: switched when installed.

2.9.2 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" or weather resistant per JIS C 8303. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, polycarbonate, UV resistant/stabilized cover plate. The cover shall have notches or other openings that allow cords and plugs to be inserted into the receptacle while the cover is closed. (reason: this is an NEC requirement, but some JIS listed products do not have this feature).

2.9.3 Ground-Fault Circuit Interrupter Receptacles

Duplex type for mounting in standard outlet box. Provide device capable of detecting current leak of 6 milliamperes or greater and tripping for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

2.10 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. JIS C 8201-5-1, JIS C 8201-4-1 and JIS C 8201-4-2, .
- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp (746 kilowatt) and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.
- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.

- i. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- j. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.
- k. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.

2.10.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type CCE/F meeting JIS C 3401, or Type CEE/F meeting JIS C 3401, and passing the flame tests included in those standards.
- b. Current transformer secondary leads: not smaller than 5.5 sqmm
- c. Control wire minimum size: 1.6mm
- d. Power wiring for 480-volt circuits and below: the same type as control wiring with 2.0mm minimum size.
- e. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

2.10.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. JIS C 8201-7-1.
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than 5.5 sqmm in size with sufficient length and space for connecting at least two indented terminals for 5.5 sqmm conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four (4) spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.

- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.10.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

2.10.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to JIS C 6436, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side.. Provide one fused secondary lead with the other lead grounded. Provide for automatic switchover and alarm upon failure of primary control circuit.

2.10.4 Enclosures for Motor Controllers

JIS C 8462-1.

2.10.5 Multiple-Speed Motor Controllers and Reversible Motor Controllers

Across-the-line-type, electrically and mechanically interlocked. Multiple-speed controllers: include compelling relays and multiple-button, station-type with pilot lights for each speed.

2.10.6 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.10.7 Pilot and Indicating Lights

Provide LED cluster lamps.

2.11 MANUAL MOTOR STARTERS (MOTOR RATED SWITCHES)

Single pole designed for surface mounting with overload protection.

2.11.1 Pilot Lights

Provide yoke-mounted, seven element LED cluster light module. Color: in accordance with JIS C 8201-4-1 and JIS C 8201-4-2.

2.12 LOCKOUT REQUIREMENTS

Provide disconnecting means capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with MLIT ESS. Comply with requirements of Division 23, "Mechanical" for mechanical isolation of machines and other equipment.

2.13 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.14 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. JIS K 6911.
- b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.
- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 3 mm thick, white with black center core.
- e. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.
- f. Minimum size of nameplates: 25 by 65 mm.
- g. Lettering size and style: a minimum of 6.35 mm high normal block style.

2.15 WARNING SIGNS

Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer in accordance with NFPA 70E and JIS Z 9101 for switchboards, panelboards, industrial control panels, and motor control centers. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.16 WIREWAYS

Material: steel galvanized 16 gauge for heights and depths up to 150 by

150 mm, and 14 gauge for heights and depths up to 305 by 305 mm. Provide in length required for the application with screw- cover enclosure per indoor IP rated enclosure per JIS C 8462-1.

2.17 SURGE PROTECTIVE DEVICES

Provide surge protective devices (SPD) which comply with JIS C 5381-11 and JIS C 5381-12 at the service entrance , MCC . Provide surge protectors in a indoor IP rated enclosure per JIS C 8462-1. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker.

2.18 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. JIS C 0920 corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.
- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.
- f. Equipment located indoors: Light Gray, .
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.19 SOURCE QUALITY CONTROL

2.19.1 Transformer Factory Tests

Submittal: include routine JEC 2200 and JIS C 61558-1 transformer test results on each transformer and also provide the results of Japanese standard "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of MLIT ESS, JIS C 0365 and to requirements specified herein.

3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

3.1.2 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

3.1.2.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 6.35 mm in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted.

3.1.3 Wiring Methods

Provide insulated conductors installed in rigid steel conduit or type G, IMC or type C, rigid nonmetallic conduit or Unplasticized Polyvinyl Chloride, or EMT or Type E, except where specifically indicated or specified otherwise or required to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Shared neutral, or multi-wire branch circuits, are not permitted with arc-fault circuit interrupters. Minimum conduit size: 16 mm in diameter for low voltage lighting and power circuits. Vertical distribution in multiple story buildings: made with metal conduit in fire-rated shafts, with metal conduit extending through shafts for minimum distance of 150 mm.

3.1.3.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 890-N force tensile strength. Leave minimum 915 mm of slack at each end of pull wire.

3.1.4 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 150 mm away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project.

3.1.4.1 Restrictions Applicable to EMT or Type E

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to severe physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT or Type E conduit.

- d. Do not use in hazardous areas.
- e. Do not use outdoors.
- f. Do not use in fire pump rooms.
- g. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.4.2 Restrictions Applicable to Nonmetallic Conduit or Type VE, HIVE, VP, HIVP

- a. PVC Schedule 40 and PVC Schedule 80 or Type VE, HIVE, VP, HIVP
 - (1) Do not use in areas where subject to severe physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, hospitals, power plants, missile magazines, and other such areas.
 - (2) Do not use in hazardous (classified) areas.
 - (3) Do not use in fire pump rooms.
 - (4) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.
 - (5) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.
 - (6) Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.4.3 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.4.4 Underground Conduit

Plastic-coated rigid steel; plastic-coated steel IMC or type C, LL or LT; PVC, Type EPC-40 or Type VE, HIVE, VP, HIVP. Convert nonmetallic conduit, other than PVC Schedule 40 or 80 or Type VE, HIVE, VP, HIVP, to plastic-coated rigid, or IMC, steel conduit before rising through floor slab. Plastic coating: extend minimum 150 mm above floor.

3.1.4.5 Conduit for Circuits Rated Greater Than 600 Volts

Rigid metal conduit or Type G only.

3.1.4.6 Conduit Support

Support conduit by pipe straps, wall brackets, threaded rod conduit hangers, or ceiling trapeze. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to

fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 40 mm in reinforced concrete beams or to depth of more than 20 mm in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Where conduit crosses building expansion joints, provide suitable watertight expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 63 mm inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction. Type E conduit shall be secured and supported at least 1 meter from terminations.

3.1.4.7 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.4.8 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required.

3.1.4.9 Flexible Connections

Provide flexible steel conduit between 915 and 1830 mm in length for recessed and semirecessed lighting fixtures. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 16 mm diameter. Provide liquidtight flexible conduit in wet and damp locations for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections.

3.1.5 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, and when specifically indicated. Boxes in other locations: sheet steel, except that aluminum boxes may be used with aluminum conduit, and nonmetallic boxes may be used with nonmetallic conduit system. Provide each box with volume required for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 100 mm square, or octagonal, except

that smaller boxes may be installed as required by fixture configurations, as approved. Boxes for use in masonry-block or tile walls: square-cornered, tile-type, or standard boxes having square-cornered, tile-type covers. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 610 mm from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.5.1 Boxes

Boxes for use with raceway systems: minimum 40 mm deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 100 mm square, except that 100 by 50 mm boxes may be used where only one raceway enters outlet. Telecommunications outlets: a minimum of. Mount outlet boxes flush in finished walls.

3.1.5.2 Pull Boxes

Construct of at least minimum size required of code-gauge aluminum or galvanized sheet steel, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

3.1.5.3 Extension Rings

Extension rings are not permitted for new construction. Use only on existing boxes in concealed conduit systems where wall is furred out for new finish.

3.1.6 Mounting Heights

Mount panelboards, circuit breakers, motor controller and disconnecting switches so height of operating handle at its highest position is maximum 1980 mm above floor. Mount lighting switches. Mount receptacles 460 mm above finished floor. Measure mounting heights of wiring devices and outlets to center of device or outlet.

3.1.7 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors 14 sqmm and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors 22 sqmm and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and

plates; or heat shrink-type sleeves. Identify control circuit terminations in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

3.1.7.1 Marking Strips

Provide marking strips in accordance with the following:

- a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.
- b. Use permanent ink for the wire numbers
- c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.
- d. Size marking strips to accommodate the two sets of wire numbers.
- e. Assign a device designation in accordance with JIS C 8201-4-1 and JIS C 8201-4-2 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.
- f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.
- g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.8 Splices

Make splices in accessible locations. Make splices in conductors 5.5 sqmm and smaller diameter with insulated, pressure-type connector. Make splices in conductors 8 sqmm and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.9 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 0.58 mm. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.10 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated walls, partitions, floors, or ceilings.

3.1.11 Grounding and Bonding

Provide in accordance with JIS Z 9290-1. Ground exposed,

non-current-carrying metallic parts of electrical equipment, access flooring support system, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, and neutral conductor of wiring systems. Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Make interconnection to the gas line on the customer's side of the meter. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system. In addition to the requirements specified herein, provide telecommunications grounding in accordance with JIS C 60364-5-54. Where ground fault protection is employed, ensure that connection of ground and neutral does not interfere with correct operation of fault protection. If a splice is made inside a metal box, then the box shall be bonded to the equipment grounding conductor.

3.1.11.1 Ground Rods

Provide cone pointed ground rods. Measure the resistance to ground using the fall-of-potential method described in JIS C 60364-6. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod, additional rods, spaced on center, not less than twice the distance of the length of the rod. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

3.1.11.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, excepting specifically those connections for which access for periodic testing is required, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

3.1.11.3 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 5 ohms, contact Contracting Officer for further instructions.

3.1.12 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and protective devices within the control circuitry are not included in this section of the specifications and are provided under the section

specifying the associated equipment.

3.1.13 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 900 mm.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Where field painting of enclosures for panelboards, load centers or the like is specified to match adjacent surfaces, to correct damage to the manufacturer's factory applied coatings, or to meet the indicated or specified safety criteria, provide manufacturer's recommended coatings and apply in accordance to manufacturer's instructions.

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 5 working days notice prior to test.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring 14 sqmm and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring to provide direct reading of resistance. All existing wiring to be reused shall also be tested.

3.5.3 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed.

3.5.4 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

DRAFT

SECTION 26 29 23

ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS
02/20, CHG 1: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

EN 61800-3 (2017) Requirements for the Control of
Electromagnetic Interference
Characteristics of Subsystems and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 519 (2022) Standard for Harmonic Control in
Electrical Power Systems

IEEE C62.41.1 (2002; R 2008) Guide on the Surges
Environment in Low-Voltage (1000 V and
Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61000-3-12 (2012) Electromagnetic Compatibility (EMC)
- Part 3-12: Limits - Limits for harmonic
currents produced by equipment connected
to public low-voltage systems with input
current >16 A and ≤75 A per phase

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment
(1000 Volts Maximum)

NEMA ICS 1 (2022) Standard for Industrial Control and
Systems: General Requirements

NEMA ICS 3.1 (2019) Guide for the Application,
Handling, Storage, Installation and
Maintenance of Medium-Voltage AC
Contactors, Controllers and Control Centers

NEMA ICS 6 (1993; R 2016) Industrial Control and
Systems: Enclosures

NEMA ICS 7 (2020) Adjustable-Speed Drives

NEMA ICS 7.2 (2015) Application Guide for AC Adjustable Speed Drive Systems

NEMA MG 1 (2021) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15 Radio Frequency Devices

UL SOLUTIONS (UL)

UL 489 (2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

UL 61800-5-1 (2016) Adjustable Speed Electrical Power Drive Systems - Part 5-1: Safety Requirements - Electrical, Thermal and Energy

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SYSTEM DESCRIPTION

1.3.1 Performance Requirements

1.3.1.1 Electromagnetic Interference Suppression

Computing devices, as defined by 47 CFR 15 and EN 61800-3 rules and regulations, must be certified to comply with the requirements for class A computing devices and labeled.

1.3.1.2 Electromechanical and Electrical Components

Ensure electrical and electromechanical components of the Adjustable Speed Drive (ASD) do not cause electromagnetic interference to adjacent electrical or electromechanical equipment while in operation.

1.3.2 Electrical Requirements

1.3.2.1 Power Line Surge Protection

IEEE C62.41.1 and IEEE C62.41.2, IEEE 519, IEC 61000-3-12 Control panel must have surge protection, included within the panel to protect the unit from damaging transient voltage surges. Surge protective device must be mounted near the incoming power source and properly wired to all three phases and ground. Fuses must not be used for surge protection.

1.3.2.2 Sensor and Control Wiring Surge Protection

I/O functions as specified must be protected against surges induced on

control and sensor wiring installed outdoors and as shown. Test the inputs and outputs in both normal mode and common mode using the following two waveforms:

- a. A 10 microsecond by 1000 microsecond waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.
- b. An 8 microsecond by 20 microsecond waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

- Schematic Diagrams; G
- Interconnecting Diagrams; G
- Installation Drawings; G
- As-Built Drawings; G

SD-03 Product Data

- Adjustable Speed Drives; G
- Wires and Cables
- Equipment Schedule

SD-06 Test Reports

- ASD Test
- Performance Verification Tests
- Endurance Test

SD-08 Manufacturer's Instructions

- Installation instructions

SD-09 Manufacturer's Field Reports

- ASD Test Plan
- Standard Products

SD-10 Operation and Maintenance Data

- Adjustable Speed Drives, Data Package 4

1.5 QUALITY ASSURANCE

1.5.1 Schematic Diagrams

Submit diagrams showing circuits and device elements for each replaceable module. Schematic diagrams of printed circuit boards are permitted to group functional assemblies as devices, provided that sufficient information is provided for government maintenance personnel to verify proper operation of the functional assemblies.

1.5.2 Interconnecting Diagrams

Show interconnections between equipment assemblies, and external interfaces, including power and signal conductors. Include for enclosures and external devices.

1.5.3 Installation Drawings

Show floor plan of each site, with ASD's and motors indicated. Indicate ventilation requirements, adequate clearances, and cable routes. Submit drawings for government approval prior to equipment construction or integration. Immediately record modifications to original drawings made during installation for inclusion into the as-built drawings.

1.5.4 Equipment Schedule

Provide schedule of equipment supplied. Schedule must provide a cross reference between manufacturer data and identifiers indicated in shop drawings. Schedule must include the total quantity of each item of equipment supplied and data indicating compatibility with motors being driven. For complete assemblies, such as ASD's, provide the serial numbers of each assembly, and a sub-schedule of components within the assembly. Provide recommended spare parts listing for each assembly or component.

1.5.5 Installation Instructions

Provide installation instructions issued by the manufacturer of the equipment, including notes and recommendations, prior to shipment to the site. Provide operation instructions prior to acceptance testing.

1.5.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6 DELIVERY AND STORAGE

Store delivered equipment to protect from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

1.7 WARRANTY

The complete system must be warranted by the manufacturer for a period of one year . Repair or replace any component failing to perform its function as specified and documented at no additional cost to the Government. Items repaired or replaced must be warranted for an additional period of at least one year from the date that it becomes functional again, as specified in FAR 52.246-21 Warranty of Construction.

1.8 MAINTENANCE

1.8.1 Spare Parts

Manufacturers provide spare parts in accordance with recommended spare parts list.

1.8.2 Operation and Maintenance Data

Provide in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Provide service and maintenance information including preventive maintenance, assembly, and disassembly procedures. Include electrical drawings from electrical general sections. Provide additional information necessary to provide complete operation, repair, and maintenance information, detailed to the smallest replaceable unit. Include copies of as-built submittals. Provide routine preventative maintenance instructions, and equipment required. Provide instructions on how to modify program settings, and modify the control program. Provide instructions on drive adjustment, trouble-shooting, and configuration. Provide instructions on process tuning and system calibration.

1.8.3 Technical Support

Provide the ASDs with manufacturer's technical telephone support in English, readily available during normal working hours.

PART 2 PRODUCTS

2.1 ADJUSTABLE SPEED DRIVES (ASD)

Provide adjustable speed drive to control the speed of induction motor(s). The ASD must include the following minimum functions, features and ratings.

- a. Input circuit breaker per UL 489 with a minimum of 10,000 amps symmetrical interrupting capacity and door interlocked external operator.
- b. A converter stage per UL 61800-5-1 must change fixed voltage, fixed frequency, ac line power to a fixed dc voltage. The converter must utilize a full wave bridge design incorporating diode rectifiers. Silicon Controlled Rectifiers (SCR) are not acceptable. The converter must be insensitive to three phase rotation of the ac line and must not cause displacement power factor of less than .95 lagging under any speed and load condition.

- c. An inverter stage must change fixed dc voltage to variable frequency, variable ac voltage for application to a standard NEMA MG 1 Part 30 motor designed for use with adjustable frequency power supplies. Switch the inverter to produce a sine coded pulse width modulated (PWM) output waveform.
- d. The ASD shall be capable of supplying 110 percent of rated full load current for one minute at maximum ambient temperature.
- e. The ASD must be designed to operate from voltage indicated on drawings, plus or minus 10 percent, three phase, 60 Hz supply, and control motors with a corresponding voltage rating.
- f. Acceleration and deceleration time must be independently adjustable from one second to 60 seconds.

Required deceleration time may be achieved using not only dynamic braking resistor but with other methods described in NEMA ICS 7.2-2015 paragraph 5.2.5.

- g. Adjustable full-time current limiting must limit the current to a preset value which must not exceed 110 percent of the controller rated current. The current limiting action must maintain the V/Hz ratio constant so that variable torque can be maintained. Short time starting override must allow starting current to reach 175 percent of controller rated current to maximum starting torque.
- h. The controllers must be capable of producing an output frequency over the range of 3 Hz to 60 Hz (20 to one speed range), without low speed cogging. Over frequency protection must be included such that a failure in the controller electronic circuitry must not cause frequency to exceed 110 percent of the maximum controller output frequency selected.
- i. Minimum and maximum output frequency must be adjustable over the following ranges: 1) Minimum frequency 3 Hz to 50 percent of maximum selected frequency; 2) Maximum frequency 40 Hz to 60 Hz.
- j. The controller efficiency at any speed must not be less than 96 percent.
- k. The controllers must be capable of being restarted into a motor coasting in the forward direction without tripping.
- l. Protection of power semiconductor components must be accomplished without the use of fast acting semiconductor output fuses. Subjecting the controllers to any of the following conditions must not result in component failure or the need for fuse replacement:
 - (1) Short circuit at controller output
 - (2) Ground fault at controller output
 - (3) Open circuit at controller output
 - (4) Input undervoltage
 - (5) Input overvoltage

- (6) Loss of input phase
- (7) AC line switching transients
- (8) Instantaneous overload
- (9) Sustained overload exceeding 115 percent of controller rated current
- (10) Over temperature
- (11) Phase reversal
- m. Solid state motor overload protection must be included such that current exceeding an adjustable threshold must activate a 60 second timing circuit. Should current remain above the threshold continuously for the timing period, the controller will automatically shut down.
- n. Include slip compensation circuit that will sense changing motor load conditions and adjust output frequency to provide speed regulation of NEMA MG 1 Part 30 designed for use with adjustable frequency power supplies motors to within plus or minus 0.5 percent of maximum speed without the necessity of a tachometer generator.
- o. The ASD must be factory set for manual restart after the first protective circuit trip for malfunction (overcurrent, undervoltage, overvoltage or overtemperature) or an interruption of power. The ASD must be capable of being set for automatic restart after a selected time delay. If the drive faults again within a specified time period (adjustable 0-60 seconds), a manual restart will be required.
- p. The ASD must include external fault reset capability. All the necessary logic to accept an external fault reset contact must be included.
- q. Provide critical speed lockout circuitry to prevent operating at frequencies with critical harmonics that cause resonant vibrations. The ASD must have a minimum of three user selectable bandwidths.
- r. Provide properly sized NEMA rated by-pass and isolation contactors to enable operation of motor in the event of ASD failure. Install mechanical and electrical interlocks between the by-pass and isolation contactors. Provide a selector switch and transfer delay timer. Motor overload and short circuit protective features must remain in use during the bypass mode.
- s. Each individual ASD must meet the following Total Harmonic Distortion (THD) requirements at the input terminals to the factory assembly of the ASD or at the load disconnecting means serving the ASD and filter assembly. These measurements should be taken with the drive set at 90 percent frequency (rpms) and the motor under a minimum of 50 percent demand.
 - (1) The Voltage THD should not exceed 2.0 percent THD.
 - (2) The Current THD should not exceed 15.0 percent THD.

- (3) If the standard factory ASD does not meet or exceed these requirements the factory must install appropriate equipment (Harmonic Traps, Filters, different Drive technology, etc.) to mitigate the distortion to assure performance of the VFD is within the limits.
- (4) These tests should be performed at the Manufacturers Laboratory facilities and submitted as part of the Product Data Submittals, in order to prevent the necessity of adding mitigation equipment in the field. If the requirements listed above are met, IEEE 519 will also be met.

t. Minimum Operating Conditions. Designed and constructed ASD's to operate within the following service conditions:

- (1) Ambient Temperature Rating: 0 to 120 degrees F.
- (2) Non-condensing relative humidity rating: less than 95 percent.
- (3) Ambient rating: Not exceed 1,006 meters.

2.1.1 ASD for HVAC Application

ASDs must have the following features:

- a. A local operator control providing the following functions:
 - (1) Remote/Local operator selection with password access.
 - (2) Run/Stop and manual speed commands.
 - (3) All programming functions.
 - (4) Scrolling through all display functions.
- b. A local operator control panel with the following data displayed:
 - (1) ASD status.
 - (2) Frequency.
 - (3) Motor RPM.
 - (4) Phase current.
 - (5) Scrolling through all display functions.
 - (6) Fault diagnostics in descriptive text.
 - (7) All programmed parameters.
- c. Standard PI loop controller with input terminal for controlled variable and parameter settings.
- d. User interface terminals for remote control of ASD speed, speed feedback, and an isolated form C SPDT relay, which energizes on a drive fault condition.
- e. An isolated form C SPDT auxiliary relay which energizes on a run

command.

- f. An adjustable carrier frequency with 16 kHz minimum upper limit.
- g. A built-in or external line reactor with 3 percent minimum impedance to protect the DC bus capacitors and rectifier section diodes, reduce power line transient voltage, line notching, DC bus over-voltage tripping and improve the inverter over-current and over-voltage conditions.
- h. Historical logging information and displays:
 - (1) Real-time clock with current time and date.
 - (2) Running log of total power versus time.
 - (3) Total run time.
 - (4) Fault log, maintaining last four faults with time and data stamp for each.
- i. The ASD must be capable of automatic control by a remote 0 to 10 VDC signal, by BACnet network command, or manually by the ASD control panel.
- j. ASDs must include the following operator programmable parameters:
 - (1) Upper and lower limit frequency.
 - (2) Acceleration and deceleration rate.
 - (3) Variable torque volts per Hertz curve.
 - (4) Starting voltage level.
 - (5) Starting frequency level.
 - (6) Display speed scaling.
 - (7) Enable/disable soft stall feature.
 - (8) Motor overload level.
 - (9) Motor stall level.
 - (10) Jump frequency and hysteresis band.
 - (11) PWM carrier frequency.
- k. ASD must have the following protective features:
 - (1) An electronic adjustable inverse time current limit with consideration for additional heating of the motor at frequencies below 45Hz, for the protection of the motor.
 - (2) An electronic adjustable soft stall feature, allowing the ASD to lower the frequency to a point where the motor will not exceed the full-load amperage when an overload ASD will automatically return to the requested frequency when load conditions permit.

- (3) A separate electronic stall at 110 percent ASD rated current, and a separate hardware trip at 190 percent current.
- (4) The ability to shut down if inadvertently started into a rotating load without damaging the ASD or the motor.
- (5) The ability to keep a log of a minimum of four previous fault conditions, indicating the fault type and time of occurrence in descriptive text.
- (6) The ability to sustain 110 percent rated current for 60 seconds.
- (7) The ability to shutdown safely or protect against and record the following fault conditions:
 - (a) Over current (and an indication if the over current was during acceleration, deceleration, or running).
 - (b) Over current internal to the drive.
 - (c) Motor overload at start-up.
 - (d) Over voltage from utility power.
 - (e) Motor running overload.
 - (f) Over voltage during deceleration.
 - (g) ASD over heat.
 - (h) Load and ground fault.
 - (h) Abnormal parameters or data in ASD EEPROM.

2.2 ENCLOSURES

Provide equipment enclosures conforming to NEMA 250, NEMA ICS 7, and NEMA ICS 6, with a heater if located outdoors. An HMCP device shall provide the disconnecting means. The operating handle shall protrude through the door, but the disconnect shall not be mounted on the door. The handle shall indicate ON, OFF, and tripped conditions. The handle shall have provisions to accommodate a minimum of three padlocks in the OFF position. Interlocks shall prevent unauthorized opening or closing of the ASD door with the disconnect handle in the ON position. The door handle interlock should have provisions to be defeated by qualified maintenance personnel.

2.3 WIRES AND CABLES

All wires and cables must conform to NEMA 250, NEMA ICS 7, NFPA 70.

2.4 NAMEPLATES

Nameplates external to NEMA enclosures must conform with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manufacturer's standard, permanent nameplates for internal areas of enclosures.

2.5 SOURCE QUALITY CONTROL

2.5.1 ASD Test Plan

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test plans.

2.5.2 ASD Test Report

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test reports.

PART 3 EXECUTION

3.1 INSTALLATION

Per NEMA ICS 3.1, install equipment in accordance with the approved manufacturer's printed installation drawings, instructions, wiring diagrams, and as indicated on project drawings and the approved shop drawings. A field representative of the drive manufacturer must supervise the installation of all equipment, and wiring.

3.2 GROUNDING

Per NEMA ICS 7.2, ASD must be solidly grounded to the main distribution.

3.3 FIELD QUALITY CONTROL

Specified products must be tested as a system for conformance to specification requirements prior to scheduling the acceptance tests. Conduct performance verification tests in the presence of Government representative, observing and documenting complete compliance of the system to the specifications. Submit a signed copy of the test results, certifying proper system operation before scheduling tests.

3.3.1 ASD Test

A proposed test plan must be submitted to the contracting officer at least 28 calendar days prior to proposed testing for approval. The tests must conform to NEMA ICS 1, NEMA ICS 7, and all manufacturer's safety regulations. The Government reserves the right to witness all tests and review any documentation. Inform the Government at least 14 working days prior to the dates of testing. Perform the ASD test with the assistance of a factory-authorized service representative.

3.3.2 Performance Verification Tests

"Performance Verification Test" plan must provide the step by step procedure required to establish formal verification of the performance of the ASD. Compliance with the specification requirements must be verified by inspections, review of critical data, demonstrations, and tests. The Government reserves the right to witness all tests, review data, and request other such additional inspections and repeat tests as necessary to ensure that the system and provided services conform to the stated requirements. Inform the Government 14 calendar days prior to the date the test is to be conducted.

3.3.3 Endurance Test

Immediately upon completion of the performance verification test, the endurance test must commence. The system must be operated at varying rates for not less than 192 consecutive hours, at an average effectiveness level of 0.9998, to demonstrate proper functioning of the complete PCS. Continue the test on a day-to-day basis until performance standard is met. The contractor is not allowed in the building during the endurance test. The system must respond as designed.

3.4 DEMONSTRATION

3.4.1 Training

Coordinate training requirements with the Contracting Officer. Provide video tapes, if available, of all training provided to the Government for subsequent use in training new personnel. Provide all training aids, texts, and expendable support material for a self-sufficient presentation shall be provided, the amount of which to be determined by the contracting officer.

3.4.1.1 Instructions to Government Personnel

Provide the services of competent instructors with minimum two-year field experience with the operation and maintenance of similar ASDs who will give full instruction to designated personnel in operation, maintenance, calibration, configuration, and programming of the complete control system. Orient the training specifically to the system installed. Instructors must be thoroughly familiar with the subject matter they are to teach. The number of training days of instruction furnished must be as specified. A training day is defined as eight hours of instruction, including two 15-minute breaks and excluding lunch time; Monday through Friday. Provide a training manual for each student at each training phase which describes in detail the material included in each training program. Provide one additional copy for archiving. Provide equipment and materials required for classroom training. Provide a list of additional related courses, and offers, noting any courses recommended. List each training course individually by name, including duration, approximate cost per person, and location of course. Unused copies of training manuals must be turned over to the Government at the end of last training session.

3.4.1.2 Operating Personnel Training Program

Provide one 2-hour training session at the site at a time and place mutually agreeable between the Contractor and the Government. Provide session to train 4 operation personnel in the functional operations of the system and the procedures that personnel will follow in system operation. This training shall include:

- a. System overview
- b. General theory of operation
- c. System operation
- d. Alarm formats
- e. Failure recovery procedures

f. Troubleshooting

3.4.1.3 Engineering/Maintenance Personnel Training

Accomplish the training program as specified. Training must be conducted on site at a location designated by the Government. Provide a one-day training session to train four engineering personnel in the functional operations of the system. This training must include:

- a. System overview
- b. General theory of operation
- c. System operation
- d. System configuration
- e. Alarm formats
- f. Failure recovery procedures
- g. Troubleshooting and repair
- h. Maintenance and calibration
- i. System programming and configuration

-- End of Section --

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SECTION 26 51 00

INTERIOR LIGHTING
05/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Japan Luminaires Association (JIL) as approved by the Contracting Officer's representative.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 0365 (2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment

JIS C 0920 (2003) Degrees Of Protection Provided By Enclosures (IP Code)

JIS C 8105-1 (2021) Luminaires - Part 1: General Requirements For Safety

JIS C 8105-2-22 (2014) Luminaires - Part 2-22: Particular Requirements - Luminaires For Emergency Lighting

JIS C 8152-2 (2019) Measurement method of white light emitting diode (LED) for lighting-Part 2: LED modules and LED light engines

JIS C 8152-3 (2013) Measurement method of white light emitting diode (LED) for lightingPart 3: Measurement of luminous flux maintenance rate

JIS C 8153 (2015) DC or AC supplied electronic control gear for LED modules -- Performance requirements

JIS C 8154 (2015) LED modules for general lighting -- Safety specifications

JIS C 8155 (2019) LED modules for general lighting -- Performance requirements

JIS C 8201-5-1 (2022) Low-Voltage Switchgear And Control Gear - Part 5-1: Control Circuit Devices And Switching Elements - Electromechanical Control Circuit Devices

JIS C 8286	(2021) Electrical accessories -- Cord sets and interconnection cord sets
JIS C 8304	(2009; R 2019) Small switches for indoor use
JIS C 5402-20-2	(2005) Electromechanical components for electronic equipment -- Basic testing procedures and measuring methods -- Part 20-2: Test 20b -- Flammability tests -- Fireproofness
JIS C 61000-3-2	(2019) Electromagnetic compatibility (EMC) -- Part 3-2: Limits -- Limits for harmonic current emissions
JIS G 3141	(2021) Cold-reduced carbon steel sheet and strip
JIS G 3302	(2022) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3547	(2015) Cold-rolled stainless steel plate, sheet and strip
JIS G 4309	(2013) Stainless Steel Wires
JIS H 8610	(1999) Electroplated-Coatings of Zinc on Iron or Steel
JIS Z 8113	(1998) Lighting vocabulary
JIS Z 9110	(2010; R 2011) General rules of recommended lighting levels
NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)	
NFPA 101	(2024) Life Safety Code
UL SOLUTIONS (UL)	
UL 2043	(2013) Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in Air-Handling Spaces
UL 924	(2016; Reprint May 2018) UL Standard for Safety Emergency Lighting and Power Equipment
UL 94	(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires or luminaire accessories are

specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, must be as defined in JIS Z 8113

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the JIS Z 9110 as applicable, for the lighting system specified. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Occupancy/Vacancy Sensor Coverage Layout

SD-03 Product Data

Luminaires; G

Light Sources; G

Drivers, Ballasts and Generators; G

LED Luminaire Warranty;

Vacancy Sensors; G

Dimming Controllers (Dimmers); G

Power Hook Luminaire Hangers; G

Exit Signs; G

Emergency Lighting Unit (EBU); G

LED Emergency Drivers; G

Occupancy Sensors; G

Ambient Light Level Sensor; G

SD-06 Test Reports

Occupancy/Vacancy Sensor Verification Tests

Energy Efficiency

1.5 QUALITY CONTROL

1.5.1 Luminaire Drawings

Include dimensions, accessories, and installation and construction details. Photometric data, including zonal lumen data, average and

minimum ratio, aiming diagram, and computerized candlepower distribution data must accompany shop drawings.

1.5.2 Occupancy/Vacancy Sensor Coverage Layout

Provide floor plans showing coverage layouts of all devices using manufacturer's product information.

1.5.3 Luminaire Design Data

Provide long term lumen maintenance projections for each LED luminaire in accordance with JIS C 8152-3. Data used for projections must be obtained from testing in accordance with JIS C 8152-3 Appendix B.

1.5.4 LED Luminaire - Test Report

Submit test report on manufacturer's standard production model luminaire.

1.5.5 LED Light Source - Test Report

Submit test report on manufacturer's standard production LED light source (package, array or module). Include all applicable and required data, as outlined under "8 Report on Test Results" in JIS C 8152-3.

1.5.6 Photometric Plan

1.5.6.1 Computer-generated Photometric Plans

Computer-generated photometric plans for each space are required to verify proposed luminaires and locations meet the required performance criteria of the design using the applicable light loss factor (LLF).

Target illumination levels are provided for each Interior Application. Depending on the application and the recommendations provided by the IES, values are given as one of the following:

- a. Minimum: No values anywhere on the calculation grid may be less than this value, within a 10 percent margin of error.
- b. Minimum Average: An average, taken over the entire task area for the application, may not be less than this value, within a 10 percent margin of error.
- c. Maximum: No values anywhere on the calculation grid may be greater than this value, within a 10 percent margin of error.
- d. Maximum Average: An average, taken over the entire task area for the application, may not be greater than this value, within a 10 percent margin of error.
- e. Uniformity: Unless otherwise noted, uniformity is calculated as a ratio of the average calculated illuminance over the minimum calculated illuminance of the calculation grid.

1.5.6.2 Schematic Photometric Plan Calculations

Schematic photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every 305 mm across

the task area.

- b. Average maintained illuminance level.
- c. Minimum and maximum maintained illuminance levels.
- d. Lighting power density (Watts per square meter).
- e. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.6.3 Final Photometric Plan Calculations

Final photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every 305 mm across the task area.
- b. Where applicable, vertical illuminance measurements at designated surface, taken at a maximum of every 305 mm across task area.
- c. Minimum and maximum maintained illuminance levels.
- d. Average maintained illuminance level.
- e. Average to minimum and maximum to minimum ratios for horizontal illuminance.
- f. Lighting power density (Watts per square meter).
- g. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.7 Occupancy/Vacancy Sensor Verification Tests

Submit test report outlining post-installation coverage and operation of sensors.

1.5.8 Test Laboratories

Test laboratories for JIS C 8152-2 and JIS C 8152-3 test reports must be a Japan Accredited Laboratory for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program for both LM-79 and LM-80 or JIS C 8152-2 and JIS C 8152-3 testing

1.5.9 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of applicable codes and standards, unless more stringent requirements are specified or indicated.

1.5.10 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for two years prior to bid opening. The two-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the two-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.10.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.10.2 Material and Equipment Manufacturing Date

Products manufactured more than six months prior to date of delivery to site must not be used, unless specified otherwise.

1.5.10.3 Energy Efficiency

Submit data indicating lumens per watt efficacy and color rendering index of light source.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 LED Luminaire Warranty

- a. Provide a written 5 year on-site replacement warranty for material, fixture finish, and workmanship. On-site replacement includes transportation, removal, and installation of new products.
 - (1) Include finish warranty to include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.
 - (2) Material warranty must include:
 - (a) All drivers.
 - (b) Replacement when more than 10 percent of LED sources in any lightbar or subassembly(s) are defective or non-starting.
- b. Warranty period must begin on date of beneficial occupancy. Provide the Contracting Officer with signed warranty certificates prior to final payment.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

Products and materials not considered to be luminaires, luminaire controls, or associated equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.2 LUMINAIRES

JIS C 8105-1, JIS C 61000-3-2 JIS C 8154 and JIS C 8155. Provide luminaires as indicated in luminaire schedule and NL plates or details on project plans. Provide luminaires complete with light sources of quantity, type, and wattage indicated. Provide all luminaires of the same type by the same manufacturer. Luminaires must be specifically designed for use with the driver, ballast or generator and light source provided.

2.2.1 LED Luminaires

Provide luminaires complete with power supplies (drivers) and light sources. Provide design information including lumen output and design life in luminaire schedule on project plans for LED luminaires. LED luminaires must meet the minimum requirements in the following table:

<u>LUMINAIRE TYPE</u>	<u>MINIMUM LUMINAIRE EFFICACY (LE)</u>	<u>MINIMUM COLOR RENDERING INDEX (CRI)</u>
LED TROFFER - 1 x 4 2 x 2 2 x 4	90 LPW	80
LED Downlight	50 LPW	90
LED Track or Accent	40 LPW	80
LED Low Bay/High Bay	80 LPW	70
LED Linear Ambient	80 LPW	80

LED luminaires must also meet the following minimum requirements:

- Luminaires must have a minimum 5 year manufacturer's warranty.
- Luminaires must have a minimum L70 lumen maintenance value of 50,000 hours as calculated by JIS C 8152-3, with data obtained per JIS C 8152-3 requirements.
- Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- Luminaires must be tested to JIS C 8152-2 and JIS C 8152-3, with the results provided as required in the Submittals paragraph of this specification.

2.3 DRIVERS, BALLASTS and GENERATORS

2.3.1 LED Drivers

JIS C 8153, JIS C 8154 and JIS C 8155. LED drivers must be electronic, constant-current type and comply with the following requirements:

- a. Output power (watts) and luminous flux (lumens) as shown in luminaire schedule for each luminaire type to meet minimum luminaire efficacy (LE) value provided.
- b. Power Factor (PF) greater than or equal to 0.9 over the full dimming range when provided.
- c. Current draw Total Harmonic Distortion (THD) of less than 20 percent.
- d. Class A sound rating.
- e. Operable at input voltage of 120-277 volts at 60 hertz.
- f. Minimum 5 year manufacturer's warranty.
- g. RoHS compliant.
- h. Integral thermal protection that reduces or eliminates the output power if case temperature exceeds a value detrimental to the driver.
- i. UL listed for dry or damp locations typical of interior installations.
- j. Fully-dimmable using 0-10V control as indicated in luminaire schedule.

2.4 LIGHT SOURCES

JIS C 8155. Provide type and wattage as indicated in luminaire schedule on project plans.

2.4.1 LED Light Sources

- a. Correlated Color Temperature (CCT) as indicated on drawings.
- b. Minimum Color Rendering Index (CRI) value of 80.
- c. High power, white light output utilizing phosphor conversion (PC) process.
- d. RoHS compliant.

2.5 LIGHTING CONTROLS

Provide lighting controls and associated equipment in accordance with drawings and applicable energy codes.

2.5.1 Toggle Switches

Provide line-voltage toggle switches as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.5.2 Dimming Controllers (Dimmers)

JIS C 8304. 120/277 V0-10 V dimmers must provide flicker-free, continuously variable light output throughout the dimming range. Provide radio frequency interference suppression integral to device. Provide dimmers utilizing pulse width modulation (PWM). Provide device with a vertical slider, paddle, rotary button, or toggle (with adjacent vertical slider) type control, with finish to match switches and outlets in same area. Provide back box in wall with sufficient depth to accommodate body of switch and wiring. Devices must be capable of operating at their full rated capacity regardless of being single or ganged-mounted, and be compatible with three-way and four-way switching scenarios. Dimmers must be capable of controlling three-wire fluorescent ballasts or LED drivers. Ensure compatibility of dimmer with separate power packs when utilized for lighting control. Dimmers and the ballasts or drivers they control, must be provided from the same manufacturer, or tested and certified as compatible for use together.

2.5.3 Sensors for Lighting Control

UL 94, JIS C 5402-20-2 and JIS C 8201-5-1.

2.5.3.1 Occupancy Sensors

Provide occupancy sensors with coverage patterns as indicated on project plans. Provide no less quantity of sensors as shown on plans, but add additional sensors when required to fulfill coverage requirement for the specific model sensor provided. Sensor must be provided with an adaptive learning function that automatically sets sensor in optimum calibration in a set period of time after installation and a non-volatile memory that saves settings after a power outage. Provide sensors designed for ceiling, wall or wall-box installation as indicated. Operating voltage must be 120 volts. Provide housing of high-impact, injection-molded thermoplastic with a multi-segmented lens for PIR and dual technology sensors. Sensor operation requires movement to activate luminaires controlled, and turns luminaires off after a set time of inactivity.

2.5.3.1.1 Passive Infrared (PIR) Sensors

Provide ceiling or wall-mounted PIR sensors meeting the following requirements:

- a. Temperature compensated, dual element sensor and a multi-element fresnel lens (Poly IR4 material).
- b. Technology to optimize automatic time delay to fit occupant usage patterns.
- c. No minimum load requirement for line voltage sensors and be capable of switching from zero to 800 W at 120 VAC, 50/60 Hz and from zero to 1200 W at 277 VAC, 50/60 Hz. Control voltage sensors must not exceed a maximum load requirement of 20 mA at 24VDC.
- d. Time delay of five to 30 minutes in increments of five minutes with a walk through and test mode set by DIP switch.
- e. LED indicator that remains active during occupancy.
- f. Built-in light level sensor that is operational from 0.8 to 18 lux.

- g. Coverage pattern tested to applicable standards.
- h. Standard five year warranty.
- i. No leakage current to load when in the off mode.

2.5.3.1.2 Ultrasonic Sensors

Provide ceiling-mounted ultrasonic sensors meeting the following requirements:

- a. Operate at an ultrasonic frequency of 26 kHz.
- b. LED on exterior of device to indicate occupant detection.
- c. Adjustable time delay period of 15 seconds to 15 minutes.
- d. Minimum five year warranty.

2.5.3.1.3 Dual Technology Sensors

Provide dual technology sensors that meet the requirements for PIR sensors and ultrasonic sensors indicated above. If either the passive infrared or ultrasonic sensing registers occupancy, the luminaires must remain on.

2.5.3.1.4 Power Packs for Sensors

UL 2043 Power packs used to provide power to one or more lighting control sensors must meet the following requirements:

- a. Input voltage - 120-277 VAC; output voltage - 24 VDC at 225 mA.
- b. Plenum-rated, high-impact thermoplastic enclosure.
- c. Utilizes zero-crossing circuitry to prevent damage from inrush current.
- d. Maximum load rating of 16 amps for electronic lighting loads.
- e. RoHS compliant.

2.5.3.2 Vacancy Sensors

Provide vacancy sensors as indicated above under paragraph OCCUPANCY SENSORS, but with requirement of a manual operation to activate luminaires controlled. Provide automatic operation to turn luminaires off after a set period of inactivity.

2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

UL 924, JIS C 8105-2-22 and NFPA 101 compliant.

2.6.1 Exit Signs

Provide exit signs consuming a maximum of five watts total.

2.6.1.1 LED Self-Powered Exit Signs

Provide in UV-stable, thermo-plastic housing with using clear

polycarbonate housing, configured for ceiling mounting. Provide 150 mm high, 19 mm stroke red lettering on face of sign. Provide chevrons on either side of lettering to indicate direction. Provide single face. Equip with automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1 1/2 hours. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver.

2.6.1.2 LED Remote-Powered Exit Signs

Provide as indicated above for self-powered type, but without battery and charger. Exit sign must contain provision for 120/277 VAC or 6-48 VDC input from remote source.

2.6.2 Emergency Lighting Unit (EBU)

Provide in UV-stable, thermo-plastic as indicated. Emergency lighting units must be rated for 12 volts, except units having no remote-mounted lamps and having no more than two unit-mounted light sources may be rated six volts. Equip units with brown-out sensitive circuit to activate battery when input voltage falls to 75 percent of normal. Equip with two LED, automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1 1/2 hours. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver.

2.6.3 LED Emergency Drivers

Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior, and fully-automatic solid-state charger, battery and inverter integral to a self-contained housing. Provide self-diagnostic function integral to emergency driver. Integral nickel-cadmium battery is required to supply a minimum of 90 minutes of emergency power at 5 watts, 10-50 VDC, constant output. Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.

2.7 LUMINAIRE SUPPORT HARDWARE

2.7.1 Wire

JIS G 3547; Galvanized, soft tempered steel, minimum 2.7 mm in diameter, or galvanized, braided steel, minimum 2 mm in diameter.

2.7.2 Wire for Humid Spaces

JIS G 4309; Annealed stainless steel, minimum 2.7 mm in diameter.

Annealed nickel-copper alloy, minimum 2.7 mm in diameter.

2.7.3 Threaded Rods

Threaded steel rods, 4.76 mm diameter, zinc or cadmium coated.

2.7.4 Straps

Galvanized steel, 25 by 4.76 mm, conforming to JIS G 3302, with a light commercial zinc coating or JIS G 3141 with an electrodeposited zinc coating conforming to JIS H 8610.

2.8 POWER HOOK LUMINAIRE HANGERS

JIS C 8105-1. Provide an assembly consisting of through-wired power hook housing, interlocking plug and receptacle, power cord, and luminaire support loop. Power hook housing must be cast aluminum having two 19 mm threaded hubs. Support hook must have safety screw. Fixture support loop must be cast aluminum with provisions for accepting 19 mm threaded stems. Power cord must include 410 mm of 3 conductor 1.2 mm Type S0 cord or power cord per JIS C 8286. Assembly must be rated 120 volts or 277 volts, 15 amperes.

2.9 EQUIPMENT IDENTIFICATION

2.9.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.9.2 Labels

Provide labeled luminaires in accordance with JIS C 8105-1 requirements. All luminaires must be clearly marked for operation of specific light sources and ballasts, generators or drivers. Note the following light source characteristics in the format "Use Only _____":

- a. Light source diameter code (T-5, T-8), tube configuration (twin, quad, triple), base type, and nominal wattage for fluorescent and compact fluorescent luminaires.
- b. Start type (programmed start, instant start) for fluorescent and compact fluorescent luminaires.
- c. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. Ballasts, generators or drivers must have clear markings indicating multi-level outputs and indicate proper terminals for the various outputs.

2.10 FACTORY APPLIED FINISH

Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of JIS C 0920 corrosion-resistance test.

2.11 RECESS- AND FLUSH-MOUNTED LUMINAIRES

Provide access to lamp and ballast from bottom of luminaire. Provide trim for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations must conform to IEEE C2 and JIS C 0365 and to the requirements specified herein. Install luminaires and lighting controls to meet the requirements of applicable codes and standards. To encourage consistency and uniformity, install luminaires of the same manufacture and model number when residing in the same facility or building.

3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature, color rendering index, and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Installation must meet requirements of applicable codes and standards. Mounting heights specified or indicated must be to the bottom of the luminaire for ceiling-mounted luminaires and to center of luminaire for wall-mounted luminaires. Obtain approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed. Recessed and semi-recessed luminaires must be independently supported from the building structure by a minimum of four wires, straps or rods per luminaire and located near each corner of the luminaire. Ceiling grid clips are not allowed as an alternative to independently supported luminaires. Round luminaires or luminaires smaller in size than the ceiling grid must be independently supported from the building structure by a minimum of four wires, straps or rods per luminaire, spaced approximately equidistant around. Do not support luminaires by acoustical tile ceiling panels. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 19 mm metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire. Provide wires, straps, or rods for luminaire support in this section. Luminaires installed in suspended ceilings must also comply with the requirements of Section 09 51 00 ACOUSTICAL CEILINGS.

3.1.3 Ballasts, Generators and Power Supplies

Typically, provide ballasts, generators, and power supplies (drivers) integral to luminaire as constructed by the manufacturer.

3.1.4 Exit Signs and Emergency Lighting Units

Wire exit signs and emergency lighting units ahead of the local switch, to

the normal lighting circuit located in the same room or area.

3.1.5 Photocell Switch Aiming

Aim switch according to manufacturer's recommendations.

3.1.6 Occupancy/Vacancy Sensors

Provide testing of sensor coverage in all spaces where sensors are placed. This should be done only after all furnishings (carpet, furniture, workstations, etc.) have been installed. Provide quantity of sensor units indicated as a minimum. Provide additional units to give full coverage over controlled area. Full coverage must provide hand and arm motion detection for office and administration type areas and walking motion for industrial areas, warehouses, storage rooms and hallways. Locate the sensor(s) as indicated and in accordance with the manufacturer's recommendations to maximize energy savings and to avoid nuisance activation and deactivation due to sudden temperature or airflow changes and usage.

3.1.7 Daylight or Ambient Light Level Sensor

Locate sensor as indicated and in accordance with the manufacturer's recommendations. Adjust sensor for 300 luxor for the indicated light level measured at the work plane for that particular area.

3.2 FIELD APPLIED PAINTING

Paint lighting equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Provide painting as specified in Section 09 90 00 PAINTS AND COATINGS.

-- End of Section --

SECTION 27 10 00

BUILDING TELECOMMUNICATIONS CABLING SYSTEM
08/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Contractor may substitute compatible Telecommunications Industry Association (TIA) or UL standards, as approved by the Contracting Officer's representative.

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 5507 (2010) LAN Twisted Pair Cable

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 3665-1-1 (2007) Tests on electric and optical fibre cables under fire conditions -- Part 1-1: Test for vertical flame propagation for a single insulated wire or cable -- Apparatus

JIS C 5964-4 (2014) Fiber optic connector interfaces -- Part 4: Type SC connector family (F04 Type)

JIS C 5964-4-100 (2018) Fiber optic connector interfaces -- Part 4-100: Type SC connector family -- Simplified receptacle SC-PC connector interfaces (F16 type)

JIS C 6011-1 (2015) Mechanical structures for electronic equipment -- Tests for IEC 60917 and IEC 60297 series -- Part 1: Environmental requirements, test set-up and safety aspects for cabinets, racks, subracks and chassis under indoor conditions

JIS C 6012-3-100 (2015) Mechanical structures for electronic equipment -- Dimensions of mechanical structures of the 482.6 mm (19 in) series -- Part 3-100: Basic dimensions of front panels, subracks, chassis, racks and cabinets

JIS C 6185-2 (2014) Optical time-domain reflectometers (OTDR) -- Part 2: Calibration of OTDR for single mode fibers

JIS C 6185-3 (2014) Optical time-domain reflectometers (OTDR) -- Part 3: Calibration of OTDR for multimode fibers

JIS C 6820	(2018) General rules of optical fibers
JIS C 6850	(2006) General rules of optical fiber cables
JIS C 6870-2	(2021) Indoor optical fiber cables -- Part 2: Sectional specification
JIS C 8435	(2022) Boxes And Box Covers Of Plastic Conduits
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 61300-2-2	(2011) Fiber optic interconnecting devices and passive components -- Basic test and measurement procedures -- Part 2-2: Tests -- Mating durability
JIS K 6911	(2006; R 2021) Thermosetting plastic general test method
JIS X 5150	(R2016) Information Technology-Generic Cabling for Customer Premises
UL SOLUTIONS (UL)	
UL 1666	(2007; Reprint Jun 2012) Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in JIS X 5150, JCS 5507, JIS C 6820, JIS C 6850, and herein.

1.2.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates. (International expression for main cross-connect (MC).)

1.2.2 Building Distributor (BD)

A distributor in which the building backbone cables terminate and at which connections to the campus backbone cables may be made. (International expression for intermediate cross-connect (IC).)

1.2.3 Floor Distributor (FD)

A distributor used to connect horizontal cable and cabling subsystems or equipment. (International expression for horizontal cross-connect (HC).)

1.2.4 Telecommunications Room (TR)

An enclosed space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling.

1.2.5 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall and continuing to the equipment room.

1.2.6 Equipment Room (ER) (Telecommunications)

An environmentally controlled centralized space for telecommunications equipment that serves the occupants of a building. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity.

1.2.7 Open Cable

Cabling that is not run in a raceway as defined by applicable codes and standards. This refers to cabling that is "open" to the space in which the cable has been installed and is therefore exposed to the environmental conditions associated with that space.

1.2.8 Open Office

A floor space division provided by furniture, moveable partitions, or other means instead of by building walls.

1.2.9 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

1.3 SYSTEM DESCRIPTION

The building telecommunications cabling and pathway system shall include permanently installed backbone and horizontal cabling, horizontal and backbone pathways, service entrance facilities, work area pathways, telecommunications outlet assemblies, conduit, raceway, and hardware for splicing, terminating, and interconnecting cabling necessary to transport telephone and data (including LAN) between equipment items in a building. The horizontal system shall be wired in a star topology from the telecommunications work area to the floor distributor or campus distributor at the center or hub of the star. The backbone cabling and pathway system includes intrabuilding and interbuilding interconnecting cabling, pathway, and terminal hardware. The intrabuilding backbone provides connectivity from the floor distributors to the building distributors or to the campus distributor and from the building distributors to the campus distributor as required. The backbone system shall be wired in a star topology with the campus distributor at the center or hub of the star. Provide telecommunications pathway systems as required, and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. The telecommunications contractor must coordinate with the 53rd SOPS concerning access to and configuration of telecommunications spaces.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications drawings; G,A0

Telecommunications Space Drawings; G, A0

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

SD-03 Product Data

Telecommunications cabling (backbone and horizontal); G, A0

Patch panels; G, A0

Telecommunications outlet/connector assemblies; G,A0

Equipment support frame; G, A0

SD-06 Test Reports

Telecommunications cabling testing;

SD-10 Operation and Maintenance Data

Telecommunications cabling and pathway system Data Package 5;

SD-11 Closeout Submittals

Record Documentation;

1.5 QUALITY ASSURANCE

1.5.1 Shop Drawings

In exception to Section 01 33 00 SUBMITTAL PROCEDURES, submitted plan drawings shall be a minimum of 279 by 432 mm in size using a minimum scale of one mm per 100 mm, except as specified otherwise. Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

1.5.1.1 Telecommunications Drawings

Provide registered communications distribution designer (RCDD) approved, drawings in accordance with applicable codes and standards. The identifier for each termination and cable shall appear on the drawings. Drawings shall depict final telecommunications installed wiring system infrastructure. The drawings should provide details required to prove that the distribution system shall properly support connectivity from the EF telecommunications and ER telecommunications, CD's, BD's, and FD's to the telecommunications work area outlets. The following drawings shall be provided as a minimum showing new work or repairs only:

- a. T1 - Layout of complete building per floor - Building Area/Serving Zone Boundaries, Backbone Systems, and Horizontal Pathways. Layout of complete building per floor. The drawing indicates location of building areas, serving zones, vertical backbone diagrams, telecommunications rooms, access points, pathways, grounding system, and other systems that need to be viewed from the complete building perspective.
- b. T2 - Serving Zones/Building Area Drawings - Drop Locations and Cable Identification (ID'S). Shows a building area or serving zone. These drawings show drop locations, telecommunications rooms, access points and detail call outs for common equipment rooms and other congested areas.
- c. T4 - Typical Detail Drawings - Faceplate Labeling, Firestopping, Americans with Disabilities Act (ADA), Safety, Department of Transportation (DOT). Detailed drawings of symbols and typicals such as faceplate labeling, faceplate types, faceplate population installation procedures, detail racking, and raceways.

1.5.1.2 Telecommunications Space Drawings (New Work or Repairs Only)

Provide T3 drawings that include telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and cabinet, rack, backboard and wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.

1.5.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, and the supervisor (if different from the installer). A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

1.5.2.1 Telecommunications Contractor

The telecommunications contractor shall be a firm which is regularly and

professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems within the past 3 years of similar scope and size. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor.

1.5.2.2 Key Personnel

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Supervisors and installers assigned to the installation of this system or any of its components shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level. Submit documentation of current BICSI certification for each of the key personnel.

In lieu of BICSI certification, supervisors and installers assigned to the installation of this system or any of its components shall have a minimum of 3 years experience in the installation of the specified copper and fiber optic cable and components. They shall have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications systems and provide the names and locations of at least two project installations successfully completed using optical fiber and copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles

in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

1.5.2.3 Minimum Manufacturer Qualifications

Cabling, equipment and hardware manufacturers shall have a minimum of 3 years experience in the manufacturing, assembly, and factory testing of components which comply with JIS X 5150, JCS 5507 and JIS C 6820 and JIS C 6850.

1.5.3 Test Plan

Provide a complete and detailed test plan for the telecommunications cabling system including a complete list of test equipment for the components and accessories for each cable type specified, 60 days prior to the proposed test date. Include procedures for certification, validation, and testing.

1.5.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.5.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site shall not be used, unless specified otherwise.

1.6 DELIVERY AND STORAGE

Provide protection from weather, moisture, extreme heat and cold, dirt, dust, and other contaminants for telecommunications cabling and equipment placed in storage.

1.7 ENVIRONMENTAL REQUIREMENTS

Connecting hardware shall be rated for operation under ambient conditions of 0 to 60 degrees C and in the range of 0 to 95 percent relative humidity, noncondensing.

1.8 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.9 MAINTENANCE

1.9.1 Operation and Maintenance Manuals

Commercial off the shelf manuals shall be furnished for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications cabling and pathway system, Data Package 5. Submit operations and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein not later than 2 months prior to the date of beneficial occupancy. In addition to requirements of Data Package 5, include the requirements of paragraphs TELECOMMUNICATIONS DRAWINGS, TELECOMMUNICATIONS SPACE DRAWINGS, and RECORD DOCUMENTATION. Ensure that these drawings and documents depict the as-built configuration.

1.9.2 Record Documentation

Provide T5 drawings including documentation on cables and termination hardware. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel layouts and cover plate assignments, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided on electronic media using Windows based computer cable management software. Provide the following T5 drawing documentation as a minimum:

- a. Cables - A record of installed cable shall be provided. The cable records shall include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility. Include manufacture date of cable with submittal.
- b. Termination Hardware - A record of installed patch panels, cross-connect points, distribution frames, terminating block arrangements and type, and outlets shall be provided. Documentation shall include the required data fields as a minimum.

PART 2 PRODUCTS

2.1 COMPONENTS

Components shall be UL or third party certified. Where equipment or materials are specified to conform to industry and technical society reference standards of the organizations, submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard. Provide a complete system of telecommunications cabling and pathway components using star topology. Provide support structures and pathways, complete with outlets, cables, connecting hardware and telecommunications cabinets/racks. Cabling and interconnecting hardware and components for telecommunications systems shall be listed or third party independent testing laboratory certified, and shall comply with applicable codes and standards and conform to the requirements specified herein.

2.2 TELECOMMUNICATIONS PATHWAY

Provide telecommunications pathways in accordance with JIS X 5150 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.3 TELECOMMUNICATIONS CABLING

Cabling shall be listed for the application and shall comply with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850. Provide a labeling system for cabling as required. Ship cable on reels or in boxes bearing manufacture date for for unshielded twisted pair (UTP) in accordance with JCS 5507 and optical fiber cables in accordance with JIS C 6870-2 for all cable used on this project. Cabling manufactured more than 12 months prior to date of installation shall not be used.

2.3.1 Backbone Cabling

2.3.1.1 Backbone Optical Fiber

Provide in accordance with JIS C 6870-2, JIS C 6820, JIS C 6850, UL 1666 and JIS C 3665-1-1. Cable shall be imprinted with fiber count, fiber type and aggregate length at regular intervals not to exceed 1 meter.

Provide the number of strands indicated, (but not less than 12 strands between the main telecommunication room and each of the other telecommunication rooms), of single-mode(OS1), tight buffered fiber optic cable.

Provide tight buffered fiber optic multimode, 50/125-um diameter laser optimized(OM3) cable as indicated.

Provide plenum (OFNP), riser (OFNR), or general purpose (OFN or OFNG) rated non-conductive, fiber optic cable as required per application. Substitution of a higher rated cable shall be permitted. The cable cordage jacket, fiber, unit, and group color shall be in accordance with JIS X 5150.

Provide plenum (OFNP) riser (OFNR), or general purpose (OFN or OFNG) rated

non-conductive, fiber optic cable as required per application. Substitution of a higher rated cable shall be permitted. The cable cordage jacket, fiber, unit, and group color shall be in accordance with JIS X 5150.

2.3.2 Horizontal Cabling

Provide horizontal cable and performance characteristics in accordance with JIS X 5150.

2.3.2.1 Horizontal Copper

Provide horizontal copper cable, UTP, 100 ohm in accordance with JCS 5507, and JIS X 5150. Provide four each individually twisted pair, minimum size 0.2 sqmm conductors, Category 6, with a blue or green thermoplastic jacket. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating (category designation) and length marking at regular intervals in accordance with JCS 5507. Provide plenum (CMP) communications rated cabling as required per application. Substitution of a higher rated cable shall be permitted. Cables installed in conduit within and under slabs shall be listed and labeled for wet locations.

2.3.3 Work Area Cabling

2.3.3.1 Work Area Copper

Provide work area copper cable in accordance with JCS 5507, with a blue or green thermoplastic jacket.

2.4 TELECOMMUNICATIONS SPACES

Provide connecting hardware and termination equipment in the telecommunications entrance facility and telecommunication equipment rooms to facilitate installation as shown on design drawings for terminating and cross-connecting permanent cabling. Provide telecommunications interconnecting hardware color coding in accordance with facility standards.

2.4.1 Equipment Support Frame

Provide in accordance with JIS C 6011-1 and JIS C 6012-3-100.

- a. Bracket, wall mounted, 8 gauge aluminum. Provide hinged bracket compatible with 482.6 mm panel mounting.
- b. Racks, floor mounted modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Provide rack with vertical and horizontal cable management channels, top and bottom cable troughs, grounding lug and a surge protected power strip with 6 duplex 20 amp receptacles. Rack shall be compatible with 482.6 mm panel mounting.
- c. Cabinets, freestanding modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Cabinet shall have removable and lockable side panels, front and rear doors, and have adjustable feet for leveling. Cabinet shall be vented in the roof and rear door. Cabinet shall have cable access in the roof and base and be compatible with 482.6 mm panel mounting. Provide cabinet

with grounding bar, rack mounted 15 cu. m fan with filter and a surge protected power strip with 6 duplex 20 amp receptacles.

- d. Cabinets, wall-mounted modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Cabinet shall have lockable front and rear doors, louvered side panels, 7 cu. m roof mounted fan, ground lug, and top and bottom cable access. Cabinet shall be compatible with 482.6 mm panel mounting. A duplex AC outletsurge protected power strip with 6 duplex 20 amp receptacles shall be provided within the cabinet.

2.4.2 Cable Guides

Provide cable guides specifically manufactured for the purpose of routing cables, wires and patch cords horizontally and vertically on 482.6 mm equipment racks and telecommunications backboards. Cable guides of ring or bracket type devices mounted on rack panels for horizontal cable management and individually mounted for vertical cable management. Mount cable guides with screws, or nuts and lockwashers.

2.4.3 Patch Panels

Provide ports for the number of horizontal and backbone cables terminated on the panel plus 25 percent spare. Provide pre-connectorized optical fiber and copper patch cords for patch panels. Provide patch cords, as complete assemblies, with matching connectors as specified. Provide fiber optic patch cables with crossover orientation in accordance with JIS C 6820 and JIS C 6850. Patch cords shall meet minimum performance requirements specified in JIS X 5150, JCS 5507 and JIS C 6820 and JIS C 6850 for cables, cable length and hardware specified.

2.4.3.1 Fiber Optic Patch Panel

Provide panel for maintenance and cross-connecting of optical fiber cables. Panel shall be constructed of 18 gauge steel or 11 gauge aluminum minimum and shall be rack mounted and compatible with a JIS C 6012-3-100 482.6 mm equipment rack. Each panel shall provide 12 multimode or single-mode adapters as duplex SC in accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic alignment sleeves. Provide dust cover for unused adapters. The rear of each panel shall have a cable management tray a minimum of 203 mm deep with removable cover, incoming cable strain-relief and routing guides. Panels shall have each adapter factory numbered and be equipped with laminated plastic nameplates above each adapter.

2.5 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES

2.5.1 Outlet/Connector Copper

Outlet/connectors shall comply with JIS X 5150 and JCS 5507. UTP outlet/connectors shall be JIS X 5150 listed, non-keyed, 8-pin modular, constructed of high impact rated thermoplastic housing and shall be third party verified and shall comply with JCS 5507 Category 6 requirements. Outlet/connectors provided for UTP cabling shall meet or exceed the requirements for the cable provided. Use shielded connectors for F/UTP cables. Outlet/connectors shall be terminated using a Type 110 IDC PC board connector, color-coded for both T568A and T568B wiring. Each outlet/connector shall be wired T568A. UTP outlet/connectors shall comply with JCS 5507 for 200 mating cycles.

2.5.2 Optical Fiber Adapters(Couplers)

Provide optical fiber adapters suitable for duplex SC in Accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic alignment sleeves, as indicated. Provide dust cover for adapters. Optical fiber adapters shall comply with JIS C 61300-2-2 for 500 mating cycles.

2.5.3 Optical Fiber Connectors

Provide in accordance with JIS C 61300-2-2. Optical fiber connectors shall be duplex SC in accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic fiber. The connectors shall provide a maximum attenuation of 0.3 dB at 850 1300 nm with less than a 0.2 dB change after 500 mating cycles.

2.5.4 Cover Plates

Telecommunications cover plates shall comply with JIS C 8435, and JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850; flush design constructed of high impact thermoplastic material ivory in color to match color of receptacle/switch cover plates specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM 302 stainless material. Provide labeling in accordance with the paragraph LABELING in this section.

2.6 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.7 FIELD FABRICATED NAMEPLATES

JIS K 6911. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with black center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm. Lettering shall be a minimum of 6.35 mm high normal block style.

2.8 TESTS, INSPECTIONS, AND VERIFICATIONS

2.8.1 Factory Reel Tests

Provide documentation of the testing and verification actions taken by manufacturer to confirm compliance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850, JIS C 6185-2 for single mode optical fiber, and JIS C 6185-3 for multimode optical fiber cables.

PART 3 EXECUTION

3.1 INSTALLATION

Install telecommunications cabling and pathway systems, including the horizontal and backbone cable, pathway systems, telecommunications outlet/connector assemblies, and associated hardware in accordance with

JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850,, and UL standards as applicable. Provide cabling in a star topology network. Pathways and outlet boxes shall be installed as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Install telecommunications cabling with copper media in accordance with the following criteria to avoid potential electromagnetic interference between power and telecommunications equipment. The interference ceiling shall not exceed 3.0 volts per meter measured over the usable bandwidth of the telecommunications cabling. Cabling shall be run with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.

3.1.1 Cabling

Install UTP, F/UTP, and optical fiber telecommunications cabling system as detailed in JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850, Appendix A and applicable codes and standards for residential cabling. Screw terminals shall not be used except where specifically indicated on plans. Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations. Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables. Provide a device to monitor cable pull tensions. Do not exceed 110 N pull tension for four pair copper cables. Do not chafe or damage outer jacket materials. Use only lubricants approved by cable manufacturer. Do not over cinch cables, or crush cables with staples. For UTP cable, bend radii shall not be less than four times the cable diameter. Cables shall be terminated; no cable shall contain unterminated elements. Cables shall not be spliced. Label cabling in accordance with paragraph LABELING in this section.

3.1.1.1 Open Cable

Use only where specifically indicated on plans for use in cable trays, or below raised floors. Install in accordance with JIS X 5150, JCS 5507 and JIS C 6820 and JIS C 6850, Appendix A. Do not exceed cable pull tensions recommended by the manufacturer. Copper cable not in a wireway or pathway shall be suspended a minimum of 200 mm above ceilings by cable supports no greater than 1.5 m apart. Cable shall not be run through structural members or in contact with pipes, ducts, or other potentially damaging items. Placement of cable parallel to power conductors shall be avoided, if possible; a minimum separation of 300 mm shall be maintained when such placement cannot be avoided.

Plenum cable shall be used where open cables are routed through plenum areas. Cable routed exposed under raised floors shall be plenum rated. Plenum cables shall comply with flammability plenum requirements per applicable codes and standards. Install cabling after the flooring system has been installed in raised floor areas. Cable 1.8 meters long shall be neatly coiled not less than 300 mm in diameter below each feed point in raised floor areas.

3.1.1.2 Backbone Cable

- a. Copper Backbone Cable. Install intrabuilding backbone copper cable, in indicated pathways, between the campus distributor, located in the telecommunications entrance facility or room, the building distributors and the floor distributors located in telecommunications rooms and telecommunications equipment rooms as indicated on drawings.
- b. Optical fiber Backbone Cable. Install intrabuilding backbone optical

fiber in indicated pathways. Do not exceed manufacturer's recommended bending radii and pull tension. Prepare cable for pulling by cutting outer jacket 250 mm leaving strength members exposed for approximately 250 mm. Twist strength members together and attach to pulling eye. Vertical cable support intervals shall be in accordance with manufacturer's recommendations.

3.1.1.3 Horizontal Cabling

Install horizontal cabling as indicated on drawings. Do not untwist Category 6 UTP and F/UTP cables more than 12 mm from the point of termination to maintain cable geometry. Provide slack cable in the form of a figure eight (not a service loop) on each end of the cable, 3 m in the telecommunications room, and 304 mm in the work area outlet.

3.1.2 Pathway Installations

Provide in accordance with JIS X 5150. Provide building pathway as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.3 Service Entrance Conduit, Underground

Provide service entrance underground as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.4 Cable Tray Installation

Install cable tray as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Only CMP OFNP type cable shall be installed in a plenum.

3.1.5 Work Area Outlets

3.1.5.1 Terminations

Terminate UTP and F/UTP cable in accordance with JIS X 5150, JCS 5507 and wiring configuration as specified. Ground F/UTP cables only at the patch panel. Terminate fiber optic cables in accordance with JIS C 6820 and JIS C 6850.

3.1.5.2 Cover Plates

As a minimum, each outlet/connector shall be labeled as to its function and a unique number to identify cable link in accordance with the paragraph LABELING in this section.

3.1.5.3 Cables

Unshielded twisted pair, F/UTP, and fiber optic cables shall have a minimum of 304 mm of slack cable loosely coiled into the telecommunications outlet boxes. Minimum manufacturer's bend radius for each type of cable shall not be exceeded.

3.1.5.4 Pull Cords

Pull cords shall be installed in conduit serving telecommunications outlets that do not have cable installed.

3.1.6 Telecommunications Space Termination

Install termination hardware required for Category 6 and optical fiber system. An insulation displacement tool shall be used for terminating copper cable to insulation displacement connectors.

3.1.6.1 Patch Panels

Patch panels shall be mounted in equipment cabinets with sufficient ports to accommodate the installed cable plant plus 25 percent spares.

- a. Copper Patch Panel and Shielded Copper Patch Panel for F/UTP Cables. Copper cable entering a patch panel shall be secured to the panel as recommended by the manufacturer to prevent movement of the cable.
- b. Fiber Optic Patch Panel. Fiber optic cable loop shall be 900 mm in length provided as recommended by the manufacturer. The outer jacket of each cable entering a patch panel shall be secured to the panel to prevent movement of the fibers within the panel, using clamps or brackets specifically manufactured for that purpose.

3.1.6.2 Equipment Support Frames

Install in accordance with JIS X 5150:

- a. Bracket, wall mounted. Mount bracket to plywood backboard in accordance with manufacturer's recommendations. Mount rack so height of highest panel does not exceed 1980 mm above floor.
- b. Racks, floor mounted modular type. Permanently anchor rack to the floor in accordance with manufacturer's recommendations.
- c. Cabinets, freestanding modular type. When cabinets are connected together, remove adjoining side panels for cable routing between cabinets. Mount rack mounted fan in roof of cabinet.
- d. Cabinets, wall-mounted modular type. Mount cabinet to plywood backboard in accordance with manufacturer's recommendations. Mount cabinet so height of highest panel does not exceed 1980 mm above floor.

3.1.7 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated wall, partitions, floors, or ceilings.

3.1.8 Grounding and Bonding

Provide in accordance with JIS C 60364-5-54 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.2 LABELING

3.2.1 Labels

Provide labeling in accordance with applicable codes and standards. Handwritten labeling is unacceptable. Stenciled lettering for voice and data circuits shall be provided using thermal ink transfer process.

3.2.2 Cable

Cables shall be labeled using color labels on both ends with identifiers.

3.2.3 Termination Hardware

Workstation outlets and patch panel connections shall be labeled using color coded labels with identifiers.

3.3 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.3.1 Painting Backboards

If backboards are required to be painted, then the manufactured fire retardant backboard must be painted with fire retardant paint, so as not to increase flame spread and smoke density and must be appropriately labeled. Label and fire rating stamp must be unpainted.

3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.5 TESTING

3.5.1 Telecommunications Cabling Testing

Perform telecommunications cabling inspection, verification, and performance tests in accordance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850. Test equipment shall conform to JIS X 5150. Perform optical fiber field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable reels from project site upon attenuation test failure.

3.5.1.1 Inspection

Visually inspect UTP and optical fiber jacket materials for UL or third party certification markings. Inspect cabling terminations in telecommunications rooms and at workstations to confirm color code for T568A or T568B pin assignments, and inspect cabling connections to confirm compliance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850, and applicable codes and standards for residential cabling. Visually confirm Category 6, marking of outlets, cover plates, outlet/connectors, and patch panels.

3.5.1.2 Verification Tests

UTP backbone copper cabling shall be tested for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has overall shield. Test operation of shorting bars in connection blocks. Test cables after termination but prior to being cross-connected.

For multimode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6850 and JIS C 6185-3 using Method A, Optical Power Meter and Light Source for multimode optical fiber. For single-mode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6185-2 using Method A, Optical Power Meter and Light Source for single-mode optical fiber. Perform verification acceptance tests.

3.5.1.3 Final Verification Tests

Perform verification tests for UTP and optical fiber systems after the complete telecommunications cabling and workstation outlet/connectors are installed.

- a. Voice Tests. These tests assume that dial tone service has been installed. Connect to the network interface device at the demarcation point. Go off-hook and listen and receive a dial tone. If a test number is available, make and receive a local, long distance, and DSN telephone call.
- b. Data Tests. These tests assume the Information Technology Staff has a network installed and are available to assist with testing. Connect to the network interface device at the demarcation point. Log onto the network to ensure proper connection to the network.

-- End of Section --

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SECTION 28 31 76

INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S3.2 (2020) American National Standard Method
for Measuring the Intelligibility of
Speech Over Communication Systems (ASA 85)

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide
<https://www.approvalguide.com/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1 (2002; R 2008) Guide on the Surges
Environment in Low-Voltage (1000 V and
Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4 (2018) Standard for Integrated Fire
Protection and Life Safety System Testing

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National
Electrical Code

NFPA 72 (2022; ERTA 22-1) National Fire Alarm and
Signaling Code

NFPA 90A (2024) Standard for the Installation of
Air Conditioning and Ventilating Systems

NFPA 170 (2024; ERTA 1 2023) Standard for Fire
Safety and Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-601-02 (2021) Fire Protection Systems Inspection,
Testing, and Maintenance

UFC 4-010-06 (2023) Cybersecurity of Facility-Related
Control Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15 Radio Frequency Devices

47 CFR 90 Private Land Mobile Radio Services

UL SOLUTIONS (UL)

UL 268 (2023) UL Standard for Safety Smoke Detectors for Fire Alarm Systems

UL 268A (2008; Reprint Aug 2023) Smoke Detectors for Duct Application

UL 464 (2023) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories

UL 497A (2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits

UL 497B (2004; Reprint Feb 2022) UL Standard for Safety Protectors for Data Communications and Fire Alarm Circuits

UL 864 (2023; Reprint Oct 2024) UL Standard for Safety Control Units and Accessories for Fire Alarm Systems

UL 1283 (2017; Reprint Feb 2024) UL Standard for Safety Electromagnetic Interference Filters

UL 1449 (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices

UL 1480 (2023) UL Standard for Safety Speakers for Fire Alarm and Signaling Systems, Including Accessories

UL 1638 (2023) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories

UL 1971 (2002; Reprint Feb 2024) UL Standard for Safety Signaling Devices for the Deaf and Hard of Hearing

UL 2017 (2008; Reprint Jan 2024) UL Standard for Safety General-Purpose Signaling Devices and Systems

UL 2572 (2016; Bul. 2018) UL Standard for Safety Mass Notification Systems

UL Fire Prot Dir UL Product IQ (updated online) at <https://productiq.ulpropsector.com/en>

1.2 RELATED SECTIONS

Section 25 05 11.21 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - FIRE AND LIFE SAFETY, applies to this section, with the additions and modifications specified herein. In addition, refer to the following sections for related work and coordination:

Section 08 71 00 DOOR HARDWARE for door release and additional work related to finish hardware.

1.3 SUMMARY

1.3.1 Scope

- a. This work includes designing and modifying the existing fire alarm and mass notification (MNS) system as described herein and on the contract drawings. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, supervising station fire alarm transmitters/mass notification transceiver, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide systems complete and ready for operation. Design and installation must comply with UFC 4-010-06.
- b. Provide equipment, materials, installation, workmanship, inspection, and testing in strict accordance with NFPA 72, except as modified herein. The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.
- c. Where an emergency generator provides standby power supply for life safety system circuits, the generator must be monitored by the FMCU and transmit emergency generator signals in accordance with NFPA 72.
- d. The fire alarm and mass notification system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

1.3.2 Qualified Fire Protection Engineer (QFPE)

Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.
- b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government functional performance testing and

performing a final installation review.

- e. Signing applicable certificates under SD-07.

1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

1.4.1 Interface Device

An addressable device that interconnects hard wired systems or devices to an analog/addressable system.

1.4.2 Fire Alarm and Mass Notification Control Unit (FMCU)

A master control unit having the features of a fire alarm control unit (FACU) and an autonomous control unit (ACU) where these units are interconnected to function as a combined fire alarm/mass notification system. The FACU and ACU functions may be contained in a single cabinet or in independent, interconnected, and co-located cabinets.

1.4.3 Remote Fire Alarm and Mass Notification Control Unit

A control unit, physically remote from the fire alarm and mass notification control unit, that receives inputs from automatic and manual fire alarm devices; may supply power to detection devices and interface devices; may provide transfer of power to the notification appliances; may provide transfer of condition to relays or devices connected to the control unit; and reports to and receives signals from the fire alarm and mass notification control unit.

1.4.4 Local Operating Console (LOC)

A unit designed to allow emergency responders or building occupants to operate the MNS including delivery of recorded messages or live voice announcements, initiate visual, textual visual, and audible appliance operation and other relayed functions.

1.4.5 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

1.4.6 Control Module and Relay Module

Terms utilized to describe emergency control function interface devices as defined by NFPA 72.

1.4.7 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

1.4.8 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Fire alarm system designer; G

Supervisor; G

Technician; G

Installer; G

Test Technician; G

Fire Alarm System Site-Specific Software Acknowledgement; G

SD-02 Shop Drawings

Nameplates; G

Instructions; G

Wiring Diagrams; G

System Layout; G

Notification Appliances; G

Initiating devices; G

Amplifiers; G

Battery Power; G

Voltage Drop Calculations; G

SD-03 Product Data

Amplifiers; G

Tone Generators; G

Digitalized voice generators; G

Manual Stations; G

Smoke Detectors; G

Duct Smoke Detectors; G

Notification Appliances; G

Textual Display Sign Control Panel; G

Textual Display Signs; G

Batteries; G

Battery Chargers; G

Supplemental Notification Appliance Circuit Panels; G

Auxiliary Power Supply Panels; G

Surge Protective Devices; G

Alarm Wiring; G

Back Boxes and Conduit; G

Terminal Cabinets; G

Environmental Enclosures or Guards; G

SD-06 Test Reports

Test Procedures; G

SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Instruction of Government Employees; G

SD-11 Closeout Submittals

As-Built Drawings

1.6 SYSTEM OPERATION

Fire alarm system/mass notification system including textual display sign control panel(s), components requiring power, except for the FMCU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

The interior fire alarm and mass notification system must be a complete, supervised, noncoded, analog/addressable fire alarm and mass notification system conforming to NFPA 72, UL 864, and UL 2572. Systems meeting UL 2017 only are not acceptable. The system must be activated into the alarm mode by actuation of an alarm initiating device. The system must remain in the alarm mode until the initiating device is reset and the control unit is reset and restored to normal. The system may be placed in the alarm mode by local microphones, LOC, FMCU, or remotely from authorized locations/users.

1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Voice, Textual)

- a. Connect alarm initiating devices to initiating device circuits (IDC) Class "B", or to signaling line circuits (SLC) Class "B" and installed in accordance with NFPA 72.
- b. Connect notification appliances to notification appliance circuits (NAC) Class "B".

1.6.2 Functions and Operating Features

The system must provide the following functions and operating features:

- a. Power, annunciation, supervision, and control for the system. Addressable systems must be microcomputer (microprocessor or microcontroller) based with a minimum word size of eight bits with sufficient memory to perform as specified.
- b. Visual alarm notification appliances must be synchronized as required by NFPA 72.
- c. Electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within the control unit.
- d. An audible and visual trouble signal to activate upon a single break or open condition, or ground fault. The trouble signal must also operate upon loss of primary power (AC) supply, absence of a battery supply, low battery voltage, or removal of alarm or supervisory control unit modules. After the system returns to normal operating conditions, the trouble signal must again sound until the trouble is acknowledged. A smoke detector in the process of being verified for the actual presence of smoke must not initiate a trouble condition.
- e. A trouble signal silence feature that must silence the audible trouble signal, without affecting the visual indicator.
- f. Program capability via switches in a locked portion of the FMCU to

bypass the automatic notification appliance circuits, fire reporting system and air handler shutdown features. Operation of this programmed action must indicate on the FMCU display as a supervisory or trouble condition.

- g. Alarm functions must override trouble or supervisory functions. Supervisory functions must override trouble functions.
- h. The system must be capable of being programmed from the control unit keyboard. Programmed information must be stored in non-volatile memory.
- i. The system must be capable of operating, supervising, and monitoring non-addressable alarm and supervisory devices.
- j. There must be no limit, other than maximum system capacity, as to the number of addressable devices that may be in alarm simultaneously.
- k. Where the fire alarm/mass notification system is responsible for initiating an action in another emergency control device or system, such as HVAC, releasing service, the addressable fire alarm relay must be located in the vicinity of the emergency control device.
- l. An alarm signal must automatically initiate the following functions:
 - (1) Transmission of an alarm signal to a remote supervising station.
 - (2) Visual indication of the device operated on the FMCU, .
 - (3) Actuation of alarm notification appliances.
 - (4) Recording of the event electronically in the history log of the FMCU.
 - (5) Operation of photoelectric smoke detectors must release the FM-200 clean agent fire extinguishing system after a 30 second time delay.
 - (6) Visual indication of the device operated on the FMCU.
 - (7) Transmission of a supervisory signal to the fire department.
 - (8) Operation of a duct smoke detector must shut down the appropriate air handler in accordance with NFPA 90A in addition to other requirements of this paragraph and as allowed by NFPA 72.
 - (9) Recording of the event electronically in the history log of the FMCU.
- m. A trouble condition must automatically initiate the following functions:
 - (1) Visual indication of the device operated on the FMCU.
 - (2) Transmission of a trouble signal to the fire department.
 - (3) Recording of the event electronically in the history log of the FMCU.

- n. System control equipment must be programmed to provide a 60-minute to 180-minute delay in transmission of trouble signals resulting from primary power failure.

1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, must be delivered, strictly in accordance with the CONTRACT CLAUSES. The fire alarm system manufacturer must submit written confirmation of this contract provision as "Fire Alarm System Site-Specific Software Acknowledgement". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted must include complete system, equipment, and software descriptions. Descriptions must show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package must also include the following:

- a. Items identified in NFPA 72, titled "Site-Specific Software".
- b. Identification of programmable portions of the system equipment and capabilities.
- c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.
- d. Provision of operational software data on all modes of programmable portions for fire alarm and mass notification.
- e. Description of Fire Alarm and Mass Notification Control Unit equipment operation.
- f. Description of auxiliary and remote equipment operations.
- g. Library of application software.
- h. Operation and maintenance manuals.

1.8 EXISTING EQUIPMENT

- a. Equipment and devices must be compatible and operable with the existing fire alarm/mass notification system and must not impair reliability or operational functions of existing supervising station fire alarm system. The supervising equipment is existing and consists of the following brands and models: supervising station control unit Monaco.
- b. Equipment and devices must be compatible and operable with the existing building fire alarm/mass notification system. Equipment must not impair reliability or operational functions of the existing system. The existing building system control unit is Monaco M2.
- c. Equipment and devices must be compatible and operable with the existing installation-wide mass notification system and must not impair reliability or operational functions of the existing system. The installation-wide mass notification system utilizes Monaco transceivers.

1.9 QUALITY ASSURANCE

1.9.1 Submittal Documents

1.9.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

1.9.1.2 Shop Drawings

Shop drawings must not be smaller than the Contract Drawings. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

1.9.1.3 Nameplates

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

1.9.1.4 Wiring Diagrams

One copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote FMCU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

1.9.1.5 System Layout

One copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candela rating of each visual notification appliance. Indicate the wattage of each speaker. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, and similar. Show/identify all acoustically similar spaces. Indicate if the environment for the FMCU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

1.9.1.6 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is

at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

1.9.1.7 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

1.9.1.8 Amplifiers

Calculations and supporting data to indicate that amplifiers have sufficient capacity to simultaneously drive all notification speakers at tapped settings plus 25 percent spare capacity. Annotate data for each circuit on the drawings.

1.9.1.9 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each control unit component, and the battery recharging period, must be included on the drawings.

1.9.1.10 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

1.9.1.11 Product Data

One copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

1.9.1.12 Operation and Maintenance (O&M) Instructions

Six copies of the Operation and Maintenance Instructions. The O&M Instructions must be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals must be approved prior to training. The Interior Fire Alarm And Mass Notification System Operation and Maintenance Instructions must include the following:

- a. "Manufacturer Data Package" as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.

- b. Operating manual outlining step-by-step procedures required for system startup, operation, and shutdown. The manual must include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.
- c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals must include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.
- d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.
- e. Software submitted for this project on CD/DVD media utilized.
- f. Printouts of configuration settings for all devices.
- g. Routine maintenance checklist. The routine maintenance checklist must be arranged in a columnar format. The first column must list all installed devices, the second column must state the maintenance activity or state no maintenance required, the third column must state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, and similar) must comply with UFC 3-601-02.
- h. A final Equipment List must be submitted with the Operating and Maintenance (O&M) manual.

1.9.1.13 As-Built Drawings

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

1.9.2 Qualifications

1.9.2.1 Fire Alarm System Designer

The fire alarm system designer must be certified as a Level IV (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

1.9.2.2 Supervisor

A NICET Level IV fire alarm technician must supervise the installation of the fire alarm/mass notification system. The fire alarm technicians supervising the installation of equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.3 Technician

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians installing the

equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.4 Installer

Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm/mass notification devices, cabinets and control units. A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system/mass notification system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.5 Test Technician

Fire alarm technicians with a minimum of eight years of experience and NICET Level IV utilized in testing and certification of the installation of the fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

1.9.2.6 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

1.9.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

1.10 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

1.11 MAINTENANCE

1.11.1 Special Tools

Software, connecting cables and proprietary equipment, necessary for the maintenance, testing, and reprogramming of the equipment must be furnished to the Contracting Officer, prior to the instruction of Government employees.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm and mass notification equipment must be listed for use under the applicable reference standards. Interfacing of UL 864 or similar approved industry listing with Mass Notification equipment listed to UL 2572 must be done in a laboratory listed configuration, if the software programming features cannot provide a listed interface control.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 72 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening.

2.2.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

a. Mass Notification Control Unit.

Nameplates must be etched metal or plastic, permanently attached by screws to control units or adjacent walls.

2.2.3 Keys

Keys and locks for equipment, control units and devices must be identical.

2.2.4 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the frame in a conspicuous location observable from the FMCU. The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions must also include procedures for operating live voice microphones. The instructions and their mounting location must be approved by the Contracting Officer before being posted.

2.3 MASS NOTIFICATION CONTROL UNIT

- a. The system must have the capability of utilizing an LOC with redundant controls of the FMCU. Notification Appliance Circuits (NAC) must be provided for the activation of strobe appliances. Audio output must be selectable for line level. A hand-held microphone must be provided

and, upon activation, must take priority over any tone signal, recorded message or PA microphone operation in progress, while maintaining the strobe NAC circuit activation.

- b. The Mass Notification functions must override the manual or automatic fire alarm notification. Other fire alarm functions including transmission of a signal(s) to the fire department must remain operational. When a mass notification announcement is disengaged and a fire alarm condition still exists, the audible and visual notification appliances must resume activation for alarm conditions. The fire alarm message must be of lower priority than all other messages (except any "test" messages) and must not override any other messages. If not manually ended, any mass notification announcement will automatically end after 10 minutes.
- c. Messages must be recorded professionally utilizing standard industry methods, in a professional female voice. Message and tone volumes must both be at the same decibel level. Messages recorded from the system microphone must not be accepted. A 1000 Hz tone (as required by NFPA 72) must precede messages and be similar to the following unless Installation or Facility specific messages are required:
 - (1) Carbon Monoxide: "May I have your attention please. May I have your attention please. Carbon monoxide has been detected in the building. Please walk to the nearest exit and leave the building." (Provide a 2 second pause.) "May I have your attention please, (repeat the tones and message on a continuous loop)."
 - (2) Fire: "May I have your attention please. May I have your attention please. A fire emergency has been reported in the building. Please leave the building by the nearest exit." (Provide a 2 second pause.) "May I have your attention please, (repeat the tones and message on a continuous loop)."
 - (3) Test: "May I have your attention please. May I have your attention please. This is a test of the building mass notification system. Please continue your normal duties. This is only a test." (Provide a 2 second pause.)
 - (4) All Clear: "May I have your attention please. May I have your attention please. An all clear has been issued, resume normal activities." (Provide a 2 second pause.)
- d. Auxiliary Input Module must be designed to be an outboard expansion module to either expand the number of optional LOC's, or allow a telephone interface.

2.4 AMPLIFIERS, PREAMPLIFIERS, TONE GENERATORS

Any amplifiers, preamplifiers, tone generators, digitalized voice generators, and other hardware necessary for a complete, operational, textual audible circuit conforming to NFPA 72 must be housed in a remote FMCU, terminal cabinet, or in the FMCU. Individual amplifiers must be 100 watts maximum.

2.4.1 Operation

The system must automatically operate and control all building speakers.

2.4.2 Construction

Amplifiers must utilize computer grade components and must be provided with output protection devices sufficient to protect the amplifier against any transient up to 10 times the highest rated voltage in the system.

2.4.3 Inputs

Equip each system with separate inputs for the tone generator, digitalized voice driver and control unit mounted microphone . Microphone inputs must be of the low impedance, balanced line type. Both microphone and tone generator input must be operational on any amplifier.

2.4.4 Tone Generator

The tone generator must produce a three-pulse temporal pattern and must be constantly repeated until interrupted by either the digitalized voice message, the microphone input, or the alarm silence mode as specified. The tone generator must be single channel with an automatic backup generator per channel such that failure of the primary tone generator causes the backup generator to automatically take over the functions of the failed unit and also causes transfer of the common trouble relay. The tone generator must be provided with securely attached labels to identify the component as a tone generator and to identify the specific tone it produces.

2.4.5 Protection Circuits

Each amplifier must be constantly supervised for any condition that could render the amplifier inoperable at its maximum output. Failure of any component must cause illumination of a visual "amplifier trouble" indicator on the control unit, appropriate logging of the condition in the history log, and other actions for trouble conditions as specified.

2.5 MANUAL STATIONS

Provide metal or plastic, surface mounted, double-action, addressable manual stations, that are not subject to operation by jarring or vibration. Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

2.6 SMOKE DETECTORS

2.6.1 Spot Type Detectors

Provide addressable photoelectric smoke detectors as follows:

- a. Provide analog/addressable photoelectric smoke detectors utilizing the photoelectric light scattering principle for operation in accordance with UL 268. Smoke detectors must be listed for use with the FMCU.
- b. Provide self-restoring type detectors that do not require any readjustment after actuation at the FMCU to restore them to normal operation. The detector must have a visual indicator to show actuation.

- c. Vibration must have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or airborne materials. The screen must not inhibit the movement of smoke particles into the chamber.
- d. Provide twist lock bases with screw terminals for each conductor. The detectors must maintain contact with their bases without the use of springs.
- e. The detector address must identify the particular unit, its location within the system, and its sensitivity setting. Detectors must be of the low voltage type rated for use on a 24 VDC system.

2.6.2 Duct Smoke Detectors

Duct-mounted addressable photoelectric smoke detectors must consist of a smoke detector, as specified in paragraph Spot Type Detectors, mounted in a special housing fitted with duct sampling tubes. Detector circuitry must be mounted in a metallic or plastic enclosure exterior to the duct. Detectors must be listed for operation over the complete range of air velocities, temperature and humidity expected at the detector when the air-handling system is operating. Detectors must be powered from the FMCU.

- a. Sampling tubes must run the full width of the duct. The duct detector package must conform to the requirements of NFPA 90A, UL 268A, and must be listed for use in air-handling systems. The control functions, operation, reset, and bypass must be controlled from the FMCU.
- b. Lights to indicate the operation and alarm condition must be visible and accessible with the unit installed and the cover in place. Remote indicators must be provided where required by NFPA 72. Remote indicators as well as the affected fan units must be properly identified in etched plastic placards.
- c. Detectors must provide for control of auxiliary contacts that provide control, interlock, and shutdown functions specified in Section 23 09 00 to INSTRUMENTATION AND CONTROL FOR HVAC. Auxiliary contacts provide for this function must be located within 1 m of the controlled circuit or appliance. The auxiliary contacts must be supplied by the fire alarm system manufacturer to ensure complete system compatibility.

2.7 NOTIFICATION APPLIANCES

2.7.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of UL 464. Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted red. Recessed audible appliances must be installed with a grill that is painted red.

2.7.1.1 Speakers

- a. Speakers must conform to the applicable requirements of UL 1480. Speakers must have six different sound output levels and operate with audio line input levels of 70.7 VRMs and 25 VRMs, by means of selectable tap settings. Interior speaker tap settings must include taps of 1/4, 1/2, 1, and 2 watt, at a minimum. Exterior speakers must

also be multi-tapped with no more than 15 watt maximum setting. Speakers must incorporate a high efficiency speaker for maximum output at minimum power across a frequency range of 400 Hz to 4,000 Hz, and must have a sealed back construction. Speakers must be capable of installation on standard 100 mm square electrical boxes. Where speakers and strobes are provided in the same location, they may be combined into a single unit. All inputs must be polarized for compatibility with standard reverse polarity supervision of circuit wiring via the FMCU.

- b. Provide speaker mounting plates constructed of cold rolled steel having a minimum thickness of 1.519 mm or molded high impact plastic and equipped with mounting holes and other openings as needed for a complete installation. Fabrication marks and holes must be ground and finished to provide a smooth and neat appearance for each plate. Each plate must be primed and painted.
- c. Speakers must utilize screw terminals for termination of all field wiring.

2.7.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "Alert" in letters of contrasting color. The light pattern must be dispersed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of 15 candela based on the UL 1971 test. Strobe must be surface mounted.

2.7.3 Textual Display Signs

Textual display signs must be LED scrolling and must not exceed 400 mm long by 150 mm high by 75 mm deep with a height necessary to meet the requirements of NFPA 72. The text display must spell out the word "EVACUATE" or "ANNOUNCEMENT" and the remainder of the emergency instructions as appropriate. The design of text display must be such that it cannot be read when not illuminated.

LCD or LED scrolling text displays must meet the following requirements at a minimum:

- a. Two lines of information for high priority messaging.
- b. Minimum of 20 characters per line (40 total) displayed.
- c. Text must be no less than height requirements and color/contrast requirements of NFPA 72.
- d. 32K character memory.
- e. Display must be wall or ceiling mounted.
- f. Mounting brackets for a convenient wall/cubicle mount.
- g. The system must interface with the textual display sign control panel to activate the proper message.

2.8 ELECTRIC POWER

2.8.1 Primary Power

Power must be 120 VAC 60 Hz service for the FMCU from the AC service to the building in accordance with NFPA 72.

2.9 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power must be automatic and must not cause transmission of a false alarm.

2.9.1 Batteries

Provide sealed, maintenance-free, sealed lead acid batteries as the source for emergency power to the FMCU. Batteries must contain suspended electrolyte. The battery system must be maintained in a fully charged condition by means of a battery charger. Provide an automatic transfer switch to transfer the load to the batteries in the event of the failure of primary power.

2.9.1.1 Capacity

Battery size must be the greater of the following two capacities. This capacity applies to every control unit associated with this system, including supplemental notification appliance circuit panels, auxiliary power supply panels, fire alarm transmitters, and Base-wide mass notification transceivers. When determining the required capacity under alarm condition, visual notification appliances must include both textual and non-textual type appliances.

- a. Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.
- b. Sufficient capacity to operate the mass notification for 60 minutes after loss of AC power.

2.9.1.2 Battery Power Calculations

- a. Verify that battery capacity exceeds supervisory and alarm power requirements for the criteria noted in the paragraph "Capacity" above.
 - (1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each control unit component, and compliance with UL 864.
 - (2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.
 - (3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components. Calculations must be performed using the minimum

rated voltage of each component.

- b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing devices or appliances remote from the power sources.

2.9.2 Battery Chargers

Provide a fully automatic, variable charging rate battery charger. The charger must be capable of providing 120 percent of the connected system load and must maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger must recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

2.10 SURGE PROTECTIVE DEVICES

Surge protective devices must be provided to suppress all voltage transients which might damage fire alarm control unit components. Systems having circuits located outdoors, communications equipment must be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, must have surge protection circuits installed at each end. The surge protective device must wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrestor must be installed directly adjacent to the power panel where the FMCU breaker is located.

- a. Surge protective devices for nominal 120 VAC must be UL 1449 listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device must also meet IEEE C62.41.1 and IEEE C62.41.2 category B tests for surge capacity. The surge protective device must feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing must be externally accessible.
- b. Surge protective devices for nominal 24 VAC, fire alarm telephone dialer, or ethernet connection must be UL 497B listed, meet IEEE C62.41.1 and have a maximum response time of 1-nanosecond. The surge protective device must feature multi-stage construction and be self-resetting. The surge protective device must be a base and plug style. The base assembly must have screw terminals for fire alarm wiring. The base assembly must accept "plug-in" surge protective module.
- c. All surge protective devices (SPD) must be the standard product of a single manufacturer and be equal or better than the following:
 - (1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed, series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.

- (2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.
- (3) For alarm telephone dialers: UL 497A listed, series connected, 130-volt, 150 mA rated with self-resetting fuse, dialer circuit protector with modular plug and play.
- (4) For IP-DACTS: UL 497B listed, series connected, 6.4-volt, 1.5A rated with 20 kA/pair surge current, data network protector with modular plug and play.

2.11 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

2.11.1 Alarm Wiring

IDC and SLC wiring must be solid copper or stranded copper cable in accordance with the manufacturers requirements. Copper signaling line circuits and initiating device circuit field wiring must be No. 18 AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 16 AWG size conductors at a minimum. Speaker circuits must be copper No. 16 AWG size twisted and shielded conductors at a minimum. Wiring for textual notification appliance circuits must be in accordance with manufacturer's requirements but must be supervised by the FMCU. Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

2.12 AUTOMATIC FIRE ALARM TRANSMITTERS

2.12.1 Radio Transmitter and Interface Panels

Transmitters must be compatible with proprietary supervising station receiving equipment. Each radio alarm transmitter must be the manufacturer's recognized commercial product, completely assembled, wired, factory tested, and delivered ready for installation and operation. Transmitters must be provided in accordance with applicable portions of NFPA 72, Federal Communications Commission (FCC) 47 CFR 90 and Federal Communications Commission (FCC) 47 CFR 15. Transmitter electronics module must be contained within the physical housing as an integral, removable assembly. The proprietary supervising station receiving equipment is Monaco and the transmitter must be fully compatible with this equipment. At the contractors option, and if listed, the transmitter may be housed in the same control unit as the FMCU. The transmitter must be narrowband radio, with FCC certification for narrowband operation and meets the requirements of the NTIA (National Telecommunications and Information Administration) Manual of Regulations and Procedures for Federal Frequency Management.

2.12.1.1 Operation

Operate each transmitter from 120-volt ac power. In the event of 120-volt ac power loss, the transmitter must automatically switch to battery operation. Switchover must be accomplished with no interruption of protective service, and must automatically transmit a trouble message. Upon restoration of ac power, transfer back to normal ac power supply must also be automatic.

2.12.1.2 Battery Power

Transmitter standby battery capacity must provide sufficient power to operate the transmitter in a normal standby status for a minimum of 72 hours and be capable of transmitting alarms during that period.

2.12.1.3 Transmitter Housing

Use NEMA Type 1 for housing. The housing must contain a lock that is keyed identical to the fire alarm system for the building. Radio alarm transmitter housing must be factory painted with a suitable priming coat and not less than two coats of a hard, durable weatherproof enamel.

2.12.1.4 Antenna

Antenna must be omnidirectional, coaxial, halfwave dipole antennas for radio alarm transmitters with a driving point impedance to match transmitter output. The antenna and antenna mounts must be corrosion resistant and designed to withstand wind velocities of 161 km/hour. Do not mount antennas to any portion of the building roofing system. Protect the antenna from physical damage.

2.12.2 Signals to Be Transmitted to the Base Receiving Station

The following signals must be sent to the base receiving station:

- a. Sprinkler waterflow
- b. Manual pull stations
- c. Smoke detectors
- d. Duct smoke detectors
- e. Fire extinguishing system
- f. Sprinkler valve supervision
- g. Combustion engine drive fire pump running
 - (1) Selector switch in position than automatic
 - (2) Engine over-speed
 - (3) Low fuel
 - (4) Low battery
 - (5) Engine trouble (for example, low oil, over temp)

2.13 ENVIRONMENTAL ENCLOSURES OR GUARDS

Environmental enclosures must be provided to permit fire alarm/mass notification components to be used in areas that exceed the environmental limits of the listing. The enclosure must be listed for the device or appliance as either a manufactured part number or as a listed compatible accessory for the component is currently listed. Guards required to deter mechanical damage must be either a listed manufactured part or a listed accessory for the category of the initiating device or notification appliance.

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

3.2.1 Battery Cabinets

When batteries will not fit in the FMCU, locate battery cabinets below or adjacent to the FMCU. Battery cabinets must be installed at an accessible location when standing at floor level. Battery cabinets must not be installed lower than 300 mm above finished floor, measured to the bottom of the cabinet, nor higher than 900 mm above the floor, measured to the top of the cabinet. Installing batteries above drop ceilings or in inaccessible locations is prohibited. Battery cabinets must be large enough to accommodate batteries and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be provided in a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface mounting provisions. The cabinet must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must indicate the control unit(s) the batteries power and must not be less than 25 mm high.

3.2.2 Manual Stations

Locate manual stations as required by NFPA 72 and as indicated on the drawings. Mount stations so they are located no farther than 1.5 m from the exit door they serve, measured horizontally. Manual stations must be mounted at 1219 mm measured to the operating handle.

3.2.3 Notification Appliances

- a. Locate notification appliance devices as required by NFPA 72 where indicated and to meet the intelligibility requirements. Where two or more visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring. Audible and visual notification appliances mounted in acoustical ceiling tiles must be centered in the tiles plus or minus 50 mm.

- b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.
- c. Speakers must not be located in close proximity to the FMCU or LOC so as to cause feedback when the microphone is in use.

3.2.4 Smoke and Heat Detectors

Locate detectors as required by NFPA 72 and their listing and as indicated on the drawings on a 100 mm mounting box. Install heat detectors not less than 100 mm from a side wall to the near edge. Heat detectors located on the wall must have the top of the detector at least 100 mm below the ceiling, but not more than 300 mm below the ceiling. Smoke detectors are permitted to be on the wall no lower than 300 mm from the ceiling with no minimum distance from the ceiling. In raised floor spaces, install the smoke detectors to protect 21 square meters per detector. Install smoke detectors no closer than 1 m from air handling supply diffusers. Detectors installed in acoustical ceiling tiles must be centered in the tiles plus or minus 50 mm.

3.3 SYSTEM FIELD WIRING

3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and professional manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

- a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote fire alarm/mass notification control units, initiating circuits, switches, relays and terminals.
- b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 200 mm by 200 mm. Only screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 50 mm lettering, on the front

of the terminal cabinet.

3.3.3 Alarm Wiring

- a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.
- b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the FMCU.
- c. T-tapping using screw terminal blocks is allowed for Class "B" signaling line circuits.
- d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.
- e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the Designated Fire Protection Engineer (DFPE).

3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 19 mm in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

- a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.
- b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.
- c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.
- d. Flexible metal conduit is permitted for initiating device circuits, notification appliance circuits and signaling line circuits 19 mm in length or less. Use liquid tight flexible metal conduit in damp and wet locations.
- e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.
- f. Exterior wall penetrations must be weathertight. Conduit must be

sealed to prevent the infiltration of moisture.

3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FMCU and the LOC must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FMCU, and remote FMCU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

3.4 DISCONNECTION AND REMOVAL OF EXISTING SYSTEM

Maintain existing fire alarm/mass notification equipment fully operational until the new equipment has been tested and accepted by the Contracting Officer. As new equipment is installed, label it "NOT IN SERVICE" until the new equipment is accepted. Once the new system is completed, tested, and accepted by the Government, it must be placed in service and connected to the supervising station. Remove tags from new equipment and tag the existing equipment "NOT IN SERVICE" until removed from the building.

- a. After acceptance of the new system by the Contracting Officer, remove existing equipment not connected to the new system, remove unused exposed conduit, and restore damaged surfaces. Remove the material from the site and dispose.
- b. Disconnect and remove the existing fire alarm/mass notification and smoke detection systems where indicated and elsewhere in the specification.
- c. Control units and fire alarm devices and appliances disconnected and removed must be turned over to the Contracting Officer.
- d. Properly dispose of fire alarm outlet and junction boxes, wiring, conduit, supports, and other such items.

3.5 CONNECTION OF NEW SYSTEM

The following new system connections must be made during the last phase of construction, at the beginning of the pre-Government tests. New system connections must include:

- a. Connection of new relays to existing magnetic door hold-open devices.
- b. Connection of new elevator recall relays to existing wiring and conduit.
- c. Connection of new system transmitter to existing installation fire reporting system.

Once these connections are made, system must be left energized. Report immediately to the Contracting Officer, coordination and field problems resulting from the connection of the above components.

3.6 PAINTING

- a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.
- b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 19 mm wide at 3-meter centers and at each side of a floor, wall, or ceiling penetration.
- c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

3.7 FIELD QUALITY CONTROL

3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level IV Fire Alarm Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, signaling line devices and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

- a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC), Voice Notification System Circuits (NAC Audio), and Signaling Line Circuits (SLC).
- b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.
- c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.
- d. Identify all test equipment and personnel required to perform each test including equipment necessary for smoke detector testing. The use of magnets is not permitted.
- e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.7.2 Pre-Government Testing

3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

- a. NFPA 72 Record of Completion.
- b. NFPA 72 Record of Inspection and Testing.
- c. Fire Alarm and Emergency Communication System Inspection and Testing Form.
- d. Audibility test results with marked-up test floor plans.
- e. Intelligibility test results with marked-up floor plans.
- f. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE) or Contracting Officer's Representative (COR). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system have been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.

- b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.
- d. Loop resistance test results.
- e. Complete program printout including input/output addresses.
- f. Copy of pre-Government Test Certificate, test procedures and completed test data forms.
- g. Audibility test results with marked-up floor plans.
- h. Intelligibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer, Qualified Fire Protection Engineer (QFPE). At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.8 MINIMUM SYSTEM TESTS

3.8.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

- a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests must be witnessed by the Contracting Officer and test results recorded for use at the final Government test.
- b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.
- c. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.
- d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.
- e. Carbon Monoxide Detector Tests: Carbon monoxide detectors must be tested in accordance with NFPA 72 and the manufacturer's recommended calibrated test method.
- f. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.
- g. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.

- h. Determine that the system is operable under trouble conditions as specified.
- i. Visually inspect wiring.
- j. Test the battery charger and batteries.
- k. Verify that software control and data files have been entered or programmed into the FMCU. Hard copy records of the software must be provided to the Contracting Officer.
- l. Verify that red-line drawings are accurate.
- m. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.
- n. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
- o. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors must be conducted using real smoke or the use of canned smoke which is permitted.
- p. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.
- q. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

3.8.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBa over ambient with a maximum of 110 dBa in any occupiable area. The provisions for audible notification (audibility and intelligibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

3.8.3 Intelligibility Tests

Intelligibility testing of the System must be accomplished in accordance with NFPA 72 for Voice Evacuation Systems, and ASA S3.2. Following are the specific requirements for intelligibility tests:

- a. Intelligibility Requirements: Verify intelligibility by measurement after installation.
- b. Ensure that a CIS value greater than the required minimum value is provided in each area where building occupants typically could be found. The minimum required value for CIS is .8. Rounding of values is permitted.
- c. Areas of the building provided with hard wall and ceiling surfaces (such as metal or concrete) that are found to cause excessive sound reflections may be permitted to have a CIS score less than the minimum required value if approved by the DFPE, and if building occupants in

these areas can determine that a voice signal is being broadcast and they must walk no more than 10 m to find a location with at least the minimum required CIS value within the same area.

- d. Areas of the building where occupants are not expected to be normally present are permitted to have a CIS score less than the minimum required value if personnel can determine that a voice signal is being broadcast and they must walk no more than 15 m to a location with at least the minimum required CIS value within the same area.
- e. Take measurements near the head level applicable for most personnel in the space under normal conditions (e.g., standing, sitting, sleeping, as appropriate).
- f. The distance the occupant must walk to the location meeting the minimum required CIS value must be measured on the floor or other walking surface as follows:
 - (1) Along the centerline of the natural path of travel, starting from any point subject to occupancy with less than the minimum required CIS value.
 - (2) Curving around any corners or obstructions, with a 300 mm clearance there from.
 - (3) Terminating directly below the location where the minimum required CIS value has been obtained.

Use commercially available test instrumentation to measure intelligibility as specified by NFPA 72 as applicable. Use the mean value of at least three readings to compute the intelligibility score at each test location.

3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

- a. The drawings must be prepared electronically and sized no less than the contract drawings.
- b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.
- c. Include a riser diagram and drawings showing the as-built location of devices and equipment.
- d. Provide Operation and Maintenance (O&M) Instructions.

In existing buildings, the transfer of devices from the existing system to the new system and the permission to begin demolition of the old fire alarm system will not be permitted until the as-built drawings and O&M manuals are received.

3.10 INSTRUCTION OF GOVERNMENT EMPLOYEES

3.10.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided. The instructor must train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. The instructor must be thoroughly familiar with all parts of this installation. The instructor must be trained in operating theory as well as in practical O&M work. Submit the instructors information and qualifications including the training history.

3.10.2 Required Instruction Time

Provide 8 hours of instruction after final acceptance of the system. The instruction must be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The training must allow for rescheduling for unforeseen maintenance and fire department responses.

3.10.3 Technical Training Manual

Provide, in manual format, lesson plans, operating instructions, maintenance procedures, and training data for the training courses. The operations training must familiarize designated government personnel with proper operation of the installed system. The maintenance training course must provide the designated government personnel adequate knowledge required to diagnose, repair, maintain, and expand functions inherent to the system.

3.11 EXTRA MATERIALS

3.11.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system must be available for a period of 10 years after the date of final acceptance of this work by the Contracting Officer. During the warranty period, the service technician must be on-site within 24 hours after notification. All repairs must be completed within 24 hours of arrival on-site.

During the warranty period, the installing fire alarm contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72 and the system manufacturers. Installing fire alarm contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

3.11.2 Spare Parts

Spare parts furnished must be directly interchangeable with the corresponding components of the installed systems. Spare parts must be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts must be delivered to the Contracting Officer at the time of the Government testing and must be accompanied by an inventory list.

3.11.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

- a. As-built shop drawings
- b. Product data sheets
- c. Design calculations
- d. Site-specific software data package
- e. All documentation required by SD-06.

-- End of Section --

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SECTION 31 00 00

EARTHWORK
08/08

PART 1 GENERAL

1.1 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1102	(2014) Method of Test for Sieve Analysis of Aggregates
JIS A 1103	(2014) Method of Test for Amount of Material Passing Test Sieve 75 μ m in Aggregates
JIS A 1205	(2009) Test Method for Liquid Limit and Plastic Limit of Soils
JIS A 1210	(2009) Test Method for Soil Compaction Using a Rammer
JIS A 1211	(2009) Test Methods for the California Bearing Ratio (CBR) of Soils in laboratory
JIS A 1214	(2013) Test Method for Soil Density by the Sand Replacement Method
JIS A 1218	(2013) Test Methods for Permeability of Saturated Soils
JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete

MINISTRY OF THE ENVIRONMENT GOVERNMENT OF JAPAN (MOE)

Notification No.46	(2001) Environmental Quality Standards for Soil Contamination
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JAPANESE GEOTECHNICAL SOCIETY (JGS)

JGS 0051	(2000) Method of Classification of Geomaterials for Engineering Purposes
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JGS 1614

(2003) Test Method for Soil Density Using
Nuclear Gauge

U.S. DEPARTMENT OF DEFENSE HEADQUARTERS, U.S. FORCES JAPAN (USFJ)

JEGS

(Apr 2024) Japan Environmental Governing
Standards

1.3 DEFINITIONS

1.3.1 Satisfactory Materials

Satisfactory materials comprise any materials classified by JGS 0051 as G, G-M, G-C, GM, GC, S, S-M, S-C, SM, SC, ML, MH, CL, CH. Satisfactory materials for grading comprise stones less than 200 mm, except for fill material for pavements and railroads which comprise stones less than 75 mm in any dimension.

1.3.2 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; and material classified as satisfactory which contains root and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.3.3 Cohesionless and Cohesive Materials

Cohesionless materials include materials classified in JGS 0051 as G, G-M, G-C, S, S-M, S-C. Cohesive materials include materials classified as GC, SC, ML, CL, MH, CH. Materials classified as GM, SM will be identified as cohesionless only when the fines are nonplastic. Perform testing, required for classifying materials, in accordance with JIS A 1205, JIS A 1102 and JIS A 1103.

1.3.4 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in JIS A 1210 (Methods C,D,E - non repeating method) abbreviated as a percent of laboratory maximum density (Modified Proctor Test).

1.3.5 Overhaul

Overhaul is the authorized transportation of satisfactory excavation or borrow materials in excess of the free-haul limit. Overhaul is the product of the quantity of materials hauled beyond the free-haul limit, and the distance such materials are hauled beyond the free-haul limit, expressed in station meters.

1.3.6 Topsoil

Material suitable for topsoils obtained from excavations is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than 25 mm diameter, brush, weeds, toxic substances, and other material detrimental to plant

growth. Amend topsoil pH range to obtain a pH of 5.5 to 7.

1.3.7 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 75 mm in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.3.8 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 0.375 cubic meter in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling that is performed merely to increase production.

1.3.9 Unstable Material

Unstable materials are too wet to properly support the utility pipe, conduit, or appurtenant structure.

1.3.10 Select Granular Material

1.3.10.1 General Requirements

Select granular material consist of materials classified as GW, GP, SW, or SP by JGS 0051 where indicated. The liquid limit of such material must not exceed 35 percent when tested in accordance with JIS A 1205. The plasticity index must not be greater than 12 percent when tested in accordance with JIS A 1205, and not more than 35 percent by weight may be finer than 75 micrometers sieve when tested in accordance with JIS A 1103.

Provide a minimum coefficient of permeability of 0.01 mm per second when tested in accordance with JIS A 1218.

1.3.10.2 California Bearing Ratio Values

Bearing Ratio: Provide Design CBR ratio in accordance with JIS A 1211 for a laboratory soaking period of not less than 4 days. Conform the combined material to the following sieve analysis:

Sieve Size	Percent Passing by Weight
2-1/2 inches	100
No. 4	40 - 85
No. 10	20 - 80
No. 40	10 - 60

Sieve Size	Percent Passing by Weight
No. 200	5 - 25

1.3.11 Initial Backfill Material

Initial backfill consists of select granular material or satisfactory materials free from rocks 300 mm or larger in any dimension or free from rocks of such size as recommended by the pipe manufacturer, whichever is smaller. When the pipe is coated or wrapped for corrosion protection, free the initial backfill material of stones larger than 150 mm in any dimension or as recommended by the pipe manufacturer, whichever is smaller.

1.3.12 Expansive Soils

Expansive soils are defined as soils that have a plasticity index equal to or greater than 20 when tested in accordance with JIS A 1205.

1.4 SYSTEM DESCRIPTION

A geotechnical report was not provided for this project. Data about the subsurface conditions of this project were based off of a geotechnical report provided by Environmental Science Corporation, dated 13 July 2015 under MCA PN 62783 SATCOM, Contract No. W912HV-13-R-0006. These data represent the best subsurface information available; however, variations may exist in the subsurface between boring locations.

1.4.1 Classification of Excavation

No consideration will be given to the nature of the materials, and all excavation will be designated as unclassified excavation.

1.4.1.1 Common Excavation

Include common excavation with the satisfactory removal and disposal of all materials not classified as rock excavation.

1.4.1.2 Rock Excavation

Submit notification of encountering rock in the project. Include rock excavation with excavating, grading, disposing of material classified as rock, and the satisfactory removal and disposal of boulders 1/2 cubic meter or more in volume; solid rock; rock material that is in ledges, bedded deposits, and unstratified masses, which cannot be removed without systematic drilling; firmly cemented conglomerate deposits possessing the characteristics of solid rock impossible to remove without systematic drilling; and hard materials (see Definitions). Include the removal of any concrete or masonry structures, except pavements, exceeding 1/2 cubic meter in volume that may be encountered in the work in this classification. If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock excavation, uncover such material and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation and has taken cross sections as required. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow ample time for classification and cross sectioning of the undisturbed surface of such material will cause the

forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

1.4.2 Dewatering Work Plan

Submit procedures for accomplishing dewatering work.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shoring; G

Dewatering Work Plan; G

SD-06 Test Reports

Testing

Borrow Site Testing

Within 24 hours of conclusion of physical tests, submit 3 copies of test results, including calibration curves and results of calibration tests.

PART 2 PRODUCTS

2.1 REQUIREMENTS FOR OFFSITE SOILS

Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Perform tests and verify threshold values in accordance with JAPAN Ministry of the Environment Notification No.46 guidelines and JEGS. Provide Borrow Site Testing for TPH, BTEX and TCLP from a composite sample of material from the borrow site, with at least one test from each borrow site. Do not bring material onsite until tests have been approved by the Contracting Officer.

2.2 BURIED WARNING AND IDENTIFICATION TAPE

Provide metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 75 mm minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems
Gray	Compressed Air

2.2.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.08 mm and a minimum strength of 10.3 MPa lengthwise, and 8.6 MPa crosswise, with a maximum 350 percent elongation.

2.2.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.10 mm, and a minimum strength of 10.3 MPa lengthwise and 8.6 MPa crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 920 mm deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.3 DETECTION WIRE FOR NON-METALLIC PIPING

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.4 MATERIAL FOR RIP-RAP

Provide Bedding material and rock conforming to these requirements for construction indicated.

2.4.1 Bedding Material

Provide bedding material consisting of sand, gravel, or crushed rock, well graded, with a maximum particle size of 50 mm. Compose material of tough, durable particles. Allow fines passing the 75 micrometers standard sieve with a plasticity index less than six.

2.4.2 Grout

Provide durable grout composed of cement, water, an air-entraining admixture, and sand mixed in proportions of one part portland cement to two parts of sand, sufficient water to produce a workable mixture, and an amount of admixture which will entrain sufficient air, as determined by

the Contracting Officer. Mix grout in a concrete mixer. Allow a sufficient mixing time to produce a mixture having a consistency permitting gravity flow into the interstices of the rip-rap with limited spading and brooming.

2.4.3 Rock

Provide rock fragments sufficiently durable to ensure permanence in the structure and the environment in which it is to be used. Use rock fragments free from cracks, seams, and other defects that would increase the risk of deterioration from natural causes. Provide fragments sized so that no individual fragment exceeds a weight of 68 kg and that no more than 10 percent of the mixture, by weight, consists of fragments weighing 0.91 kg or less each. Provide rock with a minimum specific gravity of 2.50. Do not permit the inclusion of more than trace 1 percent quantities of dirt, sand, clay, and rock fines.

2.5 CAPILLARY WATER BARRIER

Provide capillary water barrier of clean, poorly graded crushed rock, crushed gravel, or uncrushed gravel placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below. Conform to JIS A 5005 for fine aggregate grading with a maximum of 3 percent by weight passing JIS A 1103, 75 micrometers sieve, or 37.5 mm and no more than 2 percent by weight passing the 4.75 mm size sieve.

PART 3 EXECUTION

3.1 STRIPPING OF TOPSOIL

Where indicated or directed, strip topsoil to a depth of 100 mm. Spread topsoil on areas already graded and prepared for topsoil, or transported and deposited in stockpiles convenient to areas that are to receive application of the topsoil later, or at locations indicated or specified. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 50 mm in diameter, and other materials that would interfere with planting and maintenance operations. Remove from the site any surplus of topsoil from excavations and gradings.

3.2 GENERAL EXCAVATION

Perform excavation of every type of material encountered within the limits of the project to the lines, grades, and elevations indicated and as specified. Perform the grading in accordance with the typical sections shown and the tolerances specified in paragraph FINISHING. Transport satisfactory excavated materials and place in fill or embankment within the limits of the work. Excavate unsatisfactory materials encountered within the limits of the work below grade and replace with satisfactory materials as directed. Include such excavated material and the satisfactory material ordered as replacement in excavation. Dispose surplus satisfactory excavated material not required for fill or embankment in areas approved for surplus material storage or designated waste areas. Dispose unsatisfactory excavated material in designated waste or spoil areas. During construction, perform excavation and fill in a manner and sequence that will provide proper drainage at all times. Excavate material required for fill or embankment in excess of that produced by excavation within the grading limits from the borrow areas

indicated or from other approved areas selected by the Contractor as specified.

3.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Completely drain construction site during periods of construction to keep soil materials sufficiently dry. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity or provide temporary ditches, swales, and other drainage features and equipment as required to maintain dry soils. When unsuitable working platforms for equipment operation and unsuitable soil support for subsequent construction features develop, remove unsuitable material and provide new soil material as specified herein. It is the responsibility of the Contractor to assess the soil and ground water conditions presented by the plans and specifications and to employ necessary measures to permit construction to proceed.

3.2.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. Do not permit French drains, sumps, ditches or trenches within 0.9 m of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Take control measures by the time the excavation reaches the water level in order to maintain the integrity of the in situ material. While the excavation is open, maintain the water level continuously, at least 1 m below the working level.

3.2.3 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended in the manufacturer's printed installation manual. Provide vertical trench walls where no manufacturer's printed installation manual is available. Shore trench walls more than 1.2 meters high, cut back to a stable slope, or provide with equivalent means of protection for employees who may be exposed to moving ground or cave in. Shore vertical trench walls more than 1.2 meters high. Excavate trench walls which are cut back to at least the angle of repose of the soil. Give special attention to slopes which may be adversely affected by weather or moisture content. Do not exceed the trench width below the pipe top of 600 mm plus pipe outside diameter (O.D.) for pipes of less than 600 mm inside diameter, and do not exceed 900 mm plus pipe outside diameter for sizes larger than 600 mm inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures by the Contractor. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.2.3.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and

support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 75 mm or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.2.3.2 Removal of Unyielding Material

Where overdepth is not indicated and unyielding material is encountered in the bottom of the trench, remove such material 150 mm below the required grade and replaced with suitable materials as provided in paragraph BACKFILLING AND COMPACTION.

3.2.3.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with select granular material as provided in paragraph BACKFILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

3.2.4 Underground Utilities

The Contractor is responsible for movement of construction machinery and equipment over pipes and utilities during construction. Excavation made with power-driven equipment is not permitted within 600 mm of known Government-owned utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

3.2.5 Structural Excavation

Ensure that footing subgrades have been inspected and approved by the Contracting Officer prior to concrete placement. Excavate to bottom of pile cap prior to placing or driving piles, unless authorized otherwise by the Contracting Officer. Backfill and compact over excavations and changes in grade due to pile driving operations to 95 percent of JIS A 1210 maximum density.

3.3 SELECTION OF BORROW MATERIAL

Select borrow material to meet the requirements and conditions of the particular fill or embankment for which it is to be used. Obtain borrow material from the borrow areas or from approved private sources. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Borrow material from approved sources on Government-controlled land may be obtained without payment of royalties. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval. Consider necessary clearing, grubbing, and satisfactory drainage of borrow pits and

the disposal of debris thereon related operations to the borrow excavation.

3.4 OPENING AND DRAINAGE OF EXCAVATION AND BORROW PITS

Notify the Contracting Officer sufficiently in advance of the opening of any excavation or borrow pit or borrow areas to permit elevations and measurements of the undisturbed ground surface to be taken. Except as otherwise permitted, excavate borrow pits and other excavation areas providing adequate drainage. Transport overburden and other spoil material to designated spoil areas or otherwise dispose of as directed. Provide neatly trimmed and drained borrow pits after the excavation is completed. Ensure that excavation of any area, operation of borrow pits, or dumping of spoil material results in minimum detrimental effects on natural environmental conditions.

3.5 SHORING

3.5.1 General Requirements

Submit a Shoring and Sheet piling plan for approval 15 days prior to starting work. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Finish shoring, including sheet piling, and install as necessary to protect workmen, banks, adjacent paving, structures, and utilities. Remove shoring, bracing, and sheet piling as excavations are backfilled, in a manner to prevent caving.

3.5.2 Geotechnical Engineer

Hire a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for updating the excavation, sheet piling and dewatering plans as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly written report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contracting Officer is responsible for arranging meetings with the Geotechnical Engineer at any time throughout the contract duration.

3.6 GRADING AREAS

Where indicated, divide work into grading areas within which satisfactory excavated material will be placed in embankments, fills, and required backfills. Do not haul satisfactory material excavated in one grading area to another grading area except when so directed in writing. Place and grade stockpiles of satisfactory and wasted materials as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.7 FINAL GRADE OF SURFACES TO SUPPORT CONCRETE

Do not excavate to final grade until just before concrete is to be placed. Only use excavation methods that will leave the foundation rock in a solid and unshattered condition. Roughen the level surfaces, and cut the sloped surfaces, as indicated, into rough steps or benches to provide a satisfactory bond. Protect shales from slaking and all surfaces from erosion resulting from ponding or water flow.

3.8 GROUND SURFACE PREPARATION

3.8.1 General Requirements

Remove and replace unsatisfactory material with satisfactory materials, as directed by the Contracting Officer, in surfaces to receive fill or in excavated areas. Scarify the surface to a depth of 150 mm before the fill is started. Plow, step, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so that the fill material will bond with the existing material. When subgrades are less than the specified density, break up the ground surface to a minimum depth of 150 mm, pulverizing, and compacting to the specified density. When the subgrade is part fill and part excavation or natural ground, scarify the excavated or natural ground portion to a depth of 300 mm and compact it as specified for the adjacent fill.

3.9 UTILIZATION OF EXCAVATED MATERIALS

Dispose unsatisfactory materials removing from excavations into designated waste disposal or spoil areas. Use satisfactory material removed from excavations, insofar as practicable, in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes. Submit procedure and location for disposal of unused satisfactory material. Submit proposed source of borrow material. Do not waste any satisfactory excavated material without specific written authorization. Dispose of satisfactory material, authorized to be wasted, in designated areas approved for surplus material storage or designated waste areas as directed. Clear and grub newly designated waste areas on Government-controlled land before disposal of waste material thereon. Stockpile and use coarse rock from excavations for constructing slopes or embankments adjacent to streams, or sides and bottoms of channels and for protecting against erosion. Do not dispose excavated material to obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way.

3.10 BURIED TAPE AND DETECTION WIRE

3.10.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape 300 mm below finished grade; under pavements and slabs, bury tape 150 mm below top of subgrade.

3.10.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 300 mm above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 0.9 m of wire,

coiled, remaining accessible in each manhole. Furnish insulated wire over it's entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.11 BACKFILLING AND COMPACTION

Place backfill adjacent to any and all types of structures, in successive horizontal layers of loose materia not more than 200 mm in depth. Compact to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials in accordance with JIS A 1210, to prevent wedging action or eccentric loading upon or against the structure. Backfill material must be within the range of -2 to +2 percent of optimum moisture content at the time of compaction.

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs GROUND SURFACE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

3.11.1 Trench Backfill

Backfill trenches to the grade shown. Do not backfill the trench until all specified tests are performed.

3.11.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with select granular material or initial backfill material.

3.11.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with select granular material placed in layers not exceeding 150 mm loose thickness.

3.11.1.3 Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material and compact it with approved tampers to a height of at least 300 mm above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with as shown in design drawings, except as specified herein. Compact backfill to top of pipe to 90 percent of JIS A 1210 maximum density. Provide plastic piping with bedding to spring line of pipe. Provide materials as follows:

3.11.1.3.1 Class I

Angular, 6 to 40 mm, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

3.11.1.3.2 Class II

Coarse sands and gravels with maximum particle size of 40 mm, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types G, G-M, G-C, S, S-M, S-C are included in this class as specified in JGS 0051.

3.11.1.4 Final Backfill

Fill the remainder of the trench, except for special materials for roadways, railroads and airfields, with satisfactory material. Place backfill material and compact as follows:

3.11.1.4.1 Sidewalks, Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 300 mm loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils. Do not permit compaction by water flooding or jetting. Apply this requirement to all other areas not specifically designated above.

3.12 SPECIAL REQUIREMENTS

Special requirements for both excavation and backfill relating to the specific utilities are as follows:

3.12.1 Fire Lines

Excavate trenches to a depth that provides a minimum cover of 0.75 meters from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

3.13 SUBGRADE PREPARATION

3.13.1 Proof Rolling

Finish proof rolling on an exposed subgrade free of surface water (wet conditions resulting from rainfall) which would promote degradation of an otherwise acceptable subgrade. After stripping, proof roll the existing subgrade of the chiller housekeeping pad with six passes of a 13.6 meter tons, pneumatic-tired roller. Operate the roller in a systematic manner to ensure the number of passes over all areas, and at speeds between 4 to 5.5 km/hour. When proof rolling, provide one-half of the passes made with the roller in a direction perpendicular to the other passes. Notify the Contracting Officer a minimum of 3 days prior to proof rolling. Perform proof rolling in the presence of the Contracting Officer. Undercut rutting or pumping of material to a depth of 600 mm and replace with fill and backfill material.

3.13.2 Construction

Shape subgrade to line, grade, and cross section, and compact as specified. Include plowing, disking, and any moistening or aerating required to obtain specified compaction for this operation. Remove soft or otherwise unsatisfactory material and replace with satisfactory excavated material or other approved material as directed. Excavate rock encountered in the cut section to a depth of 150 mm below finished grade for the subgrade. Bring up low areas resulting from removal of unsatisfactory material or excavation of rock to required grade with

satisfactory materials, and shape the entire subgrade to line, grade, and cross section and compact as specified. After rolling, the surface of the subgrade shall not show deviations greater than 13 mm when tested with a 4 m straightedge applied both parallel and at right angles to the centerline of the area. Do not vary the elevation of the finish subgrade more than 15 mm from the established grade and cross section.

3.13.3 Compaction

Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

3.13.3.1 Subgrade for Pavements

Compact subgrade for pavements to at least 95 percentage laboratory maximum density in accordance with JIS A 1210 for the depth below the surface of the pavement shown. When more than one soil classification is present in the subgrade, thoroughly blend, reshape, and compact the top 300 mm of subgrade.

3.14 FINISHING

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 30 mm of the grades and elevations indicated except that the degree of finish for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed from settlement or washing to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.14.1 Subgrade and Embankments

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on the finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, or frozen subgrade.

3.14.2 Capillary Water Barrier

Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.14.3 Grading Around Structures

Construct areas within 1.5 m outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.15 PLACING TOPSOIL

On areas to receive topsoil, prepare the compacted subgrade soil to a 50 mm depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of 100 mm and grade to the elevations and slopes shown. Do not spread topsoil when frozen or excessively wet or dry. Obtain material required for topsoil in excess of that produced by excavation within the grading limits from areas indicated by base personnel.

3.16 TESTING

Perform testing by a Corps validated commercial testing laboratory or the Contractor's validated testing facility. Submit qualifications of the Corps validated commercial testing laboratory or the Contractor's validated testing facilities. If the Contractor elects to establish testing facilities, do not permit work requiring testing until the Contractor's facilities have been inspected, Corps validated and approved by the Contracting Officer.

- a. Determine field in-place density in accordance with JIS A 1214.
- b. When test results indicate, as determined by the Contracting Officer, that compaction is not as specified, remove the material, replace and recompact to meet specification requirements.
- c. Perform tests on recompacted areas to determine conformance with specification requirements. Appoint a registered professional civil engineer to certify inspections and test results. These certifications shall state that the tests and observations were performed by or under the direct supervision of the engineer and that the results are representative of the materials or conditions being certified by the tests. The following number of tests, if performed at the appropriate time, will be the minimum acceptable for each type operation.

3.16.1 Fill and Backfill Material Gradation

One test per 14 cubic meters stockpiled or in-place source material. Determine gradation of fill and backfill material in accordance with JIS A 1102.

3.16.2 In-Place Densities

- a. One test per 186 square meters, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines.
- b. One test per 186 square meters, or fraction thereof, of each lift of fill or backfill areas compacted by hand-operated machines.

3.16.3 Check Tests on In-Place Densities

If JGS 1614 is used, check in-place densities by JIS A 1214 as follows:

- a. One check test per lift for each 300 square meters, or fraction thereof, of each lift of fill or backfill compacted by other than hand-operated machines.
- b. One check test per lift for each 300 square meters, of fill or backfill areas compacted by hand-operated machines.

3.16.4 Moisture Contents

In the stockpile, excavation, or borrow areas, perform tests as required, to meet compaction requirements, per type of material or source of material being placed during stable weather conditions. During unstable weather, perform tests as dictated by local conditions and approved by the Contracting Officer.

3.16.5 Optimum Moisture and Laboratory Maximum Density

Perform tests for each type material or source of material including borrow material to determine the optimum moisture and laboratory maximum density values. One representative test per 150 cubic meters of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density.

3.16.6 Tolerance Tests for Subgrades

Perform continuous checks on the degree of finish specified in paragraph SUBGRADE PREPARATION during construction of the subgrades.

3.17 DISPOSITION OF SURPLUS MATERIAL

Remove surplus material or other soil material not required or suitable for filling or backfilling, and brush, refuse, stumps, roots, and timber from Government property and delivered to a licensed/permitted facility or to a location approved by the Contracting Officer.

-- End of Section --